

The Illustrated Encyclopedia of the World's

MODERN MILITARY AIRCRAFT

A technical directory of today's major warplanes



Bill Gunston

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The speed of the development of military aircraft over the past seventy-five years, and the complexity of the systems now in use, can justly claim to be among the most staggering of Man's technological achievements. From relatively crude beginnings, military aircraft have evolved into perhaps the most lethal yet beautiful machines in the world. This superb, colorful encyclopedia offers the reader in-depth studies and data for all the major combat aircraft in service today and vividly conveys the excitement aroused by these high performance, highly potent aircraft. Bill Gunston, an internationally respected authority, has written penetrating and readable analyses of the design, development and capabilities of each aircraft and its variants. His text is complemented by a multitude of photographs, detailed cutaway drawings, color profiles and precisely drawn three-view diagrams. Every entry is illustrated and the large number of full color action photographs – over 300 in all – make this the most visually impressive book on the subject ever to appear. The aircraft are conveniently sectionalised into major roles so that the warplanes of one nation can easily be compared with those of actual or potential adversaries. Not so long ago it was predicted that military aircraft would soon 'all look alike'. How far this judgement is at variance with the truth will soon become apparent to anyone merely glancing through the 256 pages of this absorbing and magnificently produced volume.

"We certainly know of no other single volume, even at twice the price of this one, that covers the subject so fully, authoritatively and attractively."
Air International, London

"First class value, and an excellent work of reference."
Royal Air Force News





Flame spurts from the 36-barrel pods as rockets ripple from a Buccaneer S.2 built for the Royal Navy and later transferred to the RAF (see page 73).

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Saab AJ37 Viggen

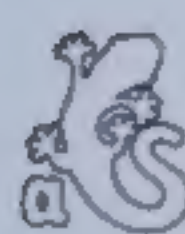
Launch of a Lockheed S-3A Viking from the foredeck catapult of the nuclear carrier USS *Enterprise* (see page 146).



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 a Salamander book

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Author's Acknowledgements

I am grateful to Salamander for the opportunity to do this book. There are many books—as there have been since about 1920—with titles similar to this; but I believe this is the first to have colour photography from beginning to end. This is partly the luck of timing, in that such pictures now exist. An equally important factor is the publisher's unrivalled expertise in managing to put together a package which is truly a Rolls-Royce, but at the price of a Mini.

As the author I would like to play down my own part. There is no great kudos in writing a book of this kind, though the grey stuff that goes round the illustrations certainly merits the space it occupies. The real effort lies in collecting the pictures. Editor Ray Bonds and his team have worked in a way that has never been surpassed in book production, and their success can be judged by simply looking at the book. I am sad that in a very few cases incessant bombardment by letter, Telex and telephone still failed to prise loose the pictures that had been promised months earlier.

It will be obvious that we have tended to choose dramatic pictures showing aircraft blasting off in fire and smoke. Such choices must always be a matter for subjective judgement. Some readers might prefer a pin-

sharp picture of the aircraft on the ground, while a few would prefer a set of engineering drawings. Our objective is to entertain, and give readers a feel for the subject, so that they can almost hear the thunder of afterburners and feel the g in a turn. And for those unfamiliar with g there is a glossary at the back.

Obviously there are several ways of arranging *Modern Military Aircraft*. The contents could be in rigorous alphabetical order, or arranged by country of origin. We have chosen to divide the contents into 'chapters' by function. The names of these duties—such as 'Fighter' or 'Attack/close support'—are not puzzling, but several aircraft were hard to fit into rigid slots. The Il-28, an old bomber, is today used chiefly as a trainer, while the main role of many A-3 Skywarriors is electronic warfare. Each basic type appears once only, so that all Canberras and B-57s appear together, as do the Intruders and Prowler. The main exception is the Boeing KC-135/707, whose complex descendent the E-3A AWACS insisted on separate treatment. For unavoidable reasons, dictated by the space needed for each entry, the alphabetical order within 'chapters' is not absolutely rigid.

Bill Gunston





Contents

Fighters	6
Attack/close support aircraft	52
Bombers	108
Reconnaissance aircraft	118
Anti-submarine and maritime patrol	130

Transports and tankers	150
Electronic warfare aircraft	180
Helicopters	192
Trainers and utility aircraft	224
Glossary of terms	253
Aircraft designations	253
Index	254
Picture credits	256

A Japanese camera records a nice left turn by two Mitsubishi T-2 supersonic trainers (see page 87).

Fighters

Before World War I aeroplanes had flown carrying manually aimed machine guns and even large-calibre automatic cannon, but the task of destroying hostile aircraft was soon almost universally assigned to fixed guns firing directly ahead, aimed by aiming the whole fighter at the enemy. From 1916 until 30 years later development was merely of degree, with guns increasing in calibre and lethality, sight systems improving fantastically, and aircraft performance and protection reflecting the fast progress in engines and structures. The only completely new factor was airborne radar, which enabled interceptions to be made in the dark or in bad weather, and temporarily introduced a special class of fighter characterized by larger size and extra crew-members. Jet propulsion gradually opened up the boundaries of flight performance and completely changed the shape of fighters, while the US Air Force led the world in developing advanced interception systems in 1948–52 which almost relegated the pilot to the role of passive machine-minder who just went along for the ride. Some suggested a fighter could have higher performance if the pilot stayed on the ground. Exasperated pilots in Korea, shaken by a totally uncomplicated fighter called a MiG-15, called for smaller and simpler fighters that would stay serviceable, climb above the enemy, fly faster and turn tighter, and thus put them back on top. The technological marvels were derided; instead of the A-1CM radar-ranged gunsight, seasoned fighter "jocks" called for a new invention called a gumsight: a piece of chewing gum fixed exactly in their forward line of sight on the windscreen. In April 1957 Britain went even further: its official view was that all fighters were obsolete.

Fighter development thus progressed in massive leaps, mistaken lurches and steps to the side or backwards. The result of the Korean experience was the F-104 Starfighter, which Lockheed called "the missile with a man in it". The Europeans went even beyond this with a flock of "light fighters" which crashed, faded entirely or were turned into light attack aircraft. The Starfighter likewise turned into an attack/reconnaissance aircraft—though the last

version of all was a fighter (F-104S), but not the kind envisaged when the F-104 was designed. By 1960, partly because of the tremendous all-round capability of the F-4 Phantom II, the wheel had turned full circle and, in contrast to light and simple fighters, the cry was for bigger and more complicated ones. The Phantom did not appeal to poorer or younger air forces, which queued up to buy Dassault's much cheaper Mirage, which looked good and went almost as fast. The Mirage also sold to some air forces that had a real job to do, notably to Israel, where the skies were blue and there was no need for long range, or for long-range weapons. Those that looked to the United States bought equally large numbers of F-5 "Freedom Fighters" for the same reasons. But the United States itself



had no interest in such aircraft; after the "TFX" (F-111) had demonstrated that, despite its unfortunate designation, this 50-ton monster was not a fighter at all, the standard fighter became the Phantom, which poured off the assembly line at St Louis in ever-increasing numbers. Then came Vietnam, where no American pilot was permitted to fire at another aircraft until he had "positive visual identification". This meant the big Phantom had to be brought within Sparrow missile range, within Sidewinder missile range, and right close to the streaking enemy. A dogfight might ensue, but the Phantom was not designed for dogfighting, and tended to stall and spin; and it had no gun. Eventually the Phantom grew a slatted wing and an internal gun, but by this time a new generation of jocks had grown

up that longed for the same old things: more climb, more altitude, more speed, tighter turns. This time the un-interceptable "enemy" was the MiG-25.

This time these needs were not confused by a wish to do away with technology, because the modern technology was reliable. But a deadly enemy, cost inflation, was making it almost impossible to do anything. One of the new fighters was the subject of a dispute between its builder and its first customer because the builder said he could not go on supplying at the price of \$16,500,000 per aircraft, and had already lost \$135 million on the first four batches (and that was a fighter developed before inflation really began to bite). This profoundly affected all Western defence, and one of its results was yet another American lurch in the direction of small and cheap fighters, leading to today's F-16. This at least appears to combine the qualities needed in a modern fighter: large SEP (specific excess power, a measure of surplus energy available for climb or manoeuvre), high T/W (thrust/weight ratio of considerably better than unity), low W/S (wing loading) obtained by reducing weight and increasing the size of the wing, and achieving the best possible flight-control system by a complex interaction of aerodynamics, system engineering, electronics and various interfaces with other systems in the aircraft and the human pilot. The F-16 has a gun, though quite an old one, and carries medium-range and short-range missiles (again old, but improved over the years).

Both the big F-15 Eagle and the smaller F-16 were planned as uncompromised fighters, to intercept and destroy hostile aircraft. Both eventually sprouted varied burdens of ordnance to fly attack missions. This role-change is hard to accomplish successfully. Two European aircraft, the Viggen and Tornado, were planned primarily for attack missions and are making the reverse transformation into fighters. The Tornado has the benefit of a swing-wing, like the multi-role F-14 Tomcat. Another swing-winger is the MiG-23, but in this case there are quite dissimilar attack and fighter versions. Designing fighters is quite difficult enough, without continuous pressure from the Soviet Union.



The pilot of a J35A Draken of the Flygvapen's F13 wing at Norrköping watches as his mate breaks away, showing the dramatic plan-shape and four Rb 324 missiles. The Draken, of which over 600 were built, was a programme of conspicuous success.



BAC Lightning

Lightning F.1 to 6 and export versions (data for F.6)

Origin
Type
Engines: Two 15 686lb (7112kg) thrust Rolls Royce Avon 302 augmented

Dimensions: span 34ft 10in (10.6m) length 55ft 3in (16.25m), height

Weights: Empty about 28,000lb

Performance: Maximum speed

Armament
Firestreak guided missiles, option of two 30mm Aden cannon in

History: First flight (P.1B) 4 April 1957, (first production F.1) 30 October 1959 (first F.6) 17 April 1964
Users: Kuwait, Saudi Arabia, UK

Development: As he had been with the Canberra, "Teddy" Petter was again moving spirit behind the award in 1947 of a study contract for a supersonic research aircraft. Later this was built and flown as the P.1 of August 1954 exceeding Mach 1 on two crude unaugmented Sapphire engines mounted one above and behind the other and fed by a plain nose inlet. In mid-1949 specification F.23/49 was issued for a supersonic fighter, and after complete redesign the P.1B was produced and flown in 1957. This had a new fuselage with a two-shock intake, the central cone being intended to house Ferranti Airpass radar. The Avon engines were fitted with primitive afterburning allowing a speed of Mach 2 to be attained on 25 November 1958. Helped by 20 pre-production aircraft, the Lightning F.1 was cleared for service in 1960. Though relatively complicated, so that the flying rate and maintenance burden were terrible in comparison with more modern aircraft, these supersonic all-weather interceptors at last gave the RAF a modern fighter with radar, guided missiles (heat-homing Firestreaks) and supersonic performance. Production was held back by the belief that all manned fighters

set forth in the Defence White Paper of April 1957) improved F.2 to be built in rather navigation. Eventually...
...but it was decided to fit...
...30mm Aden cannon. Finally in 1965...
...follow the advice of BAC and almost...
...fit the kinked and cambered wing (first...
...at much increased weights. The T.4...
...trainers equivalent to the F.2 and F.3. For...
...Alpha and Kuwait. BAC paid for development of the Lightning as a...
...interceptor and attack aircraft added F.7 to the production total to



Above: Lightning F.6, with upper side elevation showing F.1.



Left: Lightning F.6 as originally in service with No 5 Sqn, RAF, showing unpainted finish and Firestreak missiles. The flight-refuelling probe can be seen attached beneath the left wing. It does not impair the Lightning's dogfighting capability.

Right and below: The Lightning F.2A is a rebuild of the early Mk 2 almost up to F.6 standard. Both the 92 Sqn aircraft (right) and the 19 Sqn example (below, at Warton) are shown in the colour-scheme of early 1977, when both squadrons were disbanded.



Dassault Breguet Mirage 2000

Mirage 2000 (single- and two-seat versions)

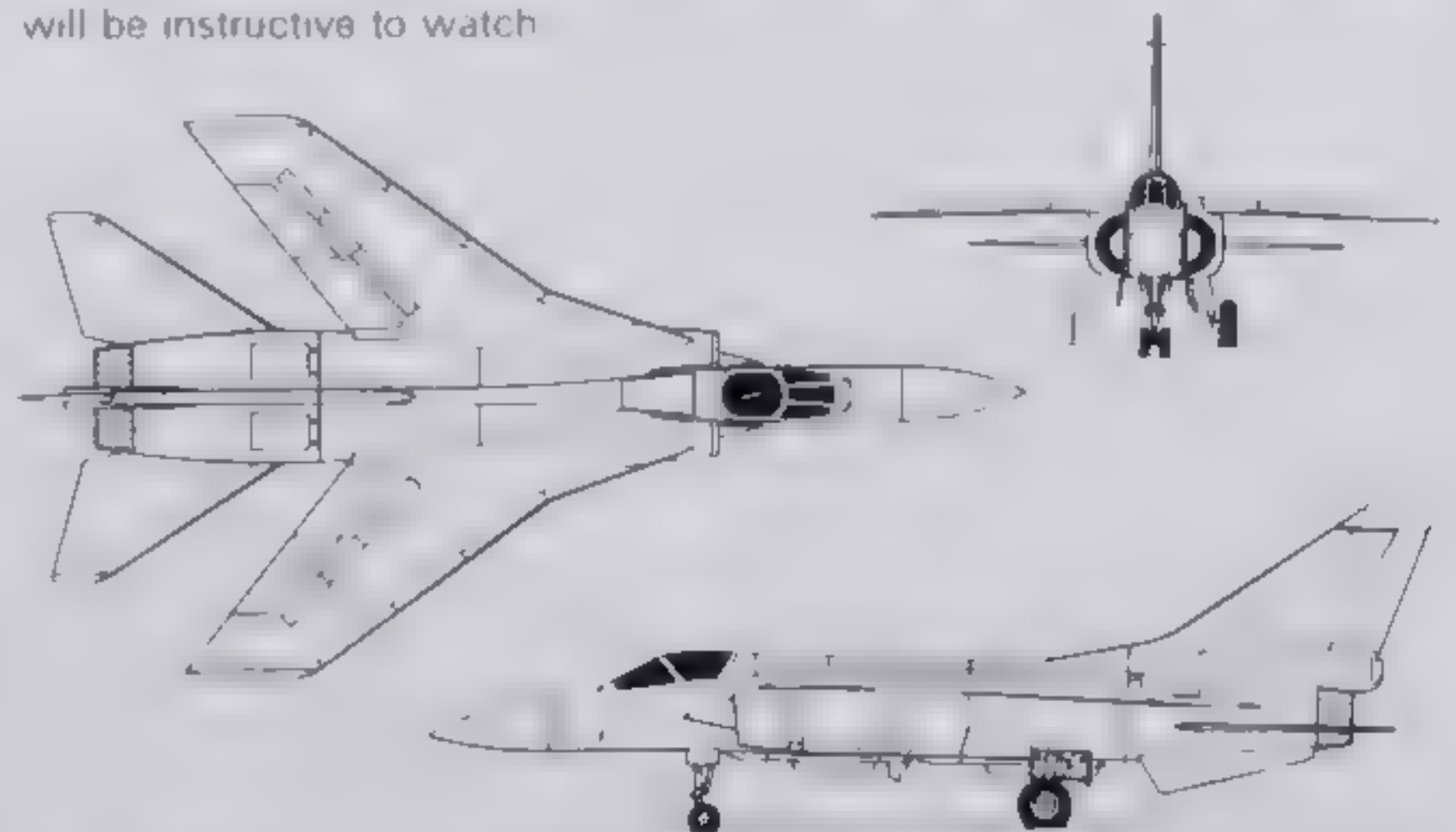
Origin: Avions Marcel Dassault/Breguet Aviation, France
Type: Originally intended to be interceptor and air-superiority fighter
Engine: One SNECMA M53 single-shaft afterburning turbofan modified in unspecified ways to achieve maximum thrust of at least 20,500lb (9300kg) dry and 25,000lb (11,340kg) with afterburner
Dimensions: Not available
Weights: Not available, but in view of manufacturer's statements loaded weight must be 24,000lb (10,886kg) or less
Performance: Maximum speed at height 1,780mph (2870km/h, Mach 2.7), time to reach 40,000ft (12,200m) from brakes-release 1min 15sec, service ceiling 65,600ft (20,000m), radius or ferry range, not available
Armament: Not disclosed, but will include further developments of the Matra Super 530 medium-range and Matra 550 Magic short range air-to-air missiles
History: Announcement of study project December 1975, first flight (prototype) 1978, available production aircraft the beginning of the 1980s
User: France (Armée de l'Air)

Development: In December 1975 the French government cancelled the Dassault-Breguet Super Mirage, which had been publicised as the Avion de Combat Futur and mainstay of the Armée de l'Air in the 1980s. In its place it announced a decision to award a study contract with Marcel Dassault for a smaller and simpler single-engined delta fighter outwardly looking very much like the Mirage III of 20 years earlier. In fact the Mirage 2000 – sometimes called the Delta 2000 – will differ significantly from the old Mirage, in aerodynamics, propulsion, structure and equipment. Aerodynamically it will be designed to incorporate American discoveries in CCV (control-configured vehicle) technology, in which aircraft are deliberately made unstable – for example, by positioning the centre of gravity much further back than usual – and using high-authority fail-safe flight-control systems to keep them under control. The result is either a smaller wing or, as in the Mirage 2000, dramatically higher manoeuvrability. Unlike the earlier Mirage deltas the 2000 will have leading-edge devices, either hinged droops or some form of slats, which will work in conjunction with the trailing-edge elevons to counteract the unstable pitching moment, or, in a tight turn, relax their effort or even help the aircraft to pitch nose-up. In the landing configuration the leading-edge devices (the French call it a "variable-camber" wing) will allow the elevons to be deflected down, adding to lift, whereas in earlier tailless deltas they have to be deflected up, effectively adding to weight just at the worst time.

Already the Mirage 2000 is being publicised as "being able to outclass combat aircraft presently being developed and produced in the Western world". It will have "fly-by-wire" multi-channel electrically signalled flight controls; composite materials, carbon fibre being mentioned, large-radius Karman fairings (a reference to area ruling of the fuselage for minimum transonic drag), an elaborate weapon system with "g.p." (general-purpose?) computer and inertial unit, and long-range digital radar. Ratio of thrust to weight is to exceed unity. Such features are what one would expect of such an aircraft, but the problems are clearly enormous, especially in a time of severe inflation and economic pressures. France has since 1975 made attempts to acquire the base of technology, especially in digital avionics, necessary to build the Mirage 2000, but has little capability as yet. Only a single French aircraft, a two-seat Mirage IIIB with Sifera system, has flown with a primitive fly-by-wire system. Thomson-CSF estimate it will take "seven to eight years" to develop a 170-km-range digital radar needed to match the developed Super 530 missile. France has little experience of advanced composite structures, and that only in small test pieces and heli-

opters. SNECMA has not announced how the M53 engine, which has limited flight-time and no other application, is going to be increased in thrust by 35 per cent. If the aircraft to fly in 1978 is truly a prototype, and not the first off a production line, it will need everything to go right to meet a in-service date of 1982 with a developed aircraft. Not least the price of Fr40 to 50 million (£4.5 to 5.5 million) will be extremely difficult to hold even in December 1975 France, because the magnitude of the development problems to France appear to have been grossly underestimated.

In the original announcement the Mirage 2000 was described as "limited to high-speed and high-level interception and reconnaissance. Attack and penetration at low levels will be undertaken by a different type" (The cancelled Super Mirage had been intended to fulfil all tactical roles.) But in December 1976 the Chief of Staff of the Armée de l'Air said he personally considered it would be necessary to build an interceptor and reconnaissance (he implied at low level) version of the Mirage 2000. It became known at this time that the new delta will apparently have nine weapon stations – which is diametrically opposed to the uncompromised high-altitude dog-fight concept announced in December 1975, and low-level use is diametrically opposed to a large area delta. The Armée de l'Air has from the start hoped to buy 200 Mirage 2000s, twice the number it judged it could afford of the Super Mirage. But future progress of the programme, helped by US industry strictly on an inter-company rather than a government basis, will be instructive to watch.



Three-view of the original Super Mirage, abandoned in 1976.

Dassault-Breguet Delta Super Mirage

Type: Long-range multi-role combat aircraft
Engines: Two SNECMA M53 turbofans each presumably to be rated at about 25,000lb (11,340kg) thrust
In January 1976 Marcel Dassault announced that, as a private venture, he was launching the Delta Super Mirage as a long-range multi-role aircraft for export. One hesitates to doubt the credibility of either the man or the company, but to fund such a programme would need many times the net worth of the company, and no consortium of overseas buyers (South Africans? Arabs? Black Africans?) appears to be conceivable. It would not be impossible for the company to finish the defunct tailed Super Mirage prototype, which was to have flown in July 1976, as an empty shell to show possible customers what the proposed Delta Super Mirage would look like. To develop it as an operational aircraft does not by any stretch of the imagination appear possible. One is left to conclude that M. Dassault either expects the French government to find the money, which is extremely unlikely, or he hopes to organise a programme involving a large number of nations prepared to share the costs and risks.

Below: Artist's impression of the Mirage 2000 in Armée de l'Air service. It will have a blended fuselage and wing, the latter having full-span elevons and leading-edge droops.



Dassault Breguet Mirage F1

Mirage F1.C

Origin The F1.C is a single-seat multi-mission fighter, designed and developed by Armaments Development and Production Corporation.

Type: Single-seat multi-mission fighter
Engine: (F1.C) 15,873 lb (7,200 kg) thrust (maximum afterburner) SNECMA Atar 9K-50 single-shaft augmented turbojet, (F1.E) 18,740 lb (8,500 kg) thrust (maximum afterburner) SNECMA M53-02 single-shaft augmented by pass turbojet

Dimensions: Span 27 ft 6 1/2 in (8.4 m), length (F1.C) 49 ft 2 1/2 in (15 m), (F1.E) 50 ft 11 in (15.53 m), height (F1.C) 14 ft 9 in (4.5 m), (F1.E) 14 ft 10 1/2 in (4.58 m)

Weights: Empty (F1.C) 16,314 lb (7,400 kg), (F1.E) 17,857 lb (8,100 kg)
Loaded (clean) (F1.C) 24,030 lb (10,900 kg), (F1.E) 25,450 lb (11,540 kg)
Maximum (F1.C) 32,850 lb (14,900 kg), (F1.E) 33,510 lb (15,200 kg)

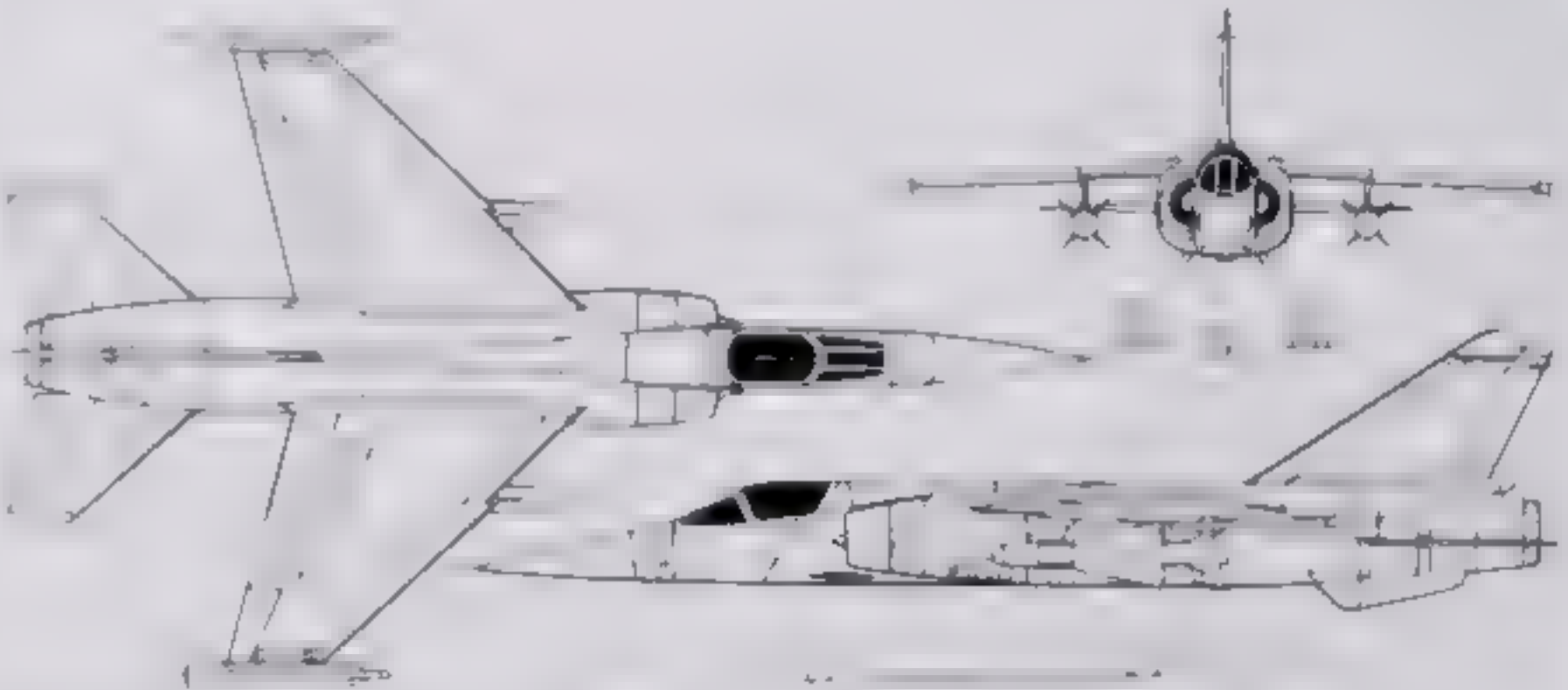
Performance: Maximum speed (clean, both versions) 915 mph (1472 km/h) (Mach 1.2) at sea level, 1,450 mph (2335 km/h) (Mach 2.2) at altitude (with modification to cockpit transparency and airframe leading edges F1.E) rate of climb (sustained to Mach 2 at 33,000 ft) (F1.C) 17,835 ft (12,780-14,580 m)/min (F1.E) above 59,000 ft (18,000 m) ceiling (F1.C) 65,600 ft (20,000 m), (F1.E) 69,750 ft (21,250 m) maximum weapons (hi to hi) (F1.C) 560 miles (900 km), (F1.E) 1,000 miles (1,600 km), ferry range (F1.C) 2,050 miles (3,300 km)

(F1.E) 2,340 miles (3,765 km)
Armament: 135 rounds five Aka universal stores pylons, rated at 4,500 lb (2,000 kg) each, 2,800 lb (1,350 kg) inner and 1,100 lb (500 kg) outer launch rails on tips rated at 280 lb (120 kg) for air-to-air missiles total 4,000 lb (4,000 kg) Typical air combat weapons, two Matra 550 Magic or Sidewinder on tips for close combat, one/two Matra 530 with infrared or radar homing and one/two Matra Super 530 for long range homing with large changes in height. Wide range of weapons for surface attack, plus optional reconnaissance pod containing cameras, SAT Cyclope infrared linescan and EMI side looking radar.

History: First flight (F1-01) 23 December 1966 (pre production F1-02) 20 March 1969 (production F1.C) 15 February 1973 (F1-M53, prototype for proposed F1.E) 22 December 1974, (F1.B trainer) 26 May 1976 service delivery (F1.C) 14 March 1973

Users: Egypt/Saudi Arabia (?), France, Greece, Iraq (?), Kuwait, Libya, Morocco (?), S Africa, Spain (?) = no firm contract announced at time of writing

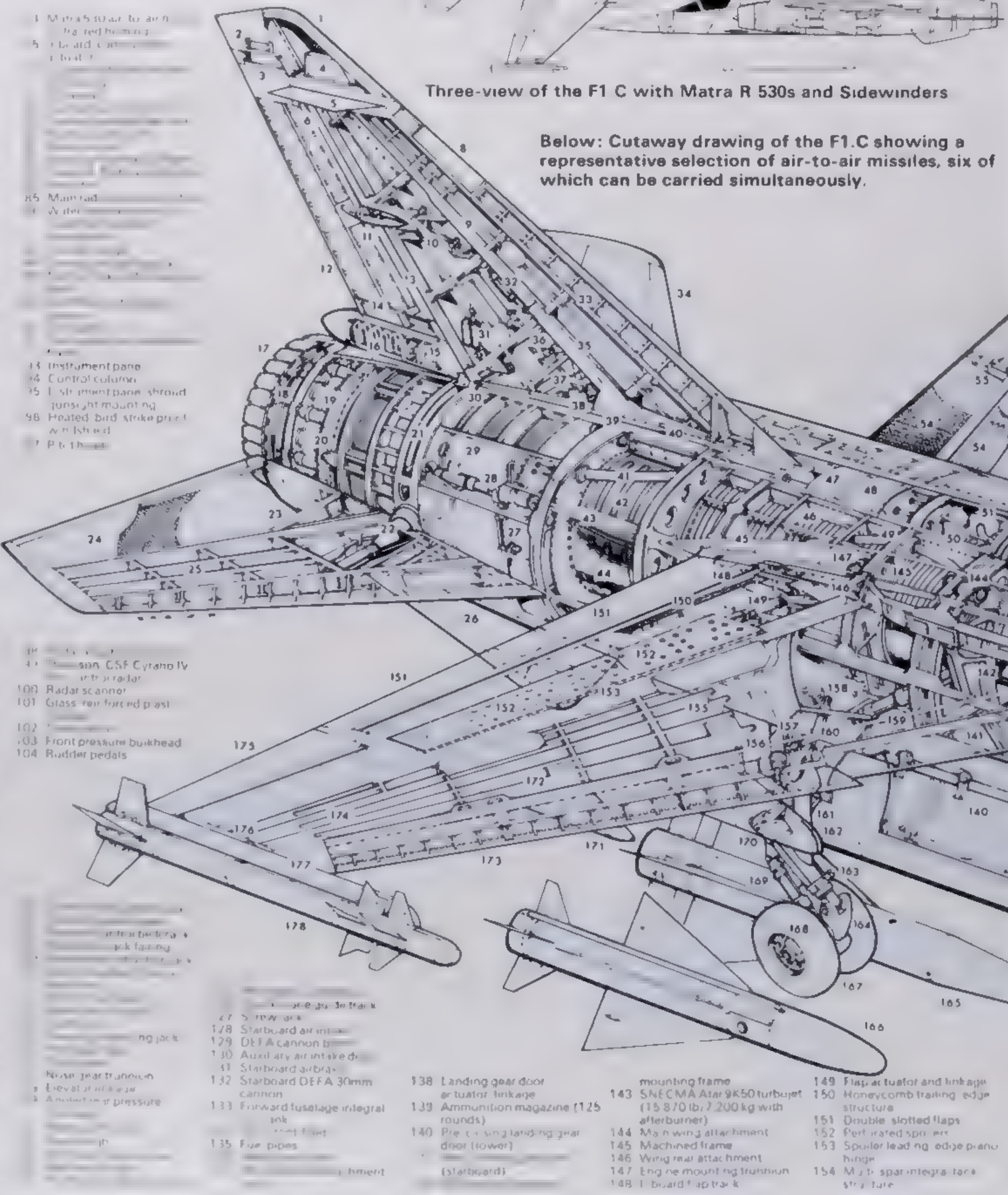
continued on page 12



Three-view of the F1.C with Matra R 530s and Sidewinders

Below: Cutaway drawing of the F1.C showing a representative selection of air-to-air missiles, six of which can be carried simultaneously.

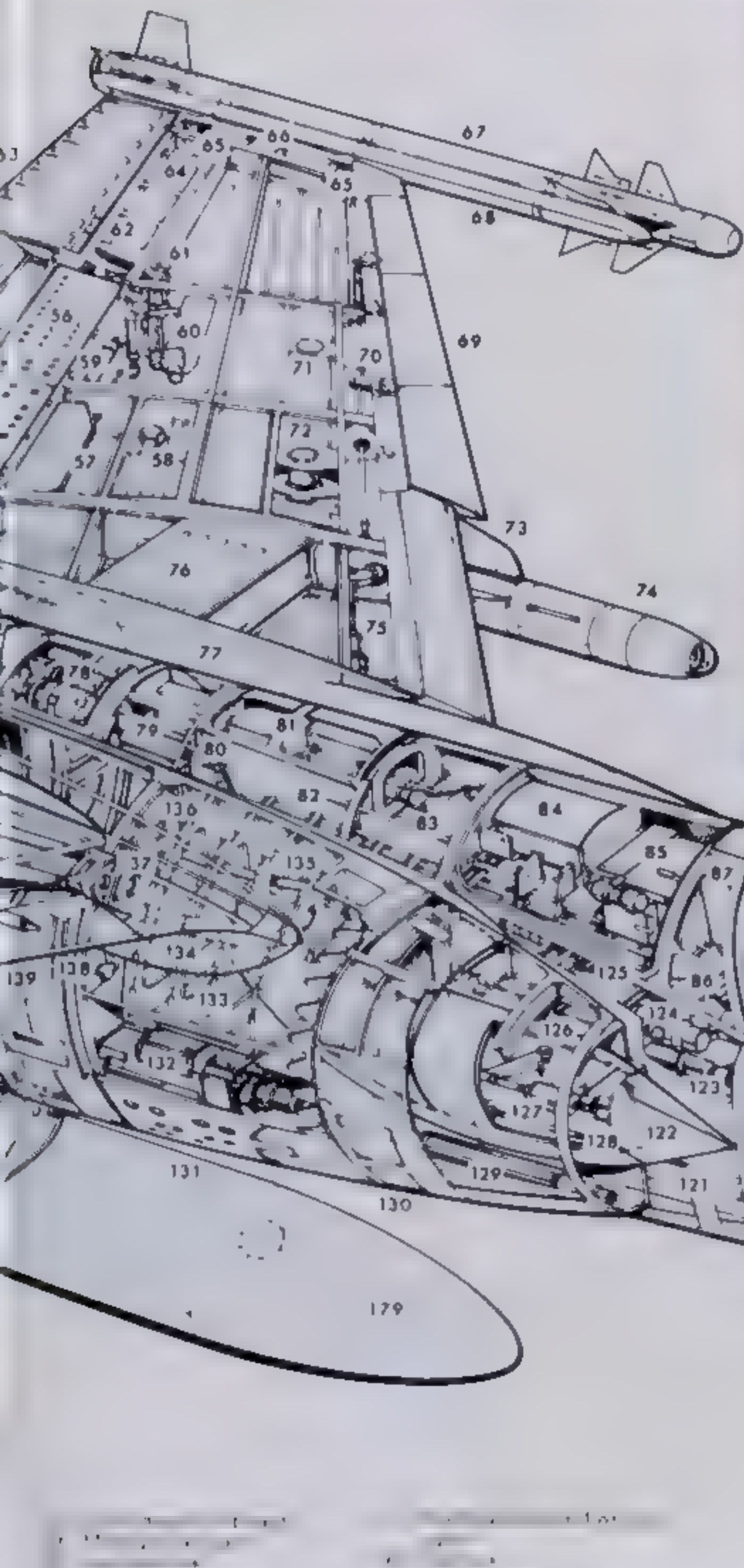
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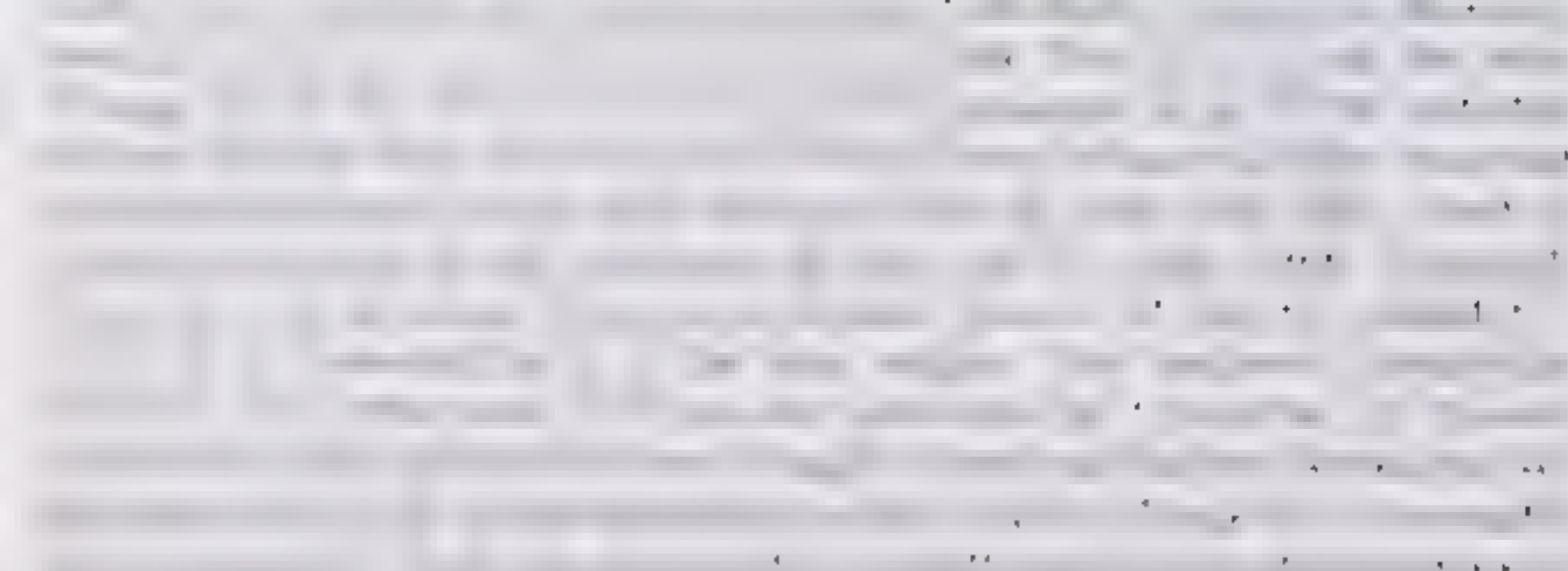
Left: This tandem-seater is the prototype F1.B, first flown on 26 May 1976. It has dual controls and full weapons capability. Most F1 customers have chosen to buy a small number of two-seaters for conversion and air-combat training. In addition to the choice of one or two seats customers can have simplified or augmented mission avionics as in the earlier delta Mirage III and 5



Above: An F1.C fighter in service with the 30th Wing.



Above: Mirage F1.C aircraft were first delivered in 1973 to the 30th wing of the Armée de l'Air at Reims. Aircraft from two escadrilles of this wing are seen here, in their combat finish of glossy grey. Today a further two squadrons of F1.Cs are operational in the 5th Wing (Escadrille) at Orange and No 12 at Cambrai and 10 at Creil are converting.



Thanks to the far higher efficiency of the new wing the field lengths and take-off and landing speeds are lower than for the delta Mirages, even though the weights are greater and the wing area much less. Increased thrust comes from the latest Atar engine and among the many less obvious advances are the Cyrano IV multi-mode radar and integral tankage for 15 per cent more fuel (trebling patrol endurance and doubling ground attack mission radius). Combat manoeuvrability in many situations was increased by as much as 80 per cent and the all-round performance of the new fighter was outstanding. Sales to Israel were prohibited, but orders were soon placed by South Africa and Spain, the former also buying a manufacturing licence. More recently the F1 was chosen by several Middle East countries and many more sales seem certain.

In 1967 the French engine company, SNECMA, began the design of a completely new engine for the Super Mirage. To test the engine the F1 was an obvious choice and the combination could not fail to be of interest to its own right. The M53 engine confers benefits in acceleration, climb, manoeuvrability and range and, to make up a more modern package, Dassault-Breguet proposed the fully modular Cyrano IV 100 radar and the SAGEM Kearfott SKN 2603 inertial navigation system, as well as the SFENA 505 digital autopilot of the F1C. The result is the F1E, which from early 1974 was strongly, but unsuccessfully, pressed on overseas customers, particularly Belgium, the Netherlands, Denmark and Norway (which agreed a common objective in replacing their F-104Gs). The Armée de l'Air did not want the F1E, but had agreed to buy a limited quantity had it been chosen by the four NATO nations. Two M53-powered prototypes were flown, but the M53-engined version was shelved in 1975. Today four versions are in production: (C) the basic aircraft, so far chosen by all customers, (E) the C with more advanced avionics (no longer offered with the M53 engine), chosen by Libya, (A) simplified avionics for low-level attack for Libya and South Africa, (B) two-seater, for Kuwait and Libya.



Top: The Ejército del Aire (Spanish Air Force) calls the Mirage F1C the C-14. The Spanish CASA company shares in some Dassault-Breguet manufacturing programmes
Above: Another F1C customer is the Hellenic (Greek) Air Force

Dassault Breguet Mirage III and 5

Mirage III and 5

Origin: Avior: Marcel Dassault/Breguet Aviation, France (actual manufacturing: 11 countries, 100,000 assembled)

Type: Single or two seat interceptor, tactical strike, trainer or reconnaissance aircraft (depending on sub-type)

Engine: (IIIC) 13,225lb (6,000kg) thrust (maximum afterburner) SNECMA Atar 9B single-shaft turbojet, (most other III and some 5) 13,670lb (6,200kg) Atar 9C, (recent III and some 5) 15,873lb (7,200kg) Atar 9K-50, (Kfir see separate entry)

Dimensions: Span 27ft (8.22m), length (IIIC) 50ft 10½in (15.5m) (IIIB) 50ft 6½in (15.4m), (5) 51ft (15.55m), height 13ft 11½in (4.25m)

Weights: Empty (IIIC) 13,570lb (6,156kg), (IIIE) 15,540lb (7,050kg) (IIIR) 14,550lb (6,600kg) (IIIB) 13,820lb (6,270kg), (5) 14,550lb (6,600kg) loaded (IIIC) 19,700lb (8,936kg), (IIIE, IIIR, 5) 29,760lb (13,500kg), (IIIB, 26,455lb (12,000kg)

Performance: Maximum speed (all models, clean) 863mph (1,390km/h) (Mach 1.14) at sea level, 1,460mph (2,350km/h) (Mach 2.2) at altitude initial climb, over 16,400ft (5,000m)/min (time to 36,090ft 11,000m, 3 min) service ceiling (Mach 1.8) 55,775ft (17,000m), range (clean) at altitude about 1,000 miles (1,610km), combat radius in attack mission with bombs and tanks (mix not specified) 745 miles (1,200km), ferry range with three external tanks 2,485 miles (4,000km)

Armament: Two 30mm DEFA 5-52 cannon, each with 125 rounds (normally fitted to all versions except when IIIC carries rocket-boost pack) three 1,000lb (454kg) external pylons for bombs, missiles or tanks (Mirage 5 seven external pylons with maximum capacity of 9,260lb, 4,200kg)

History: First flight (MD 550 Mirage I) 25 June 1955, (prototype Mirage III-001) 17 November 1956, (pre-production Mirage IIIA) 12 May 1958 (production IIIC) 9 October 1960, (IIIE) 5 April 1961, (IIIR) 31 October 1961, (IIIB) 19 July 1962, (Australian-assembled IIIO) 16 November 1963 (Swiss-assembled IIIS) 28 October 1965, (prototype 5) 19 May 1967, (Belgian-assembled 5BA) May 1970

Users: (III) Abu Dhabi, Argentina, Australia, Brazil, Egypt, France, Israel, Lebanon, Libya, Pakistan, S. Africa, Spain, Switzerland, Venezuela, (5) Abu Dhabi, Belgium, Colombia, Egypt, France, Gabon, Libya, Pakistan, Peru, Saudi Arabia, Venezuela, Zaïre

Development: The Mirage, which has come to symbolise modern aerial combat and to bring additional trade to France and incalculable prestige, especially in defence hardware, began in a most uncertain fashion. It was conceived in parallel with the Etendard II to meet the same Armée de l'Air light interceptor specification of 1952 and was likewise to be powered by two small turbojets (but, in this case, boosted by a liquid-propellant rocket engine in addition). As the small French engines were not ready, Dassault fitted the Mirage I with two British Viper turbojets and before the rocket was fitted this small delta was dived to Mach 1.15. With the rocket it reached Mach 1.3 in level flight. But Dassault had no faith in the concept of such low-power aircraft and after some work on the twin-Gabizo Mirage II took the plunge and produced a bigger and heavier Mirage III, powered by the 8,820lb thrust Atar 101G. From this stemmed the pre-production IIIA, with larger but thinner wing and completely redesigned fuselage housing the new Atar 9 engine. On 24 October 1958 Mirage IIIA-01 became the first West European aircraft to attain Mach 2 in level flight.

This clinched the decision of the Armée de l'Air to buy 100 of a slightly developed interceptor called Mirage IIIC, fitted either with guns or with a boost rocket for faster climb and better combat performance at heights up to 82,000ft. Normally the SEP 844 rocket was fitted to the IIIC, the sole armament being air-to-air missiles, such as Sidewinders and the big Matra R 530 used in conjunction with the CSF Cyrano radar, fitted to permit the new fighter to operate in all weather. Altogether 244 C models were delivered, large batches also going to South Africa and Israel (a nation which did much to develop and promote both the III and the 5). From the IIIC emerged the dual-control IIIB trainer, the longer and heavier IIIE for ground attack (with Marconi doppler radar for blind low-level navigation, new fire-control and navigation computer, and increased internal fuel) and the IIIR family of camera-equipped reconnaissance aircraft. By 1977 about 1,200 of the Mirage III family had been sold, including a fairly standard version made in Australia and an extremely non-standard version made in Switzerland after painful development problems which inflated the price and reduced the numbers bought.

In 1965 Israel suggested that Dassault should produce a special VFR (clear weather) version for ground attack in the Middle East, with the radar and fire control avionics removed and replaced by an extra 110 gallons of fuel and more bombs. The result was the Mirage 5 and Israel bought 50 of the first production batch of 60. It can be distinguished by its longer and much more pointed nose, devoid of radar unless the small Aida II is fitted. For political reasons the French refused to deliver the paid-for Mirages to Israel but more than 500 have been sold to many other countries and 106 were assembled and partly constructed, in Belgium. Largely as a result of the French action, Israel developed its own improved version of the Mirage (see IAI Kfir, separate entry).

In addition to production aircraft there have been many experimental or

unsold variants. One of the latter was the F5, a single-engine fighter proposed by Dassault and Boeing in 1965 as a simple fighter for America's allies. A single-engine fighter was fitted with retractable landing gear for better manoeuvrability (the exception being the Mirage F1, the far the biggest development programme ever undertaken for a powerful Mirage IIIV V/STOL fighter with a 19,840lb thrust SNECMA TF306 augmented turbofan for propulsion and eight 5,500lb thrust Rolls-Royce RB17-31 afterburners. The F5 was a V/STOL fighter, and the equally large F2 led to the smaller (Atar-size) F1.



Above: Three-view of Mirage 5, showing multi-sensor pods.



Above: Not a Mirage delta with a tail but a pair of IIIO ground-attack fighters of the RAAF off the Australian coast.



Above: Mirage IIICJ of Heyl Ha'Avir, Israel.



Above: Mirage IIIEP of No 5 Sqn, Pakistan Air Force.

Below: Most Mirage customers have bought dual-control trainer versions for pilot conversion to the speedy delta — with “back of the drag curve” characteristics — and weapon training. This is a Mirage 5-DV of Venezuela.



General Dynamics F-16

Model 401, YF-16, F-16A, F-16B

Origin: General Dynamics Fort Worth, USA with widespread subcontracting in Europe and European assembly of aircraft for European customers (see text)

Type: Single seat fighter bomber (B) operational trainer

Engine: One 24 000 lb (10 885 kg) thrust Pratt & Whitney F100 PW 100 (w) short afterburning turbofan

Dimensions: Span (no Sidewinders) 31 ft 0 in (9.45 m) (with Sidewinders) 32 ft 10 in (10.01 m) length (excl probe) (YF-16) 46 ft 6 in, (F-16A) 47 ft 7.7 in (14.52 m) height (F-16) 16 ft 5.2 in (5.01 m)

Weights: Empty (YF) about 12 000 lb (5443 kg) (F) about 14 800 lb (6733 kg) maximum gross (YF) 27 000 lb (12 245 kg), (F) 33 000 lb (14 969 kg)

Performance: Maximum speed Mach 1.95 equivalent to about 1,300 mph (2090 km/h) initial climb (YF) 40 000 ft (12 200 m), min service ceiling about 40 000 ft (18 300 m) range on internal fuel in interception mission about 1 300 miles (2100 km) attack radius at low level with maximum weapon load, 120 miles (193 km) attack radius with six Mk 82 bombs 339 miles (546 km)

Armament: One 20 mm M61 multi-barrel cannon on left side of fuselage nine pylons for total external load of up to 15 200 lb (6895 kg) (YF, seven pylons for total of 11 500 lb 5217 kg)

History: First flight (YF) 20 January 1974 service delivery scheduled for mid 1978

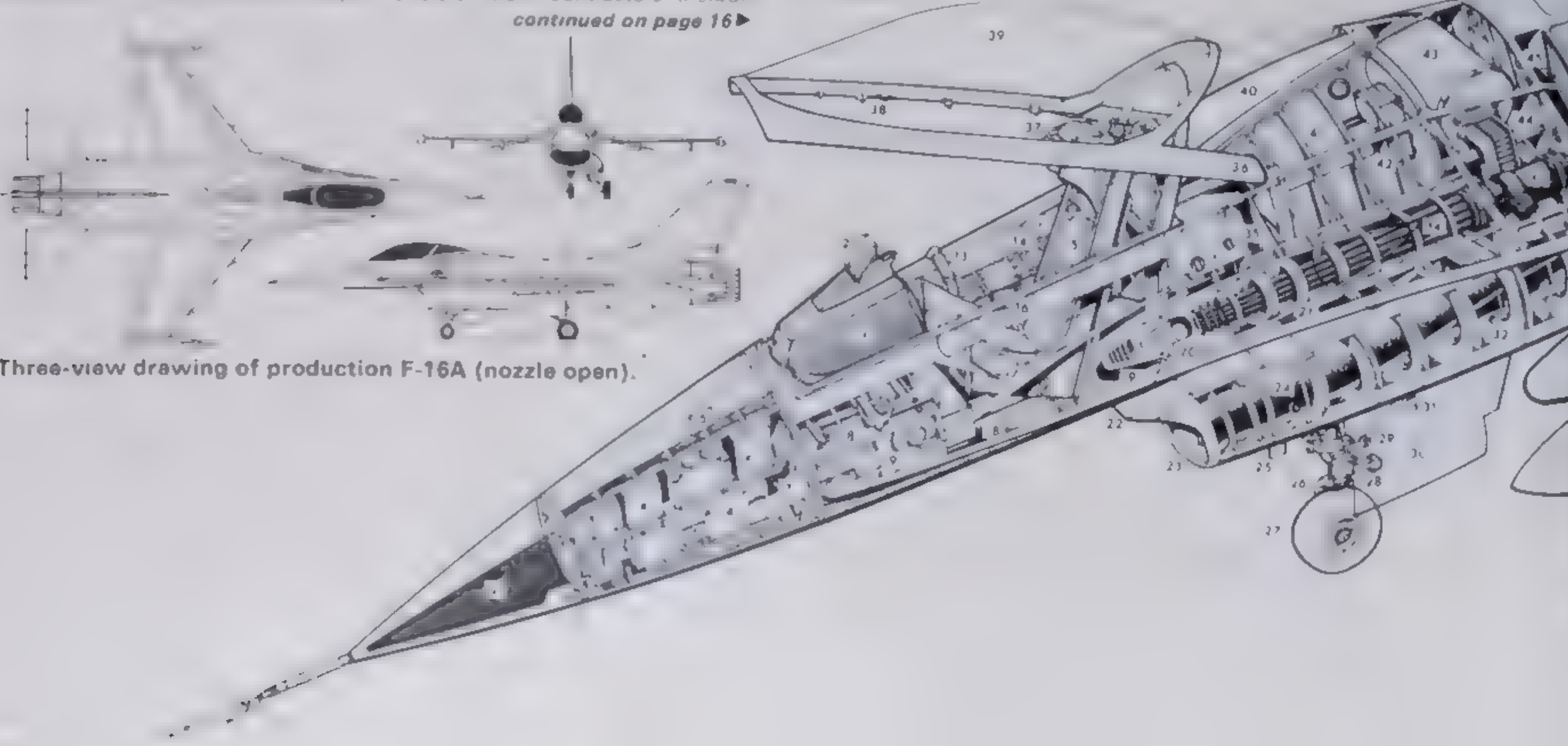
Users: Belgium, Denmark, Iran, the Netherlands, Norway, Turkey (planned see text), US Air Force

Development: One of the most important combat aircraft of the rest of the century was started merely as a technology demonstrator to see to what degree it would be possible to build a useful fighter that was significantly smaller and cheaper than the F-15. The US Air Force Lightweight Fighter (LWF) programme was not intended to lead to a production aircraft but merely to establish what was possible at what cost. Contracts for two prototypes of each of the two best submissions were awarded in April 1972, the aircraft being the General Dynamics 401 and a simplified Northrop P 530. As the YF-16 and YF-17 these aircraft completed a programme of competitive evaluation, as planned in 1974. By this time the wish of four European members of NATO — Belgium, Holland, Denmark and Norway — to replace their F-104Gs with an aircraft in this class had spurred a total revision of the LWF programme. In April 1974 it was changed into the Air Combat Fighter (ACF) programme and the Defense Secretary, James Schlesinger, announced that 650 of the winning design would be bought for the USAF with a vast support depot in Europe. In December 1974 the YF-16 was chosen as the future ACF (announced the following month) and in June 1975, after protracted and tortuous discussions it was chosen by the four European countries. As an aircraft the F-16 is exciting. It has a flashing performance on the power of the single fully developed engine (the same as the F-15) fed by a simple fixed-geometry inlet. Structure and systems are modern, with control-configured vehicle (CCV) flight dynamics, quad redundant electrically signalled controls (fly-by-wire), graphite-epoxy structures and a flared wing/body shape. Pilot view is outstanding and he lies back in a reclining Escapac seat and flies the aircraft through a sidestick controller. In the nose is an advanced pulse-doppler radar suitable for attack or interception missions and armament can be carried for both roles though the basic design was based strongly in favour of air-to-air missions in good weather at close range. It remains to be seen to what degree the F-16 can be modified to make it a better ground attack reconnaissance or all-weather interceptor aircraft. Main contractors include

continued on page 16 ▶

The second YF-16 prototype, pictured whilst undergoing weapon trials in 1976. It has just released 12 bombs each of a nominal 1,000 lb. Maximum load, with reduced fuel, is 15,200 lb.

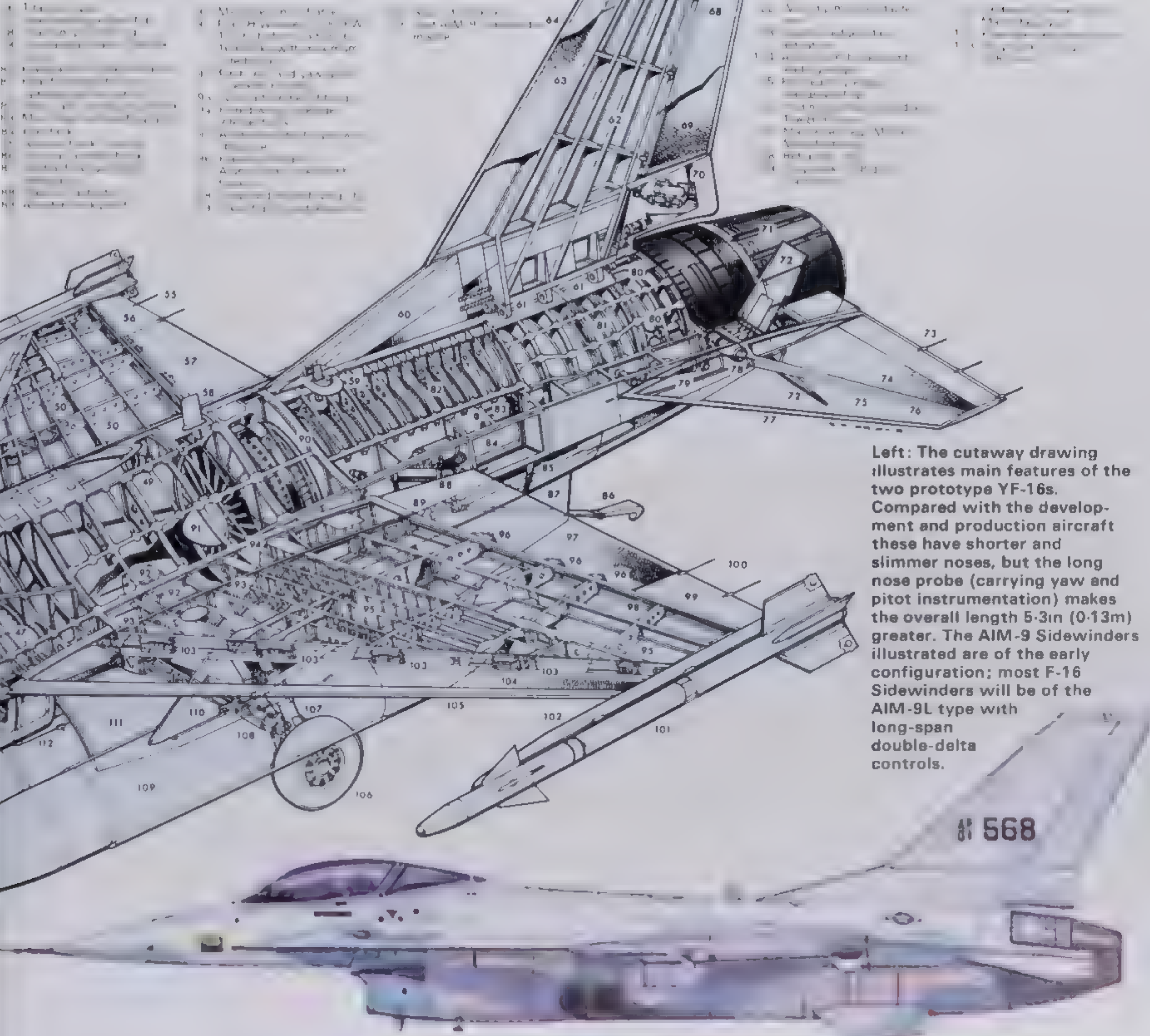
- | | | |
|--|--|--|
| 1 Air data probe (Rosemount Engineering) | 30 Nosewheel door | 62 Aluminium multi spar fin structure |
| 2 SSR-1 radar ranging system (General Electric) | 31 Door hinge | 63 Aluminium honeycomb leading edge |
| 3 Westinghouse multi mode radar (F-16) | 32 Nosewheel well (bow duct) | 64 Steel leading edge strip |
| 4 Angle of attack transducers (port and starboard) | 33 General Electric M61 20mm gun | 65 Graphite epoxy skin |
| 5 Battery and avionics compartment | 34 Cannon barrels | 66 Identification/navigation light |
| 6 Central air data computer (Sperry Flight Systems) | 35 Emergency power unit pack (Sandstrand Avionics) | 67 Static dischargers |
| 7 Air data converter | 36 Canopy hinge | 68 Graphite epoxy rudder skin |
| 8 Forward pressure bulkhead | 37 Headrest | 69 Aluminium honeycomb rudder |
| 9 Rudder pedals | 38 Canopy lock | 70 Empennage flight controls |
| 10 Control wiring (fly by wire) junction | 39 Frameless bubble canopy (Birtac Inc) | 71 Fully variable articulated nozzle |
| 11 Raised hoist rest line | 40 Aft glazing | 72 Split trailing edge airbrake (upper and lower surfaces) |
| 12 Instrument panel shroud | 41 Forward fuselage fuel tank | 73 Static dischargers |
| 13 Head up display unit (Marconi Elliott Avionics Systems HUDWAS) | 42 Accelerometers (General Electric) | 74 Aluminium honeycomb tailplan |
| 14 Side stick controller | 43 Ammunition drum | 75 Graphite epoxy skin |
| 15 Starboard instrument console | 44 Ammunition feed and link return chutes (General Electric) | |
| 16 Lightweight ejection seat (30 deg tilt back Douglass Escapac III B) | 45 Forward centre fuselage joint bulkhead | |
| 17 Arm rest | 46 Inlet duct | |
| 18 Port instrument console (power lever mounting) | 47 Hydraulic equipment bay | |
| 19 Fuselage forebody strakes | 48 Main (forward) fuselage fuel tank | |
| 20 Cooling louvers | 49 Flight refuelling receptacle | |
| 21 Gun gas suppression nozzle | 50 Multi spar wing structure | |
| 22 Gun firing frames | 51 Leading edge flap hinge line | |
| 23 Boundary layer splitter plate | 52 Leading edge manoeuvre flap | |
| 24 Fixed geometry air intake | 53 Wing tip missile adapter shape | |
| 25 Inlet duct | 54 A M-9 Sidewinder missile | |
| 26 Aerial | 55 Static dischargers | |
| 27 Nosewheel leg (Menasco Manufacturing) | 56 Fixed trailing edge section | |
| 28 Aft retracting nose gear (Goodyear Tire & Rubber) | 57 Starboard flap/eron | |
| 29 Shock absorber scissors | 58 Aerial | |
| 30 Retractor strut | 59 Forward support link | |
| | 60 Glass fibre root fairing | |
| | 61 Fin fuselage attachments | |



Three-view drawing of production F-16A (nozzle open).



Above: Under an azure Texan sky, the first F-16A development aircraft proudly wears the national flags of the first five customers. Note the enlarged nose and AIM-9L missiles.



Left: The cutaway drawing illustrates main features of the two prototype YF-16s. Compared with the development and production aircraft these have shorter and slimmer noses, but the long nose probe (carrying yaw and pitot instrumentation) makes the overall length 5.3in (0.13m) greater. The AIM-9 Sidewinders illustrated are of the early configuration; most F-16 Sidewinders will be of the AIM-9L type with long-span double-delta controls.

Above: YF-16 No 2 in 1976 development configuration (nozzle closed)

► The YF-16 is a single-engine, single-seat, delta-wing fighter aircraft. It was developed by General Dynamics and the US Air Force. The aircraft is designed for high-speed, high-altitude operations. It features a highly maneuverable design with a delta wing and a canards configuration. The YF-16 was first flown in 1978 and has since been used in various roles, including as a test aircraft and as a prototype for the F-16 Fighting Falcon. The aircraft is known for its excellent performance and its ability to operate in a wide range of environments. It has been used in a variety of conflicts and has proven to be a highly effective fighter aircraft. The YF-16 is a testament to the ingenuity and innovation of the US Air Force and General Dynamics. It is a true masterpiece of aviation engineering.



Prototype YF-16 aircraft have appeared in various colour schemes; this air-to-air picture shows the No 2 machine in low-visibility "two-tone grey".

Below: Gray smoke spews from the Gatling exhaust port of a YF-16 on a low-level firing pass during 1976 weapon trials. Fighters are no longer built without guns.



General Dynamics

F-102 Delta Dagger

F-102A, TF-102A, QF-102A and PQM-102A

Origin: General Dynamics/Convair, USA
Type: (F) single seat all weather interceptor, (TF) trainer (for remote RPV (remotely piloted vehicle), (PQM) drone target
Engine: One 17,200lb (7802kg) thrust Pratt & Whitney J57 turbojet with afterburning turbojet
Dimensions: Span 38ft 1½in (11.6m), length, (F-102A) 68ft 5in (20.85m), (TF-102A) 63ft 4½in (19.3m), height (F-102A) 21ft 2½in (6.45m), (TF-102A) 20ft 7in (6.27m)
Weights: Empty (F-102A, 19,550lb (8630kg), loaded F-102A 31,500lb (14,288kg), (maximum) 31,500lb (14,288kg)
Performance: Maximum speed (F-102A) 825mph (1328km/h, Mach 1.25), (TF-102A) 646mph (1040km/h), initial climb (F-102A) 13,000ft (3962m)/min, service ceiling 54,000ft (16,460m), range 1,350 miles (2172km)
Armament: Air-to-air guided missiles carried in internal bay, typical full load comprising three Hughes AIM-4E Falcon beam riders with active homing and three AIM-4F with infra-red homing. No armament on TF, QF or PQM
History: First flight (YF-102) 24 October 1953, (YF-102A) 20 December 1954, (TF) 8 November 1955, (QF) mid-1974, (PQM) early 1975
Users: Greece, Turkey, USA (F, TF, ANG, PQM, QF, TF, US Air Force)

Development: In 1948 Convair flew the world's first delta-wing aircraft, the XF-92A, which was part of a programme intended to lead to a supersonic fighter. This was terminated, but the US Air Force later issued a specification for an extremely advanced all-weather interceptor to carry the Hughes MX-1179 system which included radar, computer and guided missiles. For the first time the carrier aircraft became subordinate to its avionics, as a mere portion of a weapon system. The whole weapon system represented a major challenge in 1951-52 and it did not help when early flight trials showed that the specified supersonic speed could not be reached. The whole aircraft had to be redesigned to a modified shape complying with the lately discovered Area Rule, making it much longer, and fatter at the back. Once the design had been got right, 875 were delivered in 21 months, together with 63 of the subsonic side-by-side TF version. The Delta Dagger was big and impressive, painted a glossy pale grey, and had a small but comfortable cockpit where the pilot flew with two control columns, that on the left being used to adjust the sweep angle and range gate of the MG-10 radar. In the search mode the pilot flew with one hand on each stick and his eyes pressed into the viewing hood of the radar display screen. In the semi-automatic mode missiles could be extended automatically at the correct moment, as delivered the F-102 also carried FFARs (Folding Fin Aircraft Rockets) in the missile bay doors. By 1974 surviving F-102s called "Deuces" – had been assigned to the US Air National Guard and to the air forces of Greece and Turkey. Sperry converted 24 into remotely piloted QF and unmanned PQM versions for use in threat evaluation, F-15 combat and other aerial research.



Above: Three-view of F-102A with drop tanks.



Above: A colourful line-up of "Deuces"—so-called because a deuce is a two at cards—pictured in their youth when they were the last word in automatic all-weather interception.

Left: Thirsty Deuce takes on fuel from a KC-135A with high-speed boom modified for probe/drogue refuelling.



Left: One of the final users of the F-102A is the Turkish Air Force. The aircraft retain their original equipment and serial (tail) number, and are compatible with the Nadge air-defence system.

Left: A Deuce in US Air Force or Air National Guard camouflage. They have an excellent service record.

General Dynamics F-106 Delta Dart

F-106A and F-106B

Origin: General Dynamics, Cleveland, USA

Type: (F-100A) single seat all weather interceptor (F-100B) operational

Engine

Dimensions

Weights

Performance

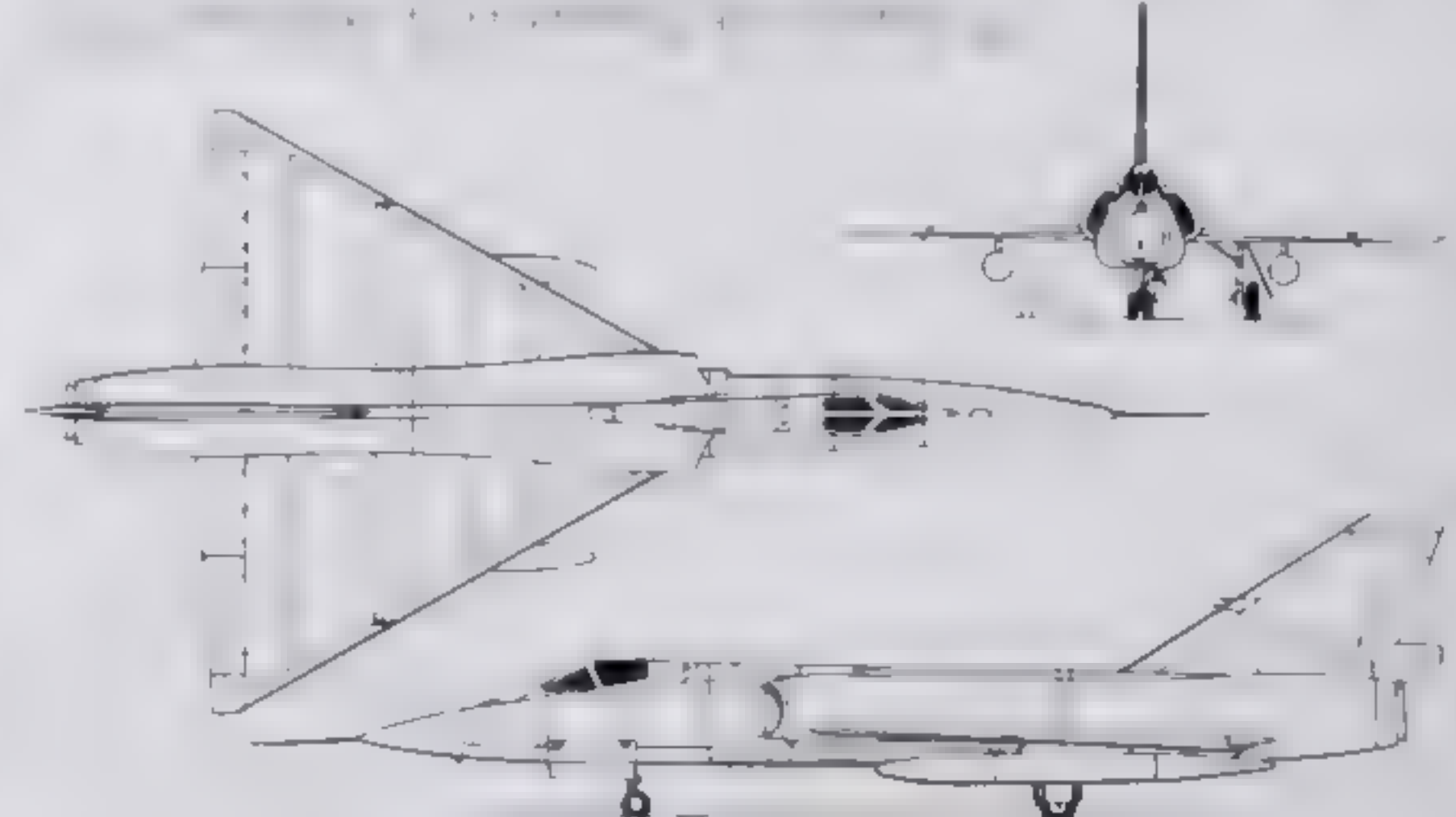
Armament

History

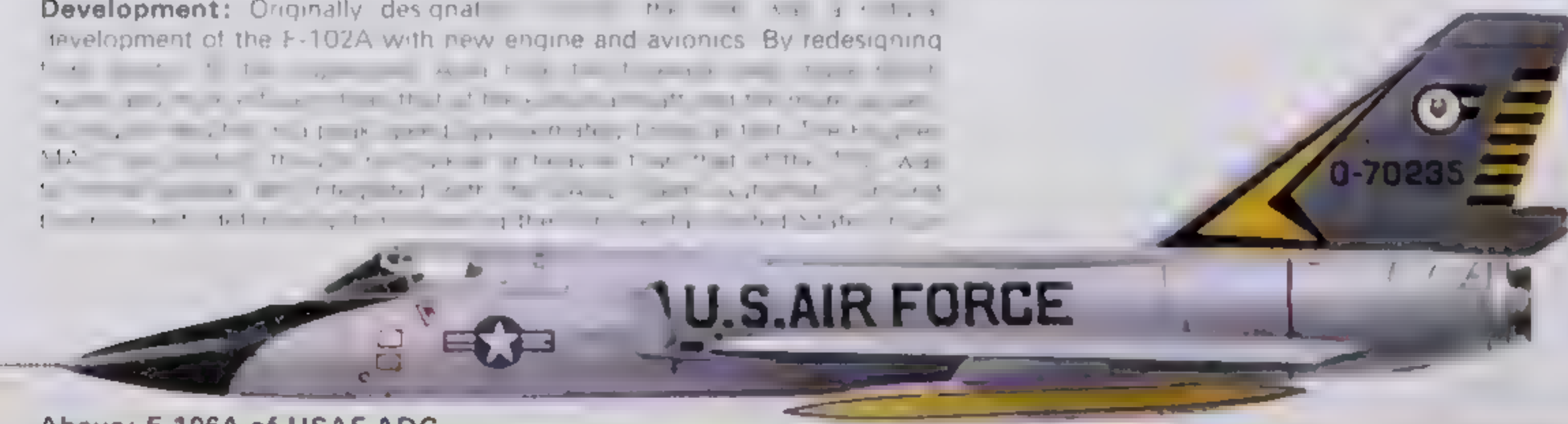
User

Development: Originally designed for the development of the F-102A with new engine and avionics. By redesigning

tively numbering originally 277 and 63) will continue until at least 1980. They have been repeatedly updated with improved avionics, infra-red



Above: Three-view of F-106A with drop tanks.



Above: F-106A of USAF ADC

Below: An F-106A Delta Dart single-seat interceptor of the 194th FIS, 144th FIG, California ANG at Tyndall AFB, Florida
Foot of page: A tandem-seat dual-control F-106B of the same Aerospace Defense Command unit (no longer active) as the single-seater shown in the drawing.



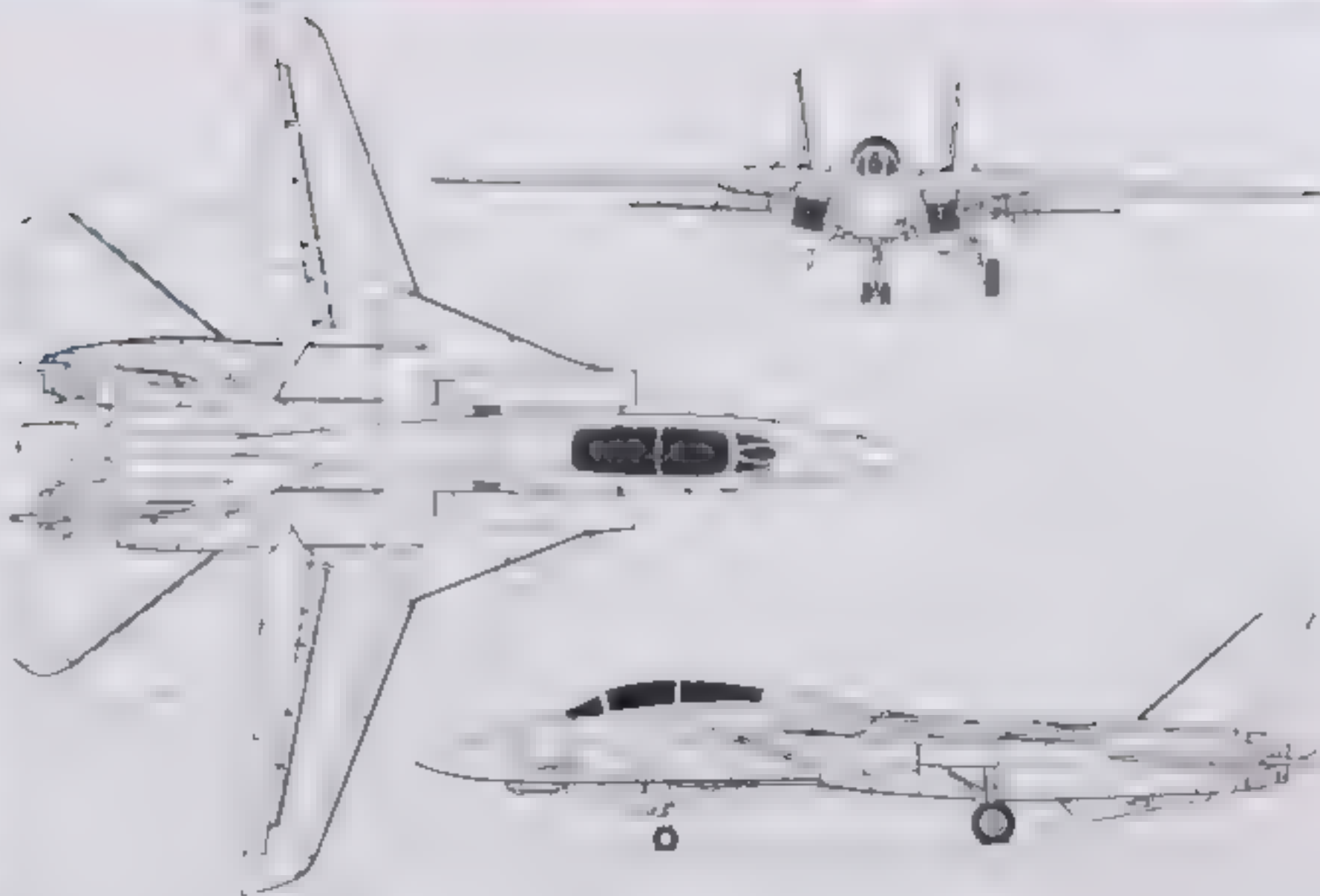
Grumman F-14 Tomcat

F-14A, B and C

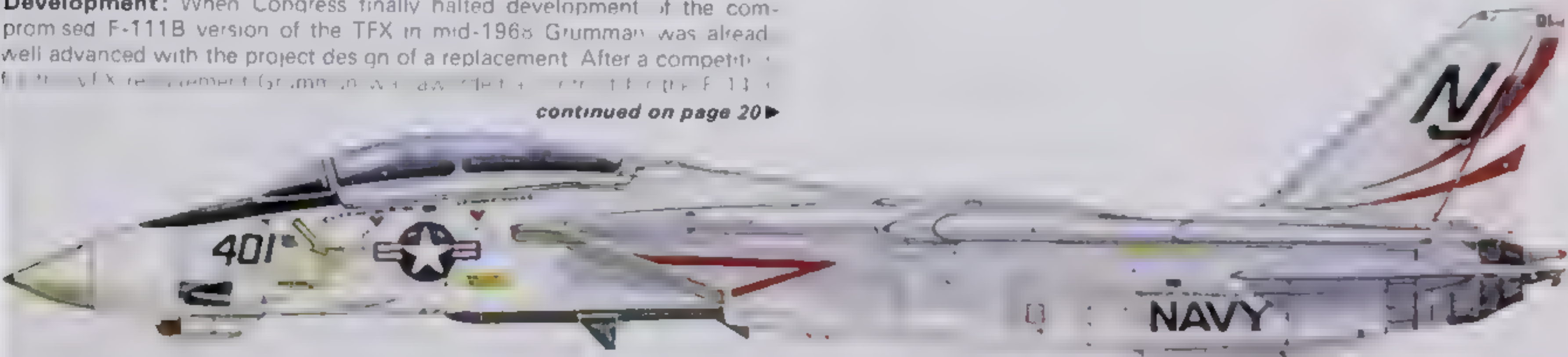
Origin: Grumman Aerospace Corp, USA
Type: Two-seat carrier-based fighter
Engines: (F-14A) two 20,000 lb (9,072 kg) thrust Pratt & Whitney F401-400 two-shaft afterburning turbofans; (B and C) two 28,000 lb (12,700 kg) thrust Pratt & Whitney F401-400 two-shaft afterburning turbofans
Dimensions: Span (68° sweep) 38ft 11in (11.87m), length 61ft 2in (18.65m), height 16ft 4in (4.97m)
Weights: Empty 37,500lb (17,010kg), (maximum) 72,000lb (32,658kg)
Performance: Maximum speed, 1,564mph (2,517km/h) at sea level, initial climb rate, 910mph (1,470km/h, Mach 1.2) at sea level, initial climb rate, 17,070m, range (fighter with external fuel) about 2,000 miles (3,218 km)
Armament: One 20mm M61-A1 multi-barrel cannon, four AIM-7 Sparrow and four or eight AIM-9 Sidewinder air-to-air missiles, or up to six AIM-54 Phoenix and two AIM-9, maximum external weapon load in surface attack role 14,500lb (6,577kg)
History: First flight 21 December 1970, initial deployment to US Navy carriers October 1972, first flight of F-14B 12 September 1973
Users: Iran (IrAF), USA (Navy, Marine Corps)

Development: When Congress finally halted development of the compromised F-111B version of the TFX in mid-1968 Grumman was already well advanced with the project design of a replacement. After a competitive bid process, Grumman was awarded the contract for the F-14.

continued on page 20



Above: Three-view of an F-14A showing (broken lines) range of wing and glove movement.



Above: F-14A as delivered in 1972 to VF-124 conversion squadron.

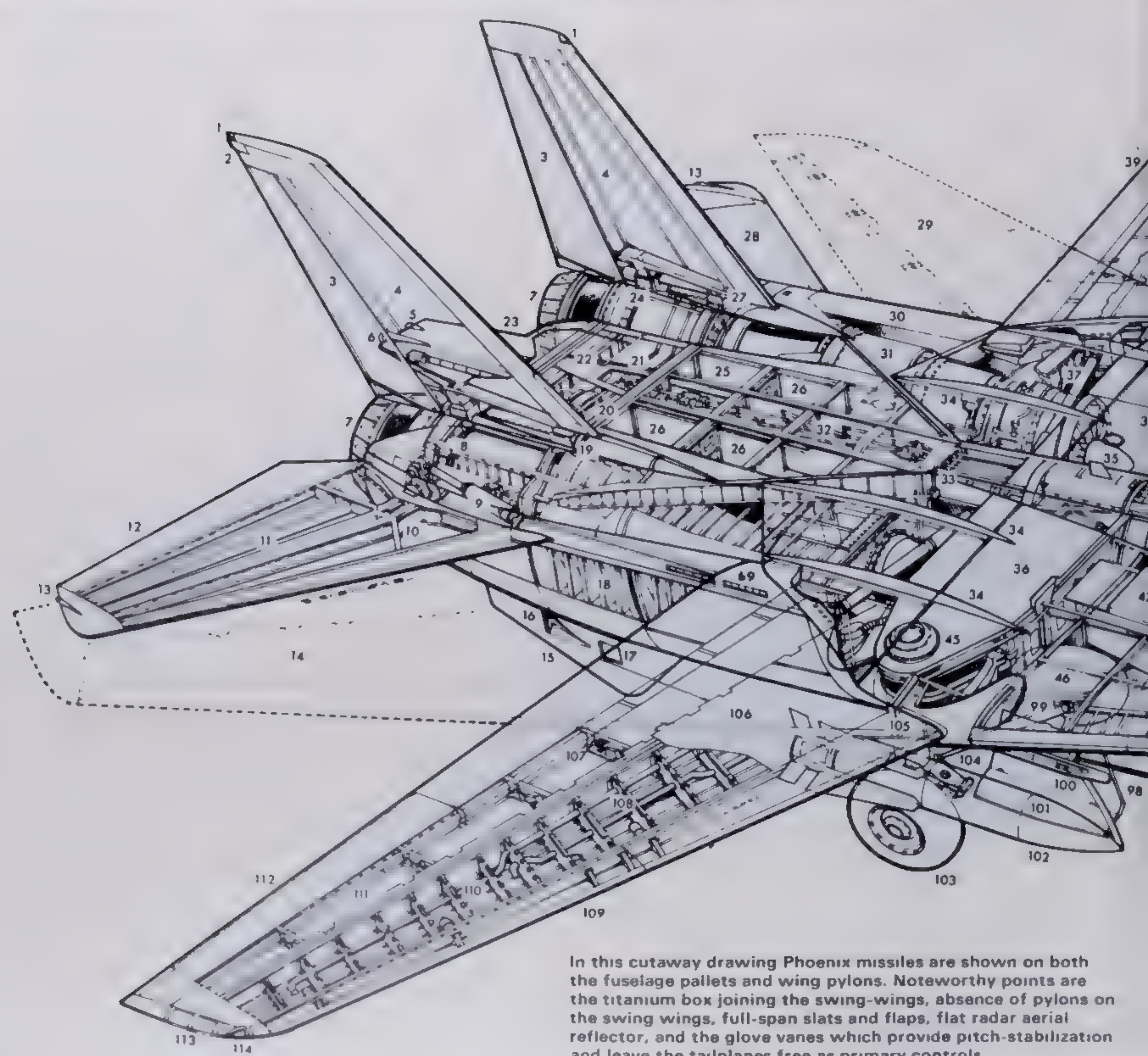
Below: A splendid head-on picture of a Tomcat from VF-211, armed with two AIM-54A Phoenix, two AIM-7F Sparrow and two AIM-9G Sidewinders. At this range the gun would be better





Above: Despite various transient problems, the Tomcat is a truly great fighter. This one comes from VF-14, embarked aboard *John F. Kennedy*

Right: The moment of motor ignition of a Phoenix, the world's longest-ranged (126+ miles) air/air missile



In this cutaway drawing Phoenix missiles are shown on both the fuselage pallets and wing pylons. Noteworthy points are the titanium box joining the swing-wings, absence of pylons on the swing wings, full-span slats and flaps, flat radar aerial reflector, and the glove vanes which provide pitch-stabilization and leave the tailplanes free as primary controls.

Hawker Siddeley

Gnat/HAL Ajeet

Fo 141 Gnat F.1, HAL Gnat and Ajeet, Hawker Siddeley Gnat T.1

Origin

Type
Engine

Dimensions

Weights

Performance

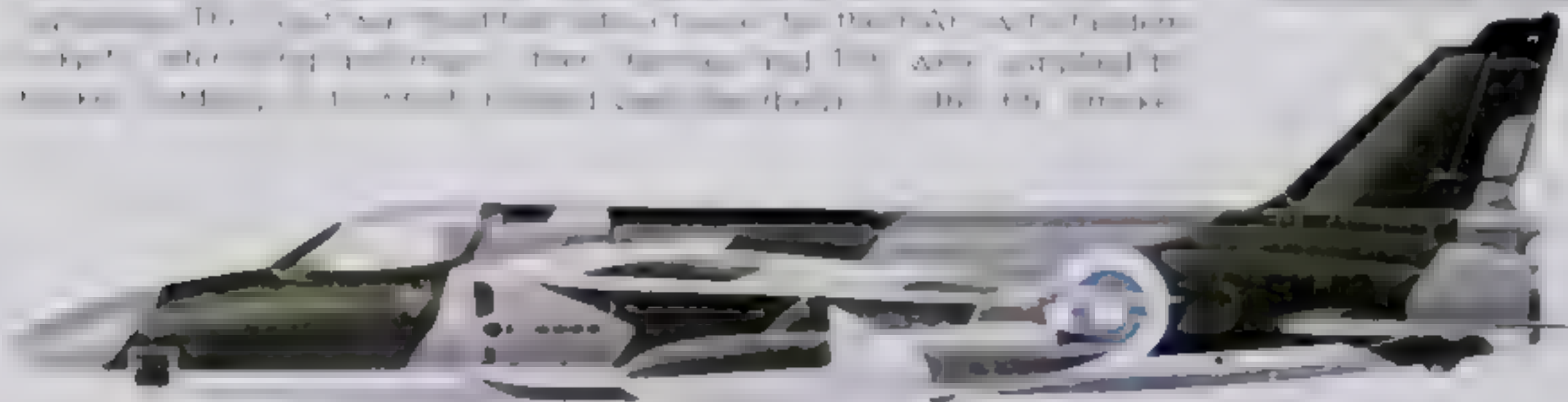
Armament

History

Users

Development

The Gnat was developed as a lightweight, high-speed fighter aircraft. It was designed to be a simple, easy-to-build aircraft that could be produced in large numbers. The aircraft was designed to be a lightweight, high-speed fighter aircraft. It was designed to be a simple, easy-to-build aircraft that could be produced in large numbers. The aircraft was designed to be a lightweight, high-speed fighter aircraft. It was designed to be a simple, easy-to-build aircraft that could be produced in large numbers.



Above This British built Gnat was a Finnish photo reconnaissance fighter.



Above: HAL Gnat (one of the last built before the Ajeet programme) photographed in company with an HJT-16 Kiran Mk1.



Above: Three-view of the Ajeet, showing four stores pylons

Gnat T1s equip the Red Arrows aerobatic team. In 1969 HAL began the development of a new aircraft which was finally agreed in 1974. The Gnat II or Ajeet (Unconquerable), it has integral-tank wings housing the same amount of fuel as a Gnat, but also underwing tanks, thus increasing its range. It also has a more powerful engine, more powerful MA. Gnats are progressively being replaced by the Ajeet in production and today the HAL Gnats have a reputation as a well-proven and effective in close combat.

Below; Close-up of a polished HAL Gnat of the Indian Air Force. To achieve a useful combat radius the Gnat has to carry underwing drop tanks, but the Ajeet now in production accommodates the same fuel load in integral wing tanks. Thus the Ajeet combines the same radius as a drop-tanked Gnat with external weapons.



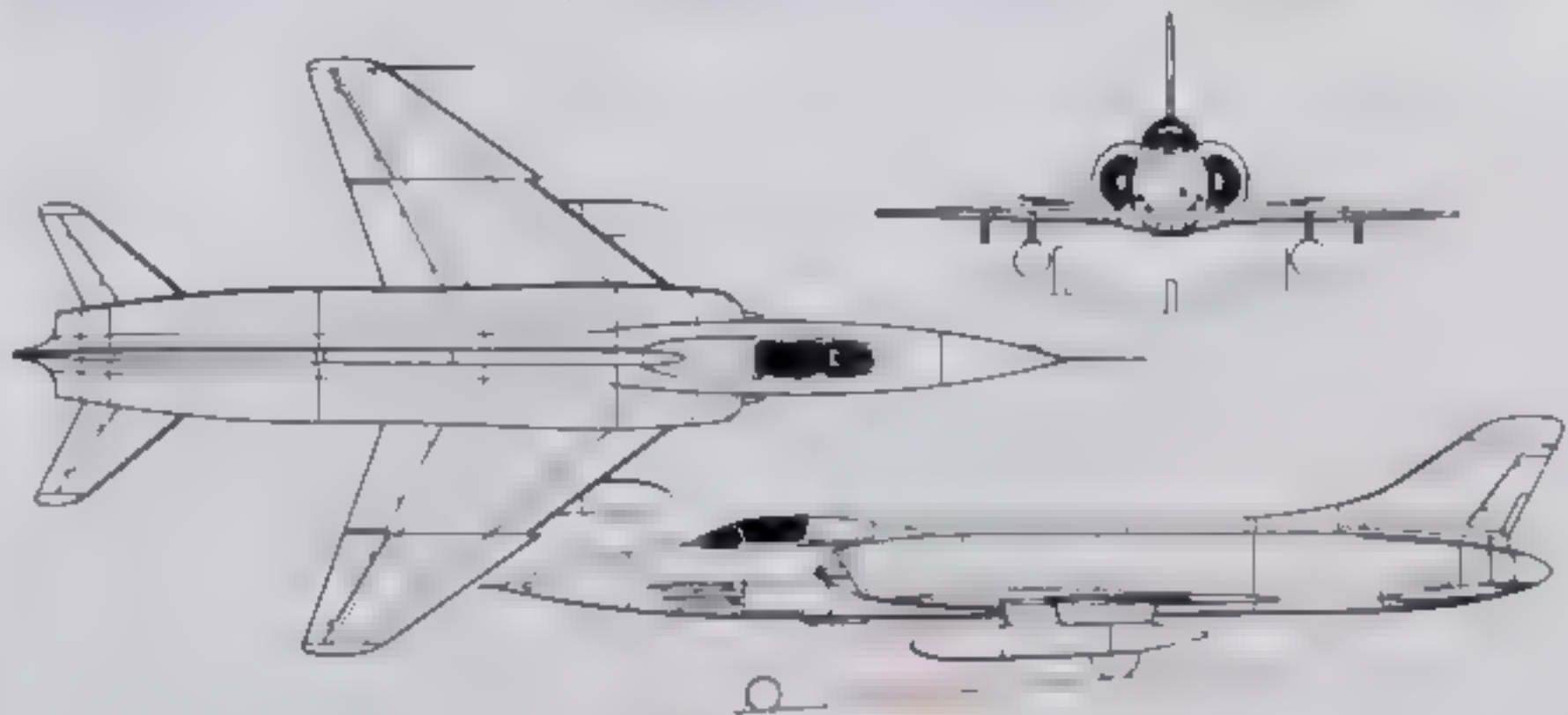
Hindustan HF-24 Marut

HAL HF-24 Mk I, IT and II

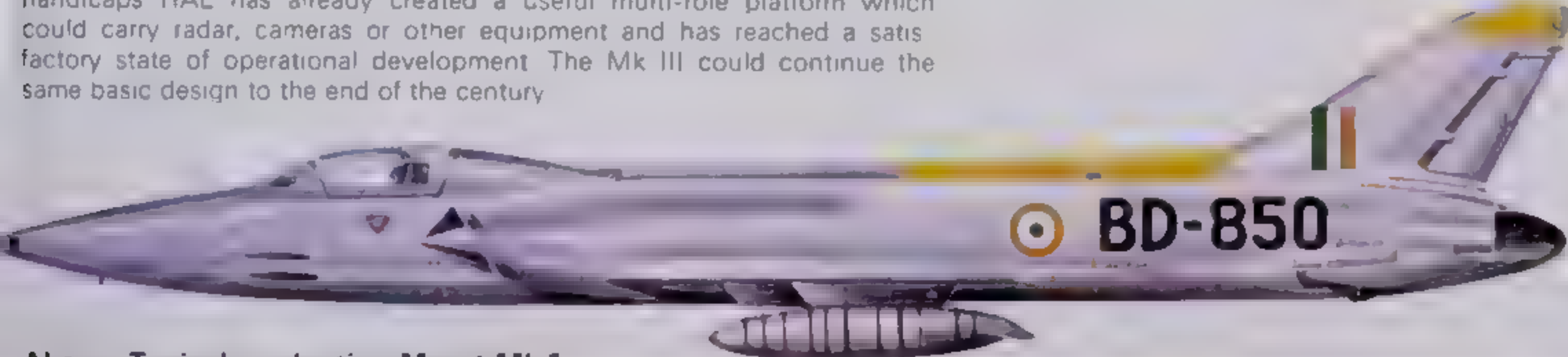
Origin: Hindustan Aeronautics, India
Type: Single-seat fighter
Engines: Two 4 850hp (3 550kW) Turbofan engines, then Bristol Siddeley Orpheus turbojets, then HAL
Dimensions: Span 26ft 6in (8.1m), height 8ft 6in (2.6m), length 11ft 9in (3.6m)
Weights: (Mk I empty 13 580 lb (6 150 kg), loaded 18 000 lb (8 150 kg) loaded maximum 24 250 lb (11 000 kg)
Performance: Maximum speed 1 210 mph (1 112 km/h, Mach 0.91, at sea level, about 675 mph (1 086 km/h, Mach 1.02) at altitude, time to climb to 40 000 ft (12 200 m) 9 min 20 sec, range 1 000 km (1 000 km)
Armament: Four 30mm Aden Mk 2 cannons with 12 rounds, retractable Matra pack of 50 SNEB 68mm rockets and 100 lb (45 kg) bombs rated at 1 000 lb (454 kg)
History: First flight 17 June 1961, (pre-production March 1963, series production) 15 November 1967 (Mk IT) 30 April 1970
User: India

Development: After 1950 the Indian government decided to accelerate development of an Indian combat aircraft and the services of Oplund Kurt Tank, the renowned Focke-Wulf designer, were secured. A new team formed by Hindustan Aircraft at Bangalore. Detail design began in 1956, the objective being to create a multi-role aircraft potentially capable of reaching Mach 2 with minimal technical risk. The prototype, powered by two of the same engines already being produced for the Gnat, proved iterative, successful, and two of the 18 pre-production Maruts ("Wind Spirit") were officially handed over (though as a token delivery) to the IAF in May 1964, the year the company reorganised and expanded into its present form as Hindustan Aeronautics. By the end of 1976 about 100 production Mk Is had been delivered, many of them being used (without loss) in the December 1971 war against Pakistan. The Mk IT has a second Martin-Baker seat in place of the rocket pack and has since 1974 also been produced in small numbers as a dual conversion and weapon trainer. In 1967 the German staff left and an Indian design team has since continued the 20 year search for a more powerful engine. HAL has tested afterburning engines and flew the Marut IBX with one Orpheus replaced by an Egyptian Brandner E-300 but the most likely solution will be the HSS-73 (Marut III) with two Turbo Union RB 199 engines in a considerably improved airframe. Despite obvious handicaps HAL has already created a useful multi-role platform which could carry radar, cameras or other equipment and has reached a satisfactory state of operational development. The Mk III could continue the same basic design to the end of the century.

Above: Flight-line preparation of Marut Mk I ground-attack fighters. These are early models, built in the late 1960s



Above: Three-view of HF-24 Marut Mk I with drop tanks.



Below: A recent Marut Mk I, with different dielectric (electronic aerial) areas from the examples above. This particular aircraft has metal fairings riveted over the two upper cannon ports. Many Mk IT trainers also have only two guns (four is usual).

Above: Typical production Marut Mk I.



McDonnell Douglas

F-101 Voodoo

F-101A, B and C and RF-101A to H

Origin: McDonnell Aircraft Co (division of McDonnell Douglas Corp), USA
Type: (A, C) day fighter-bomber, (B) all-weather interceptor, (RF) all-weather reconnaissance

Engines: Two Pratt & Whitney J57 two-shaft turbojets with afterburners (F-101B) 14 990lb (6800kg) J57-53 or -55 (others) 14 880lb (6750kg) 157-13

Dimensions: Span 39ft 8in (12.09m) length 41ft 2.55in (12.59m) height 69ft 3in (21.11m)

Weights: Empty (typical of all) 28 000lb (12 700kg) maximum loaded (B) 46 700lb (21 180kg), (all versions, overload 51 000lb, 23 130kg)

Performance: Maximum speed (B) 1 220mph (1963km/h, Mach 1.85) (others, typical) 1 100mph, initial climb (B) 17 000ft (5180m)/min service ceiling 52 000ft (15 850m), range on internal fuel (B) 1 550 miles (2500km) (others) 1 700 miles (2736km)

Armament: (B) three Falcon (usually AIM-4D) air-to-air missiles, sometimes supplemented by two AIR-2A Genie nuclear rockets on fuselage pylons, (C) three 20mm M-39 cannon (provided for four, with Tacan removed) in fuselage, (RF) none. As built all A and C and derivatives fitted with centreline crutch for 1 MT tactical nuclear store and wing pylons for two 2 000lb (907kg) bombs, four 680lb (310kg) mines or other ordnance

History: First flight 29 September 1954, service delivery (A) May 1957 final delivery (B) March 1961

Users: Canada, Taiwan, USA (ANG)

Development: By far the most powerful fighter of its day, the Voodoo was based on the XF-88 Voodoo prototype flown on 20 October 1948



Above: RF-101C with (bottom) side view of RF-101G.



Above: RF-101H reconnaissance fighter (G similar).

Originally a long-range escort for Strategic Air Command, the F-101A became a tactical attack machine, 50 were followed by 47 improved C models, all of which set records for accident-free operation and were converted to unarmed RF-101G and H for the Air National Guard, augmenting 35 RF-101A and 166 RF-101C built earlier and used intensively at all levels in Vietnam. The B interceptor sacrificed fuel for a radar operator to work the MG-13 radar fire-control; 478 were built and converted to F-101F or dual-control TF-101F for Air Defense Command (now Air National Guard)

In 1961 66 ex-ADC aircraft were transferred to the RCAF as CF-101s. In 1970 the CAF exchanged the 58 survivors for 66 improved F and TF and there still serve as the only CAF all-weather fighters

Below: In 1954 the Voodoo, then the world's most powerful fighter, was rejected by the Canadian Air Staff. Today the CAF flies arduous missions with second-hand Voodoos—and they still do a respectable job.



McDonnell Douglas F-4 Phantom II

F-4A to F-4S, RF-4, QF-4, EF-4

Origin

Type: Originally carrier based, all weather

Engines: (B-G) two 17 000 b (7711kg) thrust General Electric J79-8
15 with afterburner (C-D) 17 000lb J79-15 (E-EJ-F

Dimensions: Span 38ft 5in (11.7m) length (B-C
17.7m) (E-EJ-F and all RF versions) 62ft 11in or 63ft (19.2m) (K-M

Weights: Empty (B-C, D, G, J, N) 28 000lb (12 700kg), (E, EJ, F and
RF) 29 000 b (13 150kg) (K, M) 31 000 b (14 060kg) (L)
16 000lb (C-D-G-J-K-M-N-RF) 58 000 b (26

Performance: Maximum speed with Sparrow missiles only (low) 910mph
1464km/h Mach 1.19) with J79 engines 920mph with Spey (high)
500mph (2414km/h, Mach 2.27) with J79 engines, 1.38:
altitude, typically 28 000ft (8534m) min with J79 engine
altitude min with Spey service ceiling over 60 000ft (19 685m) with J79
engines 60 000ft with Spey range on internal fuel (no weapons) about
1 750 miles (2817km) ferry range with external fuel, typically 2 300 miles
3700km) (E and variants 2 600 miles (4184km)

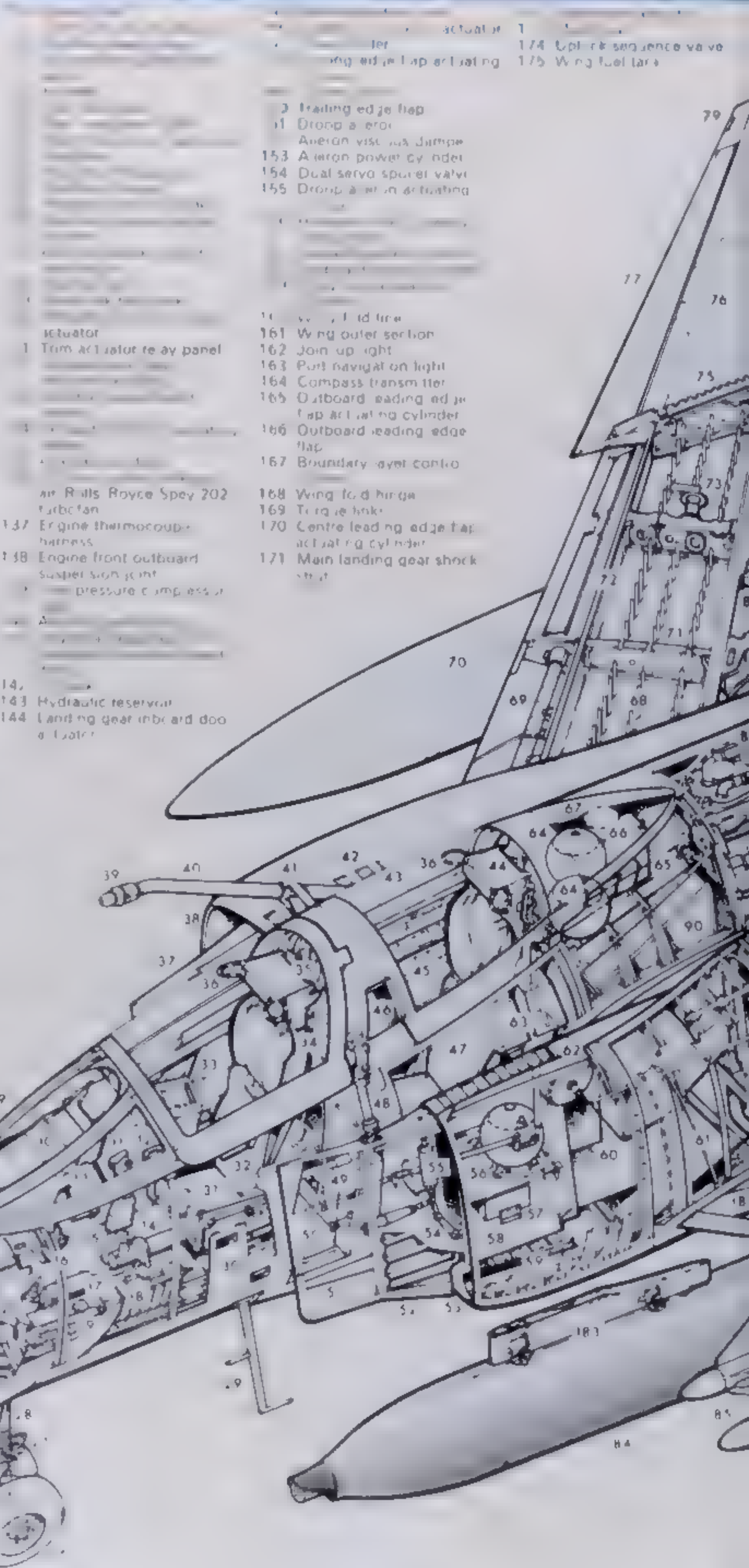
Armament: (All versions except EF, RF, QF which have no armament)
AIM-7 Sparrow air-to-air missiles recessed under fuselage inner wing
pylons can carry two more AIM-7 or four AIM-9 Sidewinder missiles in
addition all E versions except RF have internal 20mm M-61 multi barrel
gun, and virtually all versions can carry the same gun in external centreline
pod, all except RF, QF have centreline and four wing pylons for tanks
bombs or other stores to total weight of 16 000 b (7257kg)

History: First flight (XF4H-1) 27 May 1958, service delivery (F-4A)
February 1960 (carrier trials), February 1961 (inventory), first flight (Air
Force F-4C) 27 May 1963, (YF-4K) 27 June 1966 (F-4E) 30 June 1967
(EF-4E) 1976

Users: W Germany, Greece, Iran, Israel, Japan, Saudi Arabia, Singapore
(no contract announced), S Korea, Spain, Turkey, UK (RAF, Royal Navy),
JSA (Air Force, ANG, Navy, Marine Corps)

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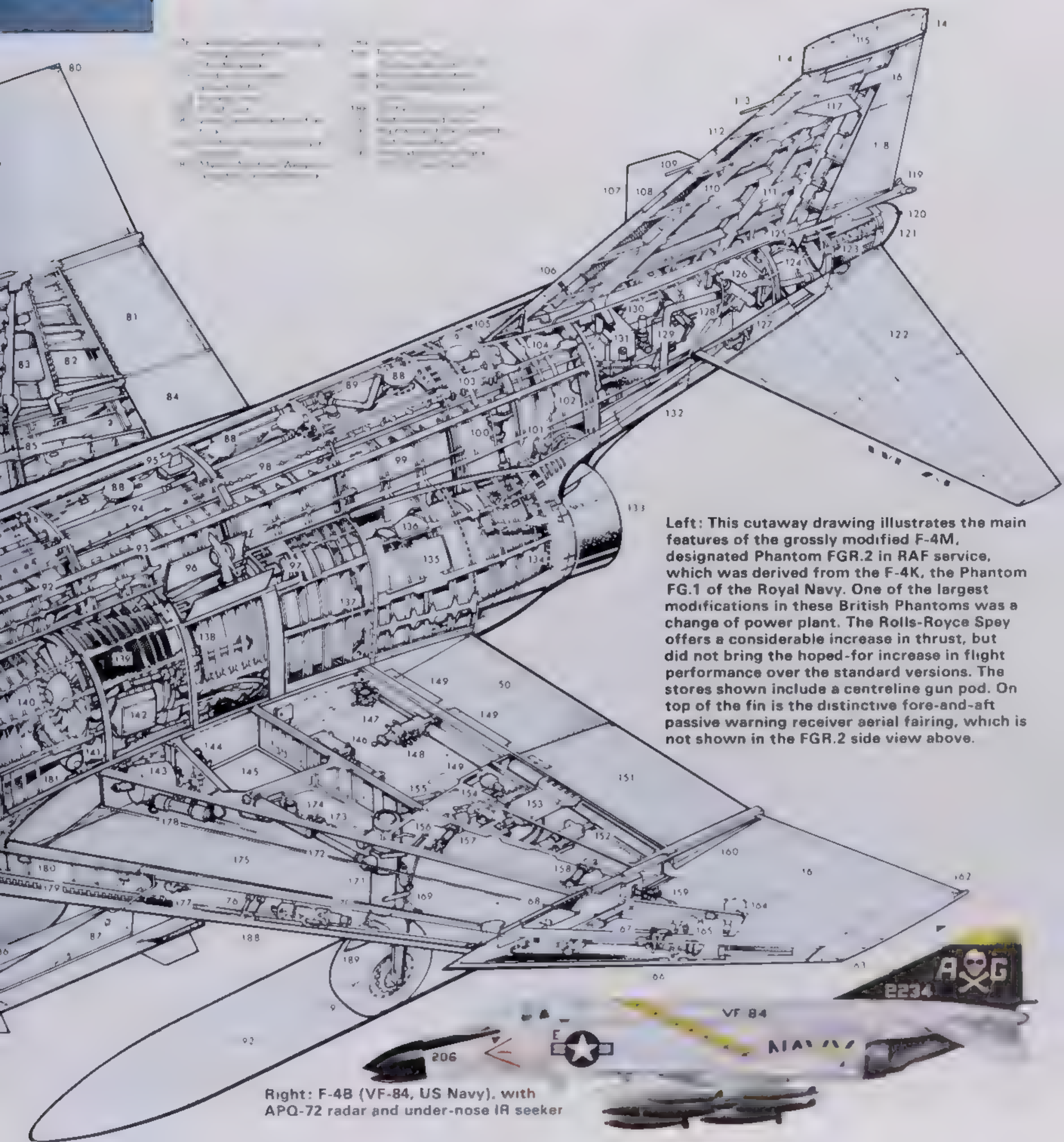
- 44 Stabilising trailing edge
- 45 Inboard rear spar assembly
- 46 Main landing gear trunnion
- 47 Main wheel well
- 48 Fuel filler access
- 49 Fuel filler
- 90 Fuselage No 1 fuel tank
- 91 Fuselage No 2 fuel tank
- 92 Bleed air
- 93 Fuselage structure
- 94 Fuselage No 3 fuel tank
- 95 Tacan antenna
- 96 Ram air turbine (extended)
- 97 Emergency generator
- 98 Fuselage No 4 fuel tank
- 99 Fuselage No 5 fuel tank



Left: The AGM-65A Maverick air/surface missile is now carried by Phantoms of Tactical Air Command. This was a test launch.

Right: F-4E, Heyl Ha'Avir (Israel), with APQ-120 solid-state radar and M61 gun.

Right, F-4M (111 Sqn, RAF), with AWG-12 radar.



Left: This cutaway drawing illustrates the main features of the grossly modified F-4M, designated Phantom FGR.2 in RAF service, which was derived from the F-4K, the Phantom FG.1 of the Royal Navy. One of the largest modifications in these British Phantoms was a change of power plant. The Rolls-Royce Spey offers a considerable increase in thrust, but did not bring the hoped-for increase in flight performance over the standard versions. The stores shown include a centreline gun pod. On top of the fin is the distinctive fore-and-aft passive warning receiver aerial fairing, which is not shown in the FGR.2 side view above.

Right: F-4B (VF-84, US Navy), with APQ-72 radar and under-nose IR seeker



Three-view of F-4E, with (top) side view of F-4M

The Phantom is a true multirole fighter, capable of performing a wide range of missions. It is the only fighter in the world that can carry a full complement of air-to-air, air-to-ground, and ground-to-air weapons. The Phantom is also the only fighter in the world that can carry a full complement of air-to-air, air-to-ground, and ground-to-air weapons. The Phantom is also the only fighter in the world that can carry a full complement of air-to-air, air-to-ground, and ground-to-air weapons.

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Above: F-4D Phantom of Imperial Iranian Air Force.



Above: The only Phantoms not assembled at St Louis are the F-4EJ series built (with major sections shipped in) by Mitsubishi.



Above: F-4K Phantom FG.1 of 892 Sqn, Royal Navy

Below: Among the most colourful Phantoms are the remotely-piloted QF-4Bs of the US Navy, here with human pilot.



McDonnell Douglas

F-15 Eagle

F-15A, TF-15A

Origin: McDonnell Douglas, Fort Worth, Texas, USA

Type: Single-engine, single-seat fighter aircraft

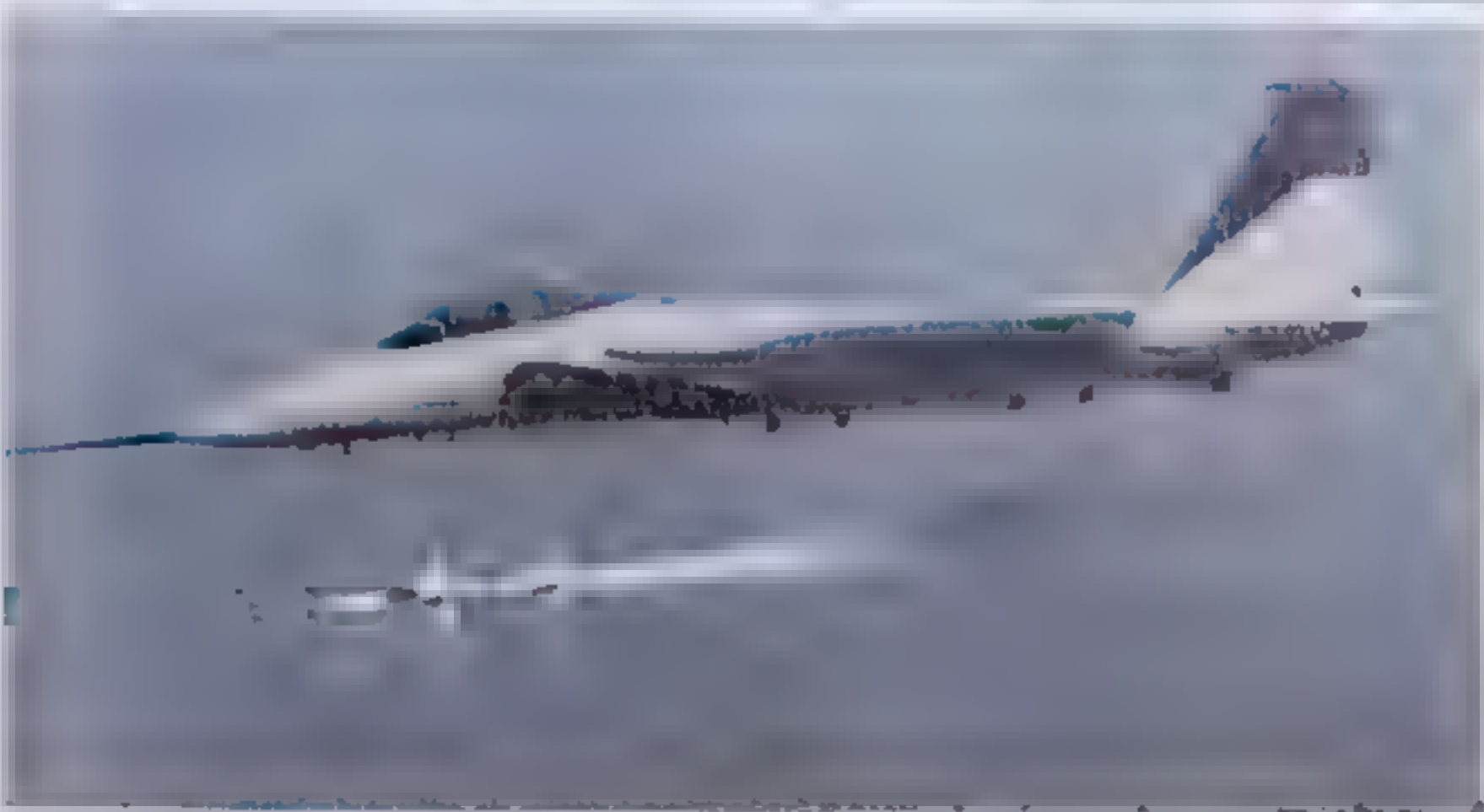
Engines: Two Pratt & Whitney F100-100 two-shaft augmented turbo-fan engines, 114,871 lb (51,440 kg) thrust each, 12,431 sq in (80,000 cm²) wet, maximum dry, 10,000 sq in (64,500 cm²) dry

Dimensions: Span 42ft 9½ in (13.05m), length 63ft 9½ in (19.45m), height 18ft 7½ in (5.68m)

Weights: Empty weight 12,500 kg (27,500 lb), gross weight (F-15A) 33,500 lb, (F-15C with four Sparrows) about 40,500 lb, (three 600gal drop tank) 54,000 lb, (three tanks and two FAST packs) 60,000 lb (27,232 kg)

Performance: Maximum speed low (over 921 mph (1482 km/h, Mach 1.22), (high) over 1,650 mph (2660 km/h, Mach 2.5), initial climb, over

continued on page 30



Launch of an AIM-7F Sparrow during weapon trials with F-15 development aircraft. Maximum Sparrow range is 28 miles

Below: 'Anything you can do, I can do better' - an Eagle lets go a stick of a dozen 1,000lb bombs in a dive-bombing attack, whilst carrying four Sparrow air-to-air missiles



Armament

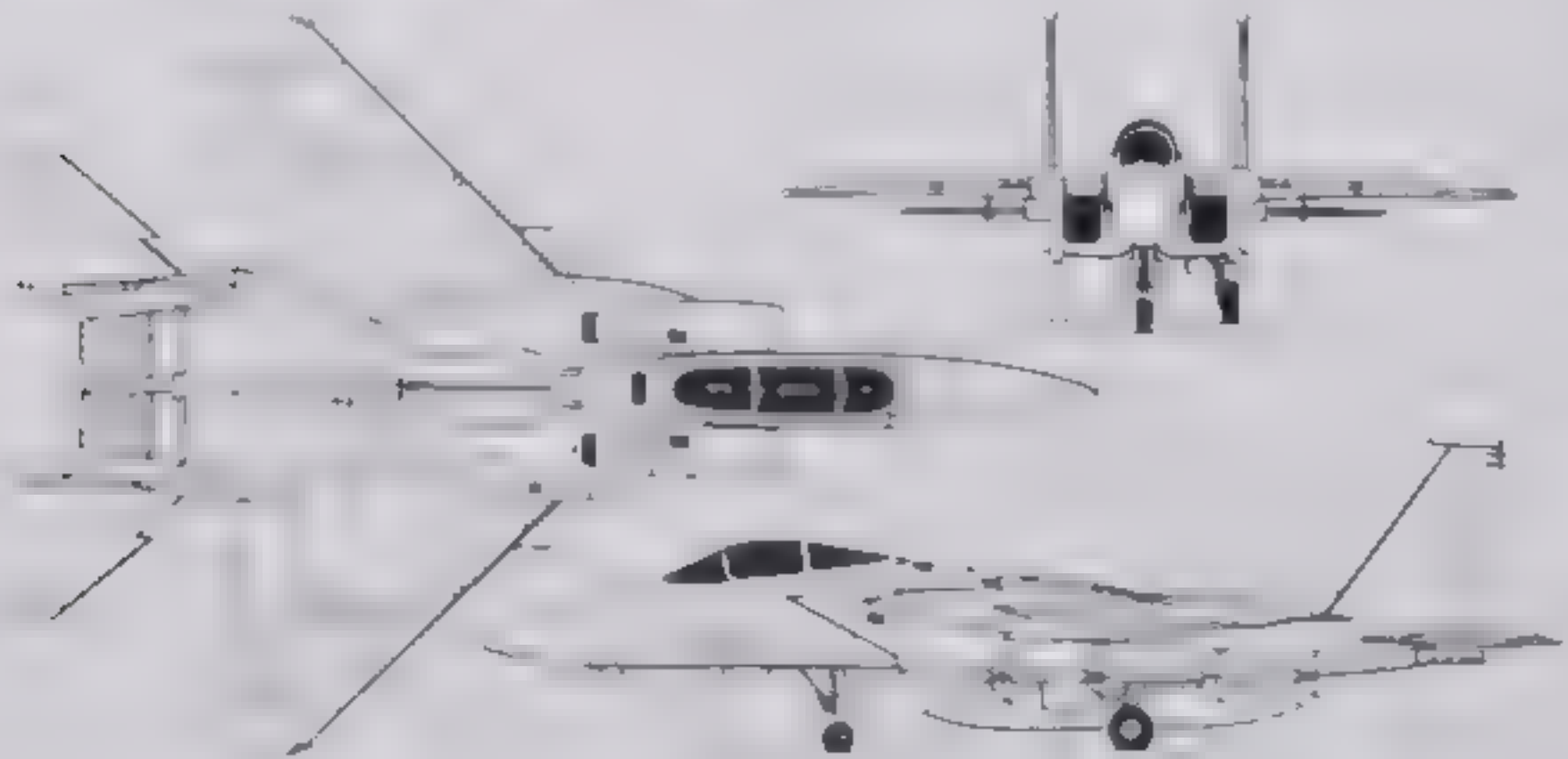
History: First flight 27 July 1972 (TF) 7 July 1973 service date

Users: Israel, USA (Air Force 1)

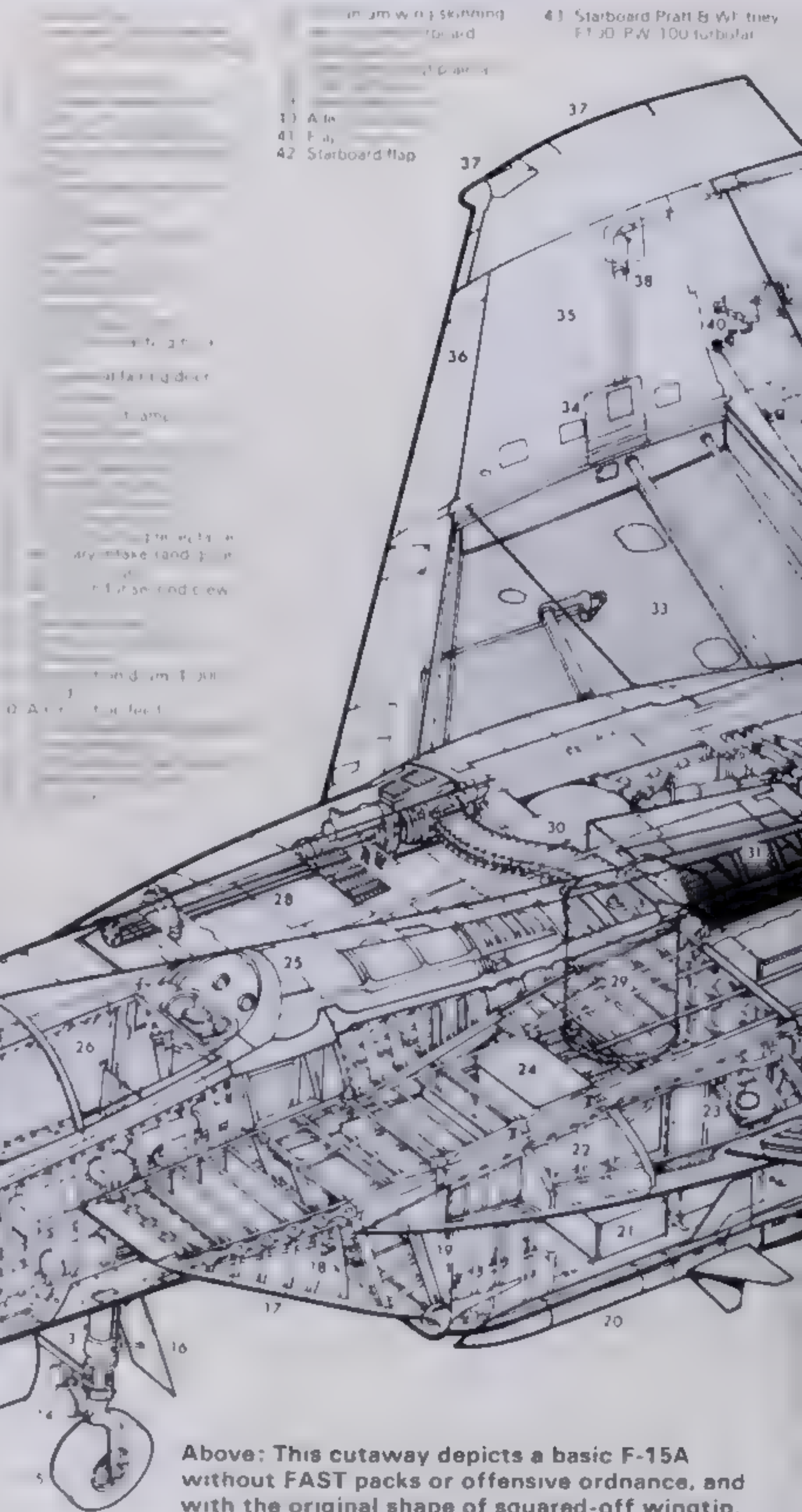
Development: Emergence of the MiG 23 and -25 in 1967 accentuated the fact that the USAF was falling behind in true fighter aircraft. The F-15 (a new air superiority fighter) were hastened and after a long development team at St Louis was selected to build the Eagle. The Eagle has emerged as probably the most powerful fighter with thrust at low levels considerably greater than the F-4. The fixed wing of no less than 530 sq ft area, a highly advanced Hughes X-band pulse doppler radar. Though a compromised machine for interception and air combat the Eagle also has formidable attack capability over intercontinental ranges. Under these attributes are its combat manoeuvrability (it can outfly any JS machine without using afterburner) and the advanced automaticity of its radar, head up display, weapon selectors and quick fire capability. Internal fuel capacity of 11 200lb can be almost trebled by adding a FAST (fuel and air tank system) pack on each side of a conformal pallet housing 10,000lb of fuel and target designators or weapons. Very extensive electronic systems for attack and defence far beyond any standard previously seen in a fighter are carried. A total USAF buy of 729 aircraft is planned, and though this has not changed since early in the programme the benefits of the "learning curve" (which reduces costs as production continues) are being much more than nullified by cost inflation. The unit price of \$7.5 million of 1975 had been more than doubled by late 1976 to over \$16.7 million, with a figure in excess of \$18 million predicted by Congress. Thus the 729 aircraft will now cost at least \$12.2 billion, a figure rising by \$500-700m each quarter. Nevertheless the outstanding qualities of this superbly capable fighter commend it to many governments. Israel has bought 21 new Eagles plus four reworked development aircraft, costing with support \$600 million, and in mid-1976 the F-15 was chosen by Japan as the FX for the Air Self-Defence Force (though as this book went to press no firm programme had been announced). McDonnell Douglas are selling hard to Australia, Canada, France, West Germany and Saudi Arabia.



In 1976 this TF-15 participated in airshows in many countries in this American Revolution Bicentennial livery. It flew the North Atlantic without air refuelling.



Above: Three view of production F-15A Eagle



Above: This cutaway depicts a basic F-15A without FAST packs or offensive ordnance, and with the original shape of squared-off wingtip. Though originally planned as an uncompromised dogfighter and all-weather interceptor, the Eagle has been proved to have outstanding capability in certain attack situations demanding very heavy bomb loads delivered at moderate speeds and heights



Arrival at an Israeli airbase of the first of 25 F-15A Eagles, still wearing USAF markings, on 10 December 1976. They cost \$600 million.



Not least of the contributors to the F-15's giant capability is made by the Hughes APG-63 radar, the world's newest in service. Here enlisted men inspect the planar-array scanner



Above: McDonnell Douglas YF-15 evaluation and test aircraft, 1972

McDonnell Douglas/Northrop

F-18 Hornet and Cobra

F-18, TF-18 and A-18

Origin: Original basic design Northrop Corp. prime contractor

Type: (F) single seat carrier based multi role fighter, (TF) single seat land based attack fighter

Engines: Two 16 000lb (7257kg) thrust General Electric F404 half augmented turbofan

Dimensions: Span (with missiles) 40ft 8½in (12.41m) height 14ft 9in (4.54m) length 56ft (17.07m)

Weights: (Provisional) empty 20 583lb (9336kg), loaded 22 152lb (10 046kg), maximum loaded (catapult limit) 50 064lb (22 708kg)

Performance: Maximum speed (clean at altitude) 1 190mph (1915km/h Mach 1.8), (maximum weight, sea level) subsonic sustained combat manoeuvre ceiling over 49 000ft (14 935m), absolute ceiling over 60 000ft (18 290m) combat radius (air to air mission, high, no external fuel) 461 miles (741km) ferry range not less than 2 300 miles (3700km)

Armament: One 20mm M61 Galling in upper part of forward fuselage nine external weapon stations for maximum load (catapult launch) of 14 000lb (6080kg), including bombs sensor pods, ECM, missiles (including Sparrow) and other stores with tip mounted Sidewinder

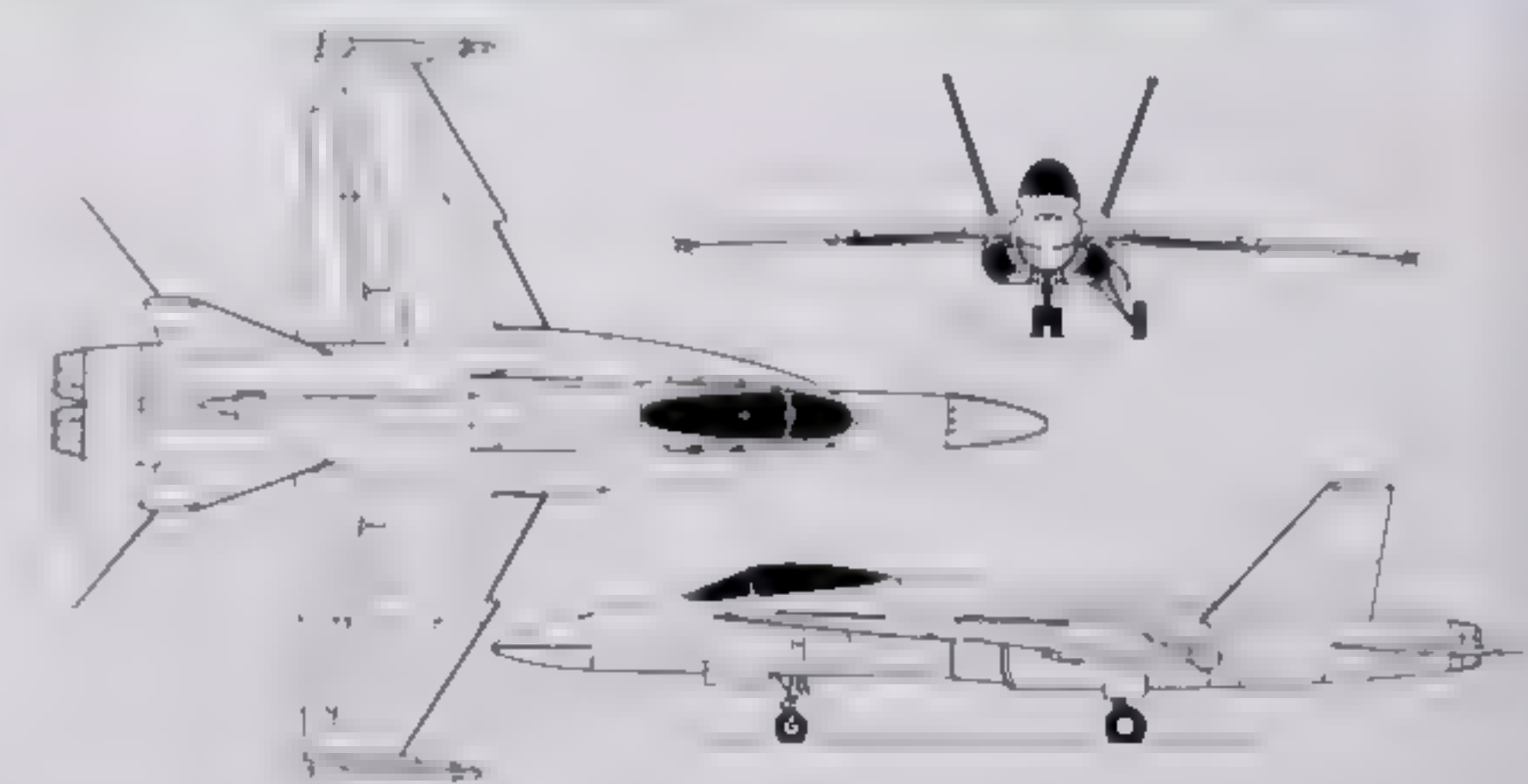
History: First flight (YF-17) 9 June 1974 (first of 11 test F-18) July 1978 (production F-18) 1980

User: USA (Navy Marine Corps)

Development: In 1971 the US Navy became concerned at the cost of the F-14 and the resulting reduced rate of procurement and total number that could be afforded. In 1973 it studied low cost versions and compared them with navalised F-15 versions and improved F-4s. In 1974 the VFX specification emerged for a wholly new and smaller fighter somewhat along the lines of the Air Force Air Combat Fighter. In May 1975 the Navy and Marine Corps announced their choice of the F-18, developed from the existing land-based Northrop F-17 by McDonnell Douglas and Northrop. In fact the F-18 will be almost twice as heavy as the original F-17 proposal but with more powerful engines, is expected to have adequate dogfight performance through the 1980s. Features include an unswept wing with large dogteeth and forebody strakes at the roots, twin canted vertical tails, simple fixed engine inlets and extensive graphite/epoxy structure. Search radar will be used in the interception and surface-attack roles, and a very wide range of weapons will be carried. In the Navy air-superiority mission the gun will be backed up by two Sparrows and two Sidewinders, and the F-18 is expected to show dramatic improvements over the F-4J in manoeuvrability, reliability and low cost. In Marine attack missions the maximum load can be 14 000lb for airfield operation, and the inertial guidance and weapon aiming are expected to offer a significant advance over the accuracy of any A-7. The Navy/Marines plan to buy 11 development aircraft plus 800 production machines by the end of 1990 at a unit cost of only \$5 924 million in 1975 dollars, without spares. About every ninth aircraft will be a



Above: Takeoff of one of the two Northrop YF-17 development aircraft. These are being used to refine a relaunched Cobra.



Above: Three-view of F-18 Hornet (A-18 to be similar).

TF mission trainer, with 500lb (227kg) less fuel, rear cockpit (without HUD) and larger canopy

In late 1976 Northrop – original designer of the YF-17 but a mere subcontractor on the F-18 – was trying to relaunch the land-based Cobra, but now as a modified F-18. Despite severe competition from the F-16 and other aircraft, Northrop aims to find worldwide sales for the Cobra replacing the F-4, F-104, A-7 and Mirage. It would have less internal fuel than the F-18, and thus even higher performance. Planned export delivery date is 1982, at \$8 million in 1975 dollars.

Below: Artist's impression of single-seat Navy F-18 Hornet.



Mikoyan/Gurevich MiG-17

MiG-17, -17P, -17F (Lim-5P and -5M, S-104, F-4), -17PF and -17PFU (NATO name "Fresco")

Origin: The design bureau of Mikoyan and Gurevich Soviet defence production as described in the text

Type: Single-seat fighter, (PF, PFU) limited all weather interceptor

Engine: (-17, -17P) one 5 952lb (2700kg) thrust Klimov VK-1 shaft centrifugal turbojet, (later versions) one 4 732/7 400 lb (2 147/3 340 kg) VK-1F with afterburner

Dimensions: Span 31ft (9.45m), length (all) 36ft 3in (11.05m) height 11ft (3.35m)

Weights: Empty (all) about 9,040lb (4100kg), loaded (F clean) 11 773lb (5340kg), maximum (all) 14 770lb (6700kg)

Performance: Maximum speed (F, clean at best height of 9 840ft 711mph (1145km/h), initial climb 12 795ft (3900m)/min, service ceiling 54,460ft (16,600m), range (high, two drop tanks) 913 miles (1470km)

Armament: (-17) as MiG-15, one 37mm and two 23mm NS-23, (all later versions) three 23mm Nudelmann-Rikter NR-23 cannon, one under right side fuselage and two under left wing hardpoint. For tank fighters 1,102lb (500kg) of bombs, packs of eight 55mm air-to-air rockets or various air-to-ground missiles

History: First flight (prototype) January 1950, service delivery, 1952 service delivery (F-4) January 1956, final delivery (Soviet Union) probably 1959

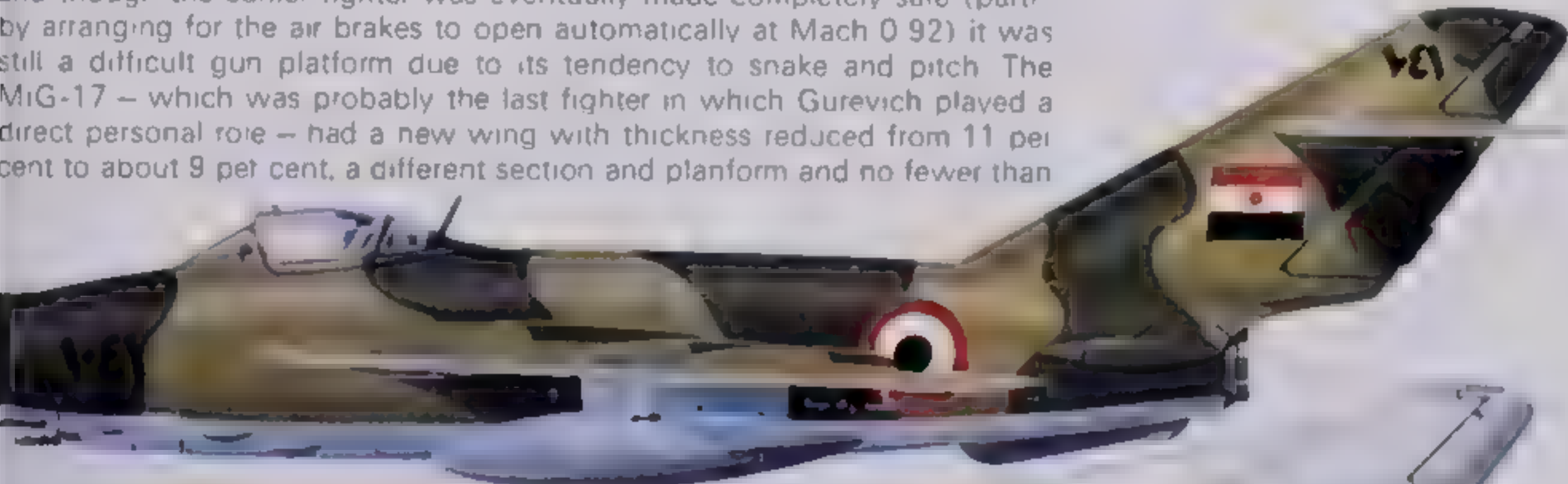
Users: Afghanistan, Albania, Angola, Argentina, Bulgaria, Cambodia, Czechoslovakia, Egypt, E Germany, Guinea, Hungary, Indonesia (in storage), Iraq, Kampuchea, N Korea, Mali, Morocco (in storage), Nigeria, Poland, Romania, Somalia, S Yemen, Soviet Union, Sri Lanka, Sudan, Syria, Tanzania, Uganda, Vietnam, Yemen Arab

Development: Only gradually did Western observers recognise the MiG-17 as not merely a slightly modified MiG-15 but a completely different aircraft. Even then it was generally believed it had been hastily designed to rectify deficiencies shown in the MiG-15's performance in Korea, but in fact the design began in about January 1949, long before the Korean war. This was because from the first the MiG-15 had shown bad behaviour at high speeds and though the earlier fighter was eventually made completely safe (partly by arranging for the air brakes to open automatically at Mach 0.92) it was still a difficult gun platform due to its tendency to snake and pitch. The MiG-17 – which was probably the last fighter in which Gurevich played a direct personal role – had a new wing with thickness reduced from 11 per cent to about 9 per cent, a different section and planform and no fewer than



Above: Three-view of typical MiG-17F (NATO name, "Fresco C")

three fences. Without taper and with inboard sweep of 47° the difference to high-Mach behaviour and in fact there are reasons. The MiG-17 can be divided to make a sonic bang. With a new tail, the rear fuselage and tail section were completely redesigned. New systems and equipment, though at first the VK-1 engine was unchanged. In 1958 the first limited all-weather version, the -17P, went into modest production with longer nose housing the same Izumrud ("Scan Odd") A radar and ranging avionics as was also in production for the MiG-19. With the introduction of an afterburning engine the airbrakes were moved aft, the wing away from the tail braked but this was not a good position and they were returned (in enlarged rectangular form) to the tail in the most important sub-type the -17F. This was made in Poland as the Lim 5P (the -5M being a rough-field close-support version with larger tyres and drag chute), in Czechoslovakia as the S-104 and in China as the F-4. The PFU was the afterburning all-weather version and the final model was the PFL with guns removed and wing pylons for four beam-riding "Alkali" air-to-air missiles. Total production for at least 22 air forces must have considerably exceeded 5 000, exports from China alone exceeding 1 000. Many 17s remained in use in the mid-1970s.



Above: MiG-17F of Syrian Air Force (Federation insignia).



Above: Chinese-built F-4 of Kampuchea (formerly Cambodia) Air Force.

Bottom left: Early MiG-17 fighters in service with the AV-MF (Soviet naval air force) Black Sea Fleet, around 1953. At first Western intelligence mistakenly thought the MiG-17 merely a modified -15, the pioneer swept-wing MiG that was such a shock when encountered by Allied pilots over Korea in 1950–51. The MiG-17 did not participate in the Korean war.

Below: A dramatic picture, taken by the combat camera of a USAF F-4 Phantom, showing the last moment of a MiG-17F of the Vietnam People's Air Force. The white shape near the left tailplane is a Sidewinder missile. The MiG is in a tight turn at about 400 knots at low/medium level, typical dogfight conditions.



Mikoyan/Gurevich MiG-19

MiG-19, -19S, -19SF (Lim-7, S-105, F-6),
-19PF and -19PM; NATO name "Farmer"

Origin

Type
Engines

Dimensions

Weights

Performance

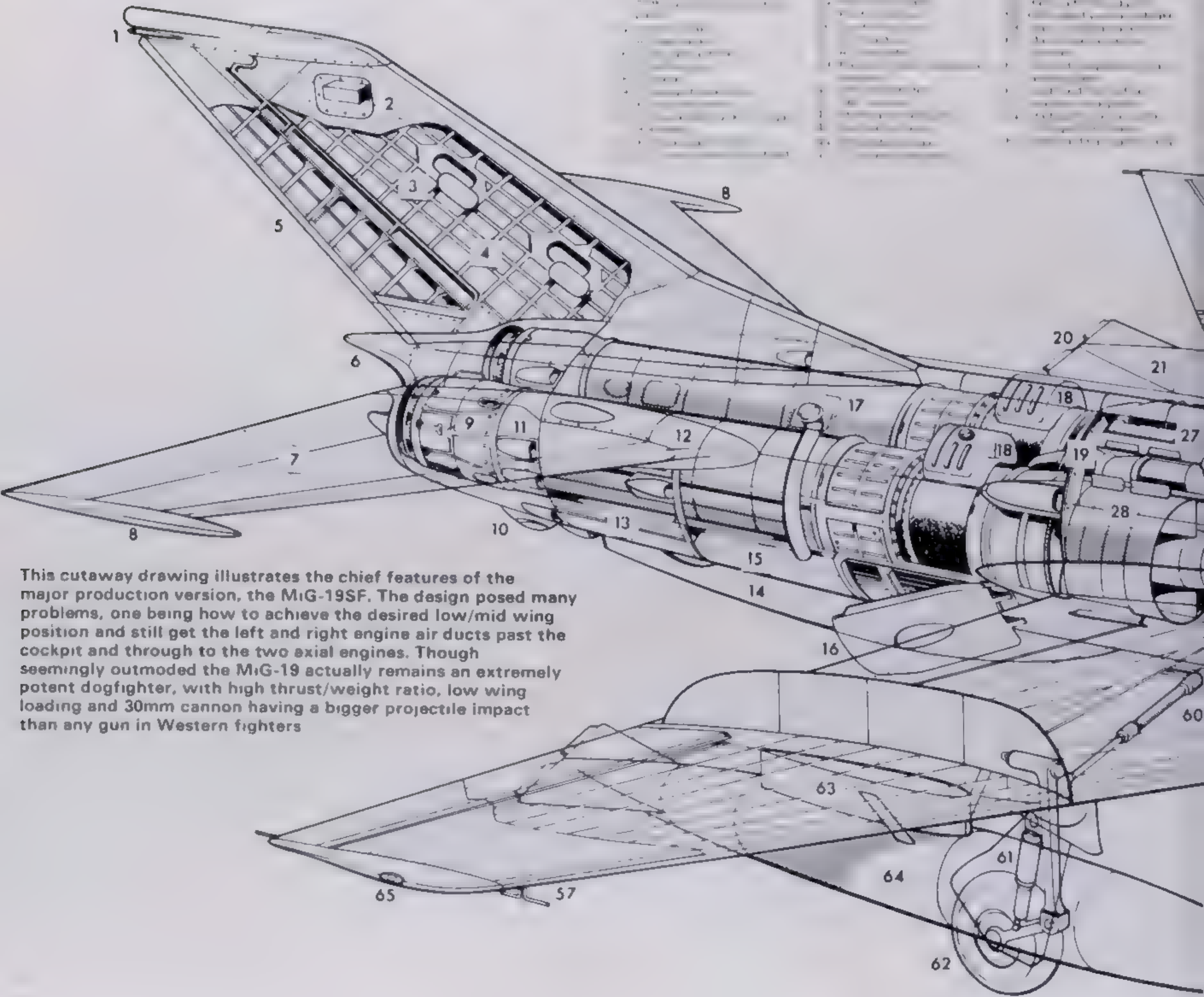
Armament
History

Users

Development: With



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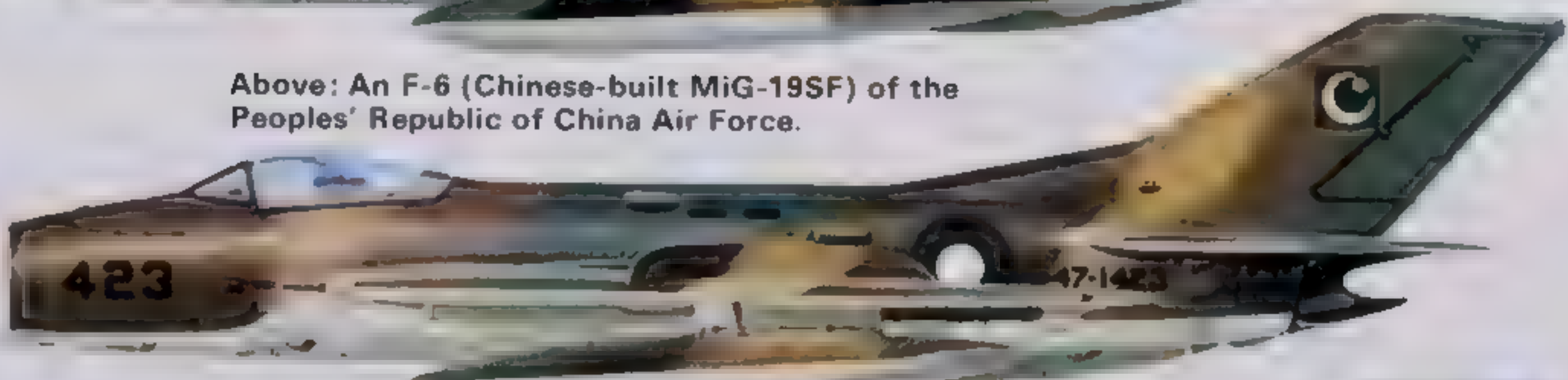
This cutaway drawing illustrates the chief features of the major production version, the MiG-19SF. The design posed many problems, one being how to achieve the desired low/mid wing position and still get the left and right engine air ducts past the cockpit and through to the two axial engines. Though seemingly outmoded the MiG-19 actually remains an extremely potent dogfighter, with high thrust/weight ratio, low wing loading and 30mm cannon having a bigger projectile impact than any gun in Western fighters



Above: A Chinese-built F-6 of the Pakistan Air Force, photographed by hand-held camera from the cockpit of an aircraft in loose formation. The F-6 gave a creditable account of itself in the warfare with India.

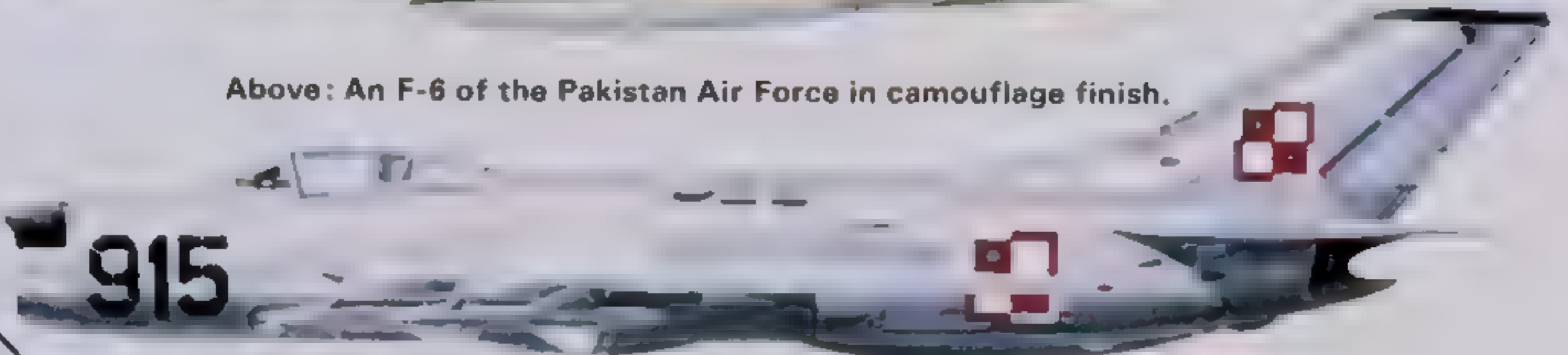


Above: An F-6 (Chinese-built MiG-19SF) of the Peoples' Republic of China Air Force.

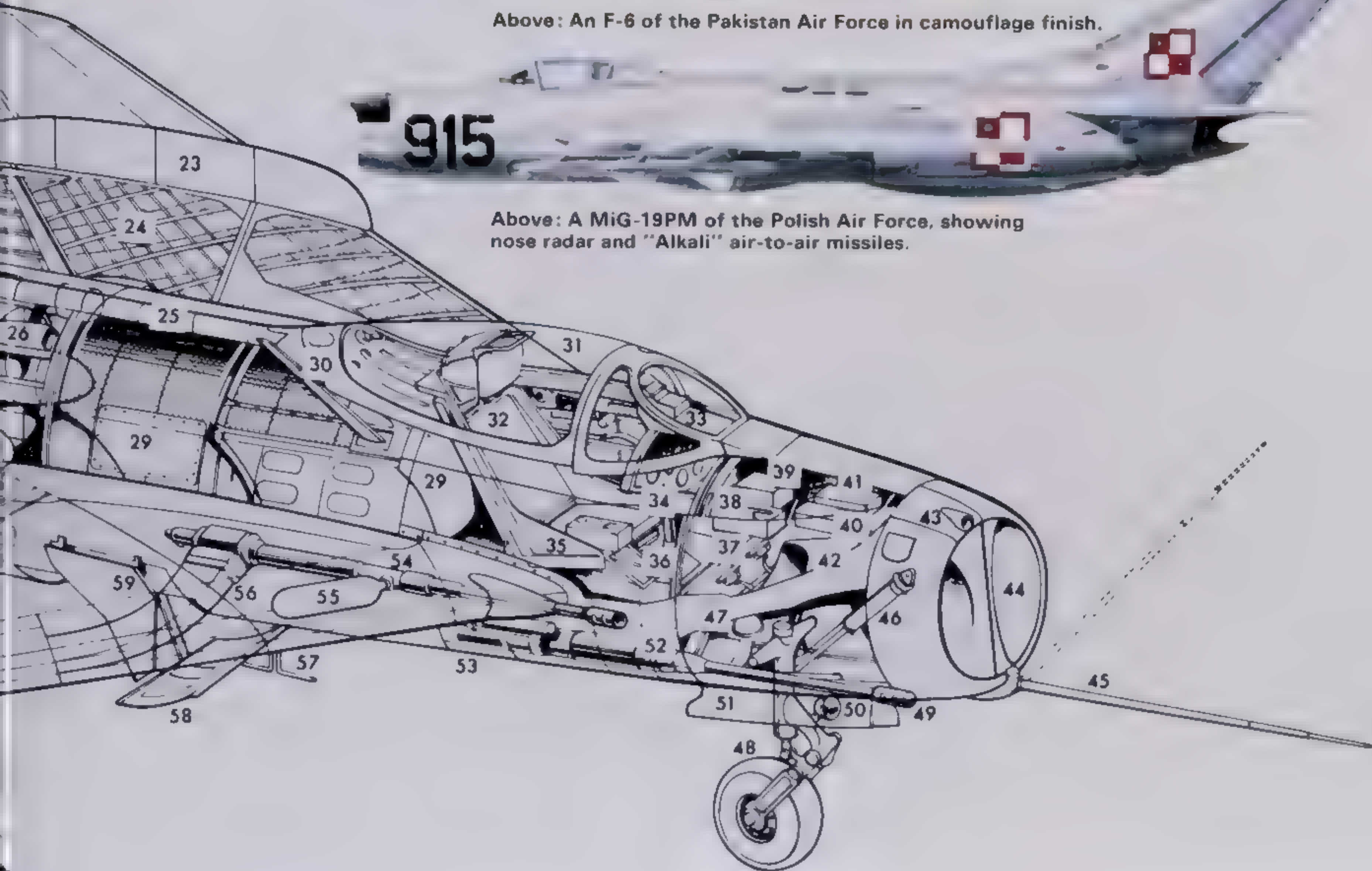


Left, above: One of the few good air-to-air photographs of a MiG-19 in Soviet IA-PVO service, this shows the PM interceptor with beam-riding "Alkali" missiles.

Above: An F-6 of the Pakistan Air Force in camouflage finish.



Above: A MiG-19PM of the Polish Air Force, showing nose radar and "Alkali" air-to-air missiles.



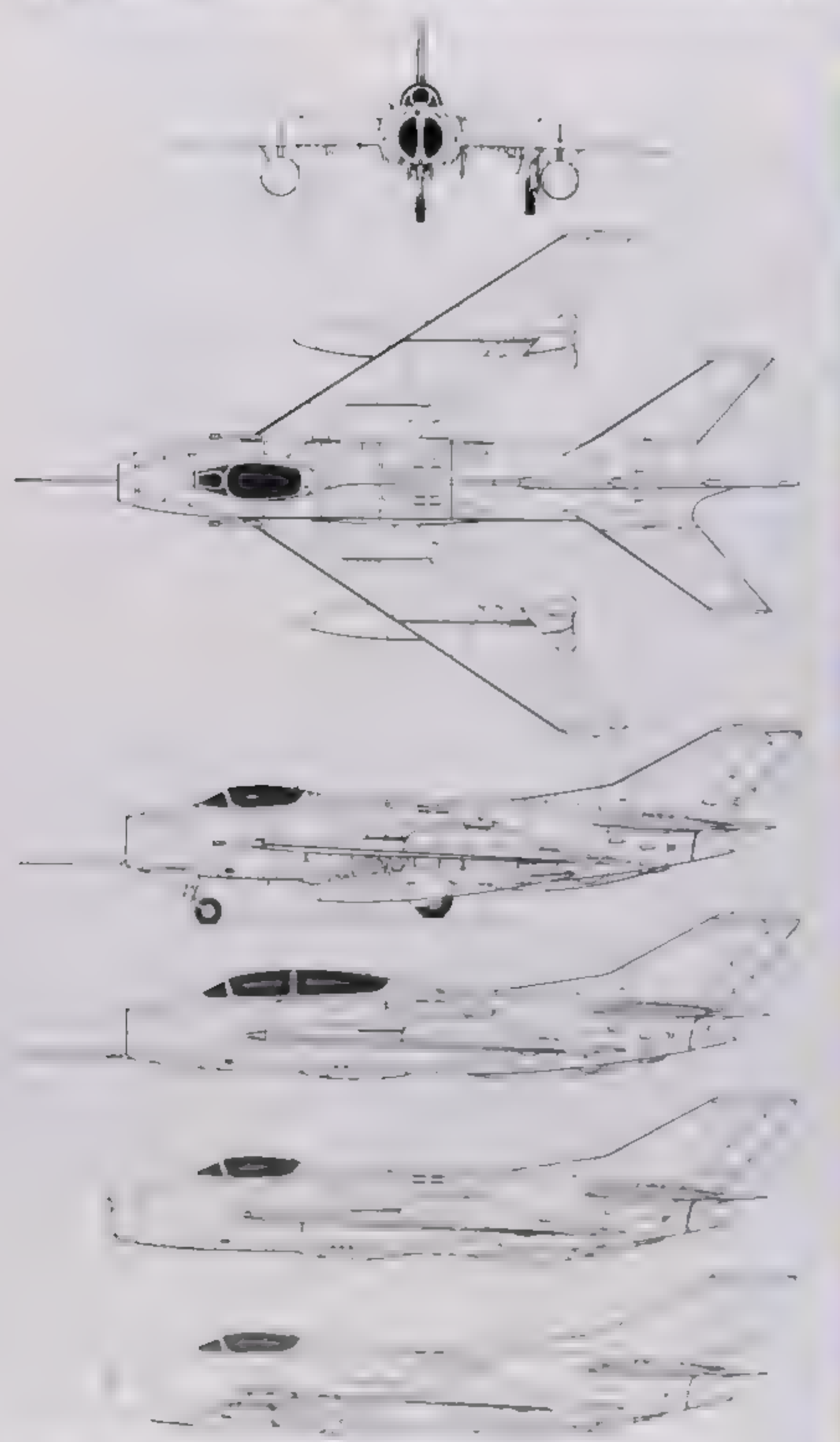


FIGURE 1: MiG-23 top three views, F-6 (-19SF) then (top to bottom) MiG-19, 19PF and 19PM with missiles

Mikoyan/Gurevich MiG-23

MiG-23, -23S and -23U ("Flogger")

Origin: The Mikoyan-Gurevich design bureau, Moscow, Soviet Union

Type: (-23S, Flogger B) single-seat all-weather interceptor with Flogger 1 export variant of unknown designation (-23U Flogger C) dual control trainer and fighter

Engine: One Tumansky afterburning turbofan of unknown type rated at about 14,500 b (6577kg) dry and 20,500lb (9300kg) with maximum afterburner

Dimensions: (Estimated) Span (72° sweep) 28ft 7in (8.7m) (11ft 10in (3.6m) length, (S, U) 16.15m), (S, U) 55ft 1½in (16.80m) height 13ft (3.96m)

Weights: (Estimated) empty 17,500lb (7940kg), loaded (clean or fighter mission) 30,000lb (13,600kg), maximum permissible 33,000lb (15,000kg)

Performance: Maximum speed, clean, 840mph (1350km/h, Mach 1.1) at sea level, maximum speed, clean, at altitude 1,520mph (2445km/h, Mach 2.2), maximum Mach number with missiles (MiG-23S) about 2.2, service ceiling about 55,000ft (16,765m), combat radius (hi-lo-hi) about 400 miles (640km)

Armament: (-23S) one 23mm GSh-23 twin-barrel gun on ventral fuselage, plus various mixes of air/air missiles which usually include AA-7 "Apex" and/or infra-red or radar-homing AA-8 "Aphid", the latter for close combat, (-23U) none reported

History: First flight, probably 1965, (first production aircraft) believed 1967, delivery, believed 1971

Users: East Germany, Iraq, Libya, Poland, Soviet Union, Syria (and a most certainly other countries in 1977)

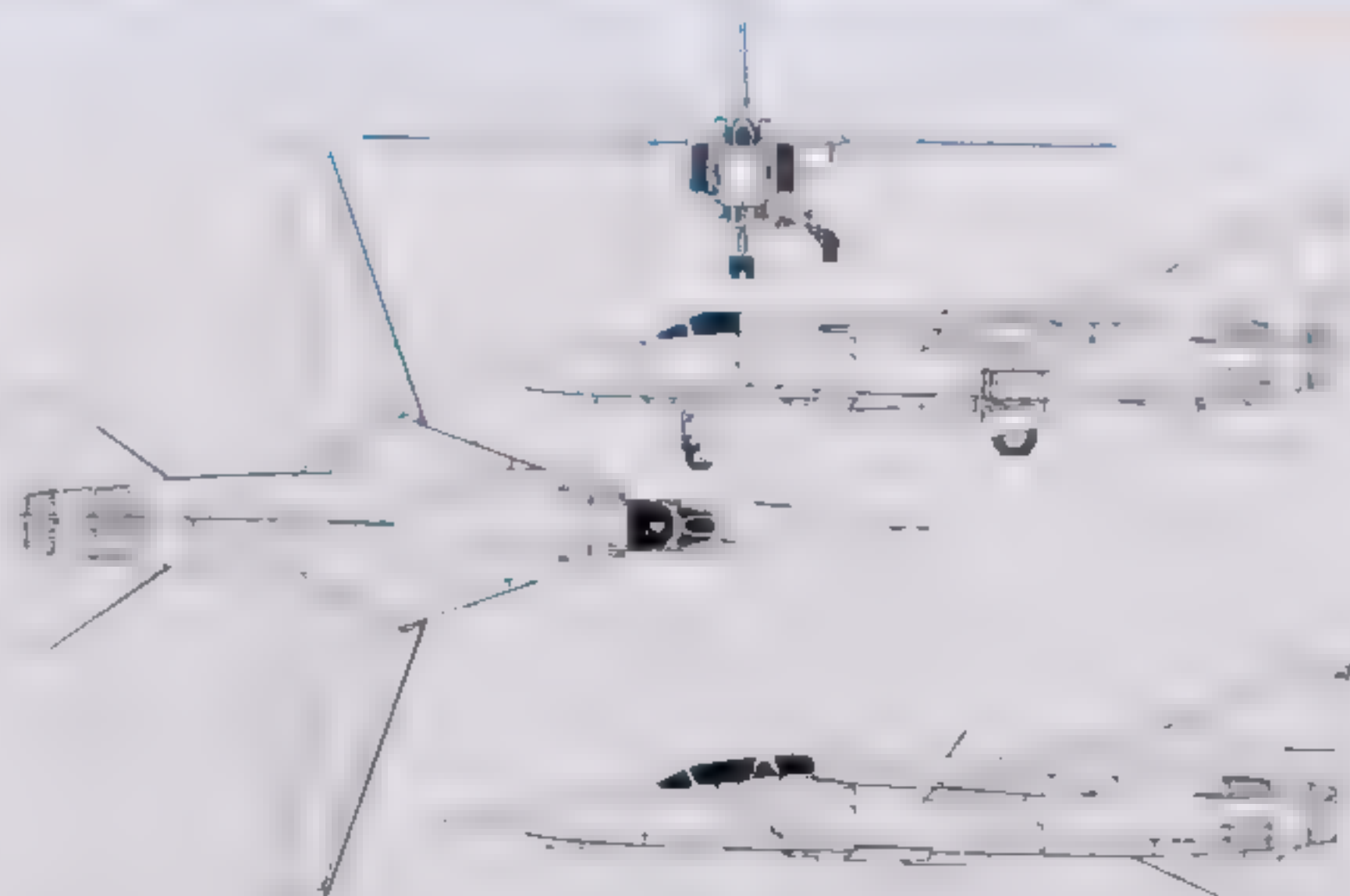
Development: Revealed at the 1967 Moscow Aviation Day, the prototype swing-wing MiG-23 was at first thought to be a Yakovlev design, though it appeared in company with a jet-lift STOL fighter having an identical rear fuselage and tail and strong MiG-21-like features (though much bigger)



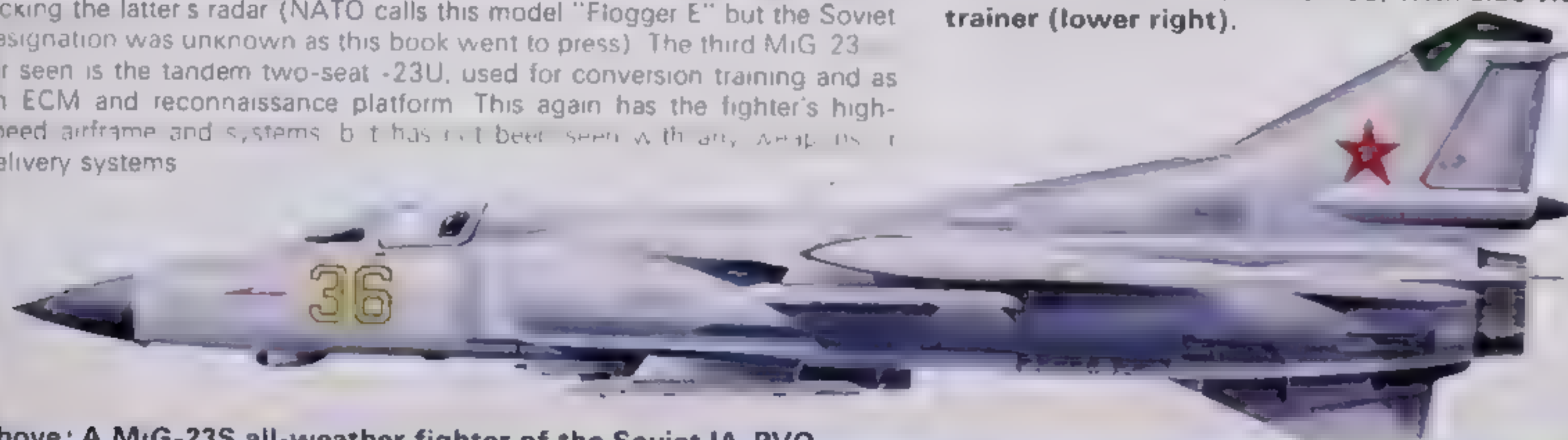
than a MiG-21). Over the past few years the MiG-23 has been developed this aircraft which is a very different aircraft from the Mirage G. By 1971 the MiG-23 was being developed as a fighter and -23U trainer, were entering service in quantity and by 1973 several hundred had been delivered to Warsaw Pact air forces and Egypt. Today Egypt is believed no longer to operate the type, but it is believed that some have been made to other countries. The MiG-27 also has been made to other countries.

There are three main versions. The first to enter service was the MiG-23S, an all-weather interceptor, with powerful highly-afterburning engines, a high "Lark" nose radar (said in 1973 by the then Secretary of the Soviet Air Force to be comparable with that of the latest Phantom") and an ECM system, a laser ranger and doppler navigator, ECM and other EW equipment, markedly superior to anything fitted in previous Soviet aircraft and apparently as good as comparable installations in Western fighters (other than the F-15).

Several hundred S models are in service with the IA-PVO and Warsaw Pact air forces, and they are replacing the Su-9 and -11 and Yak-28P. Missiles are carried on a centreline pylon (which often carries a drop-tank instead), on pylons under the inlet ducts and under the fixed wing gloves (centre section). For overseas customers a simplified sub-type is in production with the same high-Mach airframe and systems as the -23S fighter but lacking the latter's radar (NATO calls this model "Flogger E" but the Soviet designation was unknown as this book went to press). The third MiG-23 seen is the tandem two-seat -23U, used for conversion training and as an ECM and reconnaissance platform. This again has the fighter's high-speed airframe and systems but has not been seen with any weapons or delivery systems.



Above: Three-view of MiG-23S, with side view of MiG-23U trainer (lower right).



Above: A MiG-23S all-weather fighter of the Soviet IA-PVO.

Below: Called "Flogger E" by NATO, this export version of the MiG-23B is fitted with a radar of lower power, with a smaller radome. This one belongs to the Libyan Air Force, which also has the MiG-23U trainer.



Mikoyan/Gurevich MiG-21

MiG-21, 21F (S-107), 21FA, 21PF, 21FL, 21PFS, 21PFM, 21PFMA, 21M, 21R, 21MF, 21SMT, 21bis, 21U, 21US and 21UM plus countless special versions. Several versions made in China as F-8

Origin

Type

Engine

Dimensions

Weights

Performance

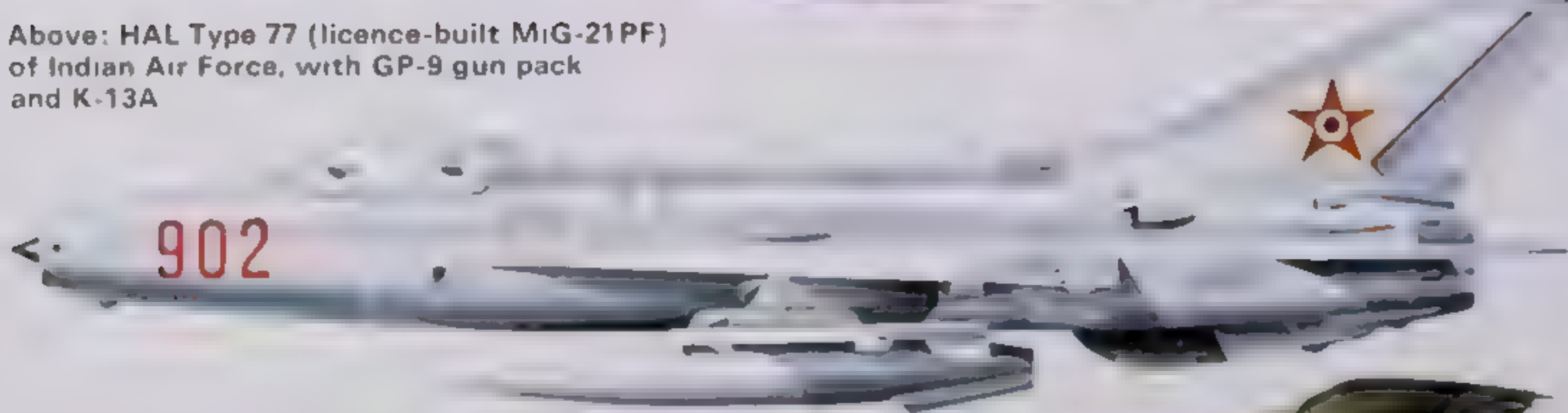
Armament
History:

Users

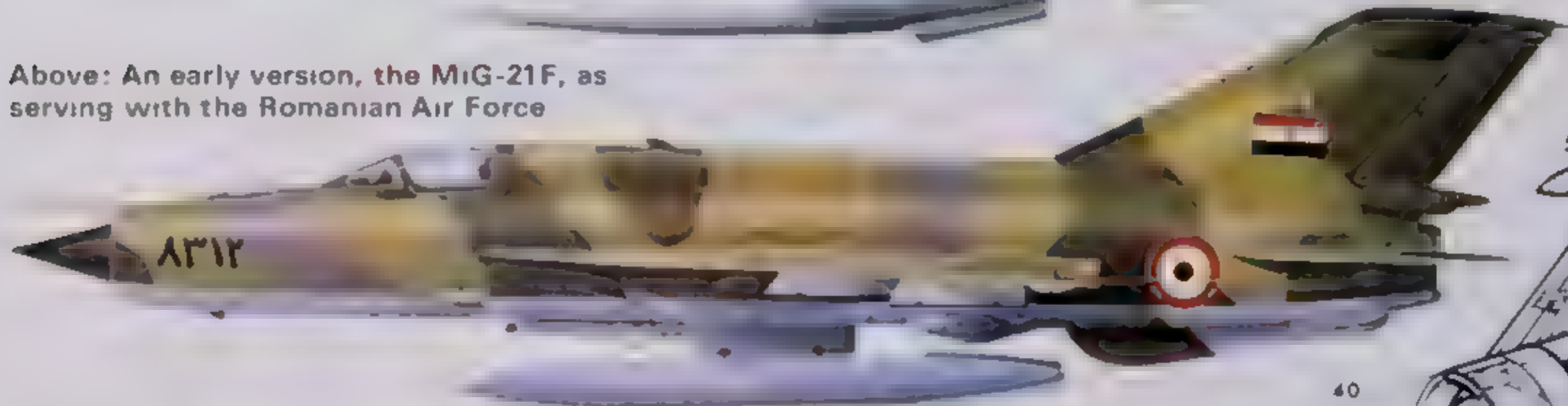
Development
world in the 19



Above: HAL Type 77 (licence-built MiG-21PF) of Indian Air Force, with GP-9 gun pack and K-13A

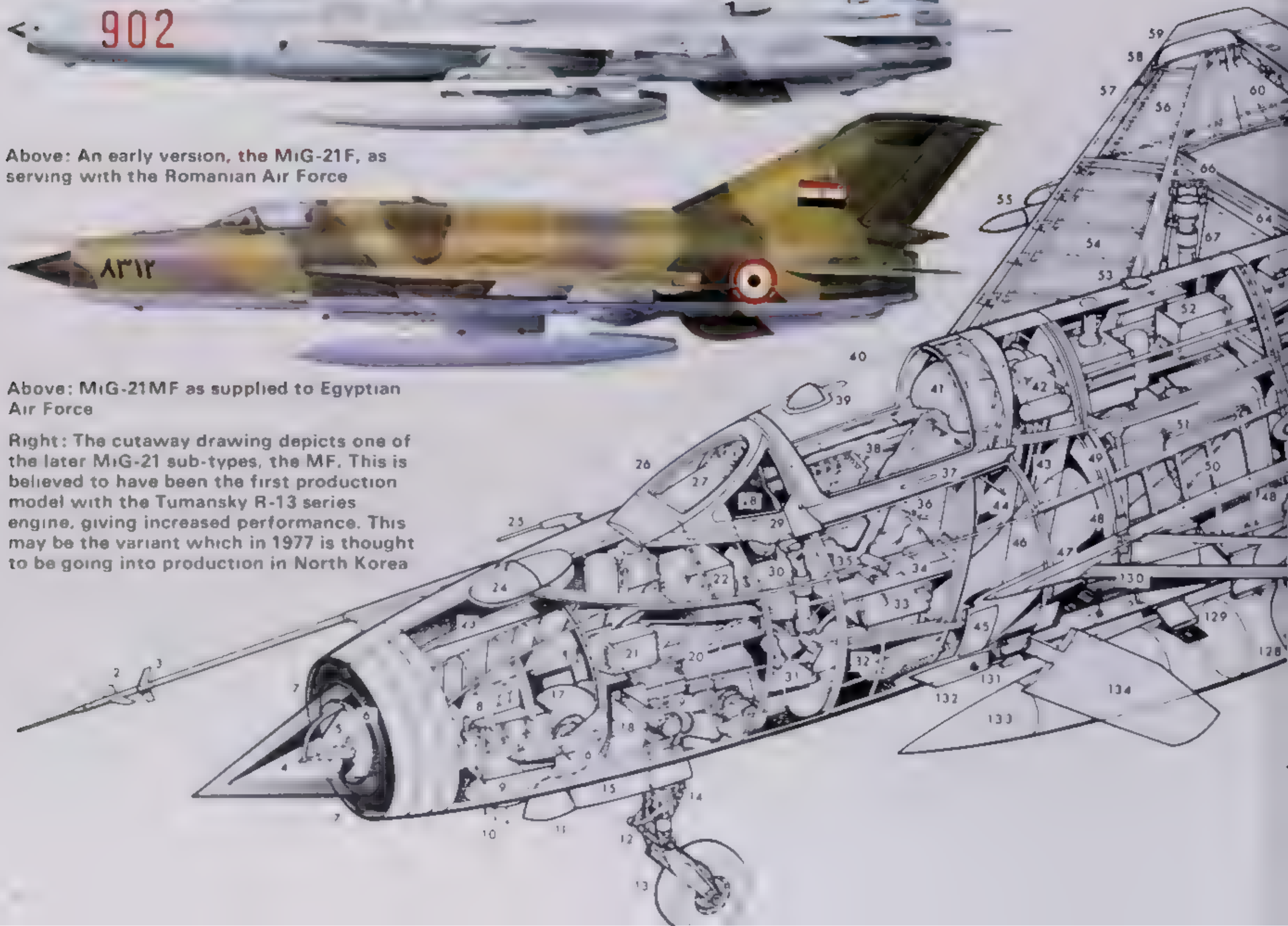


Above: An early version, the MiG-21F, as serving with the Romanian Air Force



Above: MiG-21MF as supplied to Egyptian Air Force

Right: The cutaway drawing depicts one of the later MiG-21 sub-types, the MF. This is believed to have been the first production model with the Tumansky R-13 series engine, giving increased performance. This may be the variant which in 1977 is thought to be going into production in North Korea



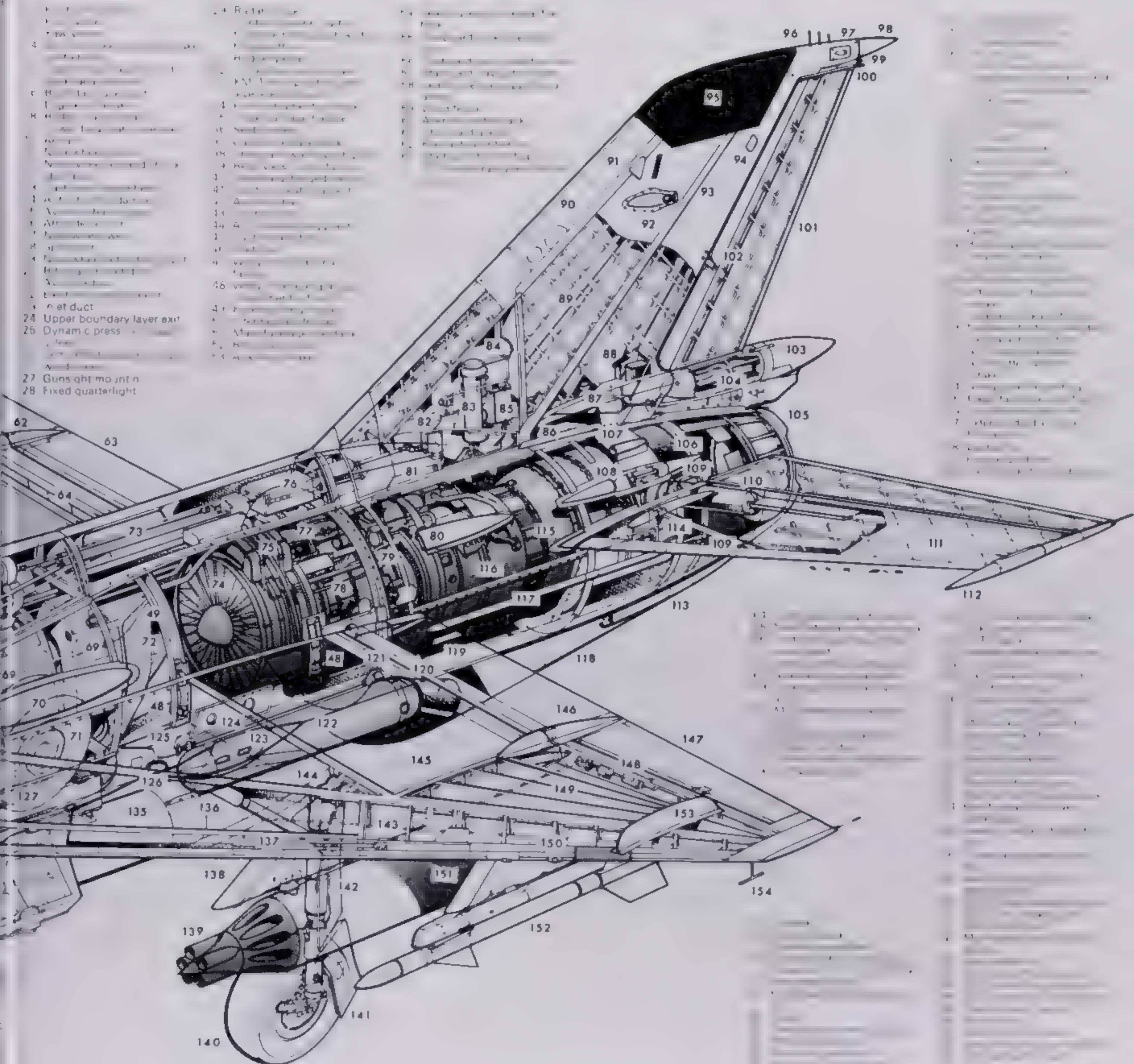
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Above: Three-view of MiG-21SMT ("Fishbed K") with four K-13A missiles.



Above: The chief sub-type of the MiG-21 in service with the Soviet IA-PVO is called "Fishbed L" by NATO, its exact Soviet designation being unknown. It is essentially an MF with a Tacan-type navigation/homing system.



Mikoyan/Gurevich MiG-25

Mig-25 ("Foxbat A"), -25R and -25U

Origin: The design bureau named for Mikoyan and Gurevich, Soviet Union

Type: "Foxbat A" (believed to be MiG 25S), all weather long range fighter, MiG 25R reconnaissance, MiG 25U tandem seat dual engine with stepped cockpit

Engines: Two Tumansky R 266 afterburning turbojets each rated at 4 500 lb (11 110kg) with full augmentation

Dimensions: Span 46ft (14 0m) length ("A") 73ft 2in (22 3m), (R) 74ft 6in (22 7m), (U) about 76ft (23 16m), height 18ft 6in (5 63m)

Weights: (Fighter) empty 44 000lb (19 960kg), normal loaded 68 350lb (31 000kg), maximum loaded with external missiles or tanks 77 000lb (34 930kg)

Performance: (Estimated) maximum speed at altitude 2 100mph (80km/h Mach 3 2), initial climb about 50 000ft (15 240m)/min, ceiling 73 000ft (22 250m) high altitude combat radius without external fuel, 700 miles (1130km)

Armament: ("A") four underwing pylons each carrying one AA 6 air to air missile (two radar, two infra red) or other store, no guns ("B") none

History: First flight (E-266 prototype) probably 1964, (production reconnaissance version) before 1969, (production interceptor) probably 1969, service delivery (both) 1970 or earlier

User: Soviet Union

Development: This large and powerful aircraft set a totally new level in combat aircraft performance. The prototypes blazed a trail of world records in 1965-67 including closed circuit speeds, payload-to-height and rate of climb records. The impact of what NATO quickly christened "Foxbat" was unprecedented. Especially in the Pentagon, Western policymakers recognised that here was a combat aircraft that outclassed everything else, and urgent studies were put in hand for a new US Air Force fighter (F-15 Eagle) to counter it. By 1971 at least two pairs of reconnaissance aircraft were flying with impunity over Israel, too high and fast for Phantoms to catch, while others have made overflights deep into Iran. This version is different in many respects, the nose having cameras instead of a "Fox Fire" radar and other sensors being carried under the large body. Both versions have twin outward sloping vertical tails, single mainwheels and a flush canopy shaped for speed rather than pilot view. From the start the main development effort has been applied to the basic MiG-25 (so-called "Foxbat A") interceptor, which has been developed in structure, systems and armament since first entering service with the PVO. In 1975 the original AA-5 missiles were supplemented, and later replaced, by the monster AA-6 "Acrid", which is easily the biggest air/air missile in service in the world. The radar-homing version has a length of about 20ft 2in (6 15m) and effective range of 28 miles (45km), the infra-red missiles have a length of just over 19ft (5 8m) and range of some 12 5 miles (20km). Another major improvement since entering service is flight-refuelling capability, not yet fitted to all MiG-25 versions. The detailed inspection of an interceptor version landed at Hakodate AB, Japan, on 6 September 1976, showed that in service pilots are forbidden to use the limits of the available flight performance, presumably to avoid thermal fatigue of the airframe, it also showed this particular machine to have early "Fox Fire" radar comparable in basic technology with the AWG-10 Phantom radar (as would be expected). Radars in current production are unquestionably solid state pulse-doppler types able to look down and track low-flying aircraft against ground clutter. Several MiG-25s, most of them MiG 25R models on ELINT missions, have been plotted by Western radars at Mach 2 8. It should be emphasized that at this speed the MiG 25 - and any other aircraft - flies in a straight line. The MiG-25 was not designed for air combat, and if it became involved in a dogfight its speed would - like any other aircraft - soon be subsonic. The MiG-25U trainer carries neither weapons nor sensors, but is needed to convert pilots to what is still, 15 years after design, a very advanced and demanding aircraft.



Above: Three-view of MiG-25 ("Foxbat A") with side view (bottom) of -25U

and not fuel and holding tankage to 572gal) and four pylons for K 13 and two 500lb bombs, four S 24 missiles and or tanks or K 13A missiles. The 21M has an internal GSh 23 and since 1973 has been built in India as Type 88. The 21R has multi sensor reconnaissance capability and in pods and wing tip ECM fairings, as do late models of the 21B. The first to have the new R 13 engine. The RF is the R 13 powered version. The latest variant still in production is the SMT (Supermaneuvering Trainer) to the spine and more comprehensive avionics including

Monqol and called Type 66 in India, the U is the tandem seat version with SPS flaps and UM the R 13 engine and four pylons. The 21M and 21R have been used to set world records. About 10 000 of all sub types have been built, and in 1977 output was continuing at perhaps three per week in the Soviet Union with a much lower rate in India. In early 1976 N Korea was said to be also in production. Many of the pilots of this neat fighter were sweet to handle and quite effective day dogfighters, but the majority of the subtypes in use have many adverse characteristics and severe limitations.

In late 1976 a new version appeared the MiG-21bis (Fishbed L), this is a cleaned up and refined MiG 21MF with Tacan type navigation and other improvements.



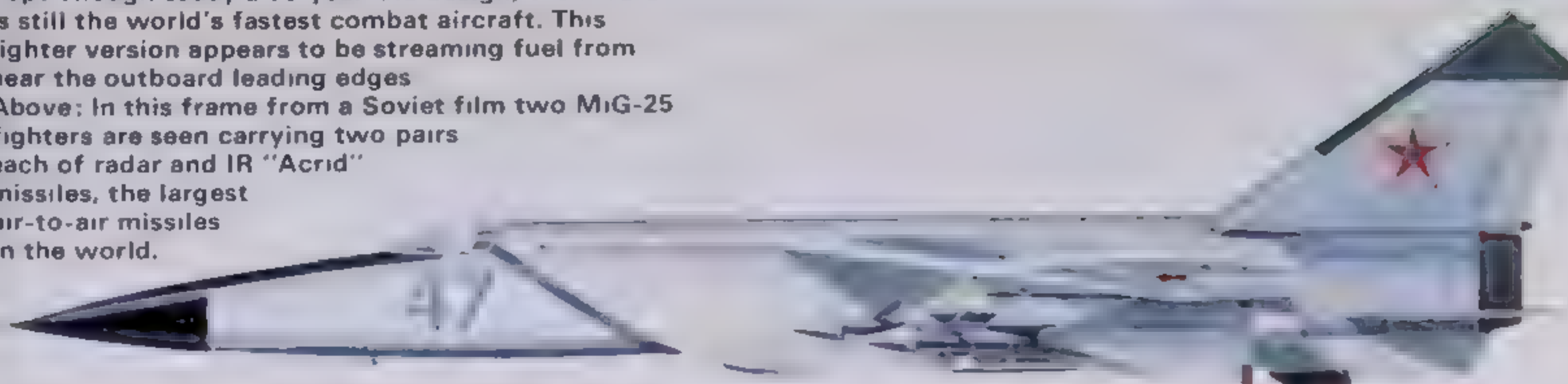
Top: This frame from a Soviet motion picture shows a MiG-21, probably a PFM, blasting off with the aid of two large JATO (jet-assisted takeoff) boost rockets, used by many versions

Above: Hardware counts for more than paint, and the variable (and crude) paintwork on these HAL-built Type 77 (MiG-21FL) does no harm. This was the first model made in India



Top: Though today a 15-year-old design, the MiG-25 is still the world's fastest combat aircraft. This fighter version appears to be streaming fuel from near the outboard leading edges.

Above: In this frame from a Soviet film two MiG-25 fighters are seen carrying two pairs each of radar and IR "Acrid" missiles, the largest air-to-air missiles in the world.



Above: Side elevation of MiG-25 interceptor, with 20ft missiles

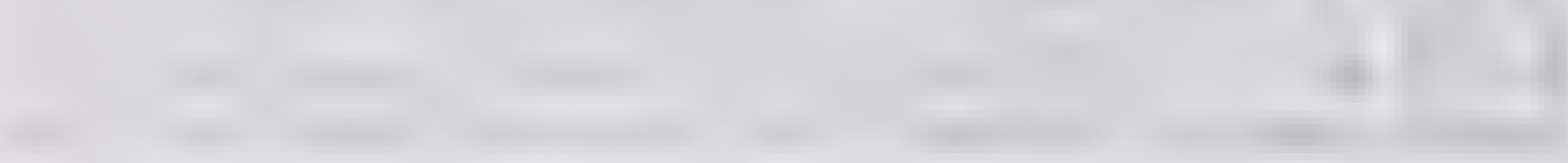
North American F-86 Sabre

F-86A to L, CL-13 (and Orenda-Sabre),
CA-26 (and Avon-Sabre)

Origin

Type: fighter, single-engine fighter, other certain versions a wealth

Engine



Dimensions: Span (m) 37ft 1 in (11.31m) (F-86F-40) and

Weight: Empty (F) 11,125lb (5,045kg) (H) 13,830lb (K) 13,367lb

Maximum loaded (F) 20,611lb (9,350kg) (H) 22,171lb (K) 20,611lb (F) 18,483lb (K) 20,171lb (Sabre 32) 18,611lb

Performance: Maximum speed (F) 6,780mph (10,910km/h) (H) 6,780mph (K) 6,780mph (Sabre 6-32) 6,780mph (peak Mach 0.9)

Altitude: 50,000ft (H) 50,000ft (K) 50,000ft (Sabre 32) 50,000ft (typical)

Armament: (A-E, F-56) 6 x 0.5 in Colt Browning M-3 usually with 26 rounds per gun, underwing hardpoints for two tanks or two stores of 100lb (45kg) each, plus eight rockets or two Sidewinders (D-L) 1 x 24 in pack of 24 2.75 in folding fin aircraft rockets (H) four 20mm (1.59) each with 150 rounds plus 1,200 lb tactical nuke or 3,000 lb of external stores or tanks (K) four 20mm M-24 cannon each with 132 rounds and two Sidewinders (Sabre 32) two 30mm Aden with 150 rounds and two Sidewinders

History: First flight (XJF-1) 27 November 1946 (XP-86) 1 October 1947 (F-86A) December 1948 first flight (YF-86D) 22 December 1949

Users: Argentina, Bangladesh, Bolivia, Burma, Ethiopia, Indonesia, Japan, S Korea, Malaysia, Pakistan, Peru, Philippines, Portugal, Saudi Arabia, S Africa, Taiwan, Thailand, Tunisia, Venezuela, Yugoslavia

Development: Certainly the most famed combat aircraft of its day, if not of the whole period since World War II, the Sabre story began with the award of Army and Navy contracts for jet fighters in 1944. The land-based prototype moved fast, with prototype contracts signed on 18 May 1944, but by 1945 the plans were badly discarded and replaced by new ones with



Three-view of North American F-86F with Sidewinder air-to-air missiles.

vept back wings and tail. The three Navy prototypes ordered on 1 January 1945, were continued and flown as conventional straight-wing aircraft. The order for 100 Furies was cut to 30, but VF-5A (later VF-51) operated it at sea and this otherwise understunged fighter was the first jet to complete an operational tour at sea. The more dramatic XP-86 set a speed of 6,180mph even with its primitive 3,750-bhp Chevrolet-built GE TG-180 engine and with the 5,000-bhp TG-190 (J47) in 1949 soon broke the world speed record at 6,710mph (without being in any way modified from the standard fighter). Many hundreds of Sabres were soon on order, and in the Korean War the F-86E, with slatted wing and powered 'flying tail', and F-86F, with extended leading edge and small fence, were brilliantly flown against the MiG-15 and established marked superiority, despite inferior climb and altitude performance. The very complex F-86D interceptor introduced the new concept of gunless collision-course interception directed by radar and autopilot, the K was a simpler stern-chase interceptor with guns which was mass produced by a consortium in Italy and Germany. USAF Sabres were completed by the powerful H model, but Navy counterparts soon appeared the FJ-2 resembling a cannon-armed F and FJ-3 and 4 (F-1C and E) having much greater power in a new air-frame. The AF-1E was the pinnacle of Sabre development with toss-bombing system, FR probe, extensive new avionics and greatly increased fuel capacity. The CL-13 series built by Canadair began as licensed E and F models (430 being supplied by Mutual Aid funds to the ailing RAF) and continued with the native Orenda engine. Australian CA-27 versions had the Avon and 30mm guns. Total production, including 300 assembled in Japan by Mitsubishi, amounted to 9,502. This handsomely exceeds the total of any other Western military aircraft since 1945, except the 'Huey' helicopter.



Above: Canadair Sabre 6 of Pakistan Air Force

Below: One of the last users of the Sabre is Japan's Air Self-Defence Force, which is phasing out 278 F-86F models after more than 20 years of service. They have been used as fighters and advanced trainers. The JASDF also formerly deployed a unique Sabre variant, the RF-86F converted by Mitsubishi as a fighter reconnaissance machine with a K-17 and two K-22 cameras in a special compartment beneath the cockpit.



Northrop F-5 Freedom Fighter and Tiger II

F-5A, B, E and F, CF-5A and D, NF-5A and B, RF-5A, E and G, and SF-5A and B

Origin: Northrop Aircraft Division, Hawthorne, USA, made or assembled under licence by partnership Canada Netherlands and by Spa

Type: (With suffix A, E, and G) single-seat fighter-reconnaissance, (with suffix B, D and F) two-seat dual fighter/trainer

Engines: (A, B, D, G) two 4,080lb (1,850kg) thrust General Electric single shaft afterburning turbojets, (E, F) two 5,000lb (2,268kg)

Dimensions: Span (A, B, D, G) 25ft 3in (7.7m), (E, F) 26ft 8in (8.1m), length (A, G) 47ft 2in (14.38m), (B, D) 46ft 4in (14.12m), (E) 48ft 3½in (14.73m), (F) 51ft 9½in (15.80m), height (A, G) 13ft 2in (4.01m), (B, D) 13ft 1in (3.99m), (E, F) 13ft 4½in (4.08m)

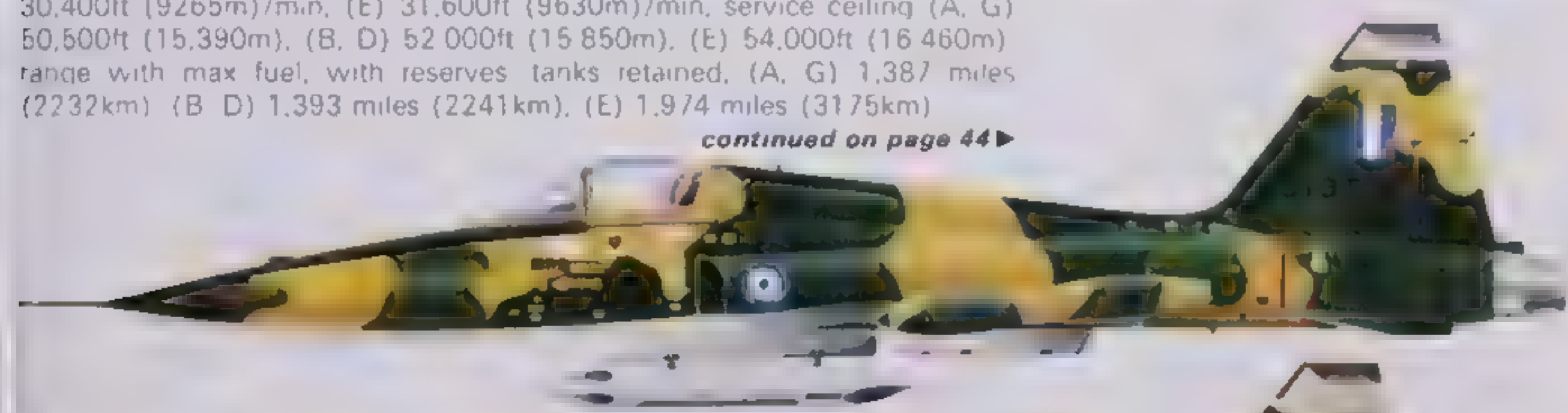
Weights: Empty (A, G) 8,085lb (3,667kg), (B, D) 8,361lb (3,792kg), (E) 9,588lb (4,349kg), (F) 9,700lb (4,400kg), maximum loaded (A, G) 20,677lb (9,379kg), (B, D) 20,500lb (9,298kg), (E, F) 24,080lb (10,922kg)

Performance: Maximum speed at altitude (A, G) 925mph (1,489km/h, Mach 1.40), (B, D) 885mph (1,424km/h, Mach 1.34), (E) 1,060mph (1,705km/h, Mach 1.60), initial climb (A, G) 28,700ft (8,760m)/min, (B, D) 30,400ft (9,265m)/min, (E) 31,600ft (9,630m)/min, service ceiling (A, G) 50,500ft (15,390m), (B, D) 52,000ft (15,850m), (E) 54,000ft (16,460m), range with max fuel, with reserves tanks retained, (A, G) 1,387 miles (2,232km), (B, D) 1,393 miles (2,241km), (E) 1,974 miles (3,175km)

continued on page 44 ▶



Three-view of F-5E Tiger II; the tandem-seat F-5F is 42in longer.



Above: F-5A of Hellenic (Greek) Air Force.



Above: F-5A of Imperial Iranian Air Force.



Above: Canadair-built CF-5A of Canadian Armed Forces.

Below: Dive-bombing attack by a Northrop F-5A Freedom Fighter, with "Coke-bottle" tip tanks (not carried by F-5E Tiger II). Though lacking sophisticated navigation and weapon-delivery systems, the F-5 has sold in large numbers to at least 28 nations. All versions carry two nose-mounted cannon, and in most there is a lead-computing optical gunsight with roll-stabilized manually depressible reticle for air-to-ground delivery, as here. A few customers, such as Saudi Arabia, have advanced F-5 versions with an inertial navigation system, and most customers have bought a small number of tandem-seat dual-control versions. Here, nominal 2,000lb bombs are being released in sequence by the simple weapon-aiming system. The F-5 cannot be used for blind first-pass attack on ground targets.



► Armament

History

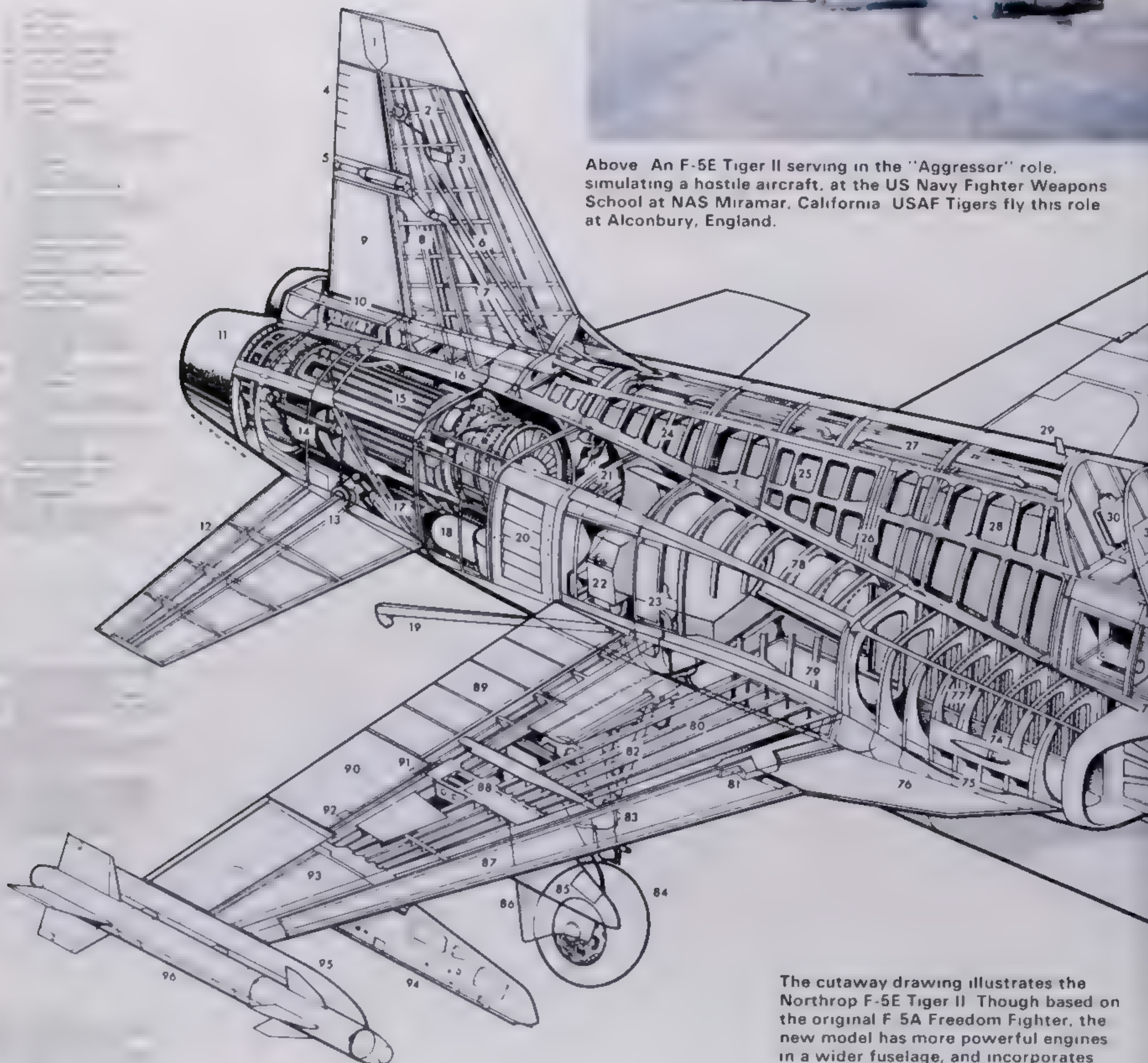
Users

Development

The F-5E Tiger II is a single-engine, single-seat fighter aircraft developed by Northrop. It is a derivative of the F-5A Freedom Fighter, but with a wider fuselage to accommodate two engines. The aircraft is designed for dogfighting and is capable of carrying a variety of air-to-air and air-to-ground weapons. It has been used by several countries, including the United States, Taiwan, and the Philippines. The F-5E is known for its excellent maneuverability and its ability to perform a variety of roles, including as an aggressor aircraft.



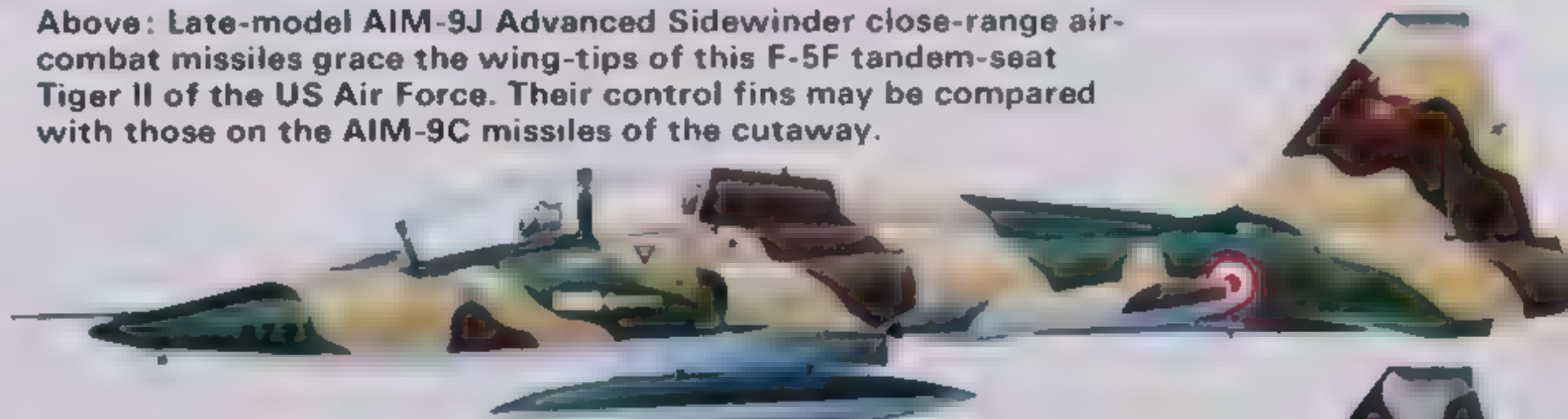
Above An F-5E Tiger II serving in the "Aggressor" role, simulating a hostile aircraft, at the US Navy Fighter Weapons School at NAS Miramar, California. USAF Tigers fly this role at Alconbury, England.



The cutaway drawing illustrates the Northrop F-5E Tiger II. Though based on the original F-5A Freedom Fighter, the new model has more powerful engines in a wider fuselage, and incorporates important changes (such as combat manoeuvring flaps and revised wing roots) to enhance dogfight capability.



Above: Late-model AIM-9J Advanced Sidewinder close-range air-combat missiles grace the wing-tips of this F-5F tandem-seat Tiger II of the US Air Force. Their control fins may be compared with those on the AIM-9C missiles of the cutaway.



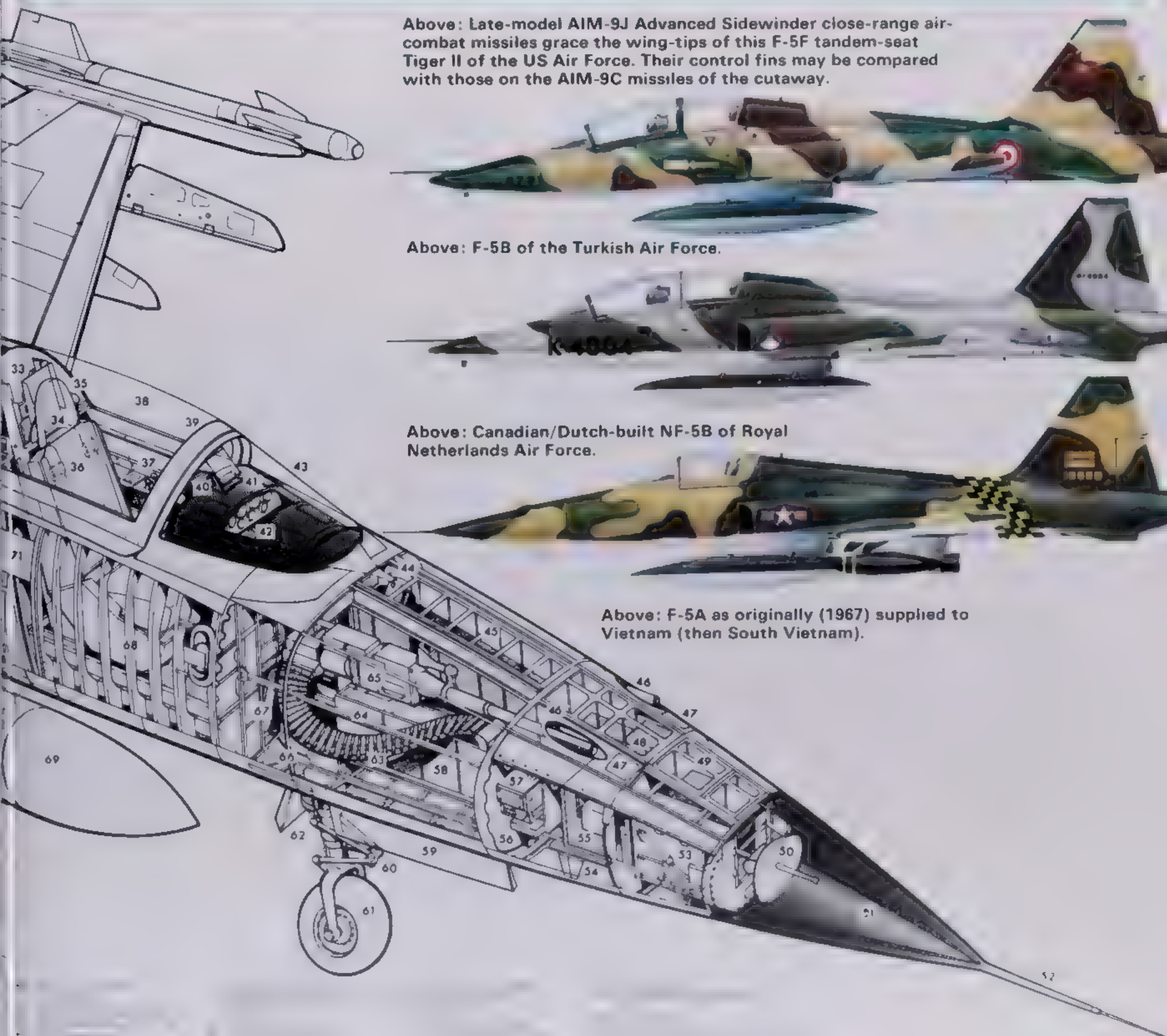
Above: F-5B of the Turkish Air Force.



Above: Canadian/Dutch-built NF-5B of Royal Netherlands Air Force.



Above: F-5A as originally (1967) supplied to Vietnam (then South Vietnam).



Saab 35 Draken

J35A, B, D and F, Sk35C, S35E and export versions

Origin
Type: (J35) single seat all weather fighter

Engine: One Svenska Flygmotor RM6

Dimensions

Weights: Empty (D) 16 017 lb (I

Performance: Maximum

each 1.4) initial climb (D onwards, clean) 34.45

Armament: (A) two 30mm Aden M 55 (side under) missiles (B) as A plus attack ordnance of 2 200lb (1000kg) (C) none (D) as B (E) usually none but provision as A (F) one plus two Rb27 Falcon (radar) and two Rb28 Falcon (infra-red) or four Rb324 (F-35) two 30mm Aden plus nine stores each rated at 1,000lb (454kg) all usable simultaneously, plus four Rb324

History: First flight 25 October 1955, (production J35A) 15 February 1958 final delivery (35XS) 1975, (Danish TF-35) 1976

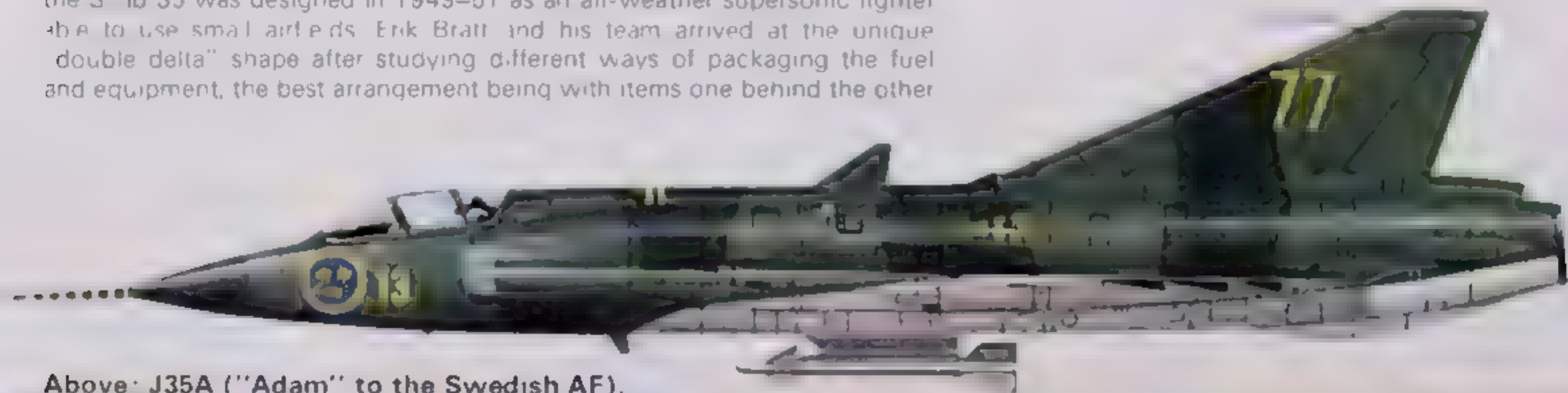
Users: Denmark, Finland, Sweden (RSAF)

Development: Again in advance of any other country in Western Europe the Saab 35 was designed in 1949-51 as an all-weather supersonic fighter able to use small airfields. Erik Bratt and his team arrived at the unique "double delta" shape after studying different ways of packaging the fuel and equipment, the best arrangement being with items one behind the other



Above: Three-view of the Falcon-armed J35F ("Filip" to the Swedish Air Force).

giving a long aircraft of very small frontal area. In 1960 attack wing F13 found the A (Adam) simple to fly and maintain, sensitive in pitch and yet virtually unbreakable. B (Bertil) was more complex, with S7 collision-course fire control integrated with the Swedish Stril 60 air defence environment. Most Sk35C trainers were converted Adams. D (David) was first to reach Mach 2, despite continual increases in weight mainly due to fuel capacity raised from 493 to 680 gallons. E (Erik) carries French OMERA cameras and in 1973 was updated with external British Vinten night/low-level pods. F (Filip) is an automatic interceptor with Ericsson (Hughes basis) radar of pulse-doppler type. Production was closed at 606 with 40 multi-role F-35, RF-35/TF-35 aircraft for Denmark and 12 XS for Finland assembled by Valmet O.



Above: J35A ("Adam" to the Swedish AF).

Below: A pleasing study of one of the six TF-35XD multi-role tandem-seat Drakens of the Royal Danish Air Force. Denmark also uses the most capable of all Draken variants, the single-seat 35XD, and the RF-35 with the Red Baron night reconnaissance pod.

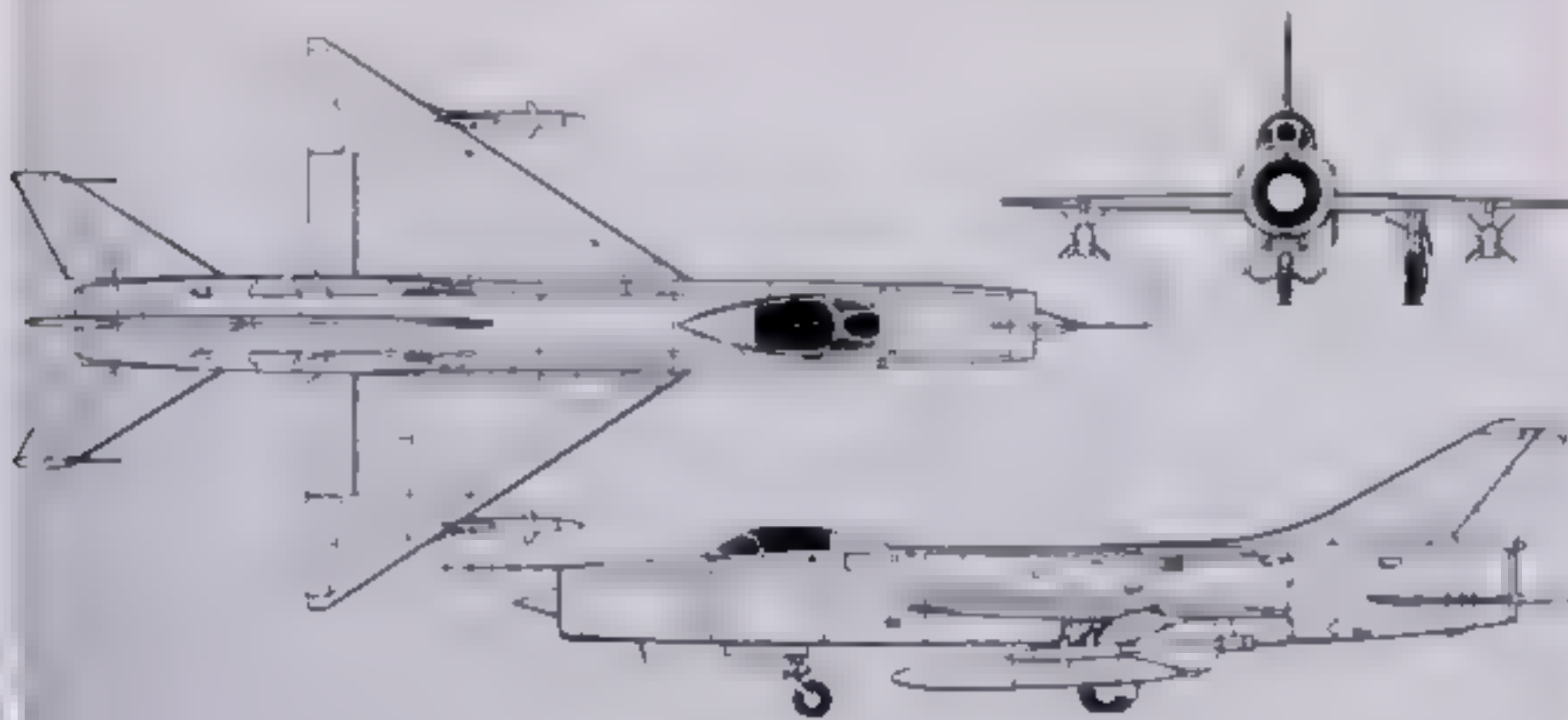


Sukhoi Su-9 and Su-11

Su-9 "Fishpot B", Su-9U "Maiden" and Su-11 "Fishpot C".

Origin: The design bureau named for Pavel O. Sukhoi, Soviet Union.
Type: Single seat all-weather interceptor (Su-9U two-seat trainer).
Engine: One Lyulka single shaft turbojet with afterburner (Su-9U two engines).
AL-7F-1 rated at 22,046lb (10,000kg).
Dimensions: Span 27ft 8in (8.43m), length (-9, -9U) about 54ft (16.5m), (-11) 57ft (17.4m), height 16ft (4.9m).
Weights: (All, estimated) empty 20,000lb (9,070kg), loaded (typical mission) 27,000lb (12,250kg), (maximum) 30,000lb (13,610kg).
Performance: (-11 estimated) maximum speed (clean, optimum height) 720mph (1,160km/h, Mach 0.95), (clean, optimum height) 1,190mph (1,910km/h, Mach 1.8), (two missiles and two tanks at optimum height) 790mph (1,270km/h, Mach 1.2) initial climb 27,000ft (8,230m), service ceiling (clean) 55,700ft (17,000m) range (two missiles and tanks) about 700 miles (1,125km).
Armament: (-9) four AA-1 "Alkali" air-to-air missiles (-9U) same as -9 or not fitted (-11) two AA-3 "Anab" air-to-air missiles one armed and the other IR.
History: First flight (-9) before 1956, (-11) probably 1966, service delivery (-9) probably 1959, (-11) 1967.
User: Soviet Union (IA-PVO).

Development: When first seen, at the 1956 Tushino display, the delta-winged Sukhoi fighter had a small conical radome above the plain nose inlet, while a second had a conical centrebody. The latter arrangement was chosen for production as the Su-9, though development was rather protracted. At first sharing the same engine installation, rear fuselage and tail as the original Su-7, the Su-9 eventually came to have no parts exactly common. No gun was ever seen on an Su-9 by Western intelligence, the primitive missiles being the only armament. At least 2,000 were built, an additional number, probably supplemented by conversions, being tandem seat dual trainers with a cockpit slightly different from that of the Su-7U. The Su-11 is cleaned up in every part of the airframe, has a longer and less tapered nose with larger radar centrebody, completely different armament (still without guns) and a fuselage similar to the Su-7B with external duct fairings along the top on each side. Though much larger and more powerful than the MiG-21, these interceptors have an almost identical tailed-delta



Above: Three-view of Su-11, with "Anab" missiles.



Above: Launch of an "AA-3 Alkali" air-to-air missile from No 2 pylon of an Su-9 of the IA-PVO. Though built in very large numbers, few Su-9 interceptors remain in service today.

Below: Another picture of an Su-9, showing the shorter and more tapered nose in comparison with the later Su-11. Like all Sukhoi aircraft of this generation, the Su-9 and -11 have limited internal fuel capacity and a large Lyulka afterburning engine, and endurance without the twin drop tanks is poor. Many Su-9s may have been rebuilt as Su-11s.



Sukhoi Su-15

Versions known to the West are code-named "Flagon-A to -E"

Origin
Type
Engines

Dimensions

Weights: (Estimated) empty (A) 24 000 lb (10 900 kg), (D) 20 000 lb (9 070 kg)

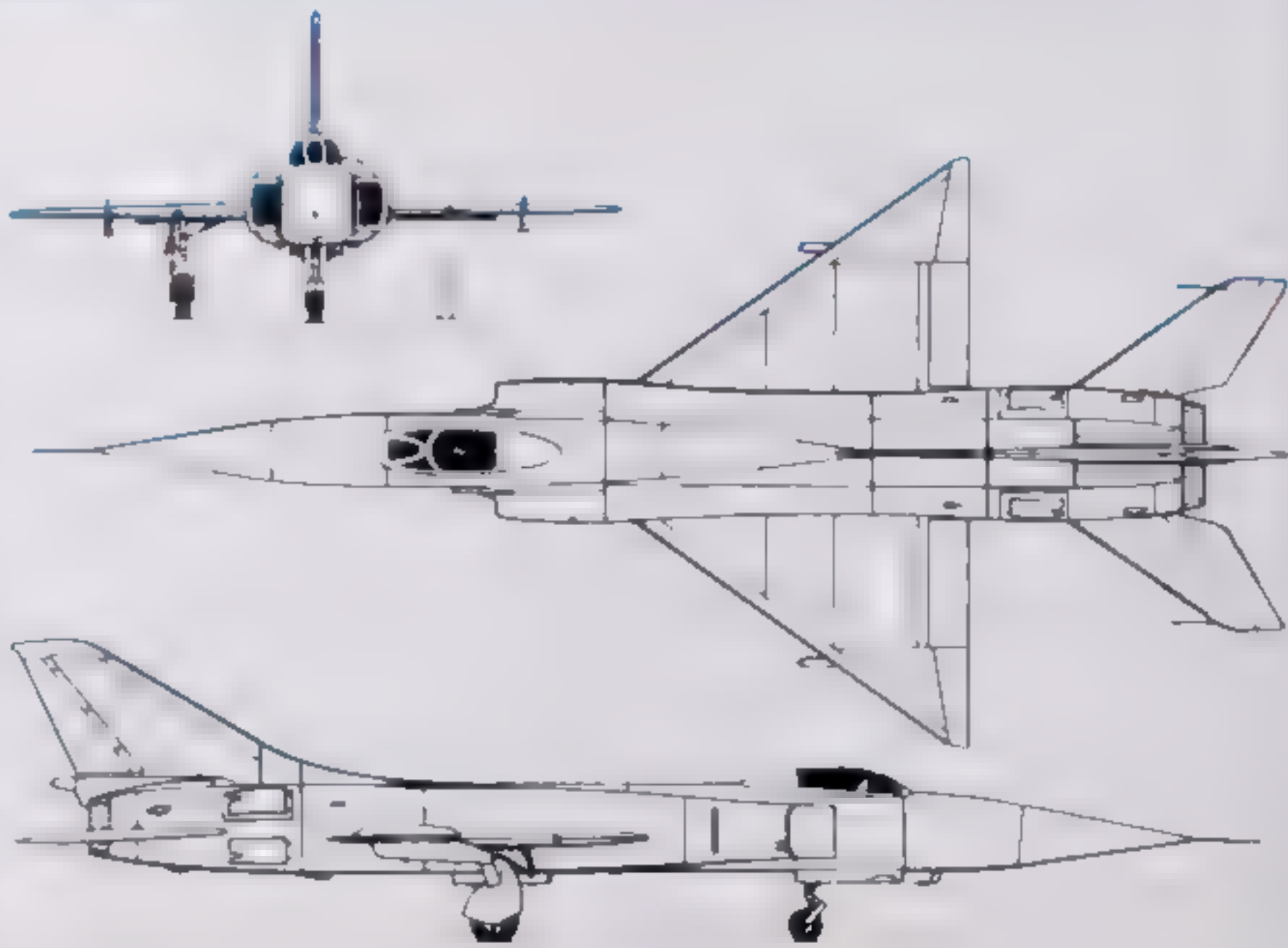
Performance: (Estimated) maximum speed at altitude 2 500 km/h (1 550 mph)

Armament: Two underwing pylons normally carry one radar "Anab" infrared "Anab" two fuselage pylons normally carry drop tanks and 2 3mm GSh-23 two-barrel cannon between them other missiles such as R-60 or AA-7 are probably now being carried (but not yet seen by the West)

History: First flight (Su-15 prototype) probably 1964 (production Su-15) probably 1966

User: Soviet Union (PVO)

Development: Following naturally on from the Su-7, the Su-15 resembles earlier aircraft in wings and tail, the latter being swept back which not only confer increased performance but also a large area for a large AI radar. The initial "Flagon-A" version entered service in 1969. "Flagon B" is a STOL rough field version with a large delta wing in the fuselage and a revised "double delta" tail. The Su-15U dual trainer "D" is basically a "B" with a large delta wing and completely updated electronics and more powerful engines. It is estimated that 400 Su-15 were in service with production of about 15 monthly. In early 1976 an estimate of PVO establishment gave the number of all Su-15 versions in combat service as 600. Though small numbers have served in Warsaw Pact countries and, in 1973, in Egypt, all Su-15s are at present believed to serve with the IA-PVO. There has been speculation in the West that later models could carry the Fox Fire radar and AA-6 "Acrid" missiles of the MiG-25.



Above: Three-view of "Flagon A" Su-15 without missiles.



Above: "Flagon B" jet-lift STOL prototype.

Below: Three "Flagon A" interceptors of the IA-PVO, with "Anabs".

Bottom: With an extremely highly-loaded wing (quite un-Russian) the Su-15 needs good airfields.

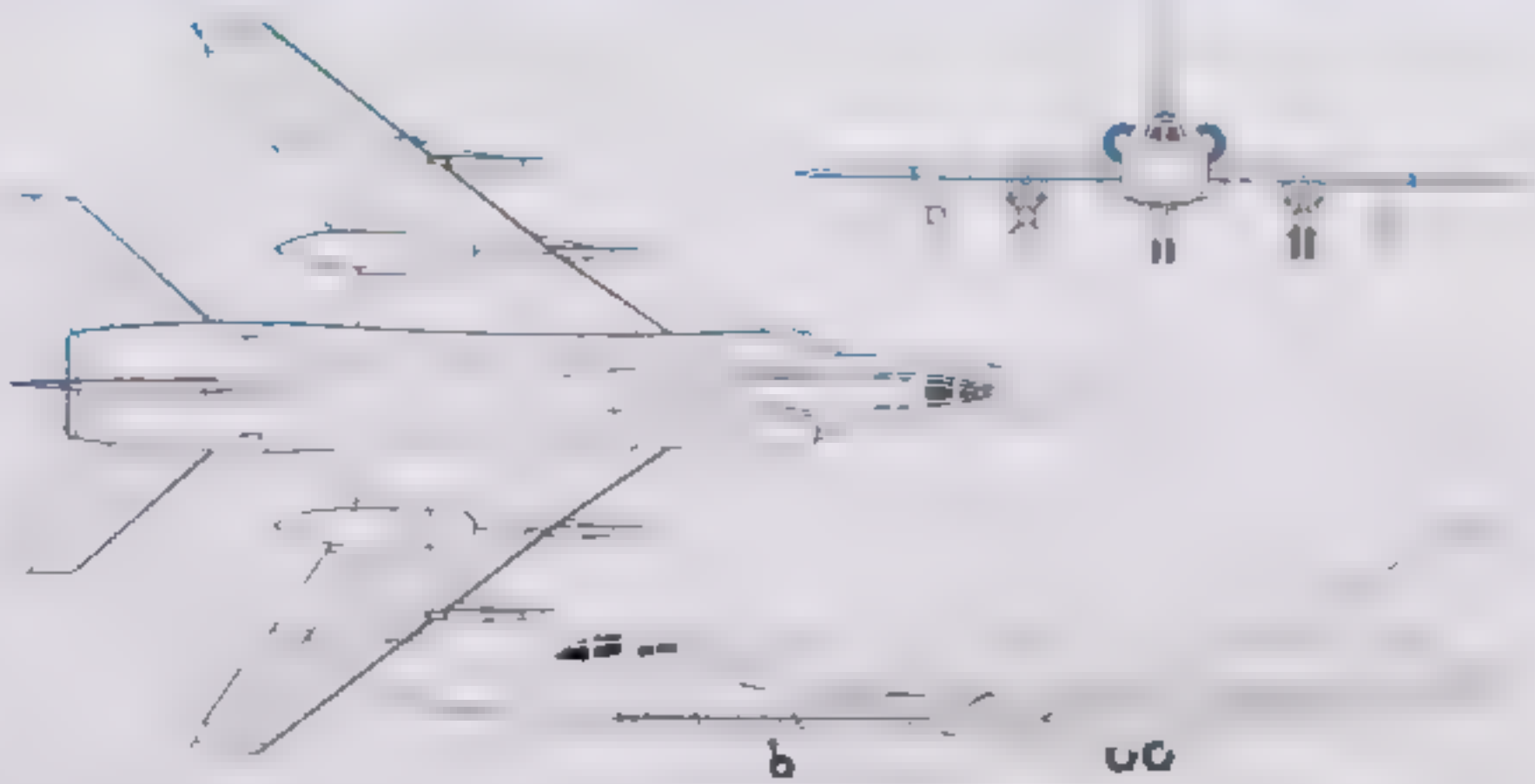


Tupolev Tu-28P

Tu-28 versions of unknown designation; Tupolev bureau, Tu-102

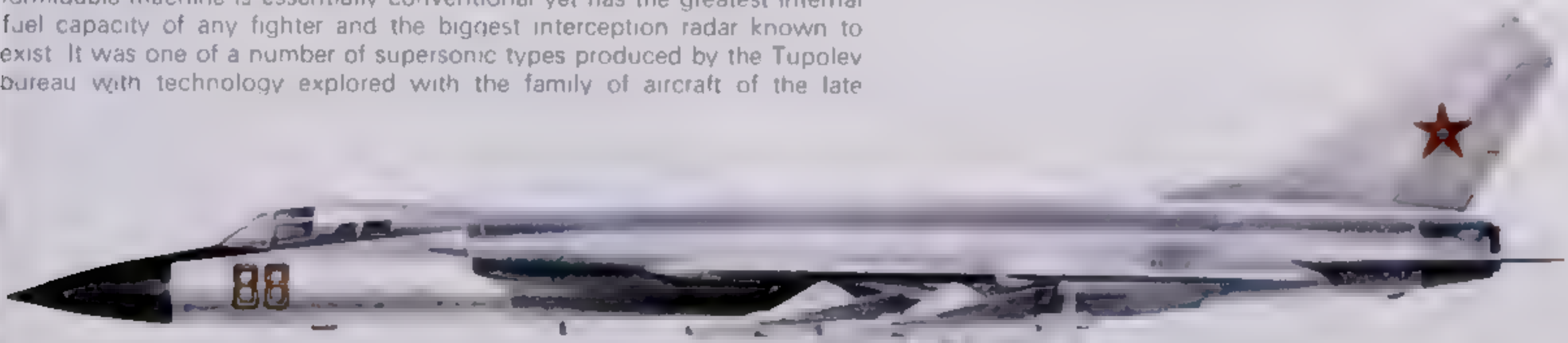
Origin: The design bureau of Andrei N. Tupolev, Soviet Union
Type: Long range all-weather interceptor
Engines: Originally, two large axial turbojets of unknown type, each with afterburning rating of about 27,000lb (12,250kg), probably similar to those of Tu-22, later versions, afterburning turbofans of about 30,000lb (13,610 kg) each, as in later Tu-22
Dimensions: (Estimated) span 65ft (20m), length 85ft (26m), height 23ft (7m)
Weights: (Estimated) empty 55,000 lb (25,000kg) maximum loaded 100,000lb (45,000kg)
Performance: (Estimated) maximum speed (with missiles, at height) 1,150mph (1,850km/h, Mach 1.75), initial climb, 25,000ft (7,500m)/min service ceiling (not gross weight) about 60,000ft (18,000m), range on internal fuel (high Patrol) about 1,800 miles (2,900km)
Armament: No guns seen in any version; mix of infra-red homing and radar-homing "Ash" air-to-air guided missiles, originally one of each and since 1965 two of each
History: First flight, believed 1957, service delivery, probably 1961
User: Soviet Union (PVO)

Development: Largest fighter known to be in service in the world this formidable machine is essentially conventional yet has the greatest internal fuel capacity of any fighter and the biggest interception radar known to exist. It was one of a number of supersonic types produced by the Tupolev bureau with technology explored with the family of aircraft of the late



Three-view of the Tu-28P long-range interceptor, with four "Ash" air-to-air missiles.

1950s known to NATO as "Backfin" (another is the Tu-22). Like the others the Tu-28P has a distinctive wing with sharply kinked trailing edge, the outer 45° panels being outboard of large fairings extending behind the trailing edge accommodating the four-wheel bogie landing gears. Two crew sit in tandem under upward-hinged canopies, and all armament is carried on wing pylons. Early versions had twin ventral fins and usually large belly fairings, but these features are absent from aircraft in current service. The Tu-28P would be an ideal strategic patrol fighter to operate in conjunction with the "Moss" AWACS.



Above: A new drawing of the latest Tu-28P, with "Ash" air-to-air missiles.

Below: Dramatic action shot of a Tu-28P "rippling" away all four of its large air-to-air missiles. Though almost 20 years old as a design, the Tu-28P is still an all-weather interceptor of great capability and in most respects remains the world's largest fighter.



Vought F-8 Crusader

Vought F-8A to F-8J, RF-8, DF-8 and QF-8

Origin
Type
Engine

Dimensions

Weights

Performance

Armament: (A-B-C) four 20mm Colt Mk 12 cannon each with 84 rounds

History: First flight (XF8J-1) 25 March 1955 (production F-8A) November

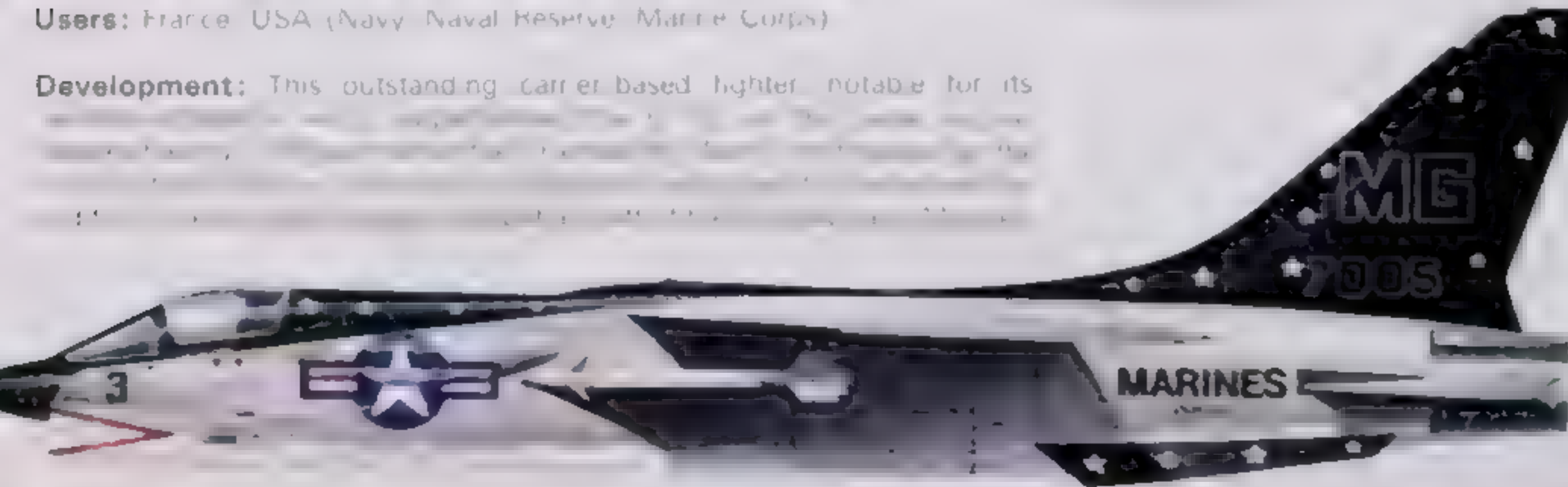
Users: France, USA (Navy, Naval Reserve, Marine Corps)

Development: This outstanding carrier-based fighter, notable for its



Three-view of F-8J, a remanufactured F-8E now in use with the US Naval Reserve

The continuous process of improvement added air-weather radar, improved engine, and other features. The F-8J, the last version, was the most advanced. It was the only Crusader to see combat service with long life ahead. Total Crusader flight time exceeded



Above: F-8L of the US Marine Corps, Squadron VMF-321.

Below: This fine air-to-air portrait of a rebuilt F-8 (probably a J) of VF-201 Sqn, US Navy, pays tribute to one of the most outstanding carrier-based aircraft of all time. It has for 20 years been a saying in the US Navy (and maybe in the Marine Corps) that "when you're out of F-8s, you're out of fighters". This did not imply criticism of the F-4 but reflected the fact that, for air combat, many "jocks" preferred the F-8.



Yakovlev Yak-28P

Yak-28 attack versions, -28P, -28R and -28U

Origin: The design of the Yak-28 was started in 1953 by the Yakovlev design bureau.

Type: 28 (unknown designation)

Engines: Two Tumansky R-11F turbojet engines, each rated at 13,120 lb (5,950 kg) thrust.

Dimensions: Length 28.5m, wingspan 11.5m, height 4.5m.

Weights: Empty (estimated typical) 14,500 lb (6,575 kg), maximum loaded (U) 30,000 lb (13,600 kg) (other versions 18,600 kg).

Performance: (Estimates based on maximum performance of R-11F) Mach 1.13, initial climb rate 8,500 ft/min (2,591 m/min), range (clean) 1,200 mi (1,932 km).

Armament: Attack versions carry 3 or 4 NR-30 30mm cannons in the fuselage or on right side of fuselage weapon bay (external stores fall toombs (estimated maximum 4,400 lb (2,000 kg) payload) between drop tank attachments and outrigger gear (for 11.5m pod of 55mm rockets), (28P) two "Anab" air-to-air missiles, (28R) radar and the other infra-red in some aircraft, two additional pylons for two K-13A ("Atoll") missiles, both "Anab" then being radar homers, (R) believed none, (U) retains weapon bay and single qu.

History: First flight, before 1961, (production attack and interceptor versions) before 1961, service delivery not later than mid-1962, final delivery before 1970.

User: Soviet Union (IA-PVO)

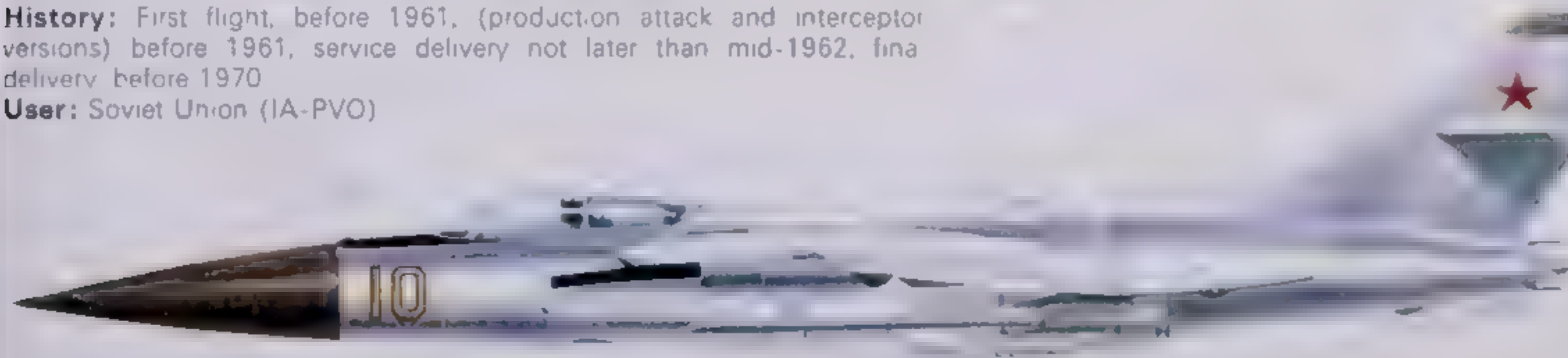


Three-view of the current sub-type of Yak-28P with pointed radome

Development: Obviously derived from the Yak-25/26/27, the Yak-28P has a high-wing, nose gear and considerably greater weight than the Yak-25/26/27. The Yak-28P was developed as a replacement for the Yak-25/26/27.

The name "Brewer 1" were built, most having been rebuilt as 1. The Yak-28P (code name "Freha") was developed as a replacement for the Yak-25/26/27. The Yak-28P was developed as a replacement for the Yak-25/26/27.

Many 28R versions ("Brewer D") with cameras and various optical sensors, may be converted attack aircraft. Flight refueling is not fitted.



Above: A long-nose Yak-28P specially drawn for this book.



Below: About to take off, a short-nose Yak-28P lines up on the runway with tailplane at negative incidence to rotate the aircraft about its rear landing-gear truck. This is the ultimate development of the Yak-25 of 1953.



Attack/close support aircraft

In terms of numbers this is the most important category of combat aircraft; yet, if the media are to be believed, the general public have never heard of it, so attack aircraft have to be called either "fighters" or "bombers". Their origin lay in aircraft of 1916-18 called ground strafers or trench fighters which were armoured to survive in close proximity to a land battle whilst doing what they could to harass the enemy land forces. In the 1930s the US Army Air Corps recognized the attack category, as did the Soviet Union, but the RAF and even the Luftwaffe concentrated on traditional fighters and bombers and flew attack missions with such lash-ups as cannon fixed to the Ju 87 or He 177 and rockets fixed to the Typhoon. During the past 30 years air forces have often regarded attack missions as suitable for obsolescent fighters, while the scene has been further confused by concentration in the 1960s on light propeller-driven attack aircraft for use in what were popularly called "brushfire wars" or Co-In (counter-insurgent) operations. To some degree these light attack machines represented a justifiable reaction against the use of totally unsuitable supersonic fighters, which were complex, hard to maintain, costly, inefficient, unable to find their targets or to use short unpaved airstrips.

The one area where attack aircraft were designed for the job was in naval warfare, but here the targets were often easy-to-find ships and the airfields were carriers. This led to excellent aircraft, which in the

case of the A-1 Skyraider stayed competitive almost 30 years longer than anyone predicted when it was new. But naval attackers needed changes to fit them for the land battle.

What kind of beast is the attack aircraft? First, it must be survivable. The notion that insurgents carry primitive weapons was predictably discredited by the appearance of shoulder-fired guided weapons with warheads more lethal than an 88mm shell. So we need plenty of armour, and a design philosophy which gives us a spare crew-member, spare engine, spare tail and might even bring back the biplane. We also need powerful ECM and plenty of decoys. There is no point in taking off if we have difficulty finding the target, so we need good navigation systems and sophisticated sensors. To destroy the target we need not just accuracy but blind first-pass accuracy, which means we can come straight across and hit it first time in a snow-blizzard at night. By no means least we need lethal ordnance, which means whopping cannon and smart bombs or guided missiles.

To carry all this demands a large margin between empty and gross weight. To give an idea of what is now possible, the Jaguar, with a wing only three-quarters the span of a Spitfire, can carry a load of fuel and weapons equal to almost *four times the total gross weight* of a Spitfire I. To make life harder, we need a large wing to lift this weight, and especially to give STOL performance; yet high-speed attack at



low level calls for the smallest possible wing, unless the crew are to be shaken to pieces (so some people cheat with swing-wings or vectored thrust). We need long landing gears with squashy soft-field tyres, and possibly drag-chutes and arrestor hooks—and all the help we can get from the concealment experts, who are today pushing their luck if they use crude camouflage netting. We also need help from our own troops, who can use laser designators to point out exactly where they want our weapons to hit; the missiles will obediently steer towards the laser light diffused or reflected from the target. Where the attack pilot can see the target he can use a TV-guided missile; to hit armour he can use a gun, especially if he happens to be strapped to an A-10.

The A-10 is the most carefully designed attack platform in history. Like the light Co-In, it offers an alternative to the highly unsuitable "fighter". Its designers accepted large size and rather low performance, and did their best to make it survivable. If they have succeeded, the A-10 will be the aircraft enemy ground troops will like least because it is designed to be based closer to the land battle (but not as close as a V/STOL), to carry more weapons, and deliver them more accurately. Nobody could call the A-10 a "fighter", so the media have labelled it a "bomber". There is no problem in fixing a simple tag to the other class of attack aircraft: they are called helicopters, and are discussed in that section, which begins on page 192.

When the McDonnell Douglas Skyhawk was designed, in 1952, the US Navy doubted it could ever be built, but today it is still in production. This A-4M, called a Camel for obvious reasons, is still an effective attack platform 23½ years after first flight



Aeritalia G91

G91R, G91T, G91PAN and G91Y

0 1 2 3

Type:

Engines

Dimensions

Weights

Performance

Armament

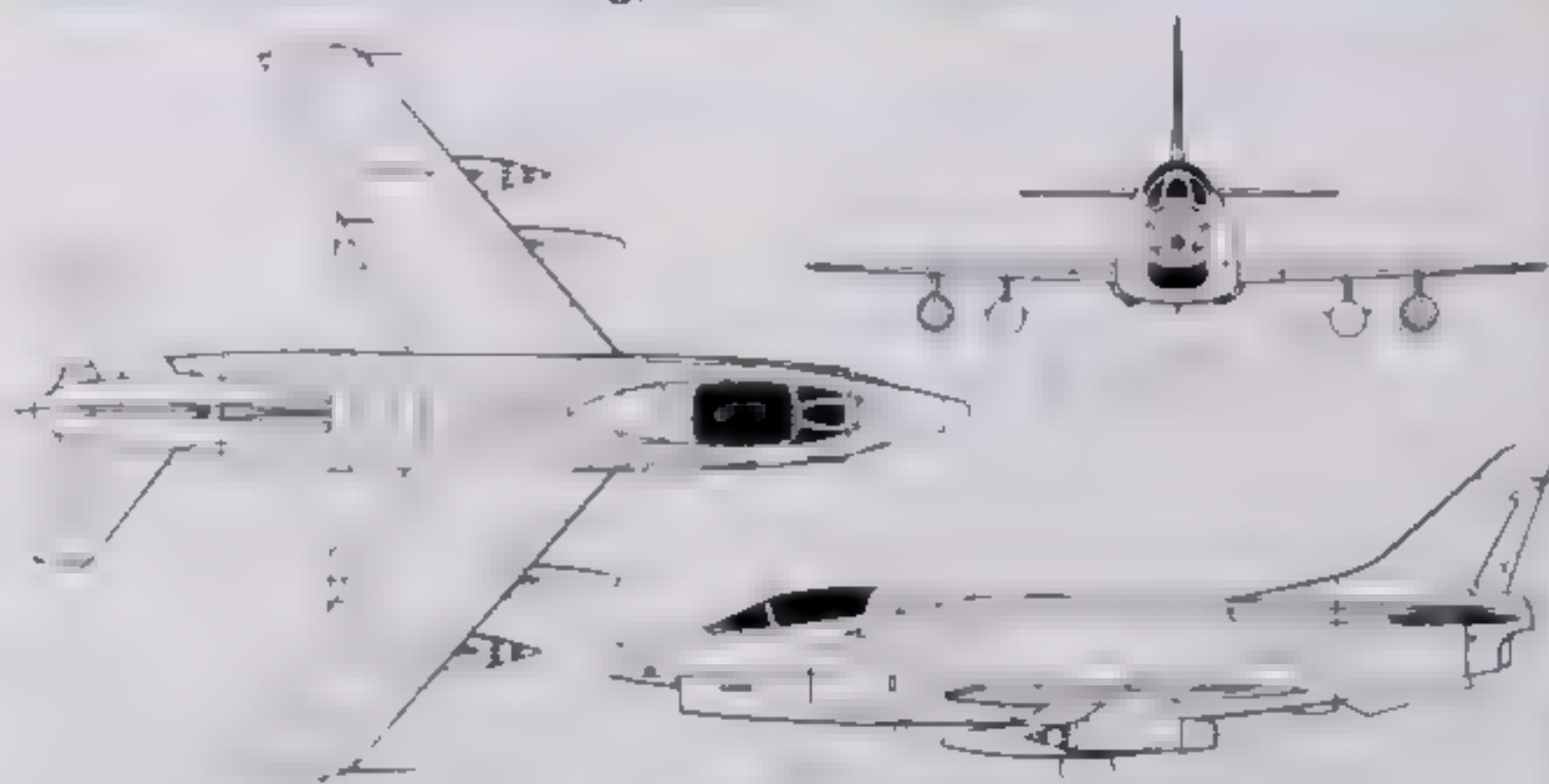
History: First flight

Users: (G91Y) Italy (earlier v)

Development

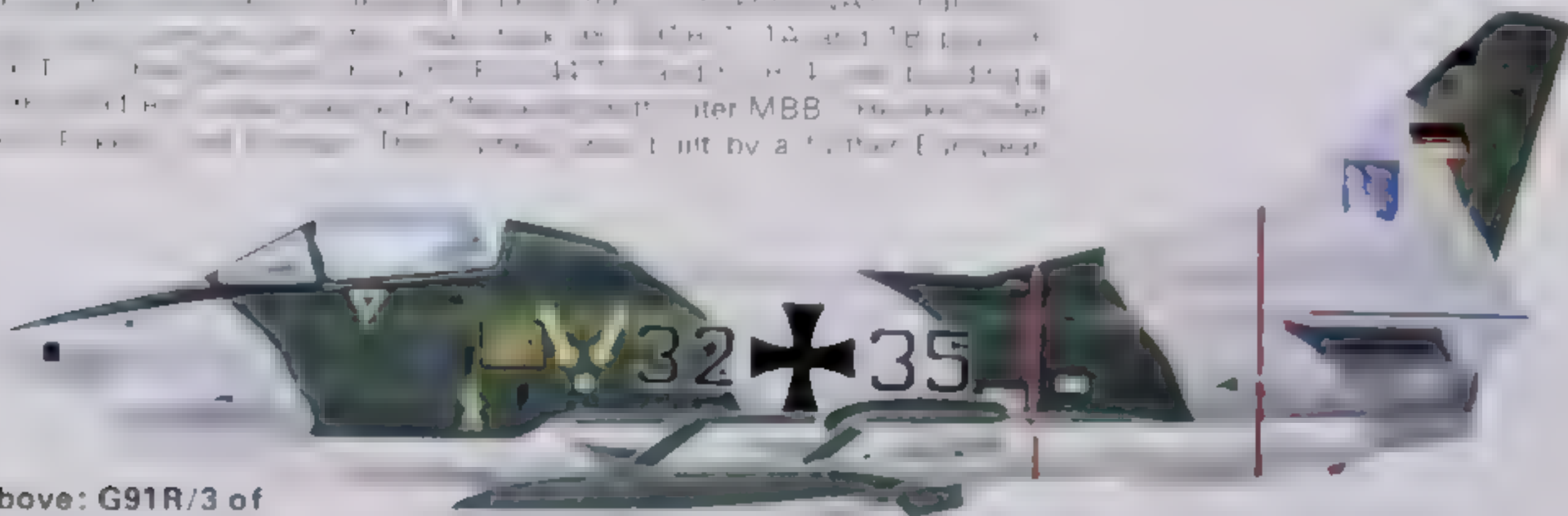


Above: Three G91R/1 light attack aircraft of the Aeronautica Militare Italiano's 2nd Wing, based at Treviso.



Above: Three-view of the twin-engined Aeritalia G91Y.

nsortium. Among many other sub-variants is the PAN version of the Pattuglia Acrobatica Nazionale. The completely redesigned G91Y has much greater thrust, better navigation aids and can fly fighter, attack or reconnaissance missions. Aeritalia, the company formed in 1969 jointly by Fiat and Finmeccanica-IRI, delivered 45 to the Regia Aeronautica in 1971-76.



Above: G91R/3 of
LeKG 43 of West German
Luftwaffe, based at Oldenburg

Below: Ear-splitting run at nought metres (slightly higher than nought feet) by a G91Y. Though often called a fighter, the G91Y has no radar and only simple air-to-air weapons (guns and unguided rockets) and is actually a light tactical attack and reconnaissance aircraft. A total of 45 was delivered in 1971-76 to the Aeronautica Militare Italiano's 32nd Wing, the original 20 development aircraft having been delivered to 8th Wing at Cervia.



BAC Strikemaster and 145

BAC 145 and Strikemaster

Origin: Hunting
Type: Two-seat light fighter
Engine: 3 410lb (1547k) thrust
Dimensions: Span 36ft

Weights: Empty 11,200lb (5,100kg), max 17,000lb (7,700kg)

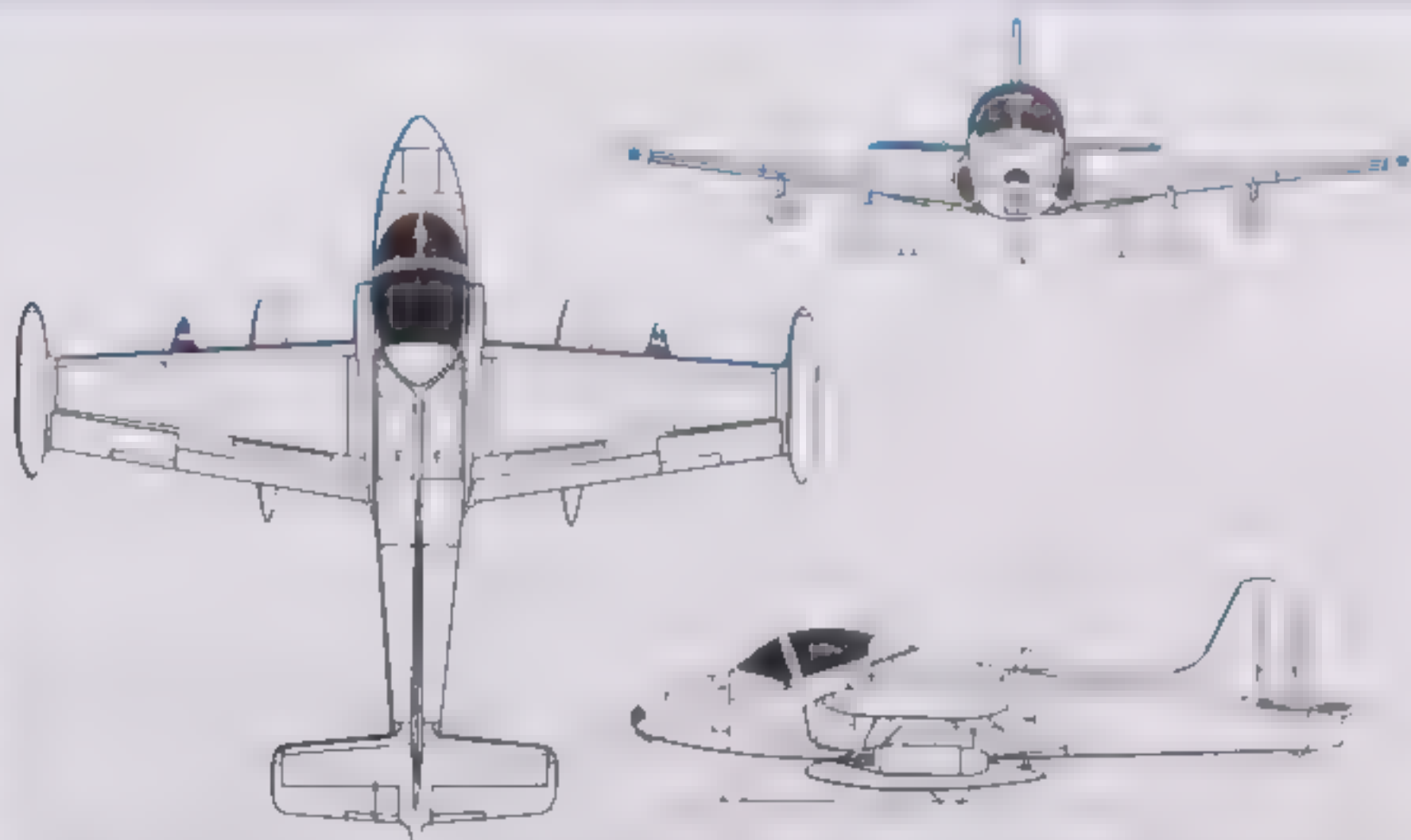
Performance: Maximum speed 481mph (774km/h), max sea level 450mph (726km/h), initial climb (max 10,000ft) 1600m/min, service ceiling 44,000ft (13,414m), combat radius 2600km

Armament: Two 30mm FN cannons, 100 rounds each, 550 rounds each, 4 underwing strongpoints

History: First flight (Jet Provost) 16 June 1954, (Strikemaster) 26 October 1967 first delivery 1968

Users: (Jet Provost) Iraq, Kuwait, Rhodesia, S Yemen, Sri Lanka, Sudan, JK, Venezuela (Strikemaster) Ecuador, Kenya, Kuwait, New Zealand, Oman, Saudi Arabia, Singapore, S Yemen

Development: The Percival Provost basic trainer was developed by Hunting then produced a jet version, and flew the first in 1954. Subsequently the Hunting (later BAC) Jet Provost became a basic trainer made in great numbers for the RAF and many overseas countries, and more powerful pressurised versions were developed. BAC then developed the BAC 145 multi-role trainer, which in turn was developed into the highly refined Strikemaster. With a more powerful Viper engine, the Strikemaster proved to be a great world-



Above: Three-view of basic BAC 167 Strikemaster with rocket pods and tanks.

wide success. It has side by side ejection seats, and can operate from the roughest airstrip whilst carrying a combat load. The typical BAC 145/167/167A/167B/167C/167D/167E/167F/167G/167H/167I/167J/167K/167L/167M/167N/167O/167P/167Q/167R/167S/167T/167U/167V/167W/167X/167Y/167Z/167AA/167AB/167AC/167AD/167AE/167AF/167AG/167AH/167AI/167AJ/167AK/167AL/167AM/167AN/167AO/167AP/167AQ/167AR/167AS/167AT/167AU/167AV/167AW/167AX/167AY/167AZ/167BA/167BB/167BC/167BD/167BE/167BF/167BG/167BH/167BI/167BJ/167BK/167BL/167BM/167BN/167BO/167BP/167BQ/167BR/167BS/167BT/167BU/167BV/167BW/167BX/167BY/167BZ/167CA/167CB/167CC/167CD/167CE/167CF/167CG/167CH/167CI/167CJ/167CK/167CL/167CM/167CN/167CO/167CP/167CQ/167CR/167CS/167CT/167CU/167CV/167CW/167CX/167CY/167CZ/167DA/167DB/167DC/167DD/167DE/167DF/167DG/167DH/167DI/167DJ/167DK/167DL/167DM/167DN/167DO/167DP/167DQ/167DR/167DS/167DT/167DU/167DV/167DW/167DX/167DY/167DZ/167EA/167EB/167EC/167ED/167EE/167EF/167EG/167EH/167EI/167EJ/167EK/167EL/167EM/167EN/167EO/167EP/167EQ/167ER/167ES/167ET/167EU/167EV/167EW/167EX/167EY/167EZ/167FA/167FB/167FC/167FD/167FE/167FF/167FG/167FH/167FI/167FJ/167FK/167FL/167FM/167FN/167FO/167FP/167FQ/167FR/167FS/167FT/167FU/167FV/167FW/167FX/167FY/167FZ/167GA/167GB/167GC/167GD/167GE/167GF/167GG/167GH/167GI/167GJ/167GK/167GL/167GM/167GN/167GO/167GP/167GQ/167GR/167GS/167GT/167GU/167GV/167GW/167GX/167GY/167GZ/167HA/167HB/167HC/167HD/167HE/167HF/167HG/167HH/167HI/167HJ/167HK/167HL/167HM/167HN/167HO/167HP/167HQ/167HR/167HS/167HT/167HU/167HV/167HW/167HX/167HY/167HZ/167IA/167IB/167IC/167ID/167IE/167IF/167IG/167IH/167II/167IJ/167IK/167IL/167IM/167IN/167IO/167IP/167IQ/167IR/167IS/167IT/167IU/167IV/167IW/167IX/167IY/167IZ/167JA/167JB/167JC/167JD/167JE/167JF/167JG/167JH/167JI/167JJ/167JK/167JL/167JM/167JN/167JO/167JP/167JQ/167JR/167JS/167JT/167JU/167JV/167JW/167JX/167JY/167JZ/167KA/167KB/167KC/167KD/167KE/167KF/167KG/167KH/167KI/167KJ/167KK/167KL/167KM/167KN/167KO/167KP/167KQ/167KR/167KS/167KT/167KU/167KV/167KW/167KX/167KY/167KZ/167LA/167LB/167LC/167LD/167LE/167LF/167LG/167LH/167LI/167LJ/167LK/167LL/167LM/167LN/167LO/167LP/167LQ/167LR/167LS/167LT/167LU/167LV/167LW/167LX/167LY/167LZ/167MA/167MB/167MC/167MD/167ME/167MF/167MG/167MH/167MI/167MJ/167MK/167ML/167MM/167MN/167MO/167MP/167MQ/167MR/167MS/167MT/167MU/167MV/167MW/167MX/167MY/167MZ/167NA/167NB/167NC/167ND/167NE/167NF/167NG/167NH/167NI/167NJ/167NK/167NL/167NM/167NN/167NO/167NP/167NQ/167NR/167NS/167NT/167NU/167NV/167NW/167NX/167NY/167NZ/167OA/167OB/167OC/167OD/167OE/167OF/167OG/167OH/167OI/167OJ/167OK/167OL/167OM/167ON/167OO/167OP/167OQ/167OR/167OS/167OT/167OU/167OV/167OW/167OX/167OY/167OZ/167PA/167PB/167PC/167PD/167PE/167PF/167PG/167PH/167PI/167PJ/167PK/167PL/167PM/167PN/167PO/167PP/167PQ/167PR/167PS/167PT/167PU/167PV/167PW/167PX/167PY/167PZ/167QA/167QB/167QC/167QD/167QE/167QF/167QG/167QH/167QI/167QJ/167QK/167QL/167QM/167QN/167QO/167QP/167QQ/167QR/167QS/167QT/167QU/167QV/167QW/167QX/167QY/167QZ/167RA/167RB/167RC/167RD/167RE/167RF/167RG/167RH/167RI/167RJ/167RK/167RL/167RM/167RN/167RO/167RP/167RQ/167RR/167RS/167RT/167RU/167RV/167RW/167RX/167RY/167RZ/167SA/167SB/167SC/167SD/167SE/167SF/167SG/167SH/167SI/167SJ/167SK/167SL/167SM/167SN/167SO/167SP/167SQ/167SR/167SS/167ST/167SU/167SV/167SW/167SX/167SY/167SZ/167TA/167TB/167TC/167TD/167TE/167TF/167TG/167TH/167TI/167TJ/167TK/167TL/167TM/167TN/167TO/167TP/167TQ/167TR/167TS/167TT/167TU/167TV/167TW/167TX/167TY/167TZ/167UA/167UB/167UC/167UD/167UE/167UF/167UG/167UH/167UI/167UJ/167UK/167UL/167UM/167UN/167UO/167UP/167UQ/167UR/167US/167UT/167UU/167UV/167UW/167UX/167UY/167UZ/167VA/167VB/167VC/167VD/167VE/167VF/167VG/167VH/167VI/167VJ/167VK/167VL/167VM/167VN/167VO/167VP/167VQ/167VR/167VS/167VT/167VU/167VV/167VW/167VX/167VY/167VZ/167WA/167WB/167WC/167WD/167WE/167WF/167WG/167WH/167WI/167WJ/167WK/167WL/167WM/167WN/167WO/167WP/167WQ/167WR/167WS/167WT/167WU/167WV/167WW/167WX/167WY/167WZ/167XA/167XB/167XC/167XD/167XE/167XF/167XG/167XH/167XI/167XJ/167XK/167XL/167XM/167XN/167XO/167XP/167XQ/167XR/167XS/167XT/167XU/167XV/167XW/167XX/167XY/167XZ/167YA/167YB/167YC/167YD/167YE/167YF/167YG/167YH/167YI/167YJ/167YK/167YL/167YM/167YN/167YO/167YP/167YQ/167YR/167YS/167YT/167YU/167YV/167YW/167YX/167YY/167YZ/167ZA/167ZB/167ZC/167ZD/167ZE/167ZF/167ZG/167ZH/167ZI/167ZJ/167ZK/167ZL/167ZM/167ZN/167ZO/167ZP/167ZQ/167ZR/167ZS/167ZT/167ZU/167ZV/167ZW/167ZX/167ZY/167ZZ



Left: Strikemaster Mk 80 of the Royal Saudi Air Force, with typical pylon loads of tanks and 18-round Matra rocket launchers. BAC delivered 25 to this customer in 1968-69, followed by ten Mk 80A. Strikemasters have proved extremely popular among Middle East air forces.

Left: Strikemaster Mk 88 of Royal New Zealand Air Force. The initial batch of ten led to a repeat order for another six used for advanced weapon training.

Left: Strikemaster Mk 87 of Kenya Air Force. Six were delivered and five are currently in use as attack/trainer aircraft. The KAF is soon to have to learn the more costly and complex F-5E.



Left: Strikemaster Mk 82 of the Sultan of Oman's Air Force, taxiing out with 1,000lb bombs and 80mm Sura rockets. The SOAF received 12, followed by eight Mk 82A, and though several have been shot down (some by SA-7 missiles) these tough and simple machines have proved ideal in helping to defeat the Dhofar rebellion, which was officially declared ended in 1976.

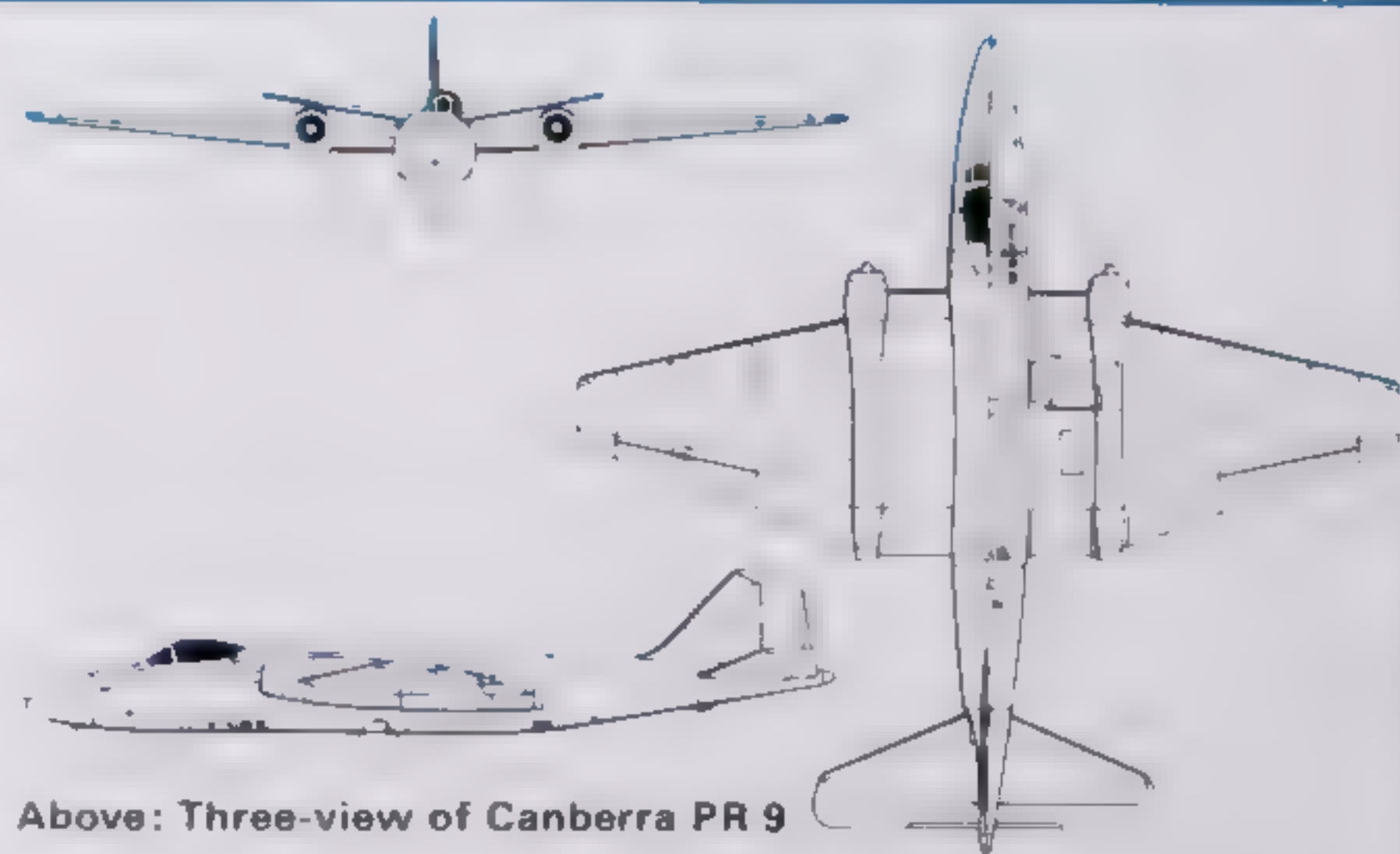
BAC Canberra and Martin/GD B-57

Canberra 1 to 24 (data for B(I).12
except where otherwise indicated)

Origin: English Electric (British Aerospace)

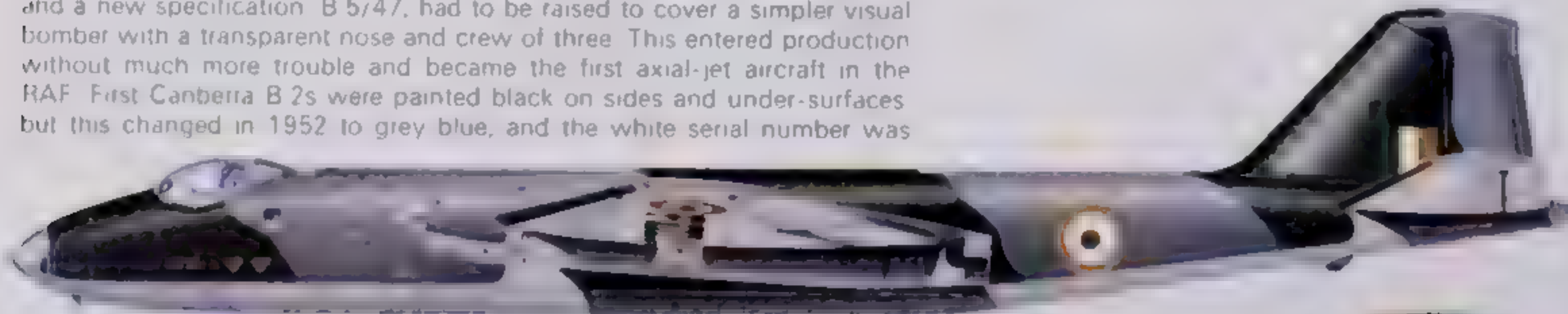
Type: Bomber
Engines: Two 7,500 b (3,402 kg) thrust Rolls Royce Avon 109 single
 is (PR 9) two 11,250 b (5,100 kg) Avon 206
Dimensions: Span 63 ft 11 in (19.5 m) (PR 9) 67 ft 10 in length 65 ft 6 in
Weights: Empty 23,173–27,950 lb (10,400–12,700 kg) loaded 43,000 lb
 (19,500 kg) maximum permissible 56,250 lb (25,515 kg)
Performance: Maximum speed 580 mph (933 km/h) at 30,000 ft (9,144 m)
 or Mach 0.83 initial climb at maximum weight 3,400 ft (1,036 m)/min
 ceiling 48,000 ft (14,630 m) range (typical mission at low level)
 800 miles (1,287 km) ferry range 1,200 miles (1,933 km)
Armament: Four 20 mm Hispano Mk II cannon (in wings)
 sixteen 4.5 in flares internally two AS 30 missiles or two 1,000 lb bombs
 or two packs of 37 rockets externally
History: First flight (prototype) 13 May 1949, first service delivery
 October 1950, first flight of B(I) series 23 July 1954
Users: Argentina Ecuador, Ethiopia, France (trials), W Germany, India
 New Zealand Peru, Rhodesia S Africa Sweden (trials), Venezuela

Development: When W E W "Teddy" Petter joined English Electric at
 Preston as chief engineer he already had a scheme for a jet bomber. To meet
 specification B 3/45 he eventually planned a straightforward unswept
 aircraft with a broad wing for good behaviour at great heights, with two of
 the new axial jet engines centred on each wing giving a total of 15,000 lb
 thrust. Like the Mosquito, the A 1 bomber was to be fast enough to escape
 interception, whilst carrying a 6,000 lb bomb load over a radius of 750 nauti-
 cal miles. It was to have a crew of two and a radar bomb sight for blind
 attacks in all conditions. The prototype amazed everyone with its low-level
 manoeuvrability and the A 1, named Canberra, was a superb flying machine
 from the start. But the radar bombing system lagged years in development
 and a new specification B 5/47, had to be raised to cover a simpler visual
 bomber with a transparent nose and crew of three. This entered production
 without much more trouble and became the first axial-jet aircraft in the
 RAF. First Canberra B 2s were painted black on sides and under-surfaces
 but this changed in 1952 to grey blue, and the white serial number was



Above: Three-view of Canberra PR 9

painted extra large to serve as a 'buzz number' visible to fighter pilots from
 a safe distance. In February 1951 a B 2 set a transatlantic record flying out to
 Baltimore to serve as pattern aircraft for the Martin B-57 programme (see
 entry). The Korean war caused a sudden jump in orders, and Canberra B 2s
 were made by EECo and by Handley Page (75), Avro (75) and Short (60).
 The PR 3 was a reconnaissance version with longer fuselage for more fuel.
 The T 4 had side-by-side dual controls. The Mk 5 prototype introduced
 Avon 109 engines and integral wing tanks, and was to be a visual target
 marker. It led to the B 6, the heavier and more powerful replacement for the
 B 2. The corresponding reconnaissance version was the PR 7, from which
 was derived the much more powerful, long span PR 9 developed and
 built by Short. Most versatile Canberra was the B(I) 8, with offset pilot
 canopy and nav/bomb position in the nose. With four 20 mm cannon (and
 ammunition for 55 seconds continuous firing) the Mk 8 also carried a wide
 range of under-wing missiles, bombs, tanks and special pods, and, like
 earlier versions, proved an export winner, particularly the B(I) 12. Until
 they were ready the B(I) 6 served in Germany in the multi-role tasks and also
 dashed to Kuwait in 1961. Later mark numbers include special trainers,
 electronic-warfare versions, target tugs, pilotless targets and, as one-off
 conversions, platforms for testing almost every British postwar engine,
 missile and airborne device. Total Canberra production was 925 in Britain,
 Australia made 49 B 20s for the RAAF.



Above: Canberra B.74 of the Indian Air Force. This service has 86
 Canberras of three distinct families

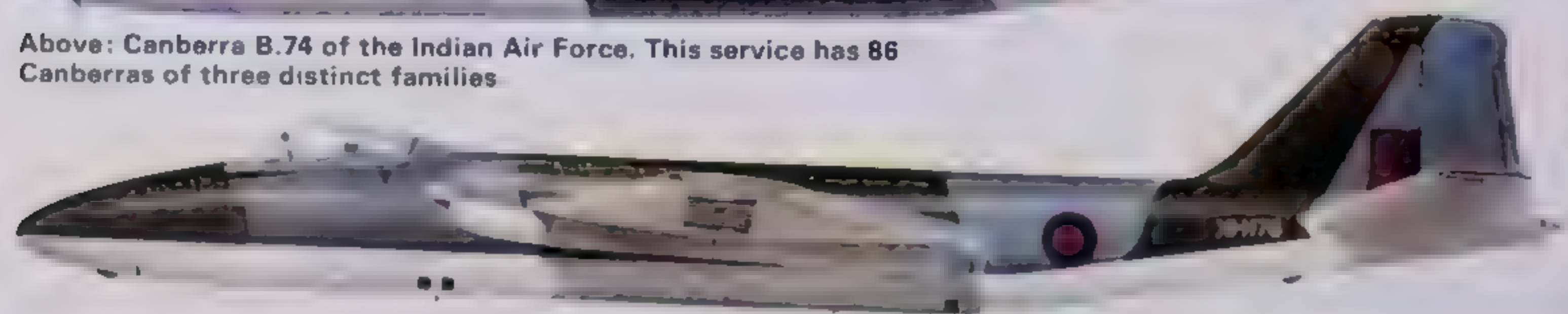


Photo below left:
 Canberra T.17 of
 No 360 Sqn, RAF
 Cottesmore. This,
 with the E.15, is a
 rebuild of earlier
 bomber versions,
 and 360 Sqn is joint-
 Services-operated.

Below: Canberra
 B 62 of 1 Escuadron
 de Bombardeo,
 Argentine Air Force.
 This again is a
 rebuild, but still a
 tactical bomber.

Above: The high altitude Canberra PR 9 equips 39 Sqn at RAF Wyton. This version was built by Shorts



B-57A to B-57G Night Intruder, RB-57A, D and F

Origin: Design, English Electric Aviation – built for US project by The Martin Co. Baltimore, USA. (RB-57F – GD F-107 version)

Type: Two-seat tactical attack and reconnaissance (RB version – strategic reconnaissance at extreme altitude)

Engines: (A, B, C, E, G) two 7,220lb (3275kg) thrust Wright J65-5 (US Sapphire) single-shaft turbojets, (D) two 11,000lb (4990kg) Pratt & Whitney J57-37A two-shaft turbojets, (F) two 18,000lb (8165kg) Pratt & Whitney TF33-11A two-shaft turbofans and two 3,300lb (1500kg) Pratt & Whitney J60-9 single shaft turbojet

Dimensions: Span (A, B, C, E, G) 64ft (19.5m), (D) 106ft (32.3m), (F) 122ft 5in (37.32m), length (A, B, C, D, E) 65ft 6in (19.96m), (G) 67ft (20.42m), (F) 69ft (21.03m), height (A, B, C, E, G) 15ft 7in (4.75m), (D) 14ft 10in (4.52m), (F) 19ft (5.79m)

Weights: Empty (A, B, C, E, typical) 26,800lb (12,200kg), (G) about 28,000lb (12,700kg), (D) 33,000lb (14,970kg), (F) about 36,000lb (16,330kg), maximum loaded (A) 51,000lb, (B, C, E, G) 55,000lb (24,950kg), (D) not disclosed, (F) 63,000lb (28,576kg)

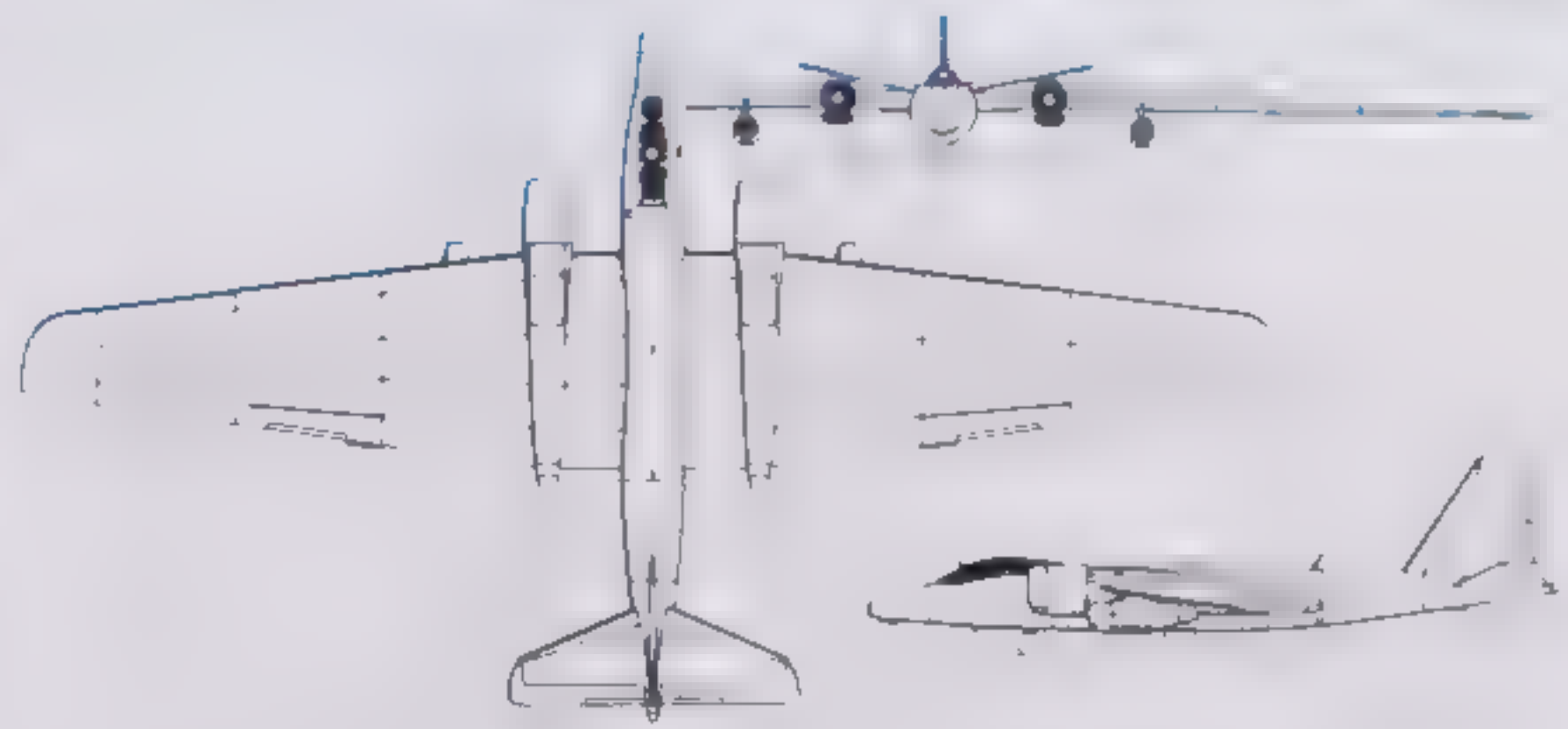
Performance: Maximum speed (A, B, C, E, G) 582mph (937km/h), (D, F) over 500mph (800km/h), initial climb (A, B, C, E, G) 3,500ft (1070m)/min, (D, F) about 4,000ft (1220m)/min, service ceiling (A, B, C, E, D) 48,000ft (14,630m), (D) 65,000ft (19,800m), (F) 75,000ft (22,860m), maximum range with combat load (high altitude) (A, B, C, E, G) 2,100 miles (3380km), (D) about 3,000 miles (4828km), (F) about 3,700 miles (5955km)

Armament: (A and all RB versions) none, (B, C, E, G) provision for four 20mm or eight 0.5in guns fixed in outer wings (very rarely, other guns fixed in forward fuselage), internal bomb load of 5,000lb (2268kg) on rotary bomb door plus eight rockets, two 500lb bombs or other stores on under-wing pylons (while retaining tip tanks)

History: First flight (Canberra in UK) 13 May 1949, (production B-57A) 20 July 1953, (B) June 1954

Users: Pakistan, Taiwan, USA (Air Force, ANG), Vietnam

Development: In October 1949 Martin flew the extremely advanced XB-51 trijet attack bomber, but this proved to be inflexible and operationally unattractive. The much less advanced British Canberra, on the other hand, proved to have precisely the qualities the US Air Force was seeking, with near-perfect operational flexibility, versatility, outstanding manoeuvrability, long range and endurance and a good weapon load. The decision to adopt this foreign combat aircraft – a step unprecedented in the US since 1918 – was swiftly followed by choice of Martin and development of the B-57A as a version built to US standards with many small modifications. The main batch comprised B-57B tandem-seaters, with dual C trainers and multi-role (tactical bomber/recce/trainer/tug) E models. Martin also made 20 grossly redesigned RB-57D reconnaissance aircraft with J57 engines on greatly extended wings. Though incapable of Canberra-style manoeuvres, nor of high speeds at low levels, the D flew many valuable multi-sensor missions over a great deal of Communist territory with the USAF and Nationalist Chinese. There were at least three D sub-types, some having counter-measures and sensing pods on the wing tips and/or tail and one version having large radomes at each end of the fuselage for strategic

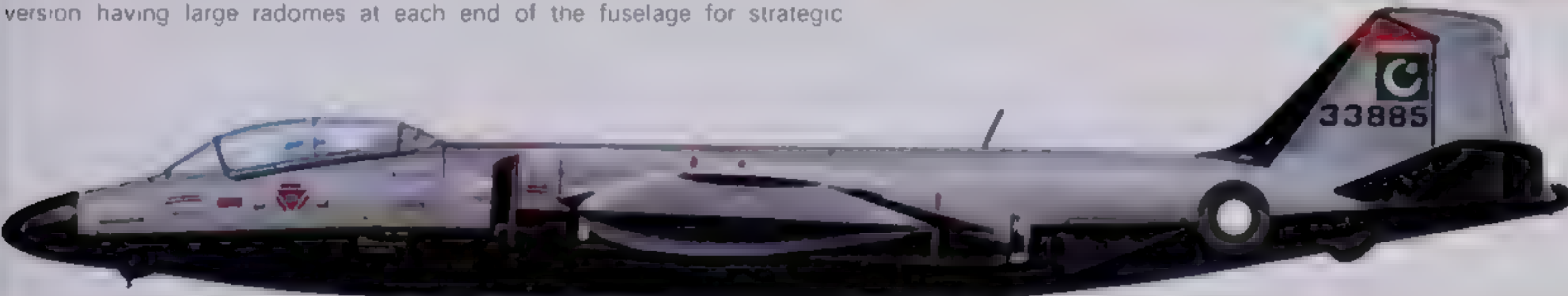


Above: Three-view of General Dynamics RB-57F.

electronic reconnaissance. Another B-57D task was to work with U-2Ds in upper-atmospheric sampling, but all of this type were grounded in 1963 as a result of structural fatigue. It was partly because of the interim nature of the D that, in 1960, General Dynamics was entrusted with the task of designing and building an even more dramatic high altitude B-57 version – the F. Though the 21 of this type were not new aircraft, little of the old is evident. The wing is entirely new, with more than double the area of the original Canberra wing and a new fatigue-resistant multi-spar structure. Most of the fuselage is new, as is the vertical tail. There are four underwing hard point for pylons, two of which are often occupied by the J60 boost engine pod supplementing the large turbofans. The nose is packed with electronic and multi-sensor equipment can be seen all over the fuselage. Various F models have operated from the United States, Europe and Middle East, Japan, Alaska, Panama, Argentina and possibly other countries. Meanwhile many of the B, C and E models have been updated by the fitment of modern night and all-weather sensing, target designation and weapon aiming systems, the rebuilt aircraft being B-57G. Major new items are low-light TV, infra-red detector and laser ranging. About half the 403 B-57s served in the night attack role in Vietnam, the G being developed just too late for the conflict. Though 10 to 15 years old, the B-57 established an outstanding record in accurate weapon delivery under the most difficult conditions.



Above: Reconditioned EB-57B of Kansas Air National Guard.



Above: B-57B of 31 Bomber Wing, Pakistan Air Force.

Below: B-57B of regular USAF unit drops eight 750lb bombs on a target in Vietnam in December 1967.



Dassault Super Mystère

Super Mystère B2 (SMB.2)

Origin: A
Type: Single seat fighter
Engine: 9 920lb
single shaft axial compressor
Dimensions:
14ft 10in (4.5m)
Weights: Empty 15,000lb (6,800kg)
Performance: Maximum speed 1,743mph (2,805km/h)
540 miles (870km)
Armament: Two 30mm DEFA
SNEB 68mm rockets two v
2 000lb (907kg)
History: First flight (Mystère IVB) 16 December 1953 (Super Mystère B2) 2 March 1955 (pre production SMB 2) 15 May 1956 (final delivery) 16 February 1957 final delivery October 1959
Users: France Israel

Development: Dassault's policy of progression by logical low-risk steps often makes it difficult to see where one type ends and another begins. The Mystère IVB, flown before 1953 was out, was a major leap ahead with tapered milled and chem-milled sheets integral tanks flush ailerons and a radar gunsight in a new nose. It also introduced the much more powerful and more highly developed Atar 101G with variable afterburner. But it proved to be only a stepping-stone to the bigger heavier and more formidable SMB 2 which introduced yet another new wing with 45° sweep and aerodynamics copied from the F-100 (but with outboard ailerons inboard flaps and a dogtooth leading edge). The flattened nose was also derived from the North American fighters of the period but Dassault kept the



Above: Three-view of Super Mystère B2, without drop tanks



Left: Regular colour scheme for the SMB.2 in Armée de l'Air service, the aircraft shown being one of No 12 Wing at Cambrai (now converting to the Mirage F1.C). The SMB.2 was the first supersonic military aircraft produced in Western Europe, and it has a great operational record.

Left: This almost dazzling Super Mystère B2 uses paint not as camouflage but to render the aircraft conspicuous. One of the aircraft formerly flown by EC 1/12, it was specially painted for the Tiger Meet at Cambrai on 13-19 June 1972, and proudly remained in its tiger skin for the remainder of its operational life.

Left: Another SMB.2 of No 12 Wing of the Armée de l'Air painted in an unusual camouflage scheme similar to that used by certain Middle East countries. Small numbers of these well-liked tactical aircraft remain in front-line service in 1977, though being replaced by Jaguar and F1.C.

Left: The Heyl Ha'Avir (Israel Defence Force/Air Force) has used several colour schemes for its combat aircraft. This SMB.2 is painted in the most common, designed for use over desert areas.

Left: The most common alternative scheme of camouflage for Heyl Ha'Avir combat aircraft is darker and shows to advantage over water and mountainous areas. Since 1958 Israel has operated this versatile fighter/bomber, but attrition and countless air combats have reduced numbers from the original 24 to about five

Dassault Etendard

Etendard IVM and IV P

Group
Type
Engine
Dimensions

Weights
Performance

Armament

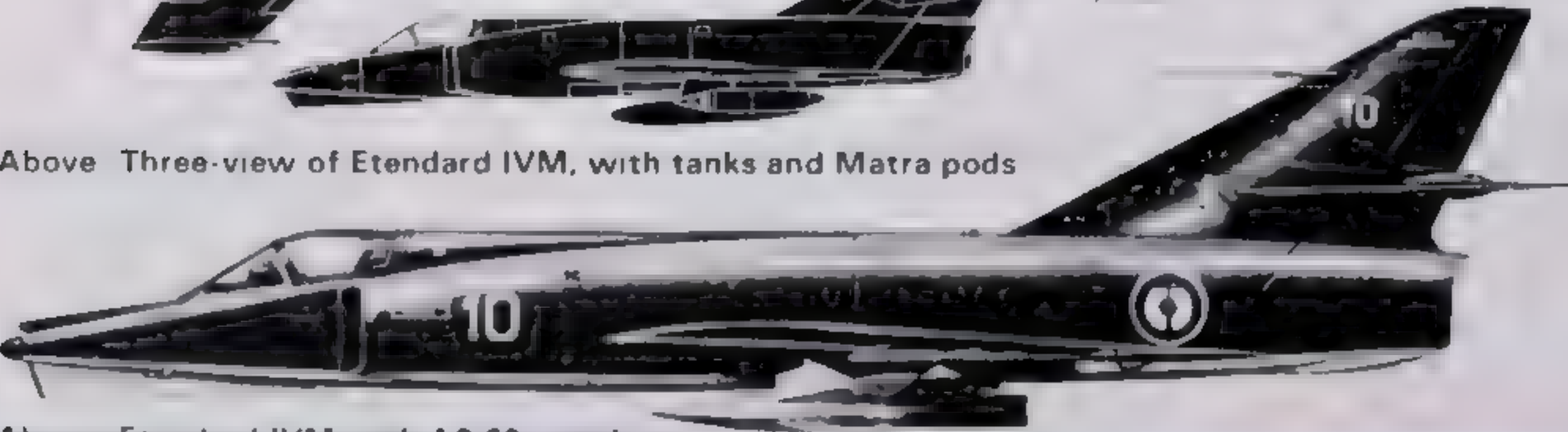
History

User

Development



Above Three-view of Etendard IVM, with tanks and Matra pods



Above Etendard IVM with AS 20 missiles.



The Etendard IVP carries five OMERA reconnaissance cameras, three in the nose (no radar) and two in the belly (no cannon)

The Etendard IVP is a reconnaissance variant of the Etendard IVM. It carries five OMERA reconnaissance cameras, three in the nose (no radar) and two in the belly (no cannon). The IVP is used for reconnaissance missions and is equipped with a fuel tank.

Below: Recovery of an Etendard IVM aboard a French carrier (both *Foch* and *Clemenceau* are still in commission). Today the Etendards are usually land-based, Flotille 11F being based at Landivisiau (together with the Etendard IVP reconnaissance squadron, 16F) and 17F being located in the Riviera at Hyères. The IVM is to be phased out from 1977.



Dassault Breguet Super Etendard

Super Etendard

Origin: Avions Marcel Dassault-Breguet Aviation, France
Type: Single-seat carrier strike fighter
Engine: 11,265lb (5110kg) thrust SNECMA Atar 8K-50 single shaft turbojet
Dimensions: Span 31ft 5½in (9.6m), length 46ft 11½in (14.31m), height 12ft 8in (3.85m)
Weights: Empty 13,889lb (6300kg), loaded 25,350lb (11,500kg)
Performance: Maximum speed 745mph (1200km/h) at sea level, Mach 1 at altitude, initial climb 24,600ft (7500m)/min, service ceiling 52,495ft (16,000m), range (clean) at altitude over 1,243 miles (2000km)
Armament: Two 30mm DEFA cannon, mission load up to 9,921lb (4500kg) carried on five pylons
History: First flight (converted Etendard) 28 October 1971, first delivery late 1977
User: France (Aéronavale)

Development: During the late 1960s it had been expected that the original force of Etendards would be replaced, in about 1971, by a specially developed version of the Jaguar, the M version with single main wheels, full carrier equipment and specially fitted for the naval strike role. A Jaguar M completed flight development and carrier compatibility, but for various reasons, mainly concerned with politics and cost, this was rejected by the Aéronavale and a search began for an alternative. After studying the A-4 Skyhawk and A-7 Corsair, the Aéronavale chose Dassault-Breguet's proposal for an improved Etendard. This has a substantially redesigned structure, for operation at higher indicated airspeeds and higher weights, a new and more efficient engine, obtained by removing the afterburner from the Atar 9K-50 of the Mirage F1C, completely new inertial navigation



Three-view of the Super Etendard (centreline pylon not shown)

system, produced mainly by SAGEM with American help, new multi-mode nose radar, produced jointly by Thomson-CSF and Electronique Marcel Dassault, with especially good performance in surface vessel detection and attack, and much greater and more varied mission load. Flight development was completed in 1974-77 with three converted Etendard IVs: the first testing the engine, the second the avionics and weapons, and the third the new wing with slats and double-slotted flaps like the Jaguar. In 1973 the Aéronavale announced it would buy 100, but this has now been cut back to 80, and budget problems have delayed service delivery until late 1977.

Below: Portrait of the first Super Etendard development aircraft, externally almost indistinguishable from the production version. This aircraft was fitted with most of the operational equipment, including the SAGEM (Singer-Kearfott) inertial navigation system and the Thomson-CSF/EMD Agave radar. Though called a "strike fighter", in fact the Super Etendard is a tactical attack machine for low-level use.



FMA IA58 Pucarà

IA 50 GII, IA 58 and Astafan Trainer

Origin	
Type	

Engines: (IA 58) two 1022ehp Turt

Dimensions: Span (A 58 and Tra

Weights: Empl 14 330 b (6500) ;

Performance: M16, 1000 ft (305 m) in 10.5 sec (0.000105 sec), 100 mph (161 km/h) (Trained) about 400 mph (643 km/h) (untrained)
 A = 3 + 1 ft (104 cm) A = 1 + 0 ft (30 cm)
 A = 2 + 10 ft (62 cm) A = 1 + 0 ft (30 cm)
 minimum (IA 1800) 1800 ft (549 m) 1800 ft (549 m)

Armament. The aircraft is capable of carrying up to 3,307 kg (7,290 lb) of stores on its fuselage and outer wings for up to 3,307 kg (7,290 lb) of stores or fuel.

History: First flight (IA 50) 23 April 1963, (IA 58) 20 August 1969
(service delivery of IA 100 November 1969)

User: Argentina

Development: The unswept wings, high-wing configuration, and the larger IA 50 GII's swept-back fuselage, tail, and tail fin and rudder. The wingspan of GII's compared to the Argentine Air Force for a transport aircraft seating for up to 15 passengers, four as photo survey aircraft with the Military Geographic Institute and one as a VIP transport for the President of Argentina. Many others were ordered later, some having ski gear for the Antarctic. The smaller IA 58 seats pilot and observer in tandem. Baker Air Force Base, Alaska, was the first delivery point for the IA 50 GII's. Deliveries began in 1961, with further batches up to a predicted total of 100 being discussed.



Three-view of IA 58 Pucará in production configuration



Above: First prototype flying with right engine feathered

Below. Production Pucara of the Fuerza Aerea Argentina, with bombs and rocket pods. Interest in this Co-In type machine has been expressed by other air forces in S America and Africa.



Fairchild Republic A-10A

A-10A

- Country
- Type
- Engines
- Dimensions
- Weights
- Performance

Armament

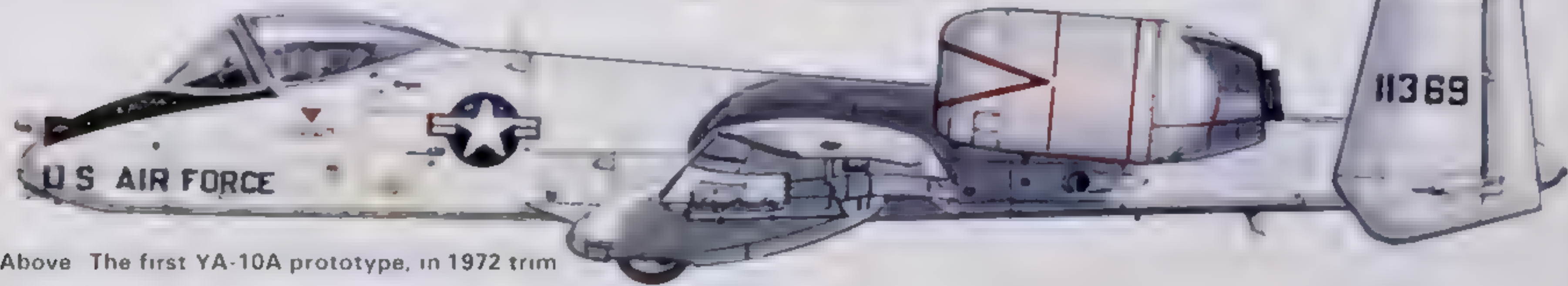
History

User: US Air Force

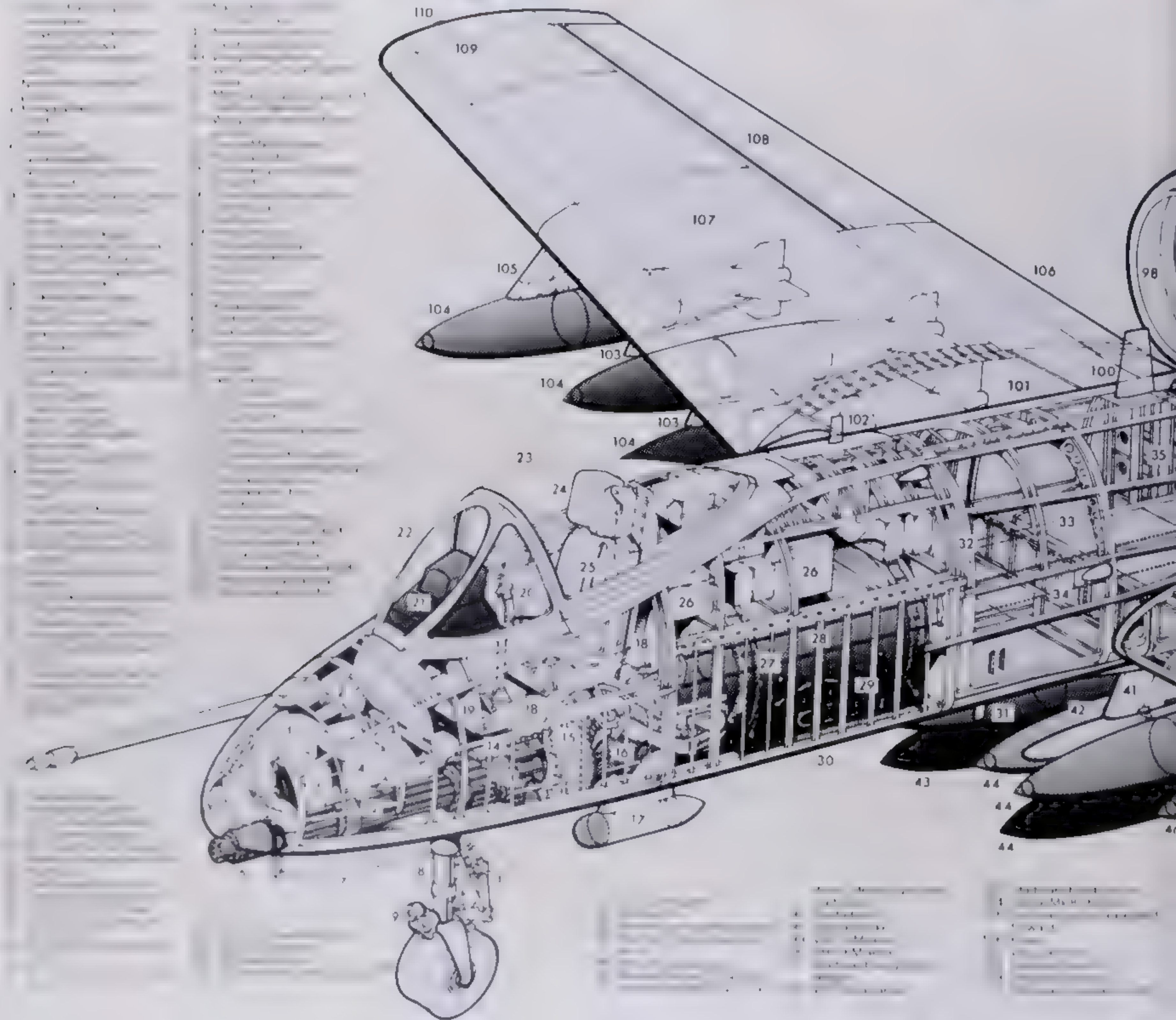
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Above: Unusual view of an A-10A as it peels away from the AT-37 chase plane to land at Edwards AFB



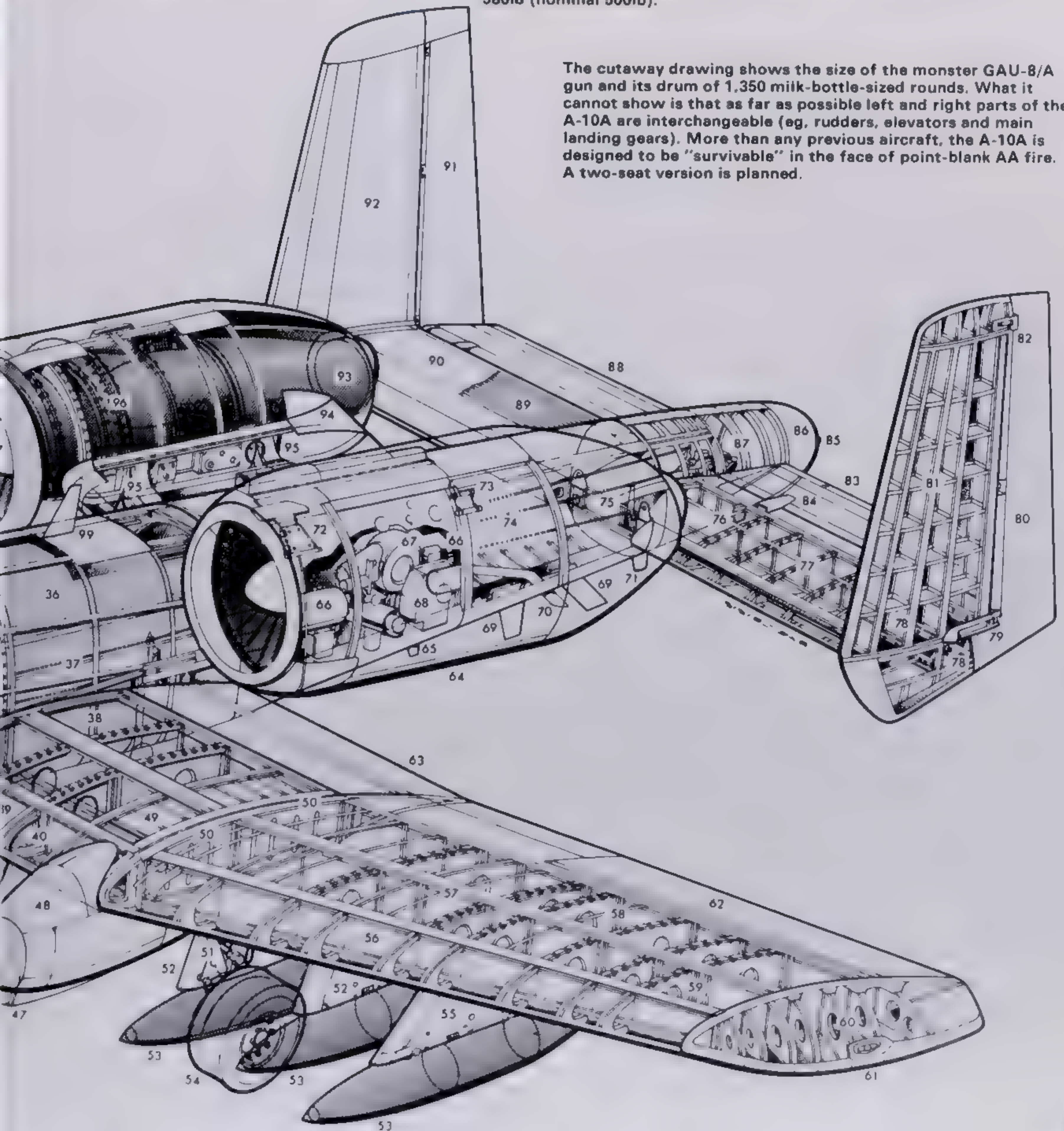
Above The first YA-10A prototype, in 1972 trim





Above: Firing the tank-killing GAU-8/A gun, with greater muzzle horsepower than any other aircraft gun in history.

Above: A production A-10A lets go a Mk 82 laser-guided bomb, one of a large family of pinpoint munitions used by the US Air Force since 1968. The A-10A can carry six of the AGM-65A Maverick missiles or 28 unguided Mk 82 bombs each weighing 580lb (nominal 500lb).



The cutaway drawing shows the size of the monster GAU-8/A gun and its drum of 1,350 milk-bottle-sized rounds. What it cannot show is that as far as possible left and right parts of the A-10A are interchangeable (eg. rudders, elevators and main landing gears). More than any previous aircraft, the A-10A is designed to be "survivable" in the face of point-blank AA fire. A two-seat version is planned.

General Dynamics F-111

**"TFX", F-111A to F-111F, EF-111A
and FB-111A**

Origin. The first record of this species was made by J. H. Mearns in 1890 from the same locality as the type of *A. fuscicornis*.

Type: Two-seat all-weather attack bomber, (EF) two-seat electronic

Engines 1A Ford 5.0 V8 1996-1997, 1B Ford 5.0 V8 1998-1999, 1C Ford 5.0 V8 2000-2001, 1D Ford 5.0 V8 2002-2003, 1E Ford 5.0 V8 2004-2005, 1F Ford 5.0 V8 2006-2007, 1G Ford 5.0 V8 2008-2009, 1H Ford 5.0 V8 2010-2011, 1I Ford 5.0 V8 2012-2013, 1J Ford 5.0 V8 2014-2015, 1K Ford 5.0 V8 2016-2017, 1L Ford 5.0 V8 2018-2019, 1M Ford 5.0 V8 2020-2021, 1N Ford 5.0 V8 2022-2023, 1O Ford 5.0 V8 2024-2025, 1P Ford 5.0 V8 2026-2027, 1Q Ford 5.0 V8 2028-2029, 1R Ford 5.0 V8 2030-2031, 1S Ford 5.0 V8 2032-2033, 1T Ford 5.0 V8 2034-2035, 1U Ford 5.0 V8 2036-2037, 1V Ford 5.0 V8 2038-2039, 1W Ford 5.0 V8 2040-2041, 1X Ford 5.0 V8 2042-2043, 1Y Ford 5.0 V8 2044-2045, 1Z Ford 5.0 V8 2046-2047, 1AA Ford 5.0 V8 2048-2049, 1AB Ford 5.0 V8 2050-2051, 1AC Ford 5.0 V8 2052-2053, 1AD Ford 5.0 V8 2054-2055, 1AE Ford 5.0 V8 2056-2057, 1AF Ford 5.0 V8 2058-2059, 1AG Ford 5.0 V8 2060-2061, 1AH Ford 5.0 V8 2062-2063, 1AI Ford 5.0 V8 2064-2065, 1AJ Ford 5.0 V8 2066-2067, 1AK Ford 5.0 V8 2068-2069, 1AL Ford 5.0 V8 2070-2071, 1AM Ford 5.0 V8 2072-2073, 1AN Ford 5.0 V8 2074-2075, 1AO Ford 5.0 V8 2076-2077, 1AP Ford 5.0 V8 2078-2079, 1AQ Ford 5.0 V8 2080-2081, 1AR Ford 5.0 V8 2082-2083, 1AS Ford 5.0 V8 2084-2085, 1AT Ford 5.0 V8 2086-2087, 1AU Ford 5.0 V8 2088-2089, 1AV Ford 5.0 V8 2090-2091, 1AW Ford 5.0 V8 2092-2093, 1AX Ford 5.0 V8 2094-2095, 1AY Ford 5.0 V8 2096-2097, 1AZ Ford 5.0 V8 2098-2099, 1BA Ford 5.0 V8 2100-2101, 1BB Ford 5.0 V8 2102-2103, 1BC Ford 5.0 V8 2104-2105, 1BD Ford 5.0 V8 2106-2107, 1BE Ford 5.0 V8 2108-2109, 1BF Ford 5.0 V8 2110-2111, 1BG Ford 5.0 V8 2112-2113, 1BH Ford 5.0 V8 2114-2115, 1BI Ford 5.0 V8 2116-2117, 1BJ Ford 5.0 V8 2118-2119, 1BK Ford 5.0 V8 2120-2121, 1BL Ford 5.0 V8 2122-2123, 1BM Ford 5.0 V8 2124-2125, 1BN Ford 5.0 V8 2126-2127, 1BO Ford 5.0 V8 2128-2129, 1BP Ford 5.0 V8 2130-2131, 1BQ Ford 5.0 V8 2132-2133, 1BR Ford 5.0 V8 2134-2135, 1BS Ford 5.0 V8 2136-2137, 1BT Ford 5.0 V8 2138-2139, 1BU Ford 5.0 V8 2140-2141, 1BV Ford 5.0 V8 2142-2143, 1BW Ford 5.0 V8 2144-2145, 1BX Ford 5.0 V8 2146-2147, 1BY Ford 5.0 V8 2148-2149, 1BZ Ford 5.0 V8 2150-2151, 1CA Ford 5.0 V8 2152-2153, 1CB Ford 5.0 V8 2154-2155, 1CC Ford 5.0 V8 2156-2157, 1CD Ford 5.0 V8 2158-2159, 1CE Ford 5.0 V8 2160-2161, 1CF Ford 5.0 V8 2162-2163, 1CG Ford 5.0 V8 2164-2165, 1CH Ford 5.0 V8 2166-2167, 1CI Ford 5.0 V8 2168-2169, 1CJ Ford 5.0 V8 2170-2171, 1CK Ford 5.0 V8 2172-2173, 1CL Ford 5.0 V8 2174-2175, 1CM Ford 5.0 V8 2176-2177, 1CN Ford 5.0 V8 2178-2179, 1CO Ford 5.0 V8 2180-2181, 1CP Ford 5.0 V8 2182-2183, 1CQ Ford 5.0 V8 2184-2185, 1CR Ford 5.0 V8 2186-2187, 1CS Ford 5.0 V8 2188-2189, 1CT Ford 5.0 V8 2190-2191, 1CU Ford 5.0 V8 2192-2193, 1CV Ford 5.0 V8 2194-2195, 1CW Ford 5.0 V8 2196-2197, 1CX Ford 5.0 V8 2198-2199, 1CA Ford 5.0 V8 2200-2201, 1CB Ford 5.0 V8 2202-2203, 1CC Ford 5.0 V8 2204-2205, 1CD Ford 5.0 V8 2206-2207, 1CE Ford 5.0 V8 2208-2209, 1CF Ford 5.0 V8 2210-2211, 1CG Ford 5.0 V8 2212-2213, 1CH Ford 5.0 V8 2214-2215, 1CI Ford 5.0 V8 2216-2217, 1CJ Ford 5.0 V8 2218-2219, 1CK Ford 5.0 V8 2220-2221, 1CL Ford 5.0 V8 2222-2223, 1CM Ford 5.0 V8 2224-2225, 1CN Ford 5.0 V8 2226-2227, 1CO Ford 5.0 V8 2228-2229, 1CP Ford 5.0 V8 2230-2231, 1CQ Ford 5.0 V8 2232-2233, 1CR Ford 5.0 V8 2234-2235, 1CS Ford 5.0 V8 2236-2237, 1CT Ford 5.0 V8 2238-2239, 1CU Ford 5.0 V8 2240-2241, 1CV Ford 5.0 V8 2242-2243, 1CW Ford 5.0 V8 2244-2245, 1CX Ford 5.0 V8 2246-2247, 1CY Ford 5.0 V8 2248-2249, 1CZ Ford 5.0 V8 2250-2251, 1DA Ford 5.0 V8 2252-2253, 1DB Ford 5.0 V8 2254-2255, 1DC Ford 5.0 V8 2256-2257, 1DD Ford 5.0 V8 2258-2259, 1DE Ford 5.0 V8 2260-2261, 1DF Ford 5.0 V8 2262-2263, 1DG Ford 5.0 V8 2264-2265, 1DH Ford 5.0 V8 2266-2267, 1DI Ford 5.0 V8 2268-2269, 1DJ Ford 5.0 V8 2270-2271, 1DK Ford 5.0 V8 2272-2273, 1DL Ford 5.0 V8 2274-2275, 1DM Ford 5.0 V8 2276-2277, 1DN Ford 5.0 V8 2278-2279, 1DO Ford 5.0 V8 2280-2281, 1DP Ford 5.0 V8 2282-2283, 1DQ Ford 5.0 V8 2284-2285, 1DR Ford 5.0 V8 2286-2287, 1DS Ford 5.0 V8 2288-2289, 1DT Ford 5.0 V8 2290-2291, 1DU Ford 5.0 V8 2292-2293, 1DV Ford 5.0 V8 2294-2295, 1DW Ford 5.0 V8 2296-2297, 1DX Ford 5.0 V8 2298-2299, 1DA Ford 5.0 V8 2300-2301, 1DB Ford 5.0 V8 2302-2303, 1DC Ford 5.0 V8 2304-2305, 1DD Ford 5.0 V8 2306-2307, 1DE Ford 5.0 V8 2308-2309, 1DF Ford 5.0 V8 2310-2311, 1DG Ford 5.0 V8 2312-2313, 1DH Ford 5.0 V8 2314-2315, 1DI Ford 5.0 V8 2316-2317, 1DJ Ford 5.0 V8 2318-2319, 1DK Ford 5.0 V8 2320-2321, 1DL Ford 5.0 V8 2322-2323, 1DM Ford 5.0 V8 2324-2325, 1DN Ford 5.0 V8 2326-2327, 1DO Ford 5.0 V8 2328-2329, 1DP Ford 5.0 V8 2330-2331, 1DQ Ford 5.0 V8 2332-2333, 1DR Ford 5.0 V8 2334-2335, 1DS Ford 5.0 V8 2336-2337, 1DT Ford 5.0 V8 2338-2339, 1DU Ford 5.0 V8 2340-2341, 1DV Ford 5.0 V8 2342-2343, 1DW Ford 5.0 V8 2344-2345, 1DX Ford 5.0 V8 2346-2347, 1DY Ford 5.0 V8 2348-2349, 1DZ Ford 5.0 V8 2350-2351, 1EA Ford 5.0 V8 2352-2353, 1EB Ford 5.0 V8 2354-2355, 1EC Ford 5.0 V8 2356-2357, 1ED Ford 5.0 V8 2358-2359, 1EE Ford 5.0 V8 2360-2361, 1EF Ford 5.0 V8 2362-2363, 1EG Ford 5.0 V8 2364-2365, 1EH Ford 5.0 V8 2366-2367, 1EI Ford 5.0 V8 2368-2369, 1EJ Ford 5.0 V8 2370-2371, 1EK Ford 5.0 V8 2372-2373, 1EL Ford 5.0 V8 2374-2375, 1EM Ford 5.0 V8 2376-2377, 1EN Ford 5.0 V8 2378-2379, 1EO Ford 5.0 V8 2380-2381, 1EP Ford 5.0 V8 2382-2383, 1

Dimensions: Span, 72 5' sweep (A, D, E, F) 31ft 11½in (9 74m), (C, FB) 33ft 11in (10 34m) span 16° sweep (A, D, E, F) 63ft (19 2m), (C, FB) 64ft 11in (19 8m) height 30ft 7in (9 4m), height 17ft 1 in (5 2m)

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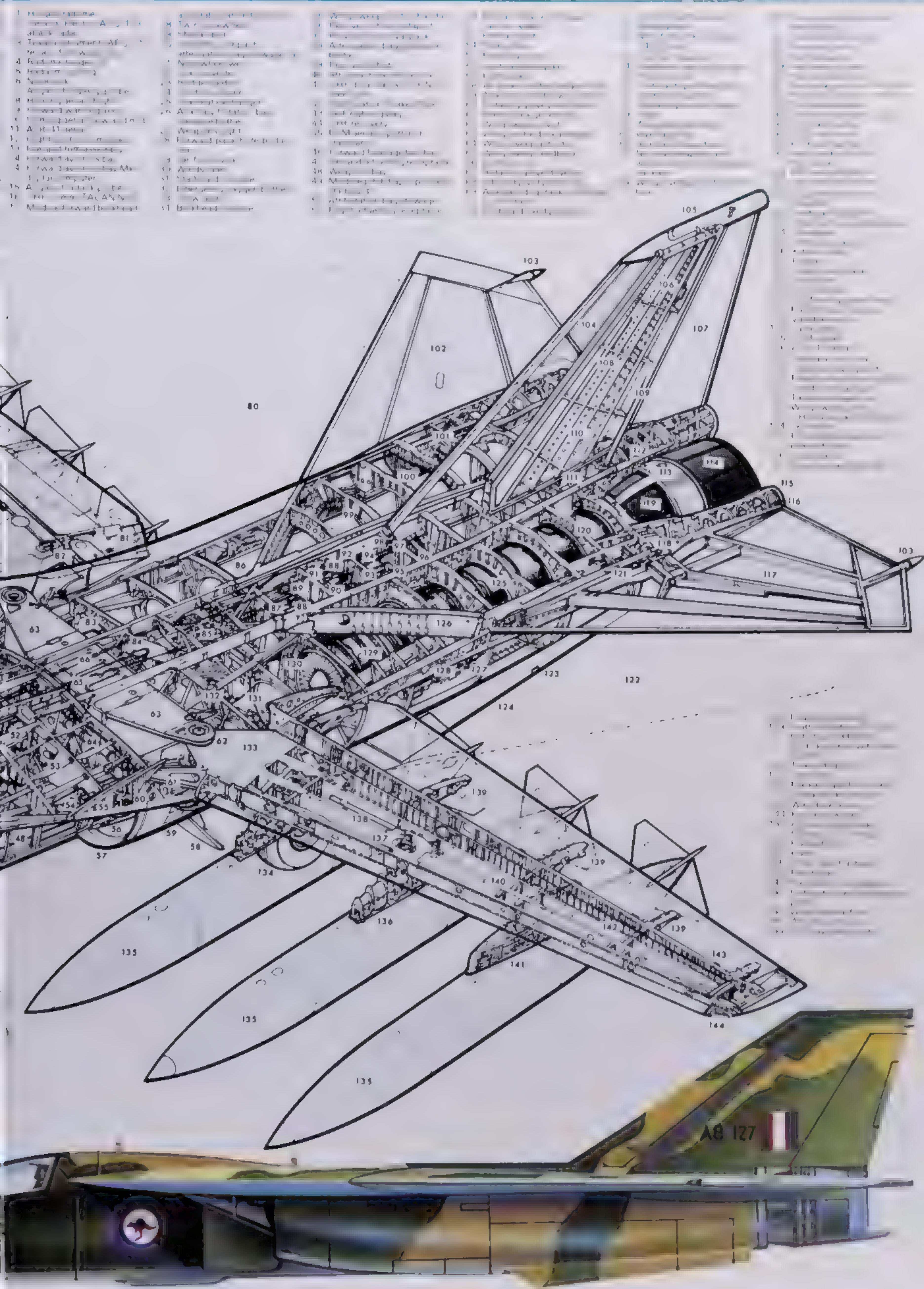
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Above: Three-view of the FB-111A strategic bomber version

Below: Destruction rains down from an F-111A development aircraft during weapon-delivery tests in 1966. Using only the two inboard (swivelling) pylons on each swing-wing, 24 Mk 82 bombs, each of a nominal 500lb mass (actual weight, 580lb), fall in a loose cluster. Maximum bomb load of most F-111 tactical versions is 28,000lb, while the FB-111A can carry 31,500lb (14,288kg).





Grumman A-6 Intruder and Prowler

Grumman A-6A,B,C,E, EA-6A and B and KA-6D

Class
Type

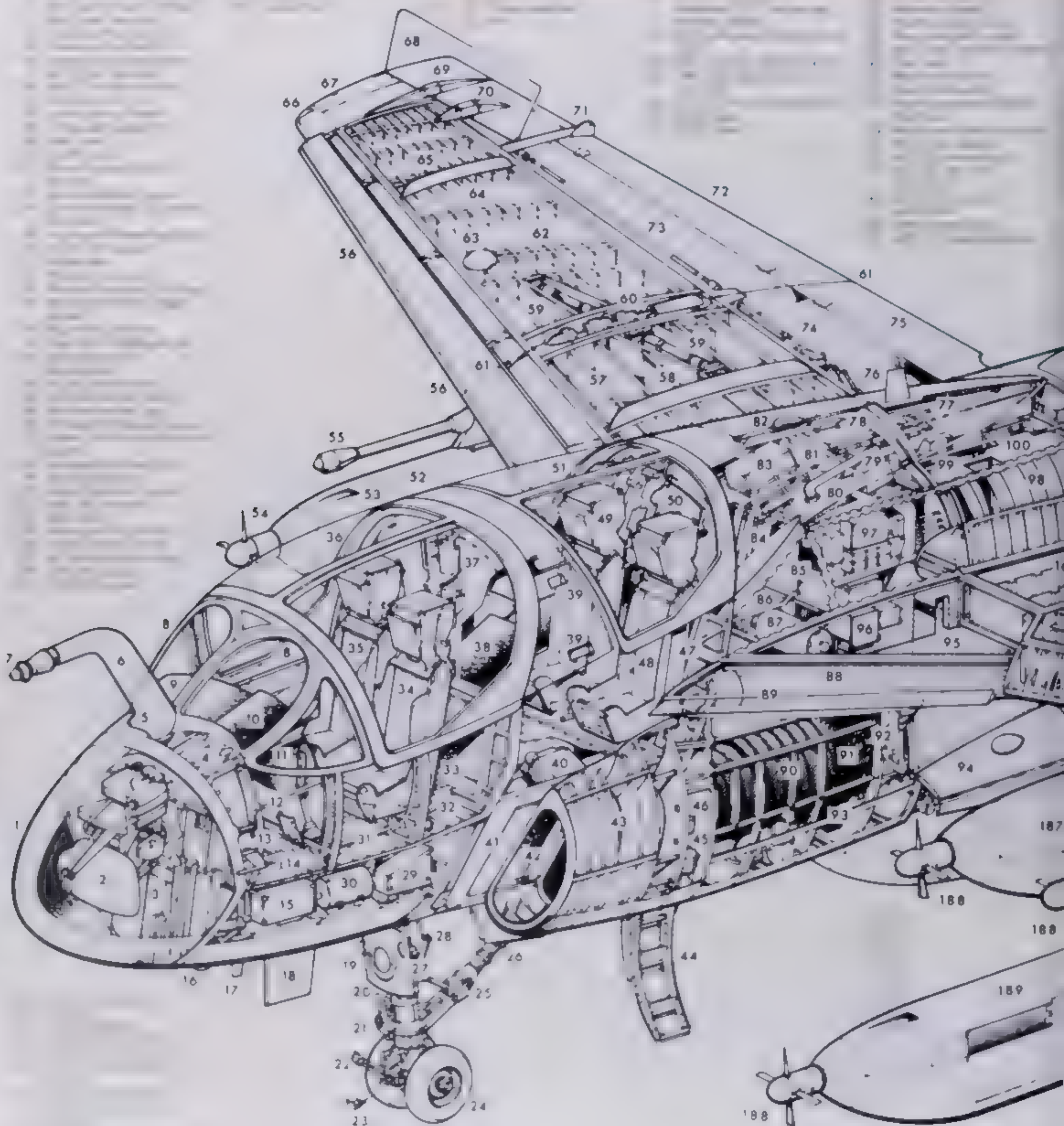
Engines

Dimensions

Weights

Performance

Armament





History

EA-6B

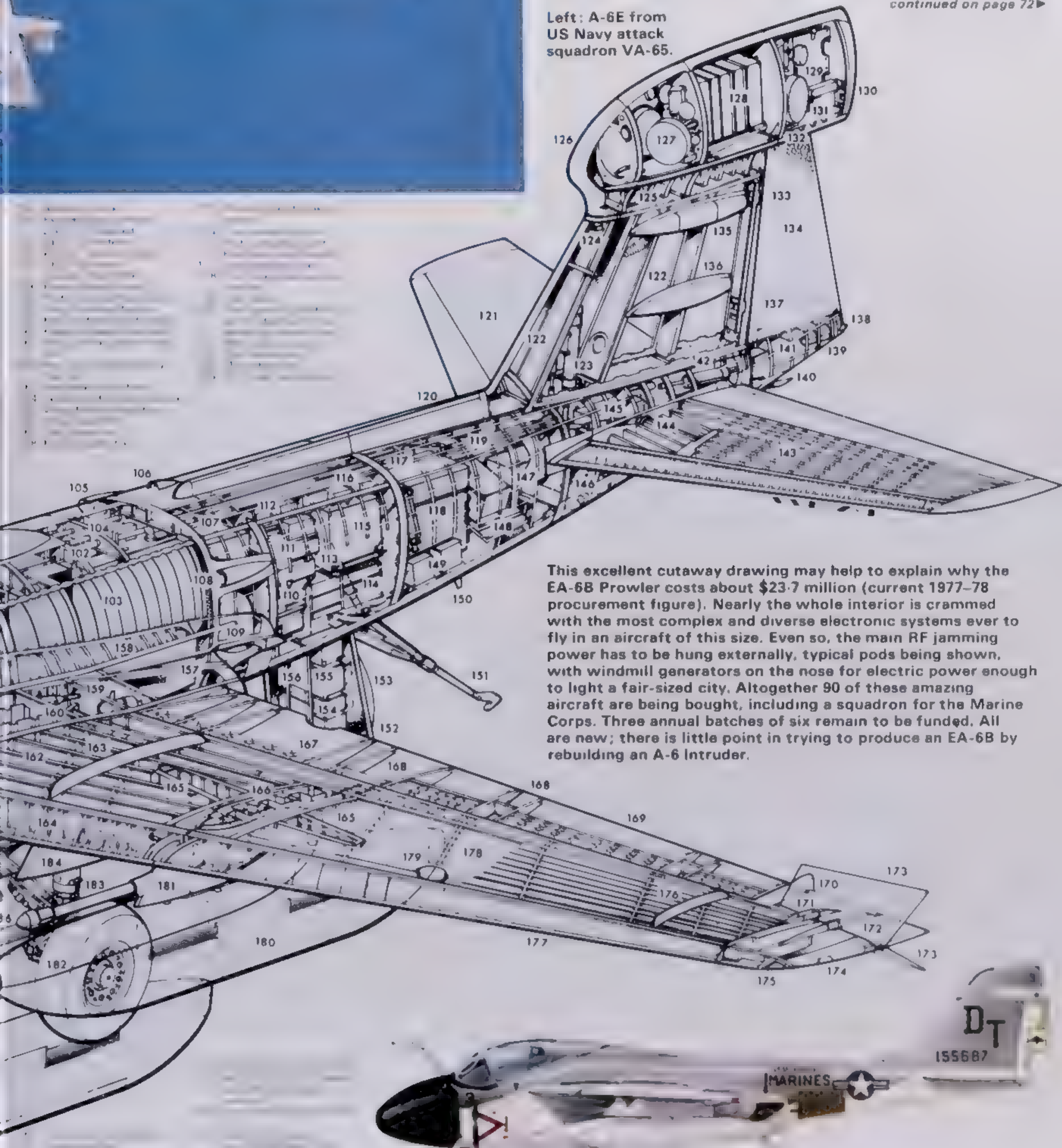
User

Development

which in the original design was intended to provide lift for STOL (short takeoff and landing) aircraft. The gross weight is much more than most of the heavy World War II fighters. The Intruder has excellent slow flying qualities with full span slats and flaps. The crew sit side by side under a broad side window canopy giving a marvelous view in all directions, the navigator having control of the extremely comprehensive navigation, radar and attack systems which are integrated

continued on page 72

Left: A-6E from US Navy attack squadron VA-65.



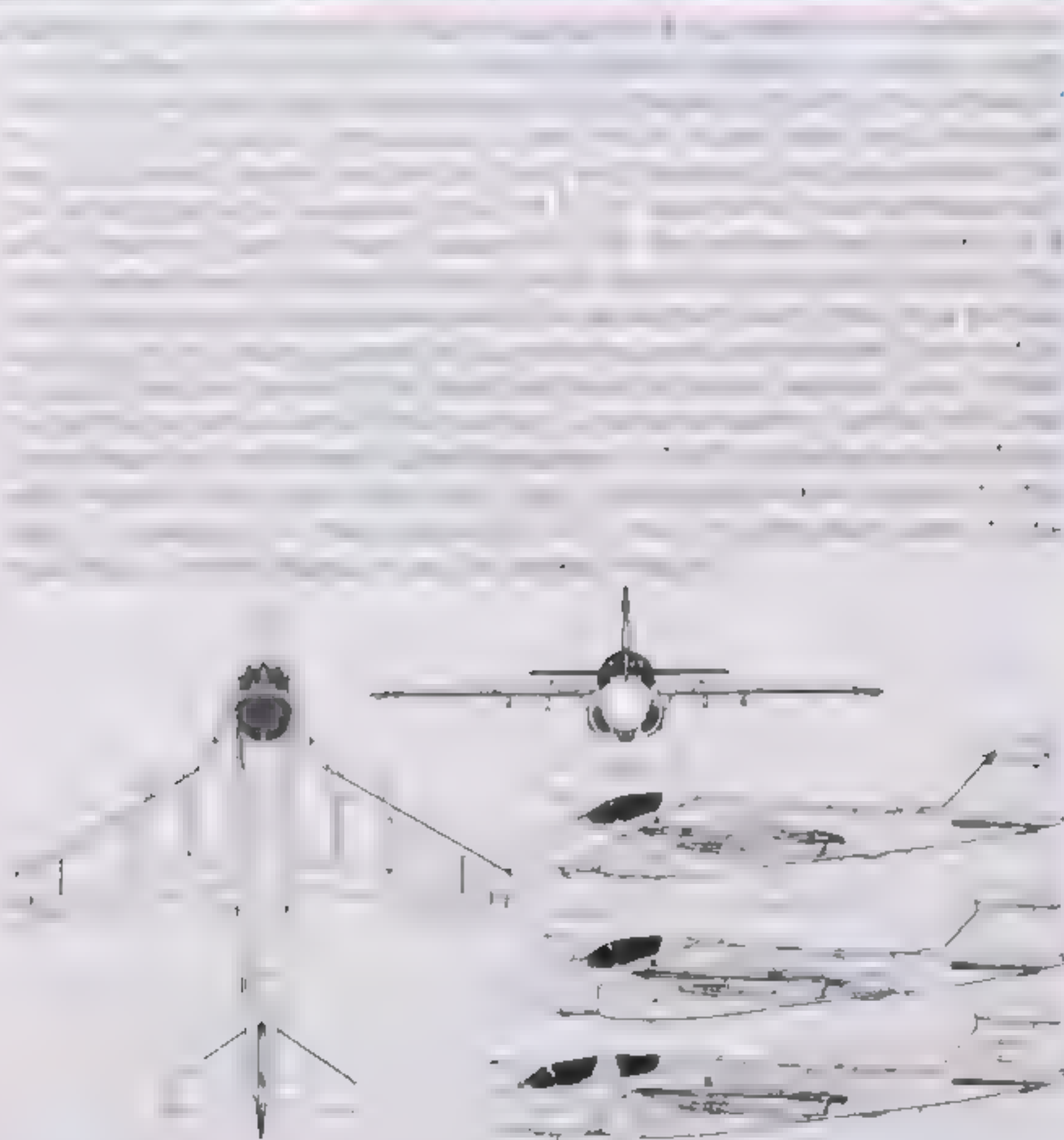
This excellent cutaway drawing may help to explain why the EA-6B Prowler costs about \$23.7 million (current 1977-78 procurement figure). Nearly the whole interior is crammed with the most complex and diverse electronic systems ever to fly in an aircraft of this size. Even so, the main RF jamming power has to be hung externally, typical pods being shown, with windmill generators on the nose for electric power enough to light a fair-sized city. Altogether 90 of these amazing aircraft are being bought, including a squadron for the Marine Corps. Three annual batches of six remain to be funded. All are new; there is little point in trying to produce an EA-6B by rebuilding an A-6 Intruder.



Above: A-6E of US Marine Corps VMA (AW)-242 "The Bats"

EA-6B (A-6E) jamming systems pod

DT
155687



Above: Three-view of A-6E, with side views of EA-6A (centre) and EA-6B (bottom)



Above: As project pilot Peter Timillo climbs down the ladder, navigator Jim Johnson inspects the TRAM (Target-Recognition Attack Multisensor) under the nose of their special test A-6E.



Below: Cadillac among combat aircraft, an EA-6B Prowler of US Navy VAQ-129 shows off its ALQ-99 high-power jamming pods with windmill generators—and staircase to get aboard



Above: Hectic launch of an A-6A strike during the involvement in Vietnam. Intruders of VA-165 aboard *Constellation*.

Left: Two Rockwell AGM-53B Condor long-range homing air/surface missiles adorn the wings of this A-6A trials aircraft of the Naval Ordnance Test Station at China Lake, California



Article closes with a note.

Origin: HALL, John, A. (c. 1920, Farnham, Surrey, British Aerospace) UK

Type: Two-seat attack and reconnaissance

Engines: (S 1) two 7,100lb (3220kg) thrust Bristol Siddeley Pegasus Mk 101 turbojets; (S 2) two 8,000lb (3630kg) Bristol Siddeley Olympus Mk 591 turbojets; (S 3) two 11,000lb (4990kg) Bristol Siddeley Olympus Mk 592 turbojets.

Dimensions: Span 42' 4" 1st 42' 2" 2nd 42' 1" 3rd 42' 1" 4th 63' 5" (19.3m) 5th 49' 1" 6th 49' 1"

Weights: Empty (1) 26 000lb (2) about 30 000lb (13 f' x 14 ft x 14 ft)
loaded (1) 46 000lb (20,865kg), (2) 62 000lb (28 123kg)

Performance: Max climb speed 4 ft/sec (1.2 m/sec), 16 ft/sec (4.9 m/sec) at sea level; initial climb rate 46 ft/sec (14.0 m/sec); 700 ft/sec (213.4 m/sec) not disclosed but over 40 000 ft (9144 m) range on typical hi lo hi strike mission with weapon load (2) 2,300 miles (3700 km)

Armament: Rotating bomb door carries four 1,000lb (454kg) bombs or multi-sensor reconnaissance pack

History. First at NA 39-3, Apr. 1958; died there, 11/23/58; 1962 report p. 5, 17/11, 1/13, 1/15, 1/16, 1/17, 1/18, 1/19, 1/20, 1/21, 1/22, 1/23, 1/24, 1/25, 1/26, 1/27, 1/28, 1/29, 1/30, 1/31, 2/1, 2/2, 2/3, 2/4, 2/5, 2/6, 2/7, 2/8, 2/9, 2/10, 2/11, 2/12, 2/13, 2/14, 2/15, 2/16, 2/17, 2/18, 2/19, 2/20, 2/21, 2/22, 2/23, 2/24, 2/25, 2/26, 2/27, 2/28, 2/29, 2/30, 3/1, 3/2, 3/3, 3/4, 3/5, 3/6, 3/7, 3/8, 3/9, 3/10, 3/11, 3/12, 3/13, 3/14, 3/15, 3/16, 3/17, 3/18, 3/19, 3/20, 3/21, 3/22, 3/23, 3/24, 3/25, 3/26, 3/27, 3/28, 3/29, 3/30, 3/31, 4/1, 4/2, 4/3, 4/4, 4/5, 4/6, 4/7, 4/8, 4/9, 4/10, 4/11, 4/12, 4/13, 4/14, 4/15, 4/16, 4/17, 4/18, 4/19, 4/20, 4/21, 4/22, 4/23, 4/24, 4/25, 4/26, 4/27, 4/28, 4/29, 4/30, 5/1, 5/2, 5/3, 5/4, 5/5, 5/6, 5/7, 5/8, 5/9, 5/10, 5/11, 5/12, 5/13, 5/14, 5/15, 5/16, 5/17, 5/18, 5/19, 5/20, 5/21, 5/22, 5/23, 5/24, 5/25, 5/26, 5/27, 5/28, 5/29, 5/30, 5/31, 6/1, 6/2, 6/3, 6/4, 6/5, 6/6, 6/7, 6/8, 6/9, 6/10, 6/11, 6/12, 6/13, 6/14, 6/15, 6/16, 6/17, 6/18, 6/19, 6/20, 6/21, 6/22, 6/23, 6/24, 6/25, 6/26, 6/27, 6/28, 6/29, 6/30, 7/1, 7/2, 7/3, 7/4, 7/5, 7/6, 7/7, 7/8, 7/9, 7/10, 7/11, 7/12, 7/13, 7/14, 7/15, 7/16, 7/17, 7/18, 7/19, 7/20, 7/21, 7/22, 7/23, 7/24, 7/25, 7/26, 7/27, 7/28, 7/29, 7/30, 7/31, 8/1, 8/2, 8/3, 8/4, 8/5, 8/6, 8/7, 8/8, 8/9, 8/10, 8/11, 8/12, 8/13, 8/14, 8/15, 8/16, 8/17, 8/18, 8/19, 8/20, 8/21, 8/22, 8/23, 8/24, 8/25, 8/26, 8/27, 8/28, 8/29, 8/30, 8/31, 9/1, 9/2, 9/3, 9/4, 9/5, 9/6, 9/7, 9/8, 9/9, 9/10, 9/11, 9/12, 9/13, 9/14, 9/15, 9/16, 9/17, 9/18, 9/19, 9/20, 9/21, 9/22, 9/23, 9/24, 9/25, 9/26, 9/27, 9/28, 9/29, 9/30, 10/1, 10/2, 10/3, 10/4, 10/5, 10/6, 10/7, 10/8, 10/9, 10/10, 10/11, 10/12, 10/13, 10/14, 10/15, 10/16, 10/17, 10/18, 10/19, 10/20, 10/21, 10/22, 10/23, 10/24, 10/25, 10/26, 10/27, 10/28, 10/29, 10/30, 10/31, 11/1, 11/2, 11/3, 11/4, 11/5, 11/6, 11/7, 11/8, 11/9, 11/10, 11/11, 11/12, 11/13, 11/14, 11/15, 11/16, 11/17, 11/18, 11/19, 11/20, 11/21, 11/22, 11/23, 11/24, 11/25, 11/26, 11/27, 11/28, 11/29, 11/30, 12/1, 12/2, 12/3, 12/4, 12/5, 12/6, 12/7, 12/8, 12/9, 12/10, 12/11, 12/12, 12/13, 12/14, 12/15, 12/16, 12/17, 12/18, 12/19, 12/20, 12/21, 12/22, 12/23, 12/24, 12/25, 12/26, 12/27, 12/28, 12/29, 12/30, 12/31, 1963 report p. 5, 17/11, 1/13, 1/15, 1/16, 1/17, 1/18, 1/19, 1/20, 1/21, 1/22, 1/23, 1/24, 1/25, 1/26, 1/27, 1/28, 1/29, 1/30, 2/1, 2/2, 2/3, 2/4, 2/5, 2/6, 2/7, 2/8, 2/9, 2/10, 2/11, 2/12, 2/13, 2/14, 2/15, 2/16, 2/17, 2/18, 2/19, 2/20, 2/21, 2/22, 2/23, 2/24, 2/25, 2/26, 2/27, 2/28, 2/29, 2/30, 3/1, 3/2, 3/3, 3/4, 3/5, 3/6, 3/7, 3/8, 3/9, 3/10, 3/11, 3/12, 3/13, 3/14, 3/15, 3/16, 3/17, 3/18, 3/19, 3/20, 3/21, 3/22, 3/23, 3/24, 3/25, 3/26, 3/27, 3/28, 3/29, 3/30, 3/31, 4/1, 4/2, 4/3, 4/4, 4/5, 4/6, 4/7, 4/8, 4/9, 4/10, 4/11, 4/12, 4/13, 4/14, 4/15, 4/16, 4/17, 4/18, 4/19, 4/20, 4/21, 4/22, 4/23, 4/24, 4/25, 4/26, 4/27, 4/28, 4/29, 4/30, 5/1, 5/2, 5/3, 5/4, 5/5, 5/6, 5/7, 5/8, 5/9, 5/10, 5/11, 5/12, 5/13, 5/14, 5/15, 5/16, 5/17, 5/18, 5/19, 5/20, 5/21, 5/22, 5/23, 5/24, 5/25, 5/26, 5/27, 5/28, 5/29, 5/30, 5/31, 6/1, 6/2, 6/3, 6/4, 6/5, 6/6, 6/7, 6/8, 6/9, 6/10, 6/11, 6/12, 6/13, 6/14, 6/15, 6/16, 6/17, 6/18, 6/19, 6/20, 6/21, 6/22, 6/23, 6/24, 6/25, 6/26, 6/27, 6/28, 6/29, 6/30, 7/1, 7/2, 7/3, 7/4, 7/5, 7/6, 7/7, 7/8, 7/9, 7/10, 7/11, 7/12, 7/13, 7/14, 7/15, 7/16, 7/17, 7/18, 7/19, 7/20, 7/21, 7/22, 7/23, 7/24, 7/25, 7/26, 7/27, 7/28, 7/29, 7/30, 7/31, 8/1, 8/2, 8/3, 8/4, 8/5, 8/6, 8/7, 8/8, 8/9, 8/10, 8/11, 8/12, 8/13, 8/14, 8/15, 8/16, 8/17, 8/18, 8/19, 8/20, 8/21, 8/22, 8/23, 8/24, 8/25, 8/26, 8/27, 8/28, 8/29, 8/30, 8/31, 9/1, 9/2, 9/3, 9/4, 9/5, 9/6, 9/7, 9/8, 9/9, 9/10, 9/11, 9/12, 9/13, 9/14, 9/15, 9/16, 9/17, 9/18, 9/19, 9/20, 9/21, 9/22, 9/23, 9/24, 9/25, 9/26, 9/27, 9/28, 9/29, 9/30, 10/1, 10/2, 10/3, 10/4, 10/5, 10/6, 10/7, 10/8, 10/9, 10/10, 10/11, 10/12, 10/13, 10/14, 10/15, 10/16, 10/17, 10/18, 10/19, 10/20, 10/21, 10/22, 10/23, 10/24, 10/25, 10/26, 10/27, 10/28, 10/29, 10/30, 10/31, 11/1, 11/2, 11/3, 11/4, 11/5, 11/6, 11/7, 11/8, 11/9, 11/10, 11/11, 11/12, 11/13, 11/14, 11/15, 11/16, 11/17, 11/18, 11/19, 11/20, 11/21, 11/22, 11/23, 11/24, 11/25, 11/26, 11/27, 11/28, 11/29, 11/30, 1

Users: S Afr ca, UK (RAF, Royal Navy)

continued on page 74 ►



Above: Spin-stabilized 68mm rockets ripple from the 18-tube launchers of a Buccaneer of the RAF

Below: Buccaneer S.2 of 809 Sqn, Royal Navy, with flight-refuelling "Buddy" Mk 20B hose-reel pod on inner right pylon

Foot of page: Low-level run by a Buccaneer S 2A of 12 Sqn, RAF Over the years the Buccaneer has been equipped with ever-better ECM and other defensive avionics, some of which show externally. Most aircraft carry wing drop tanks and a fixed FR probe.





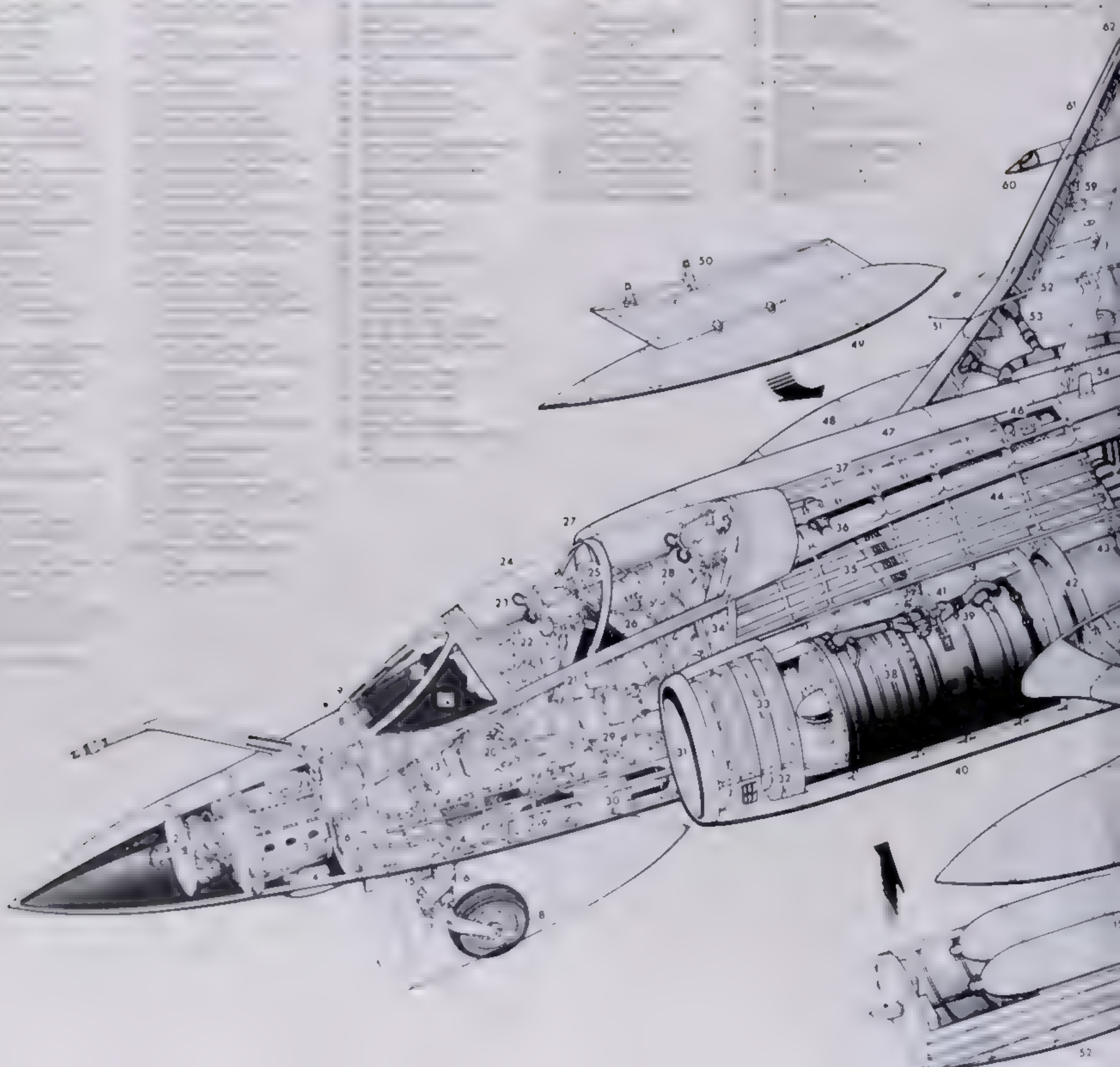
Three-view of Buccaneer S.2 with FR probe and bomb-door tank.

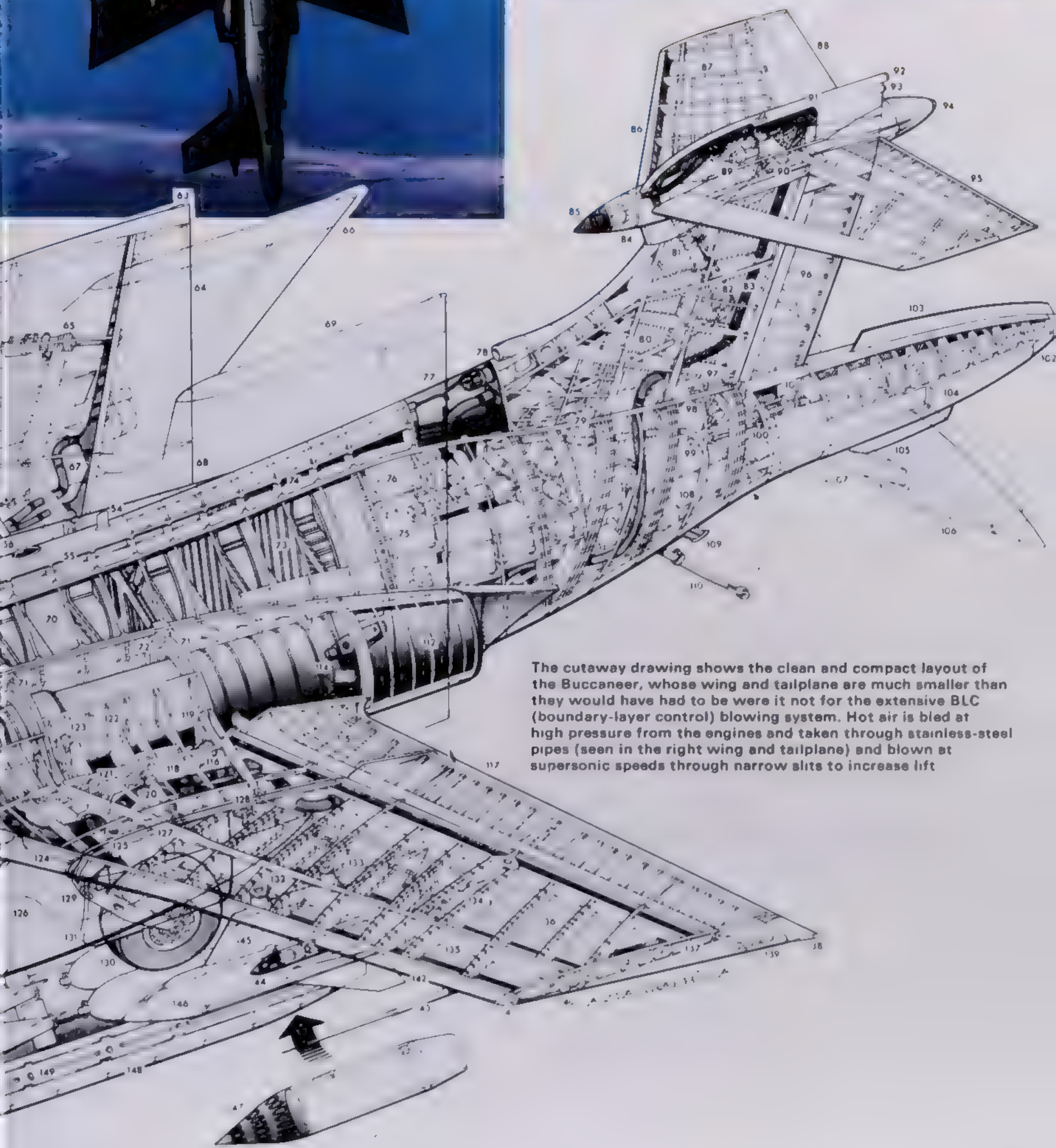


Buccaneer S.2B of 16 Sqn, Laarbruch.

Right: One of the missions of the Buccaneer is the LABS "over the shoulder" toss of nuclear weapons. Here, in vertical attitude, is an S 2A of 237 OCU.

Far right: Heading into the sunset, this S.2B of 15 Sqn, RAF Laarbruch, has bulged weapon-door fuel tank and rocket pods.





The cutaway drawing shows the clean and compact layout of the Buccaneer, whose wing and tailplane are much smaller than they would have had to be were it not for the extensive BLC (boundary-layer control) blowing system. Hot air is bled at high pressure from the engines and taken through stainless-steel pipes (seen in the right wing and tailplane) and blown at supersonic speeds through narrow slits to increase lift

Hawker Siddeley Harrier and Sea Harrier

Harrier GR.3 and T.4, AV-8A, TAV-8A and Sea Harrier FRS.1

Origin
Type: Single seat tactical attack and reconnaissance (T.4, TAV) dual

Engine

Dimensions: Span 25ft 3in (7.7m) (with bolt on tip)

Weights: Empty (GR 1) 12 200 lb (5533kg) (Sea H) 13 000lb (5897kg)

continued on page 78

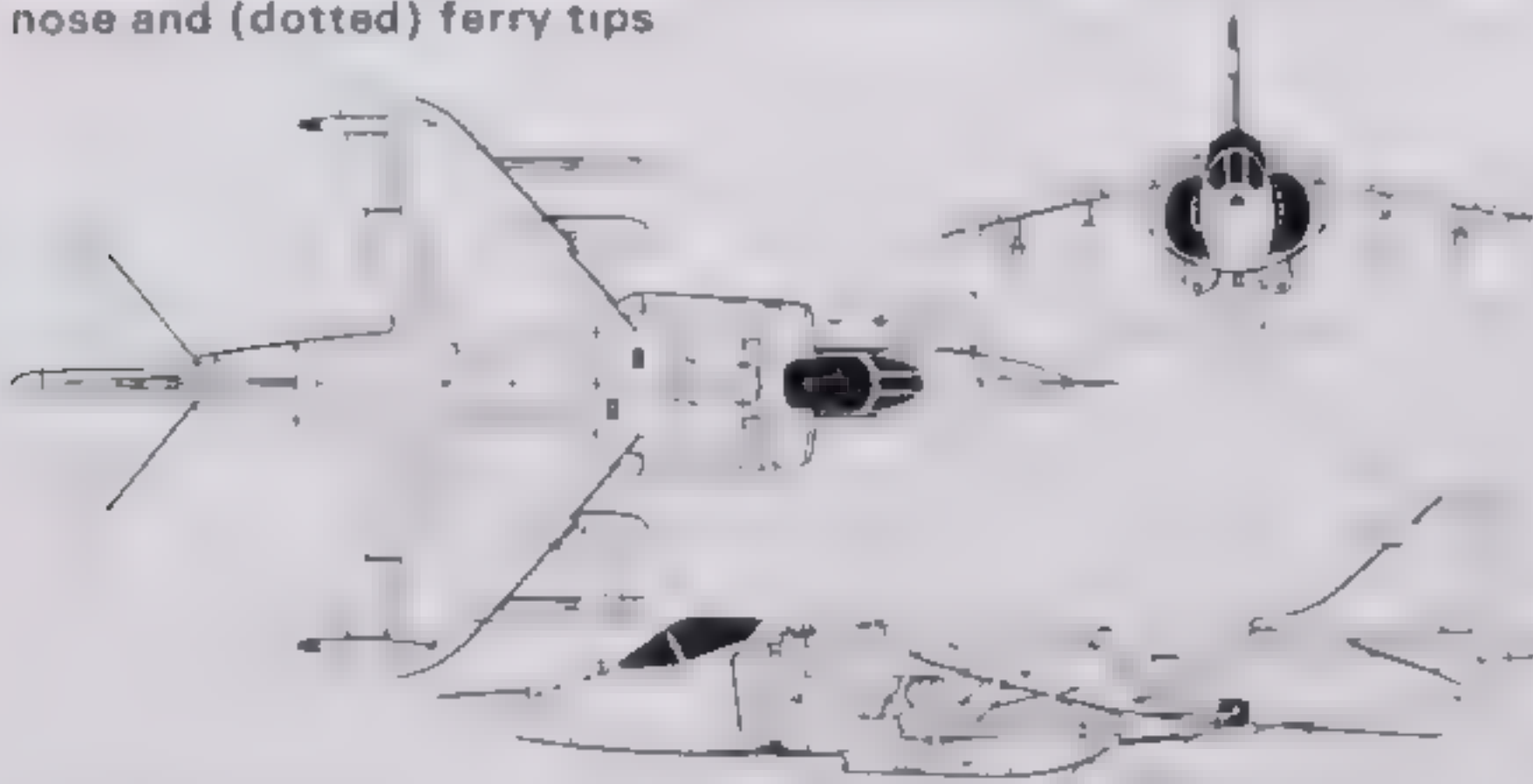


Above: Harrier GR.3 in hovering mode, modulating the airbrake to come to rest at the exact desired location. Flying the Harrier is basically simple, but is nevertheless a new technique.

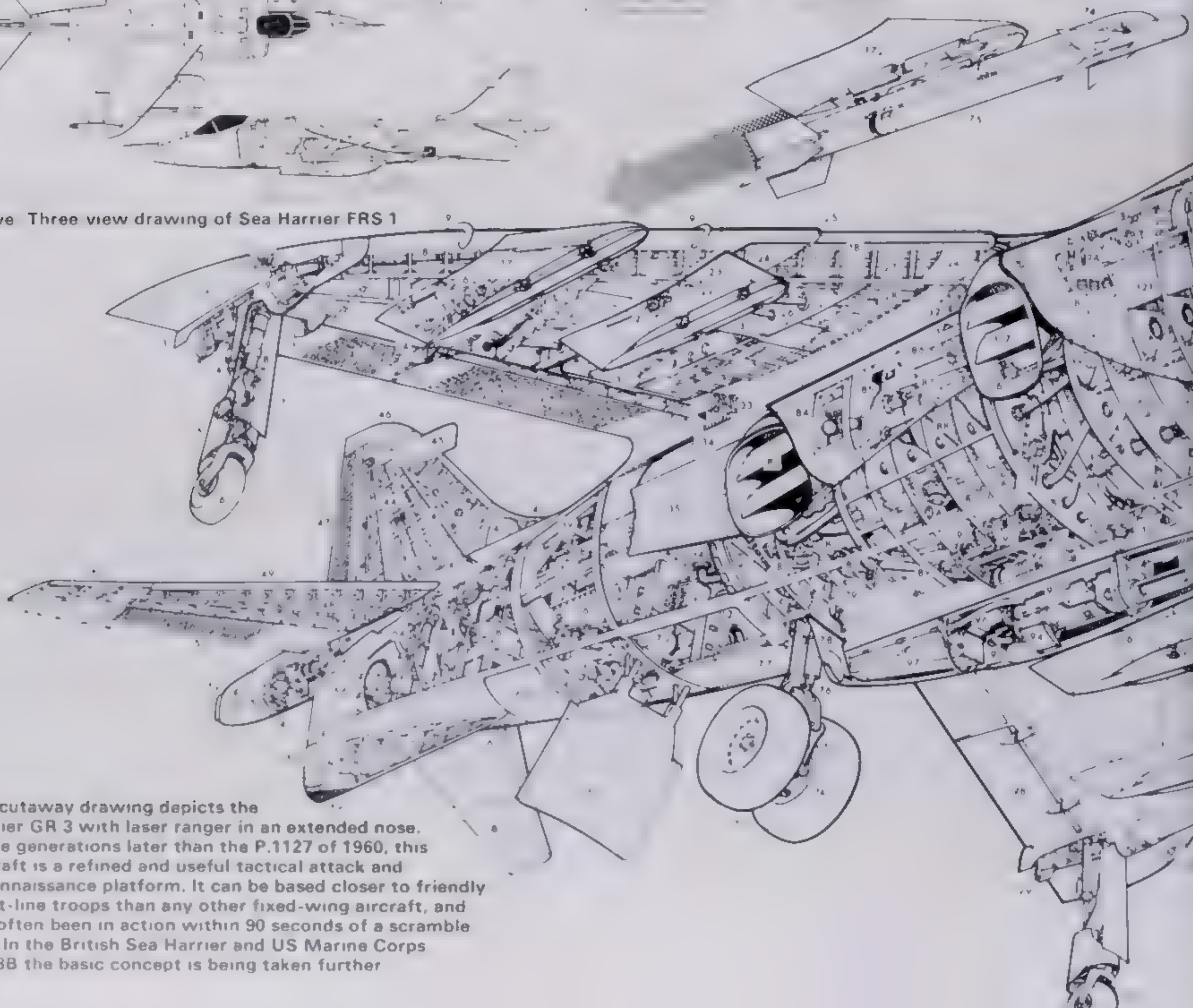
Right: A sunny pair of AV-8As of US Marine Corps VMA-231 at Cherry Point, North Carolina. This is a more basic aircraft than the RAF GR.3, with no inertial system and armed with bombs and Sidewinders.



Above: Three-view drawing of Harrier GR.3 with FR probe, laser nose and (dotted) ferry tips



Above: Three view drawing of Sea Harrier FRS 1



The cutaway drawing depicts the Harrier GR 3 with laser ranger in an extended nose. Three generations later than the P.1127 of 1960, this aircraft is a refined and useful tactical attack and reconnaissance platform. It can be based closer to friendly front-line troops than any other fixed-wing aircraft, and has often been in action within 90 seconds of a scramble call. In the British Sea Harrier and US Marine Corps AV-8B the basic concept is being taken further



- 22 Pylon spigot
- 23 Starboard inner pylon
- 24 Wing fuel tank
- 25 Wing leading edge dog tooth
- 26 Tank pressurizing air
- 27 Aileron control rod
- 28 Front spar web
- 29 Machined skin plank
- 30 Centre spar web

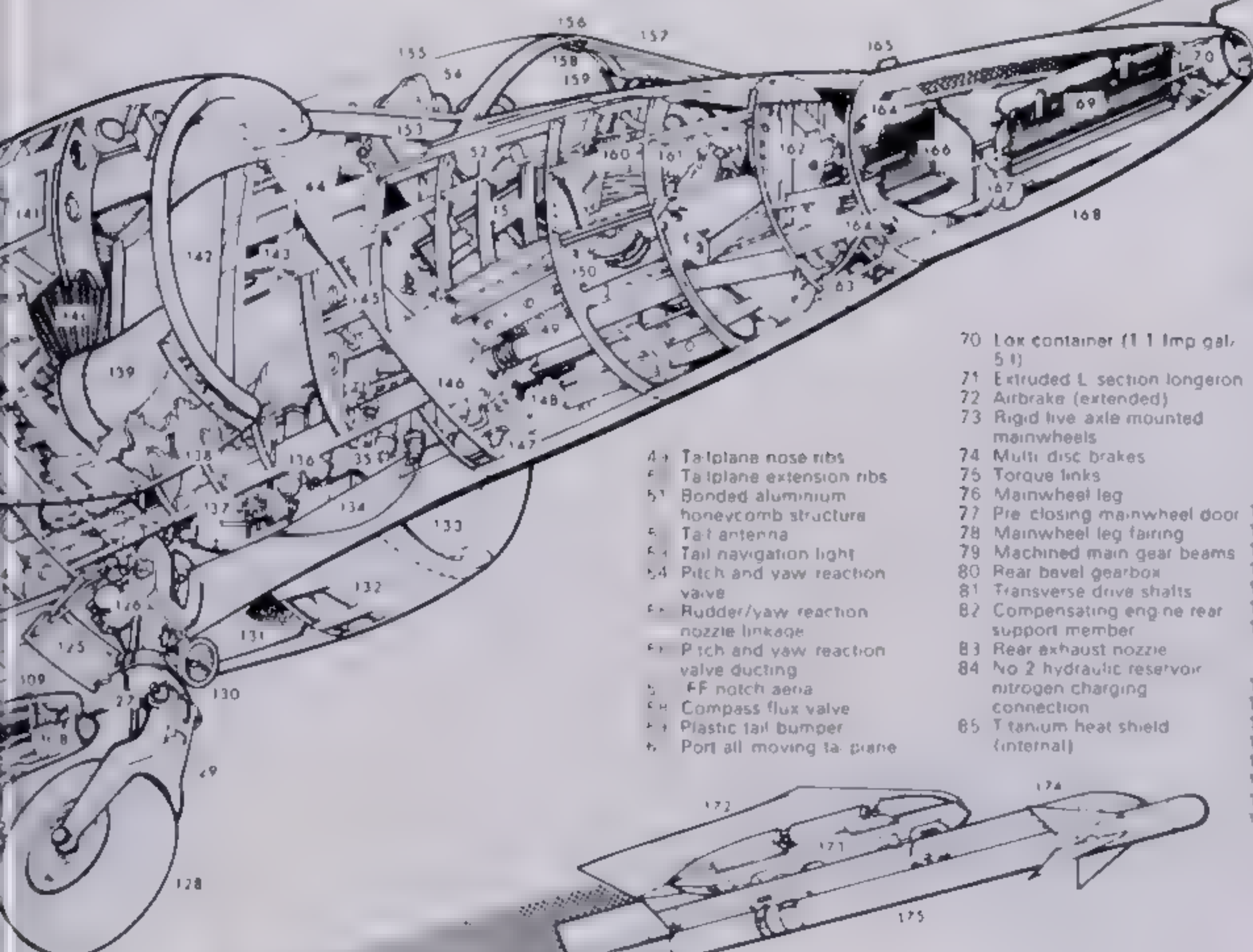
- 31 Rear spar web
- 32 Main wing attachment point
- 33 Rear spar/fuselage attachment point
- 34 Fuselage rear fuel tank
- 35 Rear nozzle heat shield
- 36 Vibration-isolating equipment rack
- 37 IFF SSR transponder
- 38 TACAN trans receiver

- 39 Ram air turbine
- 40 HF tuner
- 41 HF notch aerial
- 42 Tail fin attachment bracket
- 43 Tail fin structure
- 44 Total temperature sensor
- 45 ECM pod
- 46 VHF aerial
- 47 Rudder
- 48 Tailplane front spar

- 61 Ventral fin structure
- 62 UHF stand-by aerial
- 63 Rear fuselage access hatch
- 64 Hydraulic filter No 2 (in plane)
- 65 Tandem tailplane jack
- 66 Rudder cable tensioner
- 67 UHF stand-by
- 68 Batteries shelf
- 69 Airbrake jack

- 86 Reservoir No 2 system
- 87 Rear nozzle bearings
- 88 Starboard centre fuel tank

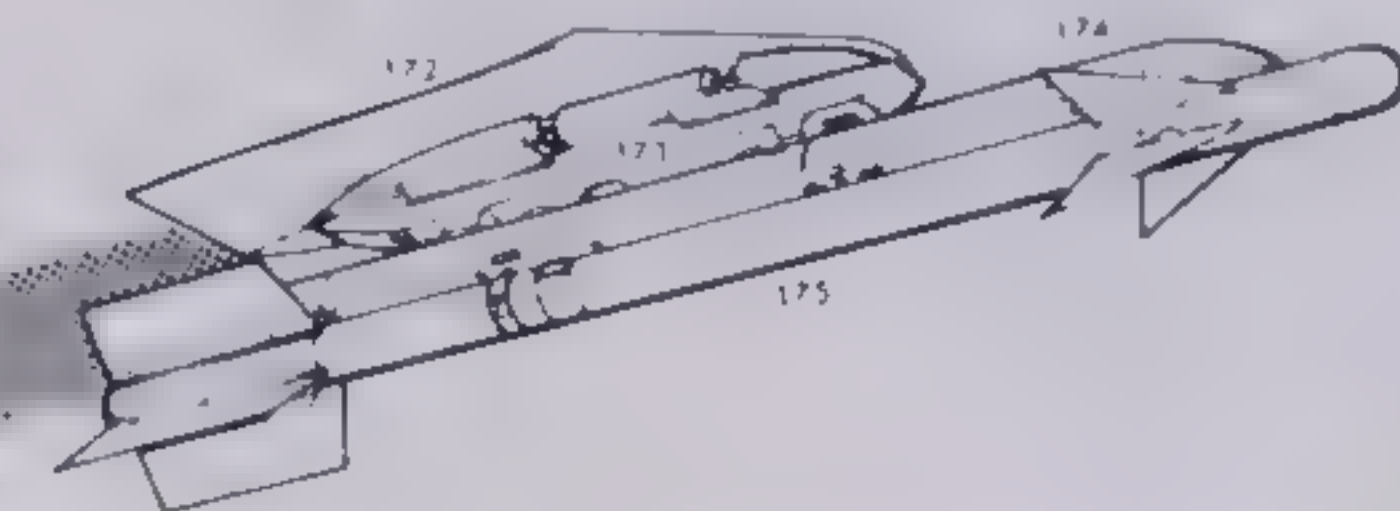
- 100 Intermediate chain
- 110 Chain and sprocket nozzle actuation
- 111 Fan air nozzle
- 112 Ground servicing points No 2 system hydraulics line and air supply external connections
- 113 GTS/APU
- 114 Venting air
- 121 Titanium heat shield (internal)
- 122 Rolls Royce Bristol Pegasus 103 engine (buried)
- 123 Starboard front fuel tank
- 124 Machined nose gear/keel beams
- 125 Nose gear fairing
- 126 Nosewheel steering motor
- 127 Shock absorber strut
- 128 Nosewheel
- 129 Nosewheel fork
- 130 Landing ramp
- 131 Port front fairing
- 132 Supplementary air doors (free floating)
- 133 Port intake
- 134 Pre-closing nose gear door
- 135 Bled air duct
- 136 Nosewheel steering hydraulics accumulator
- 137 Nosewheel input mechanism
- 138 Control cables
- 139 Intake centre body
- 140 First stage fan
- 141 Supplementary air doors (free floating)
- 142 Boundary air bleed doors
- 143 Cabin air conditioning and pressurization plant
- 144 Entry hand/foot holds
- 145 Seal mounting frame
- 146 Nozzle actuation cabin tension regulator
- 147 TACAN aerial
- 148 Bulkhead labyrinth seal
- 149 Pitch reaction valve ducting
- 150 Rudder quadrant
- 151 Starboard instrument console (TACAN and HF control panels)
- 152 Top longeron
- 153 Canopy MDC (miniature detaching cord)
- 154 Martin Baker Type 9A rocket assisted ejection seat
- 155 Canopy
- 156 Machined windscreen frame and air
- 157 Birdstrike proof wind
- 158 Rear view mirror
- 159 Head up display
- 160 Instrument panel
- 161 Red seat pan
- 162 Front pressure bulkhead
- 163 Pitch reaction valve
- 164 Nose cone attachment
- 165 IFF aerial



- 4 Tailplane nose ribs
- 5 Tailplane extension ribs
- 6 Bonded aluminium honeycomb structure
- 7 Tail antenna
- 8 Tail navigation light
- 9 Pitch and yaw reaction valve
- 10 Rudder/yaw reaction nozzle linkage
- 11 Pitch and yaw reaction valve ducting
- 12 HF notch aerial
- 13 Compass flux valve
- 14 Plastic tail bumper
- 15 Port all moving tailplane

- 70 Lox container (11 Imp gal, 5 l)
- 71 Extruded L section longeron
- 72 Airbrake (extended)
- 73 Rigid live axle mounted mainwheels
- 74 Multi disc brakes
- 75 Torque links
- 76 Mainwheel leg
- 77 Pre-closing mainwheel door
- 78 Mainwheel leg fairing
- 79 Machined main gear beams
- 80 Rear bevel gearbox
- 81 Transverse drive shafts
- 82 Compensating engine rear support member
- 83 Rear exhaust nozzle
- 84 No 2 hydraulic reservoir nitrogen charging connection
- 85 Titanium heat shield (internal)

- 90 Longitudinal drive shaft to rear nozzles
- 91 Gearbox
- 92 Master shut off (butterfly) valve (reaction control system)
- 93 Nozzle rotation air motors
- 94 Port 30 mm Aden cannon
- 95 Case ejection
- 96 Link ejection
- 97 Rigid feed chute
- 98 Ammunition box
- 99 Port aireron
- 100 Outer gear wheel fairing
- 101 Port outrigger wheel
- 102 Roll reaction outlet
- 103 Port navigation light
- 104 Port outer pylon
- 105 Port inner pylon
- 106 Ejector release unit
- 107 Port cannon fairing (starboard weapon deleted for clarity)
- 108 Blast suppressor
- 109 Frangible cap fairing
- 110 Front attachment point
- 111 Roll reaction valve ducting
- 112 Air filter
- 113 Front bevel gearbox
- 114 Transverse drive shafts
- 115 Fabricated engine front mounting frame



Hawker Siddeley AV-8A of US Marine Corps VMA-513, Beaufort, South Carolina

Armament

History: First hover (P 1127) 21 October 1960 first flight

Users

Development: In the 1950s the realisation that the thrust/weight ratio

through four nozzles which, by means of channels from a single pneumatic motor, can be swivelled to point downwards to lift the aircraft or point to the rear, for propulsion. Gradually the P 11 is transformed into the Kestrel, which equipped a UK/USA/German evaluation squadron in 1965. This was further developed into the Harrier, the much bigger Mach 2 P 1154 for the RAF and RN, having been built in 1965. Powered by a Pegasus 101 rated at 19 000lb, the GR 1 is capable of flying useful combinations of fuel and stores out of any hostile area and did more than any other aircraft to explore the advantages and problems of operational deployment of combat aircraft well away from any airfield. Numerous flights were made from a wide variety of naval vessels and record flights were made from the centre of London to the centre of New York and vice versa. The GR 1A had the 20 000lb Mk 102 engine and at this thrust the Harrier was adopted as the AV 8A by the US Marine Corps in both beach assault and defensive roles. All RAF and USMC aircraft have been re-engined with the Pegasus 103, giving a payload/range performance adequate for a wide spectrum of missions, many of which cannot be flown by any other aircraft. Using VIFF (vectoring in forward flight) the Harrier can fly "impossible" manoeuvres and has proved itself an extremely tricky customer in a dogfight. This is not its main mission, however, and the RAF Harrier GR 3 (92 built) is primarily a tactical attack platform with Ferranti INAS (inertial nav/attack system) and laser ranger. The USMC AV-8A (112, plus six for Spain named Matador) does not have either of these equipments but carries Sidewinder air/air missiles. Including two-seaters production by 1977 amounted to 231. In Britain the main effort is completing development of the redesigned Sea Harrier, which should fly in 1977. The Royal Navy will deploy 24 from throughdeck cruisers and possibly other ships, and several other navies are discussing possible orders. The Sea Harrier has a completely new nose, with raised cockpit, Blue Fox radar, much enhanced systems and equipment and weapons for surface attack, reconnaissance, anti-submarine warfare and air combat. Since 1975 talks have been held with China, which is interested in buying a large number of Harriers. The next-generation AV-8B is discussed under McDonnell Douglas.



Above: Harrier GR.3 carrying four Matra 155 launchers lets go the 19 SNEB 68mm rockets from No 3. Such missiles are of value chiefly in the air/surface role

Hawker Siddeley Hunter

Hunter 1 to 79

Origin: Hawker Aircraft, UK (now British Aerospace), licence product

Type: Single-seat fighter, fighter bomber and fighter reconnaissance

Engine: One Rolls-Royce Avon single-shaft turbojet (see text)

Dimensions: Span 33ft 8in (10.26m), length (single-seat, typical) 35ft 10in (13.98m), (two seat) 48ft 10in (14.9m), height 13ft 2in

Weights: Empty (1) 12 128lb (5501kg) (9) 13 270lb (6020kg), loaded (1) 17 750lb (8051kg) (9) maximum

Performance: Maximum speed (typical of all) 710mph (1144km/h) at sea level, Mach 0.94 at height, initial climb (Avon 100) 6000ft (1829m)/min (Avon 200 series) 8 000ft (2438m)/min, range on internal fuel 490 miles (689km), (1) 1 840 miles (2965km)

Armament: Usually one (sometimes two) 30mm Aden cannon in the cockpit floor, each with 150 rounds, single seaters underwing pylons for two 1,000lb (454kg) bombs and 24 2.75in (70mm) rockets or refurbished aircraft carrying two 230 Imp gal drop tanks

History: First flight (P 1067) 20 June 1951, (production F 1) 16 May 1953 (two-seater) 8 July 1955 final delivery from new, 1966

Users: Abu Dhabi, Chile, India, Iraq, Kenya, Kuwait, Lebanon, Oman, Peru, Singapore, Switzerland, UK (RAF, Royal Navy)

Development: Undoubtedly the most successful British post-war fighter the Hunter epitomised the grace of a thoroughbred and has always delighted its pilots. The prototype, with 6 500lb thrust Avon 100, was built to Specification F 3/48. It was easily supersonic in a shallow dive and packed the devastating four Aden cannon in a quick-release pack winched up as a unit. After being fitted with bulged cartridge boxes and a stuck-on airbrake under the rear fuselage it became a standard fighter, with Armstrong Whitworth



Three-view of Hunter FGA 9, typical of most single-seaters today.

Hawker Siddeley Sea Hawk

Hawker P.1040, Armstrong Whitworth Sea Hawk 1-6, 50 and 100-1

Origin: See text

Type: Single-seat carrier-based fighter bomber (F 1, 2, fighter)

Engine: One Rolls-Royce Nene single-shaft centrifugal turbojet, (Mks 1-4) 5 000lb (2268kg) Nene 101, (Mk 5 onwards) 5 400lb (2450kg) Nene 103

Dimensions: Span 39ft (11.89m), length 39ft 8in (12.08m), height 8ft 8in (2.79m), (Mks 100-1) 9ft 9in (3m)

Weights: Empty 9 200-9 720lb (4173-4410kg), loaded (clean) 13 220lb (6000kg), maximum 16 200lb (7355kg)

Performance: Maximum speed 599mph (958km/h) at sea level, 587mph (939km/h), Mach 0.83 at height, initial climb 5 700ft (1737m)/min, service ceiling 44 500ft (13 560m), range on internal fuel 740 miles (1191km), with drop tanks 1 400 miles (2253km)

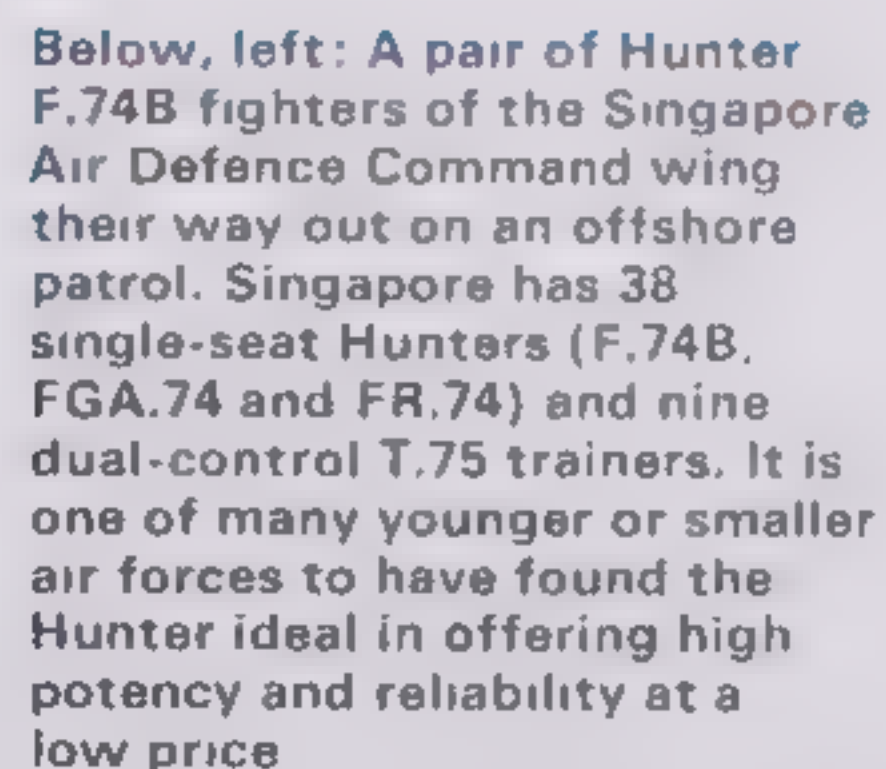
Armament: Four 20mm Hispano cannon each with 200 rounds beneath cockpit floor (Mks 3-5) underwing racks for two 500lb (227kg) bombs (Mks 4-6 and later) racks for four 500lb bombs or equivalent

History: First flight (P 1040) 2 September 1947, (N 7/46) 3 September 1948 (production F 1) 14 November 1951, squadron service, March 1953 (Mk 101) December 1956 (India) 1961

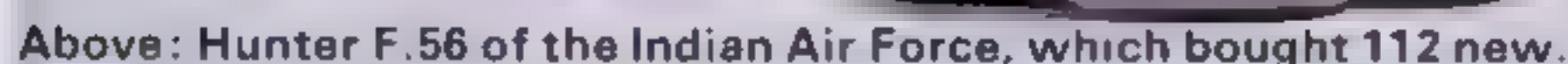
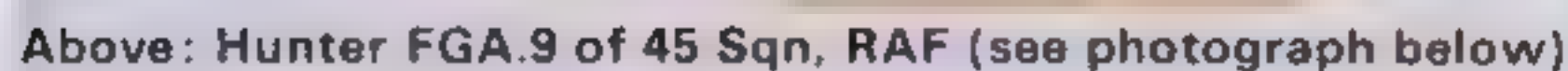
User: India (Navy)

Development: Sir Sydney Camm's first jet fighter was conventional in design with long root inlets and unswept wings and tail, but it was probably unique in that the jet pipe was split to serve two engines. This gave the aircraft a distinctive appearance compared with the conventional fighters of the time.

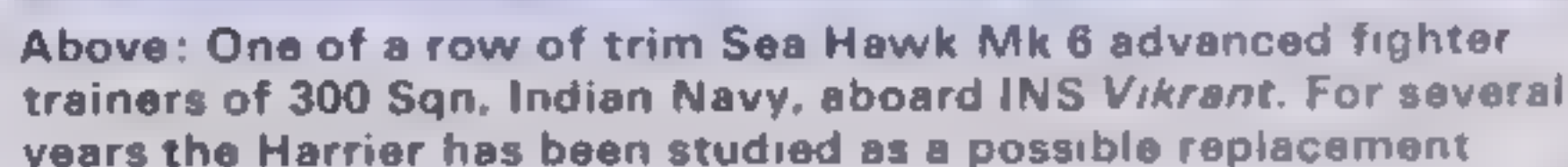
Figure 1. The effect of the concentration of the *Agrobacterium* suspension on the transformation efficiency of *Agrobacterium* strains. The concentration of the *Agrobacterium* suspension was 10⁶ cells/ml (A), 10⁷ cells/ml (B), 10⁸ cells/ml (C), and 10⁹ cells/ml (D). The concentration of the *Agrobacterium* suspension was 10⁶ cells/ml (A), 10⁷ cells/ml (B), 10⁸ cells/ml (C), and 10⁹ cells/ml (D). The concentration of the *Agrobacterium* suspension was 10⁶ cells/ml (A), 10⁷ cells/ml (B), 10⁸ cells/ml (C), and 10⁹ cells/ml (D). The concentration of the *Agrobacterium* suspension was 10⁶ cells/ml (A), 10⁷ cells/ml (B), 10⁸ cells/ml (C), and 10⁹ cells/ml (D).



Below: A scene to set the pulse racing—the start of a training sortie in FGA 9s, on an autumn afternoon. The Avon-207-powered Mk 9 incorporated all the experience gained with earlier marks, and is tougher and carries a heavier weapon load in the ground-attack mission. All were converted for the RAF from earlier sub-types, but large numbers of generally similar aircraft were built new for overseas customers. Most remanufactured Hunters have likewise begun life as Mk 4 or 6 fighters (often ex-Belgium or Netherlands).



A large, multi-engine propeller aircraft, likely a bomber, is parked on a tarmac. A person stands in the foreground, looking up at the plane.



Above: Three-view of Sea Hawk 100 with revised fin and rudder.

Lockheed F-104 Starfighter

F-104A to G, J and S, CF-104, QF-104, RF and RTF-104, TF-104
(data for F-104G)

Origin	USA
Type	Single-seat, single-engine, supersonic fighter
Engine	General Electric J65-15 turbojet, 11,000 lb (5,000 kg) thrust
Dimensions	Wingspan 25 ft 10 in (7.9 m), Length 47 ft 10 in (14.6 m), Height 12 ft 10 in (3.9 m)
Weights	Empty 14,082 lb (6,400 kg), Max 20,000 lb (9,070 kg)
Performance	Max speed 2,175 mph (3,500 km/h), Altitude 65,000 ft (19,812 m), Range 1,500 mi (2,414 km)

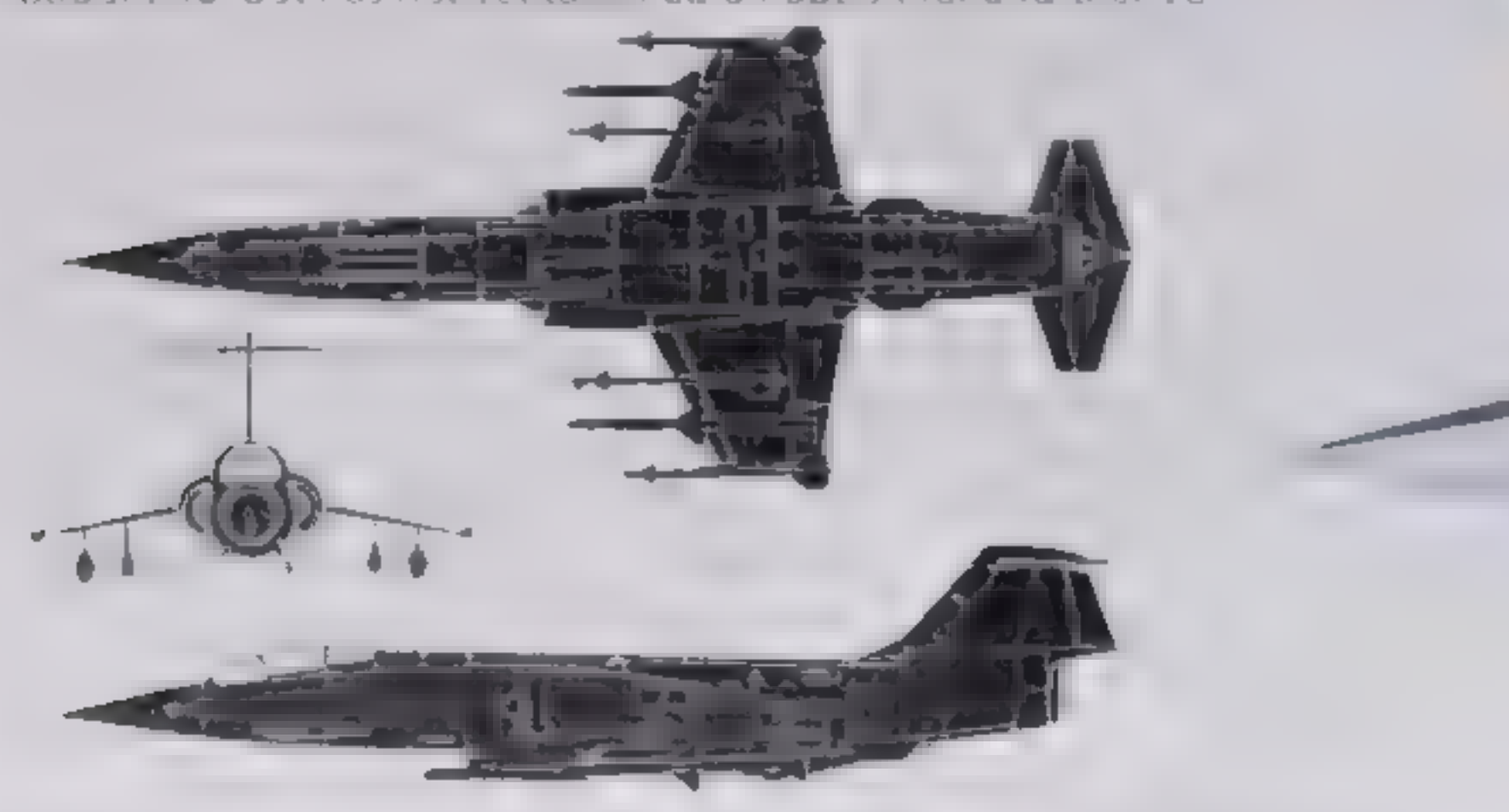
Armament: In most versions, centreline rack rated at 2,000 lb (907 kg). Typical load: two Sidewinders, two Sparrows, one 20mm M61 Vulcan cannon. Some versions have reduced fuel and one 20mm M61 Vulcan cannon.

History: First flight (XF-104) 7 February 1954, (F-104A) 17 February 1958, (F-104G) 5 October 1960, (F-104S) 30 December 1968, final delivery from United States 1964, final delivery from Aeritalia (F-104S) 1975.

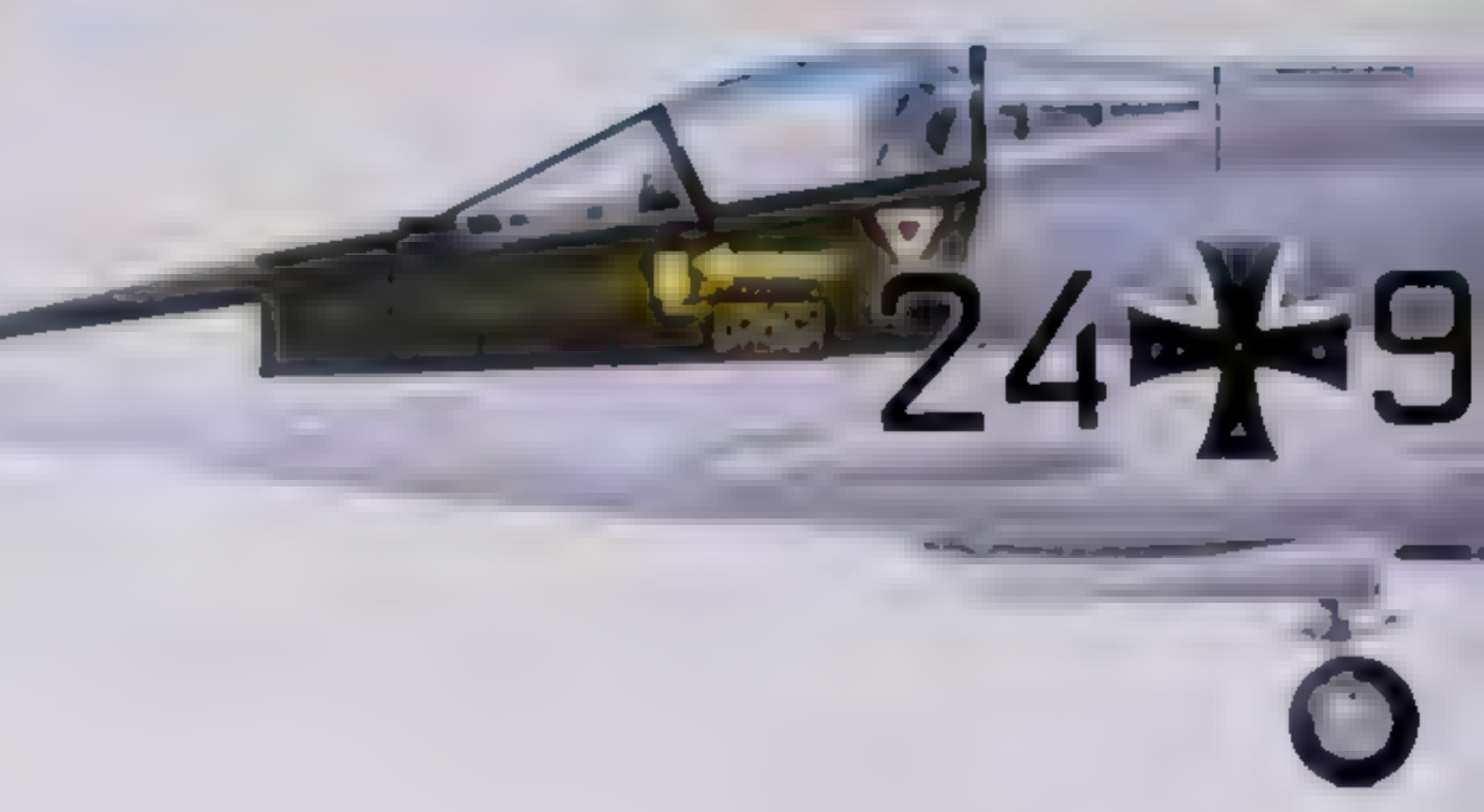
Users: Belgium, Canada, Denmark, W Germany, Greece, Italy, Japan, Jordan, Netherlands, Norway, Pakistan, Spain, Taiwan, Turkey, USA (ANG).

Development: Clarence L. ('Kelly') Johnson planned the Model 83 after talking with fighter pilots in Korea in 1951. The apparent need was for superior flight performance, even at the expense of reduced equipment and other penalties. When the XF-104 flew, powered by a 10,500 lb J65 Sapphire with afterburner, it appeared to have hardly any wing, another odd feature was the downward-ejecting seat. The production F-104A had a more powerful engine and blown flaps and after lengthy development entered limited service with Air Defense Command in 1958. Only 153 were built and after a spell with the Air National Guard, survivors again saw ADC service with the powerful GE-19 engine. Three were modified as Astronaut trainers with rocket boost, one gaining a world height record at nearly 119,000 ft in 1963. The B was a dual tandem trainer, the C a fighter-bomber for Tactical Air Command with refueling probe, the D a trainer version of the C and the DJ and F respectively Japanese and German versions of the D. The G was a complete redesign to meet the needs of the Luftwaffe for a tactical nuclear strike and reconnaissance aircraft. Structurally different, it introduced Nasarr multi-mode radar, inertial navigation system, manoeuvring flaps and other new items. Altogether 1,266 were built, including 970 by a NATO European consortium and 110 by Canadian Aeritalia built 200 basically similar CF-104s, while Japan built 207 J models closely resembling the earlier C. The German RF and RTF are multi-role sensor reconnaissance and trainer versions, while increasing numbers of all versions being turned into various QF-104 RPVs.

The only type of Starfighter built new since 1967 has been the Italian F-104S. Developed jointly by Lockheed and Fiat (Aeritalia), the S is an air priority fighter armed with two Sparrow air/air missiles (hence the suffix letter S). Built under Lockheed licence, the S has a new GE-79 engine, updated Autonetics R21G radar (with improved reliability) and several detail changes to improve performance. The secondary ground-attack capability is retained, and in recent months the Regia Aeronautica force of 205 F-104S have begun to carry the Orpheus multi-sensor reconnaissance pod carried on the centre-line. Turkey has bought 40 but has not yet taken up an option for a further 18.



Three-view of F-104S, showing Sparrows and Sidewinders





Above: The Ejercito del Aire (Spanish Air Force) was one of the recipients of the F-104G from Canadian production. The EdA squadron of 24 is no longer operational.



Far left: The Aeritalia F-104S is the only Starfighter model still in production. These examples belong to 53^a Stormo of the Aeronautica Militare Italiano, based at Cameri, used in the interceptor role.

Near left: Rockets ripple away from a CF-104 of No 1 Air Group, of Canadian Forces Europe.



RF-104G with internal camera compartment, serving with Luftwaffe AG 52

McDonnell Douglas/Hawker

AV-8B

AV-8B and proposed variants

Origin The AV-8B is a joint project between the US Navy and the US Marine Corps. It is a development of the AV-8A Harrier II, which was developed by the British Aerospace company. The AV-8B is a two-seater aircraft, with the pilot and co-pilot seated side-by-side. It is a multi-role aircraft, capable of performing a variety of missions, including ground attack, reconnaissance, and transport. The AV-8B is powered by a single turbofan engine, which provides it with the ability to take off and land vertically. The AV-8B is currently in service with the US Marine Corps, and is expected to remain in service for many years to come.

Type: A two-seater, multi-role, vertical take-off and landing (VTOL) aircraft.

Engine A single turbofan engine, providing a maximum thrust of 24,000 lb (10,700 kg).

Dimensions: Span 30ft 3½in (9.2m), height 4ft 11in (1.5m), length 11ft 3½in (3.4m).

Weights: Empty, 12,400 lb (5,600 kg); maximum take-off weight, 24,000 lb (10,700 kg).

Performance: Maximum speed, 485 mph (780 km/h); service ceiling, 11,600 ft (3,538 m); range, 1,200 miles (1,930 km); climb rate, 11,600 ft/min (3,538 m/min); turn rate, 12°/sec (0.21 rad/sec).

Armament: Two 20mm Mk 12 cannons, 16 hardpoints for missiles, bombs, and other ordnance.

History: First flight, 1978; first operational flight, 1979; first operational deployment, 1982.

Users: US Marine Corps, US Navy.

Development: The AV-8B was developed by McDonnell Douglas and Hawker Siddeley. The project was initiated in 1973, following the cancellation of the AV-8A Harrier II. The AV-8B was designed to be a more powerful and versatile version of the AV-8A, with a larger engine and a more advanced avionics suite. The AV-8B was first flown in 1978, and was first deployed to the US Marine Corps in 1982. The AV-8B has since become the primary ground attack aircraft of the US Marine Corps, and is also used by the US Navy for reconnaissance and transport missions.



Above: Three-view of AV-8B Advanced Harrier as currently planned

The AV-8B is a two-seater aircraft, with the pilot and co-pilot seated side-by-side. It is a multi-role aircraft, capable of performing a variety of missions, including ground attack, reconnaissance, and transport. The AV-8B is powered by a single turbofan engine, which provides it with the ability to take off and land vertically. The AV-8B is currently in service with the US Marine Corps, and is expected to remain in service for many years to come.

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McDonnell Douglas A-4

Skyhawk

A-4A to A-4S and TA-4 series

Origin

Type: Single seat attack bomber (TA 4)

Engine: (B, C, L, P, Q, S) one 7 700lb (3493kg) thrust Wright

Dimensions: Span 27ft 6in (8.38m) length (A) 39ft 1in, (B) 40ft 1in over FR profile (E, F, G, H, J) 40ft 1in

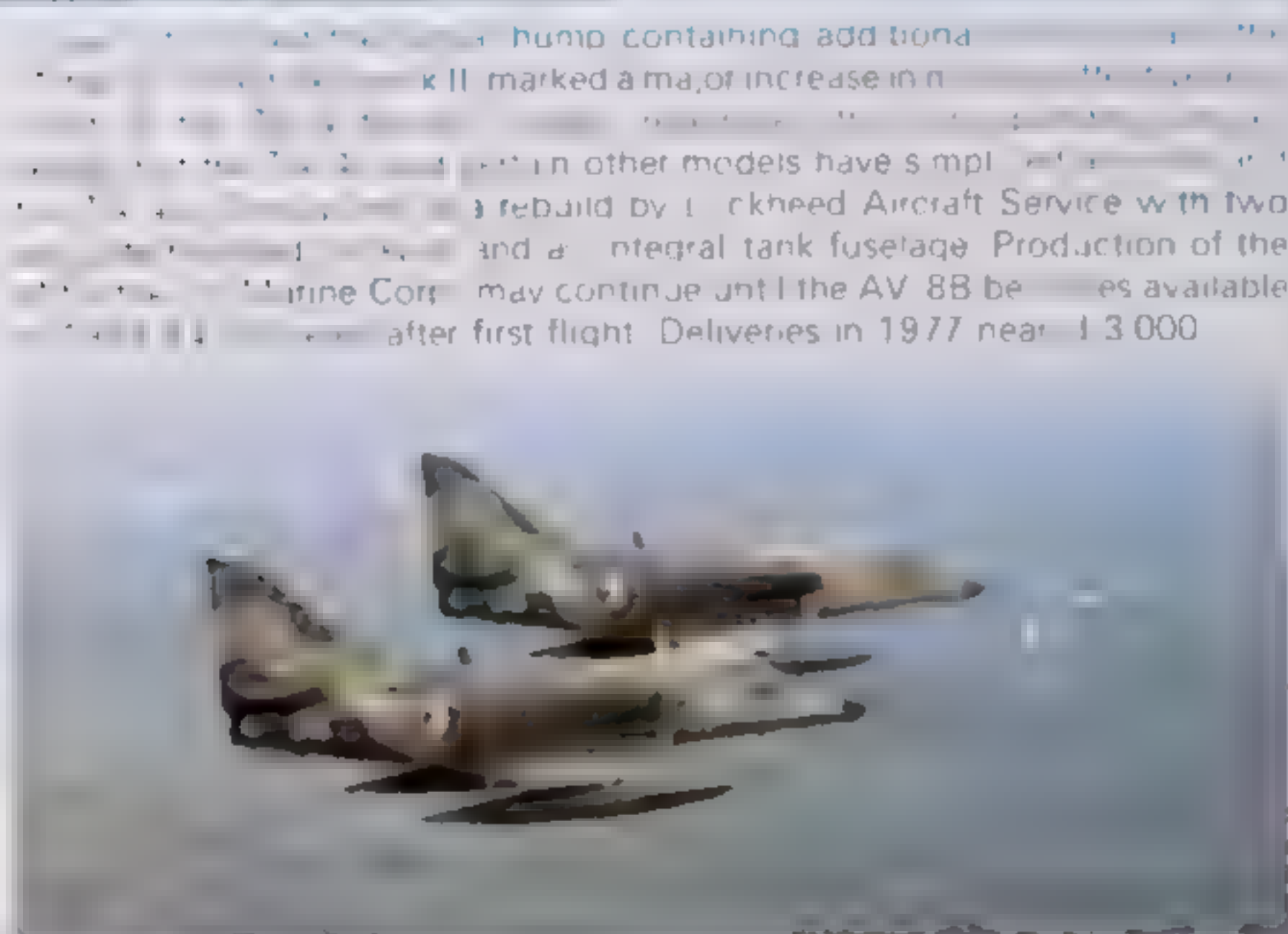
Weights: Empty (A) 7 700lb (E) 9 284lb, (typical modern single-seat) 10 465lb (4747kg), (TA 4F) 10 602 (4809kg) maximum loaded (B) 22 000lb, (all others, shipboard) 24 500lb (11,113kg) and based

Performance: Maximum speed (clean) (B) 676mph (E) 685mph, (M) 670mph (1078km/h) (TA 4F) 675mph maximum speed (4 000lb 1814kg) (F) 593mph (M) 645mph, initial climb (F) 5 620ft (1713m) min (M) 8 440ft (2572m)/min service ceiling (all clean) about 49 000ft (14935m) range (clean or with 4 000lb weapons and max fuel, all late versions) about 920 miles (1480km) maximum range (M) 2 055 miles (3307km)

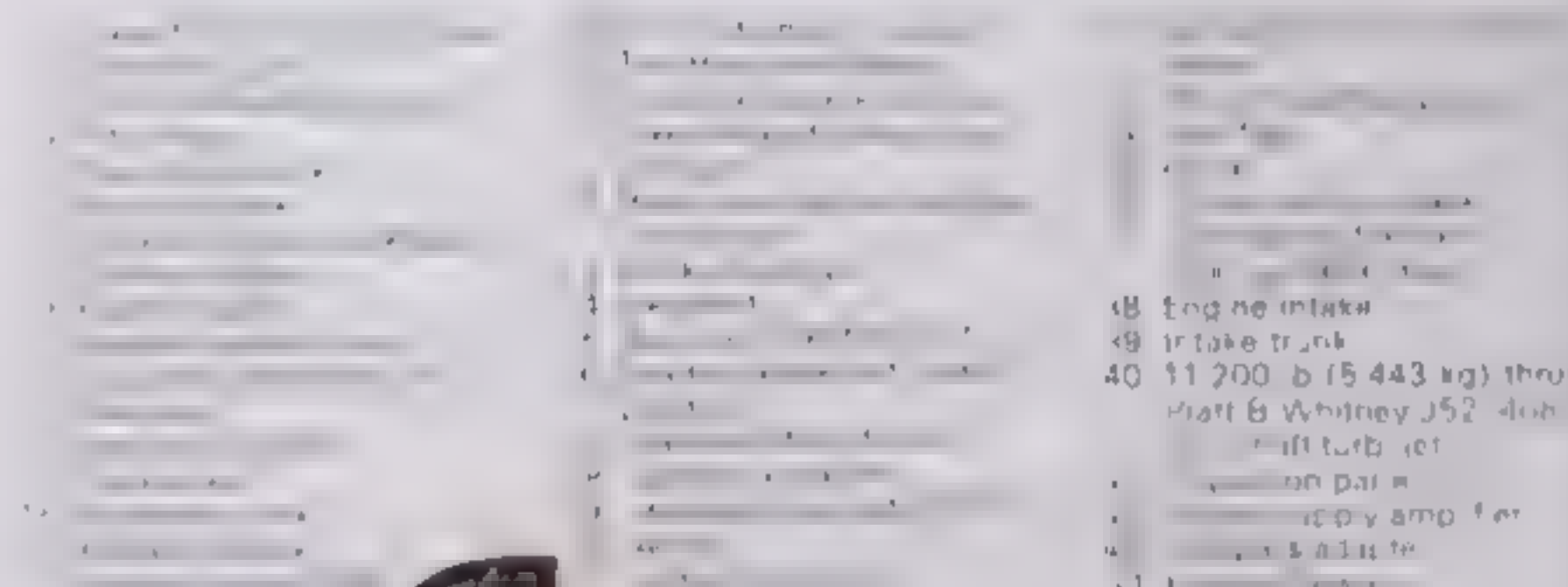
Armament: Standard on most versions two 20mm Mk 12 cannon, each with 200 rounds (H, N and optional on other export versions) two 30mm DEFA 553 each with 150 rounds Pylons under fuselage and wings for total ordnance load of (A, B, C) 5 000lb (2268kg), (E, F, G, H, K, L, P, Q, S) 8 200lb (3720kg) (M) 9 155lb (4153kg)

History: First flight (XA4D-1) 22 June 1954 (A-4A) 14 August 1954 quadron de very October 1956 (A-4C) August 1959 (A-4E) July 1961 (A 4F) August 1966, (A-4M) April 1970 (A-4N) June 1972 first of TA series (TA 4E) June 1965

Users: Argentina, Australia, Israel, Kuwait, New Zealand, Singapore USA (Air Force in SE Asia, Navy, Marine Corps)

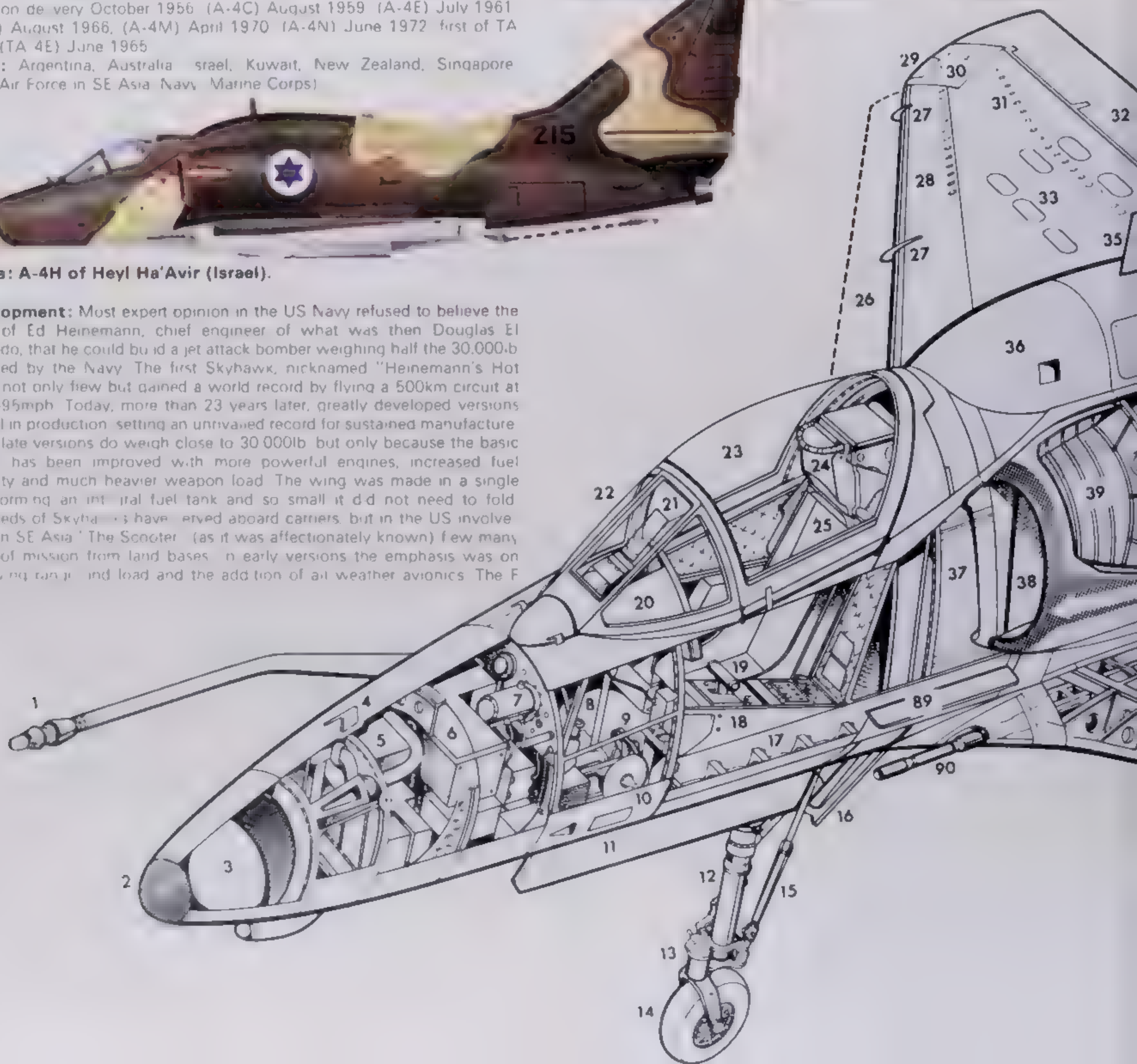


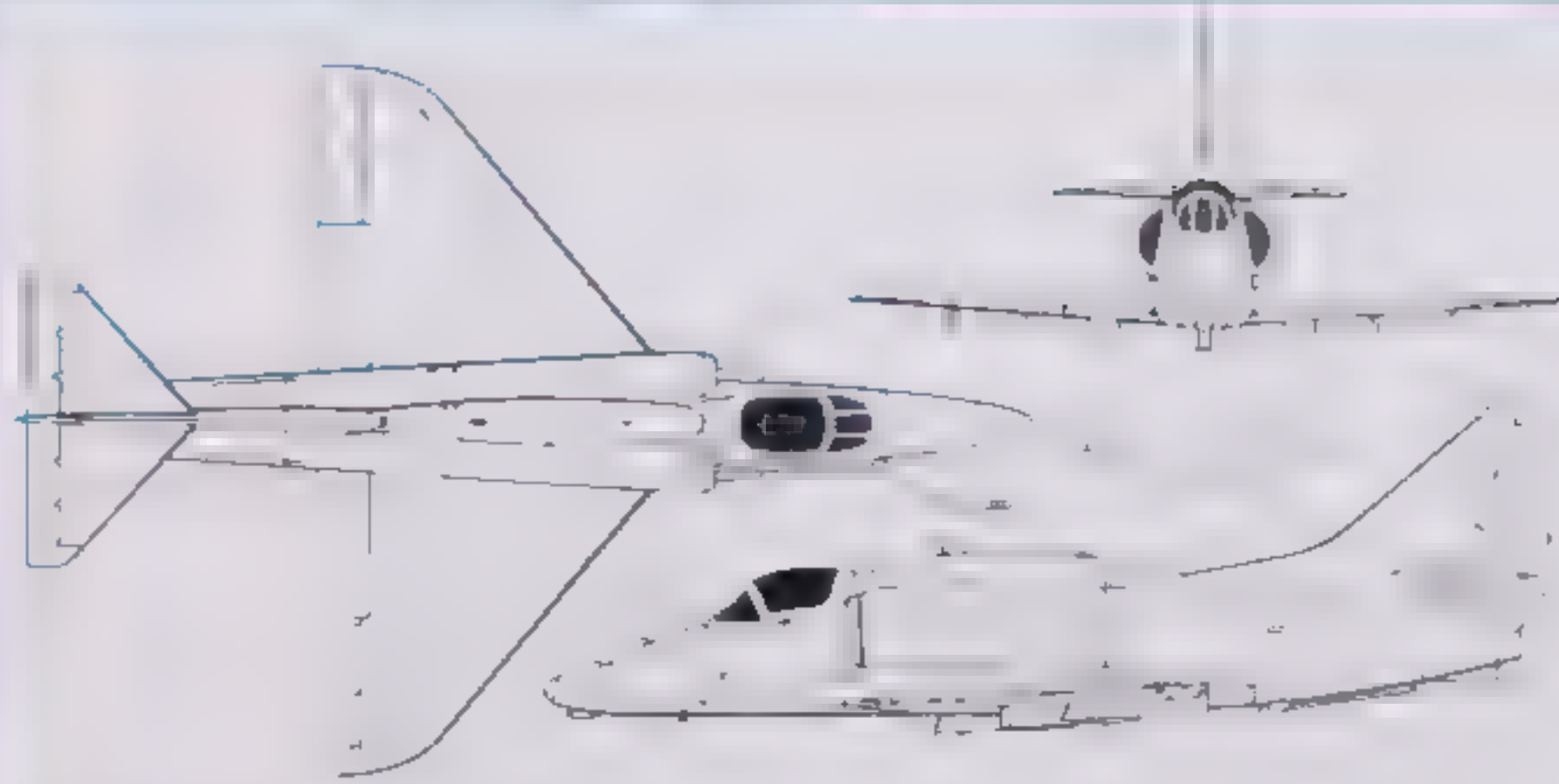
Above: Brace of A-4s (A-4B rebuilt by LAS) of Singapore, which has 40. Some Singapore A-4s have two cockpit canopies.



Above: A-4H of Heyl Ha'Avir (Israel).

Development: Most expert opinion in the US Navy refused to believe the claim of Ed Heinemann, chief engineer of what was then Douglas El Segundo, that he could build a jet attack bomber weighing half the 30,000lb specified by the Navy. The first Skyhawk, nicknamed "Heinemann's Hot Rod", not only flew but gained a world record by flying a 500km circuit at over 695mph. Today, more than 23 years later, greatly developed versions are still in production, setting an unrivalled record for sustained manufacture. These late versions do weigh close to 30,000lb, but only because the basic design has been improved with more powerful engines, increased fuel capacity and much heavier weapon load. The wing was made in a single unit, forming an integral fuel tank and so small it did not need to fold. Hundreds of Skyhawks have served aboard carriers, but in the US involvement in SE Asia 'The Scooter' (as it was affectionately known) flew many kinds of mission from land bases. In early versions the emphasis was on improving range and load and the addition of all-weather avionics. The F

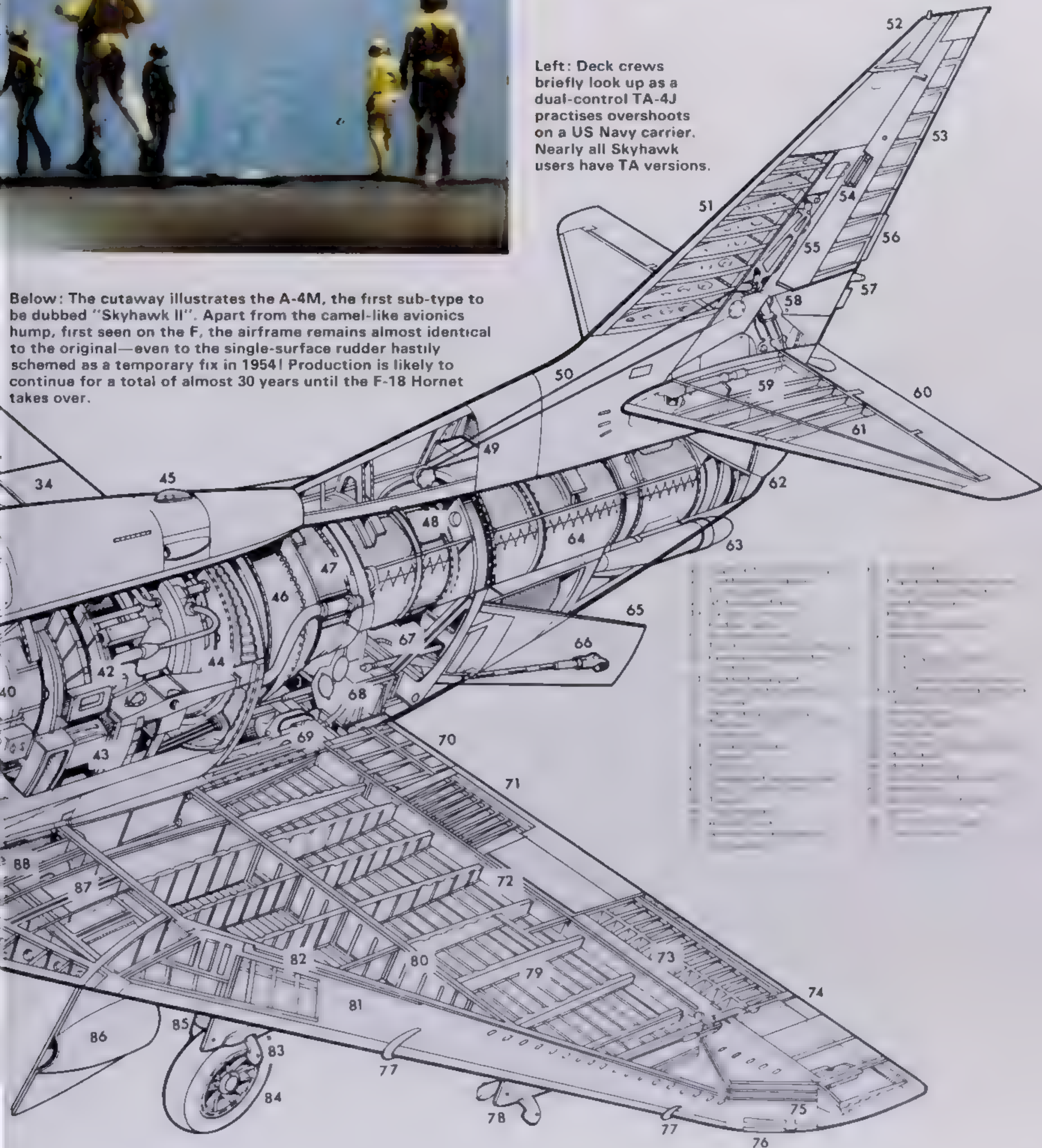




Above: Three-view of McDonnell Douglas A-4M Skyhawk II

Left: Deck crews briefly look up as a dual-control TA-4J practises overshoots on a US Navy carrier. Nearly all Skyhawk users have TA versions.

Below: The cutaway illustrates the A-4M, the first sub-type to be dubbed "Skyhawk II". Apart from the camel-like avionics hump, first seen on the F, the airframe remains almost identical to the original—even to the single-surface rudder hastily schemed as a temporary fix in 1954! Production is likely to continue for a total of almost 30 years until the F-18 Hornet takes over.



Mikoyan/Gurevich MiG-27

MiG-27 "Flogger D" and "Flogger F"

Origin
Type
Engine

Dimensions

Weights

Performance

Armament

History
Users

Development

The MiG-27 is a variable-geometry fighter aircraft, designed by the Mikoyan-Gurevich design bureau in the Soviet Union. It is the first Soviet fighter to feature a canards configuration, with a small forward canard mounted on the nose. The aircraft is designed for low-level operation and is capable of carrying a large payload of missiles and bombs. It is powered by two turbojet engines and has a maximum speed of over 1,000 mph. The MiG-27 was first flown in 1974 and entered service in 1977. It is currently in service with the Russian Air Force and the Syrian Air Force.



Above Three view of MiG-27 without swing-wing pylons



Above: An extremely revealing photograph showing one of the latest MiG-27 sub-types festooned with mission equipment.

large low-pressure tyres and special provision is made for rough-field operation. Internal fuel capacity is estimated at 1,184 Imp gallon. The aircraft is also equipped with a provision for flight refuelling. The Flogger F is the engine installation and gun of the MiG-23 with variable wings and lack the comprehensive MiG-27 avionics. These are thought to be development aircraft or an export version. Possible problems with the basic aircraft are suggested by reports that in a few months the Syrian AF has written off 13 out of 50 supplied.

Below: The MiG variable-geometry attack aircraft—almost certainly MiG-27 in FA service—is an extremely refined and cost/effective aircraft, probably being built at a higher rate than any other combat aircraft in the world.



Figure 4. (continued)

Origin: 51° 41' 10" N, 1° 45' 00" E

Engines: Two Ishikawajima Harima TF40 801A (1,100 hp) Royce/Turbomeca Adour 102) two shaft augmented turbofan maximum rating of 7,140 lb (3238 kg) (F-1) may later have

Weights: Empty (T-2) 13 668lb (6200kg), (F-1) 14,330lb (6500kg)
 loaded (T-2 clean) 21 274lb (9650kg), (F-1) 22 000lb (10 000kg)
 kg), (F-1 maximum) 30 200 lb (13 700 kg)

Performance: Maximum speed (at 10,000 ft) 1,700 km/h (Mach 1.6) initial climb 19,000 ft (5,791 m) service ceiling 50,025 ft (15,250 m), range (T-2 with external tanks) 1,785 miles

Armament

History: First f

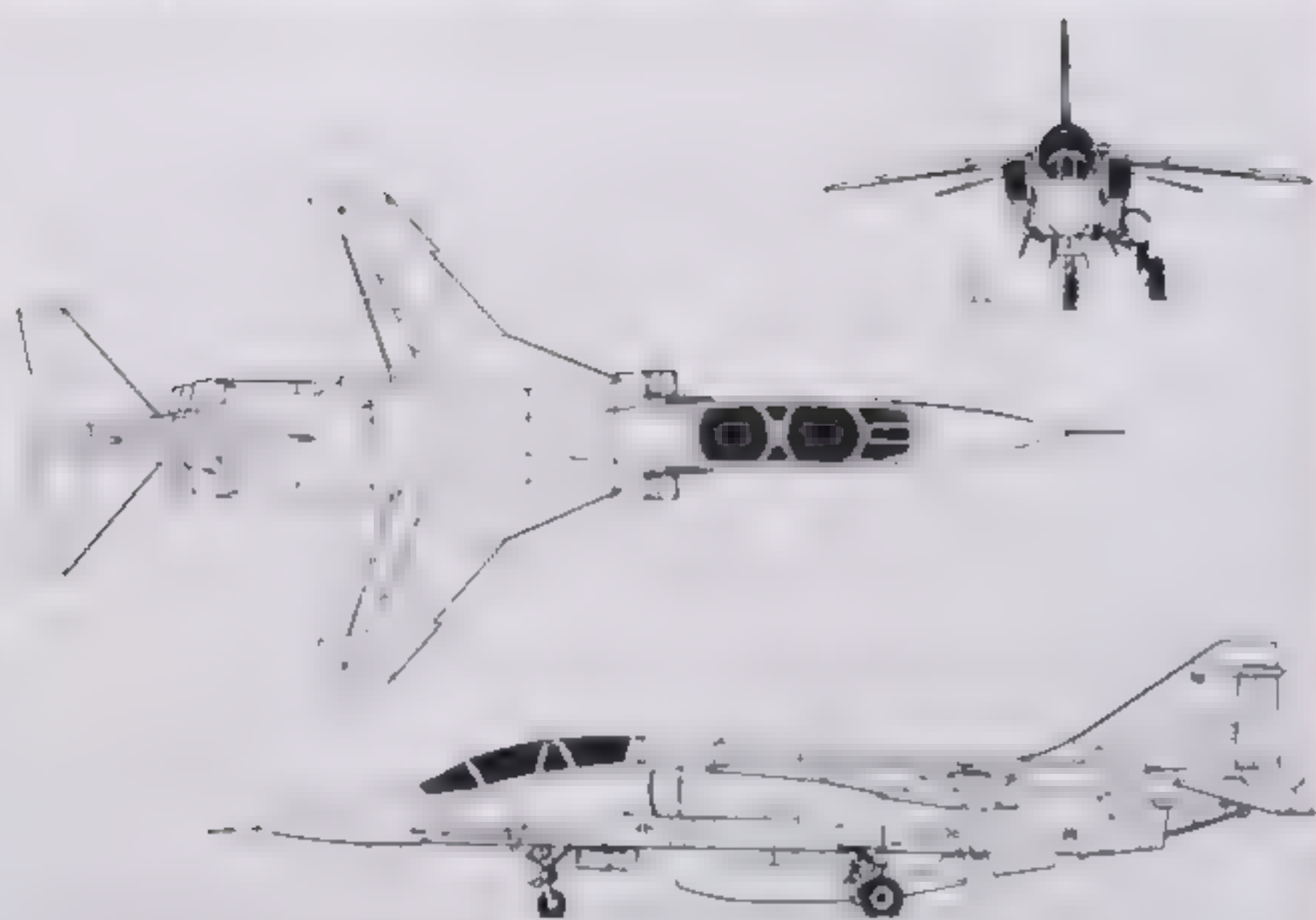
User: Japanese Air Self

Development

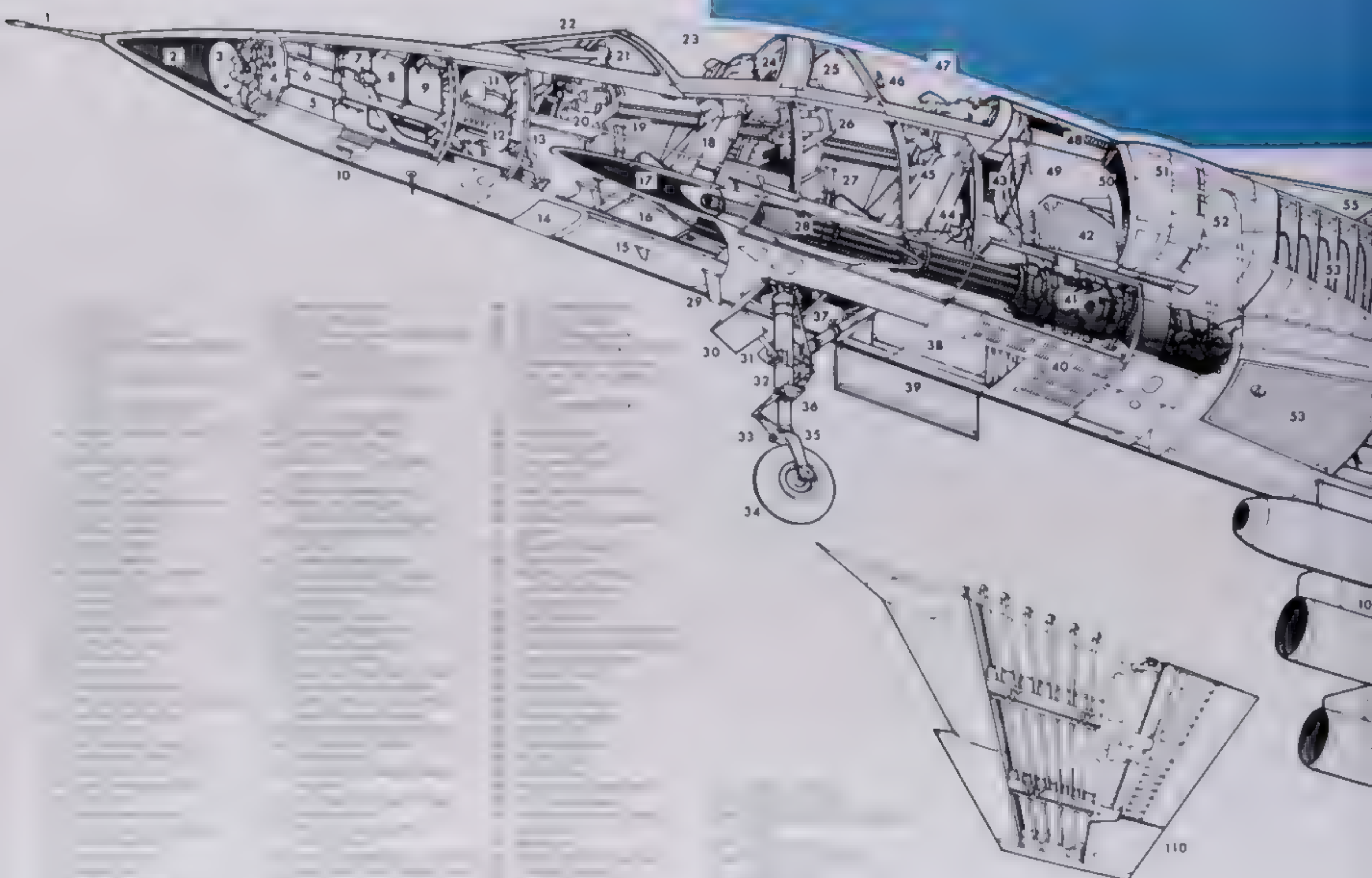
continued on page 88 ►

Below: Are Japanese air-to-air pictures especially beautiful? Certainly this unusual view of Mitsubishi T-2 tandem-seat supersonic trainers captures all the freedom and grace of jet flight.



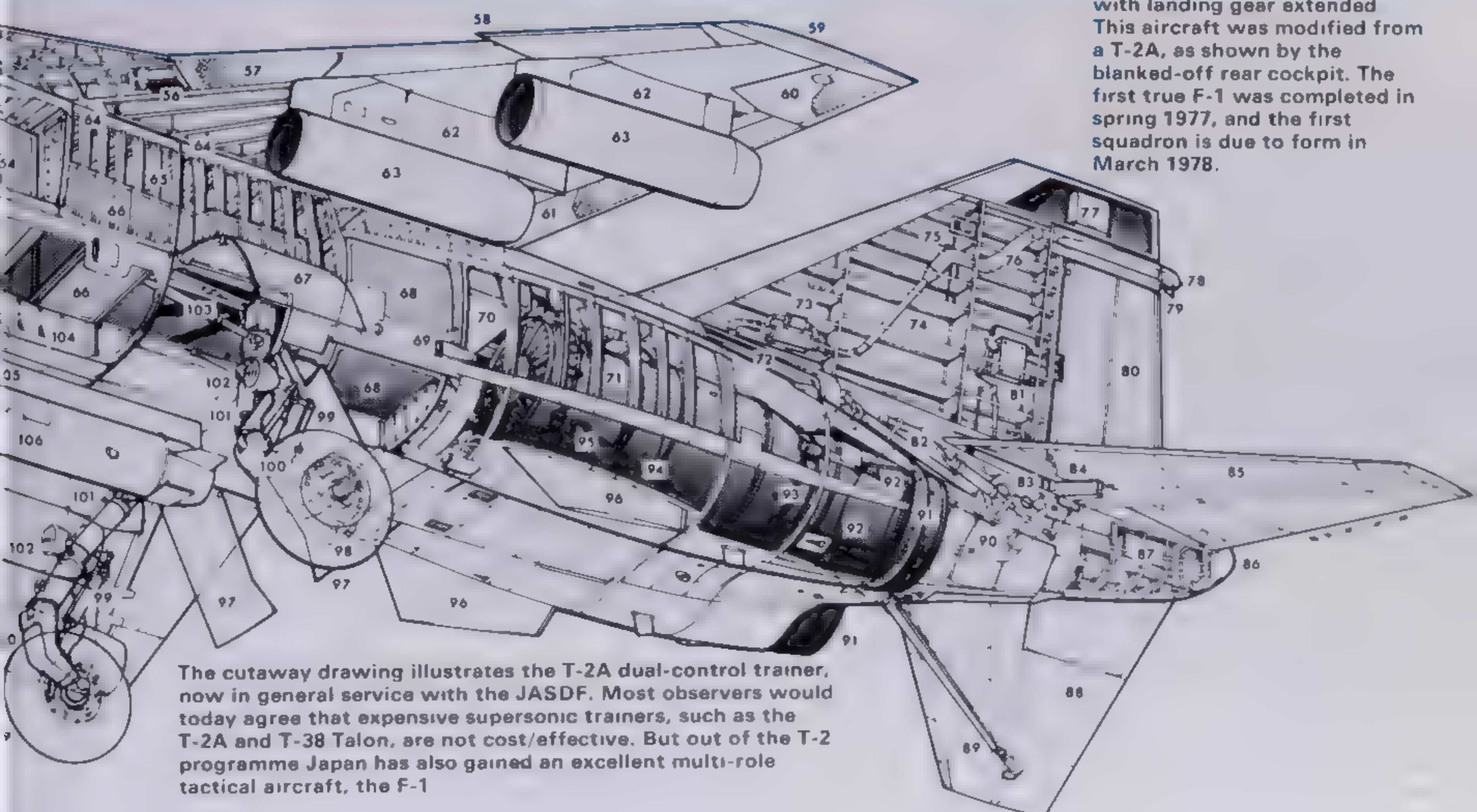


Three view of Mitsubishi T-2A without wing pylons.





Above: First prototype F-1, with landing gear extended. This aircraft was modified from a T-2A, as shown by the blanked-off rear cockpit. The first true F-1 was completed in spring 1977, and the first squadron is due to form in March 1978.



The cutaway drawing illustrates the T-2A dual-control trainer, now in general service with the JASDF. Most observers would today agree that expensive supersonic trainers, such as the T-2A and T-38 Talon, are not cost/effective. But out of the T-2 programme Japan has also gained an excellent multi-role tactical aircraft, the F-1.

North American (Rockwell)

F-100 Super Sabre

F-100A to F-100F and DF-100F

Origin
Type

Engine

Dimensions

Height (original A) 10ft 5in (3.18m); (F) 10ft 4in (3.15m)

Weights: Empty (original A) 19 700lb (C) 20 450lb, (D) 21 000lb (9525kg); (F) 22 300lb (10 115kg) maximum loaded (original A) 28 935lb (13 125kg); (F) two tanks but no weapons) 30 700lb (13 925kg)

Performance: Maximum speed (typical of all) 864mph at height (1390 km/h) Mach 1.31; initial climb (clean) 16 000ft (4900m)/min; service ceiling 45 000ft (13 720m); range (high, two 375gal tanks) 115km)

Armament: Usually four (F, only two) 20mm M-39E cannon each with 100 rounds (A) pylons for two 375gal supersonic tanks and four additional pylons (seldom used) for 4 000lb ordnance (C, D) two tanks and six pylons for 7 500lb (3402kg) ordnance, (F) two tanks and maximum of 6 000lb (2 722kg) ordnance

History: First flight (YF-100) 25 May 1953, production (A) 29 October 1953; final delivery October 1957

Users: Denmark, France (until 1977), Taiwan, Turkey, USA, etc.

Development: The success of the Sabre made it natural to attempt a successor, and in February 1949 this was planned as a larger and much more powerful machine able to exceed the speed of sound in level flight. It was not until 1950 that it might have been smaller in view of the new requirement for simple fighters with the highest possible climb and performance at extreme altitudes. Undoubtedly, the 10 per cent saving with 45° sweep, no flaps, inboard ailerons, full-span wing and a slab tail plane mounted as low as possible. Level supersonic speed was achieved for the first time with a combat aircraft, but after very rapid development the F-100 was not ready for service until 1954. Trouble due to inertia coupling between the roll and yaw axes necessitated urgent modification, the wings and fin being lengthened. Subsequently the career of the "Hun" was wholly successful, the 203 A fighters being followed by the stronger C fighter-bomber, the D with flaps and autopilot and the tandem-seat F. Total production was lower than



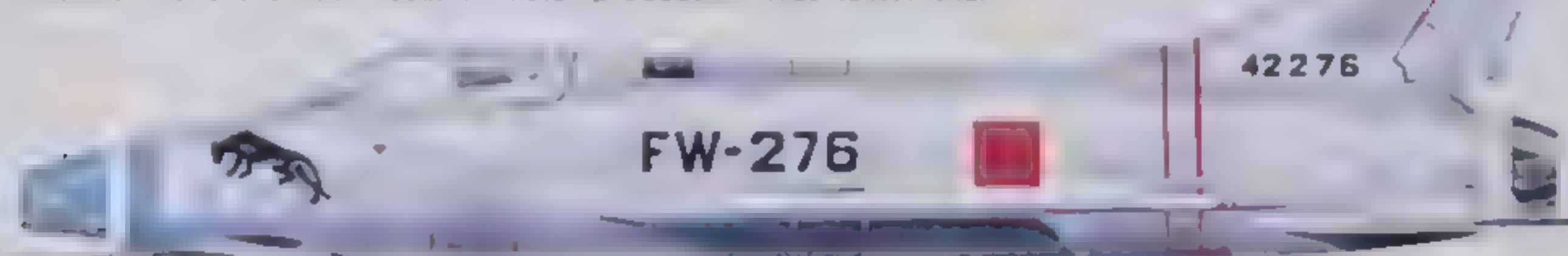
Three-view of tandem-seat F-100F in clean configuration

expected at 2,294, many being built by NAA's newly occupied factory at Columbus, Ohio. In their early years the later versions pioneered global deployment of tactical aircraft by means of probe/droque refueling, and in Vietnam they proved outstandingly good at both low attack and top cover, flying more missions than over 15,000 Mustangs flew in World War II. In 1977 the survivors of what two decades earlier had been among the world's elite warplanes were in their final months of combat duty after countless inspection, repair and modification programmes.



Above: Despite its venerable age the F-100 was one of the hardest-worked warplanes in the SE Asia conflict, flying both top cover and low-level attack. This well-used F-100D is burdened by four bombs and two supersonic tanks; and the cannon have obviously worked hard.

Below: Pop goes the drag chute as an F-100C of the Turkish Air Force recovers after a training sortie.



Above: F-100D-15 of Turkish Air Force, assigned to 1st TAF.



Panavia Tornado

Model 200 MRCA (S), (AD) and (T)

Origin: Panavia Aircraft GmbH, international company formed by British Aerospace, MBB of W Germany and Aeritalia

Type: Two seat multi-role combat aircraft, (S) optimised for strike, (AD) for air defence, (T) dual trainer

Engines: Two 14,500lb (6577kg) thrust Turbo-Union RB 199-34R three shaft augmented turbofans

Dimensions: Span (25°) 45ft 7½in (13.90m), (65°) 28ft 2½in (8.60m) length 54ft 9½in (16.70m), height 18ft 8½in (5.70m)

Weights: Empty, about 24,000lb (10,890kg), loaded (clean) about 35,000lb (15,880kg), maximum loaded, about 50,000lb (22,680kg)

Performance: Maximum speed (clean), at sea level, about 910mph (1465km/h, Mach 1.2), at height, over 1,320mph (2135km/h, Mach 2) service ceiling over 50,000ft (15,240m), range, about 1,000 miles (1610km) on internal fuel (high, wings spread), or over 3,000 miles (4830km) in ferry mode with maximum fuel

Armament: Two 27mm Mauser cannon in lower forward fuselage, seven pylons, three on the body and four on the swinging wings, for external load of up to "more than 18,000lb" (8165kg). Tornado ADV, main armament comprises two Mauser guns plus four HSD Skyflash and two Sidewinder air/air missiles recessed into fuselage

History: First flight 14 August 1974, planned service delivery 1977

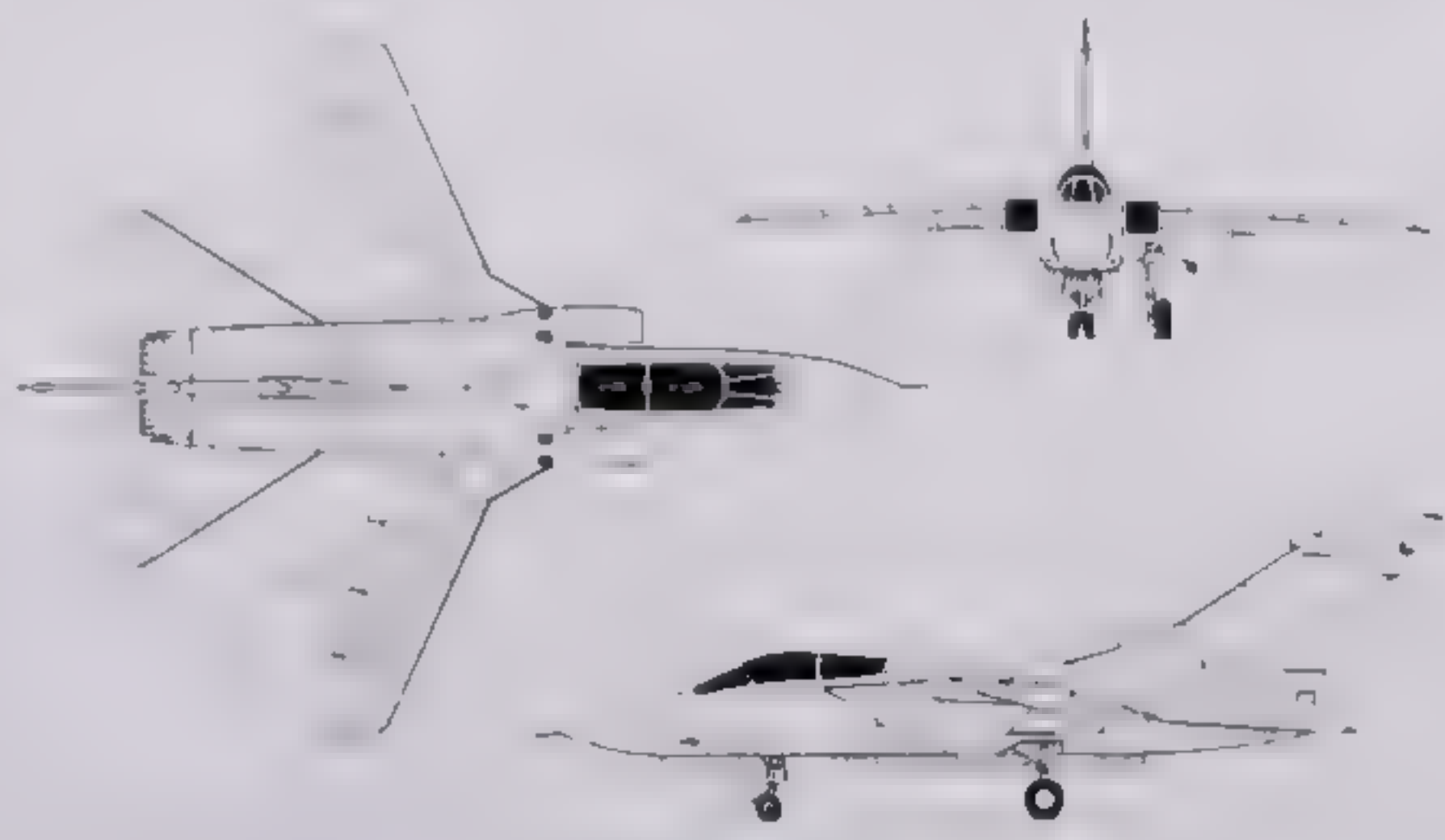
Users: W Germany (Luftwaffe, Marineflieger), Italy, UK (RAF)

Development: No combat aircraft in history has ever been planned with such care by so many possible customers. Studies began in 1967, after the French had abandoned the AFVG aircraft in the same class and decided not to participate in collaborative aircraft of this type. Panavia Aircraft was registered on 26 March 1969 in Munich as a three-nation company to manage the MRCA (multi-role combat aircraft) programme, with shares held in the ratio BAC 42½ per cent, MBB 42½ per cent and Aeritalia 15 per cent. In September 1969, after intense competition with the United States, the RB 199 was selected as the engine and a month later Turbo-Union was formed as the engine-management company with shares held in the ratio Rolls-Royce 40 per cent, MTU 40 per cent and Fiat 20 per cent. Thanks to the careful planning the MRCA programme has since demonstrated that it is possible for several nations to work together to create a modern military aircraft which promises to exceed all possible rivals in mission effectiveness, versatility and low cost, having already demonstrated better mission capability than the latest competing types designed specifically for that mission. Its design missions are close air support/battlefield interdiction, long-range interdiction/strike, naval strike, air superiority, air defence, interception, reconnaissance, training. At one time it was planned that the three nations should develop slightly different versions with either one or two seats and dry or wet wings, but all basic aircraft now in production are identical, with two seats and sealed integral-tank wings. From stem to stern the MRCA is totally modern – a fact which its many competitors

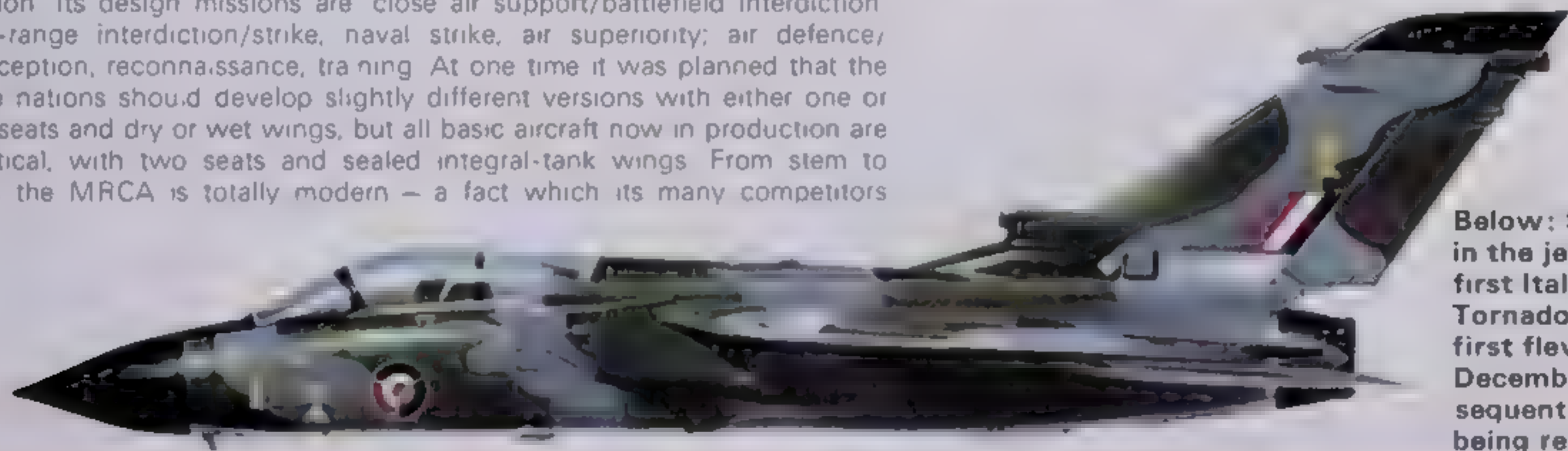
and is officially doubted as having the capability – in the European environment – to fly the interception missions of the Tornado ADV. Other combat aircraft with a single seat and non-swinging wing are grossly deficient in all roles except close-range air combat, a specialised mission for which the common version of Tornado is not intended (though its performance in this role is considerably better than a Mirage III, F-5 or F-4).

The Tornado MRCA has outstandingly efficient and compact engines of extremely advanced design, with automatically scheduled inlets and nozzles. Flight control is by large tailerons, augmented at low sweep angles by wing spoilers, the system is fully digital and signalled by quad fly-by-wire via an automatic command and stability augmentation system. For high lift at low speeds the wings have full-span slats and double-slotted flap. Other equipment includes a mapping radar, terrain following radar and computer, and laser target ranger for extreme accuracy. ECM and other penetration aids are exceptional. Planned production for the three original partners comprises 809 aircraft, of which 385 will be for the RAF, 202 for the Luftwaffe (replacing the F 104G and G91R), 122 for the Marineflieger.

continued on page 92



Three-view of Tornado IDS (interdictor/strike) variant



Above: 02 Tornado development prototype with "tri-national" markings

Below: Shock diamonds glow in the jets thrusting aloft the first Italian-assembled Tornado, prototype 05. This first flew at Turin Caselle in December 1975, but subsequently spent a long period being repaired after a heavy landing.

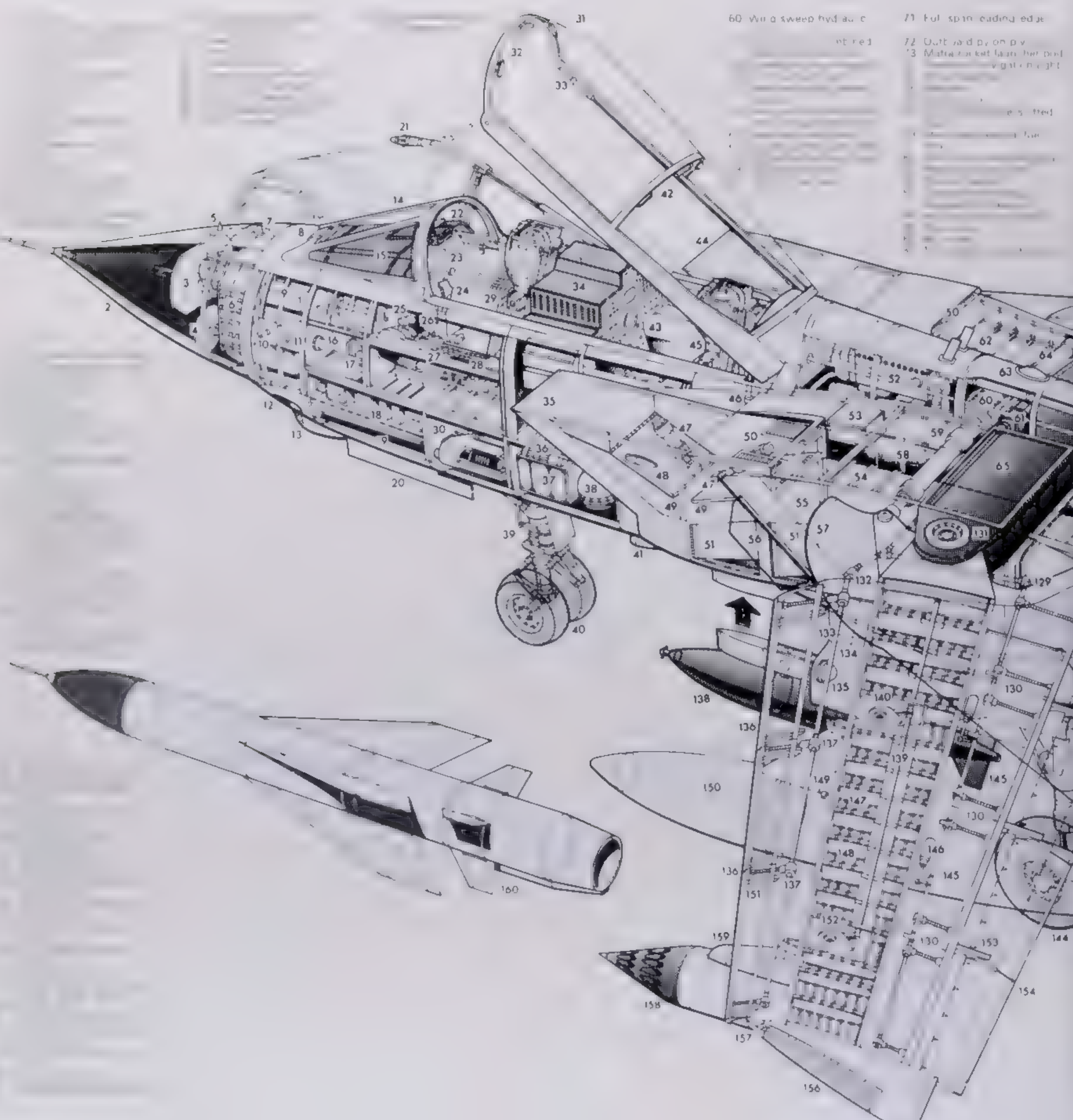




Above: Four 1,000lb (454kg) low-drag bombs fall from the fuselage pylons of Tornado 02 during the first weapon trials in 1975

about 220 will be of the common IDS (interceptor-defence) type planned to replace the Phantom in the air defence of the UK. The other 165 will be of the ADV (air-defence variant) type. The commonality with the common airframe will be very high, to give overall cost savings estimated at 80 per cent. Differences lie in systems and equipment, notably in the Marconi-Elliott/Ferranti radar, obviously of the "look-down" pulse-doppler type but with a mapping mode for the ADV's secondary attack role. The ADV will have slightly more powerful engines and its missiles will be carried recessed as in the Phantom; it will have a more powerful computer with totally different program, and other changes will be made to increase flight performance and effectiveness in the interception role. Unit cost was put at £6.492 million in August 1976 (compared with £5.29 million for the IDS version in 1976-77). There is a strong likelihood that the other Panavia nations, and other countries, will also adopt this cost-effective fighter version.

The nine prototypes of the IDS and trainer versions were assembled in 1975 and in July of 1976 the first production order was placed covering 20 aircraft for assembly in the UK, 16 in Germany and four in Italy. Panavia is, at the time of writing, being restructured to bring in Turbo-Union and Rolls-Royce so that NAMMA (the buying agency for the three customers) can do all negotiating with one "seller". A second batch of 40 was to be ordered in early 1977. Deliveries will begin later in 1977 with the six pre-series aircraft now coming off the line. These will go to RAF Cottesmore where the OCU (Operational Conversion Unit) is being set up for all customers. A tri-national weapons training unit will be at Decimomannu, Sardinia.



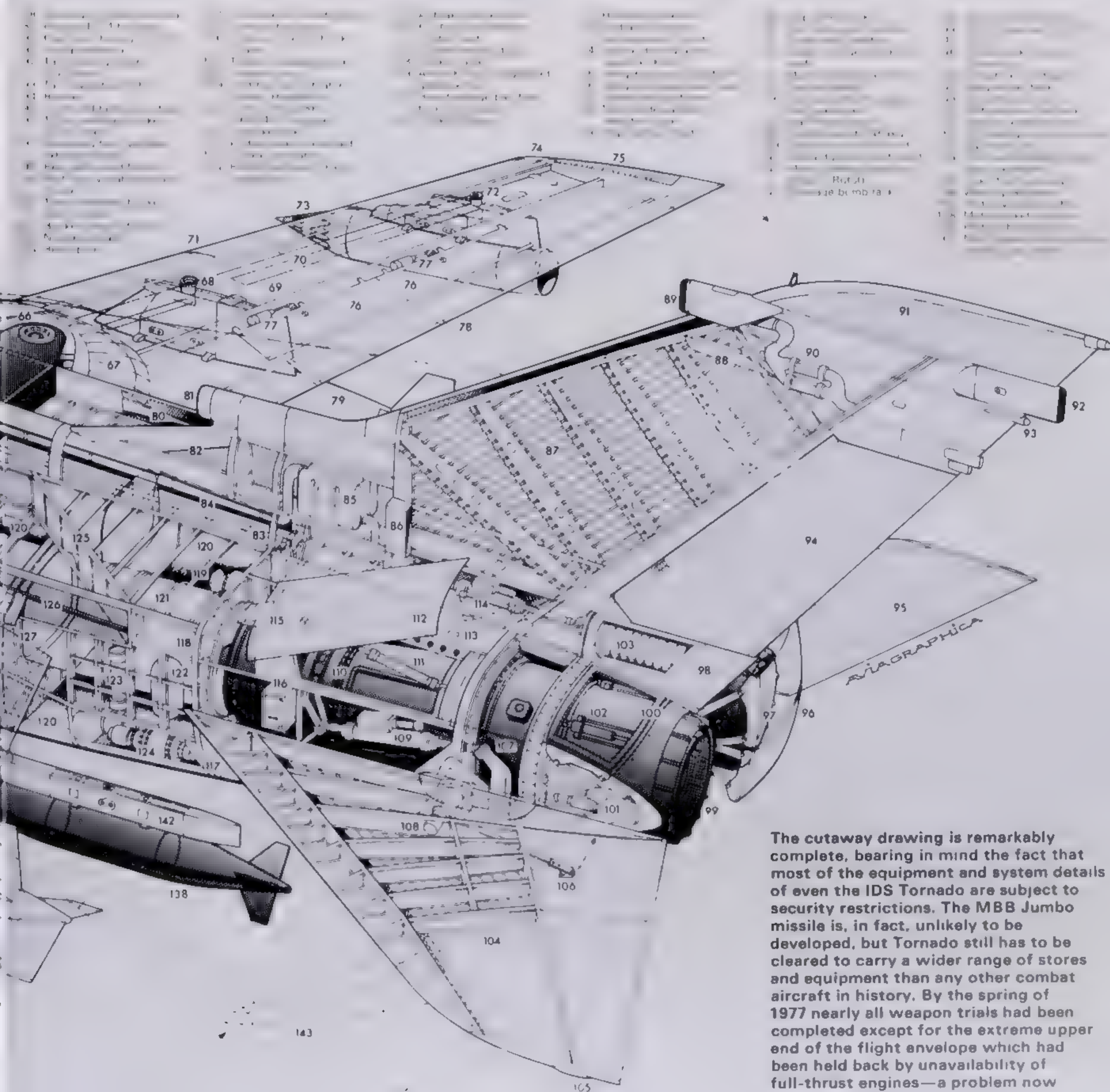
- 60 Wind sweep hydraulic
- 71 Full span leading edge
- 72 Outboard pylon pylon
- 73 Main wing fuel tank



Above: The second (04) and third (07) German-assembled Tornadoes flying from Manching with MBB crews. The second is carrying a large store on the right-hand body pylon.



Above: Prototype 06, flown by BAC at Warton in December 1975, was the first Tornado to carry the internal armament of two 27mm IKWA-Mauser cannons. Here it has supersonic drop tanks.



The cutaway drawing is remarkably complete, bearing in mind the fact that most of the equipment and system details of even the IDS Tornado are subject to security restrictions. The MBB Jumbo missile is, in fact, unlikely to be developed, but Tornado still has to be cleared to carry a wider range of stores and equipment than any other combat aircraft in history. By the spring of 1977 nearly all weapon trials had been completed except for the extreme upper end of the flight envelope which had been held back by unavailability of full-thrust engines—a problem now overcome.

Republic (Fairchild) F-105 Thunderchief

F-105B, D, F and G

Origin
Type

Engine

Dimensions: Span 34ft 11½ in (10

Weights: Empty (D) 27 500 lb (12,474 kg), (F, G) 28 393 lb (12 8

Performance

Armament: One 20mm M 61
One internal bay for ordinance
Four pylons for additional
History: First flight (YF)
1956 (D) 9 June 1959 (F)
User: USA (Air Force B

Development: The AP-63 project was a pri
Aviation to follow the F-84. Its primary mission wa
invent onal weapons in a -weathers, with very high speed and long range
Though it had only the stop-gap J57 engine the first Thunderchief exceeded
the speed of sound on its first flight, and the B model was soon in producti
for Tactical Air Command of the USAF. Apart from being the biggest single
seat, single-engine combat aircraft in history the 105 was notable for its
large bomb bay and unique swept forward engine inlets in the wing root.
Only 75 B were delivered but 600 of the advanced D were built with Nasan
monopulse radar and doppler navigation. Production was completed with
143 tandem-seat F with full operational equipment and dual controls.
Known as "the Thud" the greatest of single-engined combat jets bore a
huge burden throughout the Vietnam war. About 350 D were rebuilt during
that conflict with the Thunderstick (T-stick) all-weather blind attack
system – a few also being updated to T-stick II – with a large saddleback
countermeasures) attackers, with pilot and observer and Wild Weasel and
and Goodyear chaff pods were carried externally



Above: Two-seat F-105G Wild Weasel of 561st TFS, 35th TFW, George AFB, California.



Three-view of typical F-105D before T-stick rebuild.

Below: A tandem-seat F-105F shows its versatility by making a level bombing run whilst carrying AGM-45A Shrike anti-radiation missiles (white-painted). Shrikes were normally carried only by the "Wild Weasel" F-105G ECM aircraft



Rockwell International

XFV-12

XFV-12A prototype, constructor's designation NR-356

Origin: Rockwell International, Columbus, USA

Type: Fighter/attack technology prototype

Engine: One modified Pratt & Whitney F401-400 two-shaft afterburning turbofan rated at 16,400.b (7438kg) dry and 28,090lb (12,741kg) with maximum afterburner

Dimensions: Span (wing) 28ft 6½in (8.69m), (canard) 12ft 1½in (3.69m) length 43ft 11in (13.39m), height 10ft 4in (3.15m)

Weights: (Empty, fully equipped) 13,800lb (6259kg), (max VTOL) 19,500lb (8845kg) (max STOL) 24,250lb (11,000kg)

Performance: Maximum speed at maximum STOL weight, supersonic at low level, and in excess of Mach 2 (1,320mph, 2125km/h) at optimum height, take-off run at maximum STOL weight 300ft (91m)

Armament: Space for internal M61 gun in lower fuselage, provision for carrying air-to-surface or air-to-air weapons, with fuselage recesses for two Sparrow missiles

History: Contract award November 1972, roll-out December 1976, first flight 1977

User: US Navy

Development: In 1965-74 the US Navy studied the Sea Control Ship, an attractive form of small aircraft carrier with neither catapults nor arrestor gear but equipped on a highly automated basis for use as a base for V/STOL aircraft. Though the Soviet Union is producing ships of this type, much bigger than the projected SCS, the US Navy had to abandon the proposal through lack of money. Work continued, however, on this extremely interesting technology-demonstrator aircraft, which to save money has



Three-view of XFV-12A showing Sparrow missiles.

been designed by Rockwell's Columbus Aircraft Division in Ohio to incorporate the main wing box and parts of the inlet ducts of an F-4 Phantom and the nose, cockpit and forward fuselage, and nose and main landing gears, of an A-4 Skyhawk. Pratt & Whitney developed the large electrohydraulic diverter valve which, for V/STOL, shuts off the engine nozzle and diverts the entire efflux through large high temperature ducts. One duct feeds the canard's augmentor (ejector) flaps, while the larger duct feeds the similar flaps along the wings. Each surface has front and rear flaps across the full span, with endplates, and the ejector slits not only give direct lift and thrust but entrain an airflow 7½ times as great to give enough lift for VTOL or higher-weight STOL. All flap sections are of titanium honeycomb, fully powered to serve as control surfaces in forward flight, when the wing and canard are closed for forward speeds up to Mach 2. The fins carry small rudders above the wing but there are no leading-edge movable surfaces. There is no suggestion of producing an XFV-12A in quantity, but the technology is expected to be valuable in refining the requirements for supersonic V/STOL fighter and attack aircraft for shipboard use in the late 1980s.

Below: Artist's impression of XFV-12A fighter/attack research aircraft in jet-lift V/STOL mode.



Saab 37 Viggen

AJ37, JA37, SF37, SH37 and Sk37

Origin: Saab Svenska AB, Linköping

Type: (AJ) single-seat all-weather attack; (JA) all-weather fighter; (SF)

Engine: One Svenska Flygmotor RM8 (licence built Pratt & Whitney JT8D turbofan redesigned in Sweden for Mach 2 and fitted with mid-SK125 9700 lb (111 790 kg) RM8A

Dimensions: Span of main wing 34ft 9in (10.6m); length (AJ) 53ft 5in (16.3m); height 18ft 4in (5.6m)

Weights: Not disclosed, except AJ37 'normal armament' gross weight

Performance: Maximum speed (clean) about 1 320mph (2135km/h); Mach 1.1 at sea level; initial climb about 40 000ft (12 200m) in start of take-off run to 32 800ft-10 000m = 100sec; service ceiling 118 300m; tactical radius with external stores (not top thrust) below profile over 620 miles (1000km)

Armament: Seven pylons (option nine) for aggregate external load of 13 200 lb (6000 kg), including Rb04E or Rb05A missiles for attack, and Rb27, Rb28 and Rb324 missiles for defence. In addition the JA37 has a 30mm Oerlikon KCA gun and will carry new long- and short-range missiles for air-to-air interception. Skylark is being evaluated

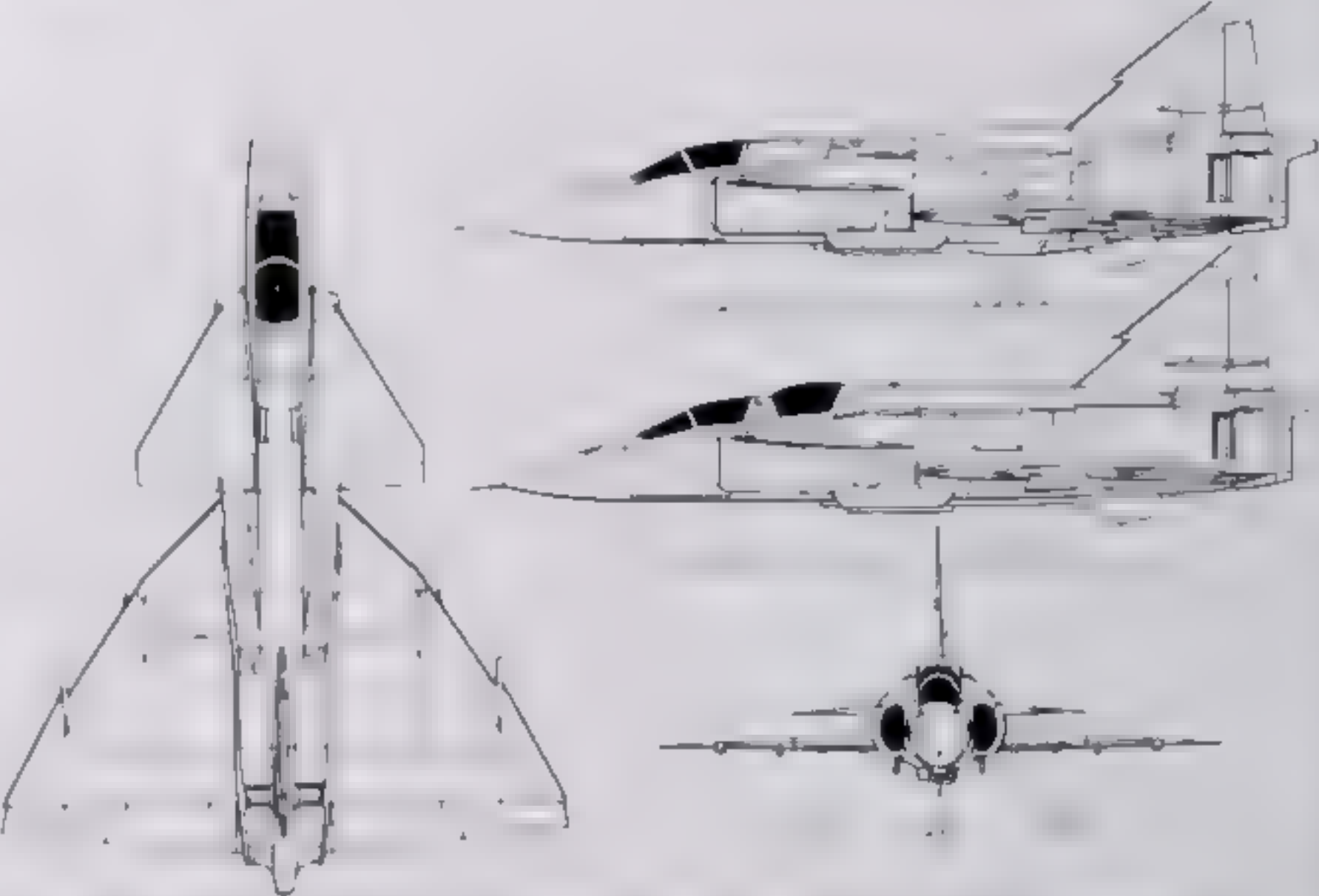
History: First flight 8 February 1967 (production AJ) 23 February 1971; private delivery (AJ) June 1971

User: Sweden (RSAF)

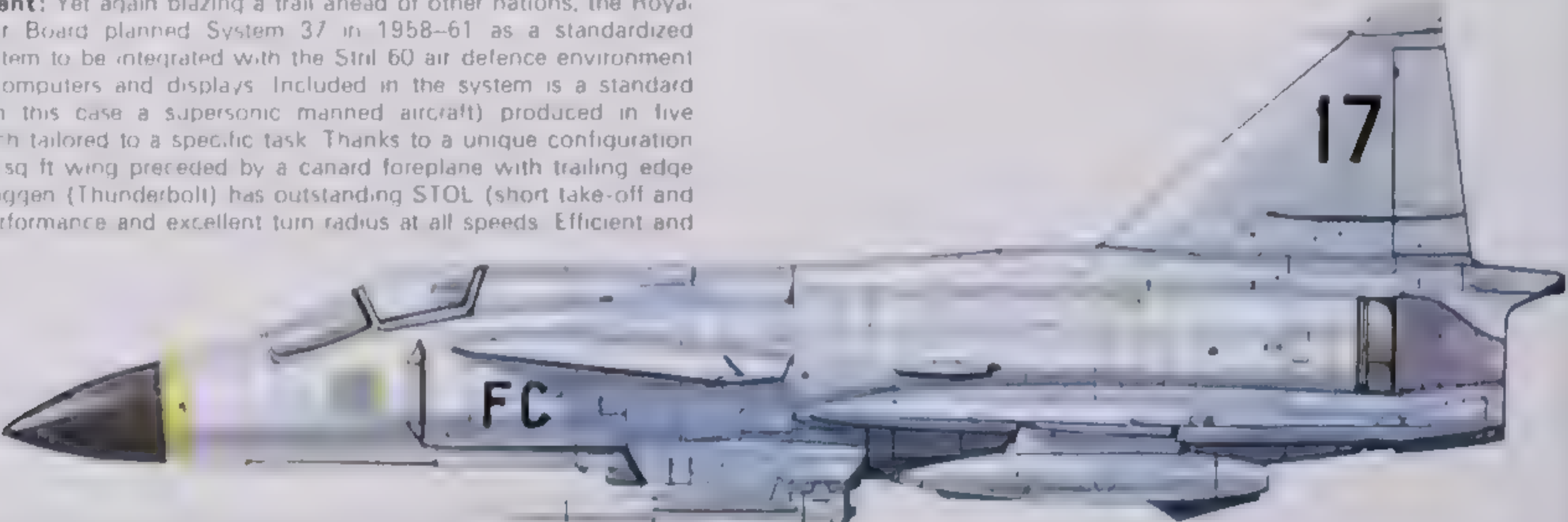
Development: Yet again blazing a trail ahead of other nations, the Royal Swedish Air Board planned System 37 in 1958-61 as a standardized air system to be integrated with the Stril 60 air defence environment of radars, computers and displays. Included in the system is a standard platform (in this case a supersonic manned aircraft) produced in five versions each tailored to a specific task. Thanks to a unique configuration with a 400 sq ft wing preceded by a canard foreplane with trailing edge flaps, the Viggen (Thunderbolt) has outstanding STOL (short take-off and landing) performance and excellent turn radius at all speeds. Efficient and

prolonged operations are possible from narrow strips 500m (1 640ft) in length, such as stretches of highway. Equipment in all versions includes headup display, autothrottle/speed control on approach, no flare landing autopilot and thrust reverser. The AJ operates camouflaged in attack wings F7, F15 and F6, with production continuing in 1977 on a mix of AJ, SF, SH and Sk models. At the beginning of the year about 145 had been delivered of the total orders for 180 of these versions. During 1976 Viggens in RSAF service were grounded until the cause of inflight structural (wing) failures

continued on page 98 ▶

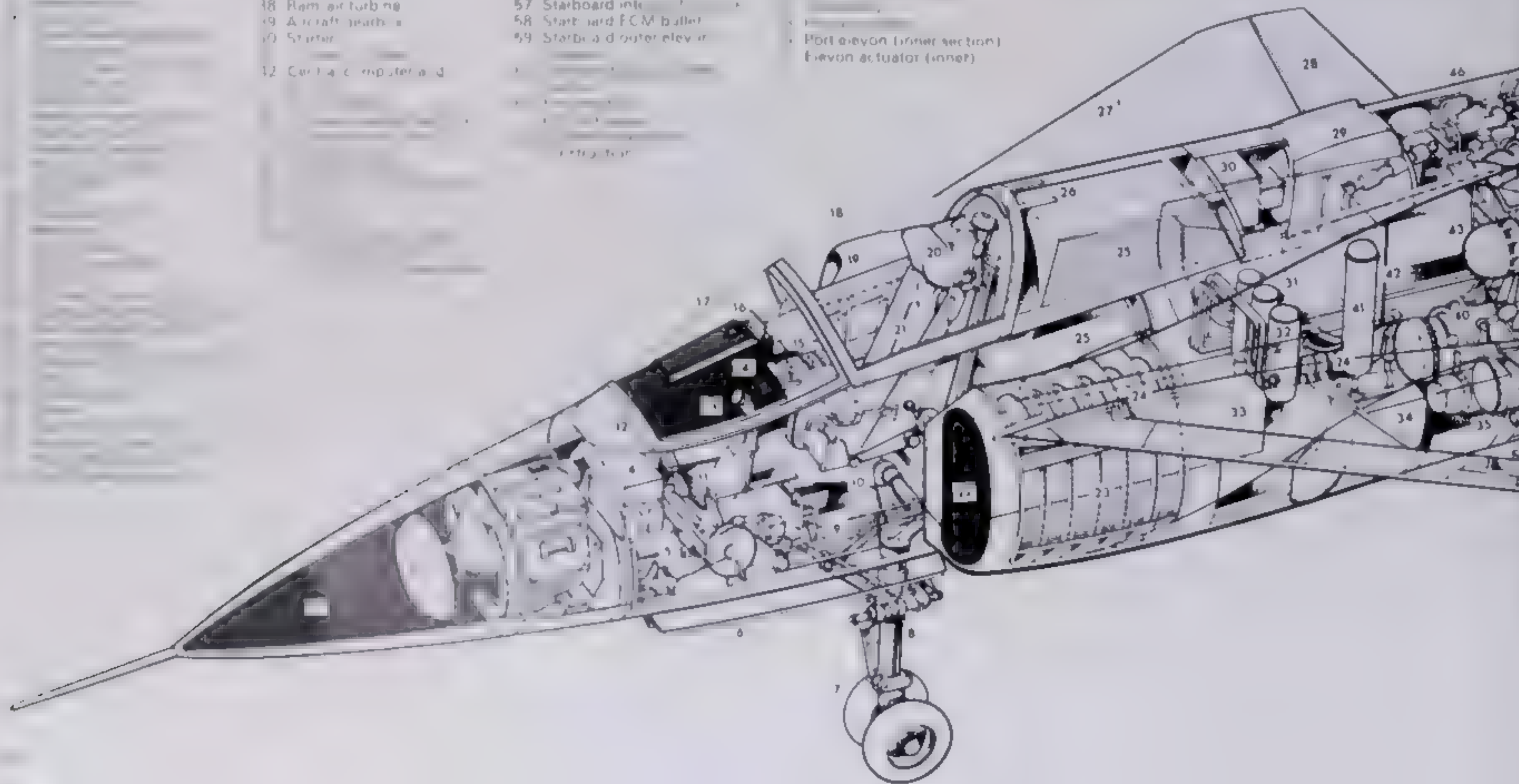


Three-view of JA37, with side view (centre) of SK37 trainer.



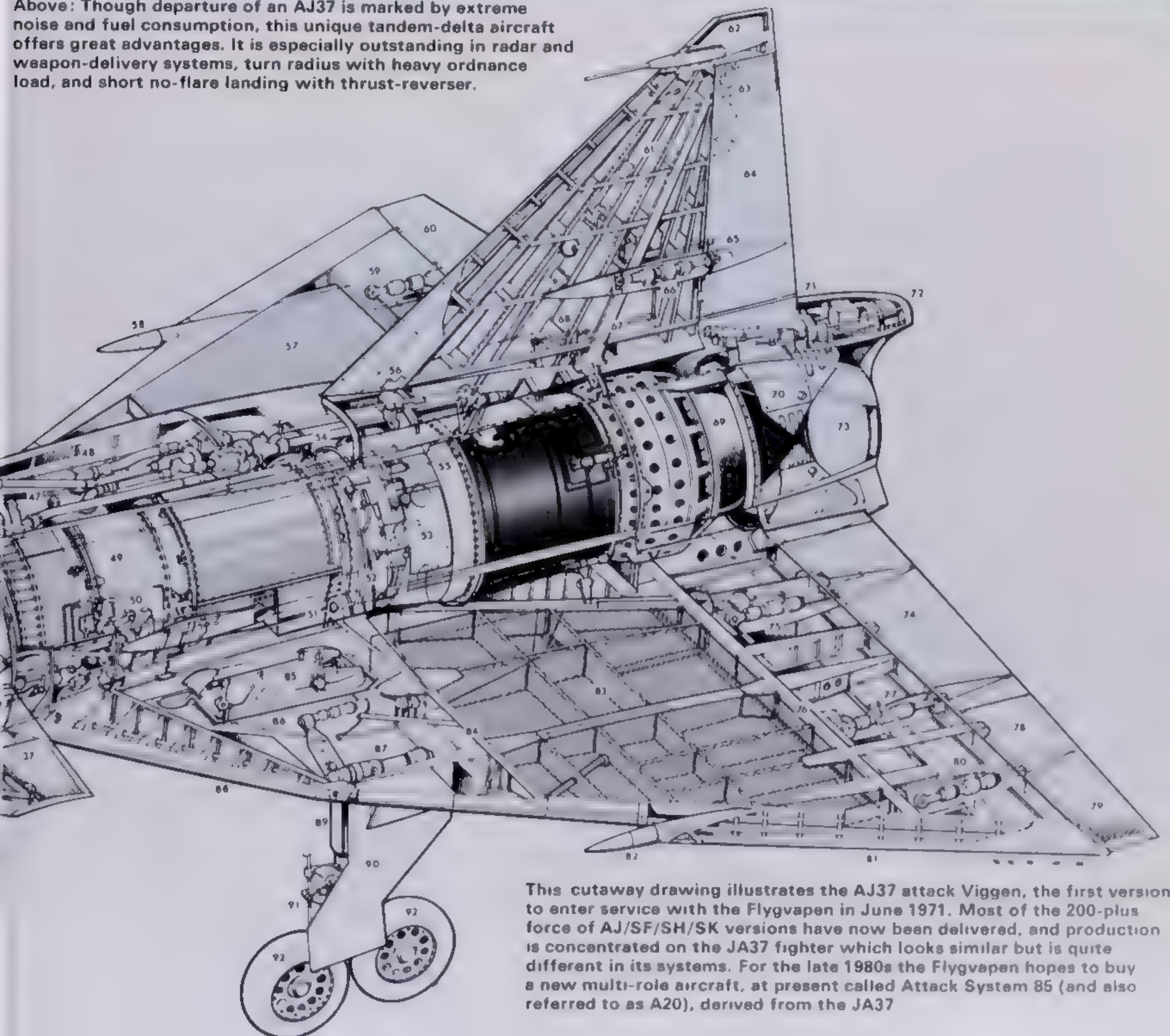
Above: Early AJ37, carrying RB05A on body pylon

- | | | | | | |
|------------------------------|------------------------------------|---|--|---------------------------------|---------------------------------|
| 1 Electric cable | 32 Central fuselage avionics | 60 Aircraft gearbox cooler | 64 Rudder | 76 Rear spar | 85 Undercarriage door (inner) |
| 2 Parabolic antenna | 33 Central fuselage avionics | 61 Main wing attachment point | 65 Rudder actuator fairing | 77 Elevon actuator (centre) | 86 Wheel well |
| 3 Ericsson PS-91 pulse | 34 Park avionics bay | 62 Fuselage saddle fuel tank (rear wall) (see 44) | 66 Actuator support | 78 Port elevon (outer section) | 87 Mainwheel leg pivot point |
| 4 Avionics bay | 35 Hydraulic tank | 63 Air brake | 67 Fin rear spar/fuselage attachment point | 79 Honeycomb construction | 88 Inner leading edge structure |
| 5 Radar avionics cooling air | 36 Forward (canard) wing structure | 64 Engine aft attachment point | 68 Control linkage | 80 Elevon actuator (outer) | 89 Mainwheel leg (Motala) |
| | 37 Honeycomb flap construction | 65 Control fins | 69 Afterburner | 81 Outer leading edge structure | 90 Mainwheel leg door (outer) |
| | 38 Ram air turbine | 66 Fin front spar/fuselage attachment point | 70 Reverser clamshears | 82 Port ECM bullet | 91 Mainwheel scissors |
| | 39 Aircraft weather | 67 Starboard intake | 71 Rear navigation light | 83 Port integral fuel tank | 92 Tandem mainwheels |
| | 40 Starter | 68 Starboard ECM bullet | | 84 Main spar | |
| | 41 Central computer and | 69 Starboard outer elevator | | | |





Above: Though departure of an AJ37 is marked by extreme noise and fuel consumption, this unique tandem-delta aircraft offers great advantages. It is especially outstanding in radar and weapon-delivery systems, turn radius with heavy ordnance load, and short no-flare landing with thrust-reverser.



This cutaway drawing illustrates the AJ37 attack Viggen, the first version to enter service with the Flygvapen in June 1971. Most of the 200-plus force of AJ/SF/SH/SK versions have now been delivered, and production is concentrated on the JA37 fighter which looks similar but is quite different in its systems. For the late 1980s the Flygvapen hopes to buy a new multi-role aircraft, at present called Attack System 85 (and also referred to as A20), derived from the JA37

Right: A tandem-seat SK37 of F15 wing at Soderhamm taxis out for a training sortie. This replaces the fuselage fuel tank and electronics by a rear cockpit, with Saab-Scania rocket-assisted seat, bulged canopy and twin periscopes. To counter the extra side area the fin has increased height and area. This dual control version can fly attack missions with any AJ37 weapons

Below: An AJ37 with some of the more common external stores. The largest is the centreline fuel tank. In the front row are low-drag bombs with stand-off fuzes. In the second row are (from the outside) RB28 Falcon air/air missile; RB05A air/surface missile (now being supplemented by RB05B with long TV seeker nose); two types of store apparently carrying multi-sensor reconnaissance equipment; two types of launcher for Bofors 135mm rockets; and RB324 Sidewinder air/air missiles. Under the wings are large RB04E anti-ship missiles



SEPECAT Jaguar

Jaguar GR.1 and T.2, Jaguar A and E, and Jaguar International

Origin: SEPECAT, consortium formed by British Aerospace and Dassault-Breget France

Type: GR.1 A multi-role fighter, (T.2 and E) trainer

Engines: Two Bristol Siddeley Pegasus turbofans (except T.2, E) or two Adour 804

Dimensions: Span 28ft 6in (8.69m), length (except T.2, E) 50ft 11in (15.52m), (T.2, E) 53ft 11in (16.42m), height 16ft 1½in (4.92m)

Weights: Empty, classified but about 15 000lb (6800kg), "normal take-off (ie. internal fuel and some external ordnance) 23 000lb (10 450kg), maximum loaded 34 000lb (15 500kg)

Performance: Maximum speed (lo. some external stores) 820mph (1320km/h, Mach 1.1) (hi. some external stores) 1 055mph (1700km/h, Mach 1.6), climb and ceiling, classified, attack radius, no external fuel 1 000 miles (1 609 km), range 2 614 miles (4 206 km)

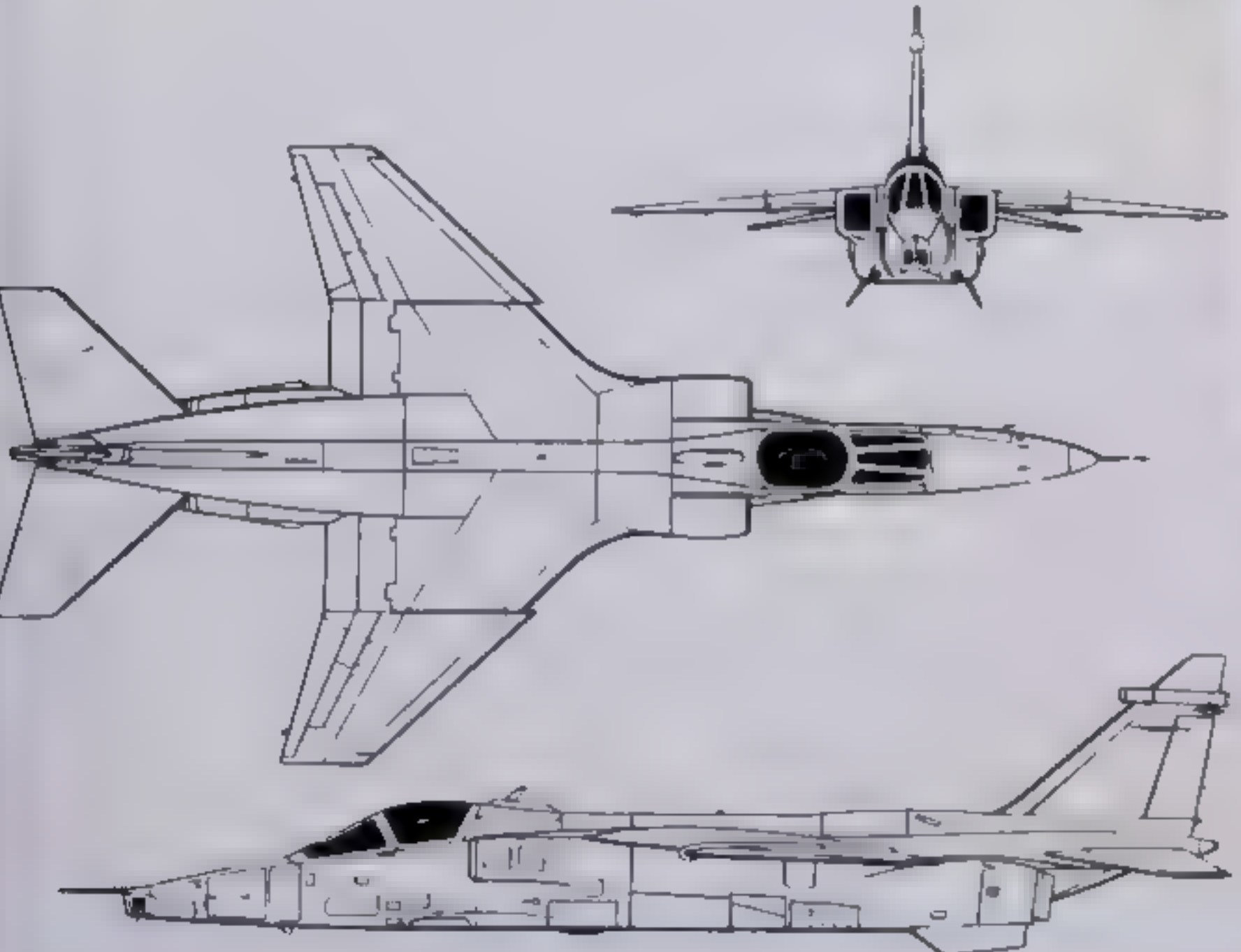
Armament: A/E two 30mm DEFA 553 cannons with 15 rounds, five pylons for total external load of 10 000lb (4536kg) (GR.1) as above but guns two 30mm Aden, (T.2) as above but single Aden. A wide range of options including increased external loads

History: First flight (E) 8 September 1968 (production E) November 1971, (production GR.1) 11 October 1972, squadron delivery (F.1) May 1972, (GR, T) June 1973

Users: Ecuador, France, Oman, UK (RAF)

Development: Developed jointly by BAC in Britain and Dassault-Breget in France, to meet a joint requirement of the Armée de l'Air and RAF the Jaguar is a far more powerful and effective aircraft than originally planned and has already demonstrated unmatched capabilities in service. The

continued on page 100



Above: Three-view of Jaguar GR.1 without stores.



Above: Takeoff of a prototype E-type trainer in France in 1969, with "dual-nationality" markings. All 40 of this simple training version have now been delivered to the Armée de l'Air, the main units being the 7th wing at St Dizier and 11th at Toul



Above: Close formation fly-past by Jaguar A-type attack aircraft and E-type trainers of the 7e wing. The Armée de l'Air deliberately bought simpler Jaguars than the RAF, though they are now being updated by optional laser and/or radar equipment.

Below: A dramatic firing picture of a Matra 550 Magic air-to-air dogfight missile being loosed from an overwing pylon. The aircraft was a BAC-modified Jaguar International, with uprated Adour 804 engines, trials with which were very successfully completed in 1976.

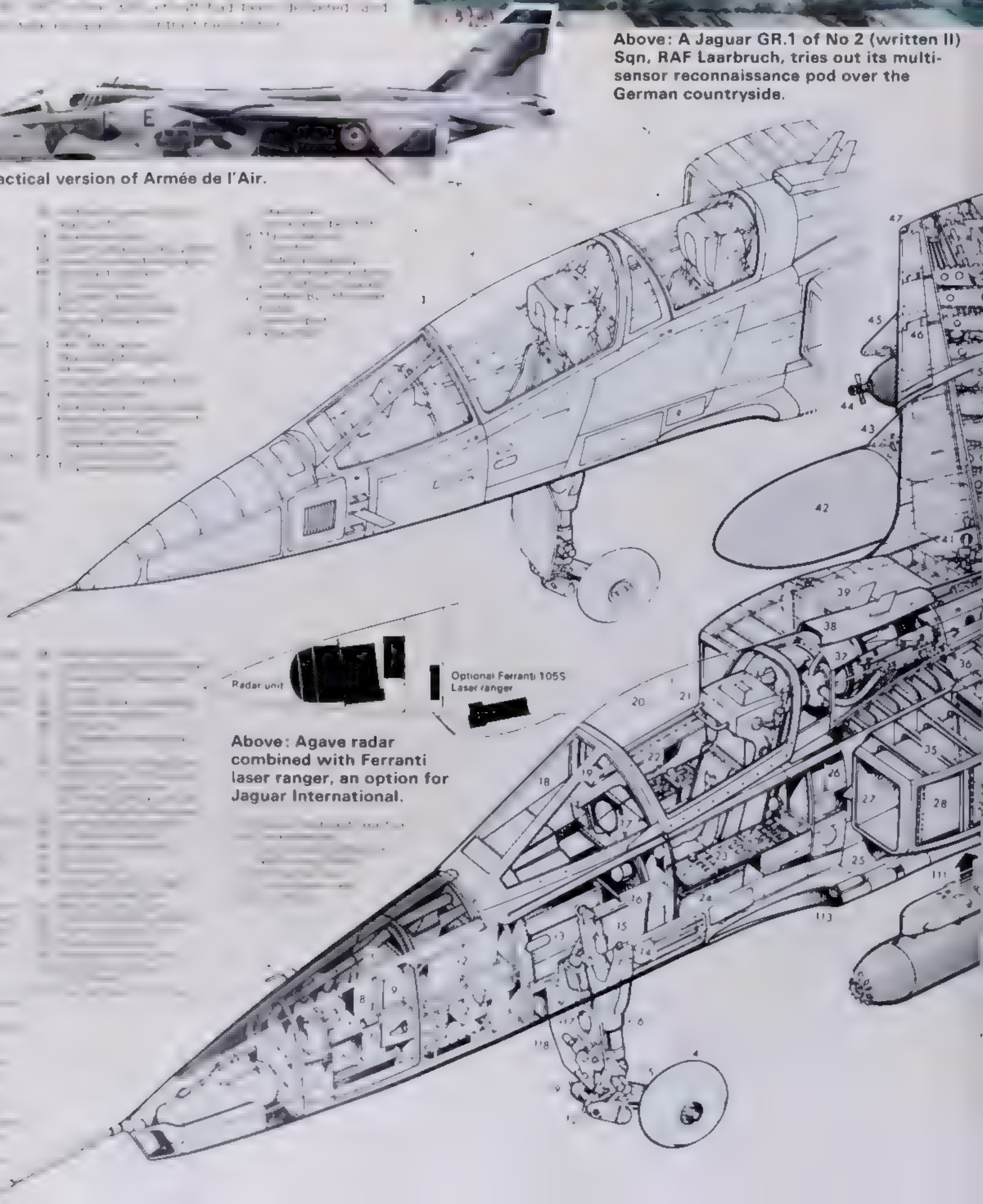


The Jaguar is a two-seater, single-engine, delta-wing aircraft. It was developed by British Aerospace (BAe) in collaboration with the French Armée de l'Air. The aircraft is designed for low-level, high-speed flight and is capable of carrying a variety of weapons, including bombs, missiles, and rockets. The Jaguar GR.1 is the reconnaissance version of the aircraft, equipped with a multi-sensor reconnaissance pod. The Jaguar A is the tactical version of the aircraft, equipped with a variety of weapons. The Jaguar is a highly versatile aircraft, capable of performing a wide range of missions. It is a testament to the skill and ingenuity of the designers and engineers who created it.



Above: A Jaguar GR.1 of No 2 (written II) Sqn, RAF Laarbruch, tries out its multi-sensor reconnaissance pod over the German countryside.

Above: Jaguar A tactical version of Armée de l'Air.



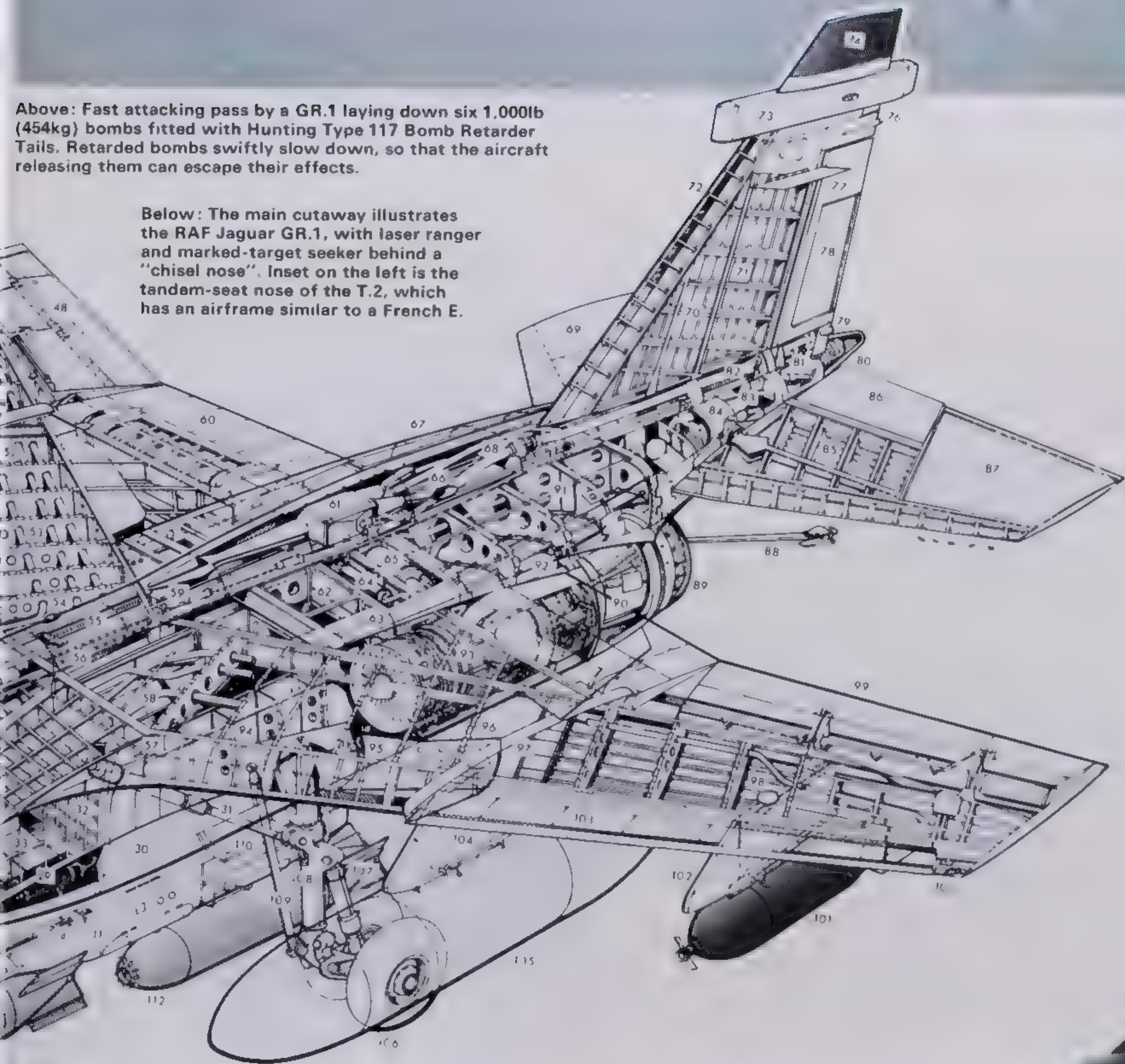
Above: Agave radar combined with Ferranti laser ranger, an option for Jaguar International.





Above: Fast attacking pass by a GR.1 laying down six 1,000lb (454kg) bombs fitted with Hunting Type 117 Bomb Retarder Tails. Retarded bombs swiftly slow down, so that the aircraft releasing them can escape their effects.

Below: The main cutaway illustrates the RAF Jaguar GR.1, with laser ranger and marked-target seeker behind a "chisel nose". Inset on the left is the tandem-seat nose of the T.2, which has an airframe similar to a French E.



Above: Jaguar GR.1 of 54 Sqn, RAF Coltishall

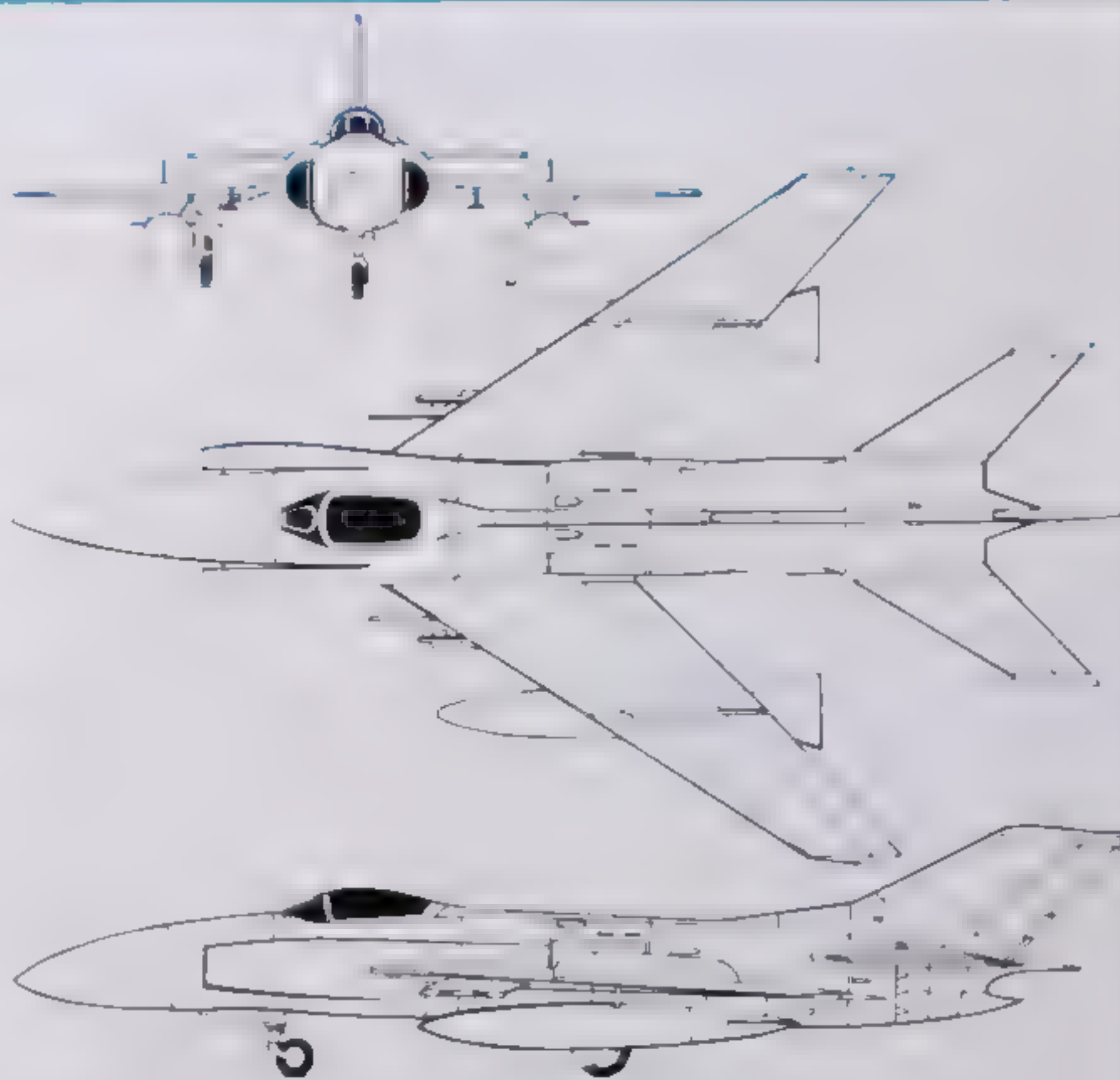
Shenyang F-9

F-9 (NATO code name "Fantan A")

Origin
Type
Engines
Dimensions (estimated) Span 33ft 5in (10.2m), length 50ft (15.2m)
Weights
Performance (estimated) Maximum speed clean (sea level)
Armament: Not known, but almost certain

History: First flight, possible 1965
Users: People's Republic of China (AI)

Development: The F-9 is derived from the F-6, the Chinese built MiG-19. The F-9 represents the first (enforced) attempt by the Shenyang plant to produce combat aircraft independently. Despite the difficulties caused by lack of industrial backing and skilled labour, production of nationalised aircraft was forced on the PRC (People's Republic of China) by its isolation from technically advanced nations and imminence of the Soviet threat. The excellent qualities of the MiG-19 basic design eventually led to the F-6 being chosen for development in preference to the F-8, the illegally manufactured MiG-21PF. During the 1960s the Shenyang F-9 took shape as an enlarged F-6 with lateral inlet ducts feeding direct to the two engines (a Chinese illustration suggests that the mid-wing has been retained, with ducts above and below), leaving the nose free for a large search radar of unknown type. The sketch referred to showed no wing cannon, but the two 30mm NR-30s of the F-6 have probably been retained in view of the great length of inlet duct ahead of the wing, interfering with pilot view. The inlets are apparently simple and non-variable, efficient at low level but limiting high-altitude Mach number.



Above: Three-view of most common F-9 (provisional, based on Chinese pictures).

The radar could be a derivative of the "Spin Scan B" as used in later North Vietnamese MiG-21PF fighters sent via China, but in the author's opinion is more likely to be a copy of the much more powerful AWG-10 or APQ-109 fitted to Phantoms of the late 1960s. Whether the PRC has also copied Sparrow and/or Sidewinder is problematical. In his Fiscal Year 1977 report the US SecDef (then Donald Rumsfeld) described the "Fantan-A" as a principal tactical aircraft of the PRC Navy.

Sukhoi Su-7

Su-7B, -7BM, -7BMK and -7U; NATO name "Fitter"

Origin: The design bureau of Pavel A. Sukhoi, Soviet Union
Type: Single-seat close-support and interdiction, (-7U) dual-control trainer
Engine: One Lyulka AL-7F turbojet rated at 15,430lb (7000kg) dry or 22,046lb (10,000kg) with maximum afterburner
Dimensions: Span 29ft 3½in (8.93m), length (all, incl probe) 57ft (17.37m), height (all) 15ft 5in (4.70m)
Weights: Empty (typical -7) 19,000lb (8620kg), maximum loaded (typical -7) 30,000lb (13,610kg)
Performance: Maximum speed, clean at altitude, (all) 1,055mph (1700km/h, Mach 1.6), initial climb (-7BM) 29,000ft (9120m)/min, service ceiling (-7BM) 49,700ft (15,150m), range with twin drop tanks (all) 900 miles (1450km)
Armament: (-7) two 30mm NR-30 cannon, each with 70 rounds, in wing roots, four wing pylons, inners rated at 1,653lb (750kg) and outers at 1,102lb (500kg) but when two tanks are carried on fuselage pylons total external weapon load is reduced to 2,205lb (1000kg)
History: First flight (-7 prototype) not later than 1955, service delivery (-7B) 1959
Users: (-7) Afghanistan, Algeria, Czechoslovakia, Egypt, Hungary, India, Iraq, N Korea, Poland, Rumania, Soviet Union, Syria, Vietnam

Development: Two of the wealth of previously unknown Soviet aircraft revealed at the 1956 Aviation Day at Tushino were large Sukhoi fighters, one with a swept wing (called "Fitter" by NATO) and the other a tailed delta (called "Fishpot"). Both were refined into operational types, losing



Above: Three-view of Su-7BMK, with side view (bottom) of -7U "Moujik"

some of their commonality in the process. The delta entered service as the Su-9 and -11, described separately. The highly-swept Su-7 was likewise built in very large numbers, optimised not for air superiority but for ground attack. As such it has found a worldwide market, and despite severe shortcomings has been exported in numbers which exceed 700. All Sukhoi combat aircraft have been made within the Soviet Union. The good points of the Su-7 family are robust structure, reasonable reliability and low cost; drawbacks are vulnerability to small-calibre fire and the impossibility of getting adequate field length, weapon load and radius of action all together. There are many variants. The original -7B was quickly superseded by the more powerful -7BM, with twin ribbon tail chutes. The most common export model is the -7BMK with low pressure tyres and other changes to improve behaviour from short unpaved strips. The -7U is the tandem dual trainer. Since 1964 many BMK have been seen with take-off rockets and four wing pylons.



Above: Flame envelops the leading edge as an Su-7B fires its cannon



earlier it was known to be in service with the FRCAP. Compared with the F-6 it should be a considerably more effective machine provided engine power has risen at least in proportion to the weight. Some Western reports suggest that the engines are the Tumansky RD-9B-811, of 8 270lb (3 750kg) maximum thrust, in the author's view an equally plausible possibility, is that the bigger and more powerful R-11 engine of the F-8 (13 120lb 5 950kg) could have been chosen. Indeed use of this engine in the F-9 might in some degree explain the early termination of Chinese production of the F-8. As this book went to press little is known of the F-9, and it could even

Above: Artist's impression of an F-9 in operational service. There is abundant evidence that Chinese arms output has increased to levels probably greater than that of any other nation.

be subject to severe problems and limitations. It should in any case not be confused with the entirely different combat aircraft (believed to be a twin-engined delta) which will be powered by the Chinese-assembled Rolls Royce Spey.



Above: Four Su-7BM tactical aircraft of the Egyptian Air Force. About 120 are in service, plus some Su-7U trainers.



Above, right: This section of Su-7Bs are carrying the twin drop tanks needed by this aircraft to supplement its limited internal capacity.

The -7BM has a number of detail differences, and in late sub-types has a low-pressure nosewheel tyre for soft fields, needing bulged doors.



Above: Well-worn Su-7B of Indian Air Force.



Above: Su-7BM of Egyptian AF



Above: Su-7B (early sub-type) of Czech Air Force.

Sukhoi Su-17 and Su-20

Su-17 "Fitter C", Su-20 and Su-22

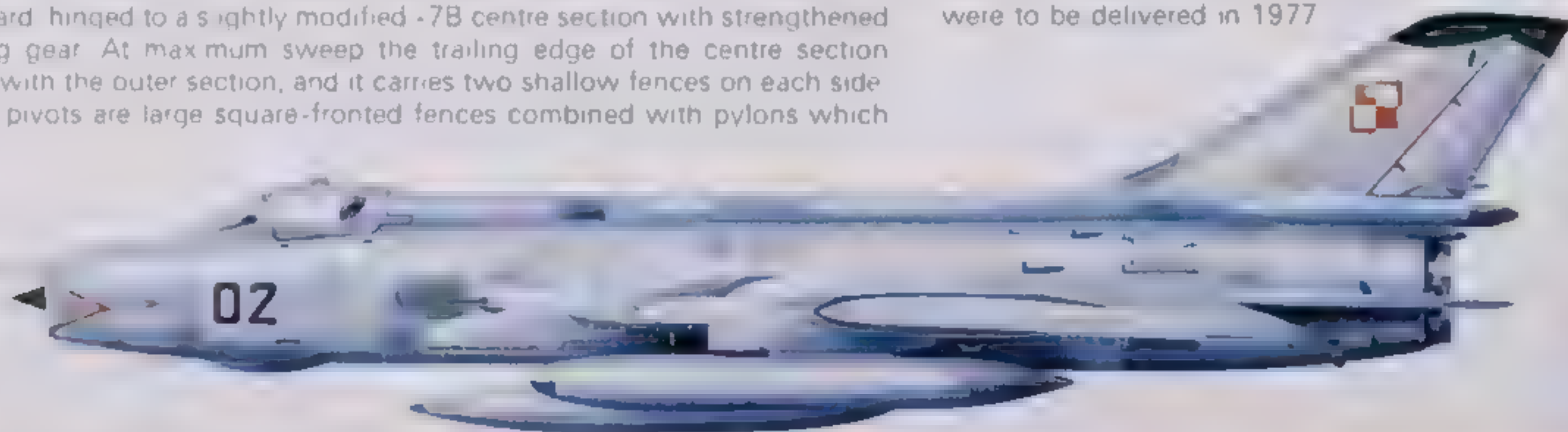
Origin: The design bureau named for Pavel O. Sukhoi, Soviet Union
Type: Single-seat attack and close support aircraft
Engine: (-17) one Lyulka AL-21F-3 single-shaft turbojet with afterburner rated at 17,200lb (7800kg) dry and 25,000lb (11,340kg) with maximum afterburner (-20/-22) believed to be AL-7F-1 rated at 22,046lb (10,000kg)
Dimensions: (a) Span (28°) 45ft 11½in (14.00m) (b2°) 34ft 9½in (10.60m) length (incl probe) 61ft 6½in (18.75m) height 15ft 7in (4.75m)
Weights: (-17 estimated -20 and -22 slightly less) empty 22,046lb (10,000kg) loaded (clean) 30,865lb (14,000kg), (maximum) 41,887lb (19,000kg)
Performance: (-17 clean) maximum speed at sea level 798mph (1284 km/h Mach 1.05) maximum speed at optimum height 1,432mph (2305 km/h Mach 2.17) climb rate 45,275ft (13,800m)/min, service ceiling 59,050ft (18,000m) combat radius with 4,410lb (2,000kg) external stores (hi lo h) 391 miles (630km)
Armament: Two 30mm NR-30 cannon, each with 70 rounds, in wing roots, eight pylons under fuselage, fixed gloves and swing-wings for maximum external load of 11,023lb (5,000kg) including the AS-7 "Kerry" air-to-surface missile (-20/-22 six pylons)
History: First public display at Domodedovo 1967, service delivery, possibly 1970 (-17) and 1972-3 (-20)
Users: Egypt (-20) Peru (-22), Poland (-20), Soviet Union (FA -17)

Development: A logical direct modification of the somewhat limited Su-7B, the Su-17 has variable-geometry "swing-wings" pivoted far outboard, hinged to a slightly modified -7B centre section with strengthened landing gear. At maximum sweep the trailing edge of the centre section aligns with the outer section, and it carries two shallow fences on each side. At the pivots are large square-fronted fences combined with pylons which



Above: Three-view of Su-20, showing range of wing sweep.

are stressed to carry 2,200lb (1,000kg) stores which in the Polish Su-20 are invariably drop tanks with nose fins. The swing-wings carry full-span slats, slotted ailerons and flaps which retract inside the centre section. Compared with the Su-7B the result is the ability to lift twice the external load from airstrips little more than half as long, and climb and level speed at all heights are much increased, even in the lower-powered Su-20 and export Su-22. Equipment in the -17 includes SRD-5M "High Fix" radar, an ASP-5ND fire-control system and comprehensive communications and IFF. Landing performance is so much better than the -7B that a braking chute is not fitted, in its place is the aft-facing aerial for a Sirena-3 radar homing and warning system at the rear of the prominent dorsal spine. Peru's 36 aircraft were to be delivered in 1977.



Below: On 6 October 1976 the Egyptian Air Force sprang a surprise on foreign observers by including this group of Su-20 variable-sweep attack aircraft in its parade commemorating the October war three years earlier.

Above: Su-20 in service with Polish Air Force (wings swept).



Sukhoi Su-19

Su-19 versions known to NATO as "Fencer"

Origin: The design bureau of Pavel O. Sukhoi, Soviet Union
Type: Two-seat multi-role combat aircraft
Engines: Two turbojet engines, each rated at 11,111 kg (24,500 lb) thrust
Dimensions: (Estimated) span (spread, about 22°) 50ft 3in (15.3m), swept (about 72°) 31ft 3in (9.53m), length 49ft 10in (15.2m), height 21ft (6.4m)
Weights: (Estimated) empty 35,000lb (15,875kg), maximum loaded 70,000lb (31,750kg)
Performance: (Estimated) maximum speed clean 950mph (1,530km/h, Mach 1.25) at sea level, about 1,650mph (2,655km/h, Mach 2.5) at 30,000ft (9,144m), climb rate 30,000ft (9,144m) in 19 seconds, range 1,500 miles (2,414km), service ceiling 50,000ft (15,240m)
Armament: One 23mm GSh 23 twin barrel cannon in lower centreline, at least six pylons on fuselage, fixed and swinging wings, for wide range of stores including guided and unguided air-to-ground or air-to-air missiles
History: First flight, probably about 1970, service delivery, 1974 or earlier
User: Soviet Union (mainly FA)

Development: First identified publicly in the West by the Chairman of the US Joint Chiefs of Staff, who described the Su-19 as "the first modern Soviet fighter to be developed specifically as a fighter-bomber for the ground-attack mission", this aircraft will probably be the chief tactical attack aircraft of the Soviet V-VS in 1980. Like the rival but much smaller MiG-27, the Su-19 is an extremely clean machine strongly reminiscent of the F-111 and Mirage G, having side-by-side seats and wing and tailplane



Three-view of Su-19, showing range of sweep (provisional)

at the same level as the Su-19, but with a different wing planform. In general, capability, the nearest Western equivalent is the F-14 Tomcat, which shows just how formidable this aircraft is. "Fencer" was on many Western lips in the 1960s, so is "Fencer" a scare-word in the 1970s. Features of the first service version include a typical Sukhoi tail, but with ventral fins, double-shock absorbers, slats and double-slotted flaps, a multi-mode attack radar, doppler laser ranger and very comprehensive EW/ECM installations.

Left: A side view of an Su-19 in service with the Soviet FA. Like the three-view this is provisional. There is some evidence that the rear fuselage and tail resembles that of the Su-15 interceptor, and the engines may be the same.

Vought A-7 Corsair II

Vought A-7A to E and TA-7C

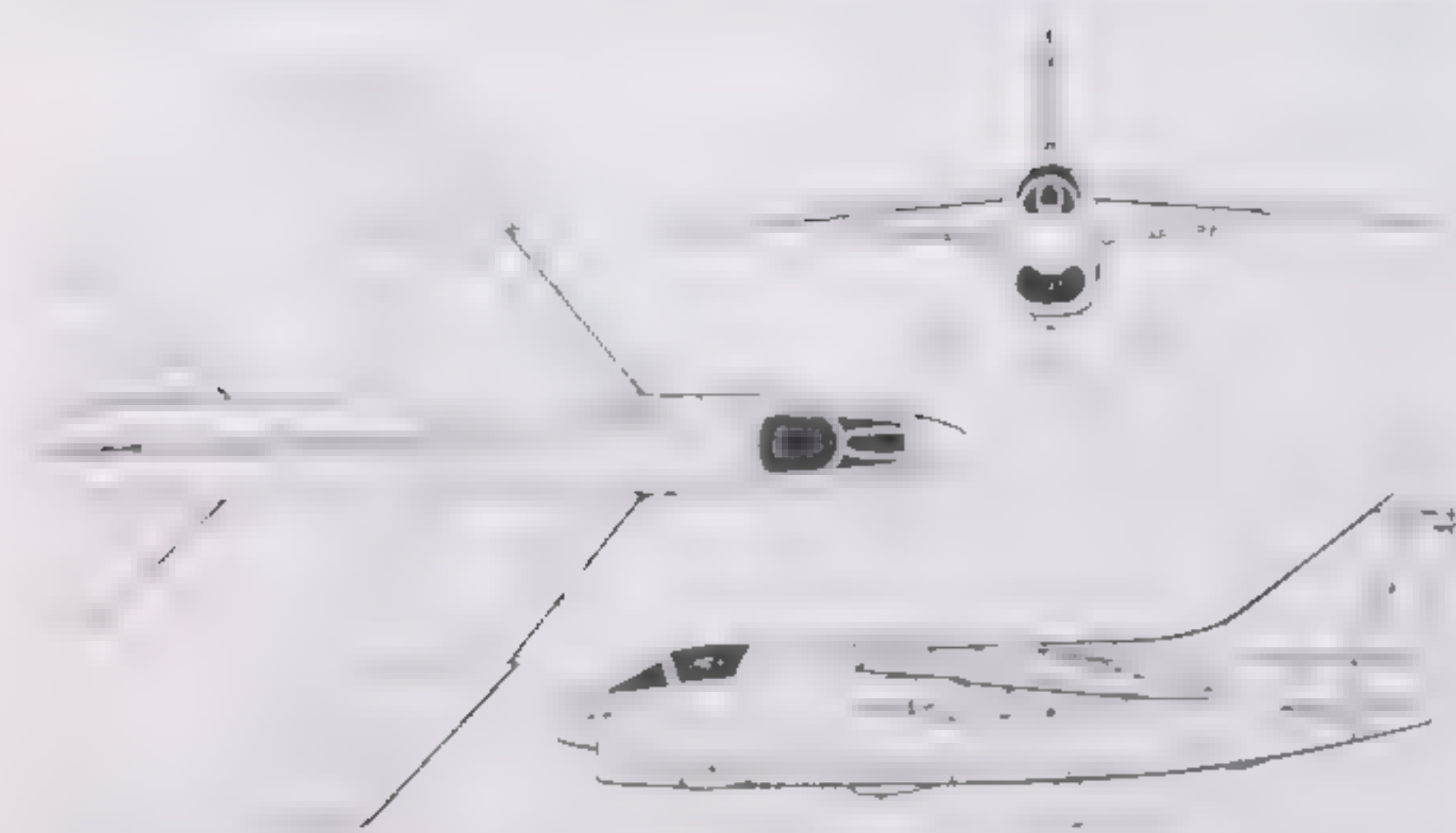
Origin: Vought Systems Division of LTV, Dallas, USA
Type: Single-seat attack bomber (carrier- or land-based), (TA) dual trainer
Engine: (A) one 11,350lb (5,150kg) thrust Pratt & Whitney TF30-6 two-shaft turbofan, (B, C) 12,200lb (5,534kg) TF30-8, (D) 14,250lb (6,465kg) Allison TF41-1 (Rolls-Royce Spey derivative) of same layout, (E) 15,000lb (6,804kg) TF41-2.
Dimensions: Span 38ft 9in (11.86m), length 48ft 1in (14.66m), (TA) 48ft 2in (14.68m), height 16ft 0in (4.88m), (TA) 16ft 5in (5.03m)
Weights: Empty, (A) 15,904lb (7,114kg), (D) 19,781lb (8,922kg), maximum loaded (A) 32,500lb (14,750kg), (D) 42,000lb (19,050kg)

Performance: Maximum speed (all single-seat versions, clean) 698mph (1,123km/h) at low level, climb and ferry rates limited by tactical radius with weapon load, typically 715 miles (1,150km), ferry range with four external tanks, typically 4,100 miles (6,600km)
Armament: (A, B) two 20mm Colt Mk 12 cannons, (C, D, E) two 20mm M61 Vulcan cannons on left side of fuselage with 1,000-round drum, external load up to theoretical 20,000lb (9,072kg)
History: First flight 27 September 1965, service delivery October 1966, first flight of D 26 September 1968
Users: Greece, Pakistan, USA (Air Force, Navy)

continued on page 106

Below: A Hughes AGM-65A (TV version of Maverick) blasts away from a Vought A-7D during weapon development tests at the USAF Armament Development and Test Center at Eglin AFB, Florida. Now, all TAC A-7 squadrons have Mavericks.





Above: Three view of the Vought A-7D (A-7E generally similar)



Above: Vought A-7A Corsair II of VA-195, US Navy (USS *Kitty Hawk*).

The 199 A-7A, used in action in the Gulf of Tonkin in 1967, followed by 196 B models. The C designation was used in 1966, retaining the TF30 engine. In 1966 the A-7D having the superior TF41 engine. Getting gun and more complete avionics for blind or automatic attack. The head up display and inertial navigation system. By late 1976 over 480 had been delivered with reduced weight. The Navy adopted the same model with an even TF41 and by late 1976 about 540 E models had been built. All over 1,400 within a decade Vought funded development of the YA-7H and is converting 81 B and C into the dual engine A-7H, similar to the D but without the ordnance carrying receptacle, at a price of \$259.2 million. The A-7H, on condition (it was reported) it did not buy a model offered by France.

Below: Four colourful A-7E attack aircraft of the US Navy look like fighters yet have more offensive punch (even discounting nuclear weapons) than twice as many four-engined bombers of World War II. The red-tailed machine comes from a unit embarked aboard USS *Coral Sea*, while the others are from the nuclear-powered super-carrier *Enterprise*.



Yakovlev Yak-36

Yak-36 "Forger A" and -36U (?) "Forger B"

Origin: The design bureau of Aleksander S. Yakovlev, Soviet Union
Type: Single-seat VTOL naval attack (and possibly, reconnaissance) aircraft ("Forger B") two-seat dual trainer

Engines: One turbojet engine (1,000 lb 771kg) mounted in the tail, one lift engine (5,600 lb 2540kg) mounted in the front of the fuselage. Estimated thrust of 5,600lb (2540kg) each

Dimensions (estimated) Span 25ft (7.6m), length (A) 49ft 3in (15.0m), (B) 58ft (17.7m), height 13ft 3in (4.0m), width with wings folded 14ft 10in (4.51m)

Weights (estimated) Empty 12,000lb (5450kg) (B slightly heavier), maximum loaded 22,050lb (10,000kg)

Performance (estimated) Maximum speed at sea level 722mph (1160 km/h, Mach 0.95), maximum level speed at optimum height 860mph (1380km/h, Mach 1.3), service ceiling about 50,000ft (15,250m). It is a high speed attack missile with an external fuel tank greater than 200 miles (320km)

Armament: Contrary to early reports there appears to be no internal gun. Four pylons under the non-folding wing centre section carry, in pairs, reconnaissance pods, ECM payloads, bombs, missiles (said to include AA-2 "Atoll" AAM and AS-7 "Kerry" ASM) and tanks. Maximum external load, about 4,000lb (1814kg) (B two-seater) none seen

History: First flight probably about 1971, service delivery, possibly, 1975

User: Soviet Union (AV-MF)

Development: At the 1967 show at Domodedovo a single V/STOL jet-lift research aircraft gave a convincing display of hovering and transitions. Called "Freehand" by NATO, it was at first thought to be the Yak-36, but this is now believed to be the service designation of the combat aircraft carried above *Kiev*, the first of the large Soviet carriers (officially classed as anti-submarine cruisers) which also carry ASW helicopters and an unprecedented array of shipboard weapons. The "Freehand" of which fewer than ten are thought to have been built, conducted trials from a specially built platform on the carrier *Moskva*. It provided information to assist the



Above: Three-view of single-seat "Forger A" (note wing fold)

design of the Yak-36. Although the aircraft is a single-seat, it is a high-speed attack aircraft. The 36" angled lift engine behind the cockpit is a turbojet engine. The lift engine is mounted in the front fuselage and the turbojet engine is mounted in the tail. The wings are rotated to about 100° to balance the rearward thrust of the lift jets. ST takeoffs are at the front of the carrier, allowing the aircraft to fly forward to release a nuclear warhead. The design is a high-speed attack aircraft, but the wings are folded to allow it to be stored on the carrier. Instead of the main engines, it is a turbojet engine. The latter has planform with a new design.

The aircraft is a high-speed attack aircraft. The supersonic speed at height is judged possible in the clean condition. Other features include Fowler flaps, large ailerons on the folding outer wing, wingtip and tail control nozzles, a ram inlet duct in the dorsal spine rear, a brake, a large vertical tail with deflection up and down, and a probably covering a small ranging radar. The "Forger" B has a completely different tandem-seat nose angled downwards and a lengthened rear fuselage to preserve directional stability. The development squadron aboard *Kiev* on her shakedown cruise from the Nikolayev yard to Murmansk flew intensively, and observers especially noted the repeated precision of take-offs and landings, indicating ship guidance. Even this aircraft is almost certainly an interim type.

Below: Single-seater making one of its characteristically precise recoveries aboard *Kiev*. A later V/STOL combat type is awaited. Foot of page: Tandem-seat (dual-control?) "Forger B".



Bombers

The bomber was the first type of military aircraft, though in November 1911 the bombs were carried in the cockpit and dropped by hand over the side. Since that time the bomber has come a very long way indeed, but it still does what is basically a trucking job, it still has to deliver accurately on target and it still has to penetrate hostile airspace. Unlike the attack aircraft it can use good paved airfields, but as its mission is over long distances against enemy heartlands it is likely to run the gauntlet of formidable defence systems. The fact that both the defence system and the means by which the bomber penetrates it are electronic was firmly established by 1943, but since then there have been major upheavals. Just on the score of basic aircraft performance, the products of one manufacturer show the extent of the change: in 1944 a typical B-24 would bomb from 28,000ft at 220mph; in 1950 a B-36D might bomb from 42,000ft at 405mph; in 1959 a B-58 could bomb from 64,000ft at 1,385mph; and the last B-58 went to the "boneyard" long ago.

While jet propulsion lifted the bomber to new heights and speeds, nuclear bombs enabled single aircraft to flatten the largest cities. Early thermonuclear bombs had a diameter of nine feet and length of 20ft, so the strategic bomber automatically became a large aircraft capable of carrying a considerable load of conventional bombs. Predictably

the technology of nuclear weapons moved at break-neck speed during the 1950s, so that by 1957 fission warheads fitted 28cm artillery shells and into small bombs carried by such aircraft as the A-4 Skyhawk, considerably smaller than contemporary fighters. Today even devastating thermonuclear devices weigh only a few hundred pounds. They can be packaged into MIRVs—multiple independently targeted re-entry vehicles—carried in the tips of ICBMs and submarine-launched missiles. Perfection of the intercontinental missile led some observers to jump to the wholly erroneous conclusion that the bomber was "an anachronism in the missile age". The full wealth of tasks a modern bomber can accomplish are breathtaking, but unfortunately classified. But a few are obvious. Its targets need not be fixed in precisely known locations. It can be retargeted instantly. It can be recalled, and so can be launched, to the safety of the sky, upon mere warning of an enemy attack. A nation with missiles but no bombers dare not fire its missiles upon a mere warning; it must wait until the enemy attack has struck home, possibly destroying all its missiles, before daring to retaliate (and then it is too late).

Today's bombers are a select bunch. The B-28 is not included because, though several air forces still use it in that role, modern airspace would not take it seriously. The B-52, the next oldest type, and



the Vulcan, inevitably find life hard. They have radar cross-sections like a barn door, and their big airframes do not take kindly to full-throttle operation at extremely low altitudes in dense air, trying to keep under the defending radar coverage. It is a measure of progress in aeronautical technology that the replacement for the Vulcan is the Tornado, which looks like a fighter (in fact in one version it is a fighter). Can the Tornado really carry the loads of a Vulcan and fly the missions of a Vulcan? The answer should be self-evident, though to be fair Britain's geographical posture is not quite what it was. Switching from the Vulcan to the Tornado does not imply one iota of degradation in mission capability—quite the reverse, though one sympathises with the redundant crew-members — and capabilities of

bombers are again being dramatically improved by the ALCM (air-launched cruise missile) which combines the small size and hard hitting power of the SRAM (short-range attack missile) with ranges of around 2,000 miles.

Such weapons take just a little of the pressure off the carrier vehicle. Shortsightedly the US Administration has used this to justify termination of the B-1 production programme. This appears to be a decision of far-reaching and potentially catastrophic consequences because the Soviet Union is supplementing its terrifying and rapidly growing superiority in ballistic missiles with an extremely versatile strategic bomber, which we so far know only as Backfire. The more one-sided strategic arms become, the greater the danger to peace.

A beautiful study of the first USAF B-1, turning gently with wings in the swept position. In a halcyon world, Americans find the need for the B-1 less obvious than its cost. Meanwhile its opposite number in the USSR rolls off the assembly lines.



Boeing B-52 Stratofortress

B-52 to B-52H

Origin
Type
Engines

Dimensions

Weights

Performance

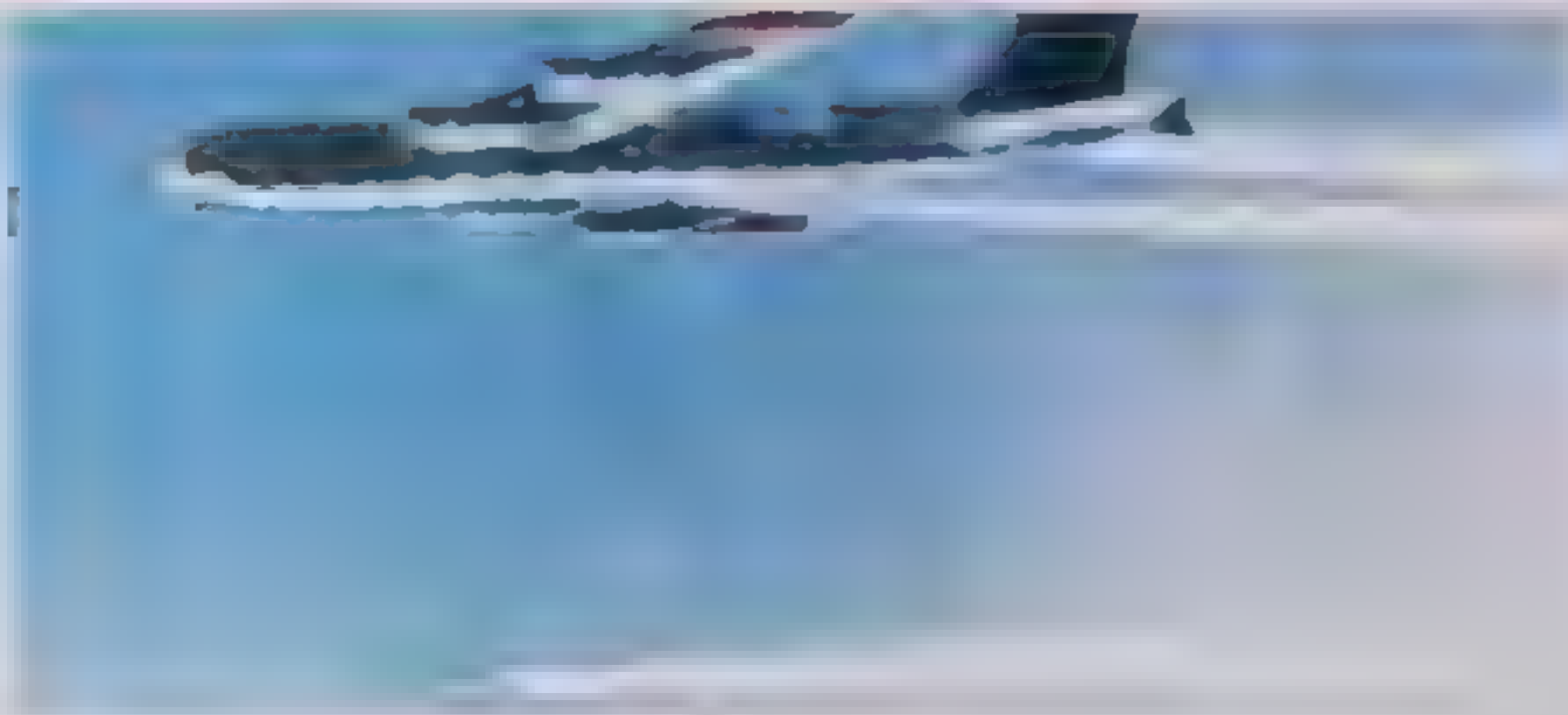
Armament: Remotely directed tail mounting for four 0.50in (B-52H)

History

User

Development

Below: Boeing B-52G showing current anti-flash colour scheme and EVS blisters



Top: First launch of powered AGM-86 (ALCM) from NB-52G in May 1976.

Above: Launch of AGM-69A SRAM from B-52H.

Below: Nose of B-52H showing ASQ-151 EVS blisters.



Above: Three-view of B-52H (without SRAMs or EVS)



Above: A black B-52D BUFF (Big Ugly Fat Fella) thunders aloft from the switchback runway at Guam for distant North Vietnam.

Below: A B-52F, a version no longer in service, rains down 750lb bombs under Loran guidance—perhaps on empty forest.



Dassault Breguet Mirage IVA

Mirage IVA

Origin
Type
Engines

Dimensions

Weights Performance



Above Three-view of Mirage IVA showing bomb but not drop tanks.

Armament: No defensive armament other than ECM, one 60 kiloton freefall bomb, or alternatively up to 16 000lb of air-to-air missiles per wings and fuselage.

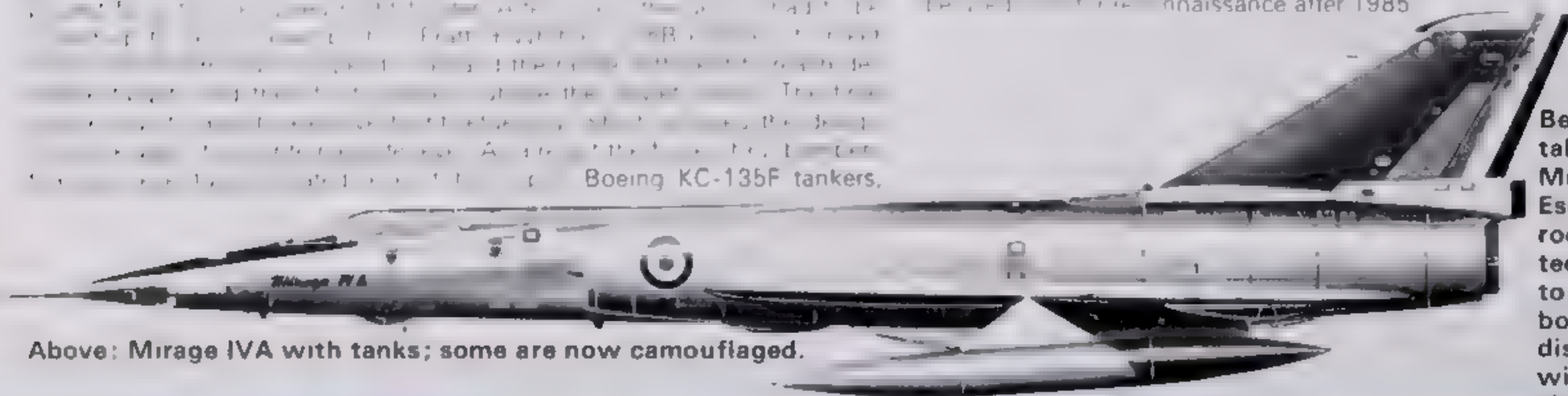
History: First flight (Mirage IV-001) 17 June 1959, (production IVA)
First delivery 1964, final delivery March 1968

User: France (Armée de l'Air CFAS)

Development: When the French government decided in 1954 to create a

This involved a long mission flown at high speed. After studying the requirements of the Vautour — a type used to form the nucleus of the force — Dassault began work on a bomber derived from a 1956

with booms fitted for probe/drogue refuelling, and also upon the "buddy technique" whereby aircraft would fly a mission in pairs, one carrying a bomb and the other spare fuel and a refuelling hose reel for transfer to its partner. Even so, the initial planning of the Commandement des Forces Aériennes Stratégiques presupposed that most missions would be one-way (or at least would not return to France). Dispersal has been maximised, with the force divided into three Escadries (91 at Mont de Marsan, 93 at Istres and 94 at Avord), which in turn are subdivided into three four-aircraft groups, two of which are always dispersed away from Escadrille HQ. Cold-weather capable and "hot" aircraft: the IVA has also been rocket blasted out of short unpaved strips hardened by quick-setting chemicals sprayed on the soil. Of the original total of 62 about 51 remain operational, 36 being at readiness as noted above and the remainder in reserve. They will be replaced by the 1980s, and their knowledge after 1985.



Above: Mirage IVA with tanks; some are now camouflaged.

Below: Practice takeoff by a Mirage IVA of 93e Escadre using ATO rockets. This technique is used to allow the bombers to be dispersed to otherwise unusable short airfields.



Hawker Siddeley Vulcan

Vulcan B.1, 1A, 2 and SR.2

Origin: A V Roe Ltd (now British Aerospace)
Type: Five-seat bomber (SR 2, strategic reconnaissance)
Engines: Four Rolls-Royce (originally Bristol the Bristol Siddeley Olympus two-shaft turbojets for details see text)
Dimensions: Span (1) 99ft (30.18m), (2) 111ft (33.83m) length (1) 97ft 1in (29.6m), (2) 105ft 6in (32.15m) (99ft 11in with probe removed) height, (1) 26ft 1in (7.94m), (2) 27ft 2in (8.26m)
Weights: Not disclosed, loaded weights probably about 170,000lb for B 1A and 250,000 lb for B 2 and SR 2
Performance: Maximum speed (1) about 620mph, (2) about 640mph (1030km/h) at height (Mach 0.97), service ceiling (1) about 55,000ft (2) about 65,000ft (19,810m) range with bomb load (1) about 3,000 miles (2) about 4,600 miles (7400km)
Armament: Internal weapon bay for conventional (21 1,000lb bombs) or nuclear bombs, (SR 2) none
History: First flight (Avro 698) 30 August 1952 (production B 1) 4 February 1955 (prototype B 2) 31 August 1957 (production B 2) 30 August 1958, final delivery 14 January 1965
User: UK

Development: Few aircraft have ever created such an impression as did the prototype Avro 698. This almost perfect triangle, painted white, demonstrated fighter-like manoeuvrability at very low levels within ten hours of first flight, despite its great size and low-powered engines (6,500lb Avons). In 1953 the second prototype was repeatedly rolled at low level. From the start the Vulcan was a winner and with a modified wing with kinked and cambered leading edge the B 1 went into service with the RAF in February 1957, powered by the 11,000lb Olympus 101. Altogether 45 Mk 1s and 1As were built, re-engined by the 12,000lb Olympus 102 and 13,500lb Olympus 104 and, in the 1A, having an extended and bulged rear fuselage full of countermeasures. Some B 1As carried refuelling probes fixed above the



Above: Three-view of Vulcan B.2 (SR.2 is similar outwardly).

huge bombing radar in the nose, all were painted in anti-radiation white. The B 2 was designed to have much better performance at great heights and matched the 17,000lb Olympus 201 with a new thinner wing of greater span and area with elevon controls. Most were equipped to carry the Blue Steel stand-off missile, but by 1966 the force of about 50 had been painted in green/grey camouflage, re-engined with the 20,000lb Olympus 301 and deployed in the tactical low-level role using gravity bombs with terrain following radar. No 27 Sqn is equipped with the SR 2 multi-sensor strategic reconnaissance conversion.



Above: Vulcan B.2 of 101 Sqn, RAF Coningsby (original three-colour markings).

Below: A Vulcan B.2 of 617 "Dam Busters" squadron deploys to Cyprus high above the cloud layer. In an operational mission this aircraft would penetrate hostile territory right "down on the deck". The Vulcan remains a fully competitive aircraft, as does its equally venerable partner the B-52.



Tupolev Tu-26 "Backfire"

Two main versions with NATO code names "Backfire-A" and "-B"

Origin
Type: heavy strategic bomber and missile platform with probable crew of three
Engines: Two afterburning turbofans, probably Kuznetsov NK-144 two-

Dimensions

Weights

Performance: (Estimated "Backfire B") maximum speed at altitude

Armament

History: First flight ("Backfire A" prototype) not later than 1969. ("Backfire B" prototype) 1973, entry to service, probably 1974
User: Soviet Union (ADD, possibly AV MF)

Development: Owing to the obvious inability of the Tu-22 to fly strategic

... of the gross wing area was fixed, just the outer ...
... is improved in other ways, though the details are still largely ...
... for conjecture. The large engines are fed through wide inlet ducts ...
... bably pass above the wing, a flight refuelling probe is fitted above ...
the nose, but even without this "Backfire-B" has an endurance of some ten ...
The Chairman of the US Joint Chiefs of Staff said in 1974 "It is ...
... ted to replace some of both the current medium and heavy bombers ...

Above: Three view of (Tu-26?) "Backfire B" (provisional)

... sparse, not even its correct Soviet bureau or service designations ha ...
surfaced publicly. It is believed to have a crew of three, a very large rad ...
extremely sophisticated ECM/EW fits. The internal bomb load ...
... (B-1 carries 6.5 times as much) and ...
the Russian insistence on having a radar-directed tail cannon is believed t ...
have prevailed over those who think such installations a waste of cost and ...
... the double-taper outer wings pivoted no ...
... ft from the centreline, another puzzle is where the landing ...
... estimated at ...
five per month, with 95 in service by the end of 1976 out of a planned tota ...

Below: Specially drawn for this book. "Backfire B" carrying AS-6, a missile with pinpoint accuracy (often only one is carried).

Below: This long-range telephoto shows a B-model climbing to altitude. Some observers calculate that much of the fuselage must be empty (for c.g. reasons), leading to lower weights and ranges.



Rockwell International B-1

B-1

Origin: Rockwell International Corp, USA

Type: Four seat strategic bomber and missile platform

Engines: Four 30 000lb (13 610kg) class General Electric F101 turbofan engines with augmented turbofans

Dimensions: Span (15°) 136ft 8½ in (41.7m), length 150ft 2½ in (45.8m), height 33ft 7 in (10.2m)

Weights: Empty, about 140 000lb (63 500kg), maximum loaded with 5000lb (2270kg) (179,170kg), (gross weight likely to settle at about 400 000lb)

Performance: Maximum speed at sea level 3000 mph (4826 km/h), maximum speed at high altitude (prototypes) 1,320mph (2135km/h, Mach 2.0), service ceiling over 60,000ft (18,300m), range with maximum weapon load, 6100 miles (9820km)

Armament: 24 Improved SRAM (AGM-69B thermonuclear missiles) internally, with provision for eight more externally, alternatively the same number of AGM 86A ALCM (air launched cruise missiles) carried on the same mounts (75 000lb (34 020kg) of free fall bombs or terrain attack bombs (40 000lb (18 144kg) externally)

History: First flight, 5 December 1974, first test flight 1977

User: Not yet announced officially

Development

The B-1 is a four-engine, variable-sweep, low-altitude, high-speed, long-range, multi-mission, multi-seat strategic bomber. It is designed to penetrate enemy air defenses and deliver a variety of weapons, including nuclear and conventional bombs, missiles, and cluster munitions. The aircraft is also capable of carrying a large number of air-to-air missiles. The B-1 is a highly advanced aircraft, and its development has been a major challenge for Rockwell International. The aircraft is designed to be able to operate in a variety of environments, and it is capable of carrying a large number of weapons. The B-1 is a highly advanced aircraft, and its development has been a major challenge for Rockwell International.

extreme parsimony. Only a single prototype was carrying the full complement of weapons, and the programme for the first flight was delayed by a number of years. The aircraft is a highly advanced aircraft, and its development has been a major challenge for Rockwell International. The aircraft is designed to be able to operate in a variety of environments, and it is capable of carrying a large number of weapons. The B-1 is a highly advanced aircraft, and its development has been a major challenge for Rockwell International.

continued on page 116

Below: Transit of the California badlands at medium altitude by the first B-1, which flew in 1974. Two more have flown (three by late 1978) and flight development is proceeding successfully despite President Carter's cancellation of the programme



...the aircraft. These... radars... confusing and... Another... feature is... an air... under... and... head at immed... for long periods so that on an alert beam... the first man... each the aircraft has only to hit a button behind the... landing gear to energise all engines and systems for immediate take-off... before the crew are completely strapped in. Yet another design feature is... a titude ride

(LARC) effected by gust sensors and operated by small canard... side of the nose and the lowest section of rudder, to guide the speeding... through turbulent air without the crew compartment suffering significant undulation. So vital is the need for the B-1 that even an economy-minded Congress, in the midst of a Presidential election, voted to... funds for production in November 1976, but President Carter... production plans in 1977. The USAF had hoped to buy at least... B-1 force to form the third leg of the strategic deterrent Triad. The requirement is for 241 aircraft, at an estimated unit price of \$77 million (including for R&D and inflation) amounting to a total investment of \$ 8,600,000.



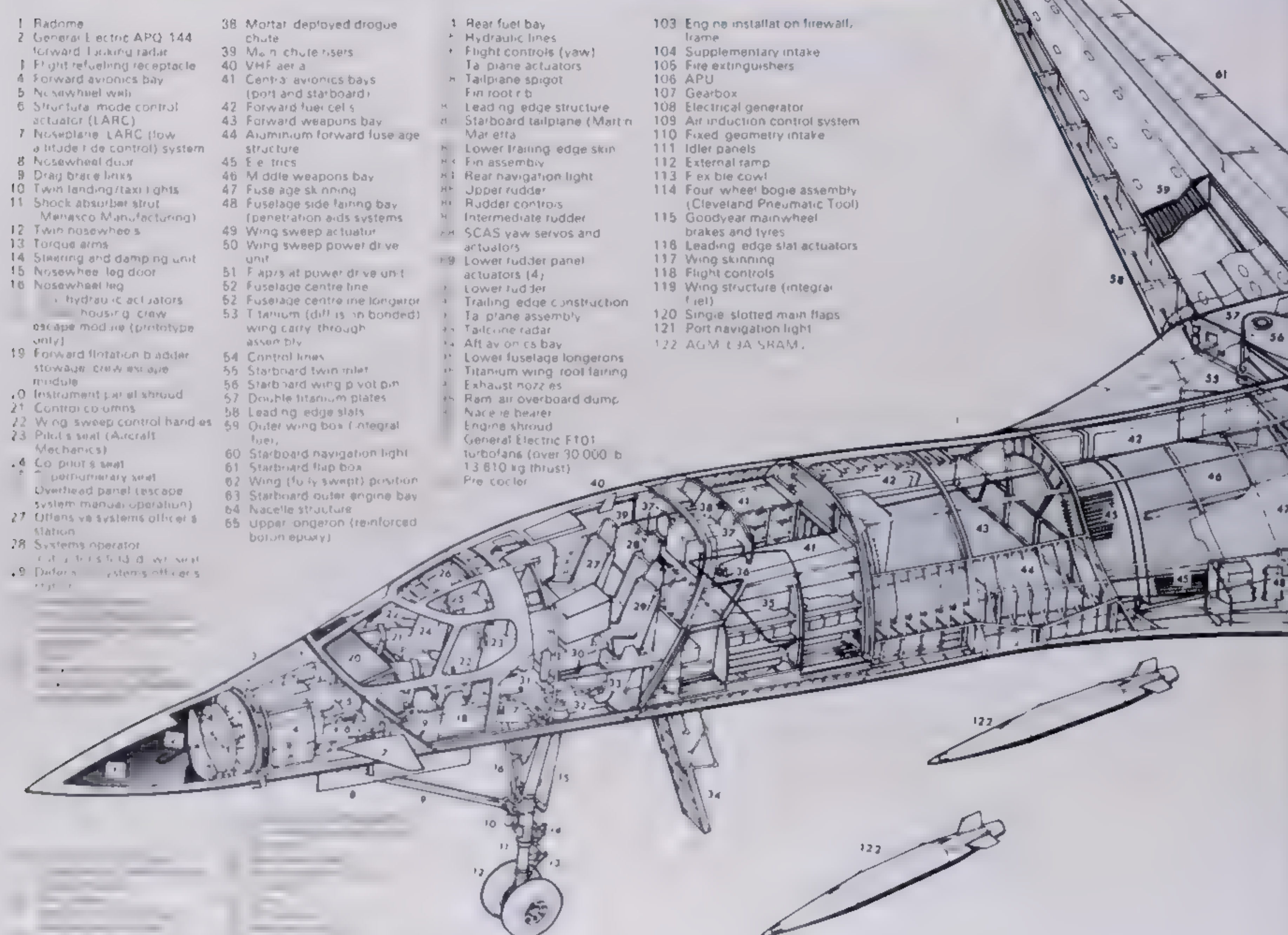
Above: Three-view of B-1 prototypes (with crew-escape pod) showing range of wing sweep.



Above: The third B-1 prototype flies with its T-38 chase-plane over the flat primeval lake beds near Palmdale. This aircraft has been flying since April 1976 and is also pictured opposite.



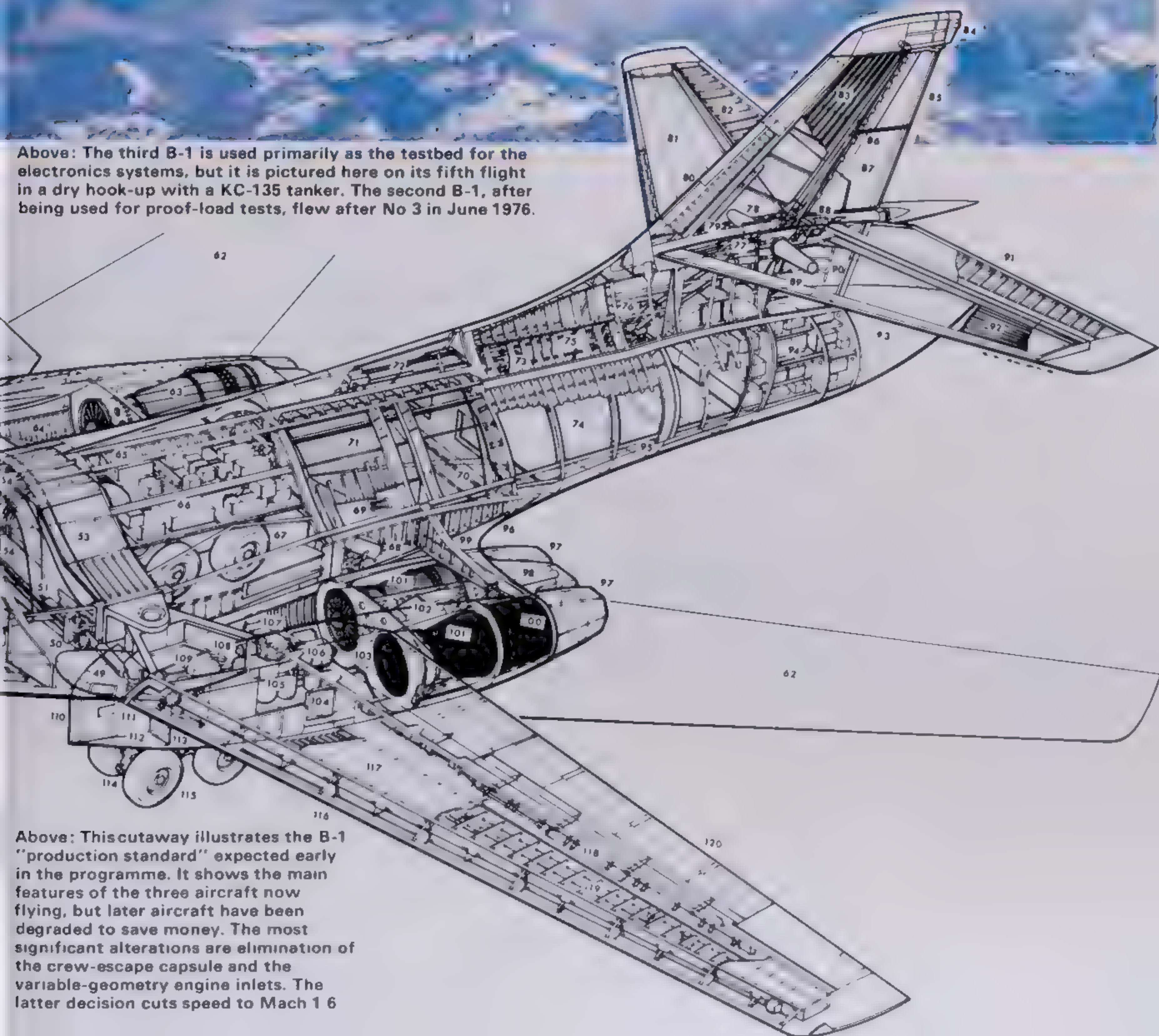
Above: B-1 first prototype in first-flight configuration.



- 1 Radome
- 2 General Electric APQ 144 forward looking radar
- 3 Flight refueling receptacle
- 4 Forward avionics bay
- 5 Nosewheel well
- 6 Structure mode control actuator (LARC)
- 7 Noseplane LARC (low altitude ride control) system
- 8 Nosewheel door
- 9 Drag brace links
- 10 Twin landing/taxi lights
- 11 Shock absorber strut (Menasco Manufacturing)
- 12 Twin nosewheels
- 13 Torque arms
- 14 Steering and damping unit
- 15 Nosewheel leg door
- 16 Nosewheel leg
 - hydraulic actuators
 - housing crew escape module (prototype only)
- 19 Forward flotation bladder stowage (crew escape module)
- 20 Instrument panel shroud
- 21 Control columns
- 22 Wing sweep control handles
- 23 Pilot's seat (Aircraft Mechanics)
- 24 Co-pilot's seat
- 25 Parachutery seat
- 26 Overhead panel (escape system manual operation)
- 27 Offensive systems officer's station
- 28 Systems operator
- 29 Defensive systems officer's station
- 38 Mortar deployed drogue chute
- 39 Main chute risers
- 40 VHF antenna
- 41 Central avionics bays (port and starboard)
- 42 Forward fuel cells
- 43 Forward weapons bay
- 44 Aluminium forward fuselage structure
- 45 Electronics
- 46 Middle weapons bay
- 47 Fuselage skinning
- 48 Fuselage side fairing bay (penetration aids systems)
- 49 Wing sweep actuator
- 50 Wing sweep power drive unit
- 51 Fap/s at power drive unit
- 52 Fuselage centre line
- 53 Titanium (diff is in bonded) wing carry through assembly
- 54 Control lines
- 55 Starboard twin inlet
- 56 Starboard wing pivot pin
- 57 Double titanium plates
- 58 Leading edge slats
- 59 Outer wing box (integral fuel)
- 60 Starboard navigation light
- 61 Starboard flap box
- 62 Wing (fully swept) position
- 63 Starboard outer engine bay
- 64 Nacelle structure
- 65 Upper longeron (reinforced boron epoxy)
- 66 Rear fuel bay
- 67 Hydraulic lines
- 68 Flight controls (yaw)
- 69 Tailplane actuators
- 70 Tailplane spigot
- 71 Fin root rib
- 72 Leading edge structure
- 73 Starboard tailplane (Martin Maritta)
- 74 Lower trailing edge skin
- 75 Fin assembly
- 76 Rear navigation light
- 77 Upper rudder
- 78 Rudder controls
- 79 Intermediate rudder
- 80 SCAS yaw servos and actuators
- 81 Lower rudder panel actuators (4)
- 82 Lower rudder
- 83 Trailing edge construction
- 84 Tailplane assembly
- 85 Tailcone radar
- 86 Aft avionics bay
- 87 Lower fuselage longerons
- 88 Titanium wing root fairing
- 89 Exhaust nozzles
- 90 Ram air overboard dump
- 91 Nacelle bearer
- 92 Engine shroud
- 93 General Electric F101 turbofans (over 30,000 lb 13,610 kg thrust)
- 94 Pre-cooler
- 103 Engine installation firewall, frame
- 104 Supplementary intake
- 105 Fire extinguishers
- 106 APU
- 107 Gearbox
- 108 Electrical generator
- 109 Air induction control system
- 110 Fixed geometry intake
- 111 Idler panels
- 112 External ramp
- 113 Flexible cowl
- 114 Four wheel bogie assembly (Cleveland Pneumatic Tool)
- 115 Goodyear mainwheel brakes and tyres
- 116 Leading edge slat actuators
- 117 Wing skinning
- 118 Flight controls
- 119 Wing structure (integral fuel)
- 120 Single slotted main flaps
- 121 Port navigation light
- 122 AGM-129 ASRAAM



Above: The third B-1 is used primarily as the testbed for the electronics systems, but it is pictured here on its fifth flight in a dry hook-up with a KC-135 tanker. The second B-1, after being used for proof-load tests, flew after No 3 in June 1976.



Above: This cutaway illustrates the B-1 "production standard" expected early in the programme. It shows the main features of the three aircraft now flying, but later aircraft have been degraded to save money. The most significant alterations are elimination of the crew-escape capsule and the variable-geometry engine inlets. The latter decision cuts speed to Mach 1.6

Reconnaissance aircraft

Reconnaissance was the only task which senior officers of armies and navies before World War I thought aeroplanes might be able to perform. When pioneer military aviators set about doing it they found the task difficult. With no opposition, and working in broad daylight, they mistook wet patches on roads for enemy transport columns, shadows of trees for marching men, and farm workers for troops advancing across fields. Even when there was no error the message had to be written on a piece of paper and dropped accurately over the right headquarters, where someone had to retrieve and correctly read it. Aerial photography had been experimented with as early as 1912, but it was not until 1916 that the art made much progress. But in World War II, thanks to the spur of one civilian—Leslie Irvin, who flew his camera-equipped Lockheed over the German fleet to show what could be done—the art and science of photo-reconnaissance made giant strides. Cameras grew larger aperture lenses and longer focal lengths, film emulsions grew faster and, in particular, much finer-grained (to stand enormous enlargement without completely losing definition), and oblique cameras were fitted to low-level fighters which could take pin-sharp pictures whilst racing across sensitive targets at full throttle.

Today we still fit batteries of even more impressive optical cameras into large bombers and into specially designed noses of fighters; but mostly it is done differently. More important still, optical photography is just one of the tools of the reconnaissance trade. Another is infra-red (IR). Linescan IR sensors can take a “picture” on which cooler areas look dark and warmer areas look light, with fine definition almost comparable with optical film. The resulting pictures show a camouflaged truck as a camouflaged truck, and they can show buried pipelines, hidden troops and countless other things, such as which aircraft on an airfield is running its engines. If a truck was parked near a wood, and driven under cover as the aircraft approached, the IR film will show the slightly warmed patch where it was parked. But IR is still just another wavelength of the electromagnetic spectrum. Some of the most important wavelengths are much longer, in radio frequencies. One of the biggest reconnaissance tasks is ESM (electronic surveillance measures),

which means detecting, measuring and analysing all radio signals and emissions in hostile territory, or emanating from ships or aircraft. The first ESM mission was courageously flown by a Wellington of the RAF in 1943, which deliberately lured a radar-equipped Ju 88 night fighter near enough for its radar signals to be linked with positive visual identification (shot to ribbons, the Wellington ditched near to Britain, with every member of its crew wounded).

Today ESM seldom leads to shots being fired, because it is all part of the ceaseless snooping war that goes on between East and West, which both sides play to the limit—sometimes, waving to each other, from cockpit, or the bridge of a so-called “trawler”, to show there are no hard feelings. Of course, if anyone violates foreign territory they do invite a shot, and not merely a warning one. In 1953–65 a sobering number of American aircraft, most of them “honest” and bearing the markings of the Air Force or Navy, were shot down or just disappeared whilst engaged in ESM and other kinds of reconnaissance in the course of which they accidentally or deliberately violated the Soviet frontier. In parallel, Lockheed built the unique U-2 for “dishonest” missions, flown by civilians with unmarked aircraft, which accomplished much at the price of highly-publicized losses over the Soviet Union, Cuba and China. Since the early 1960s “overflights” have increasingly been the duty of drones (RPVs, remotely piloted vehicles) and satellites.

Tactical reconnaissance, in association with a land campaign and involving relatively short distances, is another matter. The US Army still has a unique aircraft, the OV-1, built specially for this purpose. Lockheed developed an even stranger family of extremely quiet reconnaissance platforms intended to fly low overhead without being heard, though the aeroplane transparent to radar has not yet been invented. Most important of all, modern combat aircraft can clip on multi-sensor reconnaissance pods which can in a few minutes transform them into gatherers of intelligence of a kind and at a rate that would amaze any of the low-level reconnaissance pilots of World War II. If a fighter were to roar overhead as I write this he might well report my make of typewriter—and whether it had any electrical fault.

Almost everything about the USAF's SR-71A “Blackbird” is special. To fly 2,100 mph at 80,000 feet needs special aircrew, special flight suits and a rather fantastic flying machine in which to sit. Just how vulnerable these costly wonders are is arguable



Grumman OV-1 Mohawk

OV-1A to -1D, EV-1, JOV, RV

Origin
Type

Engines: Two 1 005shp Lycoming 1

Dimensions: Span (A-C) 42ft (12.8m) (B-D) 48ft (14.6m)

Weights: Empty (-1A) 9.937lb (4507kg) (-1B) 11.00

Performance: Max. m.p. speed (a/d) 297-310mph (480-500)
 0-60 (a/d) 11.1-11.8sec (1/4) 2.450ft (400m); m.p. (1/8) 2.250ft (716)

• 449mi range with external fuel (1 + 7)

Armament: Not normally fitted, but in South East Asia the 1A, -1B and -1C all operated with a wide variety of air to ground weapons including

History: First flight (YOV-1A) 14 April 1959 service delivery February 1961 final delivery December 19

Users: Israel (EV) US A

Development: Representing a unique class of military aircraft, the OV-10 Hawk is a specially designed battlefield surveillance machine with characteristics roughly midway between light planes and jet fighters. One of its requirements was to operate from rough forward airstrips and it has a STOL (short takeoff and landing) qualities and good low-speed control with full span slats and triple fins and rudders. Pilot and observer sit in side by side Martin Baker J5 seats and all versions have extremely good all round view and very comprehensive navigation and communications equipment. All versions carry cameras and upward firing flares for night photography. Most variants carry UAS 4 infra-red surveillance equipment and the -1B carries APS 94 SLAR (side looking airborne radar) in a long pod under the right side of the fuselage with automatic film processing giving within seconds of exposure a permanent film record of radar image on either side of the flight path. The -1D combined the functions of the two previous versions in being quickly convertible to either IR or SLAR missions. Underwing pylons can carry 150 US gal drop tanks, ECM (electronic countermeasures) pods, flare/chaff dispensers, or in the OV-10A such weapons as FFAR pods, 0.50in gun pods or 500lb (227kg) bombs - though a 1965 Department of Defense rule forbids the US Army to arm its fixed-wing aircraft! The EV-1 is the OV-1B converted to electronic surveillance with an ALQ 133 target locator system in centreline and tip



Above: In untypical maritime surroundings, the OV-1D was the final production Mohawk, with the APS-94 side-looking radar in a long external pod.



Above: Three-view of OV-1D as originally built.

Pods The RV-1C and -1D are conversions of the OV-1C and -1D for permanent use in the electronic reconnaissance role. Total production of the Mohawk was 375.

Lockheed QT-2/X-26 YO-3

QT-2, X-26B, Q-Star and YO-3A

Origin: Lockheed Missiles and Space Company, San Jose, California, USA
Type: Quiet reconnaissance

Engine: (QT-2, X-26B) one 100hp Teledyne Continental O 200A flat-four (Q-Star) originally one O-200A, then 185hp Curtiss Wright W-660 RC2 60 rotating combustion engine (YO-3A) one 210hp Teledyne Continental IO-360D flat-six piston engine

Dimensions: Span 57ft 1in (17.41m), length (QT-2 X-26B) 30ft 10in (9.40m), (Q Star) 31ft, (YO 3A) 30ft 9 14m), height, not disclosed

Weights: Empty (QT-2) 1 576lb (715kg). (Q Star) 1 969lb (894kg)
 (YO-3A) about 2 200lb (1000kg) gross (QT-2) 2.182lb (989kg) (Q Star)
 2 860lb (1297kg). (YO-3A) 3 200 lb (1452kg)

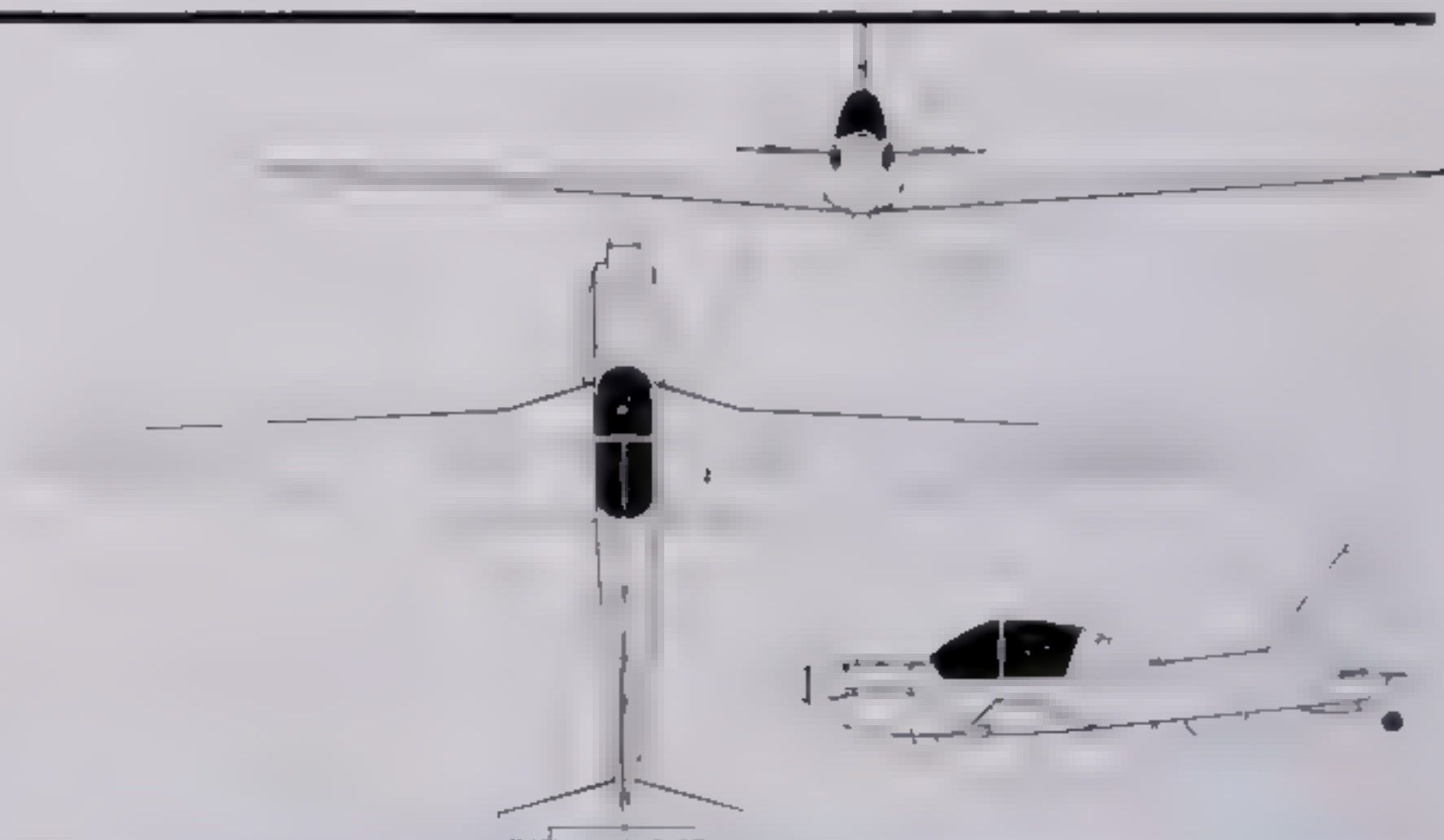
Performance: Operating speed range (a) 50 to 120kt (58-138mph 93-222km/h), quietest speed around 71mph (114km/h), endurance QT-2) 4hr (Q Star YO 3A) 6hr

Armament: None

History: First flight (QT 2) July 1967 (Q Start) June 1968 (YO 3A) May 1969 combat duty (Q 2PC) January 1968 (Y)

Users: (QT 2PC) US Army, (X 2nB) US Navy

Development is the impossibility of catching the elusive Viet Cong.



Above: Lockheed YO-3A, with sensors



Above: The second YO-3 on an early test flight, before it was fitted with sensors for deployment to Vietnam.

...n, still based on the SGS 2-32 but with extended inner chord, new fuse age with nose engine very and various sensors including radar and ...e said to make a gentle rustling ...ft they are almost impossible to detect.

Lockheed U-2

U-2A, C, D, R, CT and EPX, WU-2 and HASP U-2

Origin: Lockheed California Co, USA
Type: High-altitude photo reconnaissance, multi-sensor reconnaissance and special reconnaissance (CT) dual-engine turbojet powered, high-altitude, long-endurance, low-observable aircraft. HASP U-2 is a special reconnaissance aircraft.

Engine: U-2A) one 11,200lb (5080kg) Pratt & Whitney JT4D-10 turbojet, later replaced by two shaft turbojet. (all other models have two shaft turbojet. Pratt & Whitney JT4D-10 two shaft turbojet.

Dimensions: Span (except R) 80ft (24.38m), (U-2R) 113ft (34.44m), length (except R) 49ft 7in (15.1m), (U-2R) 63ft 19in (19.44m), height (except R) 13ft (3.96m), (U-2R) 16ft 4in (4.93m).

Weights: Empty (A) 9,920lb (4500kg), (other models except R) 11,700lb (5305kg), (U-2R) 14,990lb (6800kg), (U-2A) 14,600lb (6713kg), (others, except R, clean) 15,850lb (7190kg), (with two external wing tanks) 17,270lb (7833kg), maximum over 21,000lb (9526kg), (R) 29,000lb (13,154kg).

Performance: Maximum speed (A) 494mph (800km/h), service ceiling (A) 70,000ft (21,340m), (other models) 25,910m, maximum range (A) 2,600 miles (4,185km), (others) 4,000 miles (6,437km).

Armament: None

History: First flight 1 August 1955, service delivery, early 1956, final delivery of new aircraft Jul 1958.

Users: Taiwan, USA (Air Force, NASA).



Above: Three-view of original Lockheed U-2A.

Development: No aircraft in history has a record resembling the U-2. It was in operational use from Lakenheath (England) and Wiesbaden (Germany) in 1956, attracting the attention of spotters and amateur photographers who commented on its graceful glider-like appearance and on the odd fact that at take-off it jettisoned the small outrigger wheels under the outer wings, returning to land on small centreline wheels and the down-turned wingtips. When interest had reached fever-pitch the government blandly announced that the U-2, as it was called (a Utility designation), was used by the NACA (the National Advisory Committee for Aeronautics) for atmospheric research at heights up to 55,000ft. When on 1 May 1960 a U-2 was shot down over the Soviet Union the public were frantically kept away at gun-point and the incident was regarded as serious. In fact the U-2 had been doing the most clandestine reconnaissance over the territory of any nation. Aircraft were delivered to Watertown Strip, a remote airfield in Nevada, where CIA pilots converted and prepared for operational missions over Communist territory. Their aircraft bore no markings and operated – immune to interception and often undetected – at far above the announced 55,000ft. Other U-2s were assigned to the 4080th Strategic Wing and carried USAF markings. Most were of the more powerful U-2B type with much greater height and range. On 1 May 1960 a U-2B flown by a CIA pilot took off from Peshawar (Pakistan) to fly across the Soviet Union to Bodø (Norway), but was shot down over Sverdlovsk, presenting the Russians with unprecedented



Above: A black-painted U-2B of the US Air Force photographed over Edwards Air Force Base in 1968. Most USAF U-2 aircraft have since been rebuilt or modified as different models.



Above: This U-2D belongs to the 6512th Test Sqn at the Air Force Flight Test Centre at Edwards AFB.

material for a diplomatic incident. Later U-2s were shot down over China and Cuba, and the survivors were assigned to lawful missions involving many kinds of surveillance and sampling. The C and two-seat D make up the bulk of the fleet used today, numbering about 40 out of the 55 built. All have been extensively rebuilt, none more so than the enlarged U-2R special-reconnaissance platform which carries four times the 3,000lb (1,361kg) payload of other models. The D was the original tandem seat version, needing a second crew-member, the U-2CT dual trainer is a complete rebuild with instructor pilot seated high up behind the pupil (the U-2 needs extreme pilot skill near and on the ground), the C has a 'doghouse' housing avionics and equipment. The proposed EPX for the Navy carried APS-116 ocean surveillance radar (S-3A type) but never entered service.

Below: One of the trickiest aircraft in the world to taxi, takeoff or land, the U-2CT conversion trainer. This one with trestle supporting the wing, is 56-6692 of 100th SRW, 349th SRS at Davis-Monthan AFB.



Lockheed SR-71

A-11, YF-12A and C, SR-71A, B and C
(data for SR-71A)

Origin
Type
Engines: Two 32 500 b (14 740kg) thrust Pratt & Whitney J58 (JT)

Dimensions: Span 55ft 7in (16 95m) Length 107ft 5in (32 74m) H

Weights: Empty (typical) 60 000 b (27 215kg) • loaded 170 000 b

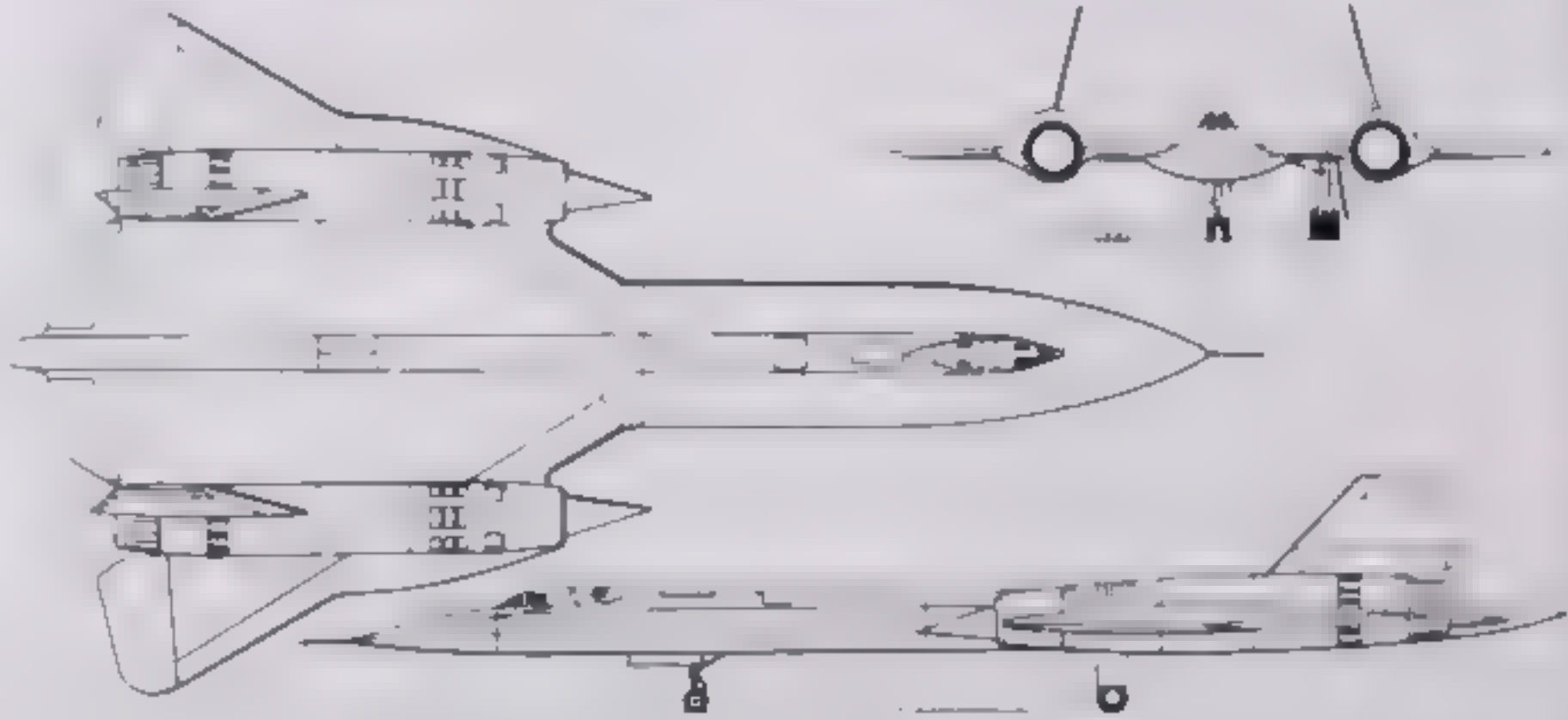
Performance: Maximum speed in excess of 2 000mph (3220km/h)
H service ceiling higher than 80 000ft (24 400m) range at Mach
24 000m) 2 982 miles (4800km) Performance limit about

Armament
History: First flight 21
1964 final delivery at
User: USA (Air Force)

Development: Despite their great size and intense noise these amazing
H were designed but test flown and put into use without a word
leaking out into public until disclosed by President Johnson in February
1964 The A 11 was originally designed as a follow-on to the U-2, capable
of flying even higher and many times faster in penetrating hostile airspace on
clandestine overflights Early in their career the three A-11s with serial
numbers 60 6934 to 6936 did overfly Communist territories in several
parts of the world Later they were completed as YF-12A research aircraft in
the Improved Manned Interceptor programme carrying Hughes ASG-18
pulse-doppler radar infra red sensors and eight Hughes AIM-47A large
long-range air-to-air missiles in an internal bay Made largely of a specially
developed alloy of titanium, the YF-12A was the most advanced aircraft of
its day and the only one to sustain a speed of Mach 3 One set a world speed
record at 2 070mph on 1 May 1965 and another beat this at 2,189mph
(3522km/h) in July 1976, in which month a 1,000km circuit was flown at
2 086mph (3356km/h) and an altitude of 86 000ft (26 212m) sustained
Surviving YF-12A variants have been stripped of weapon systems and used
in supersonic transport research flown by NASA The SR-71 strategic
reconnaissance aircraft is longer and heavier, with a fuel capacity of over
80 000lb Known as "Blackbirds" (though officially their external areas
are painted indigo blue), they equip the 9th Strategic Reconnaissance
Wing, from which detachments have operated over Vietnam, the Middle
East and many other trouble-spots Total production is at least 30, including
SR-71B and C dual trainer versions In September 1974 a standard A model



Above: The Lockheed YF-12A, the first species of Blackbird, has now been rebuilt for Mach 3 research.



Above: Three-view of the SR-71A in original form.

set a transatlantic record at almost Mach 3 with a time from passing New
York to passing London of 1hr 55min Most SR-71s in the active inventory
(some are stored) use special JP-7 high-temperature fuel, air-refueled by
specially equipped KC-135Q tankers

Below: Compared with the YF-12A the SR-71A is an appreciably
longer and heavier aircraft. This SR was photographed in 1976
about 70,000 feet below its normal mission altitude.



Myasishchev M-4

M-4 (three versions, known to West as "Bison A, B and C")

Origin: The design bureau of Vladimir M. Myasishchev, Soviet Union.
Type: (A) heavy bomber (B) strategic reconnaissance bomber (C) multi-role reconnaissance bomber
Engines: (A) four 19 180lb (8700kg) Mikulin AM-35 turbojet engines, (B and C) four 28 660 lb (13 000kg) Soloviev D-30 turbojet engines
Dimensions: (A) estimated, span 165ft 7in (50.4m), length 144ft 7in (44.1m), height 46ft (14.1m)
Weights: Estimated, empty (A) 154,000lb (70,000kg) (B, C) 164,000lb (74,400kg), maximum loaded (A) 352,740lb (160,000kg) (B, C) 375,000lb (170,000kg)
Performance: (Estimated) maximum speed (all) 560mph (900km/h), service ceiling (A) 42 650ft (13 000m), (B, C) 49,200ft (15 000m), range (all) 6 835 miles (11,000km) with 9 920lb (4500kg) of bombs or electronic equipment
Armament: (A) ten 23mm NR-23 cannon in manned turret in tail and four remotely controlled turrets above and below front and rear fuselage (two guns in each turret), internal bomb bays in tandem for at least 22 050lb (10,000kg) stores, (B, C) six 23mm cannon in two forward turrets and tail turret, internal bay for at least 10 000lb (4500kg) stores. In many versions a single 23mm gun is fixed on the right side of the nose, firing ahead.
History: First flight, probably 1953, service delivery, probably 1955 final delivery, probably about 1958
User: Soviet Union (DDA, AV-MF)

Development: A single example of this large aircraft took part in the 1954 May Day parade fly past over Moscow, its size being gauged from the escorting MiG fighters. It was expected to appear in large numbers, but



Above Three-view of M-4 of "Bison C" sub-type

The aircraft had a long service life, but its role in the Soviet Union changed after the 1954 parade. The first version, the M-4A, was a heavy bomber, but by 1955 it was being replaced by the M-4B, which was a strategic reconnaissance bomber. The M-4B was fitted with more powerful engines and its role changed from bomber to long-range overseas reconnaissance. In some cases, flight-refuelling tanker. All aircraft were equipped with probes, the rear turrets were removed and a vast amount of reconnaissance equipment fitted with from five to 17 aerostats visible all over the aircraft. In the "Bison C" sub-type a large search radar fills the entire nose, lengthening the nose by about 6ft and changing its shape. Since 1967 these now obsolescent aircraft have been frequently encountered on probing missions far over the Arctic, Atlantic, Pacific and elsewhere, at both high and low levels, the C-model having been seen most frequently.



Left: An M-4 of the sub-type called "Bison B" by NATO, photographed by an RAF Lightning over solid overcast. Most of these aircraft have been in service over 20 years.

Below: A frame from a Soviet film showing the British type of probe/drogue refuelling system used by the Soviet air forces. The receiver is an M-4 of the "Bison C" type.



Rockwell RA-5C Vigilante

A-5A, A-5B and RA-5C

Origin
Type

Engines

Dimensions

Weights

Performance

Armament History

User

Development



Above: Three-view of RA-5C with four external tanks but no ECM.

... 15 000.b
... the 57 A-5A were followed
... the airborne element of an integrated intelligence system
... the whole fleet and other forces. The RA-5C is extremely com
... equipped with multiple sensors including a side-looking
... the fuselage. These valuable aircraft have been hard-worked
... built in 1962-66. 53 A-5A and the 6 A-5B were
... standard, and in 1969-71 the production line at
... for 46 Phase II with GE-10 engines and improved

Below: Unusual view of an RA-5C Vigilante as it leaves the catapult of the nuclear-powered super-carrier USS *Enterprise*. It belongs to squadron RVAH-3.



Saab 32 Lansen

A32A, J32B, S32C

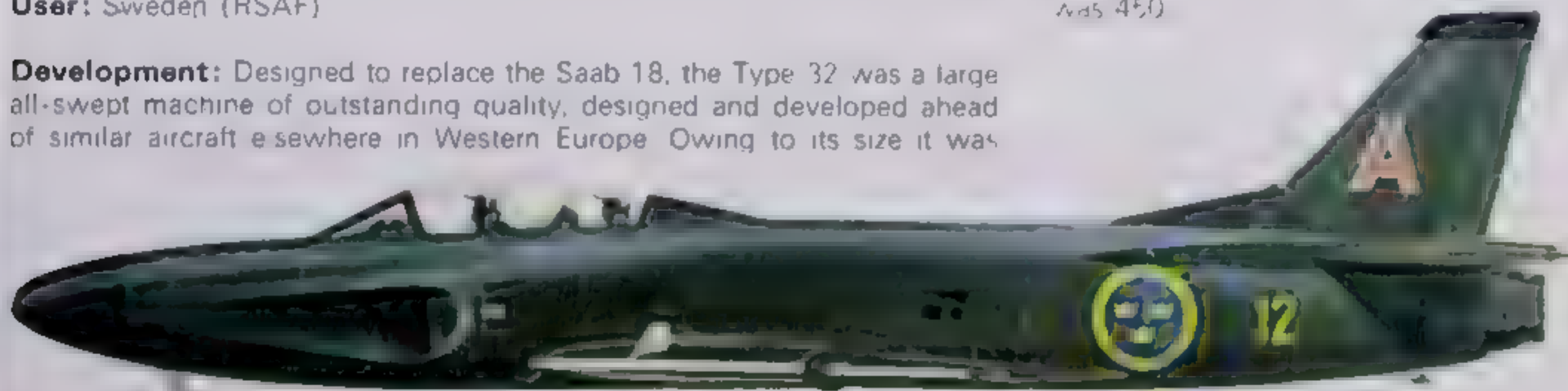
Origin: Saab-Scania AB, Linköping, Sweden
Type: (A) two-seat all-weather attack, (B) all-weather night fighter
(C) reconnaissance
Engine: One Svenska Flygmotor licence-built Rolls-Royce Avon turbojet
shaft turbojet with afterburner (A and C) 10 362lb (4700kg) thrust F15A2
(B) 15 190lb (6890kg) RM6A
Dimensions: Span 42ft 7½in (13m), length (A) 49ft 0½in (14.94m) (B)
47ft 6½in (C) 48ft 0½in height 15ft 3in (4.65m)
Weights: Empty (A) 16,398lb (7438kg), (B) 17 600lb, (C) 16,520lb
maximum loaded (A) 28 660lb (13,600kg), (B) 29 800lb, (C) typical
26 500lb
Performance: Maximum speed (all) 692mph (1114km/h Mach 0.91),
initial climb (A, C) 11 800ft (3600m)/min (B) 19 700ft/min service
ceiling (A, C) 49,200ft (15,000m), (B) 52 500ft, range with external fuel
(all, typical) 2,000 miles (3220km)
Armament: (A) four 20mm Swedish Hispano Mk V under nose wing
pylons for external load of up to 3,000lb including two Rb04C air-to-surface
missiles or two 1,320lb (600kg) bombs, (B) four 30mm Aden M 55
cannon under nose and four Rb324 (Sidewinder) missiles or FFAR pods
(C) none
History: First flight 3 November 1952 service delivery (A) December
1955, first flight (B) 7 January 1957, (C) 26 March 1957, final delivery,
(B) 2 May 1960
User: Sweden (RSAF)

Development: Designed to replace the Saab 18, the Type 32 was a large
all-swept machine of outstanding quality, designed and developed ahead
of similar aircraft elsewhere in Western Europe. Owing to its size it was



Above: Three-view of the Lansen A32A with wing pylons and ventral radar.

capable of development for three decades. It proved itself in
the long service of the Swedish Air Force. The S32C was the
shallow dive the A32A entered service with F17 and soon gained a
reputation for good handling ease, manoeuvrability and accurate
delivery. It equipped all four Swedish attack wings until the Saab A37
began to replace it in 1971. The more powerful J32B night fighter was
fitted with S6 radar fire control for lead/pursuit interception and equipped
seven squadrons in 1958-70. The S32C force was several times updated by
fitting advanced new sensors and ECM, and is in 1977 still not fully re-
placed by the SF37 and SH37. Total Lansen (meaning lance) production
was 450.



Above: J32B Lansen, formerly with F12 at Kalmar.

Below: The S32C is one of
the versions of the excel-
lent Lansen still in service
in 1977 with the Swedish
Flygvapen. The main S32C
units are two squadrons of
wing F11 at Nyköping.
There are still a few A32A
attack Lansens in F6 at
Karlsborg, and the most
powerful sub-type, the
J32C (originally a night
fighter) is a target tug.



Tupolev Tu-16

Sub-types known to West as “Badger A” to “Badger G”; Tupolev bureau, Tu-88

Origin: The design bureau of Andrei N. Tupolev, Soviet Union, built (probably, without licence) at Shenyang, China

Type: Designed as strategic bomber, see text

Engines: Believed in all versions, two Mikulin AM-3M single-shaft turbojets each rated at about 20,950lb (9,500kg)

Dimensions: Span (basic) 110ft (33.5m) (varies with FR system, ECM and other features), length (basic) 120ft (36.5m) (varies with radar or glazed nose), height 35ft 6in (10.8m)

Weights: Empty, typically about 72,750lb (33,000kg) in early versions, about 82,680lb (37,500kg) in maritime/ECM roles, maximum loaded, about 150,000lb (68,000kg)

Performance: Maximum speed, clean at height, 587mph (945km/h); initial climb, clean, about 4,100ft (1,250m)/min, service ceiling 42,650ft (13,000m); range with maximum weapon load, no missiles, 3,000 miles (4,800km); extreme reconnaissance range, about 4,500 miles (7,250km)

Armament: In most variants, six 23mm NR-23 cannon in radar-directed manned tail turret and remote-aimed upper dorsal and rear ventral barbettes; versions without nose radar usually have seventh NR-23 fixed firing ahead on right side of nose; internal weapon bay for load of 19,800lb (9,000kg), with certain versions equipped to launch missiles (see text)

History: First flight (Tu-88), believed 1952; service delivery 1954; final delivery (USSR) about 1959, (China) after 1975

Users: China, Egypt, Indonesia (in storage), Iraq, Soviet Union (AV-MF)

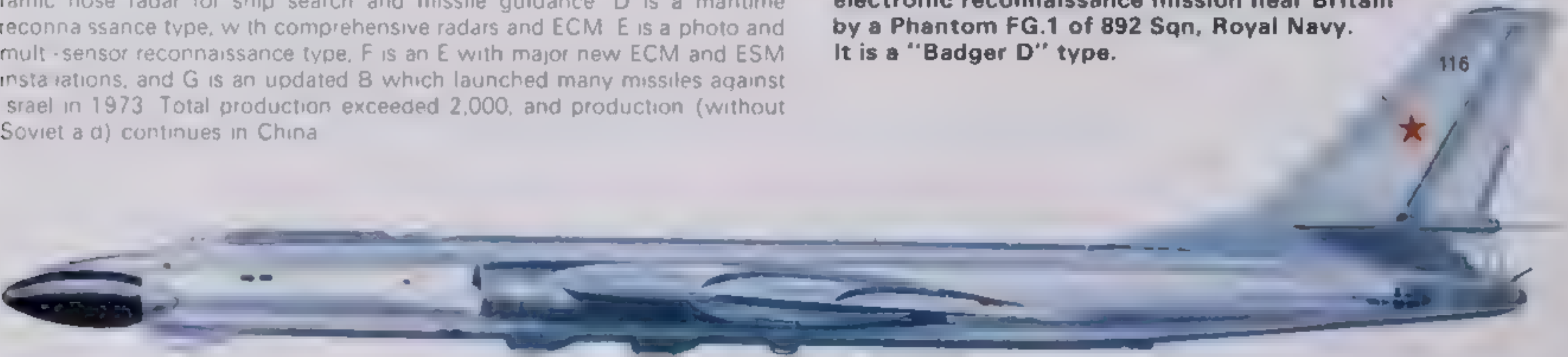
Development: Representing a simple and low-risk approach to the strategic jet bomber requirement, the Tu-88 prototype was generally in the class of the Valiant but incorporated heavy defensive armament. Technology throughout was derived directly from the Boeing B-29, which Tupolev's bureau had in 1945–53 built in large numbers as the Tu-4. The first (“Badger A”) version had blind-bombing radar and glazed nose, and a few were supplied to Egypt and Iraq. The B carried two “Kennel” cruise missiles on underwing pylons and served the AV-MF (Navy) and Indonesian AF. C carried the large “Kipper” stand-off missile on the centre-line, with panoramic nose radar for ship search and missile guidance. D is a maritime reconnaissance type, with comprehensive radars and ECM. E is a photo and multi-sensor reconnaissance type, F is an E with major new ECM and ESM installations, and G is an updated B which launched many missiles against Israel in 1973. Total production exceeded 2,000, and production (without Soviet aid) continues in China.



Above: Three-view of Tu-16 “Badger F” with side view (bottom) of “D”.



Above: Tu-16 of the AV-MF, photographed on a “Zombie” electronic reconnaissance mission near Britain by a Phantom FG.1 of 892 Sqn, Royal Navy. It is a “Badger D” type.



Above: Tu-16 of “Badger D” type, in service with the AV-MF



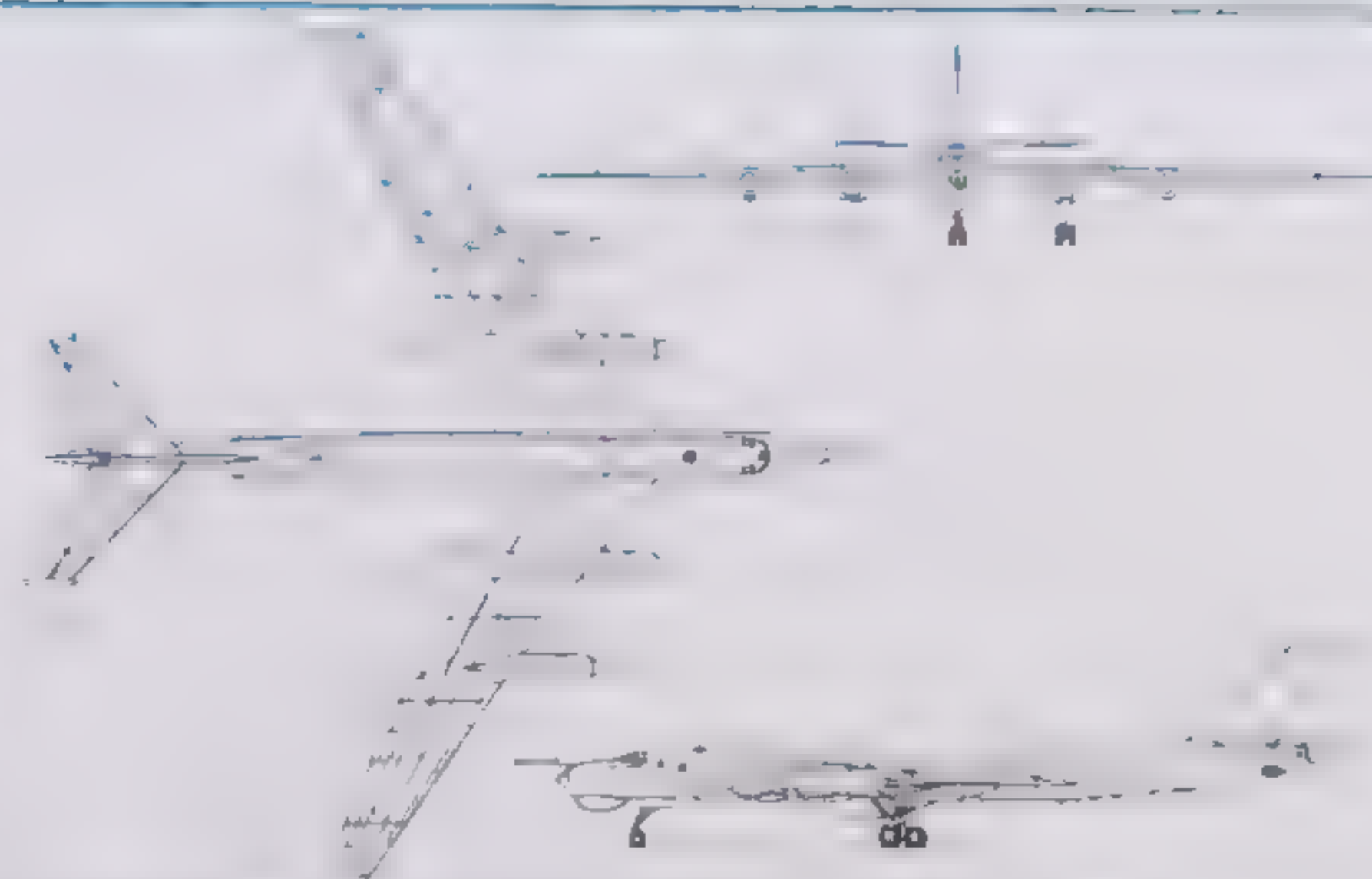
Left: A Tu-16 of the “Badger F” type, photographed by a Nimrod of the RAF as it wheels low over the North Atlantic. This type of Tu-16 is visually distinguished by its two small electronic pods carried on pylons beneath the wings. Frequently Tu-16s engaged in ESM (electronic surveillance measures) missions have worked in pairs, the “Badger D” and “Badger F” apparently being complementary. Several versions have been converted as air refuelling tankers, using an unusual wing-tip/wing-tip hook-up method.

Tupolev Tu-20

Tu-95 (Tupolev bureau designation) in versions known to West as “Bear A” to “Bear F”

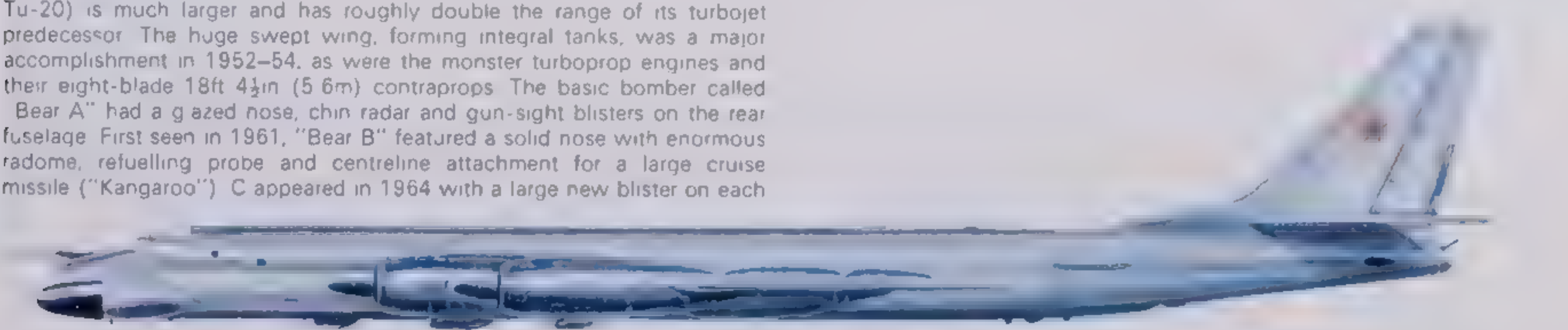
Origin: The design bureau of Andrei N. Tupolev, Soviet Union
Type: Designed as strategic bomber, see text
Engines: Four 14,795hp Kuznetsov NK-12M single-shaft turboprops
Dimensions: Span 159ft (48.5m), length 155ft 10in (47.50m) (certain versions differ by up to 6ft), height 38ft 8in (11.78m)
Weights: Empty, probably about 160,000lb (72,600kg), maximum loaded (estimate) about 340,000lb (154,000kg)
Performance: Maximum speed (typical Bear, clean) 540mph (870km/h), service ceiling, about 44,000ft (13,400m), range with 25,000lb (11,340kg) bomb load, 7,800 miles (12,550km)
Armament: Normally six 23mm NS-23 in radar-directed manned tail turret and remote-aimed dorsal and ventral barbettes (defensive guns often absent from late conversions, see below). Weapons bay for additional about 25,000lb (11,340kg)
History: First flight (prototype) mid-1954, service delivery, 1956, final delivery, probably about 1962
User: Soviet Union (ADD, AV-MF)

Development: Making use of identical systems, techniques and even similar airframe structures as the Tu-16, the Tu-95 (service designation Tu-20) is much larger and has roughly double the range of its turbojet predecessor. The huge swept wing, forming integral tanks, was a major accomplishment in 1952–54, as were the monster turboprop engines and their eight-blade 18ft 4½in (5.6m) contraprops. The basic bomber called “Bear A” had a glazed nose, chin radar and gun-sight blisters on the rear fuselage. First seen in 1961, “Bear B” featured a solid nose with enormous radome, refuelling probe and centreline attachment for a large cruise missile (“Kangaroo”). C appeared in 1964 with a large new blister on each



Above: Three-view of Tu-20 (Tu-95) of “Bear D” type.

side of the fuselage (on one side only on B), while D was obviously a major ECM/ESM reconnaissance type with chin radar, very large belly radar and from 12 to 21 avionics features visible from stem to stern. E is a multi-sensor reconnaissance conversion of A, while F is a recent further conversion with an array of ventral radars and stores bays in place of the ventral guns.



Above: Tu-20 (Tu-95) of the “Bear C” maritime surveillance type.



Above: A monster “AS-3 Kangaroo” air/surface missile drops away from its recess in the belly of a Tu-20 of the “Bear B” sub-type. This is the largest stand-off missile in the world (48ft 11in, 14.9m, long)



Left: A fine picture of a Tu-20 of the “Bear D” sub-type, taken by the pilot of an RAF Lightning north of Scotland. This model helps to guide anti-ship missiles.

Tupolev Tu-22

Tu-22 in versions known to West as "Blinder A" to "Blinder D"; Tupolev bureau, Tu-105

Origin: The design t
Type: Originally bomber, see text
Engines: Two afterburning turbojets of unknown type, each with maximum thrust estimated at 27,000 lb
Dimensions: Span 90ft 10½in (27.70m), length (most versions) 132ft 11½in

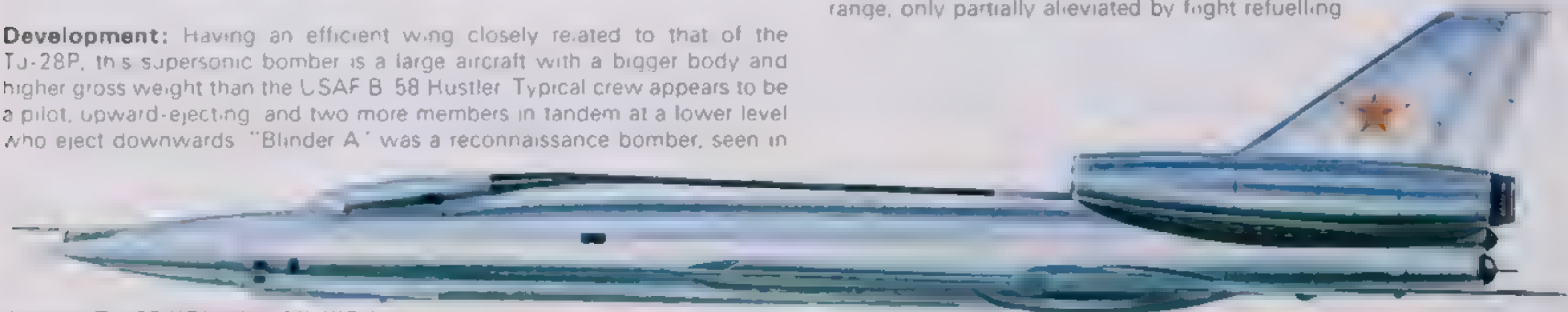
Weights: Empty, about 85,000lb (38,600kg), maximum loaded, about 185,200lb (84,000kg)
Performance: Maximum speed, about 2,100 mph (3,380 km/h); rate of climb, about 11,500ft (3,500m)/min; service ceiling, about 65,000ft (19,812m) (high, internal fuel only) 14,000 m; range, 2,000 miles (3,218 km)
Armament: One 23mm NS-23 in radar-directed barbette in tail, internal weapon bay for at least 20,000 lb (9,070kg) of free-fall bombs or other stores, or (Blinder B) one "Kitchen" stand-off cruise missile semi-recessed under centreline
History: First flight well before public display in 1961, service delivery, probably 1963
Users: Libya, Soviet Union (ADD, AV-MF)

Development: Having an efficient wing closely related to that of the Tu-28P, this supersonic bomber is a large aircraft with a bigger body and higher gross weight than the USAF B-58 Hustler. Typical crew appears to be a pilot, upward-ejecting, and two more members in tandem at a lower level who eject downwards. "Blinder A" was a reconnaissance bomber, seen in



Three-view of Tu-22 "Blinder A" with (inset) nose of "Blinder D".

small numbers. B carried the stand-off missile, had a larger nose radar and semi-flush FR probe. C is the main variant, used by Naval Aviation for over-sea ECM/ESM surveillance, multi-sensor reconnaissance and with limited weapon capability. D is a dual trainer with stepped cockpits. Recent versions appear to have later engines (probably turbofans) with greater airflow. There have been persistent reports of an interceptor version, but this seems unlikely. The abiding shortcoming of the Tu-22 has been limited range, only partially alleviated by flight refuelling.



Above: Tu-22 "Blinder A" ("Blinder C" is similar).

Below: These Tu-22 supersonic aircraft are from a unit of the AV-MF. They are probably of the "Blinder B" type with weapon bays tailored to the 37ft "AS-4 Kitchen" stand-off missile.

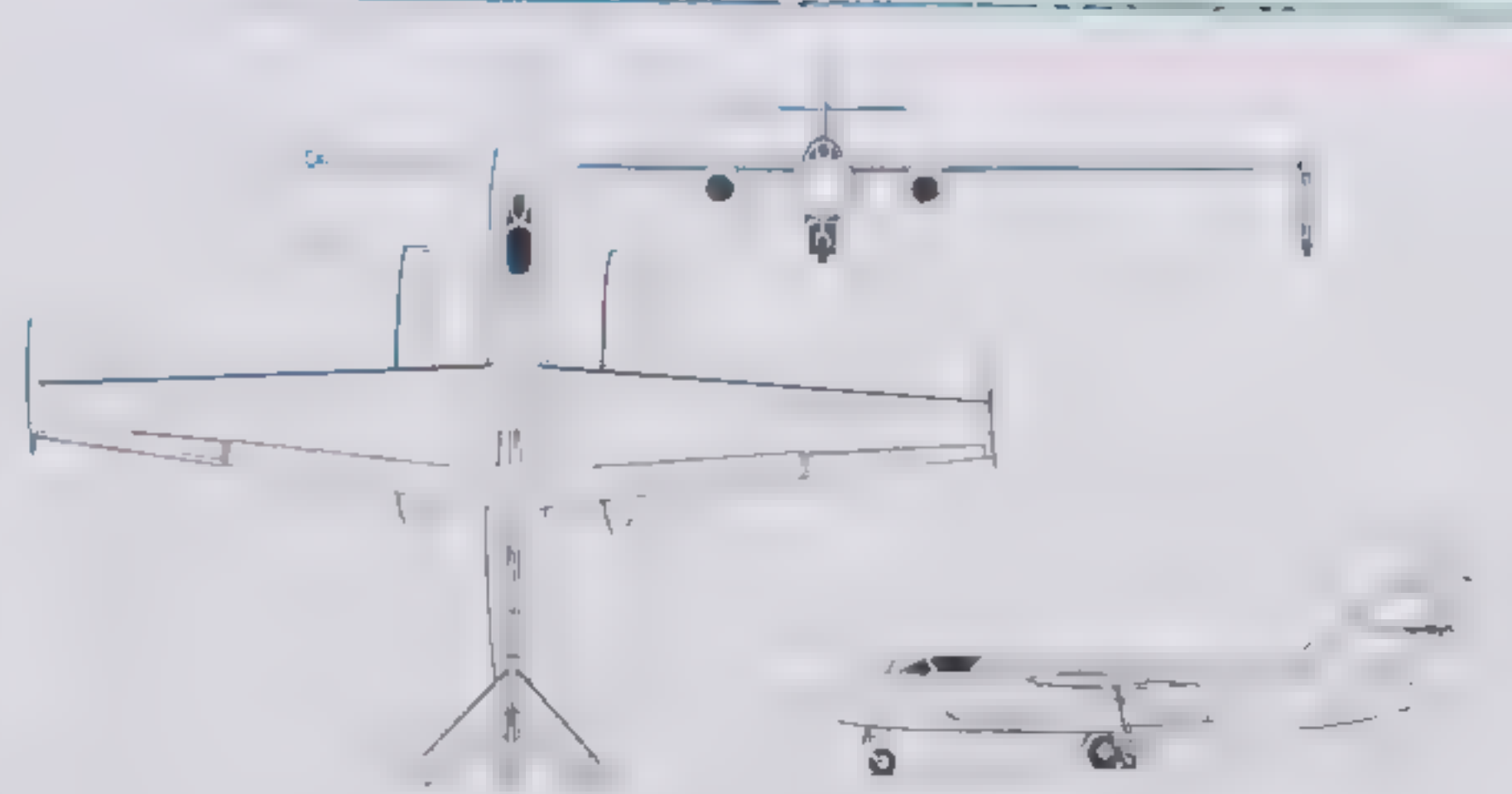


Yakovlev Yak-26

Yak-26, -27P and "Mandrake"

Origin: The design came from Alexander S. Yakovlev, Soviet designer.
Type: Low-altitude, high-speed fighter.
Engines: (26, 27P) two turbojets RD-9B (max fuel flow 1,200 lb/hr, max thrust 8,800 lb) (27P) two non-afterburning turbojets, probably RD-9 (rated at about 6,000 lb).
Dimensions: Span (26) 41.2 ft (12.5 m), (27P) 41.2 ft (12.5 m), (Mandrake) 41.2 ft (12.5 m), height (26) 20.6 ft (6.3 m), (27P) about 55 ft (16.8 m), (Mandrake) about 55 ft (16.8 m), weight (26) 14,100 lb (6,400 kg), (Mandrake) about 14,100 lb (6,400 kg).
Weights: Empty (26) about 18,000 lb (8,165 kg), maximum loaded (26) about 26,000 lb (11,800 kg), (27P) about 24,000 lb (10,900 kg), (Mandrake) possibly nearly 30,000 lb (13,600 kg).
Performance: Maximum speed (26) 27P at altitude 686 mph (1,104 km/h), Mach 0.95, (Mandrake) about 470 mph (755 km/h), initial climb about 15,000 ft (4,600 m)/min ("Mandrake", less), service ceiling (26) 27P 41,000 ft (12,500 m), ("Mandrake") about 62,000 ft (19,000 m), range at altitude (26) about 1,675 miles (2,700 km), (27P) 1,000 miles (1,600 km), (Mandrake) possibly 2,500 miles (4,000 km).
Armament: (26) one 30mm NR-30 cannon in fairing low on right side of forward fuselage, (27P) small batches with cannon and rockets, then small batches with two missile pylons, ("Mandrake" none).
History: First flight (26) before mid-1956, (27P) before mid-1956, ("Mandrake") possibly 1957.
User: Soviet Union.

Development: Stemming directly from the Yak-25, these aircraft introduced various changes, of which the most significant was afterburning engines in the Yak-26 and 27P. The leading edge was swept very sharply at the root and the trailing edge extended inboard of the longer nacelles to



Three-view of Yak-26 "Mandrake" (provisional).

meet the body at 90° and give much greater root chord. The wings were pointed (glazed on the 26 and a radome on the 27P) and in the production aircraft the outer wings were extended beyond the outrigger gear and fitted with drooped extended-chord leading edges. Only the 26 was built in large numbers (NATO name "Mangrove"). The 27P was called "Flashlight C". The high-altitude unsuited Mandrake was reported to have been not known, made overflights in Eastern Asia, the Middle East and along the borders of Communist territory in Europe before being withdrawn from most front-line units. Today a few "sneak" missions are still being made and it is probable that the aircraft making them are remotely piloted (because Mandrake and other Yak twin-jets have been important radio-controlled targets used in Soviet missile tests).

Below: The several Yak-27 sub-types (NATO name "Mangrove") were interim aircraft which led to the Yak-28 "Brewer" family—which look similar but are totally redesigned.
Foot of page: Yak-26 "Mandrake", from Soviet ciné film.



Anti-submarine and maritime patrol aircraft



During World War II the biggest problem in the war against the U-boat was building aircraft with sufficient range to remain on station in mid-Atlantic, and thus close the "aircraft gap" where Admiral Raeder's wolf-packs lurked unmolested from the sky. Today range is no great problem, but it is still a challenge to track down and kill a skilfully handled nuclear submarine, which can speed through the depths without any of the limitations imposed on conventional submarines. And the fact that submarines are today strategic weapons, able to destroy cities thousands of miles away, has added fresh impetus to the overall problem of ASW (anti-submarine warfare).

ASW is usually the main task facing the maritime patrol aircraft. It is also the primary mission of several important helicopters, as explained in that section. The helicopter is bound to be potentially faster and more agile than 10,000 tons of steel deep in the ocean, but unless he is endowed with telepathic reception the helicopter skipper cannot know in advance what his adversary intends to do next, so the chase is not one-sided. In fact, as was the case with U-boats after mid-1943, future submarines may even choose to surface and fight it out, because it is only a matter of time before they carry short-range surface-to-air missiles (such as the Vickers SLAM, submarine-launched airflight missile, which is a pop-up cluster of six Shorts' Blowpipe missiles fired from the top of the submarine fin without any other part of the vessel needing to break surface). Once at close quarters, a duel between aircraft and submarine tends to be violent. In general, the ASW aircraft presents a target that is much easier to see and easier to hit than the submarine, which is another reason why submarines are likely to carry SAMs (surface-to-air missiles).

Finding and tracking submarines is a task calling for several kinds of sensor. A submarine on the surface is almost an unfair situation, for it can be seen miles away on radar. Modern maritime radars are powerful and sensitive, and contain circuitry that automatically eliminates most of the interference

called "sea clutter" caused by reflections from the moving waves. Despite the close proximity of the sea-clutter background, such radars are specifically designed to detect such small objects as the tips of submarine fins, radio or radar aerials, schnorkels (snorts) and the other small probes the modern submarine likes to stick up into the atmosphere. A diesel-powered submarine at snort depth can be detected by radar at a range of well over ten miles, but ASW aircraft often carry an "Autolykus" sniffer that can pick them up at greater distances by detecting the concentration of a few parts per billion of diesel exhaust constituents in the atmosphere (snags here are that diesel exhaust travels only downwind, and there are lots of other diesel-powered ships going about their lawful occasions). Many ASW aircraft have a MAD (magnetic anomaly detector) either sticking out at the back or towed on a long wire to get it away from the disturbance of the aircraft itself. The MAD detects the extremely small but characteristic disturbance to the Earth's magnetic field caused by large masses of metal, such as a submerged submarine (or, to name another snag, an uncharted wreck). Probably the most important ASW sensors of all are acoustic. Sonobuoys, either passive or active, are slim cylinders dropped accurately into the sea (helicopters can cheat by "dunking", fishing the buoy out and trying in a different place). Passive buoys listen for the hydrodynamic disturbance, mostly from the screw, and machinery noise of a submerged submarine. Active buoys radiate sharp "pings" of sound and listen for reflections from an otherwise silent submarine hull. Sonics has become a subject almost as rich in measures and countermeasures as radar.

It is almost an anti-climax to report that ASW aircraft carry AS torpedoes, depth bombs, missiles and various other weapons. They also keep an eye on legitimate sea traffic, offer many kinds of assistance, conduct a great deal of reconnaissance, and in many countries help prevent smuggling, illegal immigration, violation of fishing frontiers, and even such misdemeanours as dumping tanker wastes.

With a touch of flap and its MAD boom extended, a US Navy S-3A Viking whispers across a Poseidon-missile submarine. Perhaps the most cleverly packaged aircraft in history, this carrier-based four-seater can fly ASW, ESM, reconnaissance and attack missions.



Beriev M-12 Tchaika

Be-12 Tchaika; NATO code-name "Mail"

Origin

Type

Engines

Dimensions

Weights

Performance

Armament

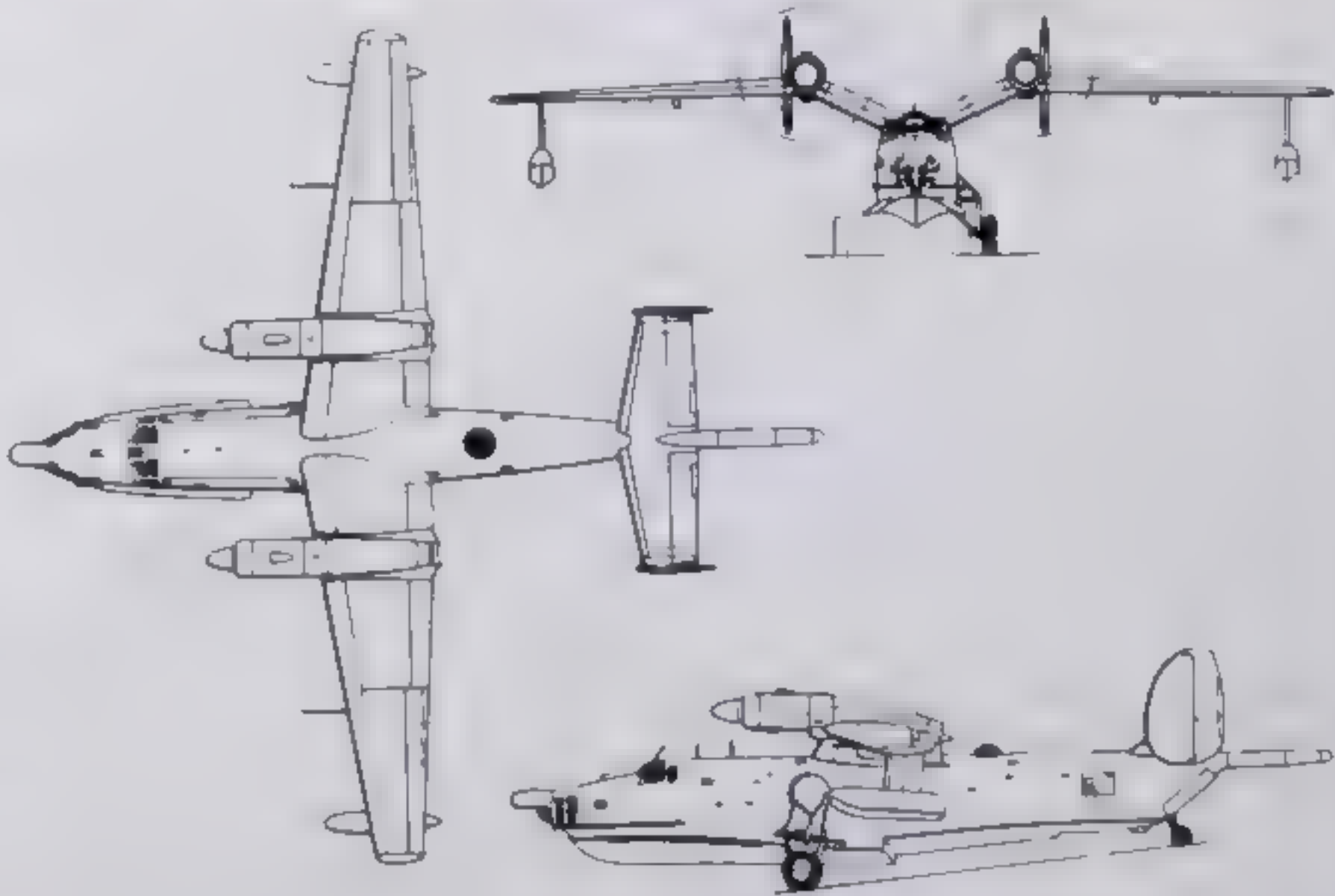
History

User

Development



Above: An M-12 Tchaika takes off from an AV-MF base (note the Tu-22 "Blinder C" in the distance)



Above: Three-view of M-12 with landing gear extended.

Below: From all accounts the M-12 is a thoroughly good aircraft on land, sea or in the air. The amphibian is a species extinct in the West.



Boeing Model 1041-133-1

Various US Navy study programmes

Origin: Model 104 manufacturers

Type: V/STOL (vertical takeoff and landing) aircraft

Engines: (Model 1041) two 8 080shp A

Dimensions 48ft 1in (14.6m) high, 111ft 6in (34m) long

Weights: $f_{\text{red}} = \frac{\alpha}{\alpha + 1}$, $f_{\text{blue}} = \frac{1}{\alpha + 1}$

Performance: Set $\alpha = 0.05$, maximum number of iterations = 100, $\epsilon = 1 \times 10^{-6}$.
 1) With 180° phase shift between the two input signals, the algorithm converges
 perfectly.

Armament: Suitable for ASW mission none in AEW

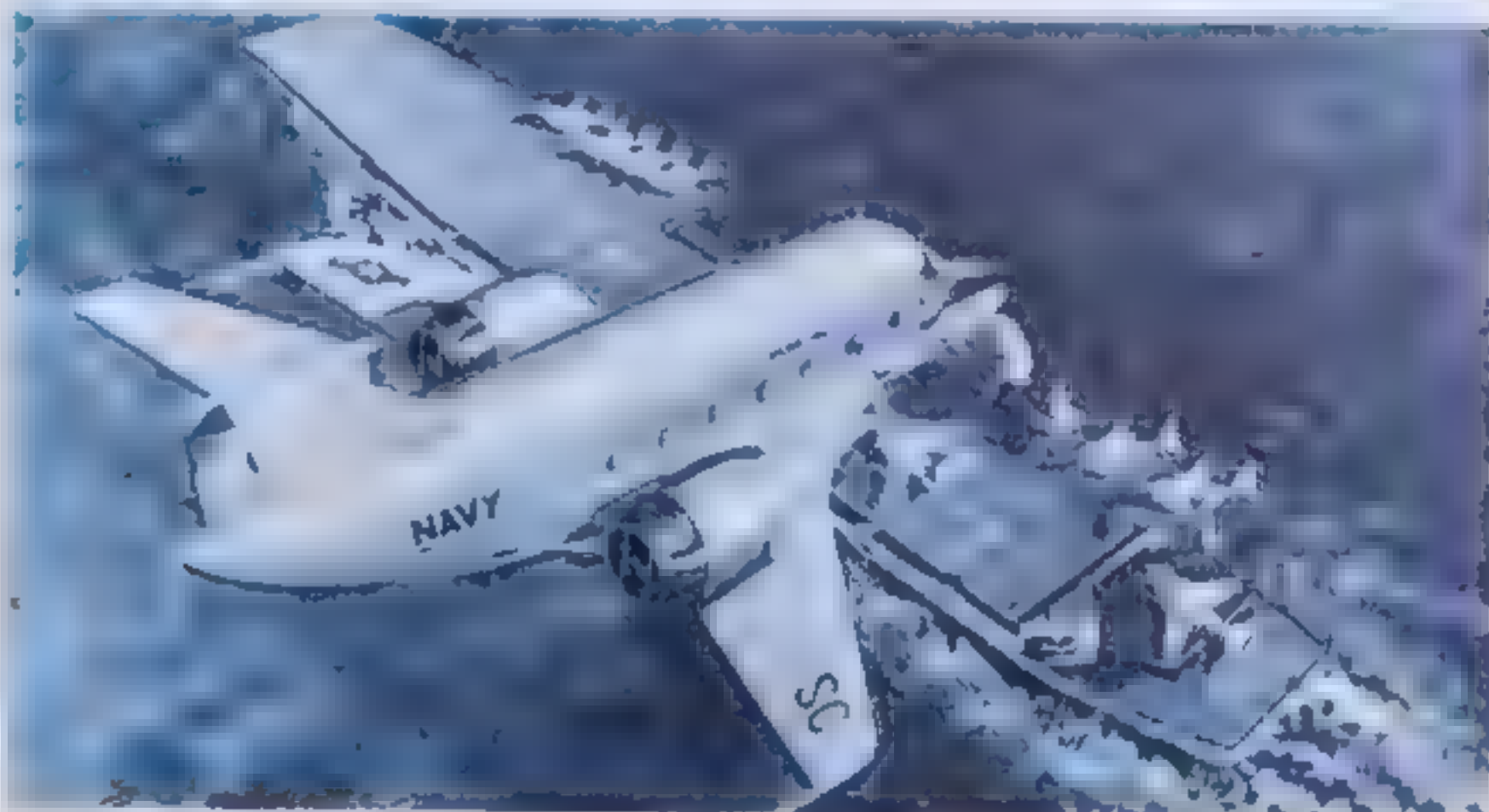
History: Study 1975-77, possible hardware funding in 1978 for first flight after 1980; no new program commitment until 1983.

User: Studies funded by the S.V.

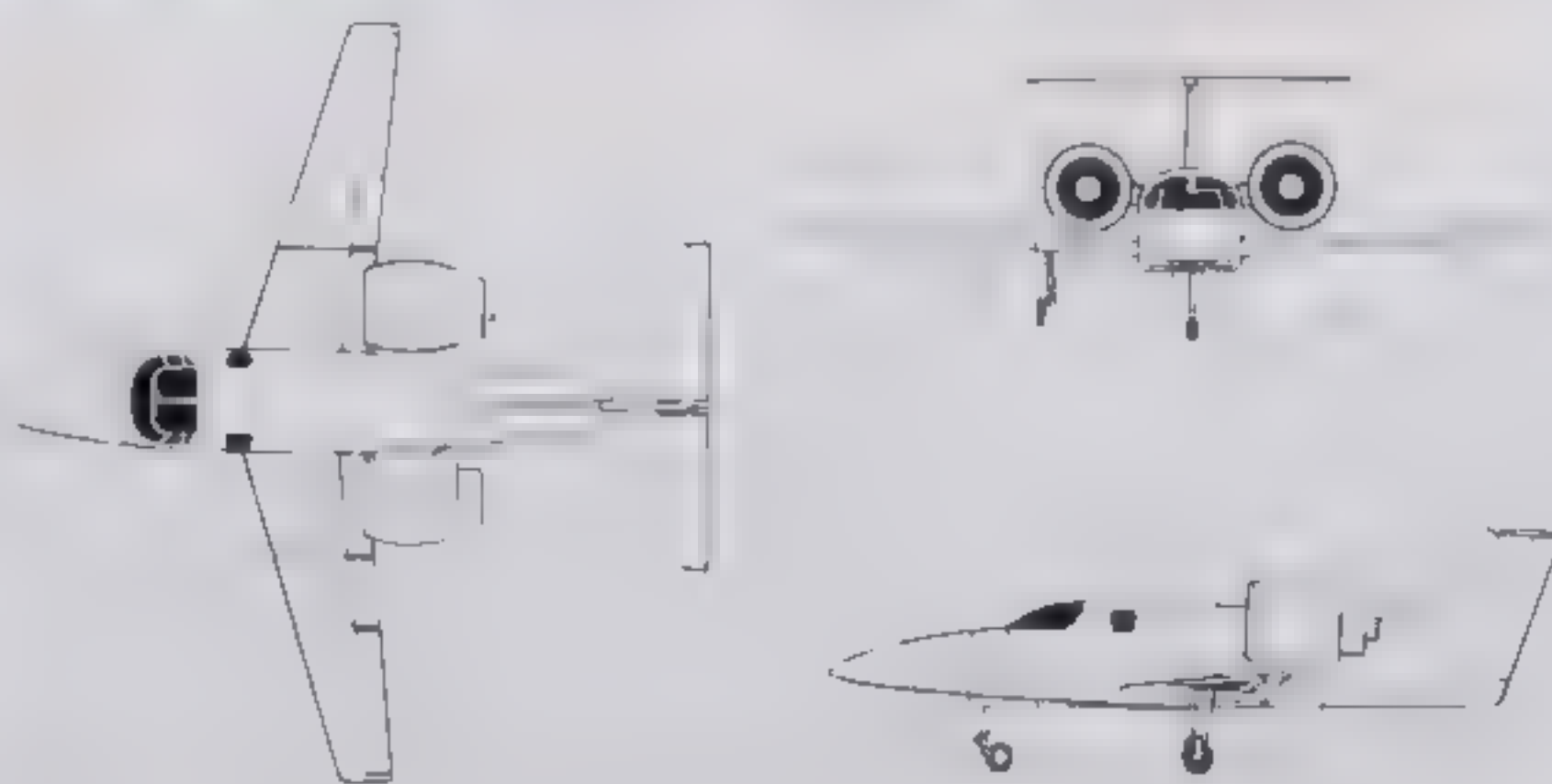
Development: The aircraft has been developed primarily for the use of those participating in a jet-fuelled assault operation. The power advantage of a full lift capability in a fast jet aeroplane demonstrated by the US Navy's F-4 Phantom II and the British Av. 8A have led to intensive study of how jet-lift technology might be applied to the full range of seaborne air power. The US Navy's fast jet strike force, several months ago, determined that the best way to exploit the ASW and the AEW aircraft would be based on a platform that has large range. The proposed B-2000, Model 1041-153-1 would have a two-compartment, stretched wing to allow 33 normal, separate and full thrust jet engines. The two turbo-shaft engines would be joined to high speed shafts to three lift fans. One would be in the nose, in the lower two floors except during jet-supported modes. The other two would be in the tail, one in the upper floor and the other in the lower floor. The two aft fuselage duct units swivelling to give lift or thrust.

There are many other attractive configurations. The McDonnell DC-10 could use the swivel-nozzle Pratt & Whitney F402 (RR Pegasus)

In February 1977 Grumman revealed their proposal for what is called the X-30. It is the M-2000, a 100,000-lb aircraft that will have a 5-3A and two pivoted geared-fan engines with a by-pass ratio of about 15. Vanes in the efflux of the engines control the aircraft in yaw and



Above: McDonnell Douglas proposes the Model 260, the subtype pictured being a Carrier On-board Delivery transport



Three-view of Boeing 1041 (engines in forward-flight mode).

FAA (Federal Aviation Agency) visitors have been impressed by a large free-flight model of this proposal. In the even longer term the Navy is interested in the possibility of a V/STOL supersonic aircraft, in which thrust for forward flight would be provided by a jet engine and VTOL by a rocket scheme could be found. The Rockwell X-12 is described in Fig. 1.

Below: Though pictured on a carrier, the point of the Boeing 1041 is that it (and similar proposals) could use small decks



Breguet Alizé

Br.1050

Origin
Type
Engine
Dimensions

Weights Performance

Armament

History: First flight (Br 90

Users: Fred

Development: In 1948 the Breguet converted the Br 960 Vultur (vulture) looked conventional but with a top speed of 559mph (900km/h) on the thrust of a Rolls Royce Merlin engine, a retractable canopy, a dorsal fin, a dorsal fin back and an AS Mamba turbopropeller. It was designed for long range and endurance. This never went into production. It was very successful and pleasant to fly. Instead, in 1954, the long range turned the Vultur into an anti submarine aircraft carrying both torpedoes and bombs. So different did it become that it was renamed Br 960 (and). The retractable radome replaced the turbojet, the main cabin moved forwards into place, a so housing sonobuoy and depth charge.



Three-view of Br.1050 Alizé (radome retracted).

was arranged for the pilot on the left, a radar operator on the right and a second sensor operator behind. Flottiles 4F, 6F and 9F still deploy most of the 75 built for the carriers *Foch* and *Clemenceau* and a further 12 applied for 310 Sqn Indian Navy aboard the carrier *Vikrant*.

Below: It is testimony to a satisfactory weapon system that no Alizé has been withdrawn from front-line service. Both 310 Sqn of the Indian Navy, ashore and embarked in INS *Vikrant*, and the carrier ASW squadrons (Flotille 4F and 6F) of the French Aeronavale, still use the Alizé exclusively, and see no replacement. This one is landing on *Clemenceau*.



Breguet (Dassault Breguet)

Atlantic

Br.1150 Atlantic

Origin: Design, Louis Breguet, France, manufacture SECBAT multinational consortium

Type: Maritime patrol and anti submarine aircraft with normal crew of 12

Engines: Two 6,106ehp Rolls-Royce Tyne 21 two-shaft turboprops

Dimensions: Span 119ft 1in (36.3m), length 104ft 2in (31.75m) height 37ft 2in (11.33m)

Weights: Empty 52,900lb (24,000kg); maximum 95,900lb (43,500kg)

Performance: Maximum speed (above 16,400ft/5000m) 409mph (658km/h), patrol speed 199mph (320km/h), initial climb at gross weight 2,450ft (746m)/min; service ceiling 32,800ft (10,000m), patrol endurance 18 hours, maximum range 5,592miles (9000km)

Armament: Unpressurised weapon bay carries all NATO standard bombs, 385lb (175kg) depth charges, four homing torpedoes (or nine acoustic torpedoes) and HVAR rockets. Underwing racks for up to four AS.12 Martel or other missiles with nuclear or conventional warheads

History: First flight 21 October 1961; (production aircraft) 19 July 1965 service delivery 10 December 1965

Users: France (Aéronavale), W Germany (Marineflieger), Italy, Netherlands (Navy), Pakistan

Development: The history of NATO shows the almost continual failure of its members to agree on weapon standardization, especially in aircraft. One of the few real attempts to do better was the 1958 decision to procure a standard aircraft to replace the Lockheed P-2. The French Br.1150 was selected from 25 designs submitted and in December 1959 two prototypes were ordered from NATO funds. Though most NATO members refused to have anything to do with it — often because their own design had not been



Three-view of Br.1150 Atlantic (radome retracted).

chosen — the programme was launched by France and Germany, which ordered 40 and 20 respectively. Subsequently the Netherlands Navy bought nine and the Italian Navy 18, so that finally the aircraft was being built in all four countries, with avionics partly supplied from the USA and Britain. The engines were made by a British/Belgian/German/French/Italian consortium, with assembly by SNECMA of France. The Atlantic proved a most comfortable and efficient machine, with pressurization above the floor of the double-bubble fuselage and a great amount of room. The aircraft frame is skinned mainly in metal honeycomb sandwich. Five of the German Atlantics bristle with special electronic countermeasures, and in 1977 plans were continuing to be studied for a heavier Atlantic II with M45 turbofan pods under the wings to allow operation at weights as high as 110,000lb (50,000kg).

Below: Pleasing study of an Atlantic in its element. It belongs to 30° Stormo, Italian Marina, based at Cagliari.



Canadair CP-107 Argus

Argus Mk 1 and 2

Origin
Type: Maritime patrol and anti-submarine aircraft with normal ops
Engines: Four 3 700hp (water injection rating) Wright R 3350

Dimensions: Span 142ft 3½in (43.37m) length 128ft 3in (39.1m)

Weights: Empty 81 500lb (36 967kg)
Performance: Maximum speed 470 mph (756 km/h)
Climb rate 1 190 ft/min (362 km/h) at low level 5 900 ft/min (1 580 m/min)

Armament: Two 18ft (6m) internal bays accommodate a normal complement of other stores. Total weight of 8 000lb (3 629 kg). Additional 345 kg.
History: First flight 29 March 1957. (33rd and last aircraft) 13 July 1960
User: Canadian Armed Forces

Development: In the early 1950s the Royal Canadian Air Force examined the possibility of replacing the Lancaster MR 10 and Catalina as standard ocean patrol aircraft. It needed great endurance and the ability to carry comprehensive ASW (anti-submarine warfare) search and attack equipment. It eventually adopted the solution of Canadair Ltd. which like its US parent General Dynamics was anxious to produce developed versions of the British Britannia turboprop airliner. To meet the RCAF need the CL 28 was produced by replacing the Britannia's turboprops by Turbo Compound engines giving it a long range and endurance at low level together with a new unpressurised fuselage incorporating weapon bays ahead of and behind the wing. Comprehensive equipment was installed including APS-20 radar in the nose, desks for seven sensor operators and in the rear fuselage batteries of sonobuoys dropped or fired through dispensers. MAD equip-



Three-view of CP-107 Argus Mk 2.

ment resulted in the long tail "stinger". At the 13th aircraft production stage it was fitted to the Mk 2 with a more advanced British radar with smaller antennas and completely rearranged communication and navigation equipment. The new radar required two small dorsal ailerons. The wing pylons were modified for the defunct Petrel missile and have seldom been used. Argus aircraft have served well for almost 20 years with four squadrons of the Canadian Armed Forces and will eventually be replaced by the far more modern Lockheed CP-140 Aurora.



Above: CP-107 Argus Mk 2 (all Mk 1 now converted) of CAF 415 Sqn.

Below: After logging many thousands of hours in harsh environments the very first CP-107 Argus still looks fresh. Here seen crossing the wake of a submarine, it will have to go on filling a vital role until the CP-140 Aurora becomes available in 1980. By then this particular Argus will have been in service almost 23 years.



Grumman S-2 Tracker and derivatives

S-2A to -2E Tracker, DHC CS2F, C-1A Trader and E-1B Tracer

Origin: Grumman Aerospace, USA, licence-built in Canada
Type: (S-2) carrier ASW aircraft, (C-1) COC, (E-1) EAWC

Engines: Two 1525hp Wright R-3600-15

Dimensions: Span (S-2A to -2C, C-1 and E-1) 69ft 8in (21.3m) (S-2D to -2G) 43ft 6in (13.26m) length (S-2A to -2C) 42ft 3in (12.8m) (S-2D to -2G) 16ft 7in (5.06m)

Weights: Empty (S-2A) 17357lb (7873kg) (S-2E) 18750lb (8500kg) loaded (S-2A) 26300lb (11929kg) (S-2E) 29150lb (13220kg) (C-1A, E-1B) 27000lb (12240kg)

Performance: Maximum speed (S-2A) 287mph (462km/h) (S-2E) 270mph (434km/h) (C-1A) 290mph (467km/h) climb rate (S-2A) 1920ft (586m)/min, (S-2E) 1390ft (425m)/min, service ceiling (S-2A) 23000ft (7010m) (S-2E) 21000ft (6400m) range (S-2A) 9000mi (1448km) (S-2E) 1300mi (2093km)

Armament: Weapon bay accommodates two electric acoustic homing torpedoes, two Mk 101 depth bombs (some versions one) or 1000lb (453kg) depth charges, six underwing pylons for 500lb (227kg) or 250lb (113kg) bombs, or for ferrying torpedoes (two on each wing) (S-2E) as above, with provision for Betty nuclear depth charge, AS-12 or other guided missiles and 7.62mm Miniguns. (C-1 and E-1) no armament

History: First flight (XS2F-1) 4 December 1952, production S2F-1 10 April 1953, combat service February 1954, final delivery February 1968

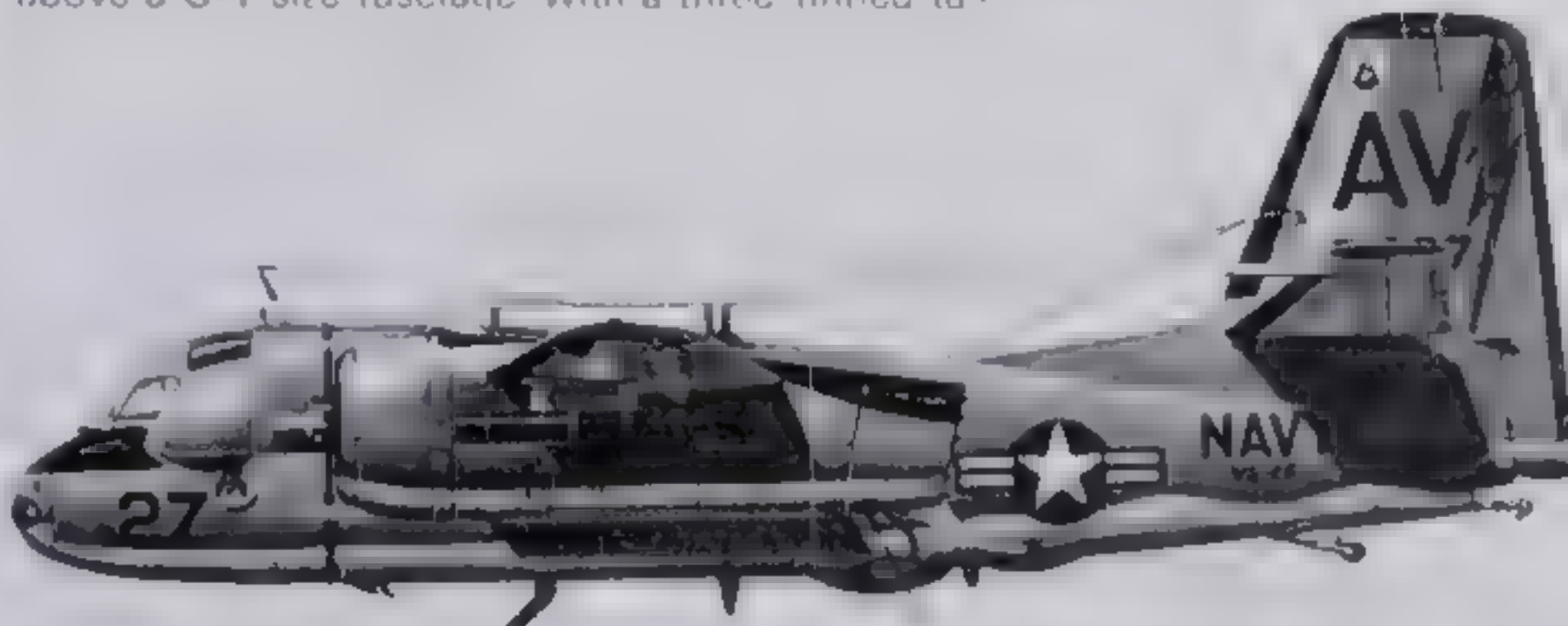
Users: Argentina, Australia, Brazil, Canada, Italy, Japan, Peru, Taiwan, Thailand, Turkey, Uruguay, USA, Venezuela

Development: This very ordinary-looking piston aircraft entered a world of advanced gas-turbine machines and has outlasted nearly all of them. It emerged from the belief that an aircraft compatible with carrier operations should be made to combine the two roles of anti-submarine warfare (ASW) search and strike, previously accomplished by one aircraft equipped with sensors and another equipped with weapons. Grumman developed the Tracker extremely rapidly, despite the need to package an extraordinary diversity of equipment into a small space. Its low-speed handling stemmed from the long span, with almost full-span slotted flaps and fixed slots on the outer folding wings. In the nose were seats for pilot, co-pilot/navigator and two radar plotters. In the S-2A (formerly S2F-1) the radar was the APS-38, in a retractable ventral bin. A magnetic anomaly detector (MAD) boom extended behind the tail, an upper search radar had its scanner above the cockpit, a 70 million candlepower searchlight was hung under the right wing and the engine nacelles were full of sonobuoys ejected through tubes to the rear. The S-2A was an ASW carrier and the CS2F-1 and -2 were built by de Havilland Canada. The S-2B was equipped with the Julie acoustic (explosive) echo-ranger and associated hydrophone sensor. The S-2C had a guided asymmetric weapon bay and larger tail, most being converted into US-2C utility or RS-2C photo aircraft. The S-2D, ordered in 1958, was physically larger, with improved accommodation and equipment. The S-2E incorporated further new equipment. The total of these versions was 1,281, including 100 Canadian-built, and several hundred have been modified to the S-2F and G versions for service until late 1977 with many of the 12 countries that have used Trackers. The C-1A



Three-view of the E-1B Tracer. The C-1A has a similar fuselage.

The S-2A was a carrier ASW carrier and the CS2F-1 and -2 were built by de Havilland Canada. The S-2B was equipped with the Julie acoustic (explosive) echo-ranger and associated hydrophone sensor. The S-2C had a guided asymmetric weapon bay and larger tail, most being converted into US-2C utility or RS-2C photo aircraft. The S-2D, ordered in 1958, was physically larger, with improved accommodation and equipment. The S-2E incorporated further new equipment. The total of these versions was 1,281, including 100 Canadian-built, and several hundred have been modified to the S-2F and G versions for service until late 1977 with many of the 12 countries that have used Trackers. The C-1A



Above: Grumman S-2E Tracker, as formerly used by VS-26 aboard USS *Randolph*.

Below: Canada's CP-121 Trackers were built at Toronto by de Havilland Canada. They remain operational.



Hawker Siddeley Nimrod

HS.801 Nimrod MR.1 and 2 and R.1

Origin: Hawker Siddeley Aviation (British Aerospace), UK
Type: (MR) maritime reconnaissance and anti-submarine aircraft with operating crew of 12, with several secondary roles, (R) electronic reconnaissance and countermeasures
Engines: Four 11,995lb (5441kg) thrust Rolls-Royce Spey 250 two-shaft turbofans
Dimensions: Span 114ft 10in (35m), length 126ft 9in (38.63m) (R.1 118ft), height 29ft 8½in (9.08m)
Weights: Empty (MR, typical) 92,000lb (41,730kg) loaded 192,000lb (87,090kg)
Performance: Maximum speed 575mph (926km/h), economical transit speed 490mph (787km/h) climb and ceiling, not disclosed, mission endurance 12 hours, ferry range, typically 5,755 miles (9,265km)
Armament: 50ft (15m) weapon bay capable of carrying very wide range of stores, including AS torpedoes, mines, depth bombs, nuclear weapons, conventional bombs and fuel tanks, two (optionally four) underwing pylons for air-to-surface missiles such as Martel or AS.12
History: First flight (aerodynamic prototype) 23 May 1967, (production MR.1) 28 June 1968, service delivery 2 October 1969
User: UK (RAF)



Above: Three-view of Nimrod MR.1 or 2 (R.1 has short tailcone).

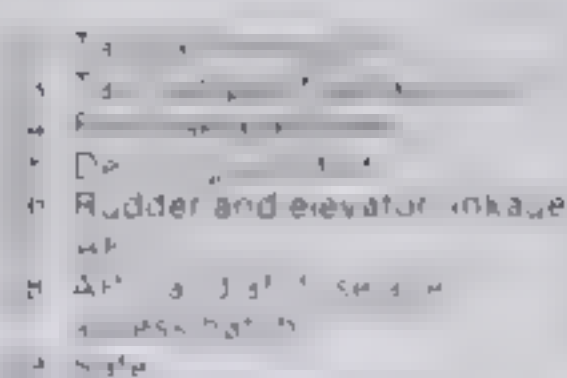
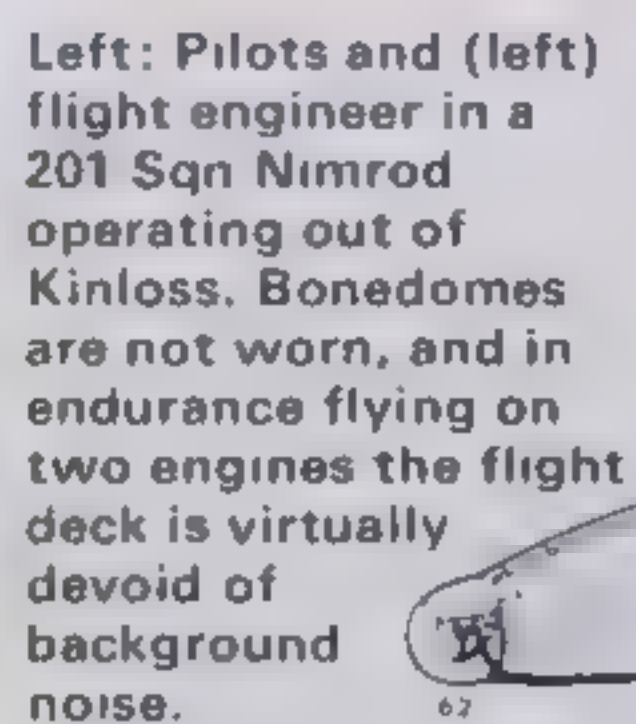
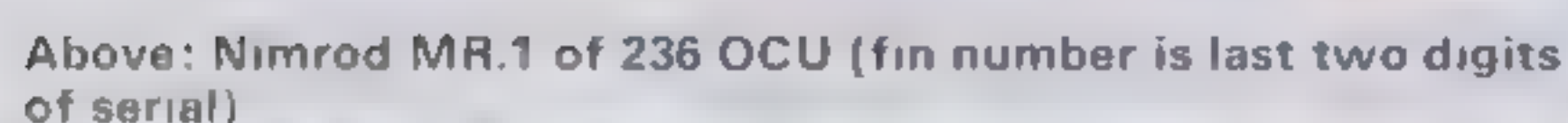
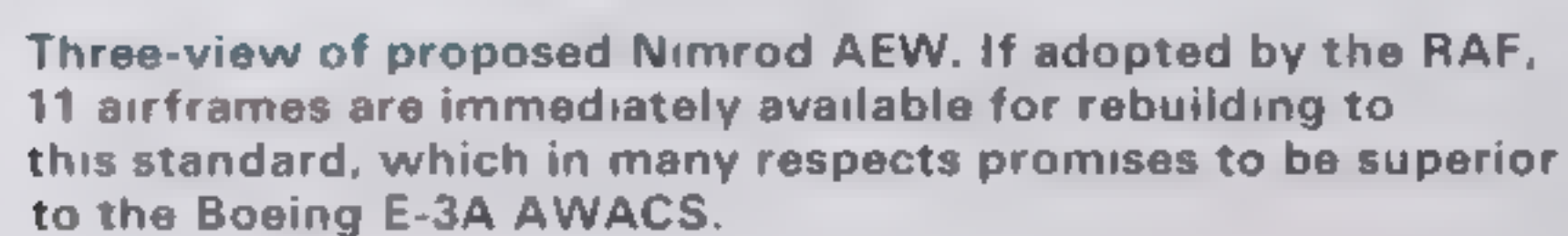
Development: Though it is derived from the Hercules, the Nimrod has proved an outstandingly successful aircraft. The aircraft was tailored to an equipment standard which, though not quite short of the ideal, but still combined very complete (air-to-surface warfare) sensing systems with digital and analog computers, advanced
continued on page 140



Above: Few aircraft have been worked harder than the Nimrod MR.1 squadrons of the RAF. This aircraft taking off from Gibraltar is from 206 Sqn, RAF Kinloss, Morayshire (Scotland). It was participating in NATO Exercise Open Gate.

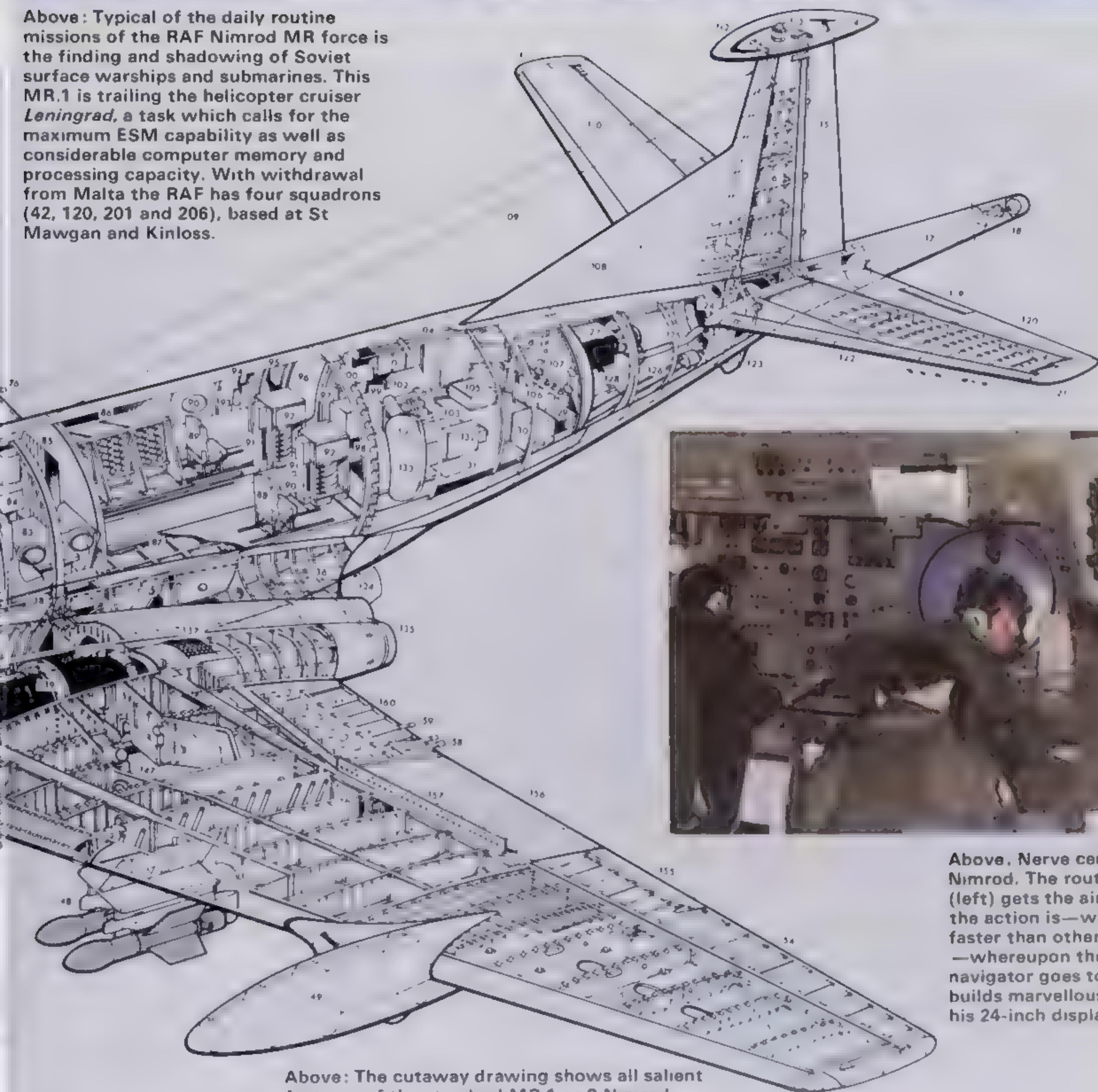
Below: On any number of engines from one upwards, the Nimrod is a splendid aircraft. No other ocean patroller offers such effortless performance, and in 1977-79 the RAF MR.1 force is being updated internally to improve combat effectiveness still further.







Above: Typical of the daily routine missions of the RAF Nimrod MR force is the finding and shadowing of Soviet surface warships and submarines. This MR.1 is trailing the helicopter cruiser *Leningrad*, a task which calls for the maximum ESM capability as well as considerable computer memory and processing capacity. With withdrawal from Malta the RAF has four squadrons (42, 120, 201 and 206), based at St Mawgan and Kinloss.



Above: The cutaway drawing shows all salient features of the standard MR.1 or 2 Nimrod, which in practice seldom carry external weapons such as the small wire-guided AS.12. All MR.1 Nimrods are being recycled back through the Woodford factory to emerge in updated MR.2 form, with EMI Searchwater radar with Ferranti FM 1600D computer, a 920C central processor with 256K words, and the new Marconi-Elliott AQS-901 acoustic system (compatible with Barra sonobuoys) with two more 920C computers

0 Aeron tab
 6 Flap outer section
 7 A stroke 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000



Above. Nerve centre of a Nimrod. The routine navigator (left) gets the aircraft to where the action is—which it can do faster than other ASW aircraft—whereupon the tactical navigator goes to war and builds marvellous pictures on his 24-inch display

Ilyushin Il-38

Il-38

Origin
Type
Engines

Dimensions: Span 122ft 8in (37.4m), length 129

Weights: Empty, approximately 90 000 lb (40.820kg), maximum loaded

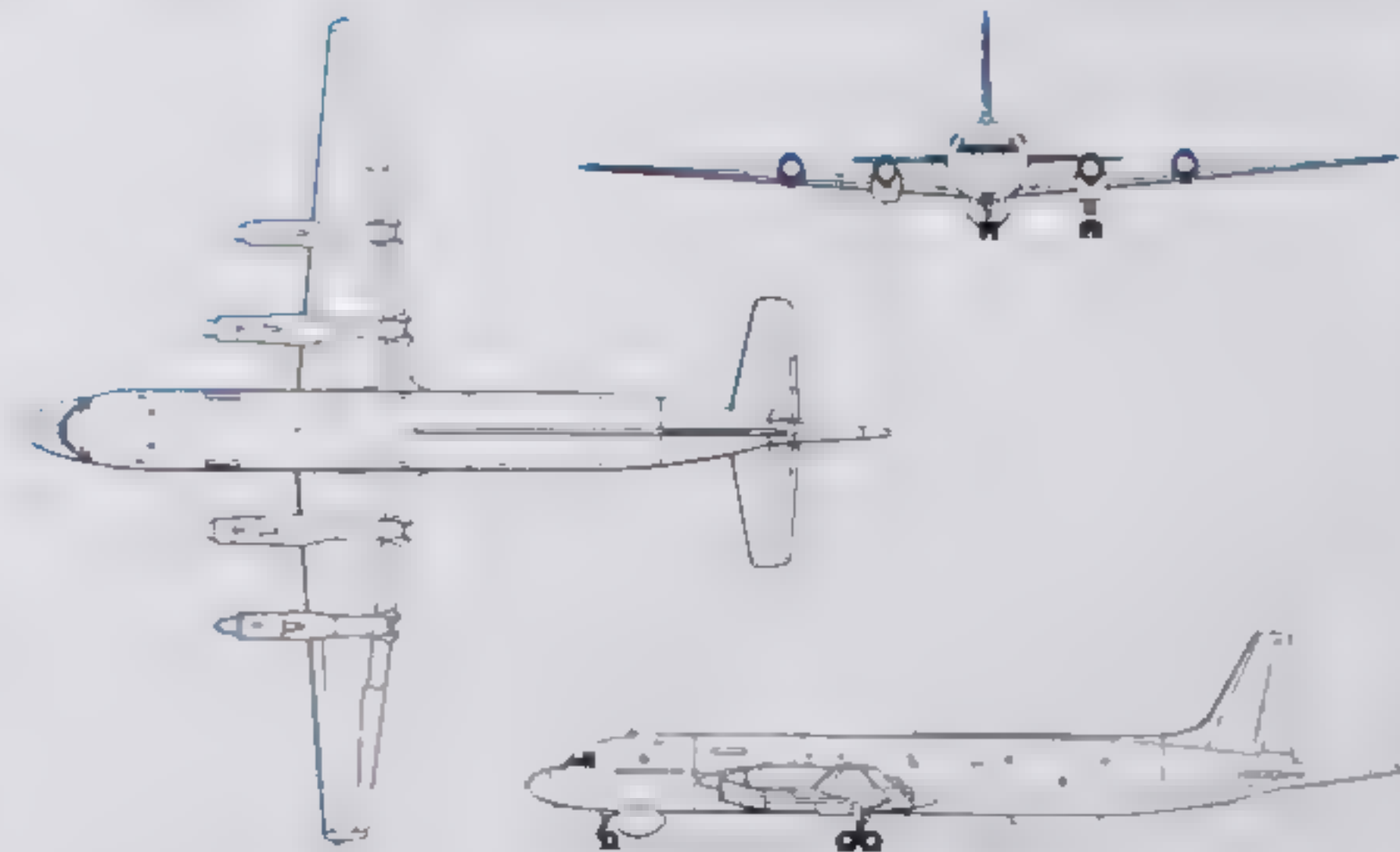
Performance: Maximum speed, about 450mph (724km/h), maximum
climb about 400mph (644km/h), range with typical mission
about 1 000 miles (7240km), endurance about 15hr

Armament: Internal weapon bay ahead of and behind wing accommodat

History: First flight (Il-18 transport) July 1957, first disclosure of Il-38,
1974, by which time it was well established in operational service

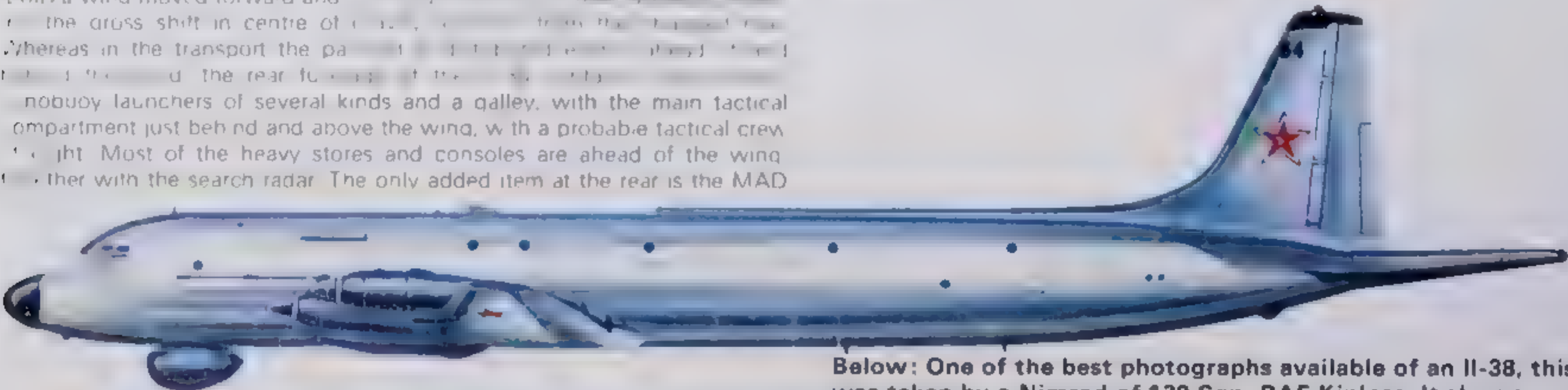
Users: Egypt (or Soviet detachments in Egypt) until 1976, India, Soviet
Union (AV-MF)

Development: Following the example of the US Navy and Lockheed with
the Electra/P-3 Orion transformation, the Soviet Naval Air Arm (AV-MF)
used the Il-18 transport as the basis for the considerably changed Il-38,
known to NATO by the code name of "May". Compared with the transport
it has a wing moved forward and a fuselage extended forward, resulting
in the gross shift in centre of gravity from the transport.
Whereas in the transport the passenger cabin was behind the wing,
in the Il-38 it is moved forward, the rear fuselage is extended
to accommodate the rear fuel tank, the sonobuoy launchers of several kinds and a galley, with the main tactical
compartment just behind and above the wing, with a probable tactical crew
of eight. Most of the heavy stores and consoles are ahead of the wing,
together with the search radar. The only added item at the rear is the MAD



Above: Three-view of Il-38 (NATO name "May").

(magnetic anomaly detector) stinger, not a heavy item. So far little
known of the Il-38 and photographs show few of the items one would
expect to see. There is no weapon bay below the wing and pressurized
fuselage, as in the Nimrod and P-3, no major sensor outlets and aerials and
no apparent external stores pylons. On the other hand the Il-38 is potentially
as good as a P-3, though probably not up to the sensor and computer
standard of the P-3C.



Above: As used by the Soviet AV-MF the Il-38 is a notably
clean aircraft.

Below: One of the best photographs available of an Il-38, this
was taken by a Nimrod of 120 Sqn, RAF Kinloss. It shows a
drogue-stabilized sonobuoy being dropped from medium altitude
by an Il-38 taking part in Atlantic exercises.



Kawasaki P-2J

GK-210, P-2J

Origin: Kawasaki Heavy Industries Gifu, Japan (licence-built under licence from Lockheed Aircraft)

Type: Maritime patrol and anti-submarine aircraft

Engines: Two 2,850hp General Electric T64-10 single-shaft turbo-prop nacle (developed by Ishikawama Harima Heavy Industries Ltd), Ishikawama Harima 137C single-shaft turbojets each rated at 3,085lb (1,400kg) static thrust.

Dimensions: Span over tip tanks 101ft 3½in (30.87m), length 95ft 10¾in (29.32m), height 29ft 3½in (8.93m)

Weights: Empty 42,500lb (19,277kg), maximum loaded 75,000lb (34,019kg)

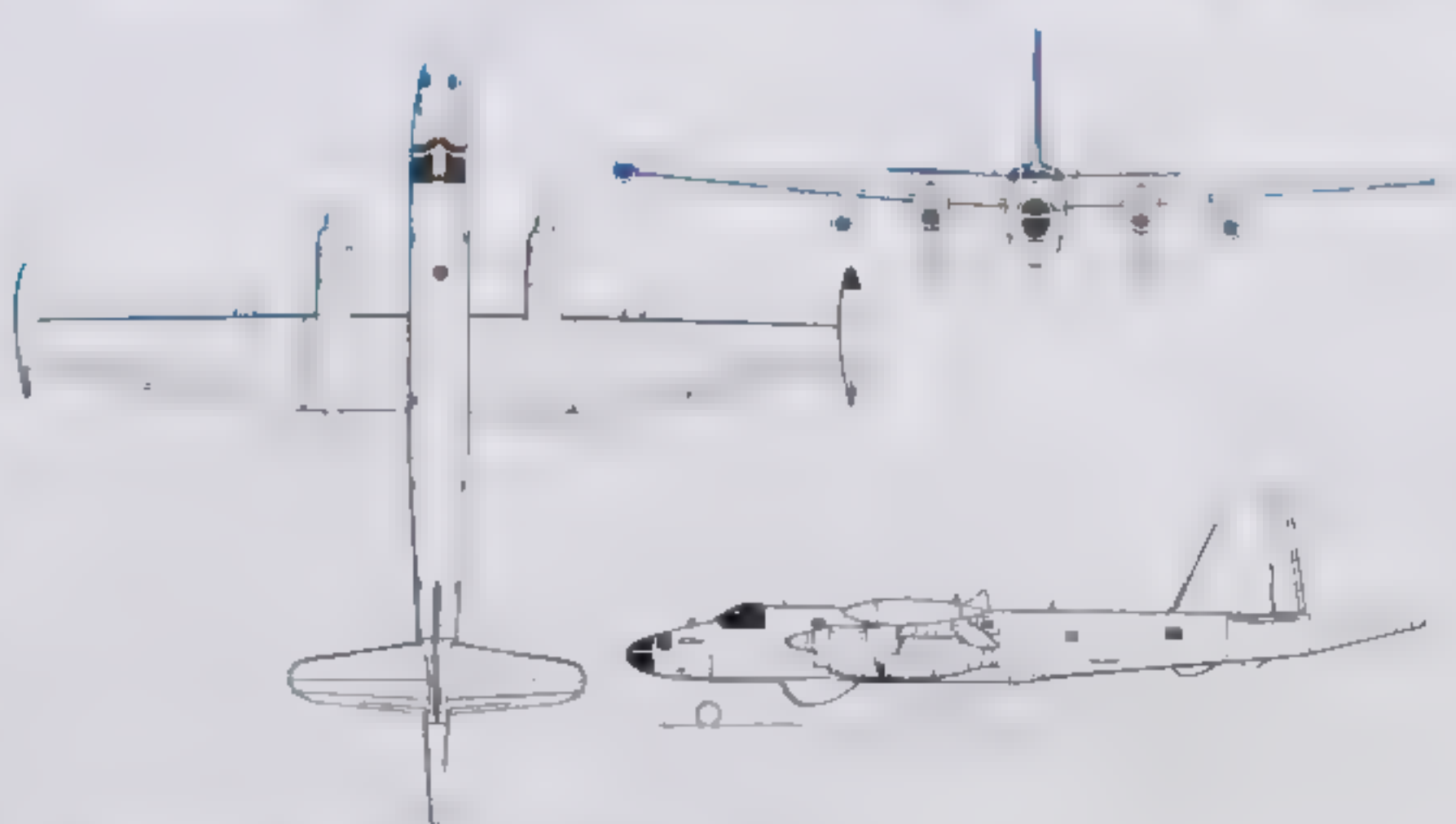
Performance: Maximum dash speed with jets 403mph (649km/h), typical cruising speed 230mph (402km/h), maximum rate of climb 1,800ft (550m)/min, range with maximum fuel 2,765 miles (4450km)

Armament: Internal weapon or sensor load up to 8,000lb (3,630kg) underwing attachments for 16 rockets

History: First flight of original P-2 Neptune 17 May 1945 (P-2J) 21 July 1966 (production P-2J) 8 August 1969 squadron delivery, October 1969 final delivery 1977

User: Japanese Maritime Self-Defence Force

Development: In 1959-65 Kawasaki licence-built 48 Lockheed P-2H Neptunes for the JMSDF, and it was to replace these that the company developed the GK-210 to meet a JMSDF requirement for a more modern machine. In fact the design is based on the P-2H, but the new turboprops were substituted for Turbo-Compound engines in new nacelles housing twin-wheel main gears; the fuselage is lengthened; the vertical tail broadened,



Above: Three-view of Kawasaki P-2J.

a VariCam trimming surface is incorporated in the horizontal tail and the capacity increased. Electronics are greatly updated, though still far below standard by modern yardsticks. Altogether 89 P-2J aircraft have been delivered, and the last may have flown before this book appears. In the 4th Defence Build-up Programme the electronics and mission equipment are being updated, though still using old radar, the outdated Jezebel-Jaweecho ranging system, and navigation, communications and computer technology of the 1960 era.

Below: Most of the planned force of 89 P-2J aircraft have been delivered; about 75 were in service by March 1977. This one belongs to No 3 Sqn, 4 Wing, of the JMSDF.



Lockheed P-2 Neptune

P-2A to H

Origin: Lockheed California
Type
Engines: (P2V-1) 1

me plus two 3 400 b (1540kg) thrust Westinghouse J34 3t
ets (-7, P-2H) same jets but 3 700hp R 3350-32W Turbo Compound
Dimensions: Span (early versions) 100ft (30.48m) (H) 103ft
length (1) 75ft 4in (2 to 4 D) 77ft 10in (23.7m) (E) 81ft
(H) 91ft 8in (27.94m) height 28ft 1in (8.58m), (H) 29ft 4in (8.94m)
Weights: Empty (1) 31 000lb (14 061kg) (3) 34 875lb (15 831kg)
(1) 49 935 lb (22 650kg) maximum loaded (1) 61 153 lb (27,740kg)
64 100lb (29 075kg) (H) 79 895lb (36 240kg)
Performance: Maximum speed (1) 303mph (484km/h)
103mph (648km/h, with jets) initial climb (alt) about 1 200ft (366m)/min,
H 1 800ft/min with jets), service ceiling 25 000-30 000ft depend
range (early models, typical) 4 000 miles (6440km) (H) about
00 miles (4000km)

Armament: See text

History: First flight (XP2V-1) 17 May 1945 service delivery (-1) March
147, (-7, H) 26 April 1954

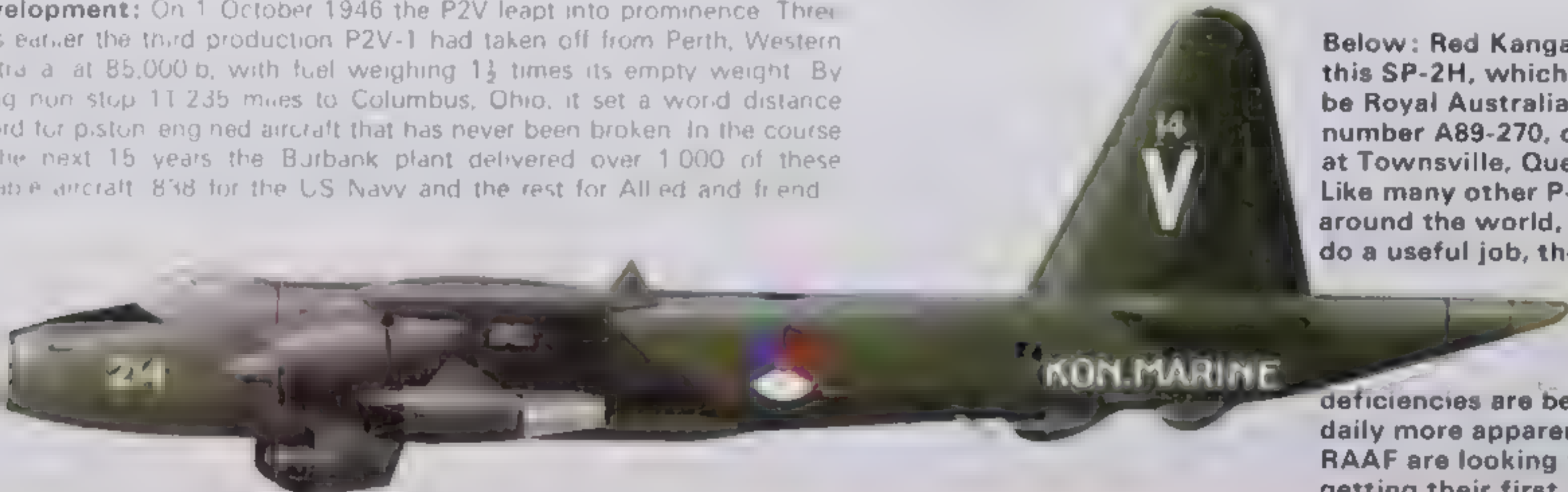
Users: Argentina, Brazil, Chile, France, Japan, Netherlands, Portugal,
USA (Naval Reserve)

Development: On 1 October 1946 the P2V leapt into prominence. Three
days earlier the third production P2V-1 had taken off from Perth, Western
Australia at 85,000 lb, with fuel weighing 1½ times its empty weight. By
flying non-stop 11 235 miles to Columbus, Ohio, it set a world distance
record for piston-engined aircraft that has never been broken. In the course
of the next 15 years the Burbank plant delivered over 1 000 of these
capable aircraft: 838 for the US Navy and the rest for Allied and friend



Above: Three-view of Lockheed SP-2H Neptune.

customers. Early versions had up to six 20mm cannon firing ahead, two
more in a power tail turret and two 0.5-in in the dorsal turret. Gradually
defensive or offensive gun armament was removed, later models having
instead many tons of extra anti-submarine sensing systems, as well as
retaining the 8 000 lb (3629kg) weapon bay. The final models, no longer in
US Navy first line use, were the anti-submarine SP-2H and Antarctic
ski-equipped LP-2H. Various models are used by the countries listed, and
the Japanese P-2J is described separately.



Below: Red Kangaroos adorn
this SP-2H, which happens to
be Royal Australian Air Force
number A89-270, of 10 Sqn,
at Townsville, Queensland.
Like many other P-2 versions
around the world, it can still
do a useful job, though the

deficiencies are becoming
daily more apparent. 10 Sqn,
RAAF are looking forward to
getting their first P-3C Update-
Two Orions before the end of 1977.

Above: SP-2H of 320 Sqn, Royal Netherlands Navy.



Lockheed P-3 Orion

P-3A, -3B and -3C with derivatives

Origin: Lockheed-California Co USA

Type: Maritime reconnaissance and anti-submarine patrol aircraft with a crew of five and tactical crew of five variants see text

Engines: Four Allison T56 single-shaft turboprops, (P-3A) T56-14, 4,500ehp with water injection, (remainder) T56-14, 4,910ehp

Dimensions: Span 99ft 8in (30.37m), length 116ft 10in (35.61m), height 33ft 8½in (10.29m)

Weights: Empty (typical B, C) 61,491lb (27,890kg), maximum loaded 142,000lb (64,410kg)

Performance: Maximum speed 473mph (761km/h), initial climb 1,950ft (594m)/min, service ceiling 28,300ft (8,625m), range 4,800 miles (7,725km)

Armament: Very varied load in bulged unpressurized weapon bay ahead of wing and on ten wing pylons, maximum internal load 7,252lb (3,290kg) can include two depth bombs, four Mk 44 torpedoes, 87 sonobuoys and many other sensing and marking devices, underwing load can include six 2,000lb (907kg) mines or various mixes of torpedoes, bombs, rockets or missiles. Maximum expendable load 20,000lb (9,071kg)

History: First flight (aerodynamic prototype) 19 August 1958, (YP-3A) 25 November 1959, (production P-3A) 15 April 1961, (P-3C) 18 September 1968

Users: Australia, Canada, Iran, New Zealand, Norway, Spain, USA (Navy, Naval Reserve)

Development: In August 1957 the US Navy issued a requirement for an "off the shelf" anti-submarine patrol aircraft derived from an established type, and this was met in April 1958 by Lockheed's proposal for a conversion of the Electra turboprop airliner. The third Electra was quickly modified as an aerodynamic prototype and deliveries of production P-3As began in August 1962. From the 110th aircraft the Deltic system was fitted with more sensitive sensors and improved displays. Four early A models were converted as WP-3A weather reconnaissance aircraft, while others became EP-3A flying special electronic missions. Three As were supplied ex-USN to Spain. The B model introduced more powerful engines without water alcohol injection and many were sold to Australia, New Zealand and



Above: Three-view of Lockheed P-3C Orion

Norway or modified as EP-3B electronic reconnaissance and counter measures with huge "canoe radars" above and below and a radar under the forward fuselage. The completely different P-3C packages into the same airframe a new and more modern tactical system with sensors and weapons controlled by a digital computer. Derivatives include the P-3I for Iran and a variant for Australia. Other versions include the RP-3D for mapping the Earth's magnetic field and special reconnaissance and transport conversions. Production in 1977 was centred upon the P-3C Update with computer memory increased from 65,000 to 458,000 words, a new computer language, receiver for the global Omega navigation system, increased acoustic-sensor capability, tactical displays for two of the sensor stations and other improvements. Canada is buying yet another new version, the CP-140 Aurora, with the ASW systems of the S-3A Viking. Deliveries exceed 450, and output is being increased to deliver to the US Navy 12 in 1977, rising to 16 in 1980 and 24 in 1981.



Above: Lockheed P-3F Orion of Imperial Iranian Air Force.

Below: The P-3C is the current production model of Orion, the favourite maritime-patrol choice of the Western world largely on account of adequate capability and a price below that of the otherwise extremely attractive Nimrod. Despite its propellers the P-3C is a good performer; in 1971 one flown by a US Navy crew set a record at 500.89mph.



Lockheed S-3A Viking

S-3A Viking, US-3A

Origin: Lockheed Co
Type

Engines: 1

Dimensions: Span 68ft 8in (20.93m)

Weights: 1
12 500lb (19 277kg)

Performance: Max

Armament: Six internal weapon bays can house four Mk 46 torpedoes or Mk 82 bombs

Links of 1

History: First flight 21 January 1978

Operational use (VS 41) 20 February 1979

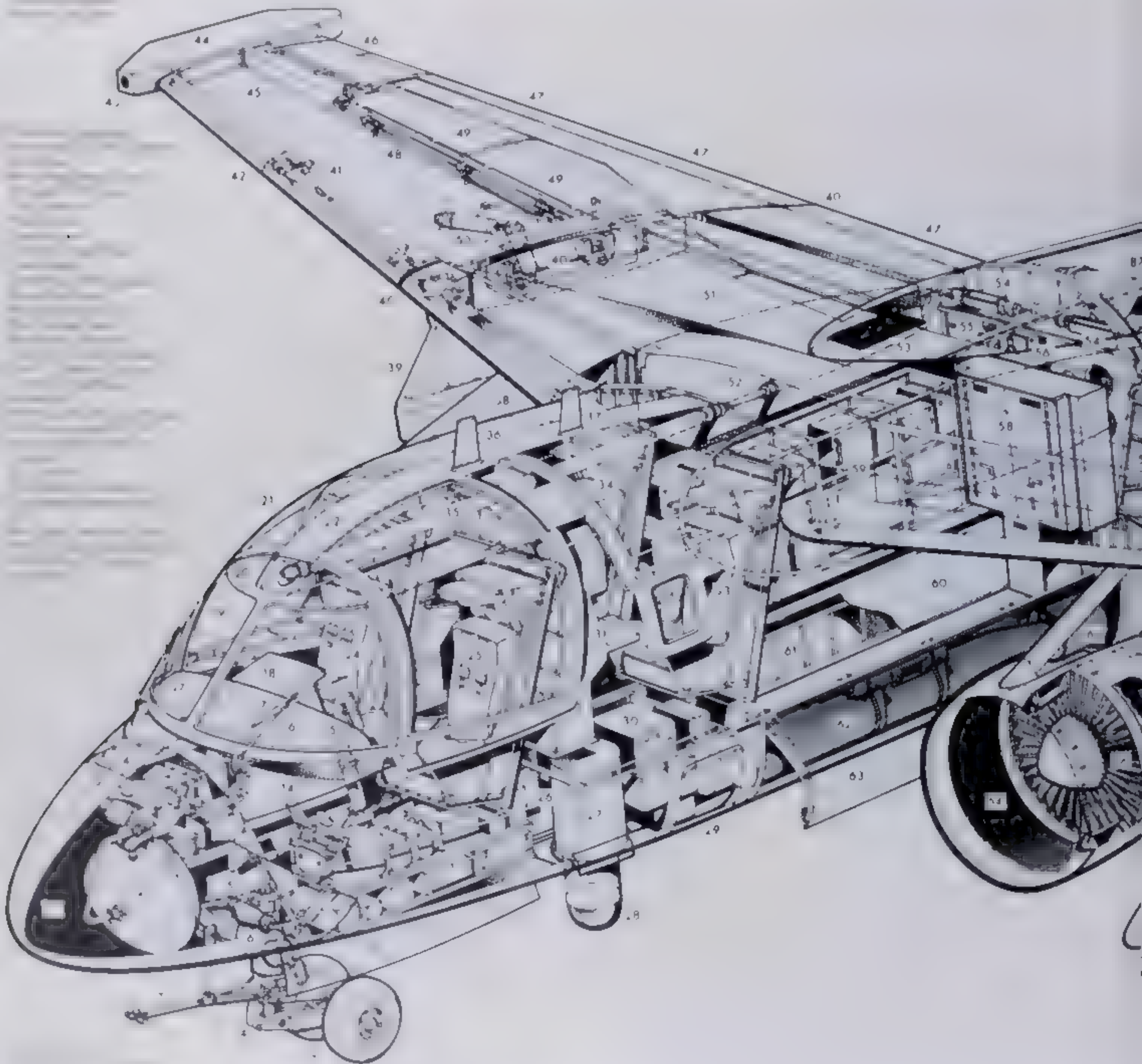
User: USA (Navy) W Germany negotiat

Development: Designed to replace the evergreen Grumman S-2, the S-3 is perhaps the most remarkable example in packaging in the history of aviation

continued on page 148

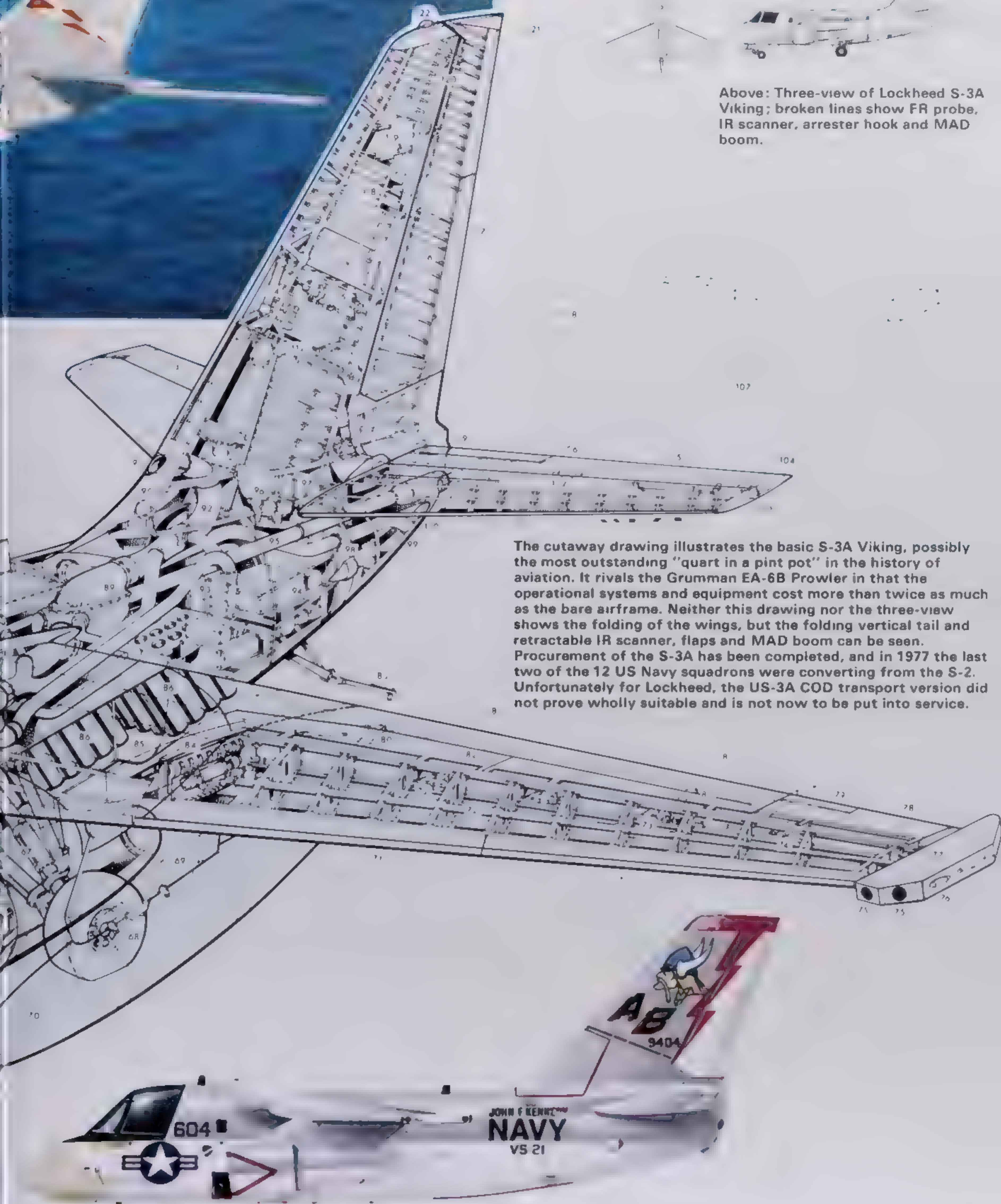


Above: Sniffing its way across the Atlantic rollers, this S-3A Viking belongs to US Navy VS-22 (anti-submarine squadron 22). The squadron is embarked aboard USS *Saratoga*, CVA-60.

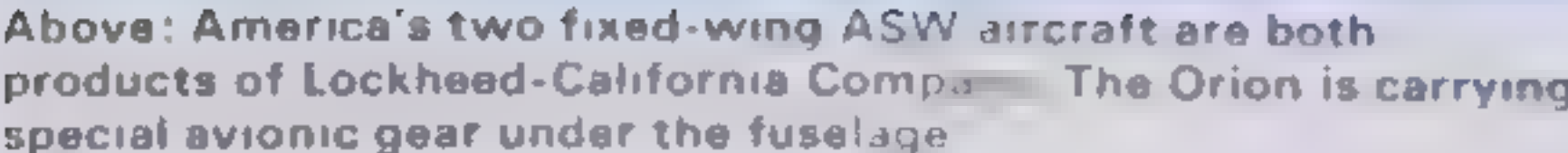
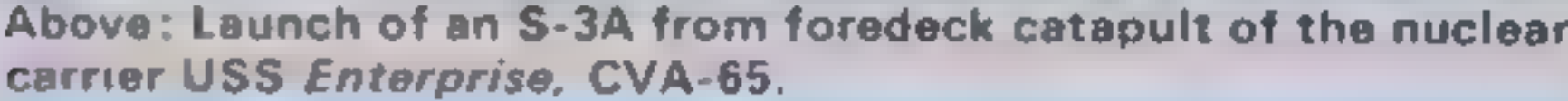
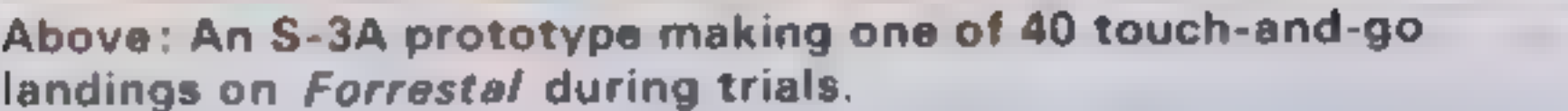




Above: Three-view of Lockheed S-3A Viking; broken lines show FR probe, IR scanner, arrester hook and MAD boom.



Above: An S-3A Viking of VS-21 (USS *John F. Kennedy*).

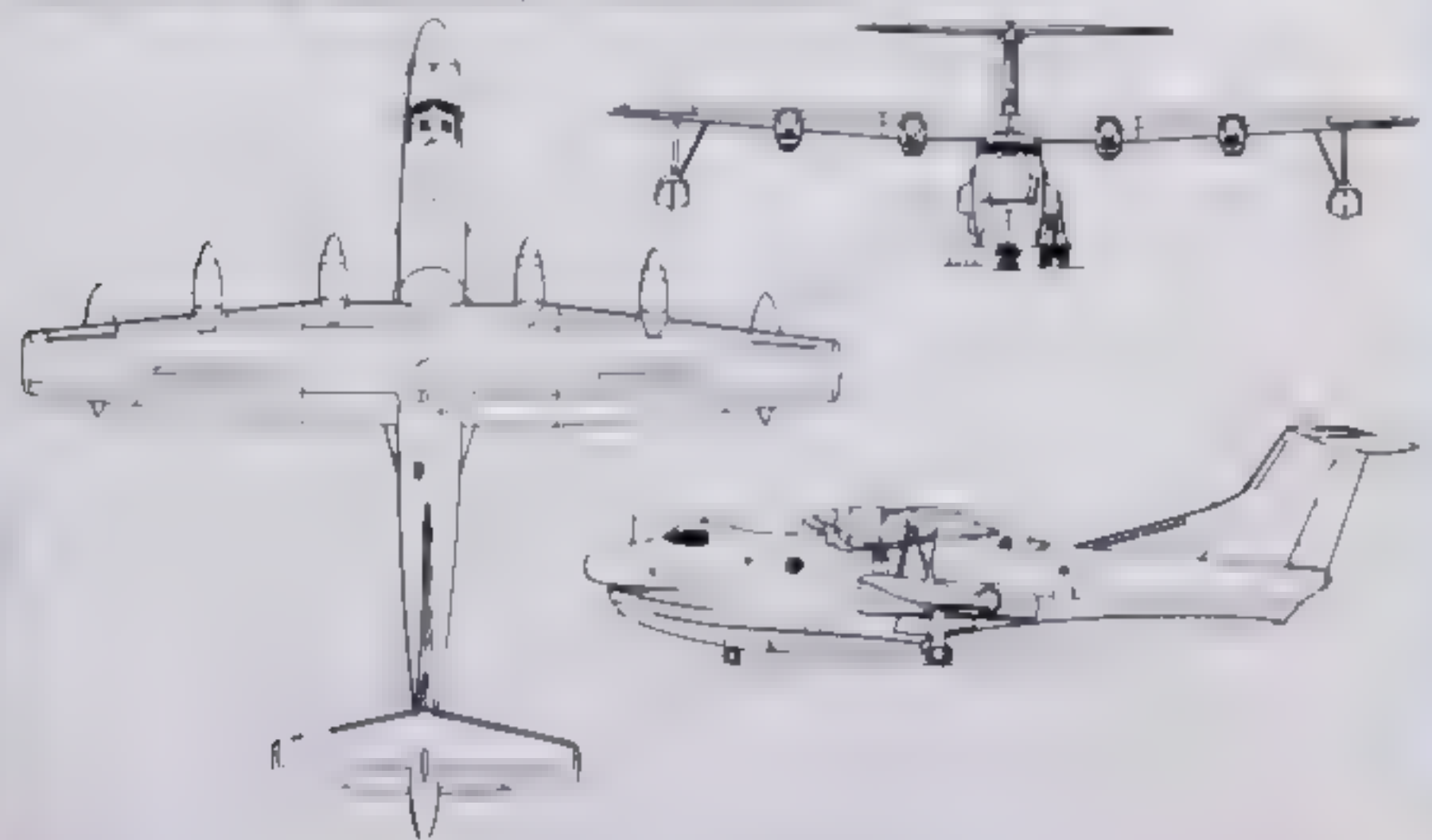


PX-S, SS-2 (PS-1) and SS-2A (PS-1 Kai)

Development: After seven years of study Shin Meiwa were awarded a contract for the prototype of a novel ASW (anti-submarine warfare) flying boat in January 1966. The resulting PX-S demonstrated the technique of flying not much faster than submerged nuclear submarines, using extensive banking over flaps and tail, and of tracking its quarry by repeatedly alighting



The US-1 (PS-1 Kai) is a fixed-wing amphibious aircraft developed by Shin Meiwa. It is a modification of the PS-1, which was developed by the Japanese Maritime Self-Defense Force (JMSDF). The PS-1 was developed in the 1960s and was the first Japanese amphibious aircraft. The US-1 (PS-1 Kai) is a more modern version, developed in the 1970s. It is a fixed-wing amphibious aircraft, meaning it can take off and land on both water and land. It is used for a variety of missions, including search and rescue, transport, and surveillance. The US-1 (PS-1 Kai) is a single-engine aircraft, with a maximum speed of 250 knots. It has a range of 1,000 nautical miles. It is currently in service with the JMSDF.



Above: Three-view of US-1 (PS-1 Kai).

Below: Colourful and quiet, the Shin Meiwa US-1 (PS-1 Kai) is the only fixed-wing air/sea rescue aircraft in production. This example is serving with JMSDF No 71 Sqn at NAS Atsugi.



Above: The two prototypes flying in formation off the Japanese coast. In the background is No 5801, the original PS-1 flying boat. No 9071 is the first PS-1 Kai (PS-1 modified) air/sea rescue amphibian, called SS-2A or US-1 by Shin Meiwa. By March 1977 17 PS-1 and three PS-1 Kai were in service.



Transports and tankers

Both these important categories of military aircraft emerged in the years immediately following World War I, in which transport aircraft were conspicuously absent. Possibly the first military transports were the Vernons delivered to the infant RAF in 1922 after Trenchard and his team—fighting a political war of survival against the older services—had demonstrated that lawless parts of the far-flung Empire could be policed at lowest cost by using small numbers of troops backed up by bomber-transport aircraft. In fact the bomber-transports seldom bombed and the last of this species was the Bombay of 1935, though at the very end of the European part of World War II more than 2,500 heavy bombers of the RAF and USAAF served briefly in the transport role flying out food and bringing back prisoners of war. The US Army Air Service also bought its first transport in 1922; this was the T.2 built by enterprising Anthony Fokker, and in May 1923 it flew non-stop from New York to San Diego, taking longer than a day and a night. Three months later another US Army crew completed the first rudimentary trials in flight refuelling.

During World War II the chief transports were merely derived from airliners, but credit is due to the Germans for building the Me 323 and Ar 232, both of which were members of the totally different family that are tailored to the military role. Fairchild built the C-82, but this missed the war. Douglas built the very large civil DC-7, but PanAm eventually turned it down and it became the C-74 Globemaster (much later there was a different DC-7). The C-74 was pressurized, but for the military role more volume was needed, and the result was the C-124 Globemaster II which had a cavernous unpressurized fuselage whose harsh comforts did not go well with 15hr sectors. The truly great advance was the C-130 Hercules of 1954, which combined adequate space and payload with pressurization, full-section rear loading or para-dropping door, and high speed with turboprop propulsion. Today we are so cosseted we even carp at the comfort of a Herky-bird, but thanks to aggressive "product improvement" by Lockheed-Georgia Company this old-stager "keeps acting newer and newer" and King Hussein just said he would buy some more "because they are such good workhorses". Some 41 other air forces think the same. Loads that will not fit a C-130 can go by a C-5A or An-22 Antei, and there is not much that these big birds cannot carry. Where we go from there depends partly on oil prospecting, opening up Siberia and similar quasi-civil tasks, because air forces do not have the money for development of much larger transports. Who knows, they might turn out to be airships?

Compared with large civil jets the military transport tends to have bigger doors (sometimes a full-section door at each end), less wing sweep and more wheels. This is because it carries bulkier loads and may need to drop them by parachute, has to operate from shorter airstrips, need not cruise so fast, and must be able to land and taxi on unpaved

surfaces. A glimpse of the future is afforded by the AMST YC-14 and YC-15, which were the last word in military transports and have no sweep at all. On the other hand the most numerous fleet of jet transports is the USAF's force of C-135 variants, and these are mainly KC-135s built as tankers and forming an intermediate stage between Boeing's company-funded prototype and the civil 707. Boeing had previously built 888 piston-engined tanker/transports designated C-97 or KC-97 (K being the prefix for a tanker), and these flew too low and too slow to be compatible with jet bombers. Accordingly a mighty force of 732 KC-135 jet tankers was built, with 35° swept wing and, originally, mostly with turbojet propulsion (today often changed to turbofan). Boeing were thereby enabled to go ahead



A Victor K.2 of 232 OCU, RAF, feeds fuel through No 1 hose to a dual-control Harrier T.4. This method of inflight refuelling is applicable to almost any combat aircraft, and an increasing number of air forces are now studying its cost effectiveness.

with the civil 707, which is bigger and heavier. Today large numbers of KC-135s are serving in many special roles, while air forces have also bought civil 707s for duty as transports and tankers. In December 1977 a version of the DC-10 was selected as the USAF's Advanced Tanker/Cargo Aircraft to supplement and replace the KC-135 fleet. Initially, 20 are being bought.

Inflation is a good reason for getting more work out of each aircraft by equipping it to fly more kinds of mission. Canada studied the 707 as a long-range ocean patrol aircraft, overland surveillance platform, transport, tanker and anti-submarine warfare aircraft, before deciding to use 707 transport/tankers and ocean patrol Orions. Unlike the Orion, the British Nimrod can do a substantial transport job with minimal interior rearrangement. Iran has even

put refuelling pods on military 747s. Except for certain parts of the US Air Force, which uses the Flying Boom method in which a boom operator fires a telescopic pipe into a receptacle on the receiver aircraft, military forces all use the British-developed probe/drogue method of flight refuelling which among other advantages allows one tanker to top-up three receiver aircraft simultaneously. A secondary role for transport-type aircraft is to serve as parent for launching and controlling drones and RPVs (remotely piloted vehicles). There are many other duties; for example, some transports are equipped (or can be equipped in a few hours) as a complete surgical hospital, with helicopter to collect casualties, the whole hospital and staff being able to fly at once to the scene of any large natural disaster.



Aeritalia G222

G222

Origin Italian military transport aircraft

Type Military transport aircraft

Engines 4x Turbo-propeller engines

Dimensions 27.5m (90ft 6in) wingspan, 20.5m (67ft 3in) length, 10.5m (34ft 6in) height

Weights: Empty 32 165lb (14 590kg), maximum useful load 19 000lb (8 600kg)

Performance Maximum speed 350kph (217mph), range 2 500km (1 550mi), service ceiling 10 000m (32 800ft), climb rate 10 000ft/min (3 000m/min)

Armament None

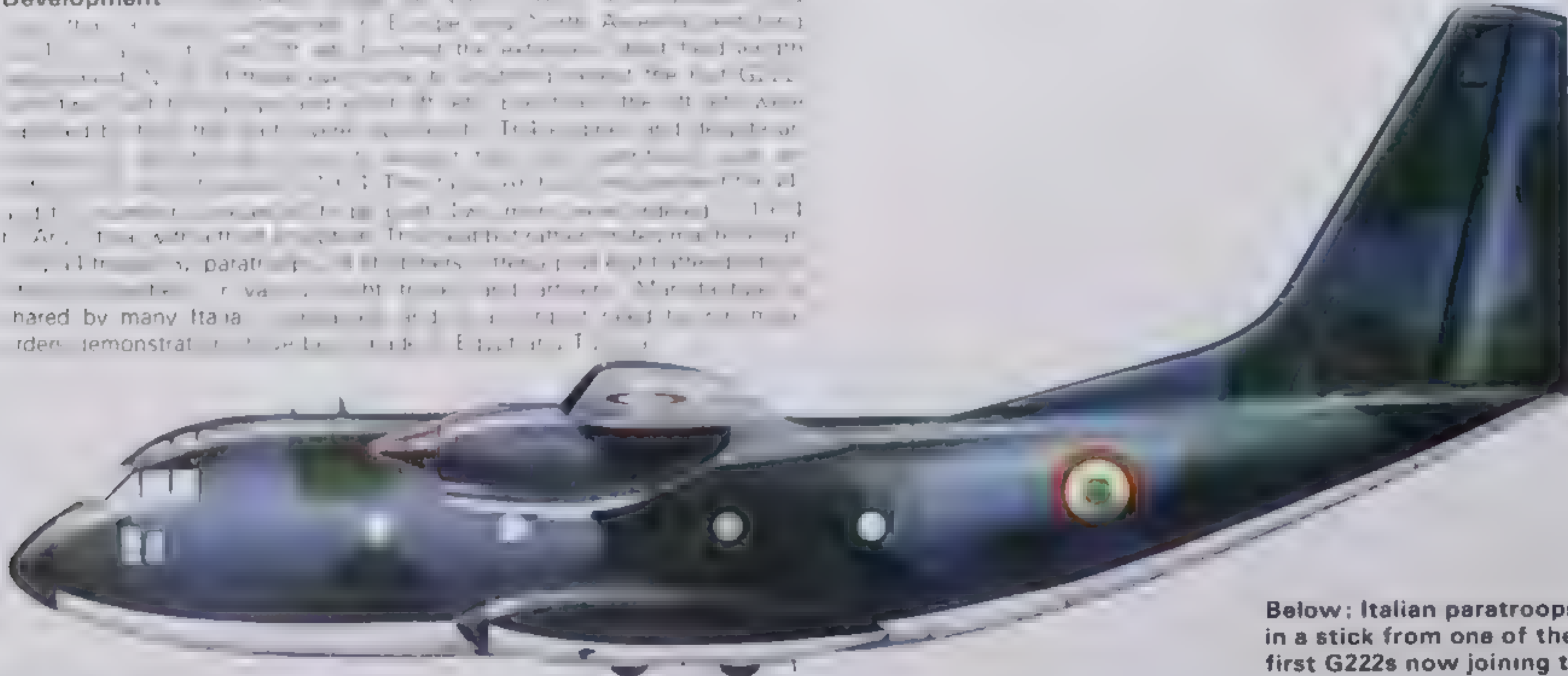
History: First flight 18 July 1970, service delivery December 1970

Users: Argentina, Dubai, Italian Air Force

Development The G222 was developed by Aeritalia, an Italian aircraft manufacturer, in cooperation with the Italian Air Force. It was designed as a military transport aircraft, capable of carrying troops and cargo, and was intended to replace the C-130 Hercules in the Italian Air Force. The G222 was first flown in 1970 and entered service in 1971. It was the first Italian aircraft to be developed and built entirely in Italy. The G222 was designed to be a versatile aircraft, capable of operating from short, unimproved runways. It has a high-wing configuration, which allows it to operate from rough, unimproved runways. The G222 has a large cargo door in the rear fuselage, which can be opened to allow cargo to be loaded or unloaded. It also has a large cargo door in the side of the fuselage, which can be opened to allow troops to enter or exit the aircraft. The G222 is a very successful aircraft, and it has been used by many Italian military units. It has also been used by other countries, including Argentina and Dubai. The G222 is a very versatile aircraft, and it is capable of performing a wide range of tasks. It is a very reliable aircraft, and it has a long service life. The G222 is a very important aircraft for the Italian Air Force, and it is a very successful aircraft.

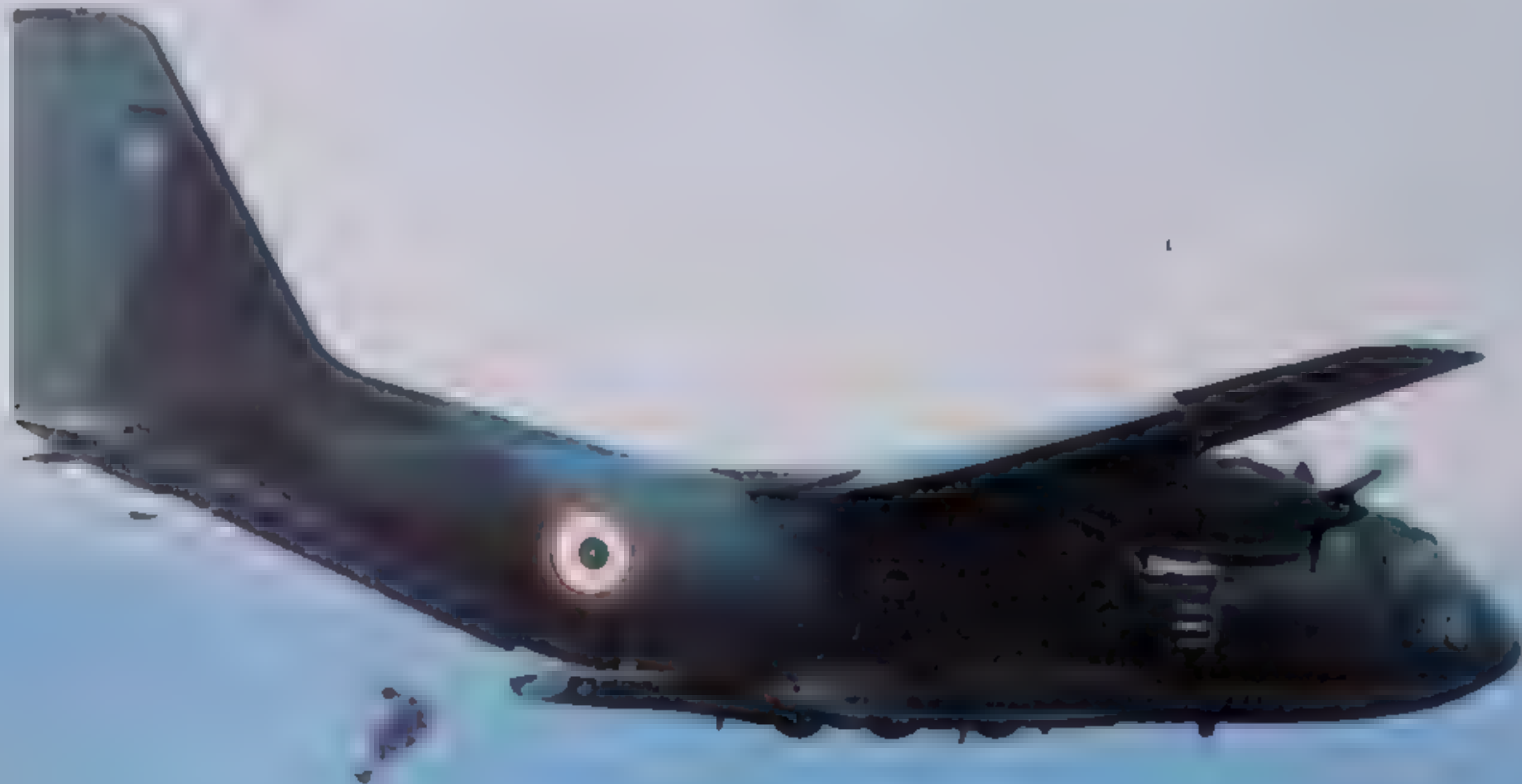


Above: Three-view of G222 production aircraft.



Above: G222 of Aeronautica Militare Italiano.

Below: Italian paratroops leave in a stick from one of the first G222s now joining the Aeronautica Militare Italiano.

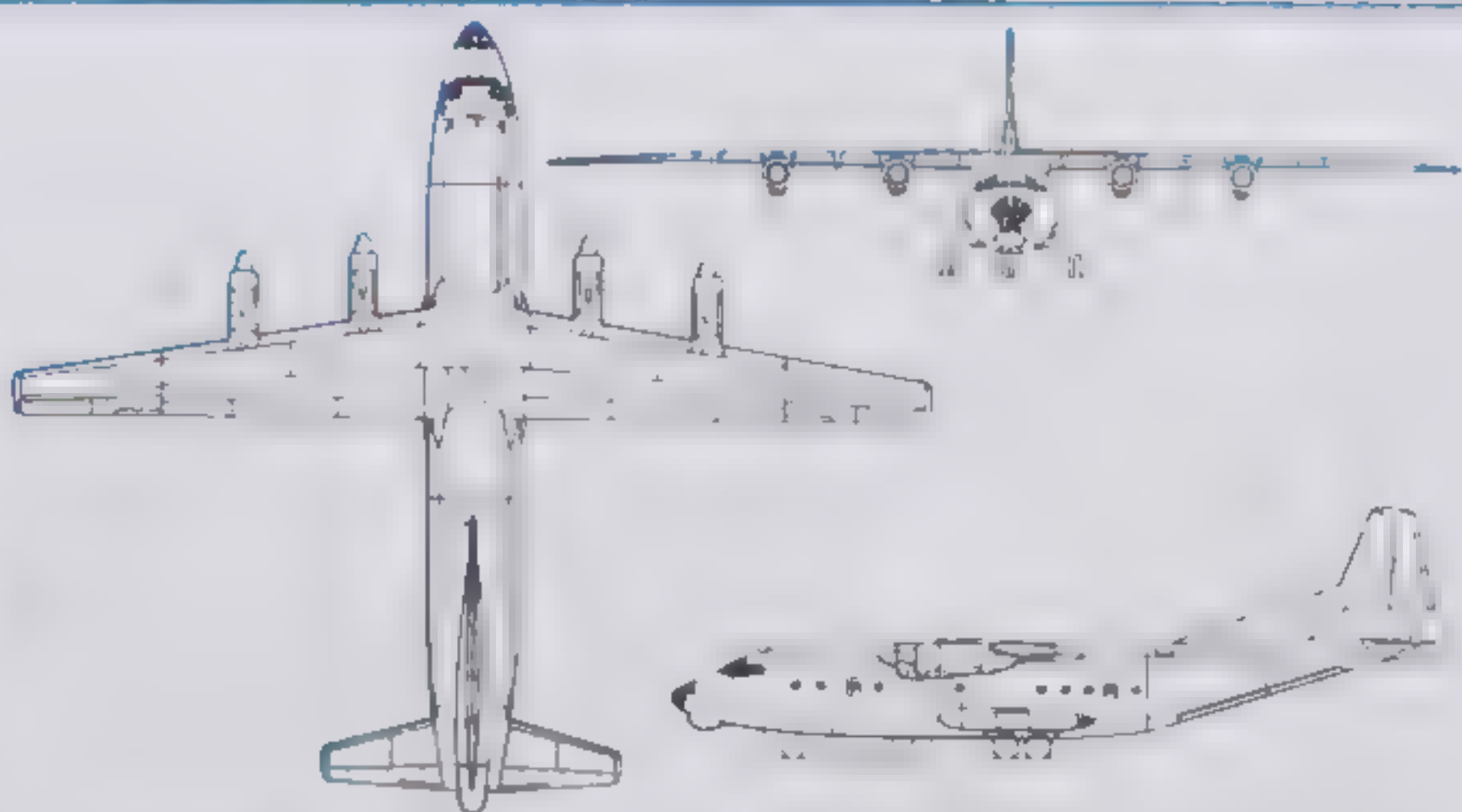


Antonov An-12

An-12; NATO code name "Cub"

Origin: Design bureau of Oleg K. Antonov, Kiev, Soviet Union
Type: Paratrooper, cargo and freight transport
Engines: Four 4000hp Ivchenko AI-26K turbo-propellers
Dimensions: Span 44ft 6in (13.5m) length 121ft 4in (37.3m) height 32ft 3in (9.83m)
Weights: Empt, 61,750lb (28,000kg) loaded 121,475lb (55,100kg)
Performance: Maximum speed 482mph (777km/h) maximum cruise speed 416mph (668km/h) maximum rate of climb 1,377ft (420m) per min service ceiling 35,600ft (10,900m) range 4,000 miles (6,437km) 2,300 miles (3,700km)
Armament: Powered tail turret with two 23mm NR-23 cannon
History: First flight (civil An-10) 1957, (An-12) believed 1958
Users: Algeria, Bangladesh, Egypt, India, Indonesia, Iraq, Poland, Soviet Union, Sudan, Yugoslavia

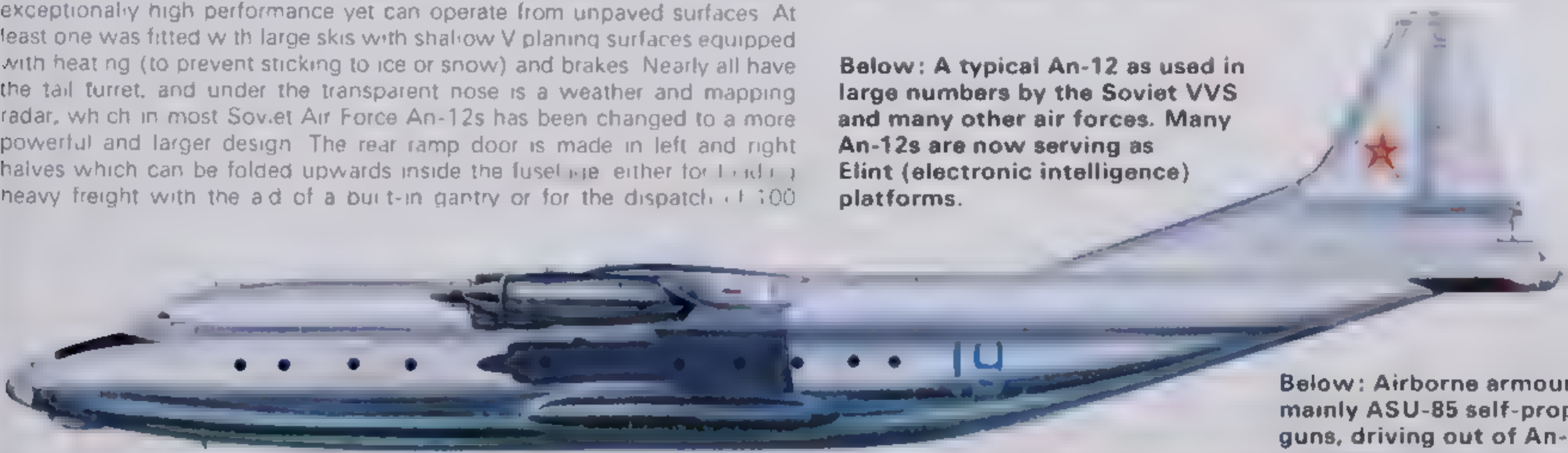
Development: In 1958 Antonov flew a large twin-turboprop which owed something to German designs of World War II and the C-130. From this evolved the An-10 airliner and the An-12, which since 1960 has been a standard transport with many air forces. Fully pressurised, the An-12 has an exceptionally high performance yet can operate from unpaved surfaces. At least one was fitted with large skis with shallow V planing surfaces equipped with heating (to prevent sticking to ice or snow) and brakes. Nearly all have the tail turret, and under the transparent nose is a weather and mapping radar, which in most Soviet Air Force An-12s has been changed to a more powerful and larger design. The rear ramp door is made in left and right halves which can be folded upwards inside the fuselage either for loading heavy freight with the aid of a built-in gantry or for the dispatch of 100



Above: Three-view of Antonov An-12 transport version

paratroops in less than one minute. Typical freight load is 44,090 (20,000kg) and the An-12 can carry all Soviet APCs, the ASU-85 SP-gun and such anti-aircraft vehicles as the ZSU-23-4 and SA-6 missile carrier. In manoeuvres of Warsaw Pact forces as many as 30 of these capable aircraft have landed at one airstrip in simulated battle conditions.

Below: A typical An-12 as used in large numbers by the Soviet VVS and many other air forces. Many An-12s are now serving as Elint (electronic intelligence) platforms.



Below: Airborne armour, mainly ASU-85 self-propelled guns, driving out of An-12 transports during an exercise



Antonov An-22 Antei

An-22 Antei (Antheus) (NATO name "Cock")

Origin: The design bureau of Oleg K. Antonov, Soviet Union
Type: Heavy, logistic transport
Engines: Four 15,000shp Kuznetsov NK-12MA single shaft turboprops
Dimensions

Weights

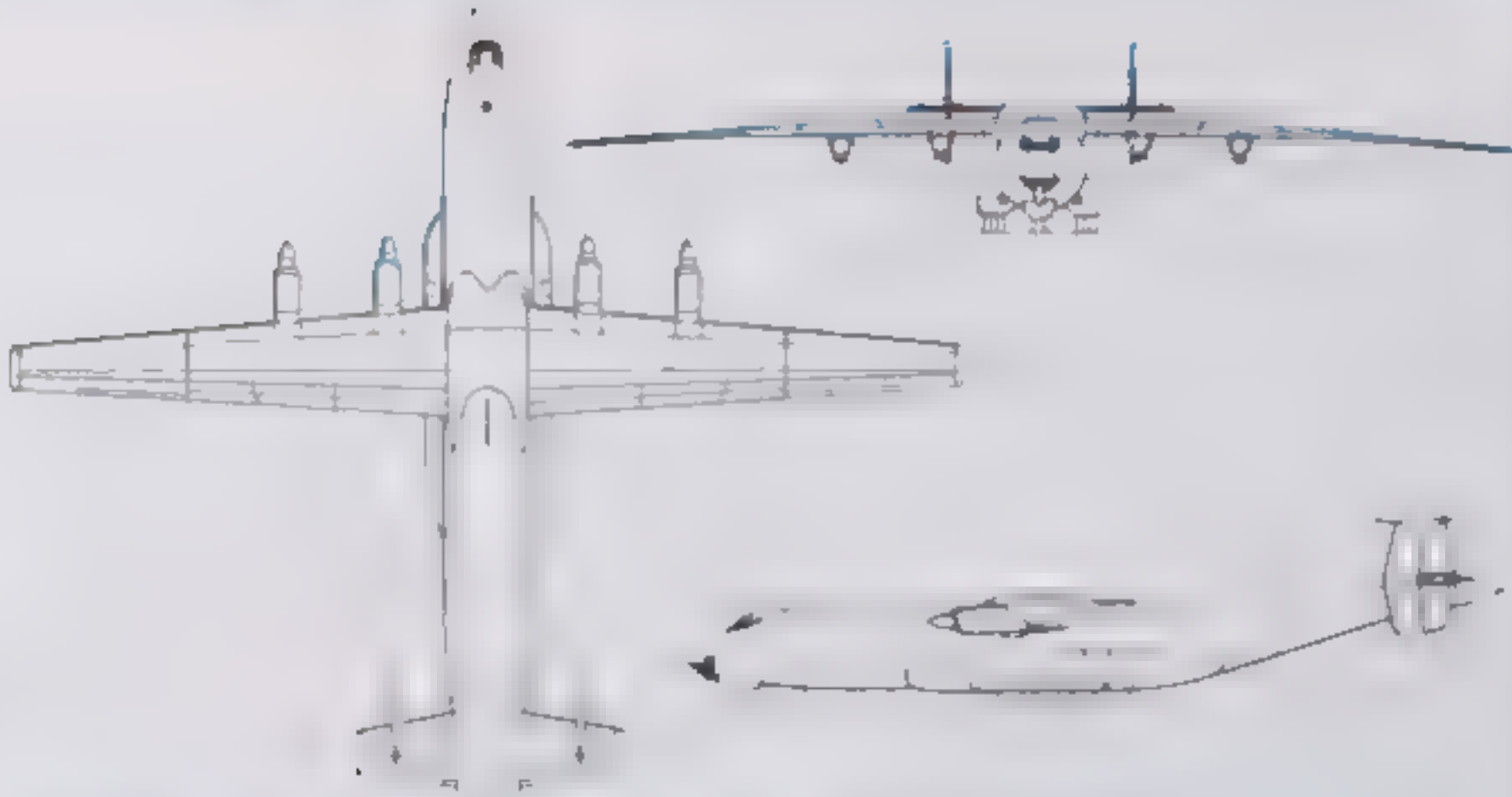
Performance

History: First flight 27 February 1975
User

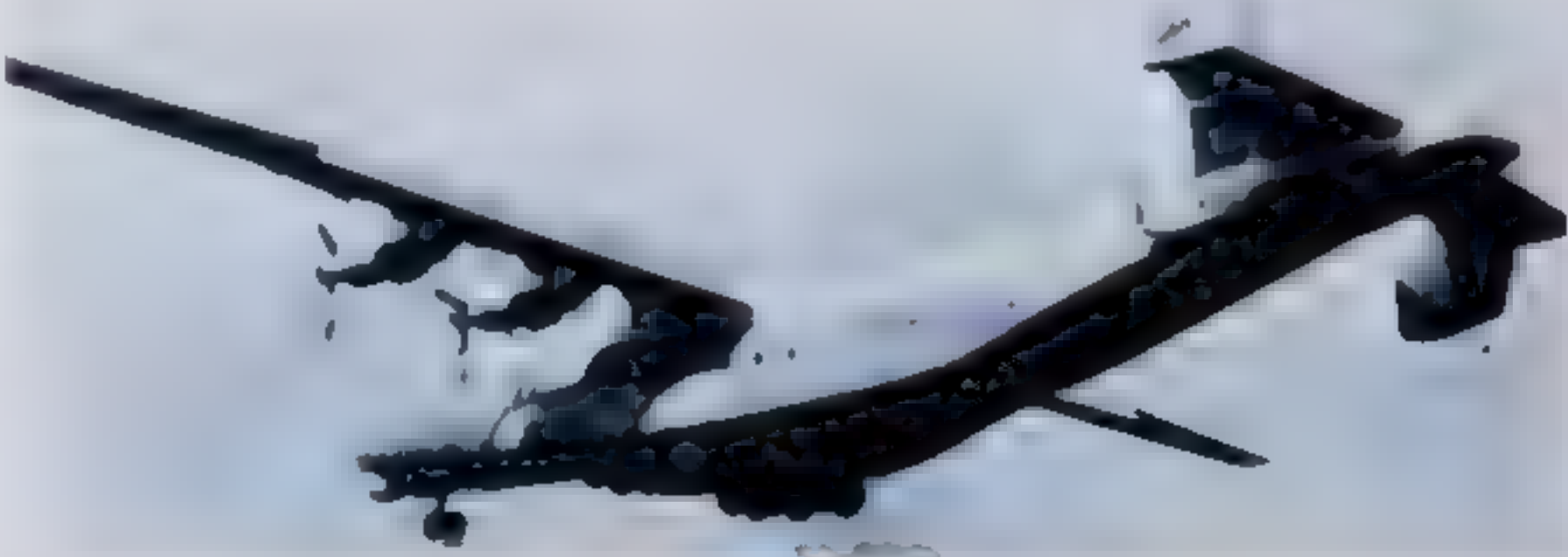
Development

The aircraft is designed to carry up to 29 passengers aft of the flight deck, 1ft 5in (4.40m) square in section, with

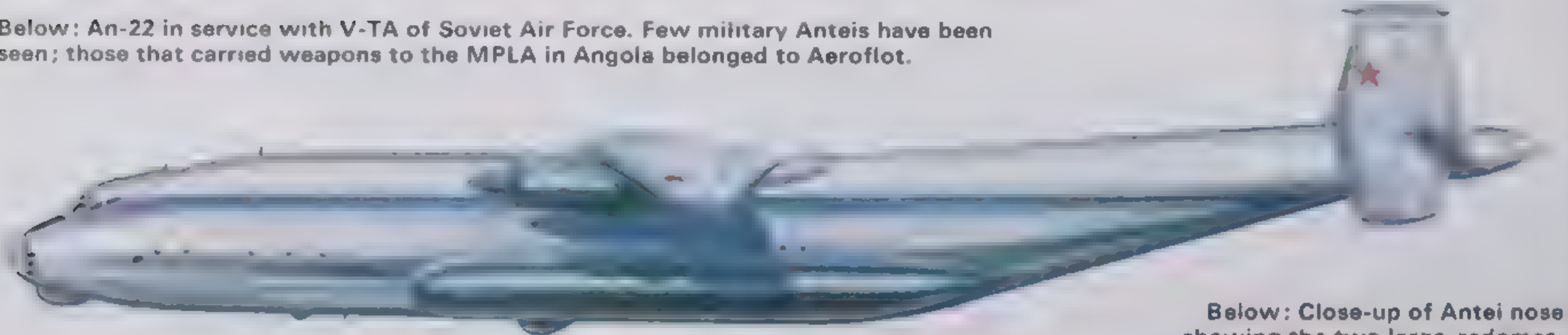
Below: An-22 in service with V-TA of Soviet Air Force. Few military Anteis have been seen; those that carried weapons to the MPLA in Angola belonged to Aeroflot.



Above Three-view of An-22 (note wing anhedral)



Above: Civil An-22 of Aeroflot seen on final approach. It is not a STOL aircraft.



Below: Close-up of Antei nose showing the two large radomes.



Antonov An-26

An-24V, 24RV, 24T, 26 and 30
(NATO names: (24) "Coke", (26) "Curl")

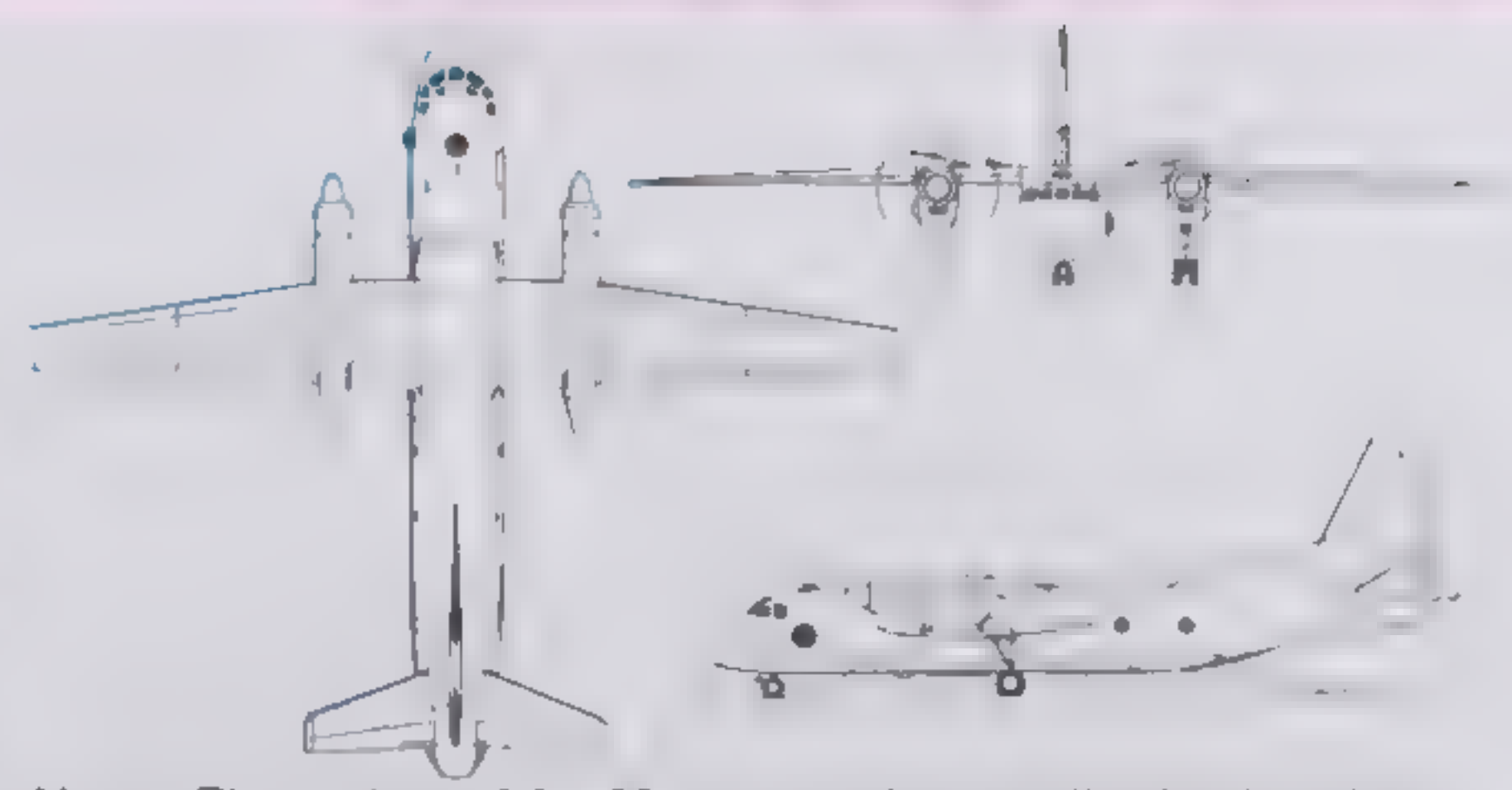
Origin: The design bureau of Oleg K. Antonov, Soviet Union
Type: Low-wing, four-engine turboprop transport aircraft

Engines: Two or four turboprop engines (An-24V, 24RV, 24T, 26 and 30 have two engines; An-24RV, 24T, 26 and 30 have four engines)
Dimensions: Span 95ft 9½in (29.20m), length overall (24) 77ft 2½in (23.53m), (26) 78ft 1in (23.80m), (30) 79ft 7in (24.26m), height (24, 30) 27ft 5½in (8.39m), (26) 28ft 1½in (8.575m)

Weights: Empty (24V) 29,320lb (13,300kg), (26) 33,115lb (15,000kg), maximum gross (24V, T, 4n3) 22,000kg (48,500lb), (26) 24,000kg (52,900lb), (30) 24,000kg (52,900lb)
Performance: Cruise speed (24) 341 mph (549 km/h), max payload of 12,125lb (5,500 kg), reserves (24V) 341 mph (549 km/h), (24T) 397 miles (640km), (26) about 400 miles (645km)

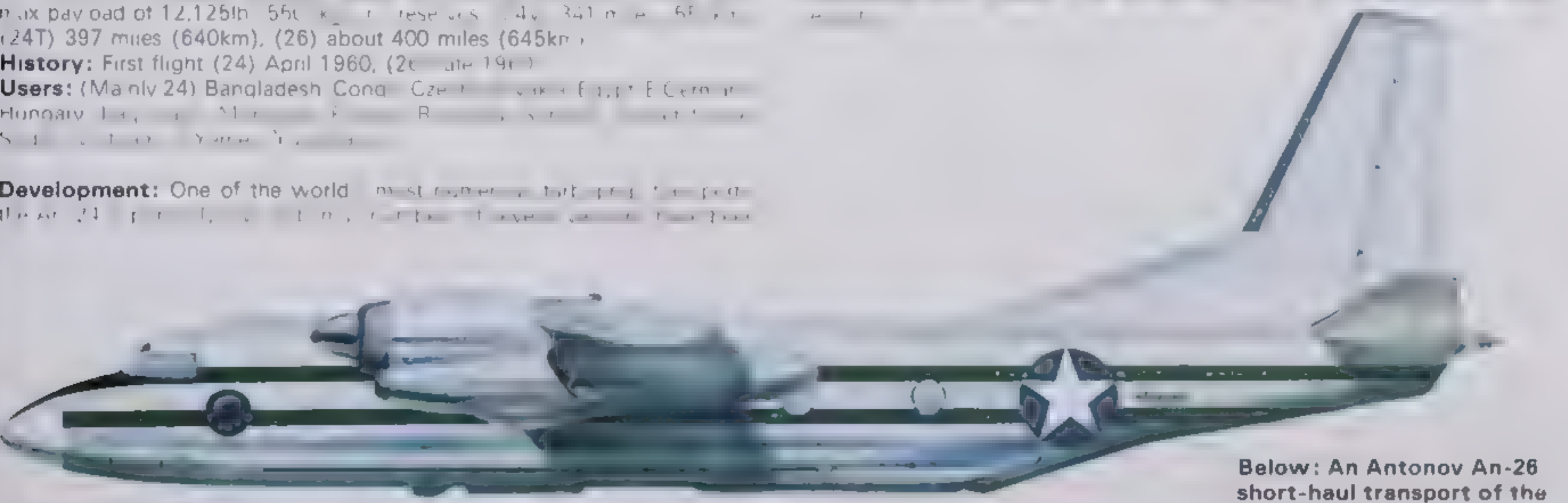
History: First flight (24) April 1960, (26) late 1960
Users: (Mainly 24) Bangladesh, Congo, Czech Republic, East Germany, Hungary, Iraq, Iran, Libya, Romania, Soviet Union, Syria, Vietnam, Sudan, Yemen, Yugoslavia

Development: One of the world's most common turboprop transport aircraft, the An-26 is a four-engine, turboprop transport aircraft.



Above: Three-view of An-26; note engine nacelles in plan view

The An-26 is a four-engine turboprop transport aircraft. It is a low-wing, four-engine turboprop transport aircraft. The aircraft is designed for short-haul transport and is capable of carrying up to 20 passengers. It is also capable of carrying up to 12,125 lb (5,500 kg) of payload.



Above: An An-26 in service with the Somali Aeronautical Corps.

Below: An Antonov An-26 short-haul transport of the Yugoslav Air Force.



Boeing C-135 family and military 707

C-135 Stratolifter family, KC-135 Stratotanker family (data KC-135A)

Origin
Type
Engines: Four 13 750.b (6238kq) thrust (water injection rating) P

Dimensions: Span 130ft 10in (39.7m) length 136ft 3in (41.0m) height

Weights: Empty 109 000.b (49 442kq) loaded 297 000lb (134 715kq)
Performance: Maximum speed 600mph (960km/h) cruising speed 400km/h service ceiling 40 000ft (12 192m) typical range

Armament
History: First flight (civil 367-80) 15 July 1954 (KC-135) 31 August 1956 final delivery (KC-135) mid 1961
Users: (C-135/variants) France US Air Force (C-137 and other 707-320 variants) Argentina Canada Egypt W Germany Iran Israel Portugal Saudi Arabia Taiwan US Air Force

Development: Boeing jet transports all stemmed from a company-funded prototype, the 367-80 flown in July 1954. After evaluation of this aircraft the US Air Force decided in October 1954 to buy 29 developed versions to serve in the dual roles of tanker for Strategic Air Command and strategic transport for MATS (later Military Airlift Command). In fact, though all KC-135 Stratotankers can serve as transports, they have seldom had to do so, because production of C-135 Stratolifters has been considerable. The Boeing 717 tanker introduced a new high-speed Boeing Flying Boom mounted under the rear fuselage. This is aimed by the aircraft then the boom is extended and fuel is pumped to the receiver aircraft. Total capacity is about 26 000 Imp gal (8 000 litres) of cargo. Between June 1954 and January 1965 no fewer than 732 were delivered to SAC. Another 88 Model 739 aircraft have been supplied as various models of C-135, most having 18 000 b (8165kq) thrust TF33 fan engines. At least 28 species of modifications are in USAF service, including VC-135 VIP aircraft, command posts and more than 21 types of 135 used for electronic intelligence, counter-

refuel the nuclear Mirage IVA *force de frappe* of the Armée de l'Air, these having booms modified for probe-drogue refueling. Though not armed with guns or bombs, the huge and varied fleets of 135s are very much at the heart of aircraft in the context of an electronically-based form of warfare and will continue to pry, probe and spoof – and feed fuel to thirsty fighters below bingo – for many years.

In comparison with C-135 models the 707-320 is longer and fatter fuselage, bigger wing, bigger engines. The USAF flies various models of 707-320, the 89th Military Airlift Wing at Travis AFB, California, as the personal long-range transport of the President. The Canadian Armed Forces CC-137 fleet is the only one refueling by boom. The 12 707-319Cs have optional boom installation on the rear fuselage.



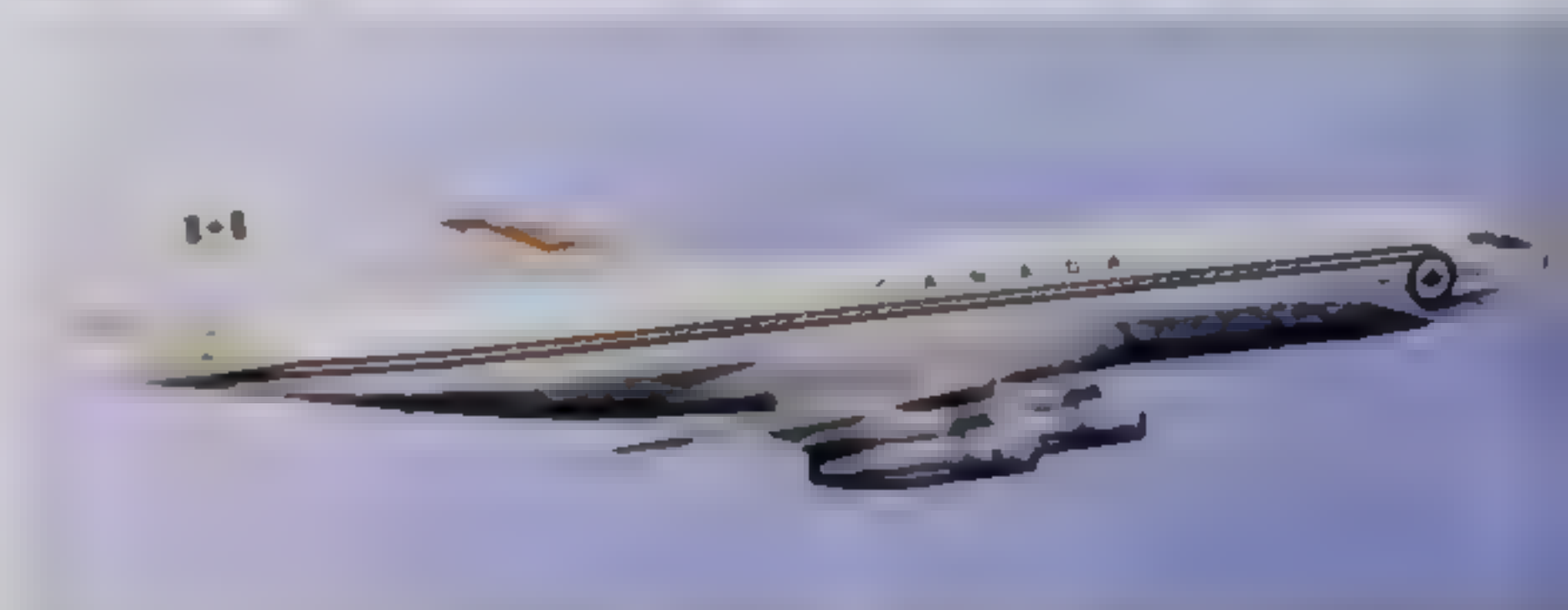
Above: Three view of KC-135B with TF33 turbofans



Above: During the SE Asia involvement, F-4E Phantoms cluster round a nice fat KC-135A like minnows round a whale.



Above: This VC-135B is a special-missions VIP transport but smaller than "Air Force One", the US President's VC-137C.



Above: This CC-137 of the Canadian Armed Forces is the only example pictured here of the larger and heavier 707-320 family.



Above: Selected from numerous special electronic and test versions, this is an EC-135N used for space range instrumentation.

Facing page, upper: One of the 12 KC-135F tanker transports with boom/drogue systems used by the French Armée de l'Air

Facing page, lower: The third KC-135, built in 1956, is serving as an EC-135A command post with the 9th Strategic Recon Wing (SR-71) at Beale AFB. Here it lands at Kadana AFB, Japan



Boeing C-97 Stratofreighter

C-97 to KC-97L (data for -97G)

Type
Engines

Dimensions

Weights
Performance

Armament
History

Users

Development

The Boeing C-97 Stratofreighter is a four-engine, propeller-driven military transport aircraft. It was developed by Boeing in the late 1940s and entered service with the United States Air Force in 1951. The aircraft is a modified version of the Boeing 707 commercial jetliner, with the fuselage stretched to accommodate a larger cargo load. It features a high-wing configuration, a T-tail, and four Pratt & Whitney R4800 turboprop engines. The C-97 is capable of carrying up to 119 passengers or 158 paratroopers, and it has a maximum range of over 3,000 miles. It was widely used during the Vietnam War and the Korean War, and it remains in service with several air forces around the world.



Three-view of the KC-97G differs little from the 1944 prototype

The KC-97G is a four-engine, propeller-driven military transport aircraft. It was developed by Boeing in the late 1940s and entered service with the United States Air Force in 1951. The aircraft is a modified version of the Boeing 707 commercial jetliner, with the fuselage stretched to accommodate a larger cargo load. It features a high-wing configuration, a T-tail, and four Pratt & Whitney R4800 turboprop engines. The KC-97G is capable of carrying up to 119 passengers or 158 paratroopers, and it has a maximum range of over 3,000 miles. It was widely used during the Vietnam War and the Korean War, and it remains in service with several air forces around the world.

Below: The jet-assisted KC-97L tanker/transport of the Air National Guard often deploy to overseas bases. This one belongs to Illinois ANG (previously assigned to Minnesota)

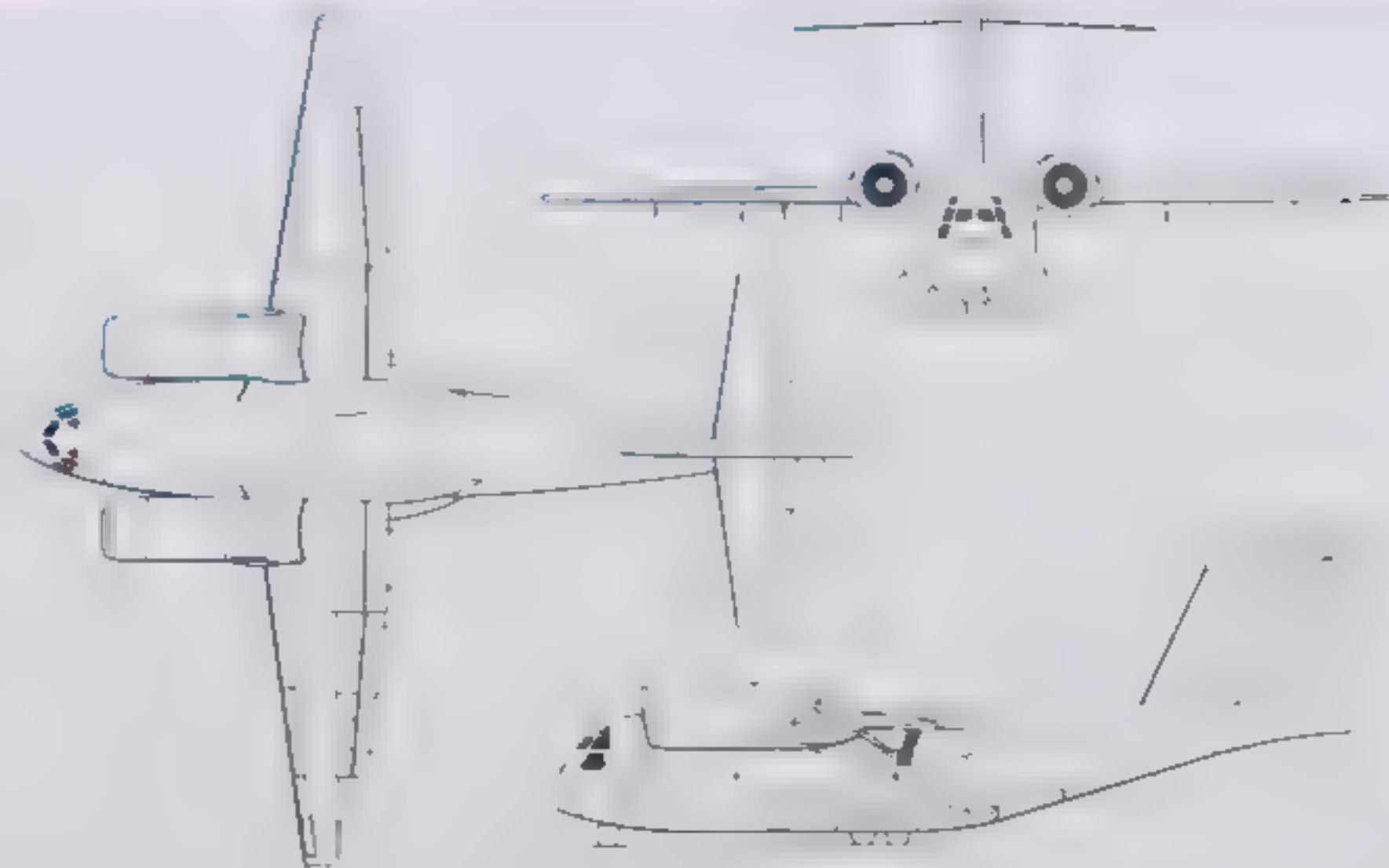


Boeing YC-14

YC-14

Origin: The Boeing Company
Type: Advanced military transport
Engines: Two 51,000lb (23,133kg) thrust General Electric F103 turbofans
Dimensions: Span 129ft (39.32m) length 131ft 8in (40.13m) height 48ft 8in (14.83m)
Weights: Empty about 85,000lb (38,555kg), loaded (STOL) 172,000lb (78,020kg), loaded (long runway) 216,000lb (97,975kg)
Performance: Maximum speed 460mph (740km/h) at 30,000ft (9,144m) takeoff and landing to and from 50ft (15m) 2,000ft (610m), range approximately 1,250 miles (2,000km) with STOL payload of 27,000lb (12,247kg) or conventional payload of 65,000lb (29,500kg)
Armament: None though gunship versions may be developed
History: First flight 9 August 1976, possible initial operational deployment 1979, but terminated by President Carter on 2 February 1978
User: Not at present funded for service

Development: In early 1972 the USAF requested proposals for an Advanced Medium STOL (short take off and landing) Transport (AMST), primarily as a replacement for the C-130 but also to gather all the diverse new technology of increasing wing lift at low airspeeds and the possibility of integrating wings and propulsion systems to achieve better short field performance. The two finalists were the McDonnell Douglas YC-15 and Boeing YC-14. Boeing chose to use a challenging but outstandingly promising method of powered lift called upper surface blowing (USB). The two large turbofans discharge through a flattened nozzle above the upper surface of the wing. In the high-lift regime the inboard flaps are depressed to 85°, with sliding link plates forming a smooth and continuous upper surface, while vortex generators emerge above the wing. The engine efflux passes across the wing and curves down over the flaps, giving lift almost equal to the net engine thrust. The outboard flaps are large double-slotted units and the entire leading edge is a variable-camber slat blown by air bled from the engines. To fly the YC-14 Marconi-Elliott of Britain supplied an extremely advanced electronic flight control system, with fibre-optical data-links between three lanes (each with its own computer) and electrical "fly by



Above: Three-view of Boeing YC-14 AMST.

wire signalling to the surface power units. The flight deck is also totally new, with EADI (electronic attitude display indicators) which superimpose flight guidance symbols on a TV picture of the ground ahead. The whole aircraft abounds with other advanced features, whilst having a simple airframe to meet the stringent customer requirement that the 300th aircraft should cost not more than \$5 million (1972 dollars excluding R&D). Obvious advantages of the YC-14 configuration are the wing is left clear underneath for payloads, ground vehicles cannot hit the engines, which can be left running for quick turnarounds, noise and infra-red signature are minimised, reverse thrust pushes the aircraft downwards, making the wheelbrakes more effective, and flight performance is extremely high (even on one engine), making steep approaches and climb-outs easy. The C-14 was designed to carry 150 troops, have a flight-refueling receptacle, be able to use 2,000ft (610m) strips and fly at less than 85kt (157km/h).

Below: Even in the cruise mode the YC-14 (first aircraft shown) is clearly out of the ordinary. Boeing's giant gamble with USB has been defeated by budgetary constraints.



CASA C.212 Aviocar

C.212 (T.12)

Origin: CASA, Madrid, Spain. Licence built by Lipnur, Indonesia

Type: Utility transport

Engines: Two 776hp Garrett AiResearch TPE331-251C single shaft turbine props

Dimensions: Span 62ft 4in (19.0m), length 49ft 10½in (15.2m), height 20ft 8in (6.3m)

Weights: Empty 8 157lb (3700kg), maximum useful load 4 410lb (2000kg) maximum loaded 13 889 lb (6300kg)

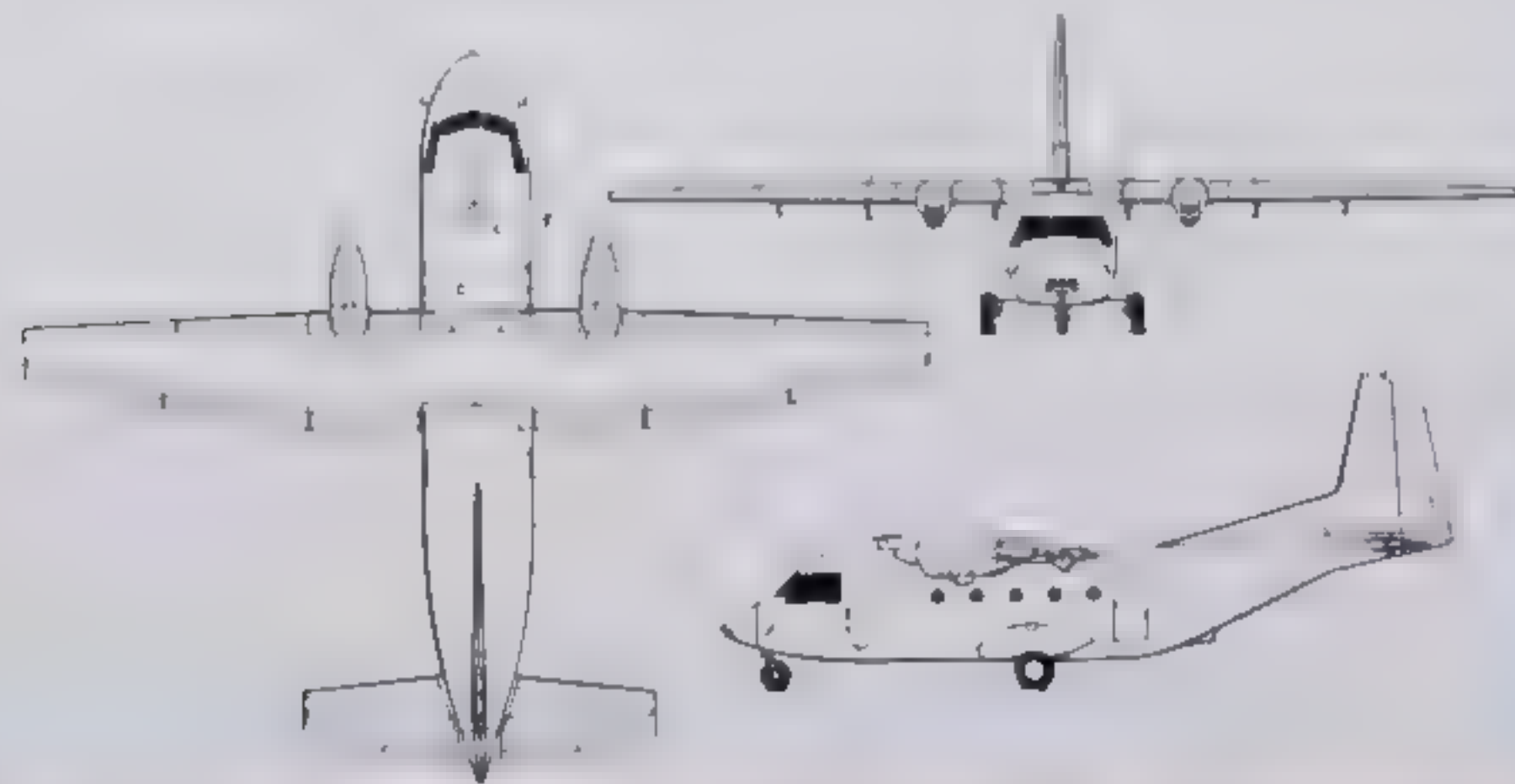
Performance: Maximum speed 230mph (370km/h) at 12 000ft economic cruising speed at 12 000ft 198mph (315km/h) initial climb at sea level 1 800ft (548m)/min, service ceiling 26 700ft (8140m), take-off or landing over 50ft (15m) 1 500ft (480m), range with maximum cargo 298 miles (480km) range with maximum fuel and 2 303lb (1045kg) load 1 093 miles (1760km)

Armament: None normally fitted

History: First flight 26 March 1971 pre-production aircraft 17 November 1972 service delivery 1973

Users: Indonesia, Jordan, Portugal, Spain, Venezuela

Development: Designed to replace the Ju 52/3m (T.2), DC-3 (T.3) and CASA 207 Azor (T.7) in the Spanish Air Force, the C.212 is designated T.12 in that service and has also been widely sold outside Spain. A simple unpressurized machine, it is intended as a 16-passenger transport, freighter (with full-section rear loading and air-dropping door), ambulance, photographic and survey aircraft and crew trainer. A civil version is also available. Deliveries began to No 461 Sqn at Gando (Las Palmas, Canary Is) and continued with 403 Sqn at Cuatro Vientos which has six photographic Aviocars. The Spanish AF is receiving eight pre-production and 32 production machines. Total orders exceed 100.



Above: Three-view of C.212 (transport, photo, nav trainer similar).



Above: In service with the Indonesian Air Force, this utility transport version of the Aviocar is designated T.12B; there is also the C.212B photo-survey version, designated TR.12A in service. The Aviocar has also sold well in foreign markets.

DH Canada Caribou

DHC-4 Caribou (C-7A and CC-108)

Origin: De Havilland Aircraft of Canada

Type: STOL tactical transport

Engines: Two 1 450hp Pratt & Whitney R 2000 D5 7M2 fourteen cylinder two-row radials

Dimensions: span 95ft 7½in (29.15m), length 72ft 7in (22.13m), height 31ft 9in (9.7m)

Weights: Empty, equipped, 18 260lb (8283kg) normal gross 28 500lb (12 928kg), maximum 31 300lb (14 197kg)

Performance: Maximum speed 216mph (347km/h), initial climb 1 355ft (413m)/min, service ceiling 24 800ft (7560m), range, 242 miles (390km) with maximum payload of 8 740lb (3965kg), 1 307 miles (2103km) with maximum fuel, take off run to 50ft (15m) and landing distance from same height, both 1 200ft (366m)

Armament: Not normally fitted

History: First flight 30 July 1958, service delivery (evaluation) August 1959 service delivery (inventory) 1961

Users: Abu Dhabi, Australia, Cameroun, Ghana, India, Kenya, Kuwait, Malaysia, Oman (aircraft for sale), Spain, Tanzania, Uganda, Zaïre, Zambia

Development: Designed to combine the carrying capacity of a C-47 (DC-3) with the STOL (short take-off and landing) performance of the company's earlier Beaver and Otter, the Caribou was a company venture undertaken in partnership with the Canadian Department of Defense Production. One was ordered by the RCAF (later Canadian Armed Forces), designated CC-108. Five, designated YAC-1, were delivered for evaluation.



Above: Personnel of the 1st Brigade, 101st Airborne Division, US Army, jump from a C-7A Caribou at Dong Ba Thien, in what was South Vietnam, in July 1965. This is just one of many duties the Caribou performs with consummate ease.



Above: The Ejercito del Aire (Spanish Air Force) calls the Caribou the T.9. The main operating unit is 372 Sqn at Villanubla, with 12 short-nose (no radar) examples. Until 1972 the EdA still operated the CASA-built Ju 52/3m, which was in many ways surprisingly similar to the T.9.

DH Canada Buffalo

DHC-5 Buffalo (C-8A, CC-115)

Origin: De Havilland Canada

Type: STOL tactical transport

Engines: Two 2,850hp General Electric T64-10 single-shaft turboprops
1,750hp (1,271 kW) (DHC-5D) T64-8, 4,400hp (3,133 kW)
at 3,000ft (914m)

Dimensions: Span 40ft 2 1/2in (12.2m) (C-8A) 7ft 4in (2.57m)
(C-115) 41ft 1in (12.4m), height 28ft 8in (8.73m)

Weights: Empty, equipped 23,157lb (10,505kg), loaded 41,000lb
(18,598kg) (DHC-5D) 49,000lb (22,223kg)

Performance: Maximum speed 271mph (435km/h), which is also
maximum cruise speed at 18,400ft (5,610m),
service ceiling 30,000ft (9,145m), range 507 miles (815km) with maximum
payload of 13,843lb (6,279kg), 2,170 miles (3,490km) with maximum fuel
and full allowances, take off and landing distances, in order of 1,100ft
(335m). The DHC-5D Improved Buffalo has slightly longer field length and
slower climb at gross weight but can airlift an 18,000lb (8,165kg) payload
691 miles (1,112km).

Armament: Not normally fitted

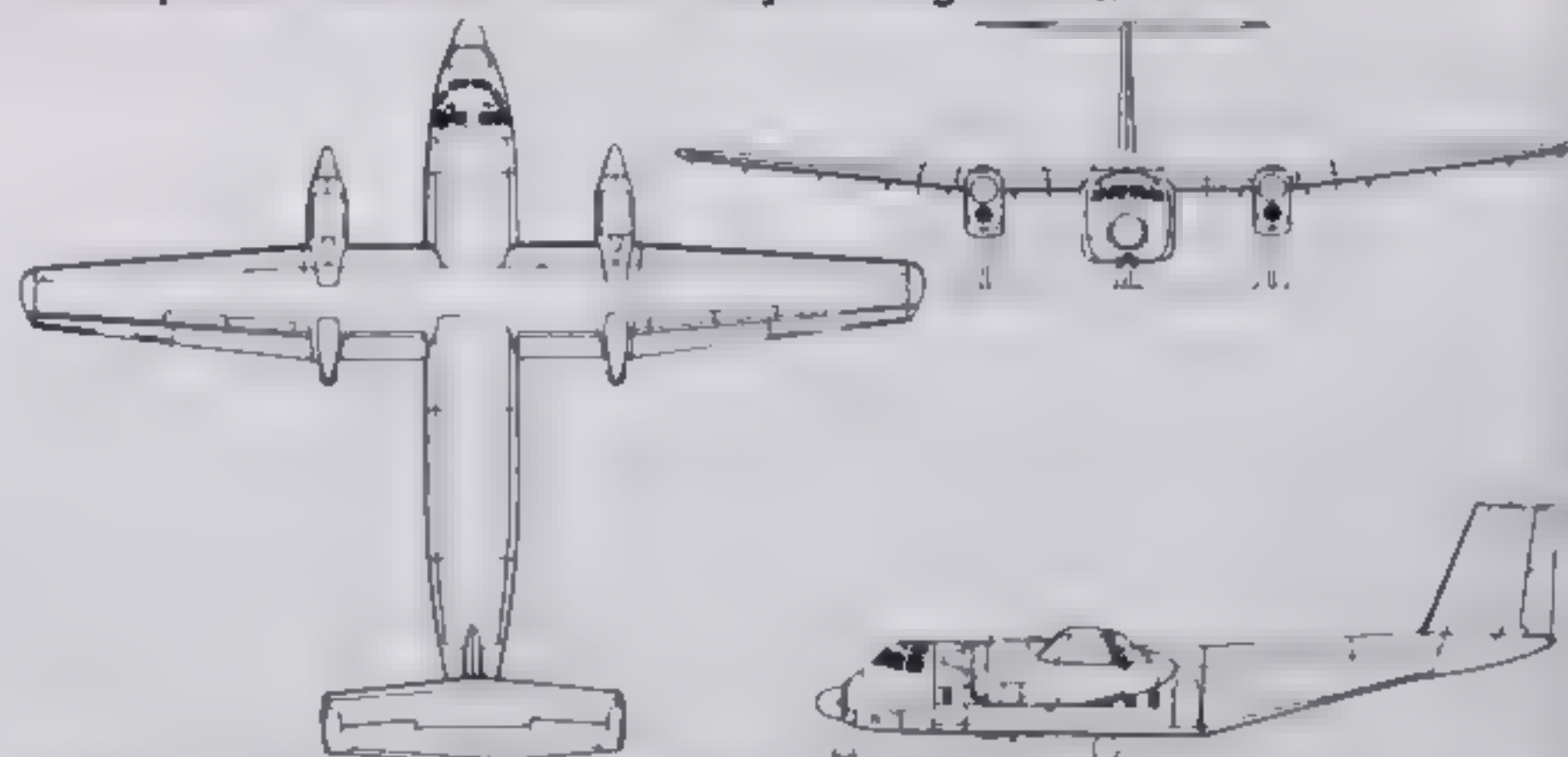
History: First flight 9 April 1964, service delivery (evaluation) April 1965,
production terminated 1972, then line reopened 1974 to deliver the Improved
Buffalo, the DHC-5D.

Users: Brazil, Canada, Ecuador, Kenya, Peru, Togo, Zaire, Zambia

Development: Unlike the Caribou, the Buffalo was designed to meet a
customer requirement, winning, against 24 other contenders, an order from
the US Army for a new STOL transport specified in May 1962. Obviously
based on the Caribou, the Buffalo has an even higher maximum lift co-
efficient, T-tail, advanced turboprop engines and considerably increased
gross weight. Development costs were shared one-third each by DHC, the
US Army and the Canadian government. Four aircraft, designated CV-7A,
were supplied to the US Army for evaluation, but a change of policy
terminated the whole programme. DHC were rescued by an order for 15
CC-115s, with uprated engines and nose radar, for the Canadian Armed
Forces. Subsequently the Brazilian Air Force bought 24 and Peru 16 and
by 1972 59 were built. One of the US aircraft, redesignated C-8A by the
US Air Force, was converted into an experimental augmentor-wing
machine powered by blown and deflected Spey turbofans. In 1974 the line
re-opened and thanks to the superior DHC-5D aircraft are now authorized
to No 102. As an assault transport the Buffalo can carry 41 troops, a 105mm
gun or Pershing missile.



Above: Two of the 16 Buffaloes operated by the STOL squadron of Grupo 41 of the Fuerza Aerea del Peru (Peruvian Air Force). Earthquake relief has been a major assignment.



Above: Three-view of typical DHC-5 with radar.

Below: Daylight shows under the wheels as an uprated C-8B (DHC-5D) storms into the air from a dirt strip. At the 1976 SBAC Show at Farnborough a -5D made the shortest fixed-wing take-off.



Fairchild C-119 Packet

C-119 to -119K Boxcar and AC-119 Shadow

Origin: Fairchild Aircraft Division (now Fairchild Industries) USA

Type: C-119 tactical transport; AC-119 multi-series armed interceptor

Engines: In most variants two 3,350hp Wright R-3350-85WA or 3,700hp R-3350-89B Turbo Compound 18 cylinder two-row radial; in later versions, two 3,250hp Pratt & Whitney R-4360-20WA Wasp Major 28 cylinder four-row radials; (AC-119K) two additional underwing booster pods each containing 2,850lb (1,293kg) thrust General Electric J85-17 turbojet (Indian AF C-119) one dorsal booster pod containing HAL Cypher 5 of 4,700lb, 2,132kg thrust or Westinghouse J34 of 3,400lb, 1,542kg thrust

Dimensions: Span 109ft 3in (34.3m), length 86ft 6in (26.36m), height 26ft 6in (8.07m)

Weights: Empty (C-119B) 37,691lb (15,981kg), (C-119K) 44,747lb (20,300kg), (AC-119K) 60,955lb (27,649kg), maximum loaded (C-119B) 74,000lb (33,600kg), (C-119K) 77,000lb (34,925kg), (AC-119K) 80,400lb (36,468kg)

Performance: Maximum speed (all versions) 243–250mph (391–402km/h), initial climb (alt) 1,100–1,300ft (335–396m)/min, service ceiling typically 24,000ft (7,315m), range with maximum payload 990–1,900 miles (1,595–3,060km)

Armament: Transport versions, none, (AC-119K) two 20mm and four 7.62mm multi-barrel rapid-fire guns firing laterally, with over 100,000 rounds total ammunition. For sensors, see text

History: First flight (XC-82) 10 September 1944, (C-119A) November 1947, service delivery (C-119B) December 1949, (AC-119) 1967

Users: Brazil (not operational), Ethiopia, India, Italy, Kampuchea, Taiwan, USA (AF/Navy, not operational), Vietnam (Civil conversions) many countries

Development: Fairchild began the design of a purpose-built cargo and troop transport – the first specialised transport ever designed for the Army Air Corps – in 1941. This flew in 1944 as the XC-82 Packet and 220 C-82s were delivered by 1948. The C-119 introduced a new nose, with the flight deck ahead of the cargo compartment instead of on top. Fuselage width was increased, engine power was greatly increased and the wings were strengthened for operation at higher weights. By 1955, when production of new aircraft ceased, 946 C-119s had been delivered to US forces, plus 141 supplied under the Mutual Defense Assistance Program to Italy, Belgium and India. Subsequently many ex-USAF C-119s were sold to Nationalist China, Brazil, Ethiopia and other countries, while those in India were fitted



Above: Three-view of C-119G Flying Boxcar.

with jet pods to improve the near zero engine climb rate of climb from hot and high airstrips. Some C-82s were converted to the C-119 configuration with a rear beaver tail operable in flight instead of a hinged rear end that could be opened only on the ground. The AC-119G and jet boosted AC-119K are extremely heavily armed interception aircraft formerly used by night in Southeast Asia. Fitted with batteries of Gatling guns, 26 AC-119Gs were delivered with night illumination systems, image intensifiers, computer fire control, various gunsights, flare launchers, crew armor and other gear. The 26 AC-119Ks, with underwing jets, added 20mm Gatling, forward looking infrared (FLIR), forward and side looking radar (SLAR), and precision nav/com equipment.

Below: In the twilight of its long career the C-119 suddenly blossomed out as a combat aircraft. The heavily armed, jet-boosted AC-119K Shadow carried enough night sensors to kill pinpoint targets even when they could not be seen visually.



Fokker-VFW F27

F27M Series 400, 500, 600 and Maritime

Origin: Fokker VFW B.V., Holland, Netherlands

Type: Transport, photographic/survey, ambulance, coastal patrol and offshore reconnaissance

Engines: Two 2,280ehp (2,140shp) Rolls-Royce Dart 532 PR turbo-prop shaft turboprops

Dimensions: Span 95ft 2in (29.0m), length 77ft 3in (23.5m), 400M 82ft 2in (25.06m), height 27ft 11in (8.5m) (500M Maritime 28ft 10in (8.71m))

Weights: Empty (400M) 23,360lb (10,596kg), (500M, Maritime) 26,000lb (11,793kg), maximum useful load (typical) 13,550lb (6,145kg), maximum loaded (all) 45,000lb (20,410kg)

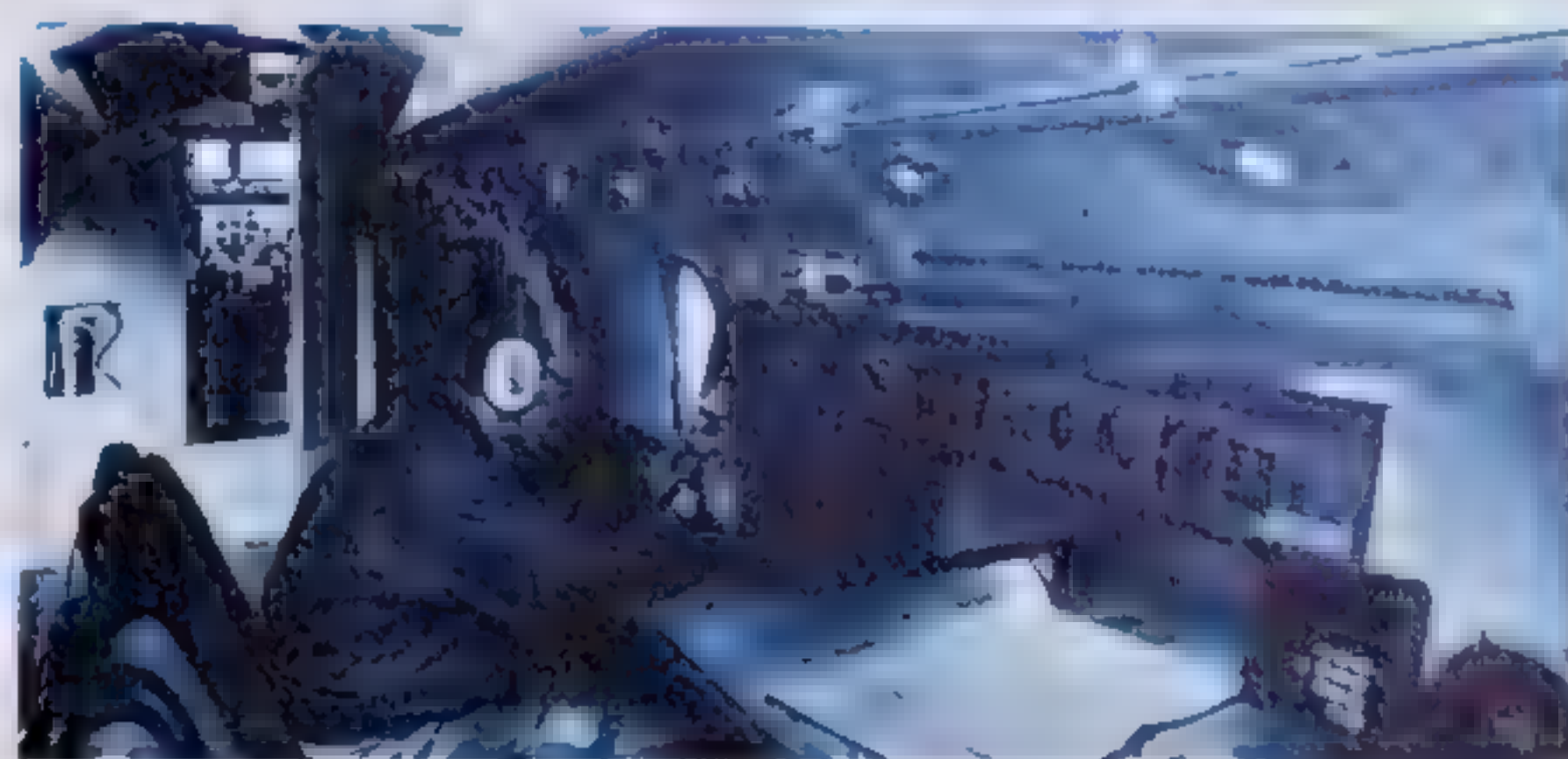
Performance: Normal cruising speed at 20,000ft (6100m) at 38,000lb (17,237kg) 298mph (480km/h), initial climb at 40,000lb (18,144kg) 1,620ft (494m)/min, field length (sea level, 40,000lb) 2,310ft (704m), range with military reserves, all-cargo, standard fuel 1,375 miles (2,213km), maximum endurance, with pylon tanks at 20,000ft (6100m) 12hr 47min

Armament: Provision on Maritime, see text

History: First flight (F27) 24 November 1955 (400M) 24 April 1965 (Maritime) 25 March 1976

Users: (Military F27) Algeria, Argentina, Ghana, Iceland, Indonesia, Iran, Ivory Coast, Netherlands, Nigeria, Pakistan, Peru, Philippines, Uruguay

Development: Originally a civil passenger and passenger/freight transport, the F27 is today widely used as a multi-role military transport, the main versions for air forces being the Mk 400M (formerly known as the Troop ship), 500M with lengthened fuselage, and the Maritime patrol aircraft. Basic aircraft accommodate 45 paratroops (50 in the 500M) or 24 stretchers (30 in the 500M) plus six or seven attendants. All military machines have a large forward freight door, and paratroop rear doors on each side. The 400M Cartographic has a comprehensive camera/mapping installation and inertial navigation. Iranian re-orders have a double underwing Marquardt target-towing installation with 25,000ft (7,620m) of wire. The Maritime was intended primarily for 200 nm fishery patrol, but is very fully equipped for military ocean surveillance. Litton provide both the LTN-72 inertial system and the APS-503F radar. Armament schemes are being developed as options.



Above: The tactical centre in the prototype Maritime, showing the neat sensor displays and navigational equipment.



Above: Three-view of Fokker-VFW F27 Maritime with tanks.

Below: An attractive paint scheme does not belie the paramilitary nature of the F27 Maritime. By 1977 orders had been placed by the Icelandic Coast Guard and Peruvian Navy



Handley Page Victor

H.P.80 Victor 1, 1A and 2

Origin: Handley Page Ltd UK, K2 rebuilt by Hawker Siddeley (now

Type: (B 1, 1A and 2) five seat strategic bomber, (K 1A and 2) four seat

Engines: (1, 1A) four 11 000lb (4990kg) thrust Rolls Royce (previously

Dimensions: Span (1) 110ft (33.53m) (2) 120ft (36.58m), length

Weights: Empty (1) 79 000lb (35 834kg), (2) 91,000lb (41.277kg)

Performance: Maximum speed (both) about 640 mph (1030km/h, Mach

Armament: No defensive armament except ECM, internal weapon bay for

History: First flight (1) 24 December 1952, (production B 1) 1 February

User: UK

Development: Designed to Specification B 35/46, the same as the

entered service, it could be intercepted by fighters or shot down by missiles

and the number ordered was so small the cost was high. To offer better

protection the B 1 was brought up to B 1A standard with much enhanced

ECM and survivors of the 50 built were in 1965-67 converted as K 1A

tankers. The much more powerful B 2, with completely redesigned airframe

and systems, offered a great increase in all-round performance, but at

height was no less vulnerable in penetrating hostile airspace and by 1964

was consigned to the low level role, carrying the big Blue Steel missile.

Several of the 30 built were converted as SR 2 strategic reconnaissance

photographic aircraft and in 1973-75 the final 20 were converted as three-

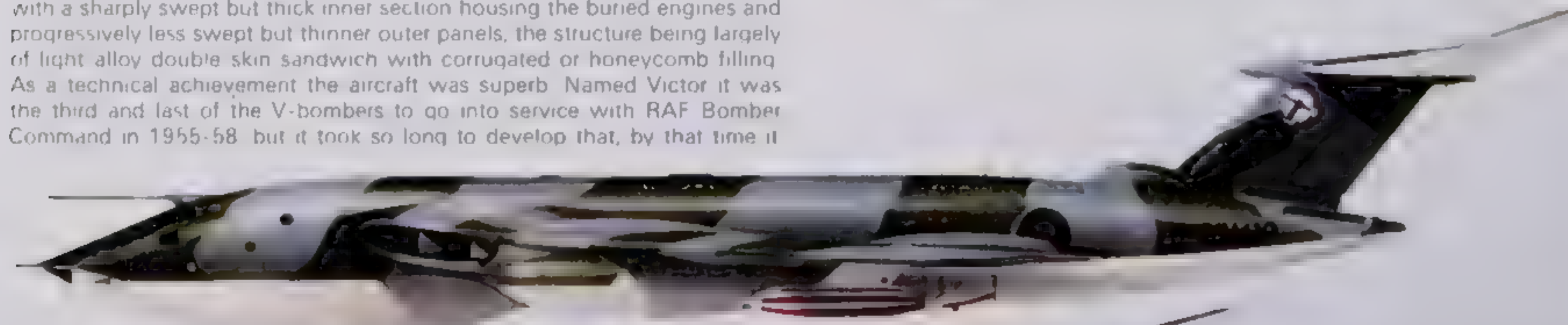
point K 2 tankers. This work was done by Hawker Siddeley at Woodford,

the old Handley Page firm having gone into liquidation in 1970. Delivery

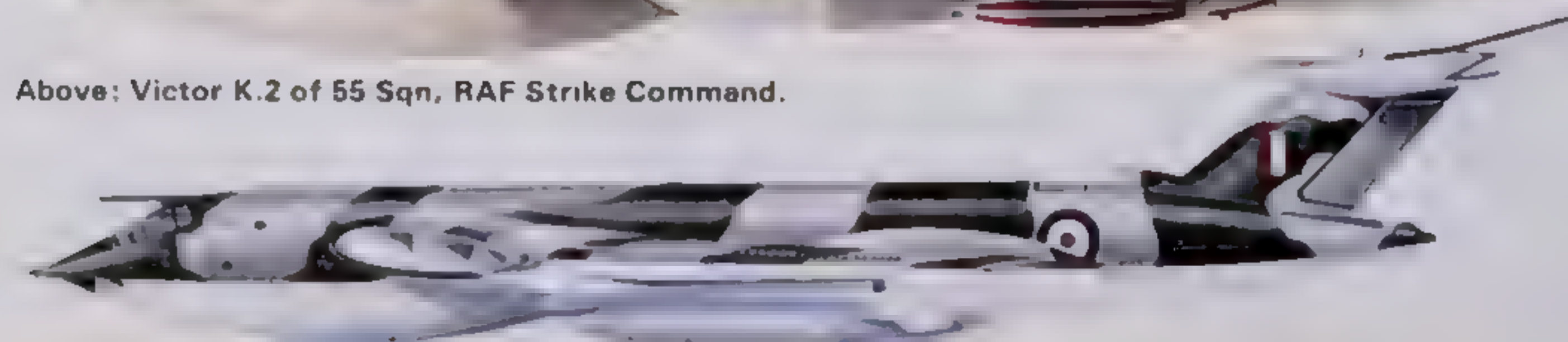
of the K 2 began in May 1974, and by 1977 the last K 1 was withdrawn



Above: Three-view of Handley Page (Hawker Siddeley) Victor K.2.



Above: Victor K.2 of 55 Sqn, RAF Strike Command.



Above: Victor B.2 (with Blue Steel missile) as formerly in service with 139 Sqn, RAF.

Below: One of the early Victor K.1A tankers, in service with 57 Sqn, pipes fuel to a thirsty Buccaneer. Though fitted with three hose-reels (some have only two) the K.1A is gradually being phased out as the much superior K.2 Victor comes into service.



Hawker Siddeley Andover, 748 and Coastguarder

HS.748, Andover, Coastguarder

Origin: Hawker Siddeley Aviation, Manchester, UK (also assembled by Hindustan Aeronautics, India)

Type: Multi-role transport, Coastguarder, maritime patrol

Engines: Two 2,280ehp (2,140shp) Rolls-Royce Dart 534-2 or 535-2 single-shaft turboprops, (Andover C 1, two 3,245ehp Dart 201)

Dimensions: Span 98ft 6in (30.02m) (Andover C 1, 98ft 3in), length 67ft 0in (20.42m) (Andover C 1, 78ft 0in, 23.77m), height 24ft 10in (7.57m) (Andover C 1, 30ft 1in)

Weights: Empty (passenger) 27,000lb (12,247kg), (military freight) 25,516lb (11,574kg), (Coastguarder) about 27,000lb, maximum loaded 46,500lb (21,092kg) (Andover C 1, 50,000lb, 22,680kg, military overload limit 51,000lb, 23,133kg)

Performance: Typical cruise at 38,000lb, 281mph (452km/h); initial climb at 38,000lb, 1,420ft (433m)/min; typical military field length at normal gross weight 3,200ft (1,000m), range with 20 per cent reserves at normal weight, 1,105 miles (1,778km) with max load, 1,646 miles (2,649km) with max fuel, radius of action at normal weight with airdrop of 12 containers of 750lb (340kg) 720 miles (1,158km), (Coastguarder has almost doubled range and endurance)

Armament: Normally none, but options available on Coastguarder

History: First flight 24 June 1960, (production Andover C 1) 9 July 1965, (Coastguarder) early 1977

Users: (Military only) Australia (AF, Navy), Belgium, Brazil, Brunei, Colombia, Ecuador, Egypt, India, S Korea, Malaysia, Nepal, New Zealand, Thailand, UK (RAF, Royal Flight), Zambia, plus undisclosed AF



Above: Three-view of the HS.748 Coastguarder.

Development: Of total sales approaching 350, almost half of all versions of this versatile STOL transport have been military. The Andover C 1 was a redesigned aircraft for the RAF, with different wing, more powerful engines driving larger propellers, a different fuselage with rear loading/dropping door, different tail and "kneeling" landing gear. These 31 machines are being sold and 10 bought by the RNZAF may later be equipped for maritime patrol. The latter is the primary duty of the Coastguarder, which has extra fuel, MAREC search radar, Decca Doppler/TANS computer and Omega VLF navigation, chute for flares and rescue dinghies sufficient for an entire 747-load of passengers. All military 748s have the option of a large freight door, strong floor and fittings for many specialized roles.



Left: The HS.748 is one of the mainstays of the Brazilian Air Force. The last six delivered have large freight doors of the kind pictured below.

Below: This Belgian 748 is fitted with a large rear freight door. The Andover, which had more powerful engines and a rear ramp/door for vehicles, was not exported.



Ilyushin Il-76

Il-76

Origin: Bureau named for Ilyushin Soviet Union

Type: Heavy freight transport

Engines: Four 26,455lb (12,000kg) thrust Soviet D-30KP-2 turbofan engines

Dimensions: Span 165ft 8in (50.5m), length 152ft 10½in (46.59m), height 48ft 5in (14.76m)

Weights: Empty, about 159,000lb (72,000kg), maximum loaded 346,125lb (157,000kg)

Performance: Maximum speed, about 560mph (900km/h), maximum cruising speed 528mph (850km/h), normal long range cruising height 42,650 ft 130,000m, range with maximum payload 18,185mi (40,000km) 3,100 miles (5,000km)

Armament: Normally none

History: First flight 25 March 1971, production deliveries 1973

User: Soviet Union (V-TA)

Development: First seen in the West at the 1971 Paris Salon, the Il-76 created a most favourable impression. Though superficially seeming to be another Ilyushin copy of a Lockheed design, in this case the C-141, in fact the resemblance is coincidental. The design was prepared to meet a basic need in the Soviet Union for a really capable freighter which, while carrying large indivisible loads, with a high cruising speed and intercontinental range, could operate from relatively poor airstrips. The result is a very useful aircraft which, though initially being used by Aeroflot in the 1971-75 and 1976-80 plans for opening up Siberia, the far north and far east of the Soviet Union, is obviously a first class strategic and tactical transport for military use. It has very powerful engines, all fitted with reversers, a high lift wing for good STOL performance and a high flotation landing gear with 20 wheels. The nose is typical of modern Soviet aircraft for 'outback' operation, and closely resembles that of the An-22. The big fuselage, usefully larger in cross-section than that of the C-141, is fully pressurized and incorporates a powerful auxiliary power unit and freight handling systems. There seems no reason why the rear clamshell doors should not be opened in flight to permit heavy dropping. In 1977 deliveries appeared to have swung in favour of the military, and a special tanker version was reported to be under development to support the Tupolev "Backfire", NATO code-name of the Il-76 is "Candid".



Above: Three-view of standard Il-76 freighter.



Above: This US Navy photograph shows several of the Il-76's good qualities, among them large body cross-section, STOL wing and "high flotation" multi-wheel landing gear. These are assets for military use.



Above: Il-76 as used by V-TA and AV-MF; tanker may be similar.

Below: Civil Il-76 photographed at Moscow during proving flights to Tyumen natural-gas and oil fields in remote Siberia.



Lockheed C-130 Hercules

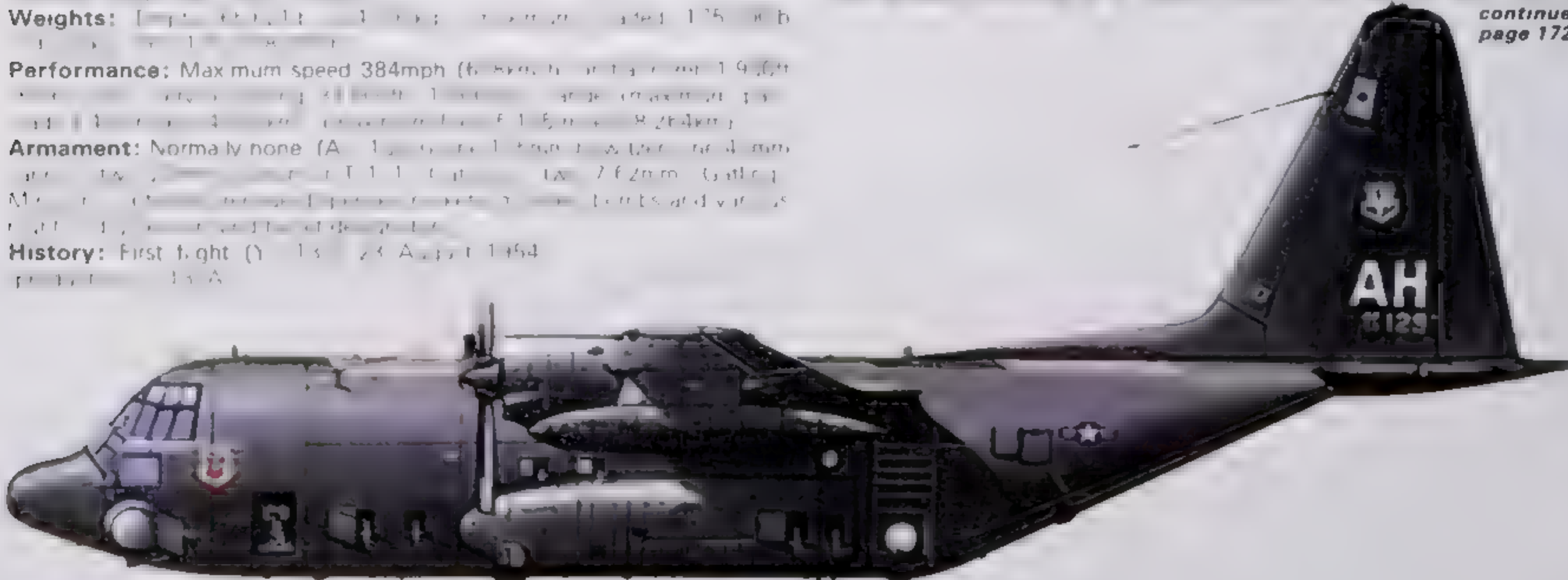
C-130 variants, see text (data for C-130H)

Origin: Lockheed
Type: Four-engine turboprop transport aircraft
Engines: Four Allison T56-A-15 turboprops
Dimensions: Length 29.83m (98ft 0in) Height 11.58m (38ft 0in) Wingspan 29.27m (96ft 0in)
Weights: Empty 11,900kg (26,250lb) Max gross 17,900kg (39,450lb)
Performance: Maximum speed 384mph (618km/h) Range 1,910km (1,187mi) Ceiling 10,668m (35,000ft) Max climb rate 1,067m/min (3,500ft/min)
Armament: Normally none (A-130H has 1 40mm cannon, 2 7.62mm machine guns, 2 2.75in rockets, 2 2.75in rockets, 2 2.75in rockets, 2 2.75in rockets)
History: First flight (Y-130) 23 August 1954
 Production 1,500

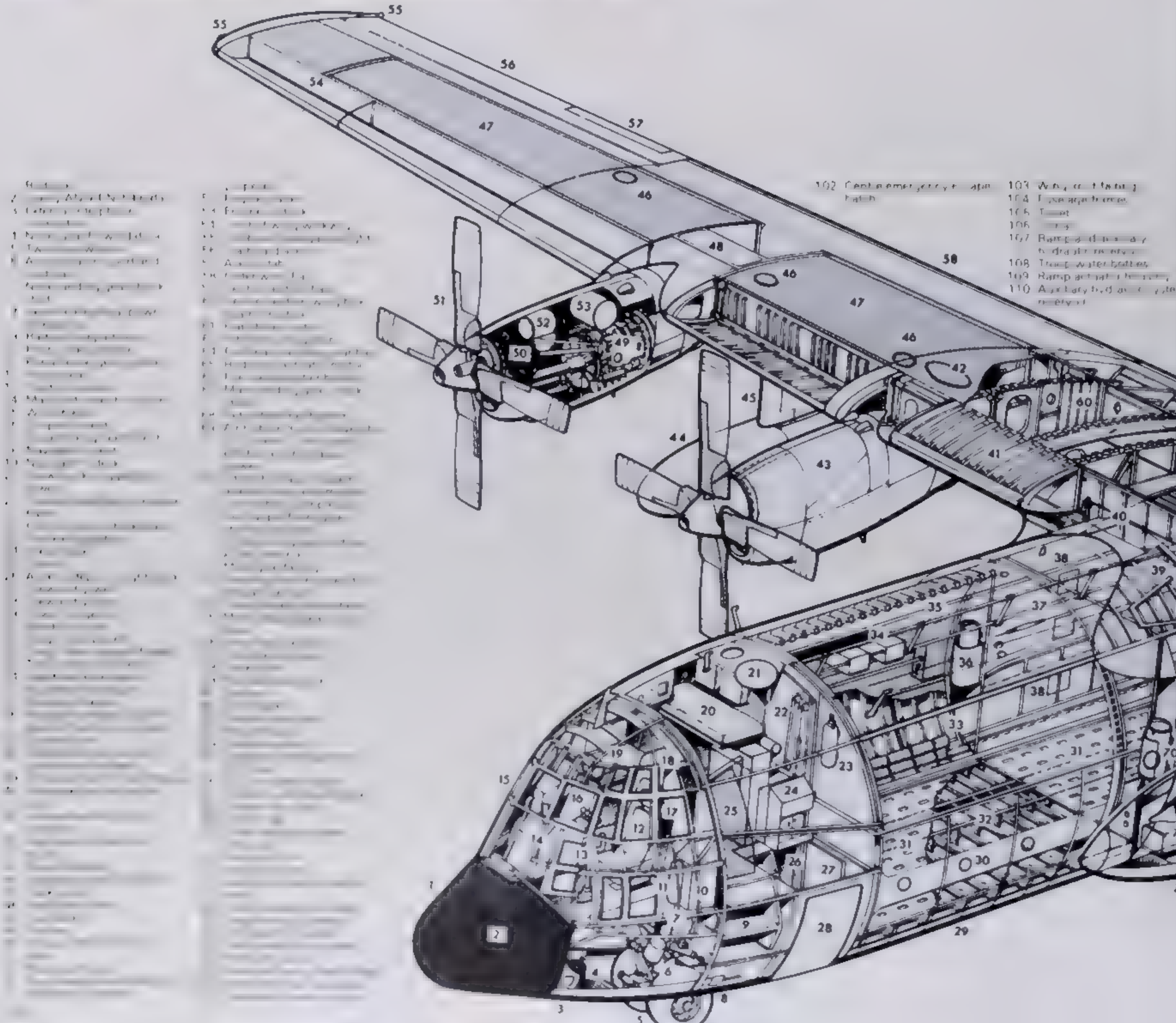
First flight 23 August 1954, delivery December 1956
Users: Abu Dhabi, Argentina, Australia, Belgium, Brazil, Canada, Chile, Colombia, Denmark, Egypt, Gabon, Greece, Indonesia, Iran, Iraq, Israel, Italy, Jordan, Kuwait, Libya, Malaysia, Mexico, New Zealand, Nigeria, Norway, Pakistan, Peru, Philippines, Saudi Arabia, Singapore, South Africa, Spain, Sweden, Switzerland, Turkey, UK, USA, USAF, Marine Corps, Navy, Venezuela, Vietnam, Zaire

Development: Though actually conventional and original and a synthesis of known techniques, the YC-130 appeared radical and bold in 1954. Among its features were a pressurized fuselage with full-section rear doors which could be opened in flight, very neat turboprop engines and rough-feld landing gear retracting into bays outside the pressure hull. Lockheed

continued on page 172



Above: AC-130H night gunship of interdiction squadrons in SE Asia, 1970-75.

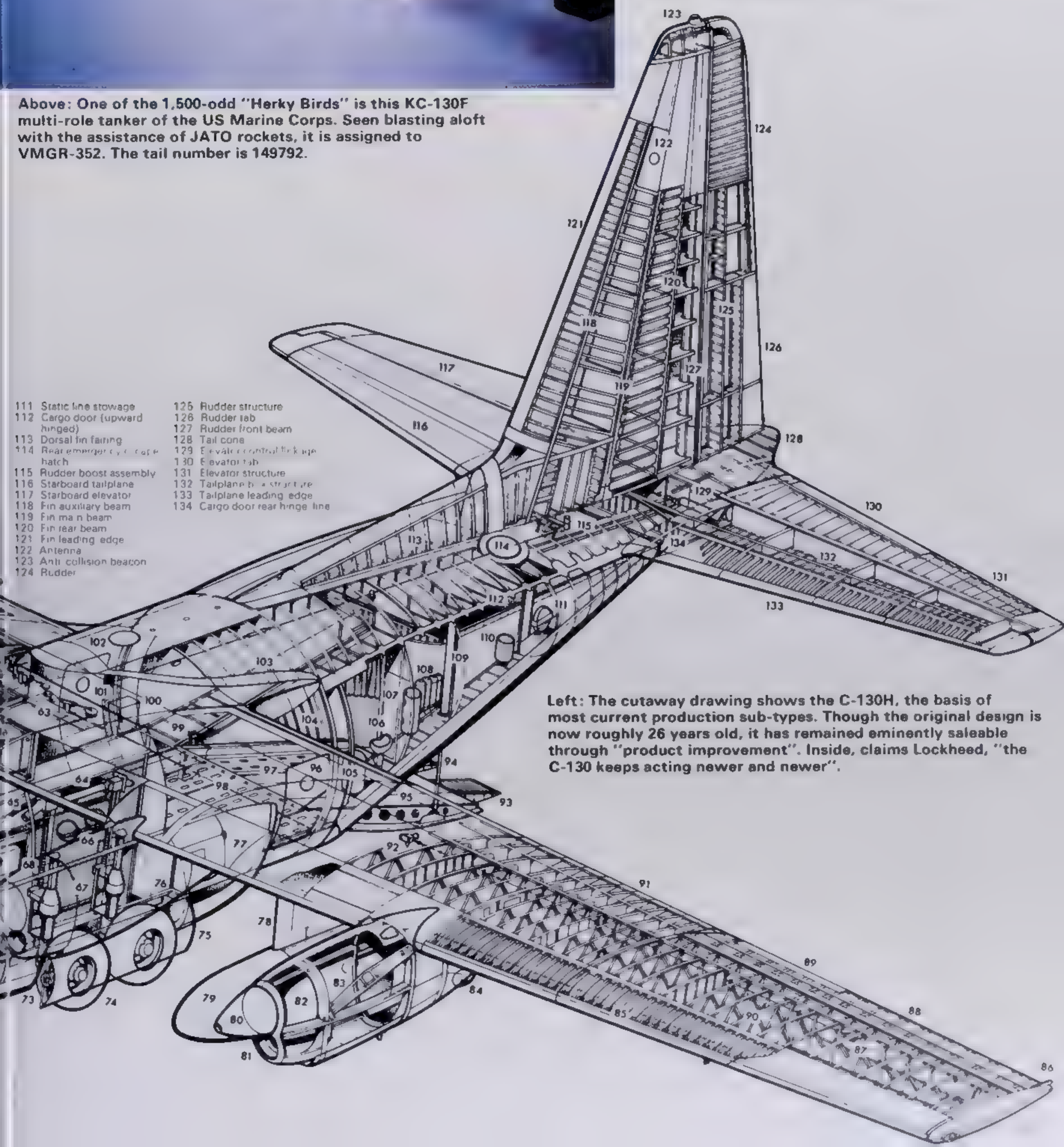




Above: One of the 1,500-odd "Herky Birds" is this KC-130F multi-role tanker of the US Marine Corps. Seen blasting aloft with the assistance of JATO rockets, it is assigned to VMGR-352. The tail number is 149792.

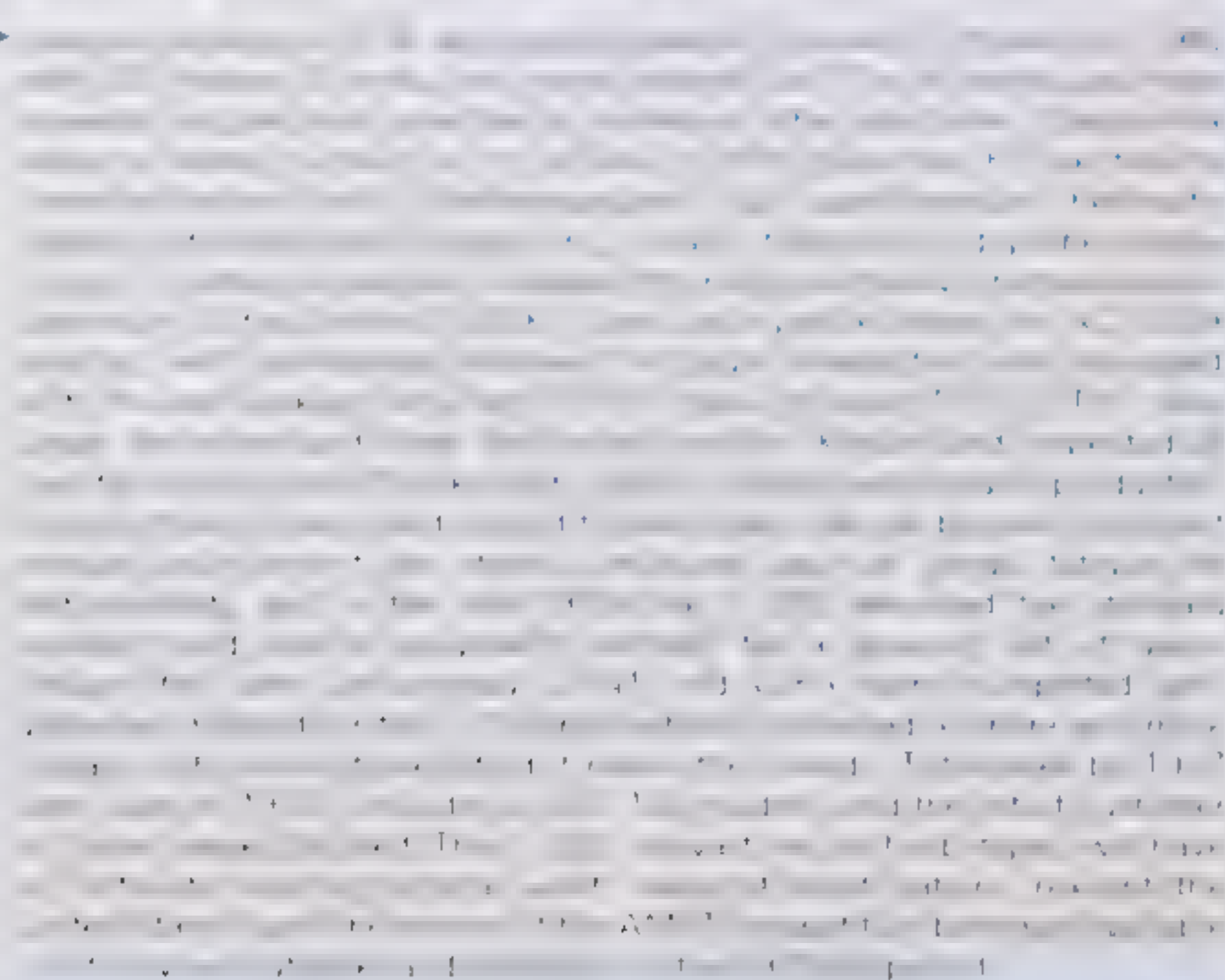


Above: Three-view of C-130E.



- | | |
|---------------------------------|--------------------------------|
| 111 Static line stowage | 125 Rudder structure |
| 112 Cargo door (upward hinged) | 126 Rudder tab |
| 113 Dorsal fin fairing | 127 Rudder front beam |
| 114 Rear emergency escape hatch | 128 Tail cone |
| 115 Rudder boost assembly | 129 Elevator control linkage |
| 116 Starboard tailplane | 130 Elevator tab |
| 117 Starboard elevator | 131 Elevator structure |
| 118 Fin auxiliary beam | 132 Tailplane hinge structure |
| 119 Fin main beam | 133 Tailplane leading edge |
| 120 Fin rear beam | 134 Cargo door rear hinge line |
| 121 Fin leading edge | |
| 122 Antenna | |
| 123 Anti collision beacon | |
| 124 Rudder | |

Left: The cutaway drawing shows the C-130H, the basis of most current production sub-types. Though the original design is now roughly 26 years old, it has remained eminently saleable through "product improvement". Inside, claims Lockheed, "the C-130 keeps acting newer and newer".



Above: Colourful and technically interesting this DC-130E is carrying on its left outer pylon a Teledyne Ryan BQM-34B (supersonic Firebee RPV) which in turn is carrying an AGM-65A Maverick TV-guided missile (white). This three-stage delivery system for air/surface weapons could become operational.



Left: XV291, a Hercules C.1 (C-130K) of RAF Strike Command, lets go a series of supply bundles during Exercise Khama Cascade to relieve famine in Nepal in March 1973. Dropping such packs at safe minimum-control speed is a fine art, especially in such "hot and high" environments.



Left: The C-130B equips No 28 Sqn of the South African Air Force, at Waterkloof, together with the C-160Z Transall.

Left: This C-130H is one delivered to 721 Sqn of the Flyvevaabnet (Royal Danish Air Force) at Vaerlose in 1975. Three Hercules were bought to replace five C-54s and eight C-47s.

Left: This C-130H is one of eight purchased by the Elliniki Aeroporia (Hellenic Air Force) to modernise the transport force of the Greek Air Materiel Command. They are part of the equipment of 355/356 Sqn at Eleusis.

Left: A Hercules C.1 (C-130K) of the RAF's Hercules Wing based at Lyneham, Wiltshire. The wing at present comprises 24, 30, 47 and 70 Sqn, plus 242 OCU, flying 45 aircraft.

Lockheed C-5A Galaxy

C-5A

Origin: Lockheed-Georgia Co, USA

Type: Strategic transport

Engines: Four 41,000lb (18,642kg) thrust General Electric TF39-1 turbo-shaft turbofans

Dimensions: Span 222ft 8½in (67.88m), length 247ft 10in (75.54m), height 65ft 1½in (19.85m)

Weights: Empty 325,244lb (147,528kg), loaded 769,000lb (348,810kg)

Performance: Maximum speed 571mph (919km/h), initial climb 1,800ft (549m)/min, service ceiling at 615,000lb, 34,000ft (10,360m), range with maximum (220,967lb, 100,228kg) payload 3,749 miles (6,033km), ferry range 7,991 miles (12,860km)

Armament: None

History: First flight 30 June 1968, service delivery 17 December 1969, final delivery May 1973

User: USA (Air Force, MAC)

Development: On some counts the C-5A is the world's largest aircraft though it is surpassed in power and weight by late models of B 747. Compared with the civil 747 it has less sweep and is considerably slower, because part of the Military Airlift Command requirement was that it should lift very heavy loads out of rough short airstrips. To this end it has a "high flotation" landing gear with 28 wheels capable of operating at maximum weight from unpaved surfaces. During development extremely difficult aerodynamic and structural problems had to be solved and there were severe difficulties concerned with fatigue and structure weight. As a result of these problems, combined with inflation, the price escalated and eventually the production had to be cut to a total of 81, equipping four squadrons. The original requirement of carrying 125,000lb for 8,000 miles was not met but in most respects the final production C-5A is substantially above prediction and an outstanding logistic vehicle which has set an impressive record of capability and reliability. The unobstructed interior has a section 19ft wide and 13ft 6in high, not including the upper deck. Freight is normally carried in containers or on 36 standard Type 463L pallets, two M-60 battle tanks can be driven on board and there is room for three packaged CH-47 Chinook heavy helicopters. Equipment includes a refuelling boom receptacle in the roof and a Norden multi-mode radar in the nose. In 1977 the trouble with wing fatigue was finally to be solved by a complete rebuild of the wing centre and inner sections. It will cost about \$1,000 million to rebuild the 70 aircraft in the active inventory (seven others will not be rebuilt) in the years 1980-84.



Above: Three-view of C-5A Galaxy as delivered to MAC.



Above: One of the basic requirements of the C-5A was that it should use unpaved airfields, necessitating 28 landing wheels. Below: Le Bourget's fire engine and ambulance look puny as a USAF Military Airlift Command C-5A blots out the sky during a Paris international air show. Not infrequently a single Galaxy has carried the entire US participation to major overseas events.



Lockheed C-141 StarLifter

C-141A and B

Origin USA Lockheed

Type Military transport

Engines: Four 21 000 b (9525kg) thrust Pratt & Whitney TF33 turbo-propellers

Dimensions: Span 159ft 11in (48.74m) length 148ft 14.2in (45.20m) height 39ft 3in (11.99m)

Weights Empty 133 773lb (60 678kg) loaded 316 600lb (143 600kg)

Performance Maximum speed 571mph (919km/h) initial climb 3 100ft (945m)/min service ceiling 41 600ft (12 680m) range with maximum fuel 4 080 miles (6565km) payload 4 080 miles (6565km)

Armament: None

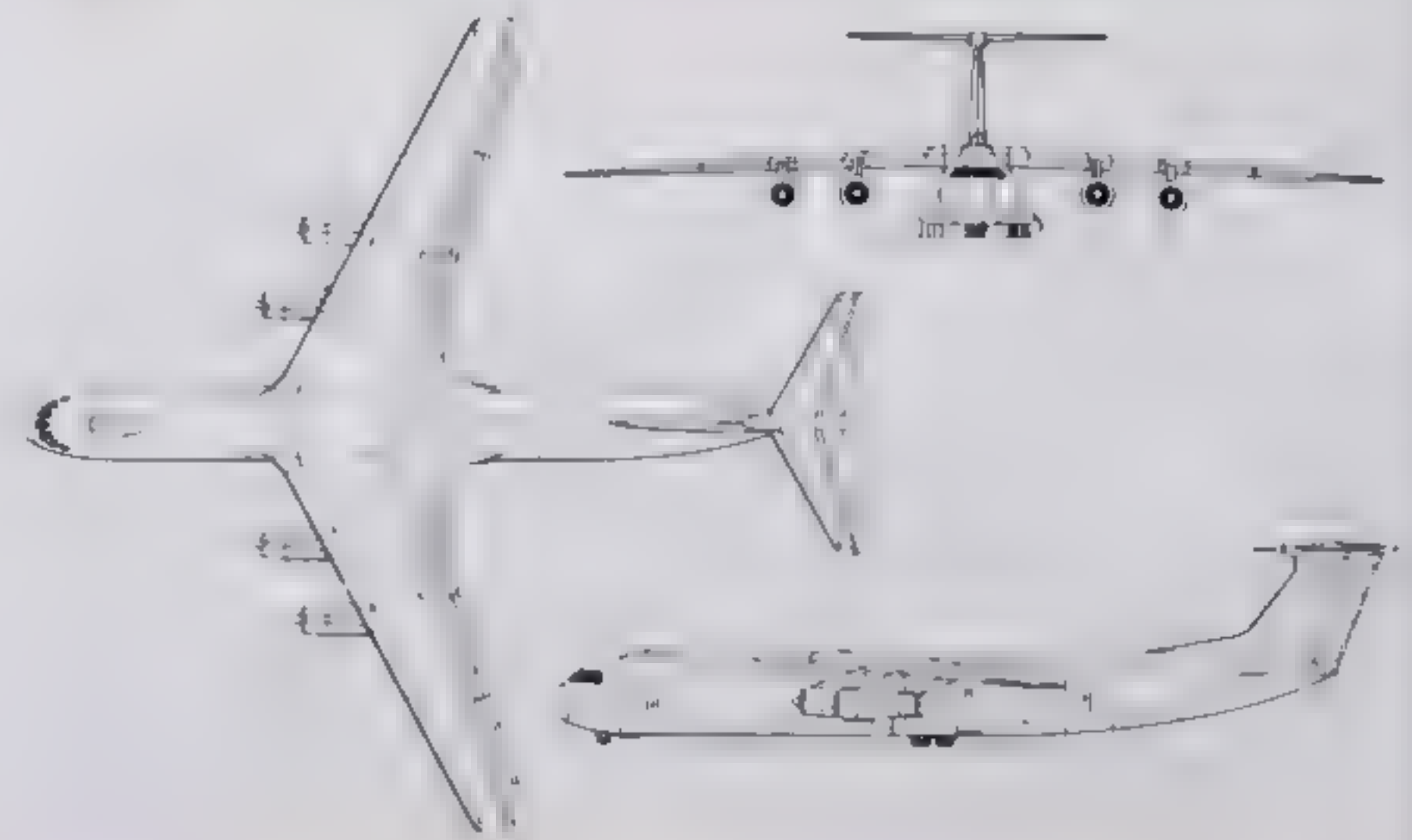
History: First flight 17 December 1963, service delivery October 1964 final delivery July 1968

User: USA (Air Force) MAC

Development: Designed to meet a requirement of USAF Military Airlift Command, the StarLifter has since been the most common of MAC's transports and has a very useful combination of range and payload. Compared with civil jet transports in the same weight category it has a less swept wing and thus lower cruising speed (typically 495mph, 797km/h) but can lift heavier loads out of shorter airstrips. The body cross-section is the same 10ft by 9ft as the C-130, and this has proved the only real shortcoming in prohibiting carriage of many bulky items (though it is said the C-141 can carry "90 per cent of all air-portable items in the Army or Air Force"). The largest item normally carried is the packaged Minuteman ICBM, for which purpose several StarLifters have had the cargo floor slightly reinforced to bear the box skids of this 86,207lb load. Alternative loads are 154 troops, 123 paratroops or 80 stretchers (litters) and 16 attendants. A C-141 holds the world record for air-dropping, with a load of 70,195lb. Total production was 285 and these aircraft equip 14 MAC squadrons. During the Vietnam war they bore the main burden of airlifting supplies westwards and casualties eastwards. All are fitted with an all-weather landing system. The user has bitterly regretted not specifying a fatter fuselage, but is funding a major rebuild programme which is adding two "plugs" to extend the fuselage 23ft 4in (7.1m) whilst adding a flight refuelling receptacle and a new wing/body fairing to reduce drag. The first rebuild was flown in March 1977, the FY78 budget includes 70 rebuilds, and eventually 274 aircraft (all in the combat inventory) may be rebuilt as C-141Bs at a cost exceeding \$700 million.



Above: One of the ubiquitous M-113 armoured vehicles emerges from a MAC C-141A. Three can be carried.



Above: Three-view of YC-141B showing 23ft 4in greater length.

Below: The 274 survivors of the original force of 284 StarLifters have flown several million hours in MAC service all over the world. Kadena AFB, Japan, is a frequent port of call.



Kawasaki C-1A

C-1

Origin: Kawasaki Heavy Industries, Gifu, Japan

Type: Medium-range transport

Engines: Two Mitsubishi (Pratt & Whitney licence) JT8D-9 two-shaft turbofans each rated at 14,500lb (6575kg)

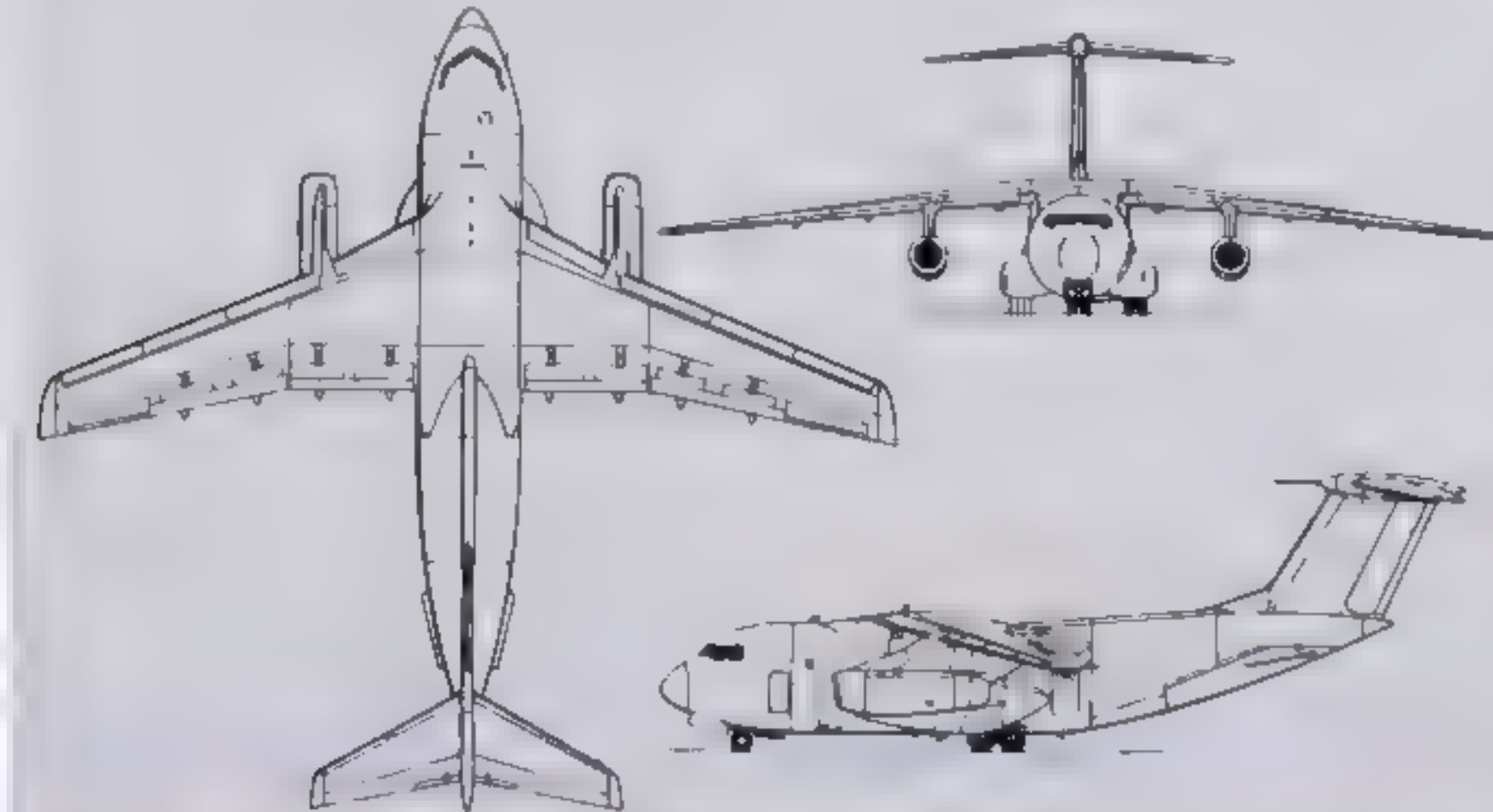
Dimensions: Span 100ft 4½in (30.6m), length 95ft 1½in (29.0m), height 32ft 9½in (9.99m)

Weights: Empty 51,410lb (23,320kg), maximum payload 17,416lb (7900kg); maximum loaded 85,320lb (38,700kg)

Performance: Maximum speed at optimum height at 78,150lb (35,450kg) 501mph (806km/h); economical cruise at same weight at greater height of 35,000ft (10,670m) 408mph (657km/h); initial climb at max weight 3,500ft (1065m)/min, service ceiling 38,000ft (11,580m); typical max weight take-off and landing over 50ft (15m) 3,000ft (914m), range with max payload 807 miles (1300km)

History: First flight 12 November 1970; handover to Japan Defence Agency for evaluation 24 February 1971, first delivery of production aircraft December 1974

User: Japan Air Self-Defence Force



Above: Three-view of Kawasaki C-1A (landing gear extended).



A production C-1A displays its strange angularities above a Japanese cloudscape. Though a competently handled exercise, the short range and small payload of the C-1 limit its value.

Development: Designed from 1966 as a replacement for the C-46 Commando, the C-1 is almost the size of a Transall but has less than half the payload and range. Though the JT8D was an available and reliable engine it is fundamentally not as well suited to STOL military transport propulsion as a turboprop or high by-pass ratio fan engine. Designed by Nihon (NAMC) the C-1 is made by a consortium of Japanese companies, with Kawasaki handling assembly and test. Typical loads include 60 troops, 45 paratroops or 36 stretcher casualties plus attendants. The wing has quad-slotted flaps, slats, spoilers and small manual ailerons, and the engines have reversers. By 1977 about 20 had been delivered of 24 on order, and further batches were planned.

Below: The modest size of the C-1A can be judged by comparing the first prototype with two Mitsubishi-assembled F-4EJ Phantom fighters. All have Air Proving Wing tail badge.



McDonnell Douglas

C-9 Nightingale/Skytrain II

C-9A Nightingale, VC-9C and C-9B Skytrain II

Origin: Douglas Aircraft Company Long Beach USA

Type: (C-9A) aeromedical airlift transport, (VC-9C) presidential transport, (C-9B) passenger transport

Engines: Two Pratt & Whitney JT12-01A turbofans

Dimensions: Span 90ft 6in (28.4m), height overall 27ft 6in (8.38m)

Weights: Empty (VC) about 57,190 lb (25,941 kg), (C-9A & C-9B) passenger configuration about 65,283 lb (29,612 kg), maximum loaded (C-9A, VC-9C) 121,000 lb (54,884 kg), (C-9B) 111,000 lb (50,353 kg)

Performance: Maximum cruise 564 mph (907 km/h), maximum range 510 mph (821 km/h), range with full payload at 15,000 lb (6,800 kg) 1,123 miles (3,095 km) (C-9B with 10,000 lb (4,536 kg) payload 1,123 miles (3,095 km))

History: First flight (DC-9) 25 February 1965, service delivery (C-9A) 10 August 1968, (C-9B) 8 May 1973, (VC) 1975

User: United States (AF, Navy, Marine Corps)

Development: In addition to well over 800 civil examples, Douglas has delivered 38 military models. The C-9A Nightingale, of which 21 are serving with Aeromedical Wings of Military Airlift Command of the USAF, carries up to 40 stretcher (litter) patients and/or more than 40 ambulatory patients, with two nurses and three technicians. It has three entrances, two with airstair, and a controlled-atmosphere special-care compartment. The US Navy's order for Skytrain II transports was raised from five to eight and finally 14. These have special Omega and inertial navigation systems and seat 107, or 45 passengers with three freight pallets, and are the longest ranged DC-9s ever built.



Above: Typical military DC-9-30 (C-9B has windows full length).



Above: Longest-ranged DC-9s in the world are the C-9B Skytrain IIs of the US Navy. This one serves with VR-30 at NAS Alameda (note Golden Gate bridge) and can reach Hawaii.



Above: A red-crossed C-9A Nightingale of the USAF, Scott AFB. Below: An immaculate C-9B Skytrain II of the US Marine Corps air transport force.



McDonnell Douglas C-133

Cargomaster

C-133A and 133B

Origin: Douglas Aircraft Co, Long Beach, USA

Type: Heavy logistic freighter

Engines: (C-133A) four 7,000ehp Pratt & Whitney T34-7 single-shaft axial turboprops, (C-133B) 7,500ehp T34-9W

Dimensions: Span 179ft 8in (54.75m), length 157ft 6½in (48.02m), height 48ft 3in (14.7m)

Weights: Empty 120,263lb (54,550kg) maximum loaded 286,000lb (129,727kg)

Performance: Maximum speed 359mph (578km/h), initial climb 1,280ft (389m)/min; service ceiling at gross weight 29,950ft (9140m), range with 44,000lb (19,960kg) payload, 4,300 miles (6920km)

Armament: None

History: First flight (first production C-133A) 23 April 1956, service delivery 29 August 1957, first flight of C-133B 31 October 1959

User: USA (USAF MAC, ANG)

Development: Designed to meet a US Air Force requirement for a logistic transport capable of carrying indivisible bulky loads which could not easily be loaded into the C-124, the C-133 was a bold, clean design with four of the powerful single-shaft turboprops developed at Pratt & Whitney since 1944. These engines had powered C-97, R7V-2 (Navy Super Constellation) and YC-124B aircraft and were fully developed in time for the C-133. No prototype was ordered, a contract for 35 being signed in 1954. While these aircraft were being delivered, from the Long Beach plant, changes were made in the shape of the rear fuselage and freight door arrangement and the final three introduced clamshell rear doors. A further 15 more powerful aircraft were then ordered as the C-133B, delivery being com-



Above: Three-view of McDonnell Douglas C-133B (133A similar).

pleted in 1961. These B-models were specially adapted to the carriage of the Thor, Jupiter, Atlas, Titan and Minuteman ballistic missiles, which they delivered to operational sites throughout the United States and to Europe. An A-model set a world record by lifting a load of 117,900lb and approximately 96 per cent of all US military equipment could be carried by these capable machines. The C-133 will be retired from Military Airlift Command before 1980, after about 20 years of service.

Below: The Golden Gate bridge gets into the act yet again, in a fine portrait of a C-133B Cargomaster in the livery of the Military Air Transport Service (now Military Airlift Command). Three USAF MAC squadrons were equipped with C-133s, but this was reduced to two squadrons as the C-5A force increased, and the Cargomasters are expected to be phased out completely before 1980.



McDonnell Douglas YC-15

YC-15 AMST

Origin: [redacted]
Type: [redacted]

Engines

Dimensions: Span (original) 110ft 4in (33.7m) 110ft 4in (33.7m) 110ft 4in (33.7m) 110ft 4in (33.7m)

Weights $\{w_1, \dots, w_{n-1}, w_n\}$ $\{1, 1, \dots, 1, 1\}$ $\{1, 1, \dots, 1, 1\}$ $\{1, 1, \dots, 1, 1\}$

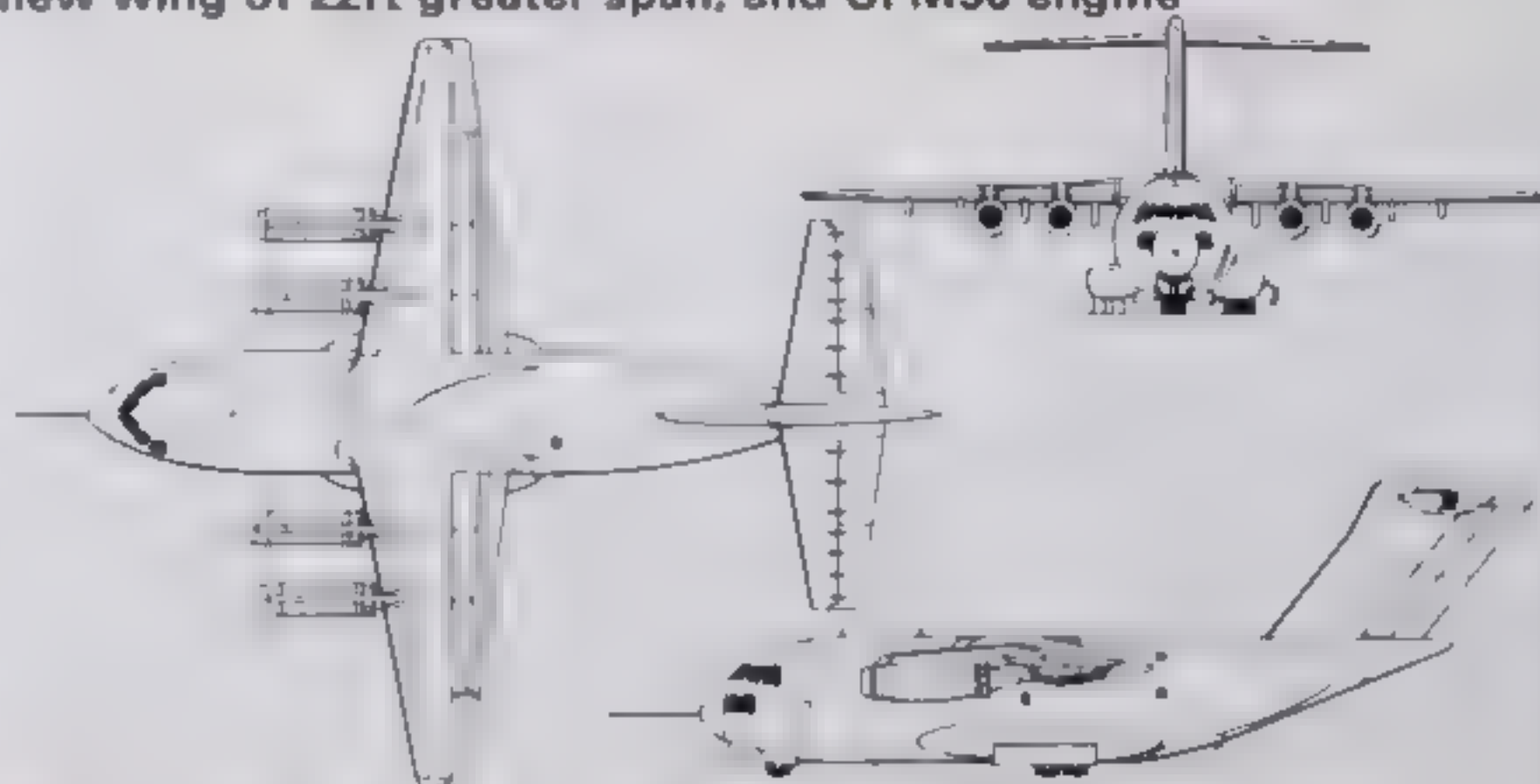
Performance: Maximum speed (brake to full throttle): 150 mph (241 km/h). 0-60 mph (0-96 km/h) time: 4.5 sec. 0-100 mph (0-161 km/h) time: 10.5 sec. 0-150 mph (0-241 km/h) time: 20.5 sec. 1/4 mile time: 13.5 sec. 1/4 mile speed: 104 mph (167 km/h). 1/2 mile time: 20.5 sec. 1/2 mile speed: 124 mph (200 km/h). 3/4 mile time: 27.5 sec. 3/4 mile speed: 134 mph (216 km/h). 1 mile time: 34.5 sec. 1 mile speed: 144 mph (232 km/h). 1.32 mile time: 40.5 sec. 1.32 mile speed: 154 mph (248 km/h). 1.64 mile time: 46.5 sec. 1.64 mile speed: 164 mph (264 km/h). 2 mile time: 52.5 sec. 2 mile speed: 174 mph (280 km/h). 2.32 mile time: 58.5 sec. 2.32 mile speed: 184 mph (296 km/h). 2.64 mile time: 64.5 sec. 2.64 mile speed: 194 mph (312 km/h). 3 mile time: 70.5 sec. 3 mile speed: 204 mph (328 km/h). 3.32 mile time: 76.5 sec. 3.32 mile speed: 214 mph (344 km/h). 3.64 mile time: 82.5 sec. 3.64 mile speed: 224 mph (360 km/h). 4 mile time: 88.5 sec. 4 mile speed: 234 mph (376 km/h). 4.32 mile time: 94.5 sec. 4.32 mile speed: 244 mph (392 km/h). 4.64 mile time: 100.5 sec. 4.64 mile speed: 254 mph (408 km/h). 5 mile time: 106.5 sec. 5 mile speed: 264 mph (424 km/h). 5.32 mile time: 112.5 sec. 5.32 mile speed: 274 mph (440 km/h). 5.64 mile time: 118.5 sec. 5.64 mile speed: 284 mph (456 km/h). 6 mile time: 124.5 sec. 6 mile speed: 294 mph (472 km/h). 6.32 mile time: 130.5 sec. 6.32 mile speed: 304 mph (488 km/h). 6.64 mile time: 136.5 sec. 6.64 mile speed: 314 mph (504 km/h). 7 mile time: 142.5 sec. 7 mile speed: 324 mph (520 km/h). 7.32 mile time: 148.5 sec. 7.32 mile speed: 334 mph (536 km/h). 7.64 mile time: 154.5 sec. 7.64 mile speed: 344 mph (552 km/h). 8 mile time: 160.5 sec. 8 mile speed: 354 mph (568 km/h). 8.32 mile time: 166.5 sec. 8.32 mile speed: 364 mph (584 km/h). 8.64 mile time: 172.5 sec. 8.64 mile speed: 374 mph (600 km/h). 9 mile time: 178.5 sec. 9 mile speed: 384 mph (616 km/h). 9.32 mile time: 184.5 sec. 9.32 mile speed: 394 mph (632 km/h). 9.64 mile time: 190.5 sec. 9.64 mile speed: 404 mph (648 km/h). 10 mile time: 196.5 sec. 10 mile speed: 414 mph (664 km/h). 10.32 mile time: 202.5 sec. 10.32 mile speed: 424 mph (680 km/h). 10.64 mile time: 208.5 sec. 10.64 mile speed: 434 mph (696 km/h). 11 mile time: 214.5 sec. 11 mile speed: 444 mph (712 km/h). 11.32 mile time: 220.5 sec. 11.32 mile speed: 454 mph (728 km/h). 11.64 mile time: 226.5 sec. 11.64 mile speed: 464 mph (744 km/h). 12 mile time: 232.5 sec. 12 mile speed: 474 mph (760 km/h). 12.32 mile time: 238.5 sec. 12.32 mile speed: 484 mph (776 km/h). 12.64 mile time: 244.5 sec. 12.64 mile speed: 494 mph (792 km/h). 13 mile time: 250.5 sec. 13 mile speed: 504 mph (808 km/h). 13.32 mile time: 256.5 sec. 13.32 mile speed: 514 mph (824 km/h). 13.64 mile time: 262.5 sec. 13.64 mile speed: 524 mph (840 km/h). 14 mile time: 268.5 sec. 14 mile speed: 534 mph (856 km/h). 14.32 mile time: 274.5 sec. 14.32 mile speed: 544 mph (872 km/h). 14.64 mile time: 280.5 sec. 14.64 mile speed: 554 mph (888 km/h). 15 mile time: 286.5 sec. 15 mile speed: 564 mph (904 km/h). 15.32 mile time: 292.5 sec. 15.32 mile speed: 574 mph (920 km/h). 15.64 mile time: 298.5 sec. 15.64 mile speed: 584 mph (936 km/h). 16 mile time: 304.5 sec. 16 mile speed: 594 mph (952 km/h). 16.32 mile time: 310.5 sec. 16.32 mile speed: 604 mph (968 km/h). 16.64 mile time: 316.5 sec. 16.64 mile speed: 614 mph (984 km/h). 17 mile time: 322.5 sec. 17 mile speed: 624 mph (1000 km/h). 17.32 mile time: 328.5 sec. 17.32 mile speed: 634 mph (1016 km/h). 17.64 mile time: 334.5 sec. 17.64 mile speed: 644 mph (1032 km/h). 18 mile time: 340.5 sec. 18 mile speed: 654 mph (1048 km/h). 18.32 mile time: 346.5 sec. 18.32 mile speed: 664 mph (1064 km/h). 18.64 mile time: 352.5 sec. 18.64 mile speed: 674 mph (1080 km/h). 19 mile time: 358.5 sec. 19 mile speed: 684 mph (1096 km/h). 19.32 mile time: 364.5 sec. 19.32 mile speed: 694 mph (1112 km/h). 19.64 mile time: 370.5 sec. 19.64 mile speed: 704 mph (1128 km/h). 20 mile time: 376.5 sec. 20 mile speed: 714 mph (1144 km/h). 20.32 mile time: 382.5 sec. 20.32 mile speed: 724 mph (1160 km/h). 20.64 mile time: 388.5 sec. 20.64 mile speed: 734 mph (1176 km/h). 21 mile time: 394.5 sec. 21 mile speed: 744 mph (1192 km/h). 21.32 mile time: 400.5 sec. 21.32 mile speed: 754 mph (1208 km/h). 21.64 mile time: 406.5 sec. 21.64 mile speed: 764 mph (1224 km/h). 22 mile time: 412.5 sec. 22 mile speed: 774 mph (1240 km/h). 22.32 mile time: 418.5 sec. 22.32 mile speed: 784 mph (1256 km/h). 22.64 mile time: 424.5 sec. 22.64 mile speed: 794 mph (1272 km/h). 23 mile time: 430.5 sec. 23 mile speed: 804 mph (1288 km/h). 23.32 mile time: 436.5 sec. 23.32 mile speed: 814 mph (1304 km/h). 23.64 mile time: 442.5 sec. 23.64 mile speed: 824 mph (1320 km/h). 24 mile time: 448.5 sec. 24 mile speed: 834 mph (1336 km/h). 24.32 mile time: 454.5 sec. 24.32 mile speed: 844 mph (1352 km/h). 24.64 mile time: 460.5 sec. 24.64 mile speed: 854 mph (1368 km/h). 25 mile time: 466.5 sec. 25 mile speed: 864 mph (1384 km/h). 25.32 mile time: 472.5 sec. 25.32 mile speed: 874 mph (1400 km/h). 25.64 mile time: 478.5 sec. 25.64 mile speed: 884 mph (1416 km/h). 26 mile time: 484.5 sec. 26 mile speed: 894 mph (1432 km/h). 26.32 mile time: 490.5 sec. 26.32 mile speed: 904 mph (1448 km/h). 26.64 mile time: 496.5 sec. 26.64 mile speed: 914 mph (1464 km/h). 27 mile time: 502.5 sec. 27 mile speed: 924 mph (1480 km/h). 27.32 mile time: 508.5 sec. 27.32 mile speed: 934 mph (1496 km/h). 27.64 mile time: 514.5 sec. 27.64 mile speed: 944 mph (1512 km/h). 28 mile time: 520.5 sec. 28 mile speed: 954 mph (1528 km/h). 28.32 mile time: 526.5 sec. 28.32 mile speed: 964 mph (1544 km/h). 28.64 mile time: 532.5 sec. 28.64 mile speed: 974 mph (1560 km/h). 29 mile time: 538.5 sec. 29 mile speed: 984 mph (1576 km/h). 29.32 mile time: 544.5 sec. 29.32 mile speed: 994 mph (1592 km/h). 29.64 mile time: 550.5 sec. 29.64 mile speed: 1004 mph (1608 km/h). 30 mile time: 556.5 sec. 30 mile speed: 1014 mph (1624 km/h). 30.32 mile time: 562.5 sec. 30.32 mile speed: 1024 mph (1640 km/h). 30.64 mile time: 568.5 sec. 30.64 mile speed: 1034 mph (1656 km/h). 31 mile time: 574.5 sec. 31 mile speed: 1044 mph (1672 km/h). 31.32 mile time: 580.5 sec. 31.32 mile speed: 1054 mph (1688 km/h). 31.64 mile time: 586.5 sec. 31.64 mile speed: 1064 mph (1704 km/h). 32 mile time: 592.5 sec. 32 mile speed: 1074 mph (1720 km/h). 32.32 mile time: 598.5 sec. 32.32 mile speed: 1084 mph (1736 km/h). 32.64 mile time: 604.5 sec. 32.64 mile speed: 1094 mph (1752 km/h). 33 mile time: 610.5 sec. 33 mile speed: 1104 mph (1768 km/h). 33.32 mile time: 616.5 sec. 33.32 mile speed: 1114 mph (1784 km/h). 33.64 mile time: 622.5 sec. 33.64 mile speed: 1124 mph (1800 km/h). 34 mile time: 628.5 sec. 34 mile speed: 1134 mph (1816 km/h). 34.32 mile time: 634.5 sec. 34.32 mile speed: 1144 mph (1832 km/h). 34.64 mile time: 640.5 sec. 34.64 mile speed: 1154 mph (1848 km/h). 35 mile time: 646.5 sec. 35 mile speed: 1164 mph (1864 km/h). 35.32 mile time: 652.5 sec. 35.32 mile speed: 1174 mph (1880 km/h). 35.64 mile time: 658.5 sec. 35.64 mile speed: 1184 mph (1896 km/h). 36 mile time: 664.5 sec. 36 mile speed: 1194 mph (1912 km/h). 36.32 mile time: 670.5 sec. 36.32 mile speed: 1204 mph (1928 km/h). 36.64 mile time: 676.5 sec. 36.64 mile speed: 1214

History: Contract award December 1972, first flight 26 August 1975

User: Not at present funded for service

[illegible]

Above: First takeoff in February 1977 of the first YC-15 with the new wing of 22ft greater span, and CFM56 engine



Above: Three-view of YC-15 with original wing and engines

Below: Takeoff at the McDonnell Douglas plant at Long Beach of the second YC-15, which is camouflaged. After extensive trials this aircraft is now flying with a JT8D-209 engine.



Transall C-160

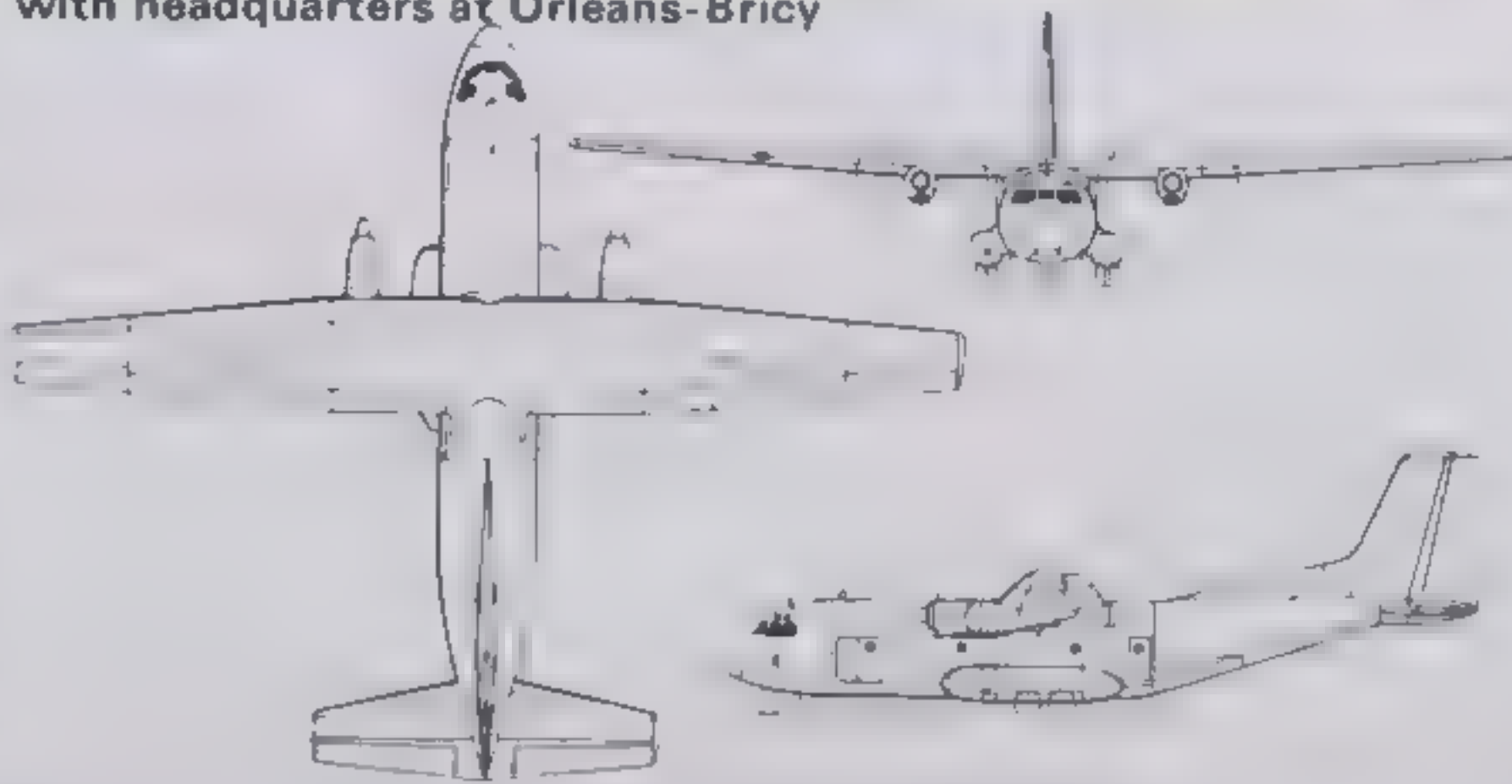
C-160D, C-160F and C-160Z

Origin: Airbus (co-developed by British Aerospace)
Type: Tactical transport
Engines: Two 6,100ehp Rolls Royce Type 2, two shaft turbo-propellers
Dimensions: Span 131ft 3in (40.20m), length 100ft 3in (30.4m), height 40ft 6 $\frac{1}{2}$ (12.36m)
Weights: Empty (equipped) 63,400lb (28,758kg), maximum payload 35,274lb (16,000kg), maximum loaded 112,435lb (51,000kg)
Performance: Maximum speed at 16,000ft (4,875m) 368mph (542km/h), economical cruise at 20,000ft (6,096m) 282mph (454km/h), climb rate 1,300ft (396m)/min, service ceiling 25,500ft (7,770m), typical take-off and landing distances over 50ft (15m) with no reverse-pitch on landing 2,900ft (884m), range with max payload and all allowances 1,056 miles (1,700km), ferry range 3,230 miles (5,200km)
History: First flight 25 February 1963, service delivery April 1968, last delivery October 1972, start of delivery of additional batch 1979
Users: France (C-160F and civil 160P for Night Mail Service), W German (160D), S Africa (160Z) and Turkey (160D)

Development: Transall is a name coined from Transporter Allianz, a consortium formed in 1959 by what are today Aérospatiale, MBB and VFW-Fokker. The objective was to produce a military transport to replace the Noratlas (which had been built by both countries) and also meet civil and export needs. Provision was made for jet lift or jet-boosted STOL versions, but in the event all the 179 production aircraft were basic freighters and the only export sale was nine to S Africa. The engines were made by a consortium (Rolls-Royce, SNECMA, MTU and FN) as were the 18ft (5.5m) propellers (HSD, Rateau-Figeac). The airframe is pressurized and has electric wing and tail anti-icing (almost unique). The 110 bought by the Luftwaffe were more than was needed, 20 were later transferred to Turkey and others are in storage. Despite this in 1976 Aérospatiale secured a commitment from the French government to build a further 75, not to meet customer demand but to maintain employment. The Armée de l'Air has agreed to take 25 of the new batch, and 50 export sales are being sought at a price subsidized to make it competitive.



Above: Takeoff of a C-160F of the 61e Escadre, Armée de l'Air, with headquarters at Orléans-Bricy



Above: Three-view of C-160 (all sub-types similar).

Below: A C-160D of the Luftwaffe, which has 76 Transalls in wings LTG 61 and 62 and 14 in an OCU at Wunstorf. Another 20 are in storage, yet Aérospatiale intends to build more.



Electronic warfare aircraft

The initials EW also stand for "early-warning", and the AEW (airborne early-warning) aircraft is the chief type discussed in this chapter. But it is a specialized type, too costly for all but a few nations—preferably a group acting in collaboration—even to consider acquiring. Far more important in terms of numbers and platforms are the EW (electronic warfare) systems that today must be carried not only by combat aircraft but also by army vehicles and naval



craft. This kind of EW involves detecting, evaluating, precisely locating and then neutralizing or penetrating hostile electronic defences. The EW inventory of a modern air force includes systems for noise and deception jamming against current and future threats, missile-fuze jamming, RF (radio frequency) warning and homing, real-time warning and reconnaissance of enemy RF emitters, physical and dispensed countermeasures (some of which are not ECM—electronic countermeasures—but IR heat sources to decoy infra-red homing missiles), and countermeasure control and display. One aircraft was specially designed for this kind of EW, and it is naturally included in this section. A few other special EW aircraft are modifications of existing combat types, notably the F-105G Wild Weasel Thunderchief, the EF-4E Phantom and the EF-111A. But the majority of EW systems in aerial warfare are carried by regular combat aircraft. Modern aircraft, such as the F-15 and B-1, carry a comprehensive "suite" of EW devices internally, betraying their presence only by small aerials flush with the

skin or projecting in the form of streamlined fairings pointing to the front or rear. Older aircraft, or aircraft which only occasionally have to penetrate hostile electronic defences, carry their EW systems packaged into self-contained standardized payloads, either in the form of a pod carried on an external weapon pylon or as a blister attached to the underside of the fuselage.

Whereas ECM is an active EW function intended to confuse or mislead the enemy's electronic defence systems, ESM (electronic surveillance measures) is a passive function. It normally comprises listening, as far as possible on every possible electronic defence frequency simultaneously, to detect all the enemy's signals. Clever airborne equipment instantly displays each new signal, giving its bearing, frequency and pulse-width, and probably also its pulse-repetition interval and signal level. This is a form of reconnaissance, and as well as being practised by many purpose-designed reconnaissance aircraft is also an important duty of maritime patrol aircraft as they probe potentially

Modern warfare is invariably based on an electronic environment. This Grumman E-2C Hawkeye of USS *Constellation* can protect the ship and its aircraft, double the number of kills made by the carrier's Tomcats and serve as a powerful data-link



dangerous warships.

But the king of the military sky is an aircraft that carries no armament yet costs more than any other aircraft in history. The AWACS, today's pinnacle of the AEW species, provides the most detailed surveillance of everything in the airspace within a radius of some 245 miles (400km). The chief reason underlying its long and costly development was the fact that, in the 1950s, combat aircraft took to flying very low. Streaking along "the nap of the Earth" they could not be detected because of the line-of-sight limitations of ground radars, and because the radars in AEW aircraft could not detect the low-flier against the massive reflection from the ground. The solution was extremely advanced radars which can discriminate between airborne targets and ground clutter. The Westinghouse radar of the Boeing E-3A is a pulse-doppler type which, by comparing the frequency of transmitted and received pulses, measures the speeds of every target that reflects back its radar energy. Digitally processed signals are sent out in succession from numerous

apertures across an aerial face about 24ft (7.5m) long and 5ft (1.52m) deep, rotating six times per minute. One of the features of this radar is that virtually all the energy goes into the main narrow beam; the sidelobes, present in all normal radars, are extremely small, almost eliminating "clutter" from ground reflections. Part of the E-3A's cost of some \$100 million is accounted for by computers, handling some 740,000 operations each second, and comprehensive displays for the command and control of an entire theatre of operations. One can expect the Tu-126 ("Moss") to have similar capabilities.

Heaviest of all military aircraft, the Boeing E-4A is the successor to the EC-135 as the carrier of the National Military Command System. Whether one can class these as combat aircraft is problematical, though they are certainly military. They are merely a way of ensuring that, should the US seat of government be destroyed, a complete national military command system will survive in the safety of the sky.

Boeing E-3A (AWACS)

E-3A (AWACS)

Origin
Type
Engines

Dimensions

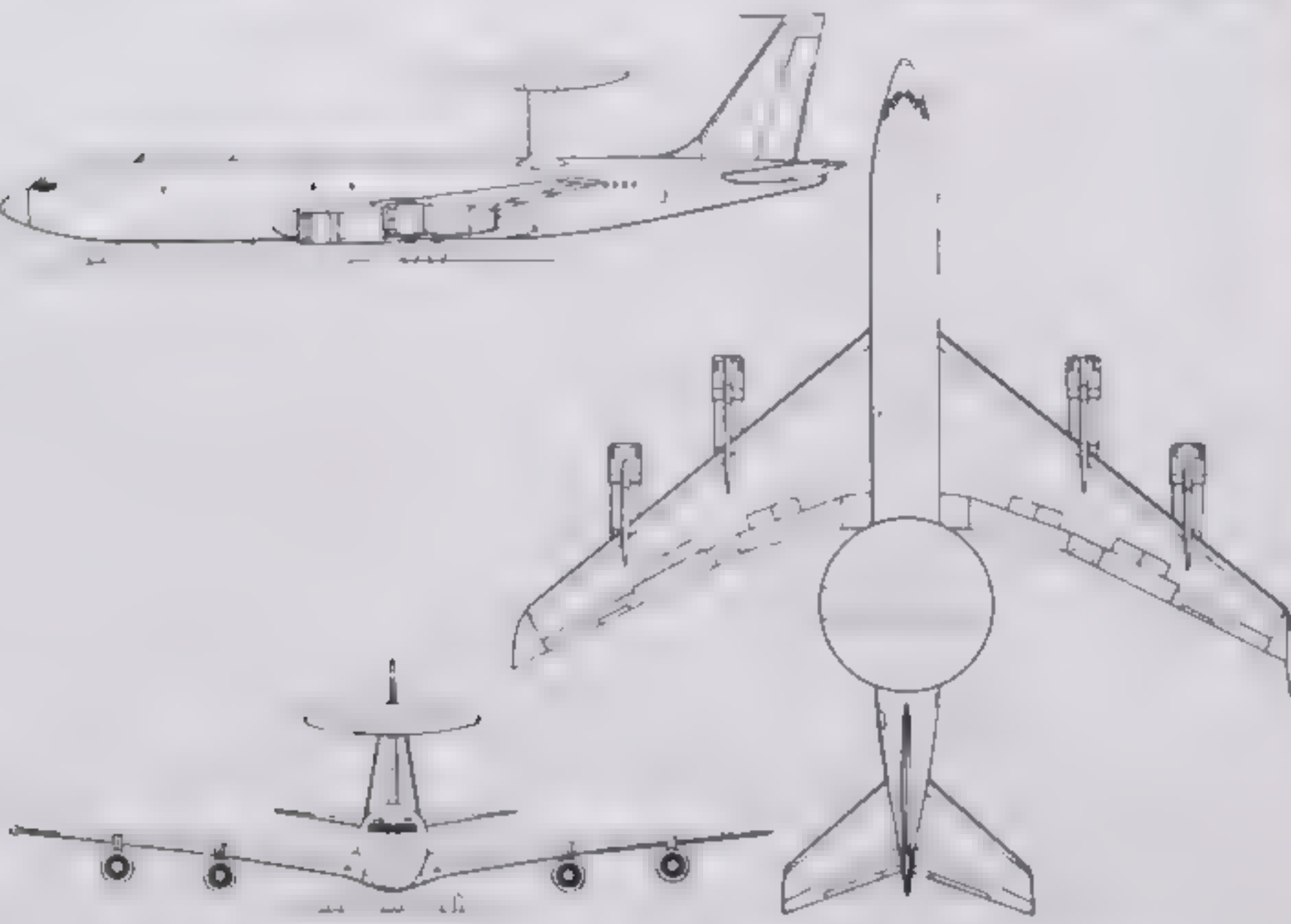
Weights

Performance: Maximum speed, about 600mph

Armament

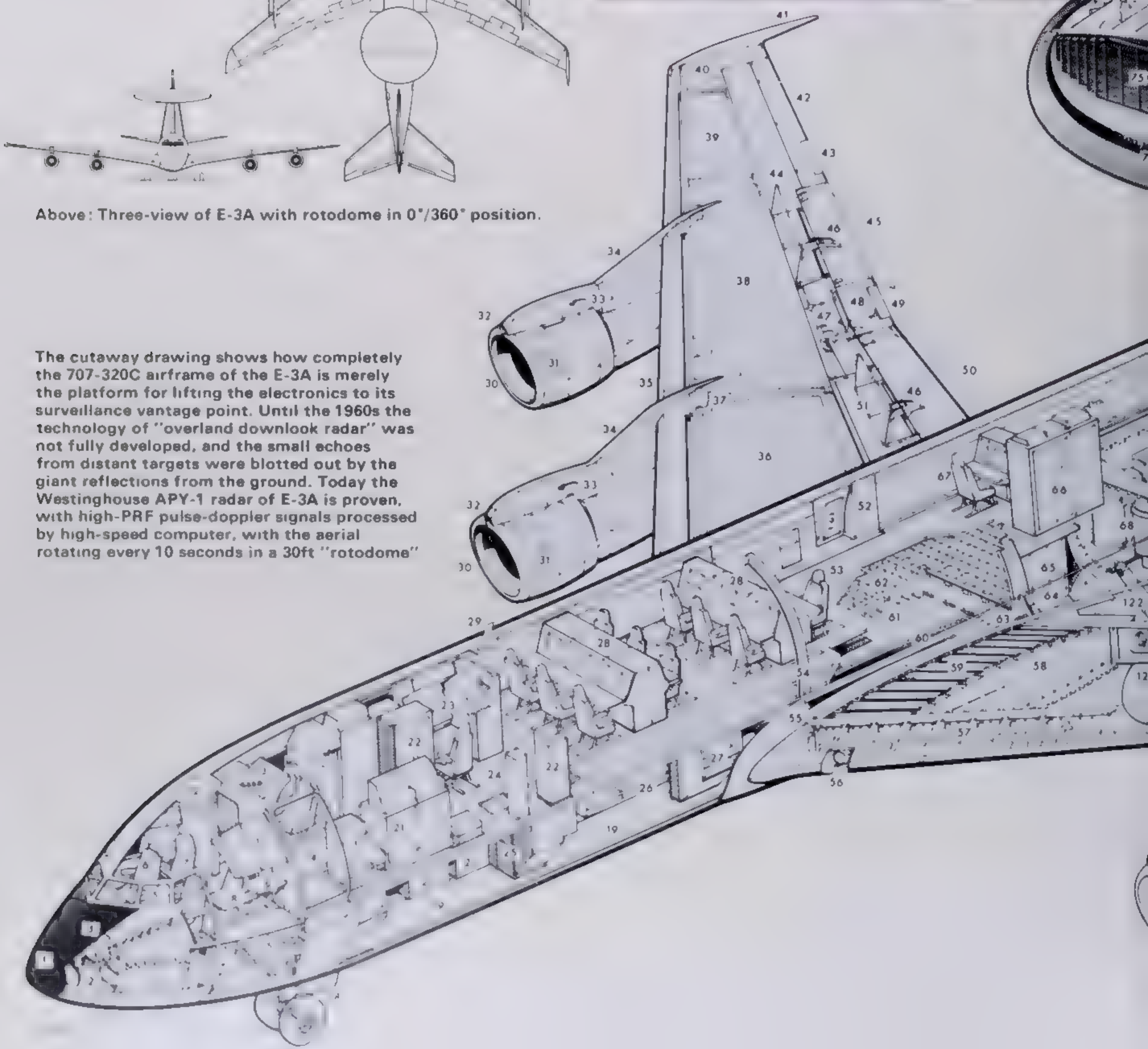
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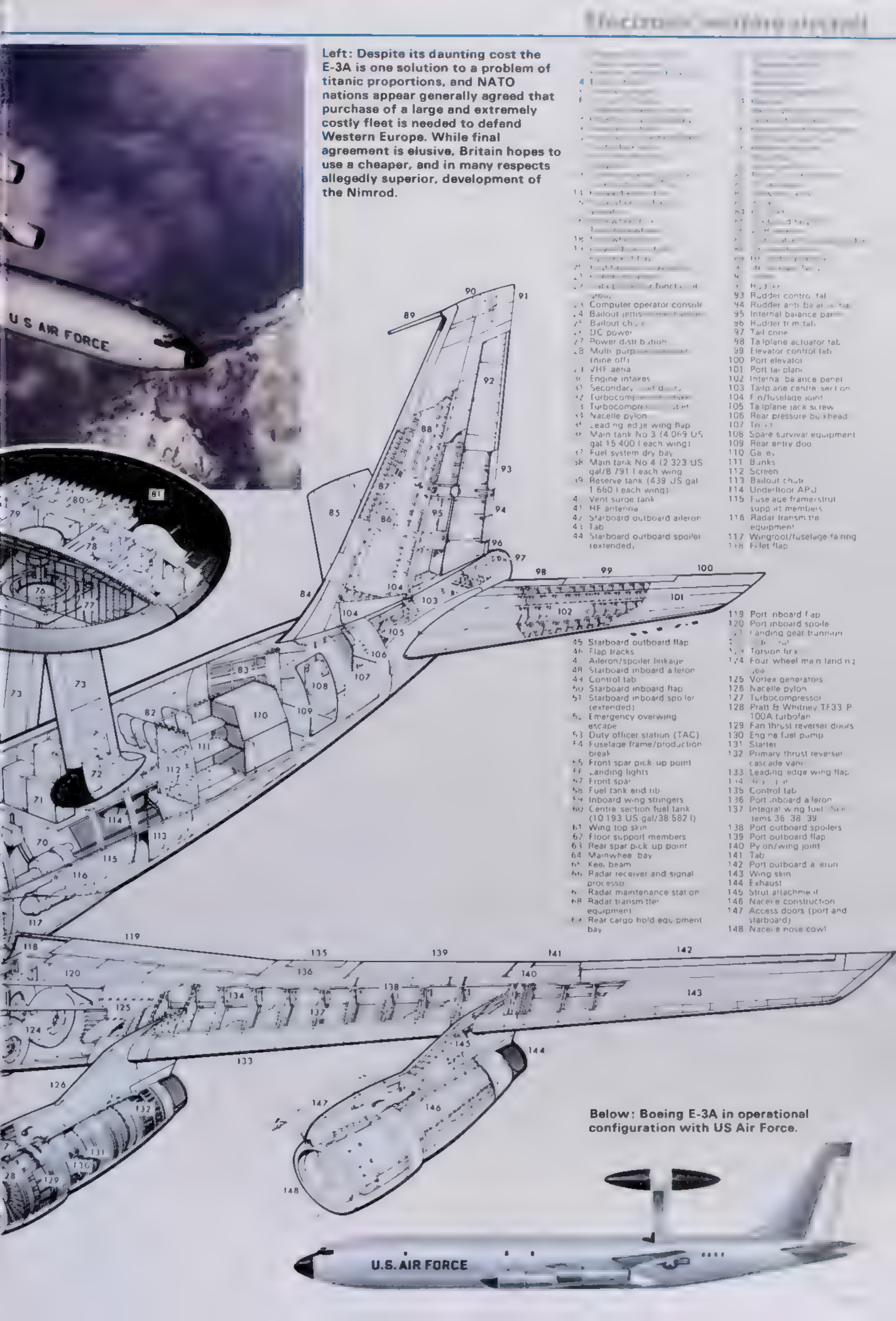
continued on page 184▶



Above: Three-view of E-3A with rotodome in 0°/360° position.

The cutaway drawing shows how completely the 707-320C airframe of the E-3A is merely the platform for lifting the electronics to its surveillance vantage point. Until the 1960s the technology of "overland downlook radar" was not fully developed, and the small echoes from distant targets were blotted out by the giant reflections from the ground. Today the Westinghouse APY-1 radar of E-3A is proven, with high-PRF pulse-doppler signals processed by high-speed computer, with the aerial rotating every 10 seconds in a 30ft "rotodome"





Left: Despite its daunting cost the E-3A is one solution to a problem of titanic proportions, and NATO nations appear generally agreed that purchase of a large and extremely costly fleet is needed to defend Western Europe. While final agreement is elusive, Britain hopes to use a cheaper, and in many respects allegedly superior, development of the Nimrod.

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Below: Boeing E-3A in operational configuration with US Air Force.

► **User:** Air Force (Air Force Defense Command) and TAC

Development: Developments in radars and airborne data processing led first to the early warning aircraft exemplified by the Douglas EA-109 and EC-119, and then around 1965 to the concept of the Airborne Warning And Control System (AWACS). This is a flying surveillance and management station from which the entire air situation of a small sector or part of a hemisphere can be obtained. Carrying an extremely powerful microwave radar, a mass of sensing and data processing systems and advanced displays both for the crew and for transmission to the ground, an AWACS can maintain perfect watch on every kind of aerial vehicle, hostile and friendly, over a radius exceeding 200 miles (322km). This facilitates the most efficient handling of every situation right down to air traffic control at a beach head or the best route for helicopters to rescue a friendly pilot. The first AWACS is the Boeing E-3A. Derived from the 707-320B airliner, it was to have had eight TF34 engines but owing to their fantastic overall cost these aircraft now retain the TF33 and have emerged somewhat less complex than once planned. Trials with rival Hughes and Westinghouse radars were held with EC-137D test aircraft: the Westinghouse was chosen, with aerial rotating at six times a minute in a 30ft (9.14m) dome high above the rear fuselage. Except for the E-4A the E-3A is a most certain, the most expensive aircraft ever to enter military service. Initial operational service in 1977-78 is to be followed by a total of 34 planned aircraft for efficient protection of US airspace. By the time this book appears the European NATO nations may also have decided to buy the E-3A. In late 1976 discussion had centred around 27 European-based aircraft costing \$2.273 million.



Above: Multi-Purpose Display Consoles (MPDC) are the operational heart of the AWACS. They provide the mission crew with the enormously comprehensive display and control interfaces to carry out the surveillance, command and weapons-direction tasks. This aft-looking view shows the radar receiver in the background.



Above: Takeoff of one of the EC-137D test aircraft in 1972. These were commercial 707-320 airframes converted to test "brass board" models of the two competing radars. Westinghouse won

Boeing E-4A and 747

E-4 (AABNCP)

Origin: The Boeing Company, USA

Type: Airborne command post, with crew/staff of 28-60

Engines: (E-4A) four 45,500lb (20,639kg) thrust Pratt & Whitney F105-100 (JT9D) turbofans, (E-4B) four 52,500lb (23,815kg) thrust General Electric F103-100 turbofans

Dimensions: Span 195ft 8in (59.64m), length 231ft 4in (70.5m), height 63ft 5in (19.33m)

Weights: Empty, probably 380,000lb (172,370kg), loaded 803,000lb (364,230kg)

Performance: Maximum speed 608mph (978km/h) at 30,000ft (9144m), maximum Mach number 0.92, normal operational ceiling 45,000ft (13,715m), normal unrefuelled range about 6,500 miles (10,460km)

Armament: None

History: First flight (747) 9 February 1969, (E-4) January 1973

User: US Air Force (see text), basic 747 is used by Iran (see text)

Development: Since the late 1950s the United States has created a growing fleet of various kinds of EC-135 aircraft as airborne command posts. Operated by the National Military Command System and SAC, these are the platforms carrying the national strategic and economic command and decision-taking machinery, with perfect unjammable communications to all government and military organizations and the capacity to survive even a nuclear war. Since 1965 the EC-135 has become restrictive, and to meet



future needs and provide for a larger staff in greater comfort, with the capability of more flexible action and response, the 747B airframe was adopted to carry the Advanced Airborne National Command Post (AABNCP)



Above: Three-view of typical JT9D-powered 747.



Above: E-4A before conversion to E-4B (same aircraft pictured below as E-4B).

The first two were fitted with Pratt & Whitney JT9D engines, but the standard was later adopted as that of the fourth aircraft, bought for \$39 million in December 1973 with GE engines and later equipment. Much of the special equipment has been taken from the EC-135A, H, J, K, L and P force. There are three decks, the main deck being a mixture of executive offices and luxurious living quarters and the rest being packed with advanced electronics. The first three (E-4A) aircraft are based at Andrews AFB near Washington, eventually, without cost to the Air Force, they will be retrofitted with F103 engines and they will also be provided with E-4B equipment. It is planned to base the E-4B and two further E-4B aircraft at Offutt AFB, headquarters of SAC in Nebraska. The Imperial Iranian AF uses a fleet of 747 transports, most of which have boom and pod installations to serve as tankers.

Below: The world's most powerful military aircraft, the E-4B is a much more advanced and comprehensive aircraft than the E-4A. Programme cost for the force of six amounts to more than \$850 million. It carries a larger battle staff, and unique SHF/LF/VLF communications radio systems.



Grumman E-2 Hawkeye

E-2A, B and C Hawkeye and C-2A Greyhound

Origin
Type
Engines
Dimensions

Weights

Performance

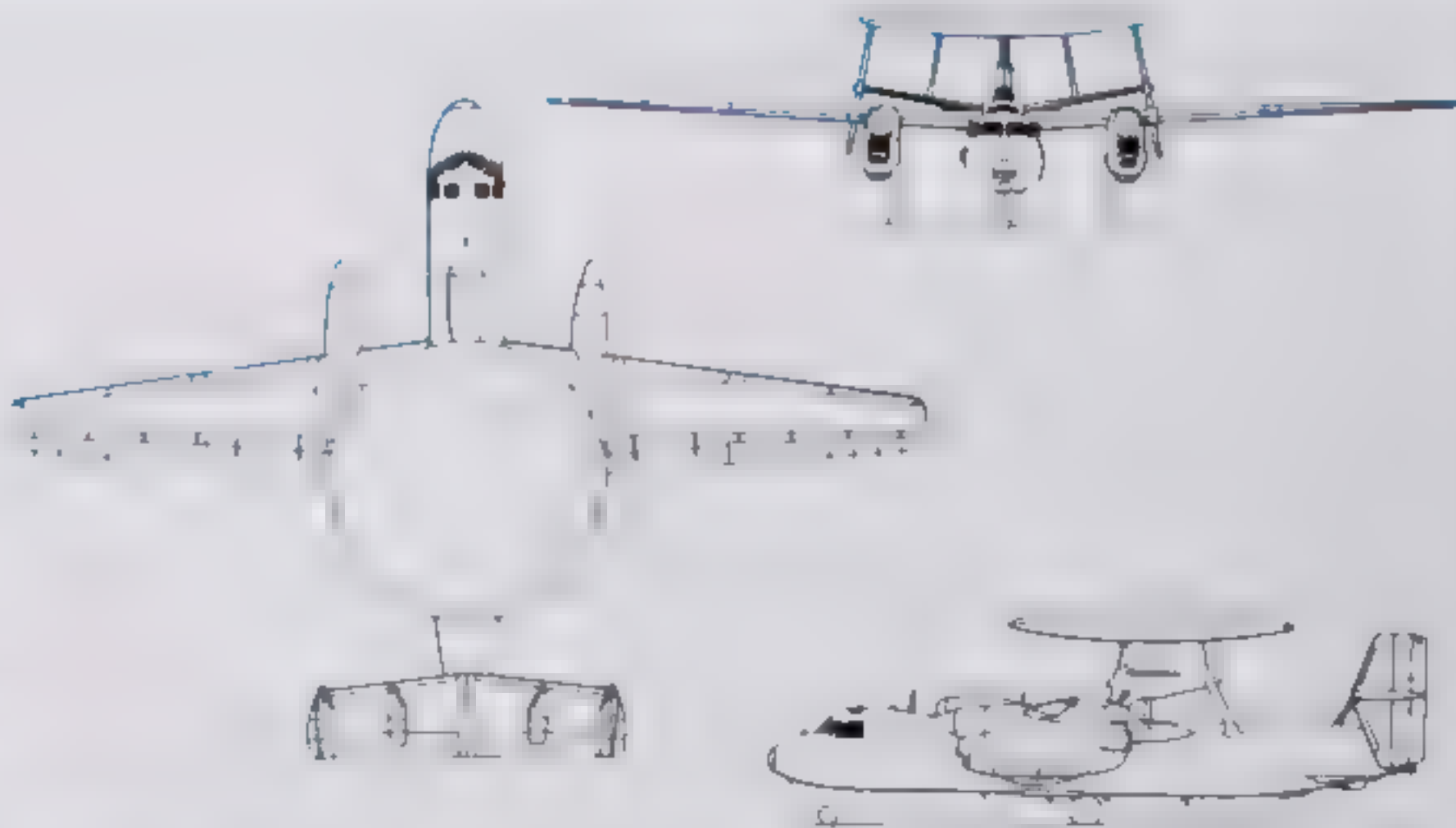
Armament: None

History: First flight (W2F-1) 21 October 1960, (production E-2A) 19 April 1961 (E-2B) 20 February 1969 (E-2C) 20 January 1971 (E-2C) 19 April 1971

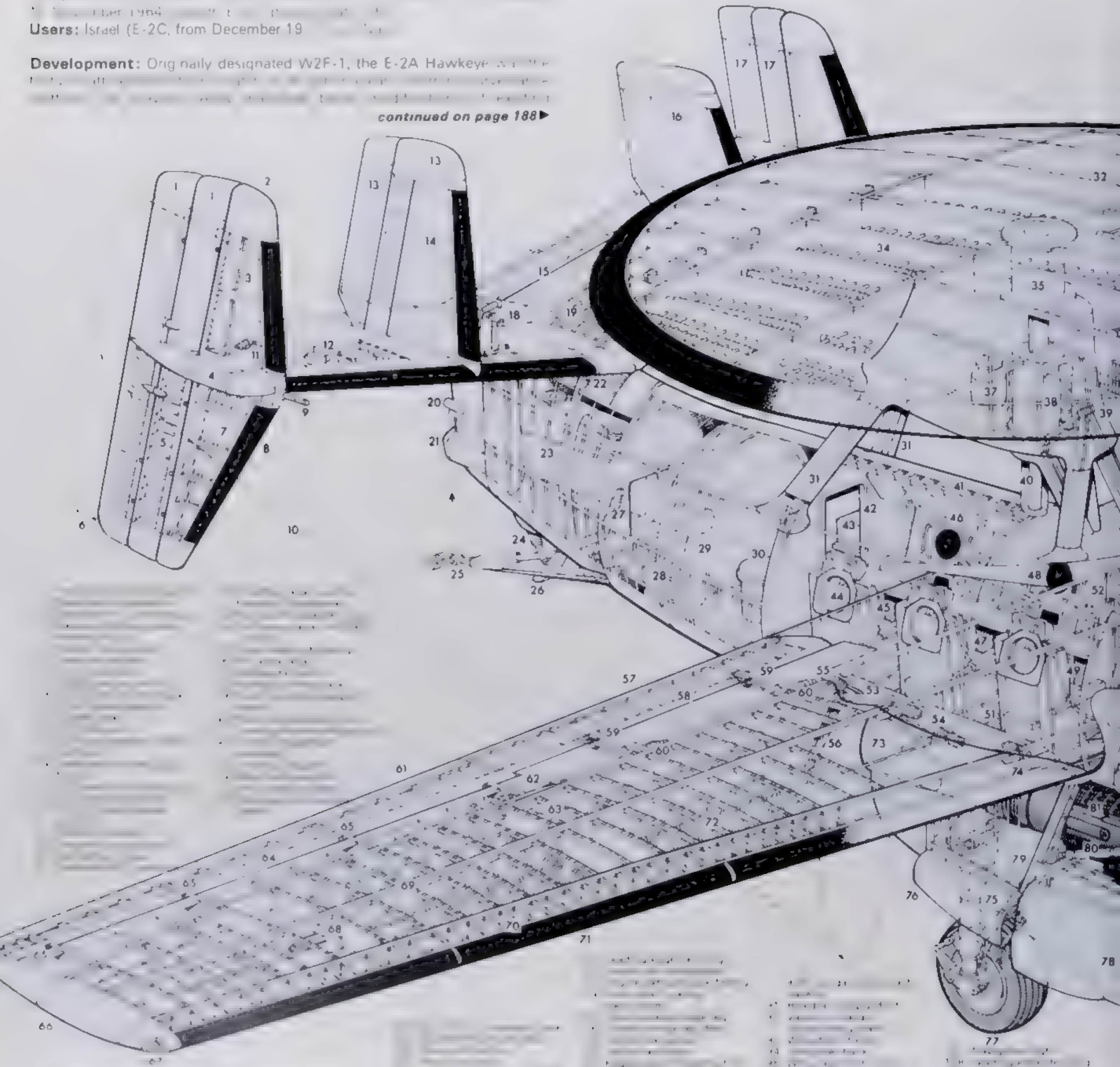
Users: Israel (E-2C, from December 1971)

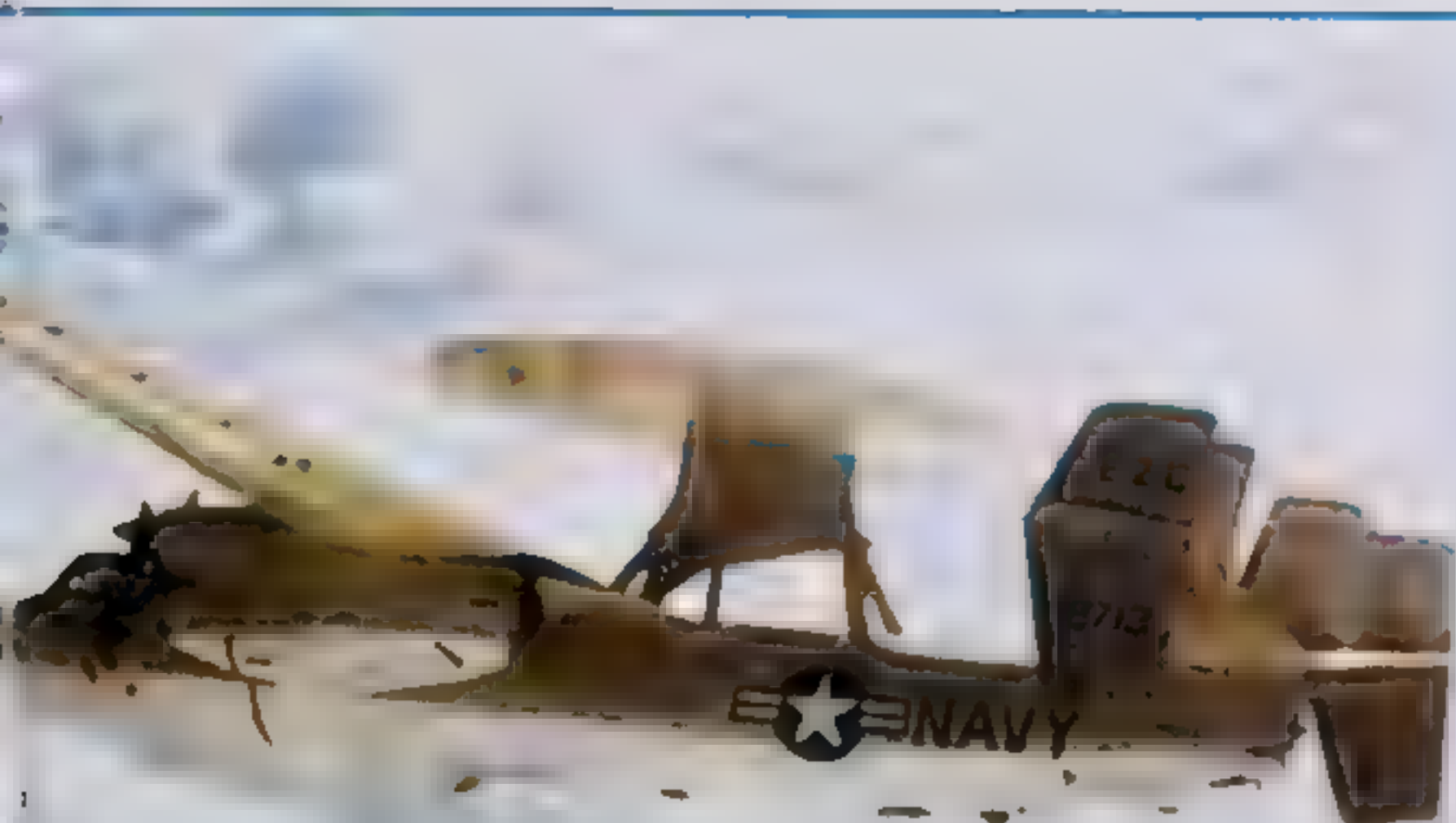
Development: Originally designated W2F-1, the E-2A Hawkeye was the first of a series of aircraft designed to provide early warning and command and control for the US Navy's fleet.

continued on page 188



Above Three-view of E-2C Hawkeye

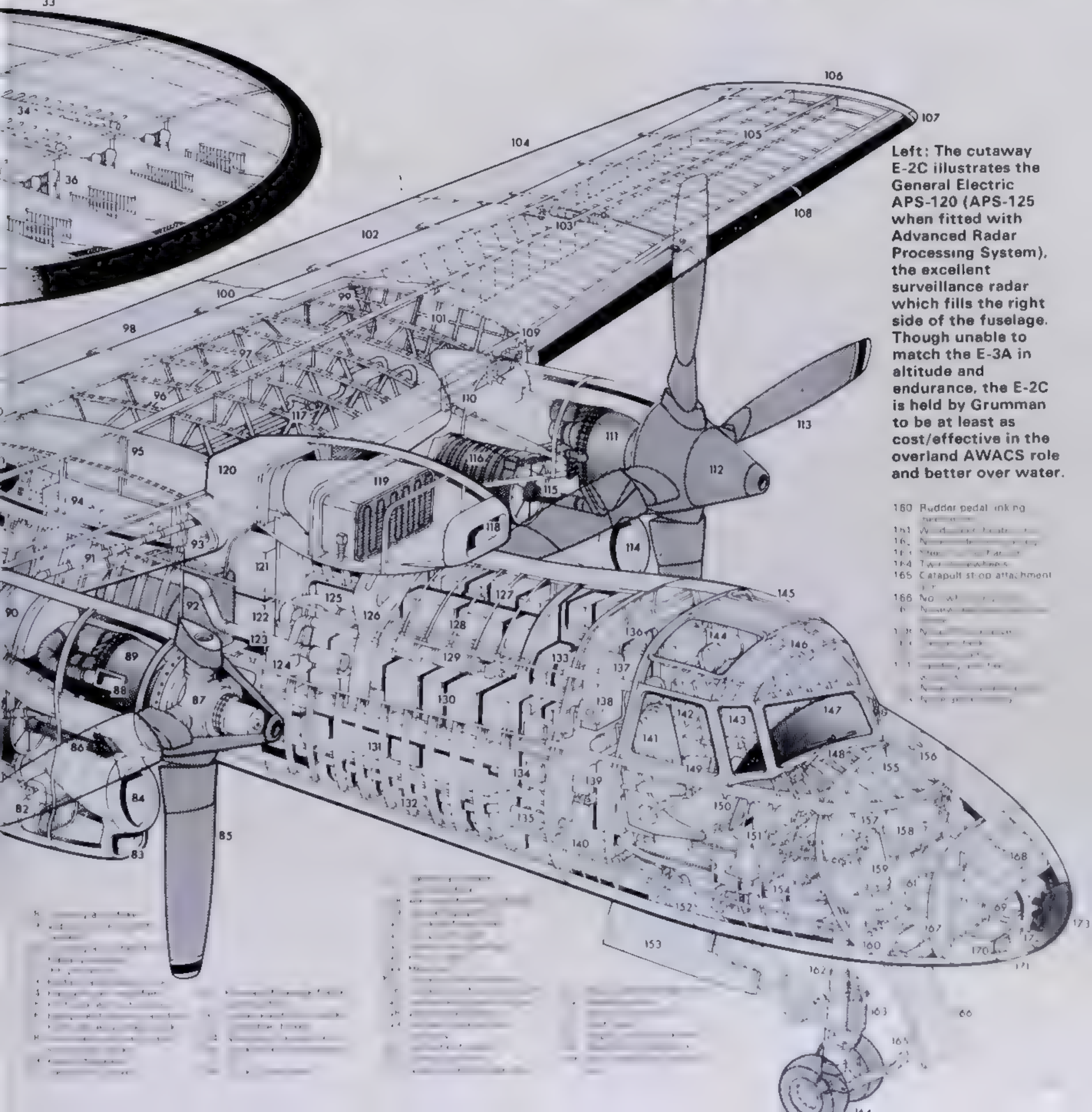




Above: Test-flying an E-2C development aircraft. The aircraft itself was familiar, but the new radar, computers and displays meant exhaustive airborne debugging.



Above: The fully proven aircraft, outwardly not greatly dissimilar to the one on the left apart from the unit markings. The TE-2C trainer also looks similar.



Left: The cutaway E-2C illustrates the General Electric APS-120 (APS-125 when fitted with Advanced Radar Processing System), the excellent surveillance radar which fills the right side of the fuselage. Though unable to match the E-3A in altitude and endurance, the E-2C is held by Grumman to be at least as cost/effective in the overland AWACS role and better over water.

- 160 Rudder pedal linkage
- 161 Wind tunnel heater
- 162 Nose landing gear
- 163 Starboard fuel tank
- 164 Tail boom
- 165 Catapult stop attachment
- 166 No. 1 engine
- 167 No. 2 engine
- 168 No. 3 engine
- 169 No. 4 engine
- 170 No. 5 engine
- 171 No. 6 engine
- 172 No. 7 engine
- 173 No. 8 engine

►Types) Equipped with an APS 96 long range radar with scanner rotating 360° in 10 minutes inside a 24ft diameter radome the E-2A has a flight crew of two and three controllers seated aft in the Airborne Tactical Data System (ATDS) compartment, which is constantly linked with the Naval Tactical Data System (NTDS) in Fleet HQ or the appropriate land base. The E-2A can handle an entire air situation and direct all friendly air operations in attacking or defensive missions. From the E-2A were derived the E-2B, with microelectronic computer, and the C-2A Greyhound COD (carrier on board delivery) transport, able to make catapult takeoffs and arrested landings with 39 passengers or bulky freight. The final version was the E-2C, with APS 120 radar and APA 171 aerial system, with OL 93 radar data processor serving a Combat Information Center (CIC) staff with complete knowledge of all airborne targets even in a land clutter environment. Though it has an advanced and costly air frame, more than three quarters of the price of an E-2C is accounted for by electronics. This version entered service with squadron VAW 123 at NAS Norfolk, Virginia, in November 1973. In 1977 Grumman is nearing the completion of E-2C manufacture, and with US government help is making a strong sales effort within NATO and in Middle East countries, emphasising the low price (about \$13 million in 1975 US dollars). Four aircraft were bought by Israel, but the NATO nations appear to prefer the Boeing E-3A. Grumman hope to sell the Growth E-2C with up to 50 per cent greater radar range, increased computer capacity and fuel in the outer wings.



Above: Thoroughly popular with the US Navy aircrew who fly it, the Hawkeye is a reliable and relatively forgiving aircraft. Here, one approaches to an arrested landing.

Below: An E-2C Hawkeye of VAW-125 (USS John F. Kennedy).

Bottom: An E-2C Hawkeye of VAW-125, embarked aboard USS *Constellation*. This aircraft is one of the early deliveries being retrofitted with the Advanced Radar Processing System.



Lockheed EC-121

Warning Star

EC-121C to EC-121T

Origin: Lockheed California Company, Burbank, USA
Type: Electronic reconnaissance and specialist missions
Engines: Four 3,250hp Wright R-3350-91 Turbo-Compound 18 cylinder radials
Dimensions: Span over tip tanks 126ft 2in (38.45m), length (most) 116ft 2in (35.41m), height overall (most) 24ft 8in (7.52m)
Weights: Empty (typical of most) 80,611lb (36,564kg), maximum loaded 145,000 lb (65,770kg)
Performance: Maximum speed at altitude 321mph (517km/h), typical cruise 259mph (417km/h), service ceiling 20,600ft (6,279m), range 4,600 miles (7,400km); endurance limit 20 hours
History: First flight (Constellation) 9 January 1943, (prototype C-121C) 1952, final delivery of rebuilt EC-121T 1973
Users: India (Navy), US Air Force and Navy

Development: After delivery of 135 military versions of the Constellation and "stretched" Super Constellation, Lockheed began in 1953 to deliver special electronic platforms (then designated RC-121C for USAF Air Defense Command, now serving as EC-121C and later versions with Aerospace Defense Command). At least 223 of four main families of electronic Super Constellation were supplied to the US Air Force and Navy, and these have been several times re-equipped or completely rebuilt, with designation EC-121K, P, Q, R, S and T. Most served in SE Asia, their main duties being electronic reconnaissance, early warning, and acting as airborne relay stations in the Igloo White system for detecting and pinpointing hostile road traffic. Crew in these variants ranges from seven to 28. The less-specialised Warning Stars still serve with ADC and the Air Reserve 79th Sqn at Homestead AFB, Florida, operating over the Atlantic, and several highly refined EC models serve with the Navy. These aircraft are all highly sophisticated platforms and the fact their equipment is inside a basically old airframe is irrelevant. The more basic Super Constellations used for reconnaissance by the French Armée de l'Air and Indian AF have been withdrawn.



Above: Three-view of EC-121H Warning Star of USAF



Above: EC-121H photographed at Sioux City, Iowa, as basic AEW-Elint platform with R-3350-93 engines and crew of 17 to 26

Below: Lockheed NC-121K of US Navy VAQ-33, with Elint collection system, pulse analysers and Litton ALQ-124 ECM installation. Operating crew of 16 to 19.



McDonnell Douglas

EA-3 Skywarrior and EB-66

A-3 Skywarrior and B-66 Destroyer

Origin: Douglas Aircraft Co (see title)

Type: Originally (A3D) carrier-based strategic bomber, (B 66) tactical

Engines: (A-3) two 12,400 lb (5625kg) thrust Pratt & Whitney J57-11 turbo-shaft turbojets, (B 66) two 10,000 lb (4536kg) thrust Allison T71-13

Dimensions: Span 72ft 6in (22.1m) length, typical (EA 3B) 76ft 4in

Weights: Empty (A-3B) 39,409 lb (17,875kg), (B 66B) 42,369 lb (19,218kg), maximum loaded (A 3B) 82,000 lb (37,195kg), (B 66B)

Performance: (Typical) maximum speed 610mph (982km/h), initial climb rate 10,000 ft/min, service ceiling 43,000 ft (13,110m), range with maximum fuel 2,000 miles

Armament: Normally none as built, most bomber A-3s and B 66s had two remotely controlled 20mm cannon in a tail turret and internal provision for 100 500 lb (5443kg) bombs (15,000 lb 6804kg in the B 66)

History: First flight (XA3D 1) 28 October 1952, (RB 66A) 28 June 1954, first delivery (A3D 1) December 1954 (first squadron, March 1956) (RB 66B) March 1956, last delivery of new aircraft (A3D 2Q) January 1961 (WB 66D)

User: USA (A 3 Navy, B 66, Air Force, not operational)

Development: The Douglas El Segundo design team under Heinemann produced the A3D as the world's first carrier-based strategic bomber matching the design both to the predicted size and mass of future thermonuclear bombs and to the deck length and strength of the super-carriers of the *Forrestal* class in 1948. The bomb bay dictated a high wing, which in turn meant the landing gears had to retract into the fuselage. Outer wings and vertical tail folded hydraulically and the nose was filled with an advanced blind bombing radar. Unfortunately Westinghouse, supplier of both the engines and the radar, succeeded in delivering only the Aero-21B tail turret. Redesigned with the J57 engine the A3D-1 finally equipped VAH-1 (Heavy Attack Squadron 1) in 1956. Soon the Skywarriors on the catapults of giant carriers of the 6th Fleet in the Mediterranean and the 7th Fleet in the Pacific were playing a central role in the balance of power in the Cold War, and giving a new global dimension to naval air power. Eventually, though only 280 Skywarriors were built (not enough, as it turned out), the changing scene called for numerous completely different versions mainly achieved by rebuilding bombers. Major production was of the A3D-2 bomber, restyled A-38B. From this evolved the A3D-2P (RA-3B) reconnaissance, A3D-2Q (EA 3B) electronic countermeasures and A3D-2T (TA-3B) radar/nav trainer, followed by rebuilds for the KA-3B tanker, EKA-3B ECM/tanker and various special-mission and test versions. The EKA-3B is still standard equipment of tanker/EW detachments aboard attack carriers and the USNR has two squadrons of KA-3Bs.

The B-66 Destroyer was produced by the Long Beach plant to meet the needs of the US Air Force. What had begun as a minimum modification of the Skywarrior turned into a totally different aircraft. Though it looked similar, hardly a single airframe part or item of equipment was common and the

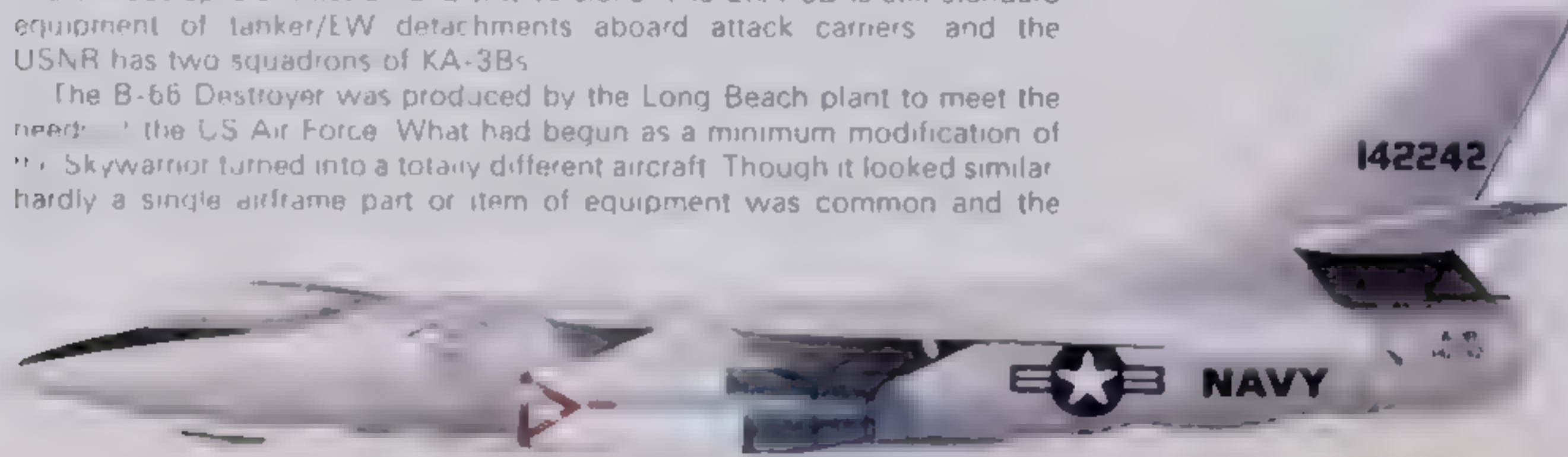


Above: The rebuilt B-66 bombers pioneered tactical ECM and ESM. This is an EB-66E, with five tons of electronics.



Above: Three-view of EKA-3B Skywarrior.

B 66 proved difficult and expensive. After building five RB-66As for indoctrination, Long Beach and a re-opened wartime plant at Tulsa built 145 RB-66B reconnaissance aircraft as well as 72 B-66B bombers. There followed 36 RB-66C electronic reconnaissance aircraft with four-man crew and 36 WB 66Ds for weather reconnaissance. Modified further over the years for many clandestine special ferreting missions, a number of EB-66E aircraft served throughout the Vietnam war, but were withdrawn from combat duty in 1974.



Above: Demilitarized A-3B as flown by the Aerospace Recovery Facility, NAS El Centro, California.

Below: Like the Hunter, the A-3 Skywarrior has had a long life in rebuilt versions, and twice the number actually built were eventually needed. One of the variants still in use is this TA-3B radar/ESM/nav trainer, with six pupils in a transport-type fuselage, seen landing at Atsugi AB, Japan. Note the refuelling probe, tail bumper, and large airbrakes on the rear fuselage. The other sub-type still in operational use is the EKA-3B (above).



Tupolev Tu-126

Design bureau Tu-126; service designation, possibly Tu-24 (NATO name "Moss")

Origin: The design bureau named for Andrei N. Tupolev, Soviet Union
Type: Airborne warning and control system

Engines: Four 14,795hp Kuznetsov NK-12MV single-shaft turboprops

Dimensions: Span 167ft 8in (51.10m), length 182ft 6in (55.48m), (with FR probe) 188ft 0in (57.30m), height overall 52ft 8in (16.05m)

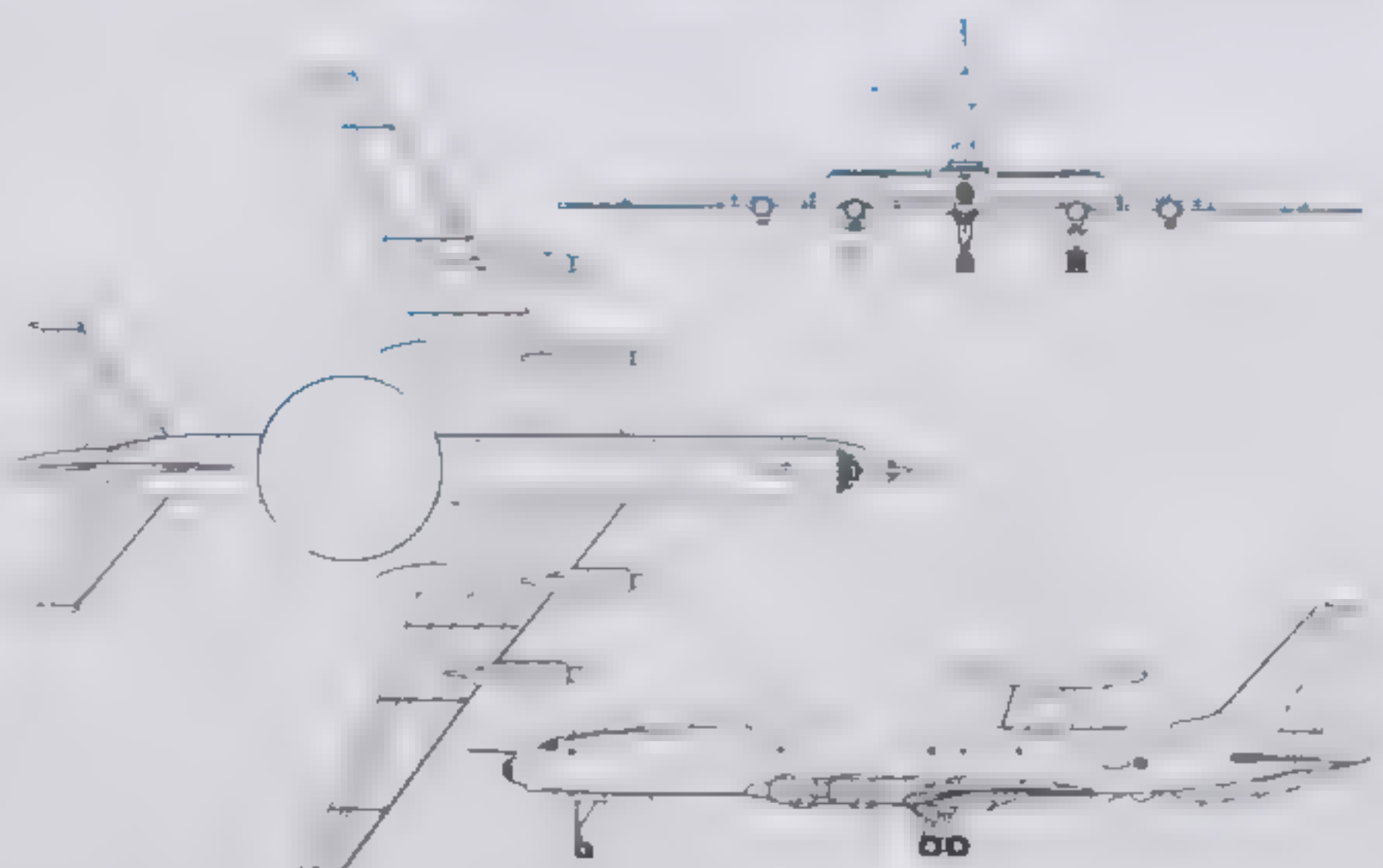
Weights: Empty (estimate) 198,400lb (90,000kg), maximum loaded 375,000lb (170,000kg)

Performance: Probable maximum speed 500mph (805km/h), on-station height 40,000ft (12,200m), normal operational endurance 18hr, ferry range at least 6,000 miles (9,650km)

History: First flight probably 1962-64, service delivery not later than 1967

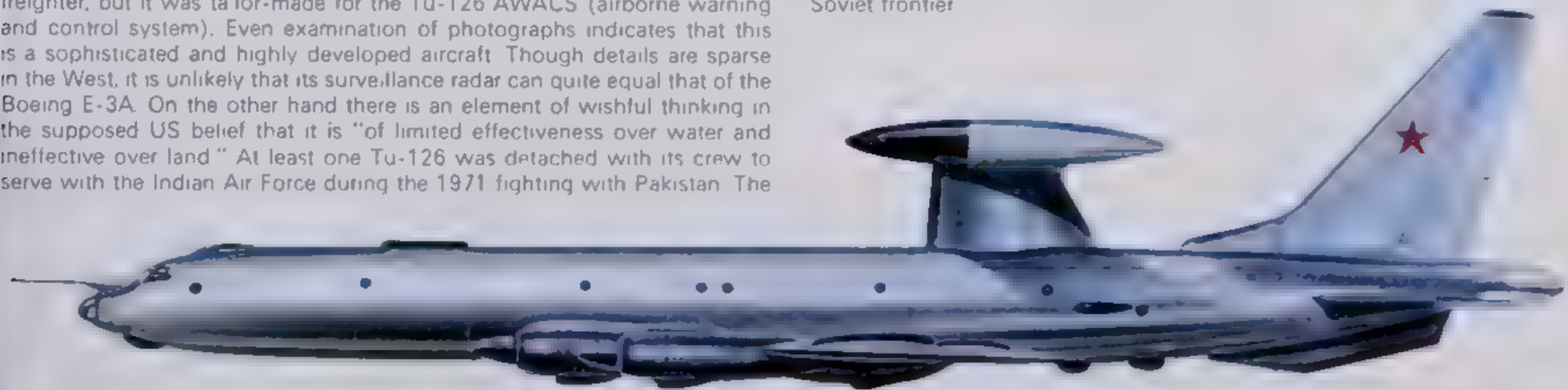
User: Soviet Union (almost certainly IA PVO)

Development: Tupolev's family of swept-wing turboprops bigger and faster than any others ever put into service began with the long-range bomber Tu-95 (service designation Tu-20). From this evolved the considerably bigger Tu-114 civil airliner and the same-size Tu-114D special transport. The Tu-114 was never used as the basis for a large military freighter, but it was tailor-made for the Tu-126 AWACS (airborne warning and control system). Even examination of photographs indicates that this is a sophisticated and highly developed aircraft. Though details are sparse in the West, it is unlikely that its surveillance radar can quite equal that of the Boeing E-3A. On the other hand there is an element of wishful thinking in the supposed US belief that it is "of limited effectiveness over water and ineffective over land." At least one Tu-126 was detached with its crew to serve with the Indian Air Force during the 1971 fighting with Pakistan. The



Three-view of Tu-126 "Moss" with landing gear extended.

number in Soviet service in 1976 was put at "at least ten" but ten times that number would be needed to patrol even the main sections of the Soviet frontier.



Above: Drawn for this book, a recent Tu-126.

Below: Enlargement from a Soviet ciné film. The "Moss" force is probably being increased by rebuilding Tu-114 transports.



Helicopters

Though the first two helicopters flew only four years after the Wright Brothers' first powered flight, the problems of rotating-wing VTOL were extremely difficult and held back the development of useful helicopters until after World War II. Today, however, the helicopter is a refined and reliable vehicle whose great versatility and unique command of the air have overcome its inherent high installed power, high initial and operating costs (compared with aeroplanes of similar payload and performance), and inability to match the speed and range of most other combat aircraft. Though there is a place in the world's air forces and navies for the light piston-engined "chopper" almost all the machines in this chapter have turboshaft engines, which have dramatically improved the helicopter's payload, per-

formance, reliability and operating cost-levels. The turboshaft has also brought other advantages to the military helicopter. It has reduced its infra-red signature, especially when this was a basic design objective, making the machine harder to hit by heat-seeking missiles. It has also reduced its noise, again especially when this was deliberate policy, though some quiet-engined helicopters still suffer from "rotor slap" which gives the enemy at least 60 seconds' warning. It has made it possible to slam the throttle(s) open and thereafter forget about the automated power plant and concentrate on the mission. It has made possible the provision of two engines when one would do, giving genuine twin-engine safety. And it has facilitated the uplift of fuel from other Service units, because, though armies



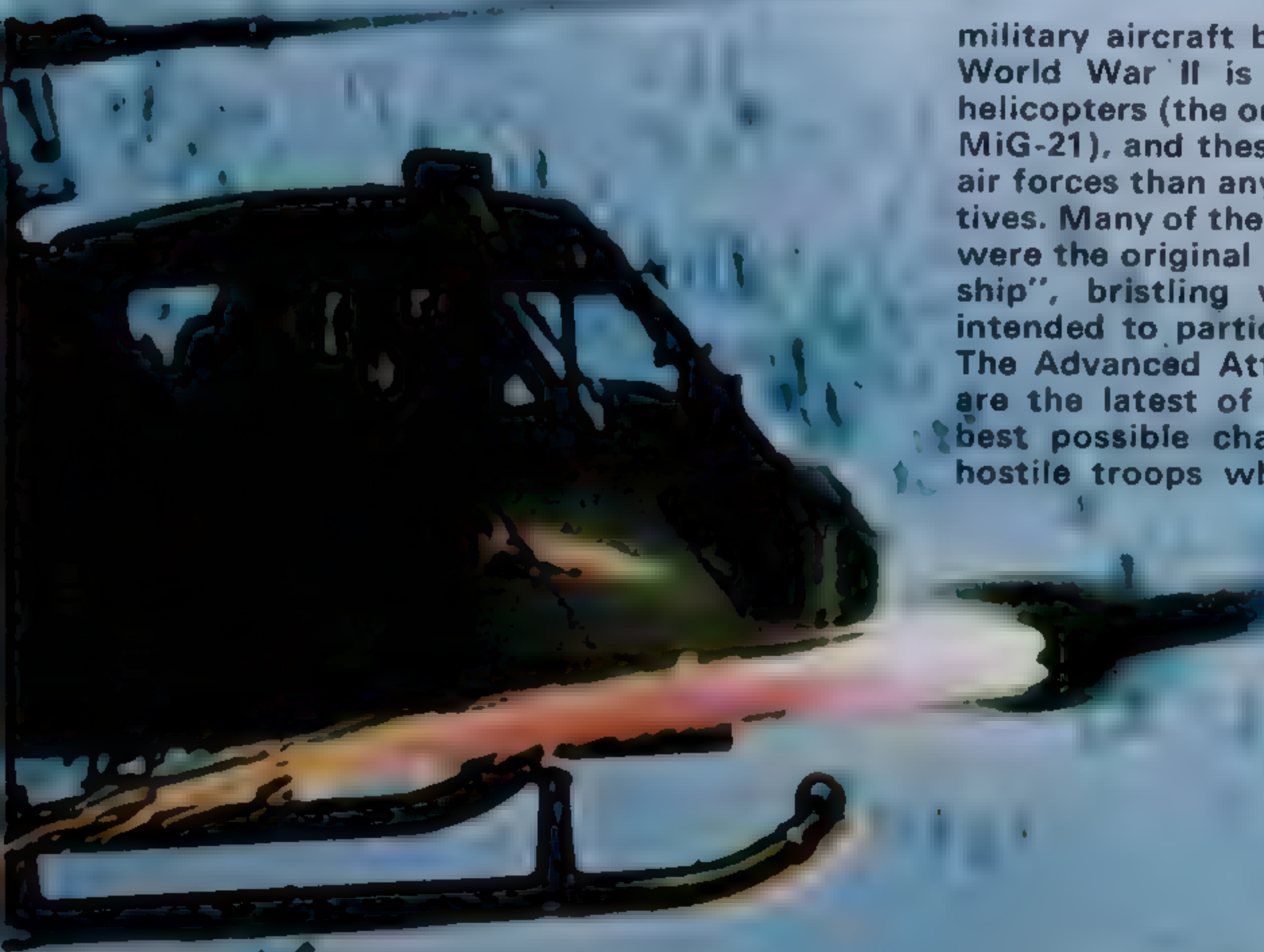
Over terrain affording little room for manoeuvre, Westland Scout AH1 (a) is shown firing a blue-tipped SS.11 missile, the rocket with burning specks appearing at the point of motor ignition. The helicopter's gyroscopically-stabilised cabin is the stabilised mount for the

and navies have gasoline, they do not have 100/130-grade suitable for aviation piston engines.

A few helicopters have been equipped to participate in air combat (for example, the US Navy Sea Sprite with Sparrow missiles), and the predecessor of the very first military helicopter, the Flettner 265, demonstrated that in 20 minutes of determined attacks with camera-guns a Bf 109 and Fw 190 could not score a single hit. Another important helicopter role is to find crashed aircraft—friendly or hostile—no matter whether or not they are in enemy-held territory, and recover them for examination and possible repair and re-use. Of course, the evacuation of casualties, including downed aircrew, was one of the very first tasks assigned to helicopters. Another duty associated with the air battle is radar surveil-

lance; 20 years ago the most powerful type of helicopter then available in the West throbbed aloft carrying a large radar to fly the AEW mission described in the last section of this book. This was a sensible idea only if the local commander had no carrier or airstrip available from which an AEW aeroplane could climb much higher and stay up much longer, and the AEW helicopter is today extinct. Apart from these roles helicopters have not played much part in air battles; most of those described on the following pages spend their life with soldiers or sailors.

Of course, the pure trainer or large transport helicopter explain themselves, and considerable numbers are used by almost every major country. Some helicopters are useful in both roles, and in many others, so it should not be surprising that the



military aircraft built in the largest numbers since World War II is probably the "Huey" family of helicopters (the only type that might exceed it is the MiG-21), and these machines are also used by more air forces than any type other than the DC-3 derivatives. Many of the Hueys are slim two-seaters which were the original production examples of the "gunship", bristling with sensors and weapons and intended to participate directly in the land battle. The Advanced Attack Helicopters of the US Army are the latest of this class, designed to have the best possible chance of survival in proximity to hostile troops who in future may have powerful

anti-aircraft weapons. Gunships cannot, however, carry troops or supplies. The Lynx typifies the versatile multi-role class that can hit enemy troops or armour, whilst trying to stay hidden itself, and can also carry sections of ten men and/or army weapons and supplies. A third battlefield family carry payloads but not weapons.

In the sea war the chief roles of the helicopter are transport for amphibious assault, transport for vertical replenishment (supplying all kinds of ships or beachheads) and ASW (dealt with in a separate section). Most of the early ASW choppers were incapable of carrying both sensors and A/S weapons, and so worked in pairs. Later the US Navy decided the primary controlling task should be performed by surface ships, using helicopters (or even remotely piloted drone helicopters, no longer in service) merely to carry sensors and weapons a few miles distant. The British boldly decided the ASW helicopter could operate as a self-contained search and attack platform, and the Westland Sea King was the first to carry a complete tactical centre as well as sensors and weapons.

Aérospatiale Alouette

SA.313 Alouette II, 315 Lama,
316 Alouette III, 318 Alouette II (Astazou)
and 319 Alouette III (Astazou).

Origin: Aérospatiale Marignane, France licence built in many countries

Type: Multi-role utility

Engine: One Turbomeca turboshaft, (SA 313B) 300 shp (220 kW), (SA 316B) 450 shp (330 kW)

Dimensions: Diameter of rotor 31ft 11in (9.75m), (III) 32ft 10in (10.03m), length of fuselage (II) 31ft 11in (9.75m), (III) 32ft 10in (10.03m)

Weights: Empt., (II) 1,014kg (2,236lb), (III) 1,014kg (2,236lb), (SA 316B) 2,520lb (1,143kg), (SA 318C) 2,520lb (1,143kg)

Performance: Maximum speed (318C) 127mph (205km/h), (316B) 130mph (210km/h), range with max payload (318C) 62 miles (100km), (316B) 300 miles (480)

History: First flight (313B) 12 March 1955, (316) 28 February 1959, (318C) 17 March 1969

Users: (II, Lama military) Argentina, Belgium, Cent Af Rep, Chile, Colombia, Congo, Dahomey, Ecuador, W Germany, India, Ivory Coast, Kenya, Laos, Lebanon, Liberia, Malagasy Rep, Mexico, Senegal, Sweden, Switzerland, Togo, Tunisia, (III) Abu Dhabi, Austria, Bangladesh, Burma, Cameroun, Congo, Denmark, Dominica, Ecuador, Ethiopia, Gabon, Ghana, Hong Kong, India, Indonesia, Iraq, Ireland, Israel, Ivory Coast, Jordan, Laos, Lebanon, Liberia, Malagasy Rep, Mexico, Netherlands, Pakistan, Peru, Portugal, Rhodesia, Romania, Rwanda, Singapore, S Africa, Spain, Switzerland, Venezuela, Yugoslavia, Zaire, Zambia

Development: By far the most successful European helicopter, the Alouette (Lark) owes its customers to the Turbomeca company which was first in the world to develop light turbine aero engines. When production of the five-seat Alouette II ended in 1975 1,305 had been built, about 1,000 being 313s. The HAL Lama is a high-altitude weight-lifter with II airframe and III engine and transmission (all III engines are de-rated to give about 570shp under hot/high conditions). It is made by Aérospatiale and HAL, the Indian Army name being Cheetah. By 1977 over 1,450 seven-seat IIIs had been built, most being 319B, also Cheetah when made in India. Options include various weapons, pontoon floats, rescue hoist and simple radar.



Above: An Aérospatiale SS.11 anti-tank missile accelerates away from an Alouette III of the ALAT, French Army.



Above: Three-view of SA 316C Alouette III (319 similar).

Below: Representing a large and diverse group of military Alouette III users around the world, this machine of the Royal Danish Navy is more colourful than most. It is one of eight which are maintained by 722 Sqn (Sea King) of the RDAF. It is typical of the basic SA 316C sub-type with seven seats, wheels, and no armament.



Above: Alouette III general-utility model of Royal Malaysian AF.



Aérospatiale Super Frelon

321 G, Ja and L

Origin: Soc Aérospatiale France

Type: (G) Anti submarine and offshore patrol helicopter (Ja L) utility transport

Engines: Three 1 630hp Turbomeca Turmo III C turboshaft

Dimensions: Diameter of main (six-blade) rotor 62ft (18.9m) length (rotors turning) 75ft 7 in (23m), height (tail rotor turning) 21ft 10in (6.6m)

Weights: Empty, equipped 14 607lb (6626kg), maximum 28 660lb (13 000kg)

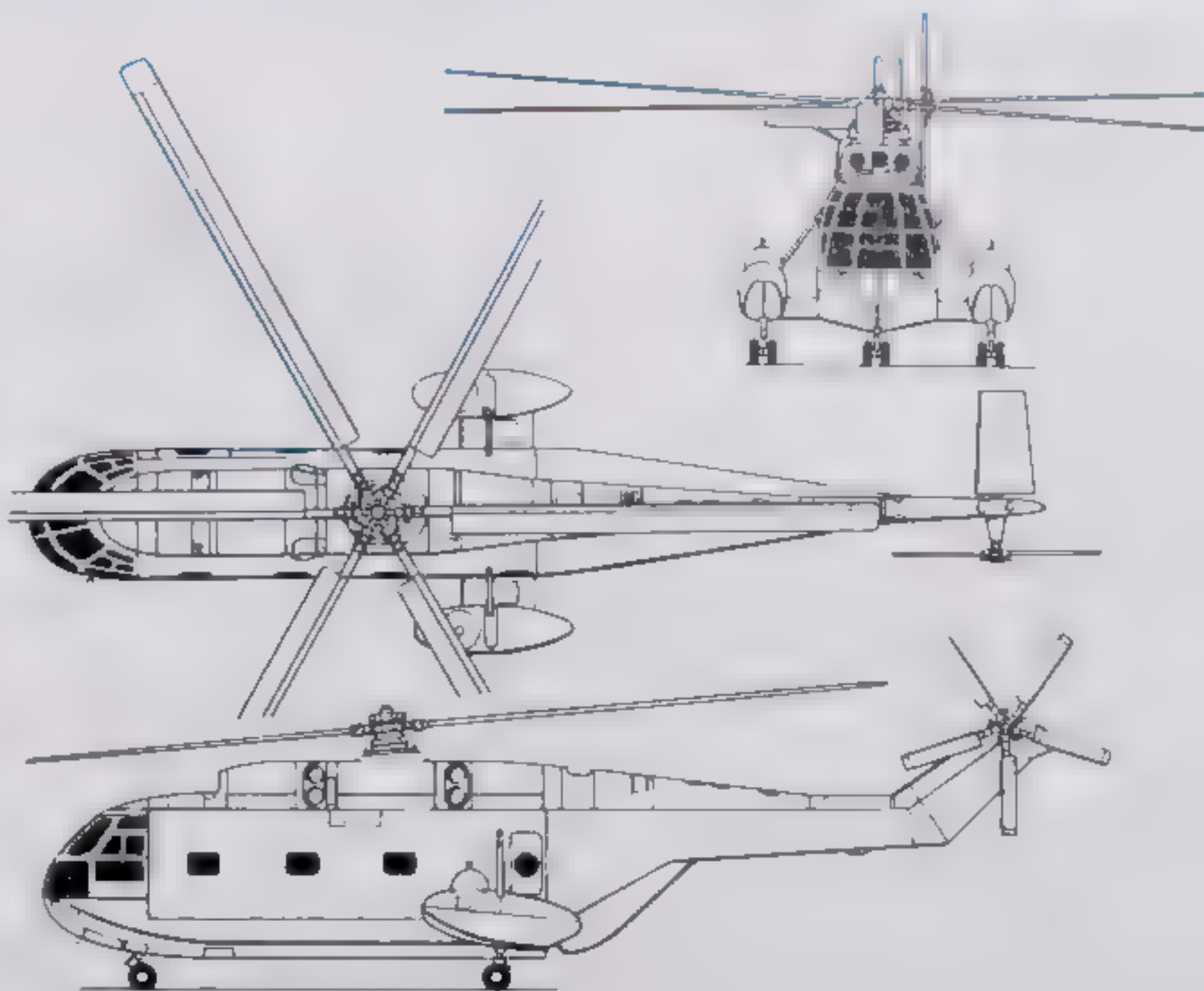
Performance: Maximum speed 171mph (275km/h) (a Super Frelon prototype in racing trim set a record at 212mph in 1963), maximum rate of climb 1,312ft (400m)/min (three engines), 479ft (146m)/min (two engines), service ceiling 10,325ft (3150m), endurance in ASW role 4hr

Armament: See text

History: First flight SA 3210 Super Frelon December 1962, SA 321G November 1965

Users: China France (Aéronavale) Iran Israel Libya, S Africa Syria reported but unconfirmed by Aérospatiale

Development: The biggest and heaviest helicopter yet produced in quantity to a West European design, the Super Frelon first flew in 1962 at the Marignane (Marseilles) plant of what was then Sud-Aviation. It was derived from the SA 3200 Frelon with the assistance of Sikorsky Aircraft whose technology and experience were used in the lifting and tail rotors and drive systems. Fiat of Italy assisted with the main gearbox and power transmission and continue to make these parts. The Super Frelon has been made in three versions, SA 321F civil airliner, SA 321G anti-submarine and SA 321Ja utility. The 321Ja is the most numerous version and has been sold to several air forces. A sub-variant called SA 321L serves in quantity with the South African Air Force, and Israel used Super Frelons to carry commando raiders to Beirut Airport. The 321G is a specialised ASW aircraft which equips Flotille 32F of the Aéronavale (French Naval Air Arm). It operates in groups, usually of four, one carrying a Sylphe panoramic radar and dunking sonar to find targets and the others each armed with four homing torpedoes. In the anti-ship role the 321G can carry two of the big Exocet long-range missiles. Another role is towing and mine-sweeping



Above: Three-view of SA 321Ja (most non-radar versions similar)

and the three powerful engines provide enough power reserve for a towing pull of 6,600lb (3000kg). All combat Super Frelons can operate from air fields, ships or from water.

Below: Though the primary mission of most of the SA 321G Super Frelon force of the Aéronavale is anti-submarine patrol in support of the French nuclear-missile submarines, this G-model is seen engaged in launch tests for the AM39 Exocet missile. Weighing 1,430lb (650kg), the air-launched version of Exocet is a formidable anti-ship missile, with speed of Mach 0.93 and range (helicopter-launched at 330ft altitude) of just over 32 miles (52km).



Aérospatiale/Westland Gazelle

SA 341B, C, D, E, F and H and SA 342K

Origin

Type
Engine

Dimensions

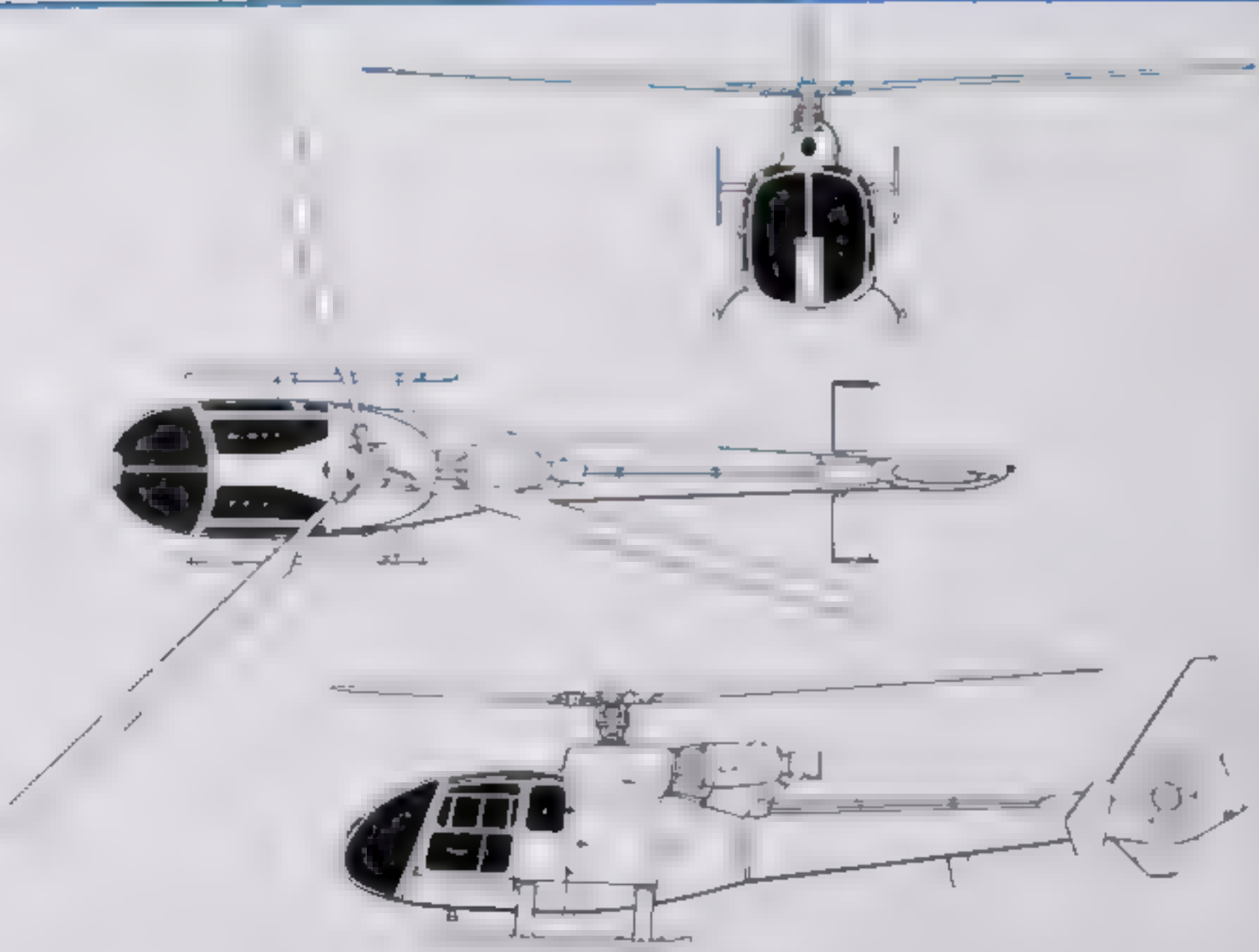
Weights

Performance

Armament

History
Users

Development



Above Three-view of typical SA 341 with Astazou IIIA, no wheels



Above Gazelle HT.3 trainer of RAF CFS, Shawbury



Below: Today the Alouette is vulnerable over the battlefield, but the much faster and more nimble Gazelle has a fair chance of survival. This Gazelle AH.1 of the British Army is scouting on manoeuvres with armour (the Chieftain Mk 3 battle tank has its turret pointing astern). Note the two blade aerals under the tail boom for Army communications.



Aérospatiale/Westland

Puma and Super Puma

SA 330B, C, E, H and L and SA 331 Super Puma

Origin: Aérospatiale (France) and Westland Helicopters (UK) association with the British Helicopter Association

Type: Assault helicopter

Engines: 330B, C, E, H and L: Turbomeca TM300 (1,800 hp), Turbomeca TM300 (1,800 hp); 331: Turbomeca TM300 (1,800 hp), Turbomeca TM300 (1,800 hp)

Dimensions: Diameter of four-blade main rotor 41 ft 11 in (12.78 m), length overall (rotors turning) 69 ft 6 1/2 in (21.29 m), height 16 ft 10 1/2 in (5.14 m)

Weights: Empty (H) 9,100 lb (4,130 kg), loaded (H) 15,430 lb (7,000 kg), (L) 16,315 lb (7,400 kg), (331) about 21,000 lb (9,500 kg)

Performance: Maximum cruise (S/L) 159 mph (257 km/h), maximum speed with standard fuel (330, typical) 360 miles (580 km)

Armament: Many customer options including weapon pods, missiles or miss'es, and various axial- or side-firing cannon or Minigun installations

History: First flight 15 April 1965, service delivery (330B) April 1969

Users: (Military) includes Abu Dhabi, Algeria, Belgium, Cameroun, Chad, Chile, Ecuador, France, Gabon, Ivory Coast, Kuwait, Morocco, Nepal, Nigeria, Pakistan, Portugal, S Africa, Togo, Tunisia, UK (RAF), Zaire

Development: Developed for the French Army ALAT (Aviation Légère de l'Armée de Terre), this fast and capable helicopter has surpassed all expecta-



Three-view of SA 330 Puma, without armament

tions in sales to both military and civil customers. By 1977 sales exceeded 500, most of them outside the Anglo-French partner countries that build the original version. There are large sliding doors on each side and loads include 20 troops or 6,600 lb (3,000 kg) slung. The technically more advanced Super Puma, likely to prove an equally successful machine, is not a collaborative project.



Above: Puma HC.1 of 230 Sqn, RAF Odiham.

Below: Winching down a heavy load of goodies to some very exposed infantry from an even more exposed Puma HC.1 of 33 Sqn, in the RAF tactical helicopter force based at Odiham. Though designed to meet the needs of the ALAT (French Army Aviation), the Puma has proved to be exactly right for military customers (and now offshore oil support operators) all over the world. SA 331 Super Puma promises to be as great a sales success in the 1980s.



Agusta A109 Hirundo

A 109A military and A 129

Origin
Type
Engines
Dimensions: Diameter of four-blade main rotor 36ft

Weights

Performance: maximum cruise 165mph (266)

Armament: (109

History: First flight 4 August 1971, (production) 1971
Users: Italian Army (evaluation)

Development: Not yet in prod



Above: Three-view of A 109A (without TOW missiles)

A 129 would have a slim tandem-seat nose similar to that of the Hughes AH-64 with pilot in the rear, and extensive weapon options. proposed engines are Allison 250 C30 derated to 450hp, with 550hp available from

Below: One of the prettiest helicopters, the Agusta A 109A is being evaluated by the Italian Army. Production could be large.



Bell 47

AB 47, KH-4, H-13 Sioux

Origin: Bell Helicopter Co (now Bell Helicopter Textron) Fort Worth, USA, licence built by Kawasaki, Japan, Agusta, Italy, Westland, UK

Type: Three-seat utility and training

Engine: Flat six piston engine with crankshaft vertical, usually 178/200hp Franklin (early models) 240hp Lycoming VO-435 or (late) 270hp TVO-435

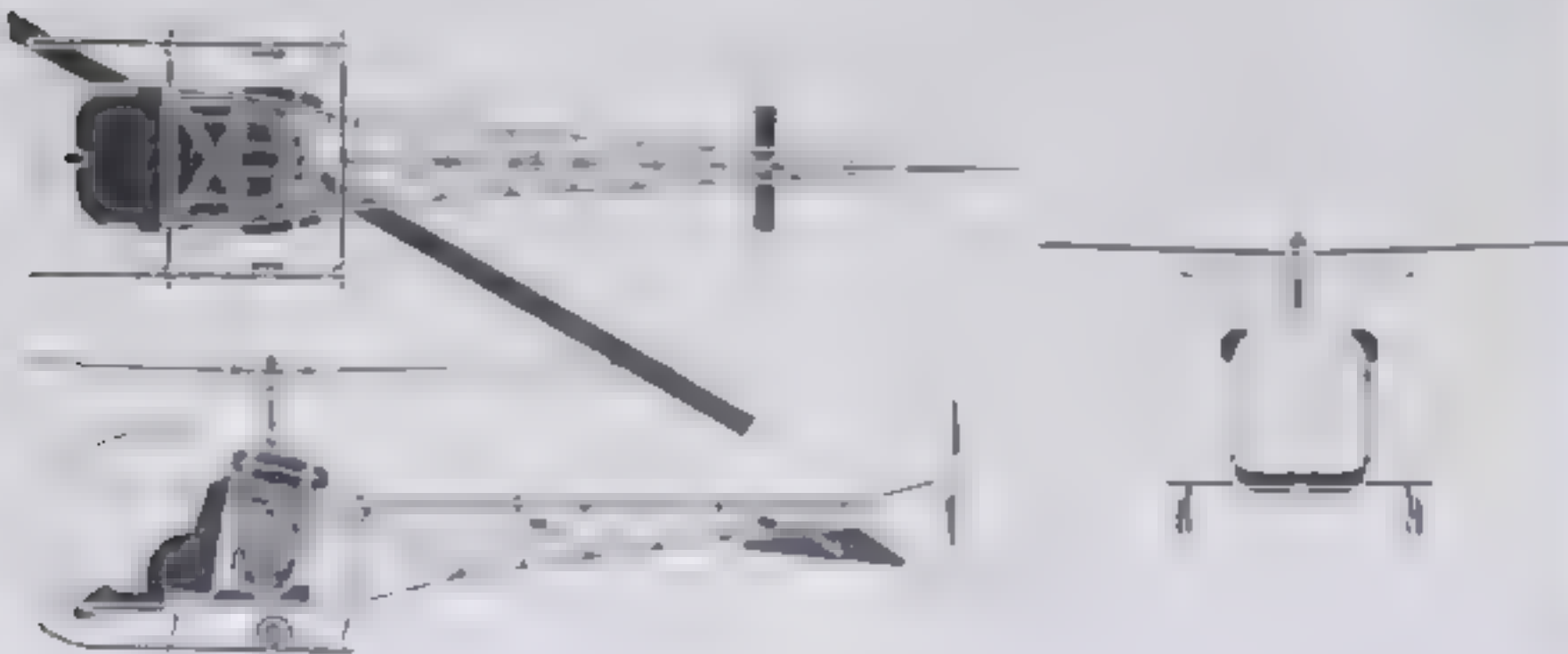
Dimensions: (Typical of late models) diameter of two-blade main rotor 37ft 1 1/2in (11.32m), length overall (rotors turning) 43ft 4 1/2in (13.20m)

Weights: (47J-3) empty 1,819lb (825kg), maximum loaded 2,950lb (1,340kg)

Performance: (Typical late model) maximum speed 105mph (169km/h), cruising speed 86mph (138km/h), range at low level, no reserve, 210 miles

Armament: Many equipped with fixed forward firing gun (LMG, GPMG)

History: First flight (prototype) 8 December 1945, service delivery of 1



Above: Three-view of typical 47G

Users: Argentina, Australia, Austria, Brazil, Burma, Cambodia (not believed operational), Canada, Chile, Dahomey, Ecuador, Greece, Guinea, Honduras, India, Indonesia, Iran, Italy, Jamaica, Japan, Kenya, Libya, Malagasy, Malaysia, Malta, Mexico, Morocco, New Zealand, Pakistan, Paraguay, Peru, Spain, Sri Lanka, Taiwan, Tanzania, Turkey, UK (Army)

Development: Larry Bell flew his first helicopter in mid-1943, and in 1946 the completely redesigned Model 47 became the first helicopter in the world to be certificated for general use. Over 5,000 were built by Bell, Kawasaki (KH-4 variants) and Westland. Agusta still builds at a low rate, with 1,200 delivered since 1954. Most military 47s are liaison and training machines seating the pilot centrally in front and two passengers behind.



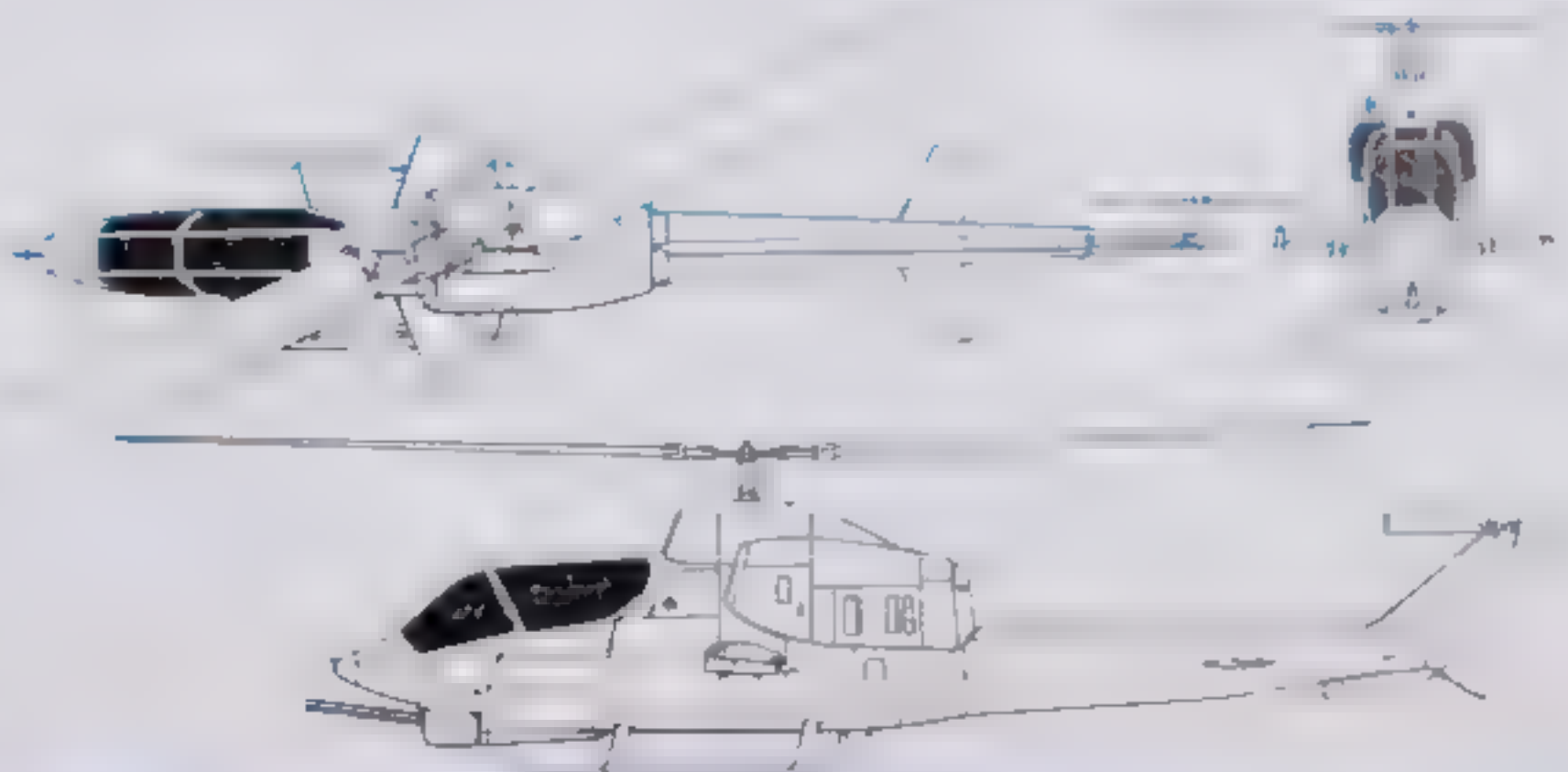
Above: 32 years after first flight thousands of Bell 47 variants still clatter people to and fro. Here, an Agusta-built Sioux AH.1 of British Army passes Schloss Marienburg, south of Hanover

Bell 209 HueyCobra

AH-1G to -1T HueyCobra (data for -1G)

Origin: Bell Helicopter Textron, USA
Type: Two-seat combat helicopter
Engine: 1,100shp Lycoming T53-L-13 turboshaft
Dimensions: Main-rotor diameter 44ft (13.4m) overall length (rotors turning) 52ft 11½in (16.14m) length of fuselage 44ft 5in (13.54m) height 13ft 5½in (4.1m)
Weights: Empty 6,073lb (2,754kg) maximum gross 9,500lb (4,308kg)
Performance: Maximum speed 119mph (52km/h) maximum climb rate 3,100ft/min (not vertical) 1,230ft (375m) min service ceiling, 11,400ft (3483m) hovering ceiling in ground effect 9,900ft (3015m) range at sea level with 8% reserve 357 miles (574km)
Armament: Typically one 7.62mm multi-barrel Minigun, one 4.2in grenade launcher, both in remote controlled turret, 2.75in x barrel, 3.5in three barrel cannon, post rotor hub gun for 70mm rocket, 2.75in x barrel or Minigun pods or 20mm cannon pods (TOWAC or eight TOW missiles in tandem tube launchers in two outer pylons, one being available for other stores)
History: First flight 7 September 1965, combat service June 1967 (TOW Cobra January 1973)
Users: Iran, Saudi Arabia, Spain (Navy), US Army, US Marine Corps

Development: First flown in 1965 after only six months of development the HueyCobra is a combat development of the UH-1 Iroquois family. It combines the dynamic parts – engine, transmission and rotor system – of the original Huey with a new streamlined fuselage providing for a quick view of the front and pilot above and behind him and for a wide range of fixed and power-aimed armament systems. The first version was the US Army AH-1G with 1,100hp T53 engine, of which 1,124 were delivered, in addition 11 to the Spanish Navy for anti-ship strike and 38 as trainers to the US Marine Corps. The AH-1Q is an anti-armour version often called TOWCobra because it carries eight TOW missile pods as well as the appropriate sighting system. The AH-1J SeaCobra of the Marine Corps and Iranian Army has twin engines, the 1,800hp UAC Twin Pac having two T400 power sections driving one shaft. Latest versions are the -1Q, -1R, -1S and -1T with more power and new equipment. All Cobras can have a great variety of armament.



Above: Three-view of AH-1J or similar Twin Pac model



Above: Bell AH-1J SeaCobra of Imperial Iranian Army Aviation Service, with twin T400 engines. Note XM197 electrically-driven turret with triple 60in (1.52m) barrels.



Above: Original Cobra, the US Army AH-1G.

Below: Probably the most important anti-tank missile in the world (pending a new one from the Soviet Union), TOW gets its name from Tube launched, Optically sighted and Wire guided. It arms the AH-1Q TOW Cobra.



Bell "Huey" family

**XH-40, UH-1 Iroquois series
(Models 204, 205, 212 and 214), CH-118
and -135, and Isfahan**

Origin: Bell Helicopter Textron, Fort Worth, USA, built under licence to Agusta (Italy), Fuji (Japan) and AIDC (Taiwan)

Type: Multi-role utility and transport helicopter

Engine: Originally one Lycoming T53 free turbine turboprop 640-shp, later rising in stages to 825, 930, 1,100

shp (me Agusta-built AB-204), Rolls Royce Gnome

800-shp P&W PT6F-3 (T400) coupled turboshafts (flat-rate

and with 900-shp immediately available from either fu

ther (214) 2,930-shp Lycoming LTC4B (T55) flat-rated at 2,050-shp

Dimensions: Diameter of twin-blade main rotor (204 UH-1B/C) 44ft 0in

(13.41m) (205, 212) 48ft 0in (14.63m) (tracking tips) 48ft 2½in (14.69m)

(214) 50ft 0in (15.24m) overall length (rotors turning) (early) 53ft 0in

(16.15m) (virtually all modern versions) 57ft 3¼in (17.46m) height

overall (modern typical) 14ft 4¾in (4.39m)

Weights: Empty (XH-40) about 4,000lb (1,814kg), (typical 205) 4,667lb

(2,116kg), (typical 212) 5,549lb (2,517kg), (214/214B) about 6,000lb

(2,722kg) maximum loaded (XH-40) 5,800lb (2,631kg) (typical 205)

9,500lb (4,309kg) (212/UH-1N) 10,500lb (4,762kg) (214B) 16,000lb

(7,257kg)

Performance: Maximum speed (all) typically 127mph (204km/h) econ

omise speed, usually same max range with useful payload typically

248 miles (400km)

Armament: See text

History: First flight (XH-40) 22 October 1956 (production UH-1) 1958

(205) August 1961 (212) 1969 (214) 1974

Users: (Bell-built) Argentina, Australia, Austria, Brazil, Brunei, Burma,

Canada, Chile, Colombia, Dubai, Ethiopia, W. Germany, Ghana, Greece,

Guatemala, Indonesia, Iran, Israel, Italy, Jamaica, Japan, Kampuchea,

S. Korea, Lebanon, Malaysia, Mexico, Morocco, Netherlands, New Zealand,

Norway, Oman, Panama, Peru, Philippines, Spain, Taiwan, Thailand,

Turkey, Uganda, Uruguay, USA (AF, Navy, Marines, Army, Coast Guard),



Above: Three-view of Bell UH-1H Iroquois, with additional side view (bottom) of UH-1N.

Venezuela, Yugoslavia, Zambia, (Agusta-built) Austria, Iran, Italy, Kuwait, Lebanon, Morocco, Netherlands, Norway, Oman, Saudi Arabia, Somalia, Spain, Sweden, Turkey, United Arab Emirates, Yemen, Yugoslavia, Zambia (Fuji-built) Japan, (AIDC-built) Taiwan



Above: Agusta Bell AB 204B of Luftstreitkräfte (Austrian Air Force)



Above: Bell 205 of Heyl Ha'Avir (Israel Defence Force Air Force).

Foot of page, left: Agusta builds unique ASW versions with radar, sonobuoys and (yellow) torpedoes. This is the latest of this family, the twin-engined 212ASW, in production for Italy and Turkey.

Below: a locally developed Bofors Bantam guided missile zips from an Agusta-Bell 204B of the Swedish Army Aviation Department.





Above: Large numbers of Huey versions served in Vietnam. Here an air gunner blasts away with a 7.62mm M60 from a new UH-1H.

Below: UH-1N twin-engined "Two-Twelve" of the US Navy, with seats for 14 passengers.

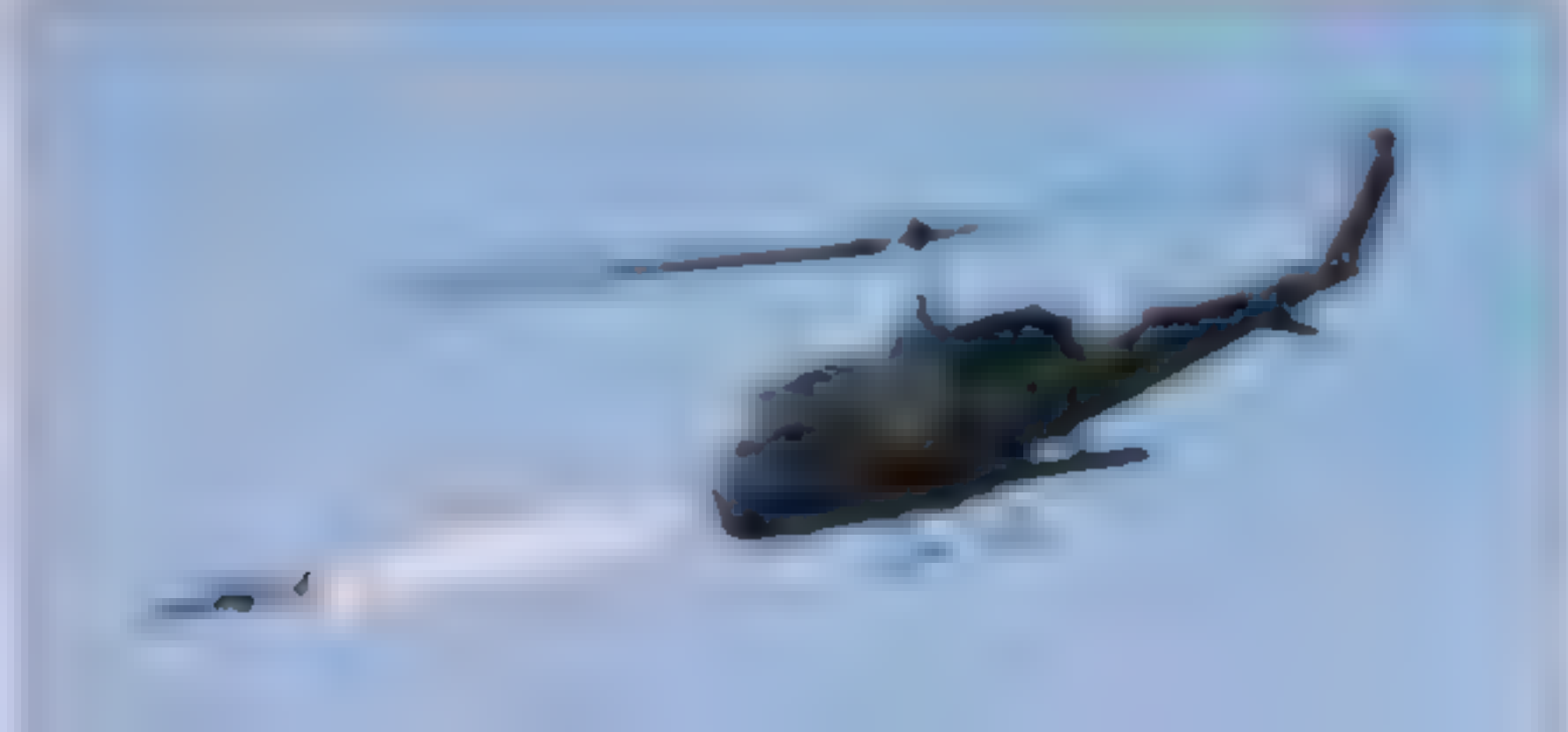
Bottom: The kind of mission that Huey pilots took as routine in SE Asia; putting down troops where landing was impossible.

Foot of page, left: Test launch of a Sistel Sea Killer Mk 1 missile from AB.204AS of Italian Navy.



Above: Bell UH-1D Iroquois (US Army).

Development: Used by more air forces, and built in greater numbers, than any other military aircraft since World War II, the "Huey" family of helicopters grew from a single prototype, the XH-40, for the US Army. Over 20 years the gross weight has been almost multiplied by three, though the size has changed only slightly. Early versions seat eight to ten, carried the occasional machine-gun, and included the TH-1L Seawolf trainer for the US Navy and the Italian-developed Agusta-Bell 204AS with radar and ASW sensors and torpedoes. The Model 205 (UH-1D, -1H &c) have more power and carry up to 15 passengers. Dornier built 352 for the West German Army, and similar versions are still in production at Agusta, Fuji and Agusta. Canada sponsored the twin-engined 212 (UH-1N, Canada CH-139), which again is made in Italy in an ASW version, with a new radar, AQS-13, variable depth sonar and two torpedoes. Model 214, H-14, H-14A, H-14B and 214B first ordered by Iran, in whose service the 214A Istahban I set several climb and altitude records. The 214 series have a new high torque transmission system "Noda Matic" vibration damping suspension, broad rotor blades allowing speed to rise to 150mph (241km/h). Many Hueys (called thus from the original "HU" designation, later changed to UH) carry guns, anti-tank missiles and special night fighting gear, but most are simple casevac and assault transports. Official US military name is "Huey". The HueyCobra "gunship" models are described on the previous page.



Bell Kiowa and JetRanger

Variants, see text.

Origin

Type
Engine

Dimensions

Weights

Performance

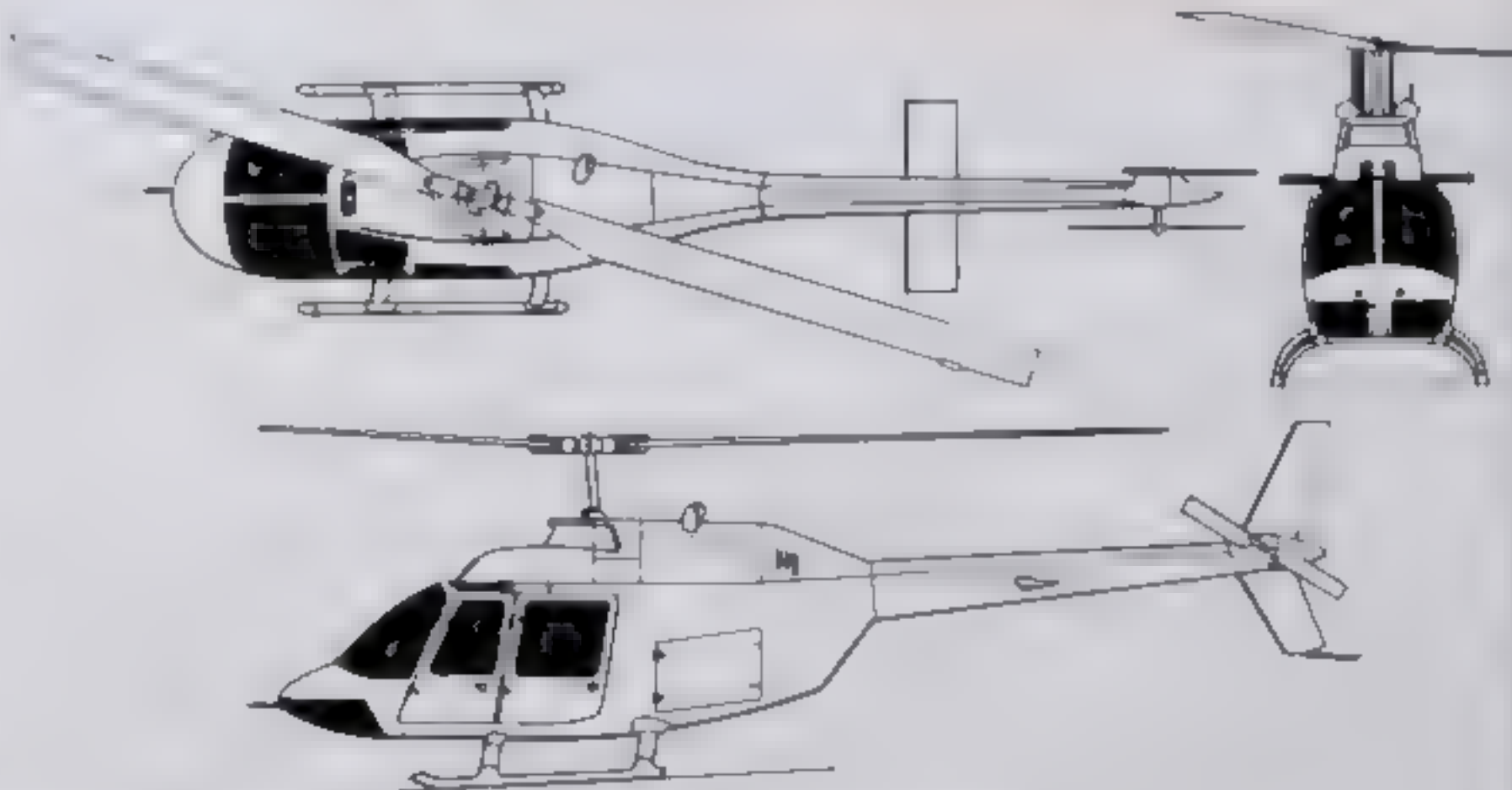
Armament History

Users

Development: First flown as the OH-63 in 1962, the AH-63 was developed as a light attack helicopter for the US Army. It was designed to be a simple, easy-to-fly aircraft that could be operated by a single pilot. The AH-63 was developed as a light attack helicopter for the US Army. It was designed to be a simple, easy-to-fly aircraft that could be operated by a single pilot. The AH-63 was developed as a light attack helicopter for the US Army. It was designed to be a simple, easy-to-fly aircraft that could be operated by a single pilot.



Above The Canadian Armed Forces have 74 brightly painted CH-136



Above: Three-view of typical 206B JetRanger II.

Below: The US Army received 2,200 OH-58A Kiowas, which have flown a seven-figure number of hours. Standard armament includes the XM27 kit based on the 7.62mm multi-barrel Minigun. The force may now be reworked to OH-58C standard.



Boeing-Vertol YUH-61A

YUH-61A (to be named if adopted)

Origin: Boeing Vertol, Philadelphia, USA

Type: Proposed LAMPS helicopter

Engines: Two 1,536shp General Electric T700 turbo shafts

Dimensions: Diameter of four-blade main rotor 49ft 0in (14.94m), overall (rotors turning) 60ft 8½in (18.50m), height (rotors stopped) 15ft 6in (4.72m)

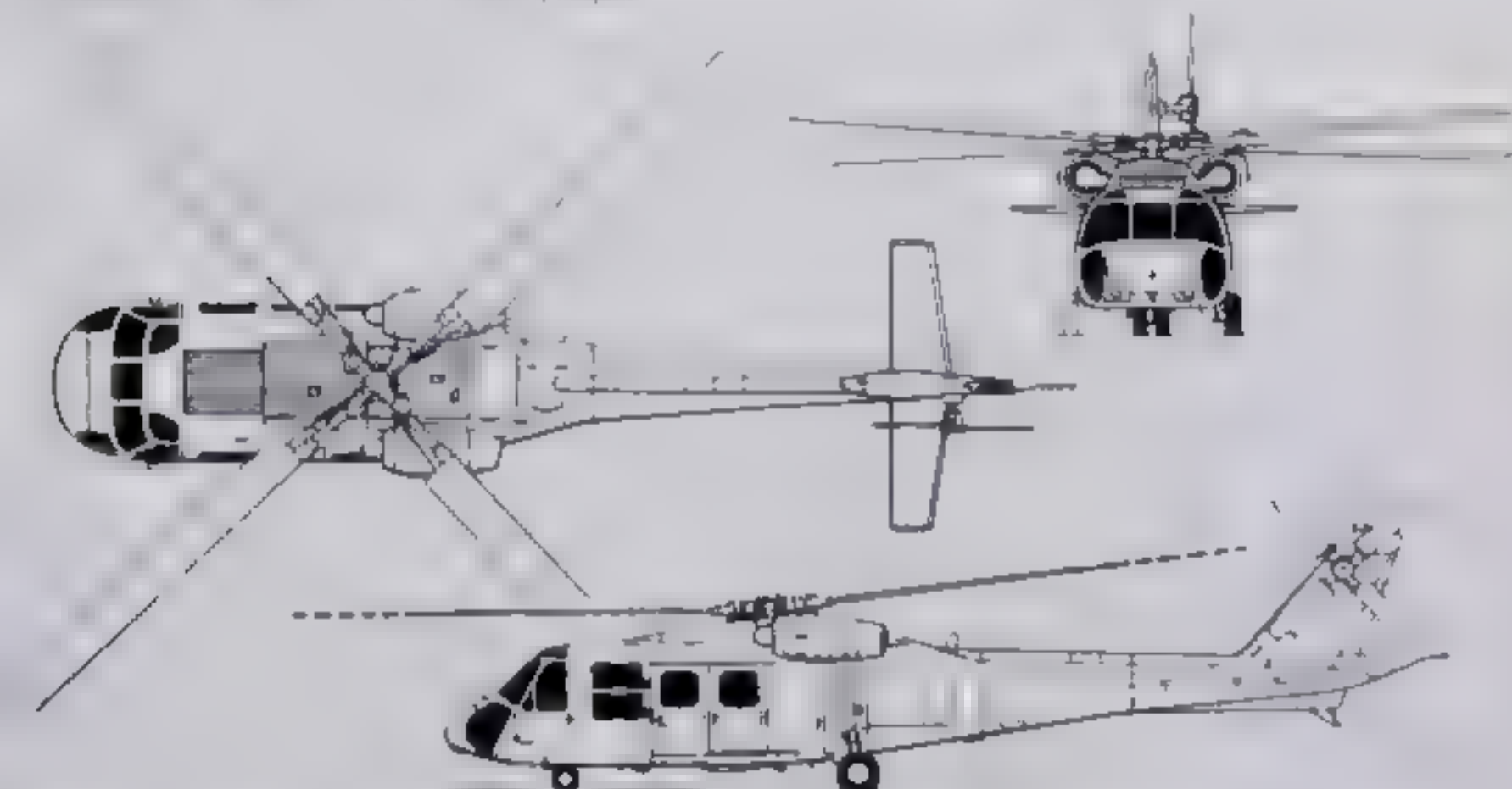
Weights: Empty 9,750lb (4422kg), maximum loaded 19,000lb (8935kg)

Performance: Cruise at 4,000ft (1220m) 167mph (269km/h), range with max 5,924lb (2687kg) payload and 30min reserve 300 miles (595km)

History: First flight 29 November 1974

User: Under evaluation for US Navy

Development: Under a \$91 million contract awarded in August 1972 Boeing Vertol designed and delivered three prototypes of this outstanding tactical transport helicopter to compete against the Sikorsky YUH-60A in the UTTAS (Joint Tactical Transport Aircraft System) programme. The YUH-61A is extremely important to Vertol even though the civil Model 179 is already a firm programme with an order for 28 from one oil company placed in 1975. The company failed to win the UTTAS programme, but it would be premature to eliminate the YUH-61A from this book because a modified version could win the important US Navy LAMPS III competition. The Light Airborne Multi-Purpose System is intended to provide DDG-963 and FFG-7 warships with helicopters to destroy hostile submarines and fire missiles at hostile surface ships.



Above: Three-view of YUH-61A (LAMPS III will differ in detail).



Above: Seen in a tight turn near the ground, the YUH-61A was competing for a potentially large US Navy order as this book went to press.



Above: For shipboard use a LAMPS III version would have a redesigned landing gear and totally different mission equipment including radar, sonobuoys, MAD "bird" and AS torpedoes

Boeing-Vertol H-46 family

CH-46 Sea Knight, UH-46, CH-113, KV-107

Origin: Boeing Vertol, Philadelphia, USA

Type:

Engines: Two 1,536shp General Electric T700 turbo shafts

Dimensions: Diameter of four-blade main rotor 49ft 0in (14.94m), overall (rotors turning) 60ft 8½in (18.50m), height (rotors stopped) 15ft 6in (4.72m)

Weights: Empty (KV-107/II 2) 9,750lb (4422kg), maximum loaded (KV) 19,000lb (8935kg) (E) 21,400lb (9706kg)

Performance: Typical cruise 120mph (193km/h), range with max 5,924lb (2687kg) payload 109 miles (175km), (2,400lb 1087kg) 63 miles (102km)

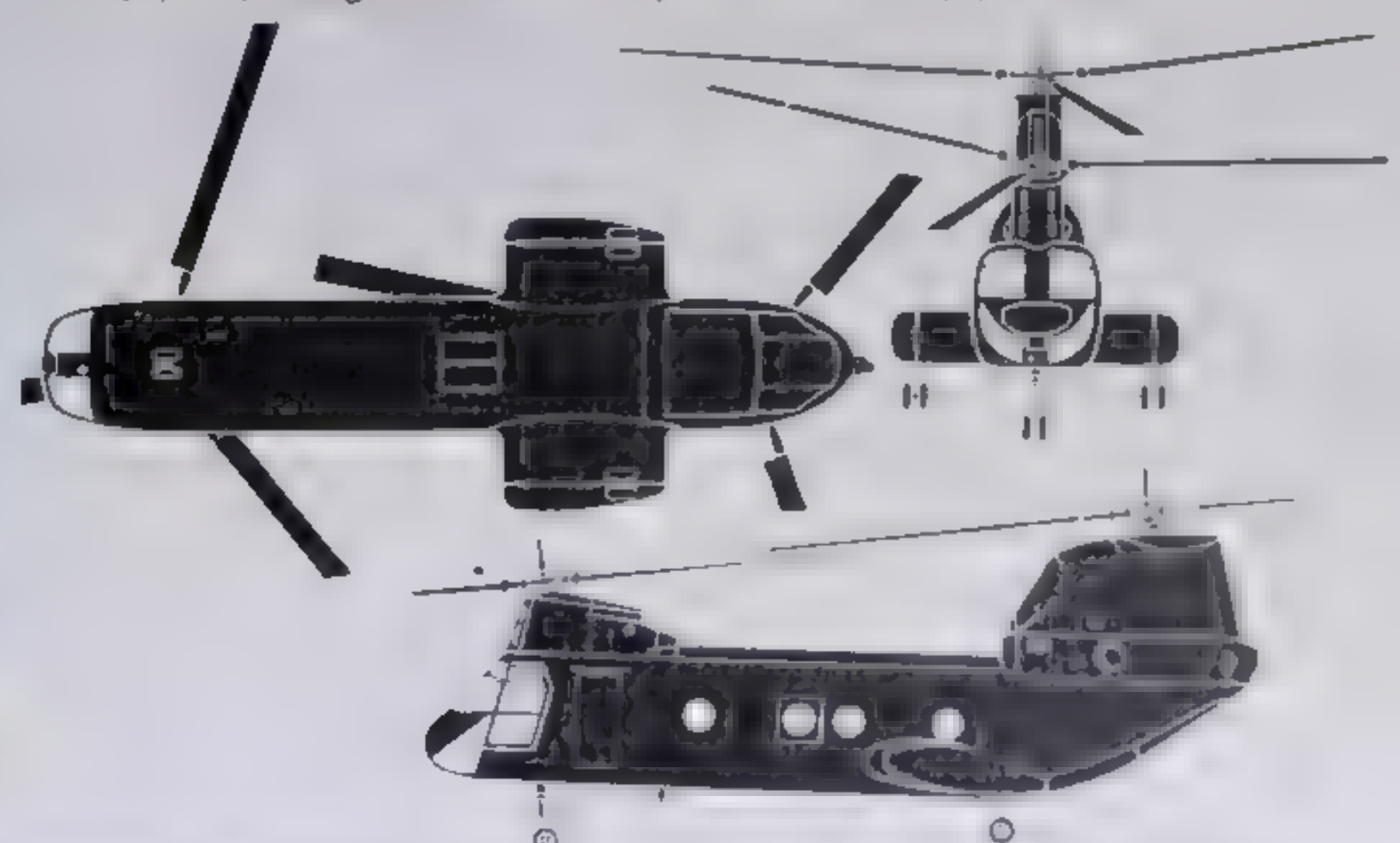
History: First flight (107) April 1958, (prototype CH-46A) 27 August 1959

Users: Canada, Japan, Sweden, USA (Marine Corps, Navy)

Development: The CH-46A Sea Knight was an assault transport carrying up to 25 equipped troops or 4,000lb (1814kg) cargo. Over 600 of these were followed by more powerful CH-46D and F versions, the Navy CH-46A for ship replenishment and the CH-46E and UH-46D with 1,870hp T58 engines. Canada uses CH-113 Labradors and 113A Voyageurs. Sweden uses the HKP-7 with Gnome engines, and in Japan Kawasaki has built nearly 100 of various KV-107 versions.



Above: Kawasaki builds several versions of the basic 107, including the KV-107/IIA-3 of the JMSDF (upper) and KV-107/II-5, designated HKP-7 by Swedish Navy (lower)



Above: Three-view of CH-46D (other versions generally similar)

Boeing-Vertol CH-47 Chinook

CH-47A, B and C Chinook (data for C)

Origin

Type

Engines

Dimensions

Weights

Empty 20 616lb (9351kg), loaded (condition 1) 33 000lb

Performance: Maximum speed (condition 1) 189mph (304km/h) (11)

Armament

History

Users: Australia, Canada, Iran*, Israel, Italy*, Spain, Thailand, US Army

Development: Development of the Vertol 114 began in 1956 to meet the



Above: Three-view of CH-47C (CH-47B similar)

proved a valuable vehicle, soon standardised a
rescuing civilians (on one occasion 147 refug
work for salvage or repair
0 million. The A model
numerous improve
Chinooks
Piasack.)



Above: Boeing-Vertol CH-47B Chinook before complete rebuild to form pictured below.

Below: These
two Boeing-
Vertol CH-47C
Chinooks are
special night
observation
attack transports
with multi-
sensor nose
installations.



Hughes TH-55 Osage

Model 300 and 300C, TH-55A Osage, NH-300C

Origin: Hughes Helicopters, Culver City, USA (NH 300C) Elicopteros, Ascoli, Italy Kawasaki, Japan

Type: Light observation helicopter

Engine: One 180hp Lycoming HIO 360 A1A (TH-55A, HIO-360 flat six (300C NH-300C) 100 hp flat six (300C)

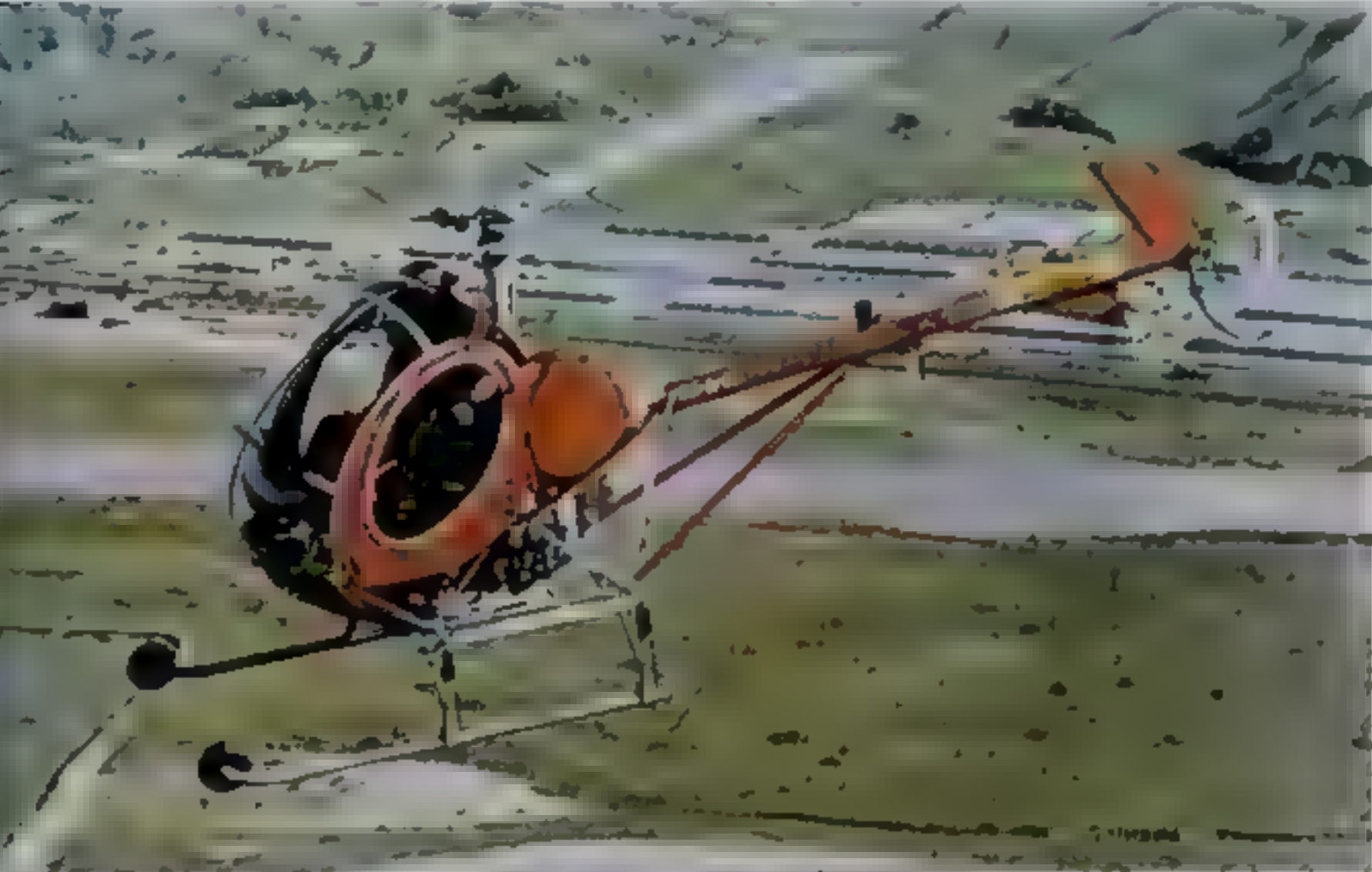
Dimensions: Diameter of three-blade main rotor 25ft 3in (7.7m), (300C) 25ft 3in (7.7m), length overall (rotors turning) 30ft 3in (9.24m), height overall 8ft 1in (2.44m)

Weights: Empty (TH) 1,008lb (457kg), (300C) 1,050lb (476kg), maximum loaded (TH) 1,850lb (839kg), (300C) 2,000lb (907kg)

Performance: Max cruise (TH) 75mph (121km/h), (300C) 75mph (121km/h) range with no reserve (TH) 204 miles (328km), (300C) 230 miles (370km)

History: First flight (Model 269) October 1956, (300) 1961, (300C) August 1969

Users: Brazil, Guyana, Italy, Japan, Nicaragua, Sierra Leone, USA



Above: Despite all the pupils can do, nearly 700 Osages are still training pilots to fly the 9,000 helicopters of the US Army.

Hughes OH-6 Cayuse and 500M

OH-6 Cayuse, 500M and NH-500M, 369HM, Defender and RACA-500

Origin: Hughes Helicopters, Culver City, USA, (NH) BredaNardi, Ascoli, Italy, (369HM) Kawasaki, Japan, (RACA) RACA, Buenos Aires, Argentina

Type: Light multi-role helicopter

Engine: One Allison turboshaft, (OH-6A) T63-5A flat-rated at 252.5shp (500M) 250-C18A flat-rated at 278shp

Dimensions: Diameter of four-blade main rotor 26ft 4in (8.03m), length overall (rotors turning) 30ft 3in (9.24m), height overall 8ft 1in (2.44m)

Weights: Empty (OH) 1,229lb (557kg), (500M) 1,130lb (512kg), maximum loaded (OH) 2,700lb (1,225kg), (500M) 3,000lb (1,361kg)

Performance: Max cruise at S/L 150mph (241km/h), typical range on normal fuel 380 miles (611km)

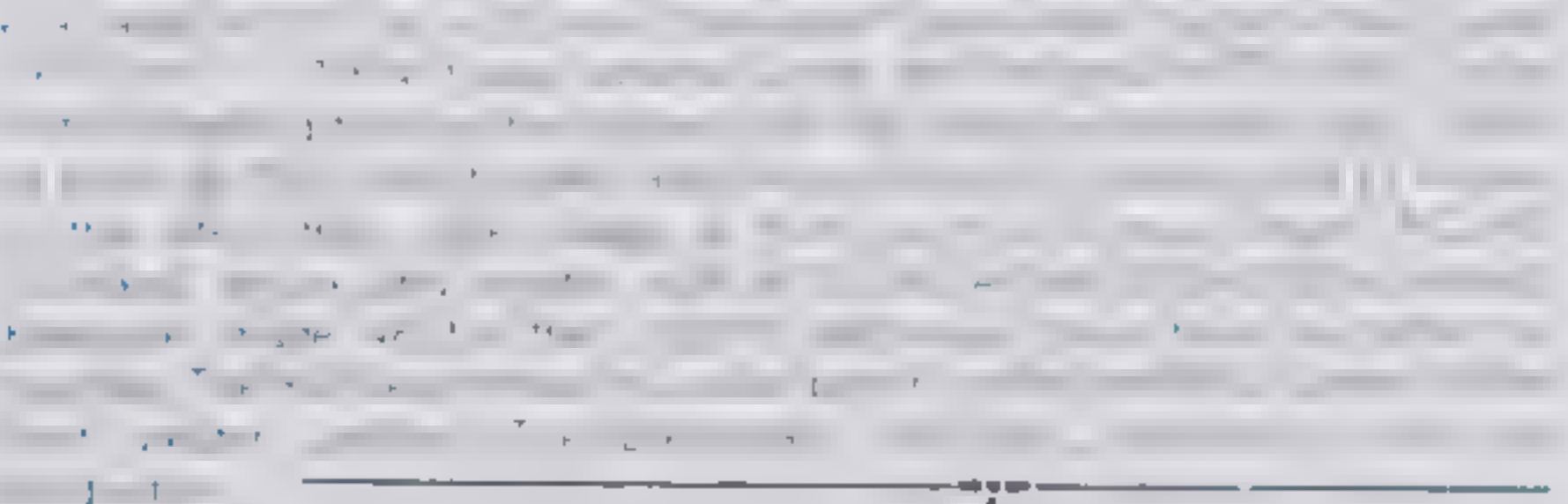
Armament: See text

History: First flight (OH-6A) 27 February 1963, (500M) early 1968

Users: (Military) Argentina, Bolivia, Colombia, Congo, Denmark, Dominican Republic, Italy, Japan, Pakistan, Philippines, Sierra Leone, Spain, Taiwan, United States

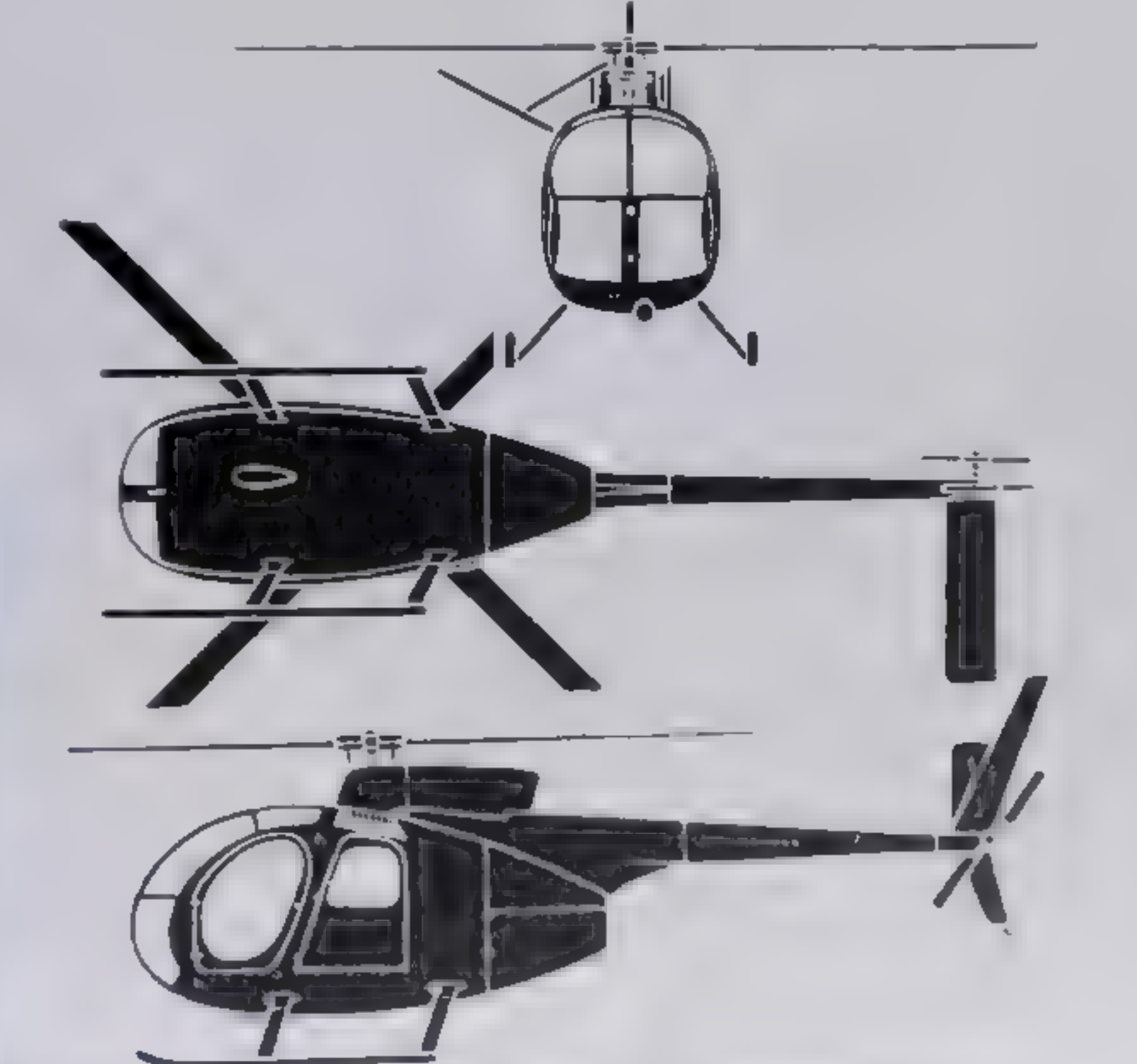


Development: The Hughes 21



Above: Three-view of TH-55A Osage (other 300 similar).

Development: Original winner of the controversial LOH (Light Observation Helicopter) competition of the US Army in 1961, the OH-6A Cayuse is one of the most compact flying machines in history, relative to its capability. The standard machine carries two crew and four equipped troops, or up to 1,000lb (454kg) of electronics and weapons including the XM-27 gun or XM-75 grenade launcher plus a wide range of other infantry weapons. The US Army bought 1,434, and several hundred other military or para-military examples have been built by Hughes or its licensees. BredaNardi is helping Pakistan get into production. The Spanish Navy 500Ms carry MAD and two Mk 44 torpedoes. The 500D M is a more powerful seven-seater. In 1977 an armed model was named Defender, carrying four TOW missiles and sight system, it is to go into production in South Korea.



Above: Three-view of OH-6A (500M and Defender similar)

Left: The tadpole-like "Loach" (popular name for it in the US Army) is nimble and carries a wealth of weapons

Hughes AH-64

Model 77, AH-64 (to be named)

Origin: Hughes Helicopters

Type: Armed tactical

Engines: Two 1 536shp General Electric T700 turboshaft

Dimensions: Diameter of four blade main rotor 48ft 0in (14.63m) length overall (rotors turning) 55ft 8½in (16.96m) height overall 12ft 1½in (3.69m)

Weights: Empty 9 500lb (4 309kg), maximum loaded 17 400lb (7 892kg)

Performance: Maximum speed at 13 200lb (5 987kg) 197 mph (317 km/h), cruise 180mph (289km/h) max vertical climb 3 000 ft/min (914 m/min) range on internal fuel 359 miles (578km)

Armament: Hughes XM 230 30mm "chain gun", 16 FFAR rockets, 16 TOW missiles and infra-red and other sensors and range of other options

History: First flight 30 September 1975, Phase 2 go-ahead November 1976

User: To be US Army

Development: First helicopter in history to be adopted after having been planned from the start as a "gunship", the AH-64 was selected over the Bell AH-63 at the end of 1976 after a tough but instructive competitive fly-off. Unlike the YAH-63 it seats the pilot behind the gunner/co-pilot, and it also has a small four-blade main rotor and tailwheel. Hughes developed the Chain Gun as a radically simpler and cheaper 30mm cannon with chain drive and rotary bolt action. Its turret is designed to fold under the armoured cockpit in a crash landing and the whole helicopter is designed to survive strikes at any point with shells up to 23mm calibre.



Above: Three-view of YAH-64 (note 60°/120° tail-rotor setting).

Below: Winner of the potentially enormous AAH competition, the first prototype YAH-64 blasts a ground target with folding-fin 2.75in rockets, 76 of which can be carried.



Kaman SH-2 Seasprite

**UH-2, HH-2 and SH-2 in many versions
(data for SH-2D)**

Origin: Kaman Aerospace Corp, USA

Type: Ship-based multirole fighter. ASW and air defense.

Engine(s): 125cc 4 stroke 10HP 12.5hp General Electric
T58 free turbine turbopropeller current version 12.5hp 125cc

Dimensions. Hull water diameter 44ft (13.41m), vert. height (base turning) 52ft 7in (16m), fuselage length 40ft 6in (12.3m), height 15ft 10in (4.84m).

Weights: Empt, t 953 b (3153k); max m, m k, t 12 301, t 12 301

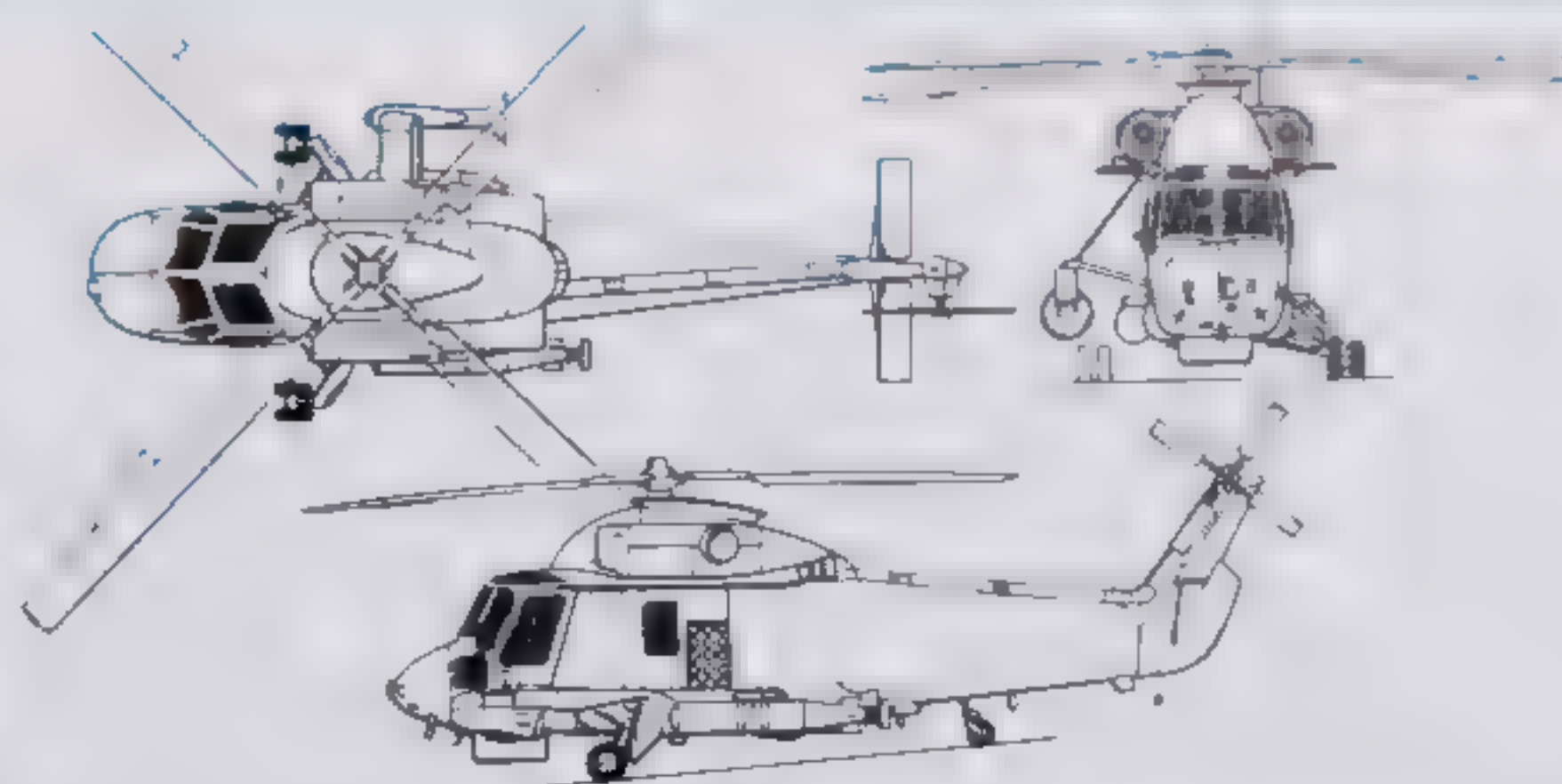
Performance Maximum speed 168mph (270km/h), maximum rate of climb (m/vert) 3 2440ft (744m), min. service ceiling 22 500ft (6860m), range 422 miles (679m)

Armament: See text

History: First flight: XH-2K 1, 2 July 1959; service delivery: XH-2K 1 later named XH-2A 18 December 1962; final delivery: new 1972 conversion in 1975.

User: LSA (Naval)

Development: Originally designated HH-2K-1 and named Seaprite, this exceptionally neat helicopter was at first powered by a single turbine engine mounted close under the rotor hub and was able to carry a wide range of loads including five passengers in its unobstructed central cabin with two crew in the nose. The main units of the tail wheel type landing gear retracted fully. About 190 were delivered and all were later converted to have two T58 engines in nacelles on each side. Some are HH-2C rescue variants, with armour and various armament including chin Minigun turret and waist-mounted machine guns or cannon, others are unarmed HH-2D. One has been used in missile firing (Sparrow III and Sidewinder).



Above: Three-view of Kaman SH 2F Seasprite (gear extended).

transformation of the fleet. A. Sampalis has a 1972 Eurofighter, converted to serve the LAMF's, plus the most modern fighter submarine and a future delivery of the Eurofighter than two times the size of equipment in the fleet. The Eurofighter has MAD gear ECM new radar and communications, plus a Mk 44 and a Mk 46 (plus a A will eventually be brought up to NATO standard with improved electronic gear, weapons and improved engines and weapons. The Eurofighter is the state of the art fighter of the future - impetus for a purpose designed LAMF's new project, greater size and power.

Below: One of the world's neatest helicopter platforms, the SH-2F is the only helicopter to meet the US Navy interim LAMPS need for ASW and anti-ship missile defence. Clearly visible are radar, sonobuoys, MAD "bird", homing torpedo and extensive avionics aeriats.



Kamov Ka-25

Ka-25

(several versions, designations unknown)

Origin

Type

Engines

Dimensions

Weights

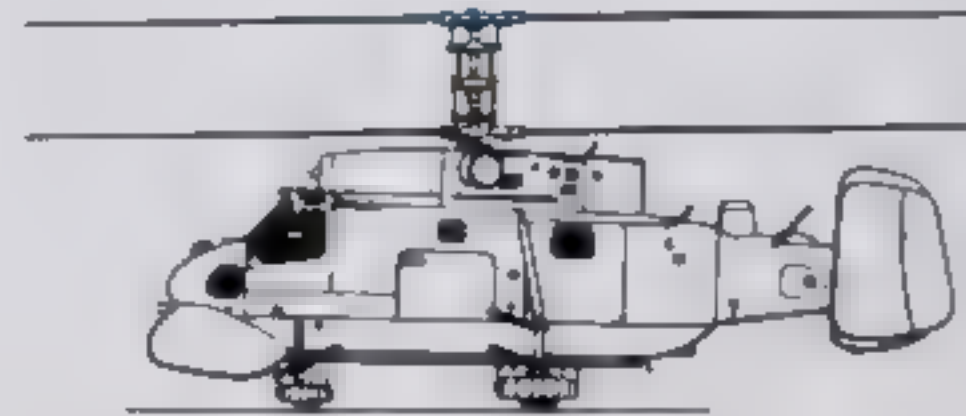
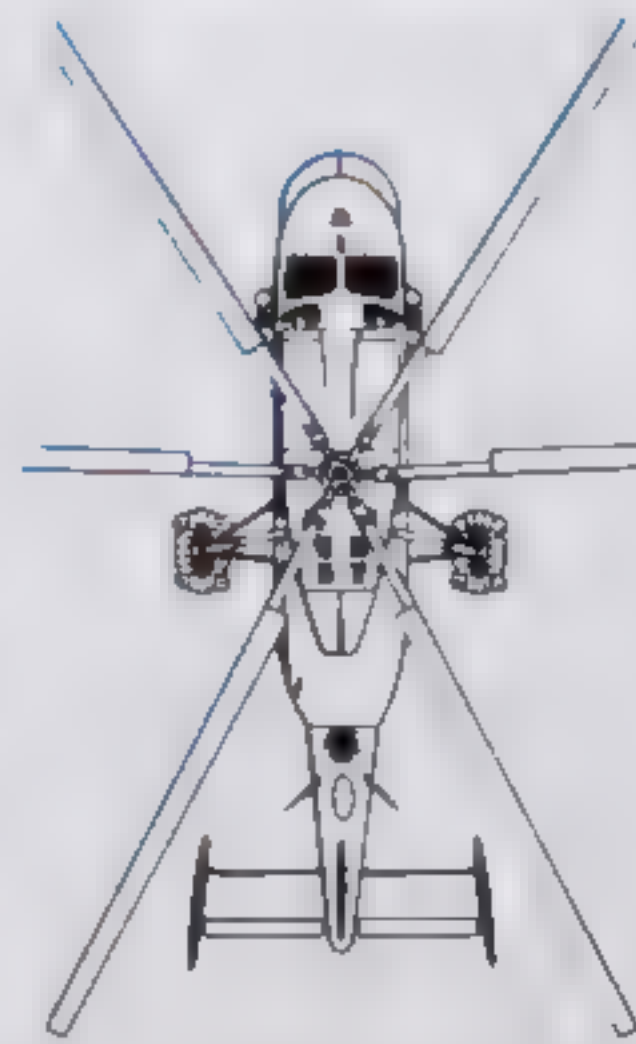
Performance

Armament: One or two 400mm AS torpedoes

History: First flight (Ka-20) probably 1960, service delivery of initial

Users

Development: Nikolai Kamov, who died in 1973, designed the first Soviet helicopter, the Kamov Ka-26, in 1947. The Ka-26 was a two-rotor helicopter with a tandem rotor configuration. It was designed for use as a transport helicopter, but it was also used for a variety of other purposes, including as a rescue helicopter and as a gunship. The Ka-26 was the first Soviet helicopter to be exported, and it was used by the Soviet Air Force and the Soviet Navy. It was also used by the Soviet Union's allies, including China and Cuba. The Ka-26 was a significant achievement for the Soviet Union, as it was the first Soviet helicopter to be designed and built from scratch. It was also the first Soviet helicopter to be exported, and it was used by the Soviet Air Force and the Soviet Navy. It was also used by the Soviet Union's allies, including China and Cuba. The Ka-26 was a significant achievement for the Soviet Union, as it was the first Soviet helicopter to be designed and built from scratch. It was also the first Soviet helicopter to be exported, and it was used by the Soviet Air Force and the Soviet Navy. It was also used by the Soviet Union's allies, including China and Cuba.



Above: Three-view of Ka-25 "Hormone A" with emergency flotation system.



Above and below: Two examples of the basic ASW version of the Ka-25, called "Hormone A" by NATO. The upper photograph was taken by the US Navy and the lower by the Royal Navy (note Buccaneer and Phantom aboard "Ark Royal").



MBB BO 105

105C and 105VBH

Origin: Messerschmitt Bolkow Blohm Munich, West Germany
production by PADC, Philippine
Type: Multi-role all-weather helicopter
Engines: Two 400shp Allison 250 turboshaft engines
Dimensions: Diameter of four-blade main rotor 38ft 10 3/4in (11.84m) tail rotor 9ft 9in (2.97m) overall (rotors turning) 38ft 10 3/4in (11.84m) height 11ft 9in (3.61m) wheel track 10ft 6in (3.19m)
Weights: Empty 2,469 lb (1,120 kg) maximum gross weight 4,400 lb (2,000 kg) maximum loaded 5,000 lb (2,270 kg)
Performance: Maximum speed 167 mph (270 km/h) cruise 144 mph (232 km/h) max climb 1,378 ft (420 m) range 300 mi (483 km) fuel no reserve (S/L) 563 miles (585 km)
Armament: Various options include: Hot or TOW missiles and stabilized sight system
History: First flight 16 February 1967, type certification October 1970
Users: (Military) W Germany, Netherlands, Nigeria, Philippine

Development: Using an advanced rotor developed with Aerospatiale's small twin-turbine helicopter is expensive but extremely flexible and stable, aerobatic and cleared for IFR operation in the most adverse conditions. By 1977 some 350 had been sold, including a small number for military use. The German Army has evaluated the 105VBH liaison and observation version, and an anti-tank 105 with six Hots and all-weather sight system. An order for about 300 had long been expected for the German Army. At the time this book went to press, and several other military sales were being negotiated. Boeing Vertol sell in the Americas and PADC assemble the basic 105C in the Philippines. The BO 106 is a wider version, seating seven instead of four.

Right: Launch of a HOT missile from the No 4 tube of a BO 105 equipped with only four tubes.

Below: The standard six-tube HOT installation, with the head of the stabilized sight projecting above the cabin.



Above: Three-view of the wide-body BO 106.

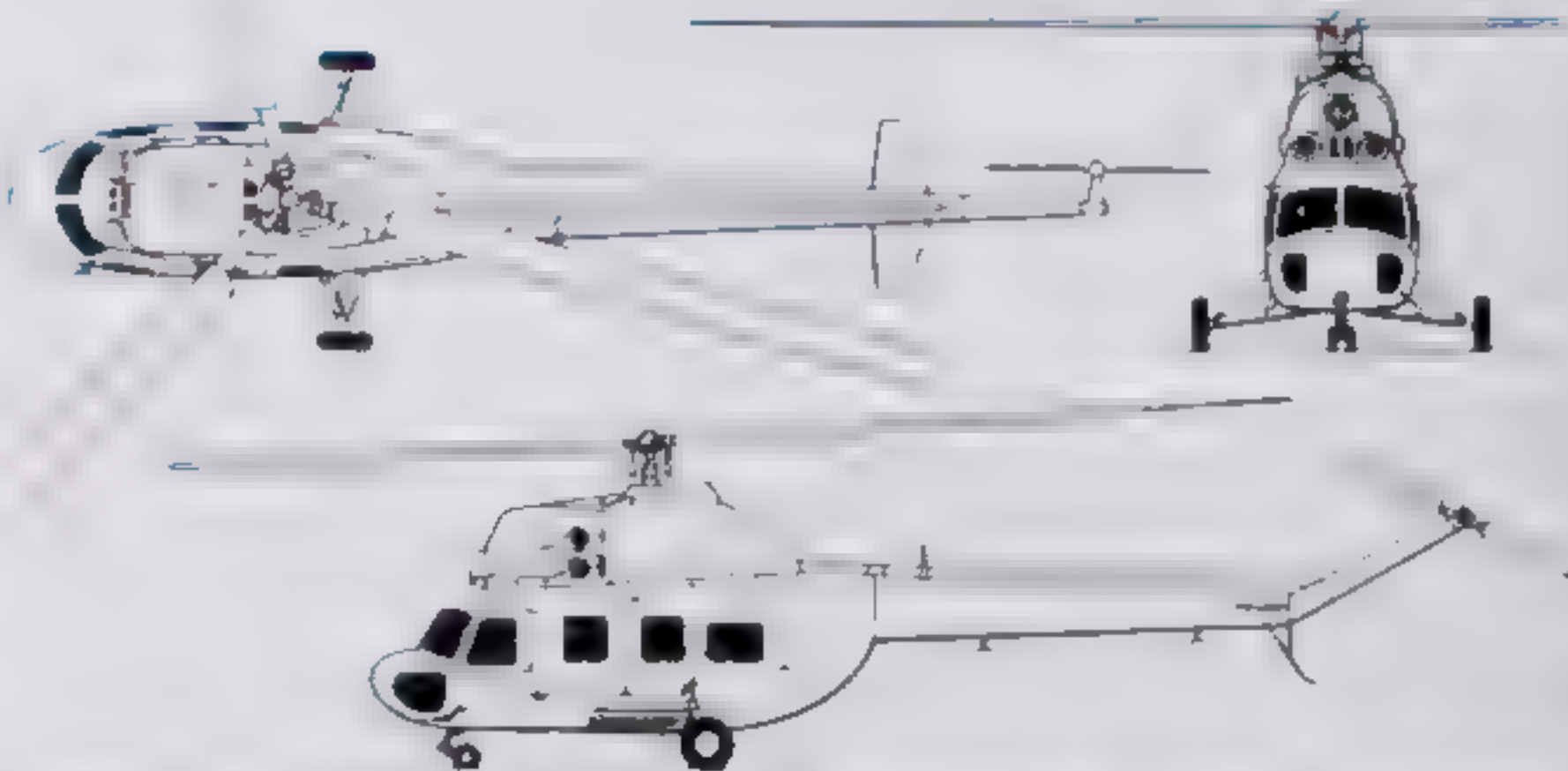


Mil (WSK-Swidnik) Mi-2

Mil Mi-2 (V-2) and Mi-2M (NATO name "Hoplite")

Origin	Soviet Union
Type	Multi-role transport helicopter
Engines	Two Mikoyan-1000 turboshafts
Dimensions	Length: 20.5m (67ft 3in) Height: 4.5m (14ft 9in) Rotor diameter: 16.5m (54ft 2in)
Weights	Empty: 2,500kg (5,512lb) Max: 4,000kg (8,818lb)
Performance	Max speed: 240km/h (149mph) Cruise: 180km/h (112mph) Range: 400km (249mi) Ceiling: 3,000m (9,843ft)
History	First flight: 1968
Users	Soviet Union, Poland, Czech Republic, Hungary, Romania, Bulgaria, Yugoslavia, Albania, Afghanistan, Iraq, Iran, Libya, Syria, Vietnam, Yemen, etc.

Development: The first production helicopter in the Soviet Union was the Mil Mi-2. It was designed by the Mil design bureau in Moscow. The first prototype flew in 1968. The first production version, the Mi-2, entered service in 1970. It was a two-engine, twin-rotor helicopter. It was designed to be a multi-role transport helicopter. It could carry up to 12 troops or 4,000kg of cargo. It could also be used as an ambulance or a search and rescue helicopter. The Mi-2 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries.



Three-view of WSK-Swidnik Mi-2 without special role equipment

The Mi-2 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries. The Mi-2 has been used in many roles, including transport, assault, and search and rescue. It has been used by the Soviet Union, Poland, Czech Republic, Hungary, Romania, Bulgaria, Yugoslavia, Albania, Afghanistan, Iraq, Iran, Libya, Syria, Vietnam, Yemen, etc. The Mi-2 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries.



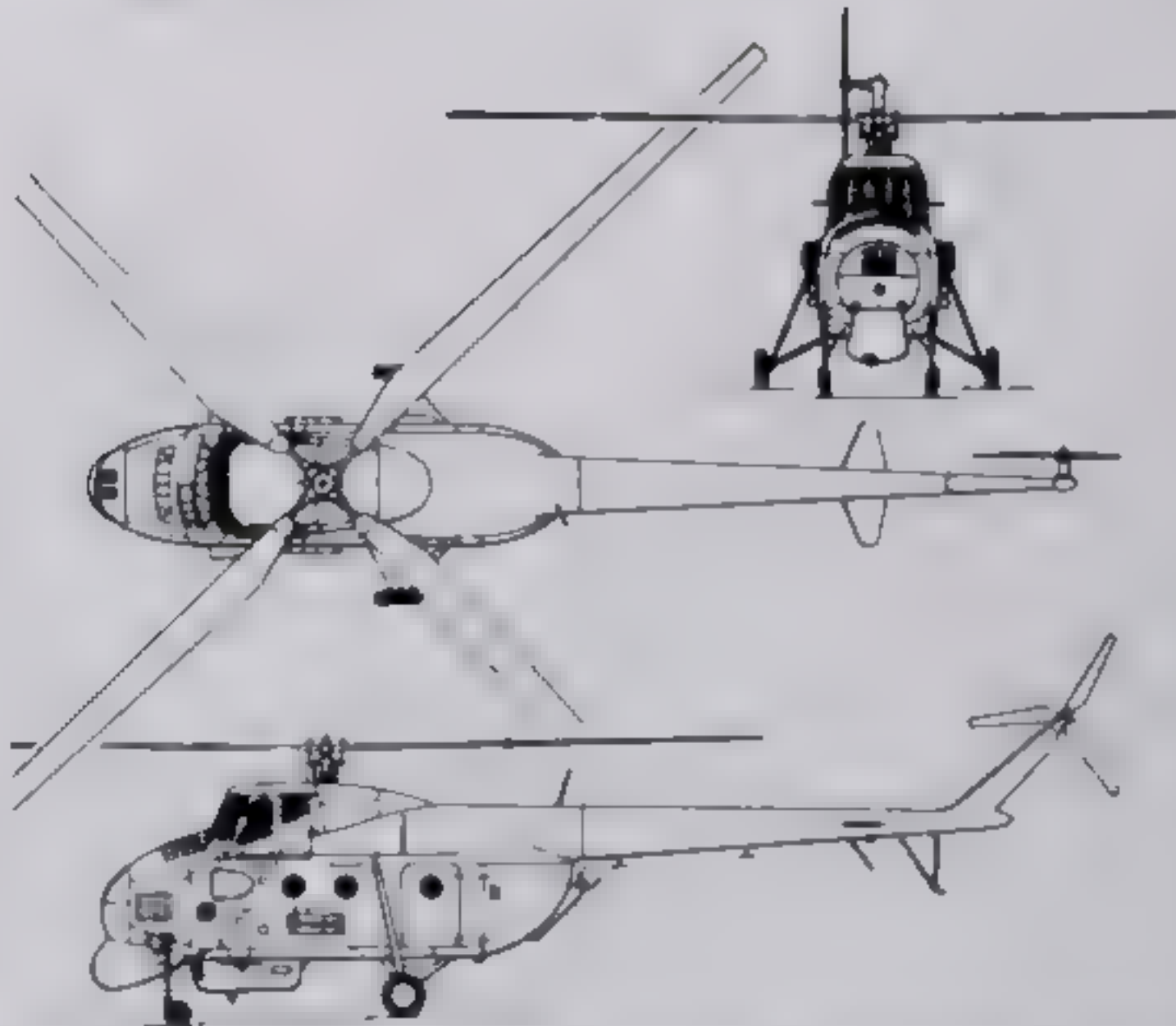
Above: A WSK-Swidnik Mi-2 of the Polish Air Force (skis often added round wheels).

Mil Mi-4

Mi-4 (sub-designations unknown) (NATO name "Hound")

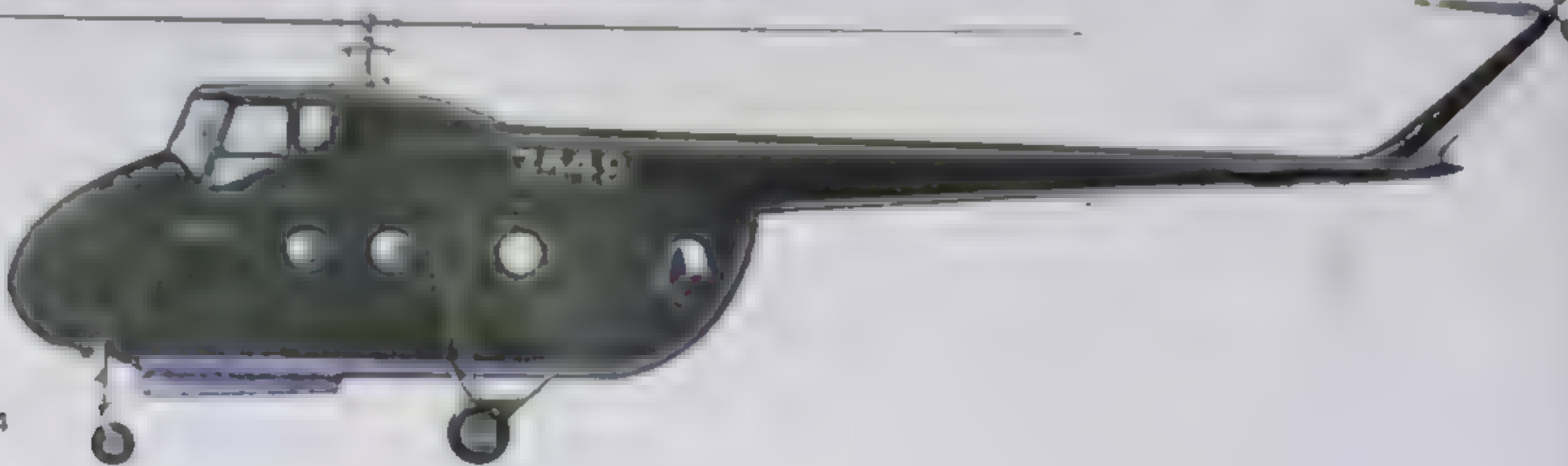
Origin	Soviet Union
Type	Multi-role transport helicopter
Engine	One Mikoyan-1000 turboshaft
Dimensions	Length: 20.5m (67ft 3in) Height: 4.5m (14ft 9in) Rotor diameter: 16.5m (54ft 2in)
Weights	Empty: 2,500kg (5,512lb) Max: 4,000kg (8,818lb)
Performance	Max speed: 240km/h (149mph) Cruise: 180km/h (112mph) Range: 400km (249mi) Ceiling: 3,000m (9,843ft)
Armament	None
History	First flight: 1968
Users	(Military) Afghanistan, Albania, Algeria, Bulgaria, China, Cuba, Czechoslovakia, Egypt, Finland, East Germany, Hungary, India, Indonesia, Iraq, Iran, Israel, Italy, Japan, Korea, Laos, Libya, Romania, Soviet Union, Syria, Vietnam, Yemen, etc.

Development: The Mi-4 was designed by the Mil design bureau in Moscow. It was a single-engine, single-rotor helicopter. It was designed to be a multi-role transport helicopter. It could carry up to 12 troops or 4,000kg of cargo. It could also be used as an ambulance or a search and rescue helicopter. The Mi-4 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries.



Three-view of Mi-4 ASW version (Soviet designation unknown)

The Mi-4 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries. The Mi-4 has been used in many roles, including transport, assault, and search and rescue. It has been used by the Soviet Union, Poland, Czech Republic, Hungary, Romania, Bulgaria, Yugoslavia, Albania, Afghanistan, Iraq, Iran, Libya, Syria, Vietnam, Yemen, etc. The Mi-4 has been used in many conflicts, including the Soviet-Afghan War, the Iran-Iraq War, and the Bosnian War. It is still in service today in many countries.



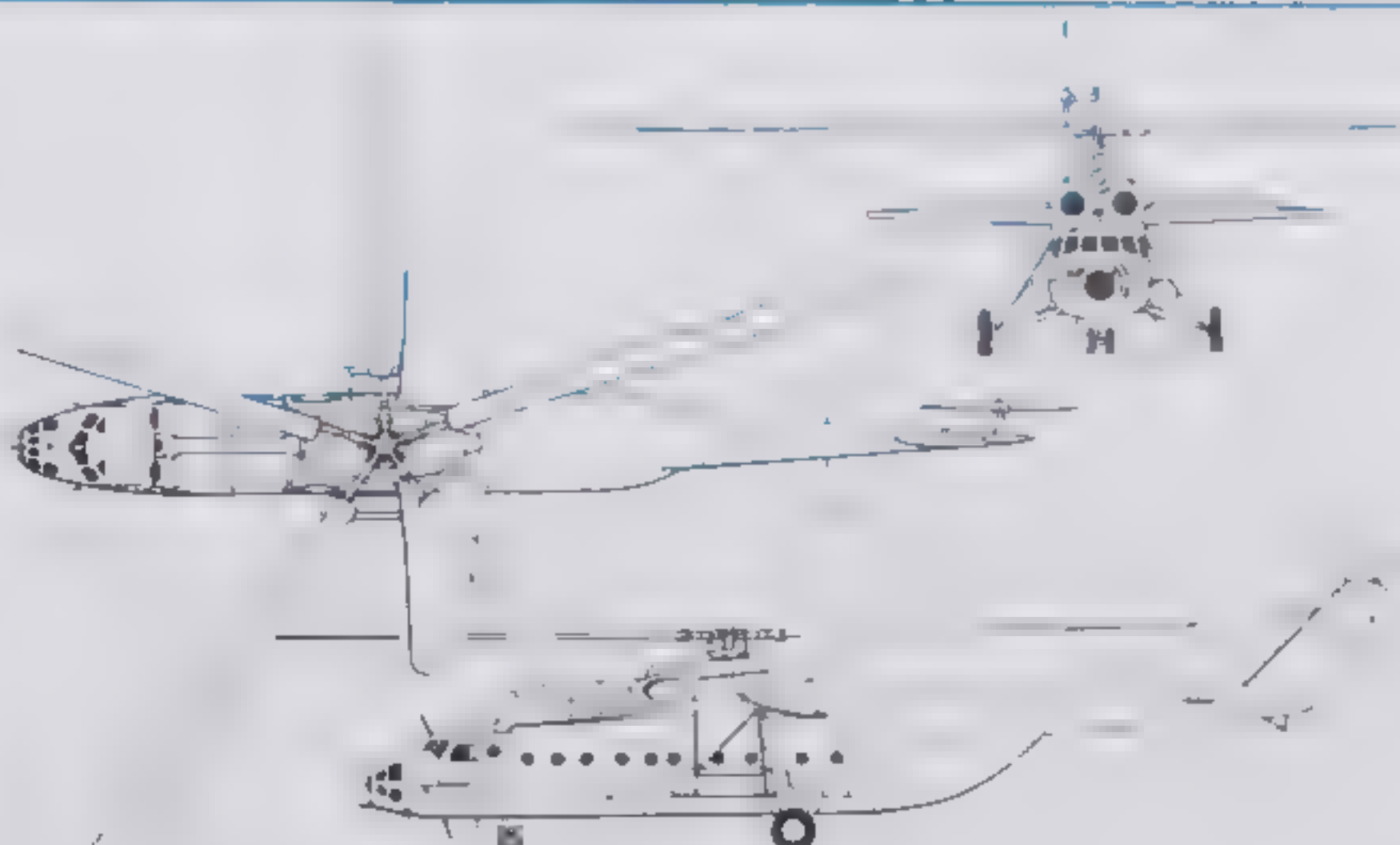
Right: Basic transport Mi-4 Czech Air Force

Mil Mi-6 and Mi-10

Mi-6 ("Hook"), Mi-10 and -10K ("Harke")

Origin: The design bureau named for Mikhail Mil, Soviet Union
Type: basic heavy transport helicopter (-10 crane), (-10K) crane helicopter
Engines: (-6, -10) two 5,500shp Soloviev TV3-117A single-shaft free-turbine engines driving common R-7 gearbox
D-25VF
Dimensions: Main rotor diameter 114ft 10in (35.0m), (10, -10K) 137ft 11in (42.0m), (10, -10K) 137ft 11in (42.0m), fuselage length (-6) 108ft 10in (33.18m), (10, -10K) 108ft 10in (33.18m), height (-6) 32ft 4in (9.86m), (10) 32ft 2in (9.8m), (-10K) 29ft 10in (9.1m)
Weights: Empty (-6, typical) 60,055lb (27,240kg), (-10) 60,185lb (27,300kg), (-10K) 54,410lb (24,680kg), maximum loaded (-6) 93,700lb (42,500kg), (-10) 96,340lb (43,700kg), (-10K) 83,776lb (38,000kg) with 5,500shp engines (90,390lb 41,000kg expected with D-25VF engines)
Performance: Maximum speed (-6) 186mph (300km/h) (set 100km circuit record at 211.36mph, beyond flight manual limit), (-10) 124mph (200km/h), service ceiling (-6) 14,750ft (4,500m), (-10, -10K) limited 9,842ft (3,000m), range (-6 with half payload) 404 miles (650km), (-10 with 12,000kg platform load) 155 miles (250km), (-10K with 11,000kg payload, 6,500shp engines) over 280 miles (450km)
Armament: Normally none, but Mi-6 often seen with manually aimed nose gun of about 12.7mm calibre
History: First flight (-6) probably early 1957, (-10) 1960, (-10K) prior to 1965
Users: (-6) Bulgaria, Egypt, Indonesia (in storage), Iraq, Soviet Union, Syria, Vietnam, Zambia, (-10) Soviet Union

Development: Development by Mikhail L. Mil's design bureau at Zaporozhye of the dynamic system (rotors and shafting) of the Mi-6 was a task matched only by Soloviev's development of the huge R-7 gearbox, which



Three-view of basic Mi-6 with wings.

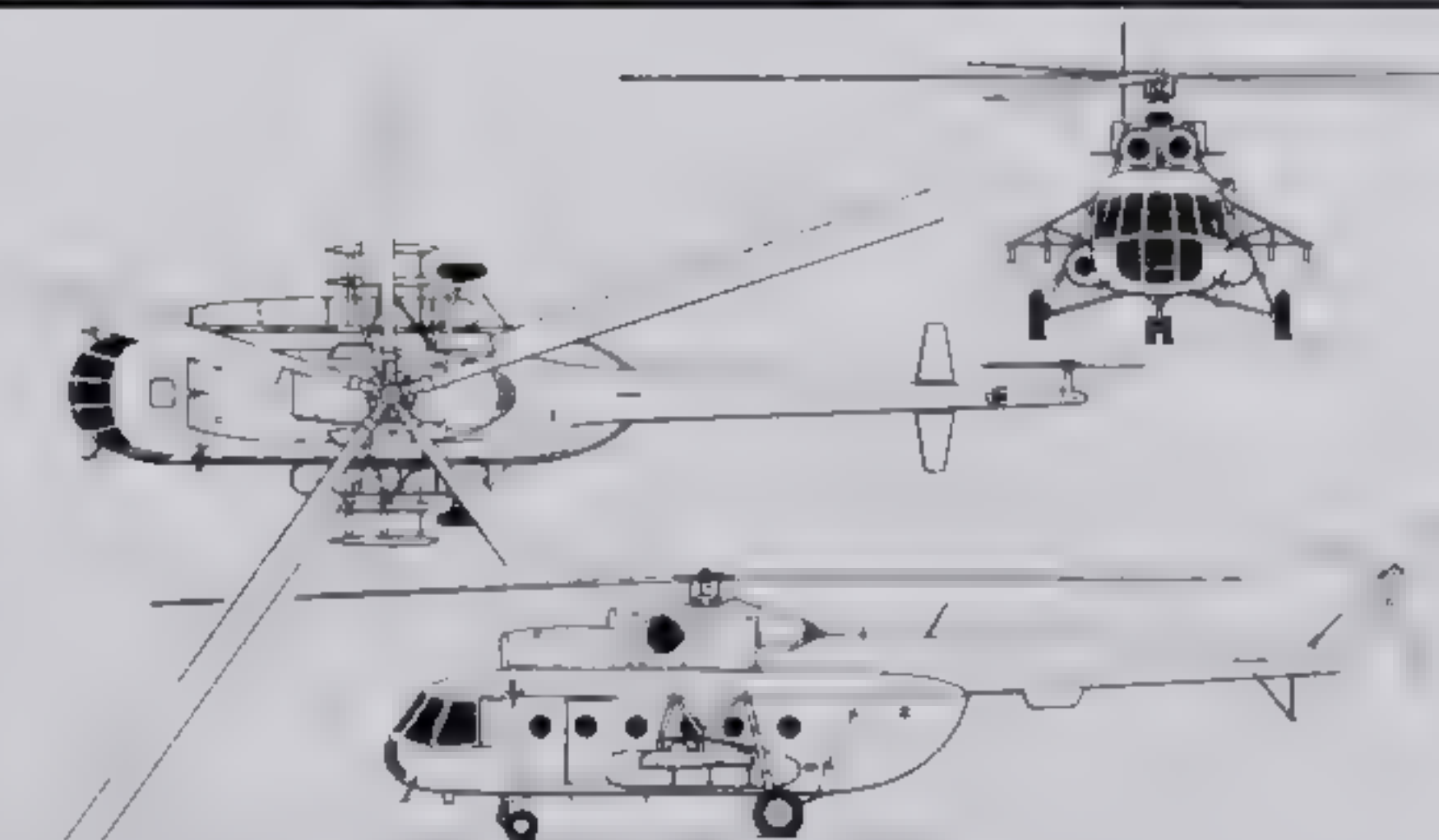
weighs 7,054lb (much more than the pair of engines). By far the biggest rotor system yet flown, this served to lift by far the biggest helicopter, the Mi-6 (NATO code name "Hook"), which quickly set world records for speed and payload, though the normal load is limited to 26,450lb (12,000kg) internally, loaded via huge clamshell rear doors, or 19,840lb (9,000kg) externally slung. About 500 have been built, possibly half being in military use. Most have the rotor unloaded in cruising flight (typical v 150mph) by a fixed wing of 50ft 2in in span. These huge helicopters have played an active role in field exercises carrying troops (typically 68) and tactical missiles or vehicles in the class of the BRDM. The Mi-10 (code name "Harke") has lofty landing gears which enable it to straddle a load such as a bus or prefabricated building 3.75m (12ft 3in) high, heavy loads weighing 33,070lb (15,000kg) and up to over 65ft in length have been flown. It uses a TV viewing system for load control, but the short-legged Mi-10K has an under-nose gondola.

Mil Mi-8

Mi-8, Mi-8T (NATO names "Hip", Haze")

Origin: The design bureau named for Mikhail Mil, Soviet Union
Type: General utility helicopter for internal loads and externally mounted weapons, "Haze", ASW
Engines: Two 1,500shp Isotov TV2-117A single-shaft free-turbine engines driving common VR-8A gearbox
Dimensions: Main rotor diameter 69ft 10in (21.29m), overall length, rotors turning, 82ft 9in (25.24m), fuselage length 60ft 0in (18.31m), height 18ft 6in (5.65m)
Weights: Empty (-8T) 15,026lb (6,816kg), maximum loaded (all) 26,455lb (12,000kg) (heavier weights for non-VTO operation)
Performance: Maximum speed 161mph (260km/h), service ceiling 14,760ft (4,500m), range (-8T, full payload, 5 per cent reserve at 3,280ft) 298 miles (480km)
Armament: Optional fitting for external pylons for up to eight stores carried outboard of fuel tanks (always fitted), typical loads eight pods of 57mm rockets, or mix of gun pods and anti-tank missiles (Mi-8 not normally used in anti-tank role)
History: First flight 1960 or earlier, service delivery of military versions before 1967
Users: Afghanistan, Bangladesh, Czechoslovakia, Egypt, Ethiopia, Finland, E. Germany, Hungary, India, Iraq, N. Korea, Libya, Pakistan, Peru, Poland, Romania, Somalia, Soviet Union, S. Yemen, Sudan, Syria, Vietnam, Yugoslavia

Development: Originally powered by a single 2,700shp Soloviev engine the Mi-8 soon appeared with its present engines and in 1964 added a fifth blade to its main rotor. It has since been the chief general utility helicopter of the Warsaw Pact powers and many other nations. By mid-1974 it was announced that more than 1,000 had been built, the majority for military purposes and with about 300 having been exported. Since then the Mi-8 has continued in production. The basic version is a passenger and troop carrier normally furnished with quickly removable seats for 28 in the main cabin. The -BT is the utility version without furnishing and with circular windows, weapon pylons, cargo rings, a winch/pulley block system for

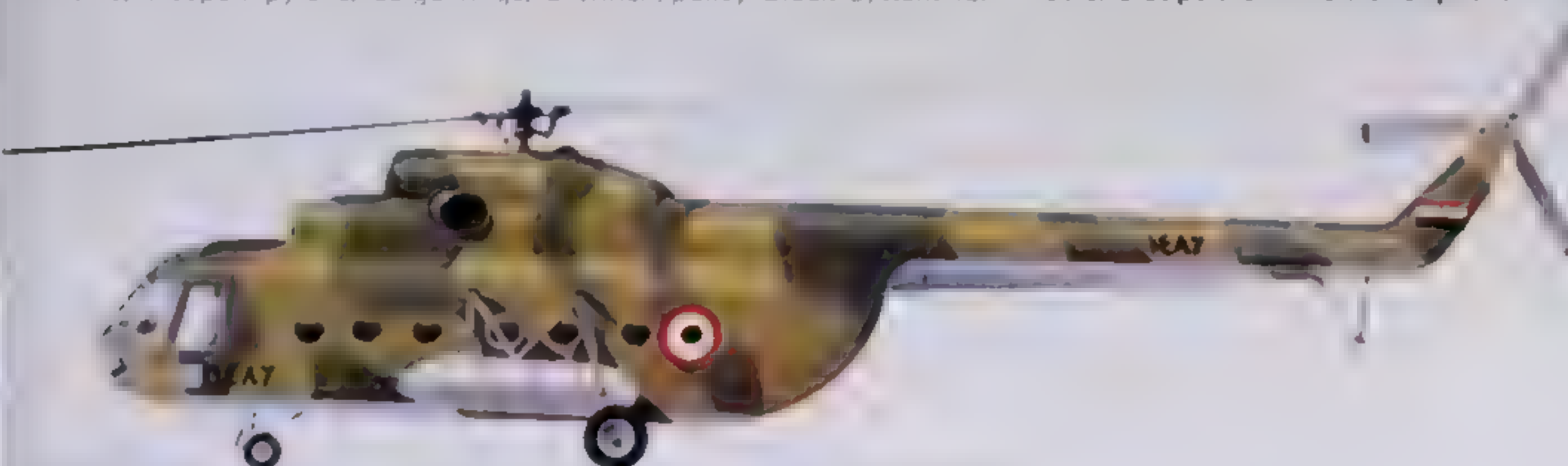


Three-view of assault Mi-8 with weapon pylons and tailboom radar.

loading and optional electric hoist by the front doorway. All versions have large rear clamshell doors (the passenger version having a stairs incorporated) through which a BRDM and other small vehicles can be loaded. "Haze" has AS radar, MAD bird and weapons.



Above: Ilmavoimat means the Finnish Air Force, which has four of the capable Mi-8 helicopters for general airlift missions.



An Mi-8 serving with the Egyptian Air Force (no radar or weapons)

Sikorsky S-58

H-34 Seabat, Seahorse, Choctaw

Origin: Sikorsky Aircraft Division, United Aircraft Corporation, United Technologies Corporation, Stratford, CT, USA (previously, Sikorsky Aircraft, Stratford, CT, USA and redesignated with turbine power, Westland Wessex)

Type: Utility, transport, ASW, search, rescue and other roles

Engine: One 1,525hp Wright Cyclone R 1820-81 nine cylinder radial

Dimensions: Diameter of four blade main rotor hub in (17'0" fuselage length, 41'1" rotors, 46ft 3in (14.25m) height, versl 14ft 3in (4.38m)

Weights: Empty, typical 7,750lb (3515kg) ASW 8,275 lb (3745kg) maximum loaded 14,000 lb (6350kg)

Performance: Typical cruise 98mph (158km/h) range with full payload reserve 280 miles (450km)

History: First flight 8 March 1954 (S-58T) 19 August 1970

Users: (Military) include Argentina, Belgium, Brazil, Canada, Central African Republic, Chile, France, W Germany, Haiti, Indonesia, Israel, Italy, Japan, Laos, Netherlands, Nicaragua, Philippines, Thailand

Development: This useful helicopter was designed to meet the need of the US Navy for an anti submarine platform (but still not capable of independent search/strike operation) and it entered service in 1955 as the HSS-1, Redesignated SH-34G and with automatic stabilization designated SH-34J, this model was named Seabat, while the Marines UH-34D and E and Executive UH-34D are Seahorses. Large numbers were built as assault transports for the Army (CH-34A and C Choctaw) while S-58 built 166 for the Algerian war. Sikorsky delivered 1,821 and still today busy converting and rebuilding many of these tough but outdated machines with the PT6T Twin Pac engine giving them a new lease of life as the S-58T. New S-58T versions are planned.



Above: Three-view of basic S-58 (-58T has different front end).

Below: There is justification for describing the S-58 as the first helicopter in the world capable of doing a really worthwhile military job (the only rival, in the same time-scale, is the larger Mil Mi-4). This UH-34D Seahorse is trucking an M-274 Mechanical Mule (with 106mm recoilless rifle) from an assault carrier to beach landing.



Sikorsky S-61

SH-3A and -3D Sea King, HH-3A, RH-3A and many other variants

Origin: Sikorsky Aircraft Division of United Technologies Corporation, Stratford, Connecticut, USA

Type: Multi-mission helicopter
Engines: Two General Electric T401 turboshaft engines, 4,200 hp (3,113 kW) each

Dimensions: Diameter of main rotor 62ft (18.9m), length 51ft 10in (15.8m), height 16ft 10in (5.1m)

Weights: Empty (simple transport versions typical) 9,763lb (4,430kg), max gross weight 15,000lb (6,800kg), max payload 4,000lb (1,800kg)

Performance: Max speed 160 mph (258 km/h), cruise 140 mph (225 km/h), range 2,200 miles (3,540 km), max altitude 15,000 ft (4,572 m), max climb rate 3,000 ft/min (914 m/min)

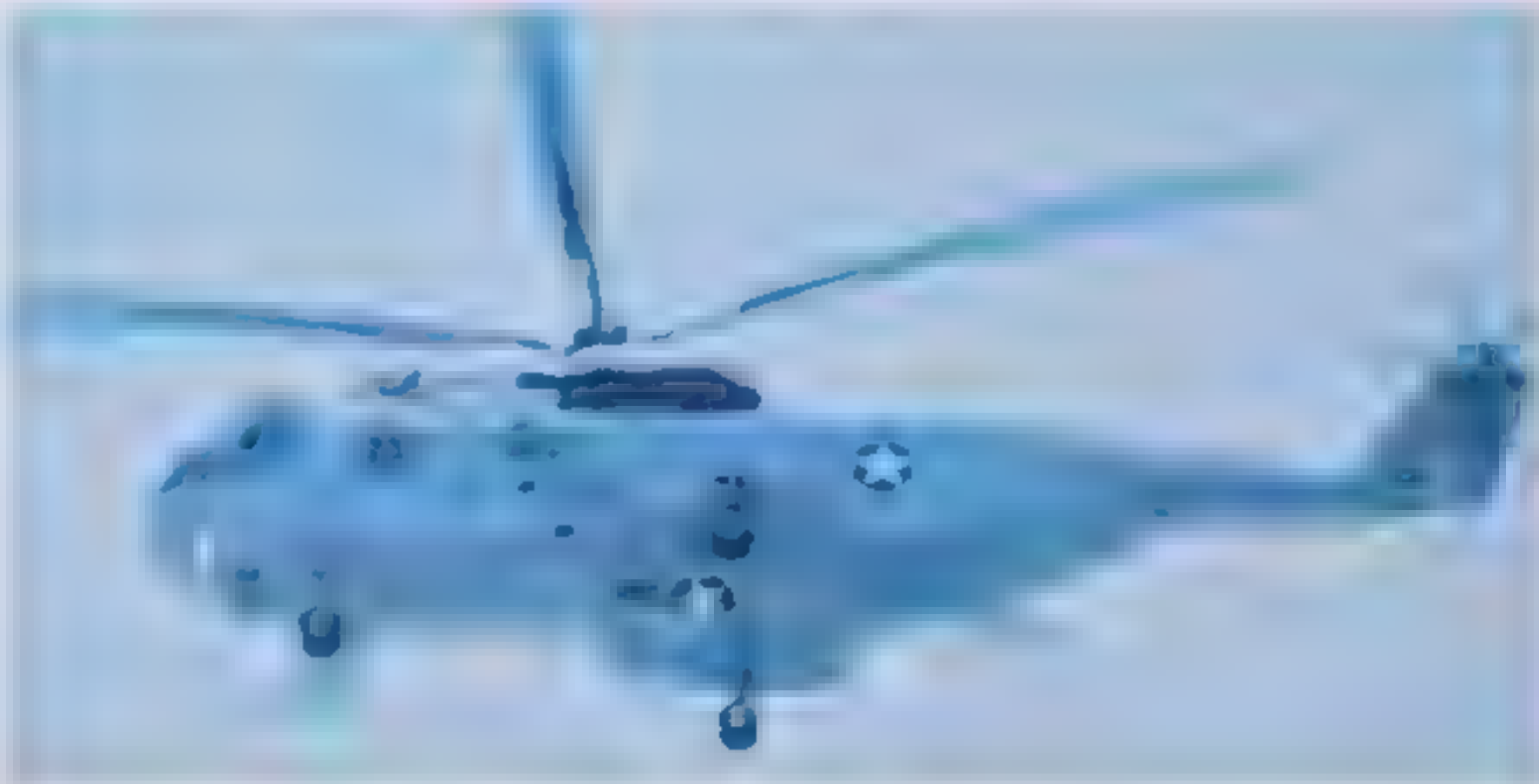
Armament: Very variable

History: First flight 11 March 1953

Users: Argentina, Australia, Belgium, Brazil, Canada, Denmark, Egypt, Greece, India, Israel, Italy, Japan, Mexico, Netherlands, Norway, Pakistan, Saudi Arabia, South Africa, Spain, Taiwan, Thailand, USA, UK, West Germany, Yugoslavia

Development: Representing a quantum jump in helicopter capability, the S-61 family soon became a staple product of Sikorsky Aircraft, founded in March 1923 by Igor Sikorsky who left Russia after the Revolution and settled in the United States. He flew the first wholly practical helicopter in 1940, and his R-4 was the first helicopter in the world put into mass production (in 1942). A development, the S-51, was in 1947 licensed to the British firm Westland Aircraft, starting collaboration reviewed on later pages. The S-55 and S-58 were made in great numbers in the 1950s for military, civil and military purposes, both now flying with various turbine engines. The S-61 featured an amphibious hull, twin turbine engines located above the hull close to the drive gearbox and an advanced flight-control system. First versions carried anti-submarine warfare (ASW) sensors and weapons but later variants were equipped for various transport duties, minesweeping, drone or spacecraft recovery (eg lifting Astronauts from the sea), electronic surveillance and (S-61R series) transport/gunship and other combat duties.

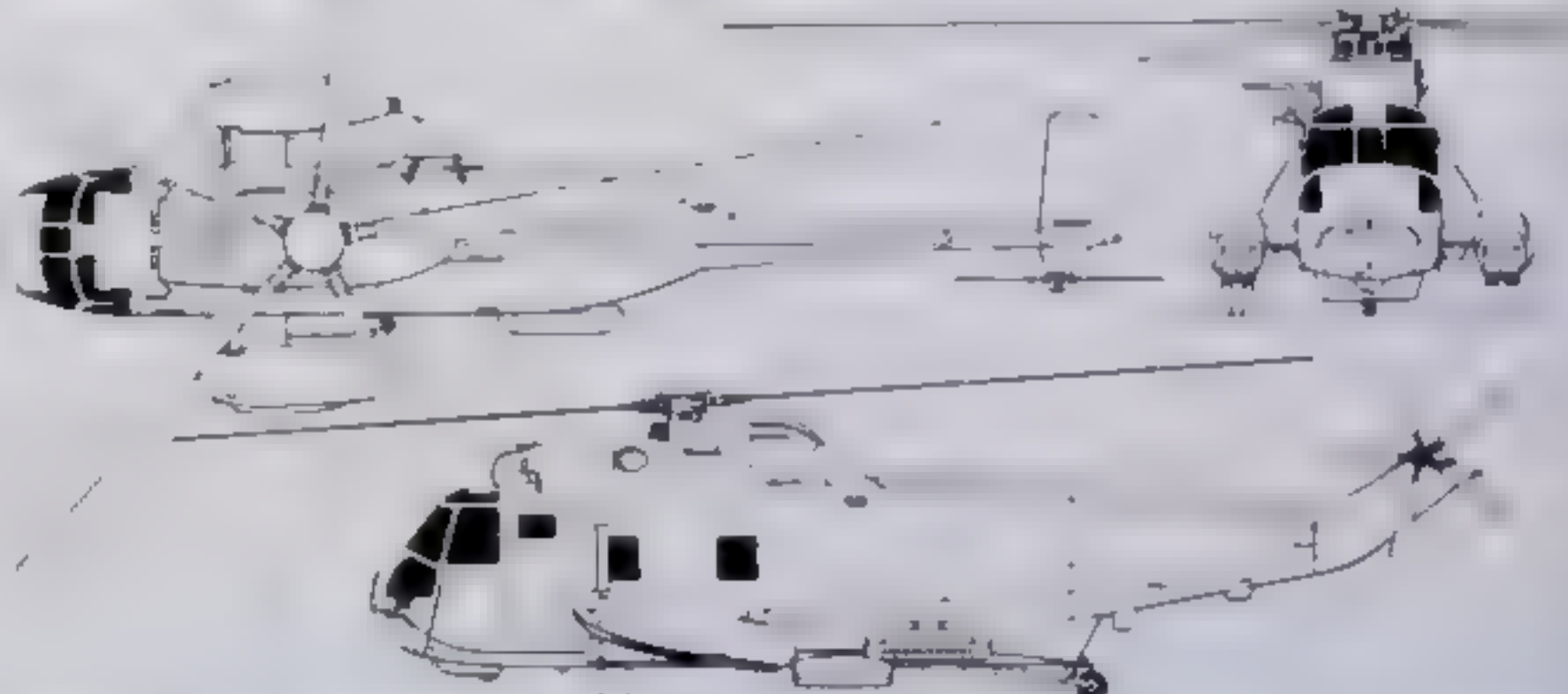
Below: One of the 80-odd SH-3D Sea King ASW helicopters serving with ten helicopter ASW squadrons (in this case, HS-2). The only obvious mission equipment is the towed MAD "bird"; soot from the engines testifies to hard flying.



Above: S-61R (Air Force CH-3E) with sponsons, retractable tricycle landing gear and rear cargo ramp



Above: Agusta-Sikorsky SH-3D of the Marina Italia (Italian Navy) launches a Sea Killer Mk 2, the delivery weapon in the Marte air-to-ship system developed by the Italian Sistel company. Effective range is 15.5 miles (25km).



Above: Three-view of SH-3H showing radar, sonobuoys and MAD.



Sikorsky S-64

S-64, CH-54A and B Tarhe

Origin: Sikorsky Aircraft, Division of United Technologies Corp., Stratford, Conn.
Type: Crane helicopter
Engines: (CH-54A) two 4500shp Pratt & Whitney T55-A1 turboprops
(CH-54B) two 4800shp T73-70s
Dimensions: Length 67ft 5in blade main rotor 72ft 0in (21.95m) height overall (rotors turning) 88ft 6in (26.97m) height overall 18ft 7in (5.67m)
Weights: Empty (A) 19,234lb (8724kg) maximum loaded (A) 42,000lb (19,050kg) (B) 47,000lb (21,318kg)
Performance: Maximum cruise 105mph (169km/h) hover (out of ground effect) 630ft (210m) range with max fuel and 10 percent reserve (typical) 230 miles (370km)
History: First flight (S-64) 9 May, 1962; service delivery (CH-54A) late 1964, (B) late 1969
User: US Army

Development: Developed from the first large US Army helicopter, the S-56, via the piston-engined S-60, the S-64 is an efficient weight lifter which in Vietnam carried loads weighing up to 20,000 lb (9072 kg). The CH-54A Tarhes used in that campaign retrieved more than 380 shot down aircraft, saving an estimated \$210 million, and carried special vans housing up to 87 combat-equipped troops. The improved CH-54B distinguished externally by twin main wheels, has lifted loads up to 40,780 lb (18,497 kg) and reached a height of 36,122 ft (11,010 m). There is no fuselage; just a structural beam joining the tail rotor to the cockpit in which seats are provided for three pilots, one facing to the rear for manoeuvring with loads. The dynamic components (rotor, gearboxes, shafting) were used as the basis for those of the S-65. With cancellation of the HLH (Heavy-Lift Helicopter) the S-64 remains the only large crane helicopter in the West.



Above: Three-view of S-64 (all versions similar).

Below: A 155mm howitzer of the M-114 family makes a reasonable load of almost six US tons for a CH-54A Tarhe of the US Army. Note main-rotor tip trails.



Sikorsky S-65

CH-53A Sea Stallion, HH-53, RH-53 and many other variants

Origin

Type
Engines

Dimensions

Weights: Empty (CH-53D) 23 485lb (10 653kg) maximum loaded

Performance: Maximum speed (most) 19

Armament
History

Users

Development: Obviously developed from the S-61, the S-65 was the first helicopter to be designed specifically for the US Marine Corps. The S-65 was developed by Sikorsky in the late 1950s, but it was not until 1968 that a standard CH-53A completed a prolonged series of loops while others set records for speed and payload. Most of the CH-53A's development was done in Germany, together with the US Marine Corps. The CH-53D (the 'Sea Stallion') was developed in the late 1960s and early 1970s, and it was the first helicopter to be designed specifically for the US Marine Corps. The CH-53D was developed by Sikorsky in the late 1960s and early 1970s, and it was the first helicopter to be designed specifically for the US Marine Corps. The CH-53D was developed by Sikorsky in the late 1960s and early 1970s, and it was the first helicopter to be designed specifically for the US Marine Corps.



Above: CH-53D of the Heer (Federal German Army).



Above: Three-view of the CH-53D (other twin-engined versions similar).

Below: Sikorsky were fortunate in being able to put together the S-64 dynamic parts and the T64 engine to produce the CH-53A for the Marine Corps. This has served as the basis for the most capable family of Western helos



Sikorsky UH-60A

S-70, YUH-60A

Origin: Sikorsky Aircraft, Division of United Technologies Corp. (Sikorsky Aircraft is a subsidiary of United Technologies Corp.)

Type: Multi-role tactical transport

Engines: TWA 1 536shp General Electric T700-700 + 1 + 1 + 1 + 1

Dimensions Overall length of four blade main rotor 53ft 8in (16.36m), length overall (rotors turning) 64ft 10in (19.76m), height overall 16ft 5in (5.00m)

Weights: Empty about 10,000lb (4536kg), maximum loaded 21,400lb (9707kg)

Performance: Maximum speed at 16,750lb (7597kg) mission weight
198mph (318km/h)

History: First flight 17 October 1974

Users' ≥ 1 in x_i



Above: Three-view of UH-60A (LAMPS III would look different).

Right and below: The YUH-60 could carry a squad of eleven troops and manoeuvre with bank angles greater than 90°.

Development

other loads include four stretchers (3175kg) (12 000lb), 5443kg, in the rear. Stowed in the front is a side-firing machine gun and the weight of the vehicle when folded it fits a C-130. It can be carried by a C-130. It would carry radar, MAD, sensors, and a communications system.

In December 1976 the UTTA choice has yet to be made.



Westland Wessex

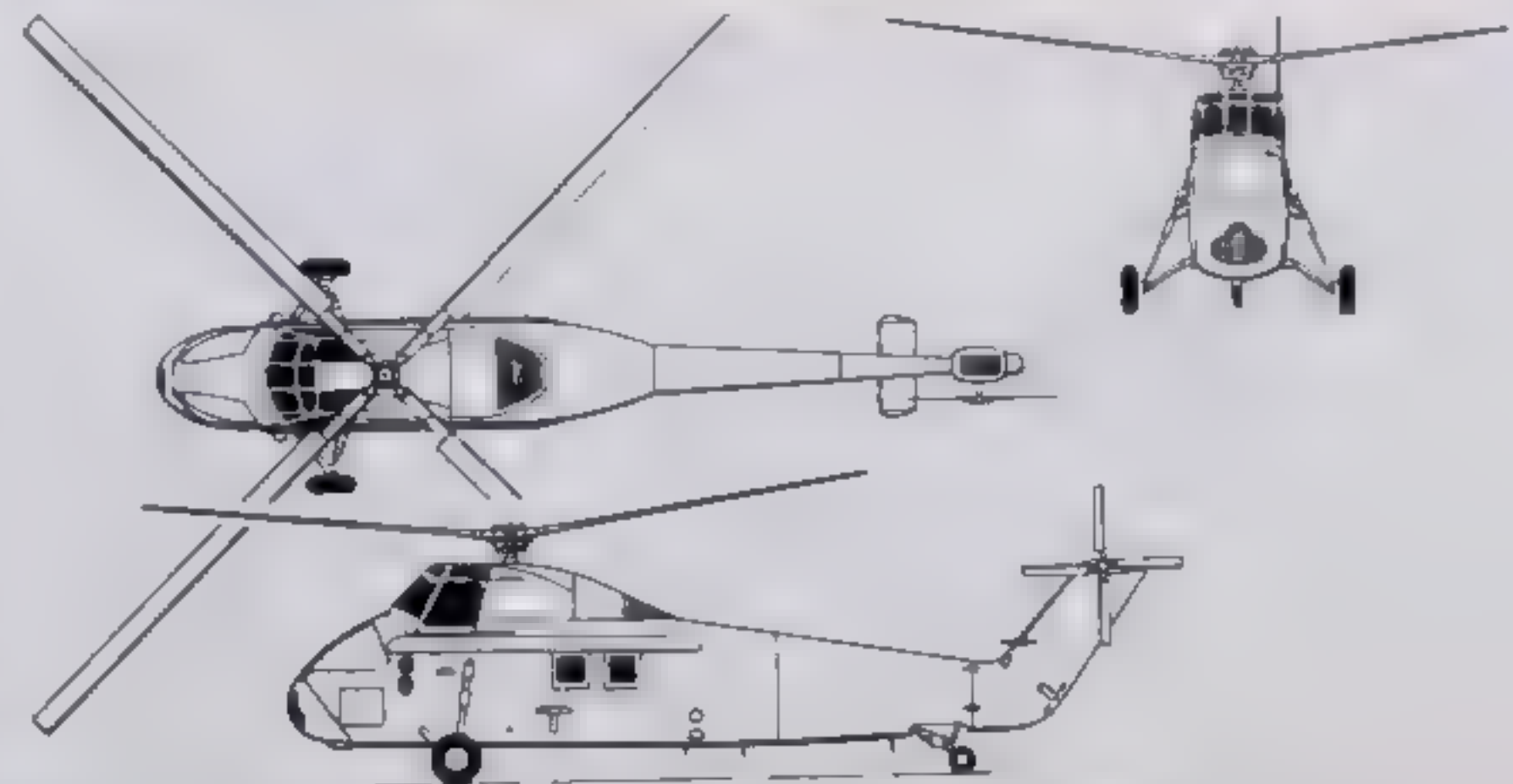
Wessex HAS.1, HC.2, HAS.3, CC.4, HU.5 and civil/export versions

Origin: Westland Helicopters, Yeovil, Somerset, England
Type: Multi-role helicopter (see text)
Engine(s): (1) one 1,450shp Rolls-Royce (Napier) Gazelle 161 free turbine turboshaft (2) Rolls-Royce Coupled Gnome 161/111, (3) two 1,350shp power sections, (3) one 1,600shp flat-rated Gazelle 165 (otherwise as Mk 2 except HAS 31 has 1,540shp Gazelle 162)
Dimensions: Diameter of four-blade main rotor 56ft (17.07m) height overall (rotors turning) 65ft 9in (20.03m) length of fuselage (2) 48ft 4in (14.74m) height overall 16ft 2in (4.93m)
Weights: Empty (1) 7,600lb (3447kg), (5) 8,657lb (3927kg), maximum loaded (1) 12,800lb (5715kg), (2) 5,311lb (2409kg), (3) 13,500lb (6120kg)
Performance: Maximum speed 133mph (214km/h) cruising speed 121mph (195km/h), maximum (not vertical) rate of climb 2,165ft (650m)/min, service ceiling 10,000–14,000ft (3048–4300m) range with standard fuel (1) 390 miles (630km)
Armament: See text
History: First flight (rebuilt S-58) 17 May 1957; service delivery (HAS 1) April 1960, final delivery (civil) 1970
Users: Australia, Brunei, Ghana, Iraq, UK (RAF, Royal Navy, Marines)

Development: In 1956 the Royal Navy dropped its plan to buy the twin Gazelle Bristol 191 and opted for a cheaper and less risky solution with a Sikorsky S-58 (US Navy HSS-1) using a single Gazelle. Development was fast, and about 150 anti-submarine versions were used by the Fleet Air Arm, the HAS 1 being supplanted by the refined HAS 3 called Camel because of its hump-backed radome. The RAF bought a version with Coupled Gnome engines, over 100 HC 2 utility versions being followed by almost 100 HU 5 Commando assault machines. Two CC 4 VIP transport serve the Queen's Flight. HMAS Melbourne is main base afloat for 27 HAS 31 of the RAN which, like those in British service, have undergone progressive updating since 1965. Many weapon fits are in use, the HAS marks carrying one or two homing torpedoes and all having capability to mount guns or a wide range of missiles and rockets.



Above: A Wessex HU.5 used as a Commando assault transport by the Royal Navy puts down on the right numbered spot



Above: Three-view of typical Gazelle-Wessex (Mk 3 or 31)

Below: Winter exercises in Norway look tough, but this RAF Wessex HC.2 finds it easier to hover at maximum weight.



Westland Sea King and Commando

Sea King HAS.1 and Mk 41-50;
Commando 1 and 2

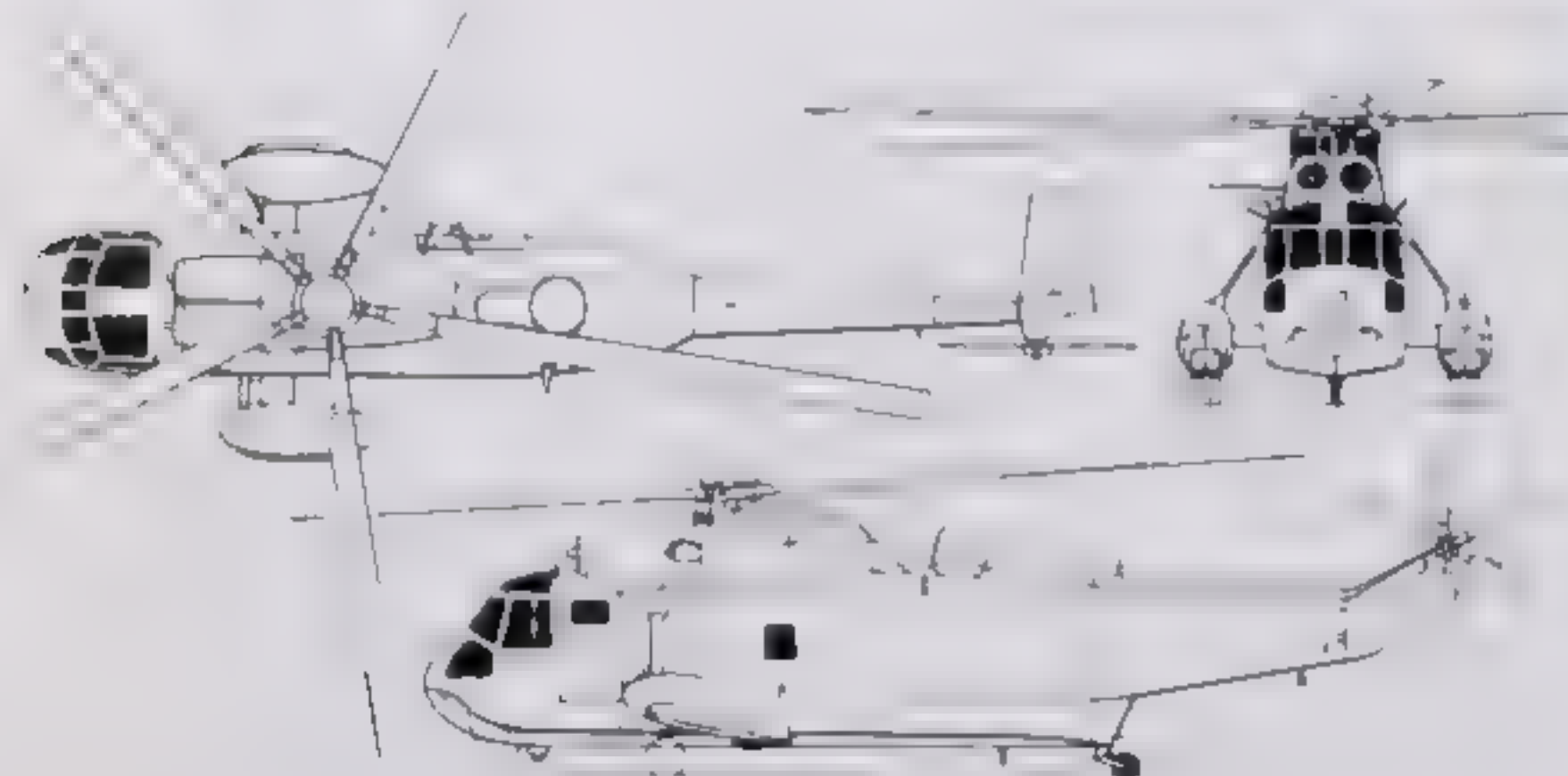
Origin
Type

Engines: Two Rolls Royce Gnome (derived from GE T58) free turbine

Dimensions: Diamet

Weights: Empt

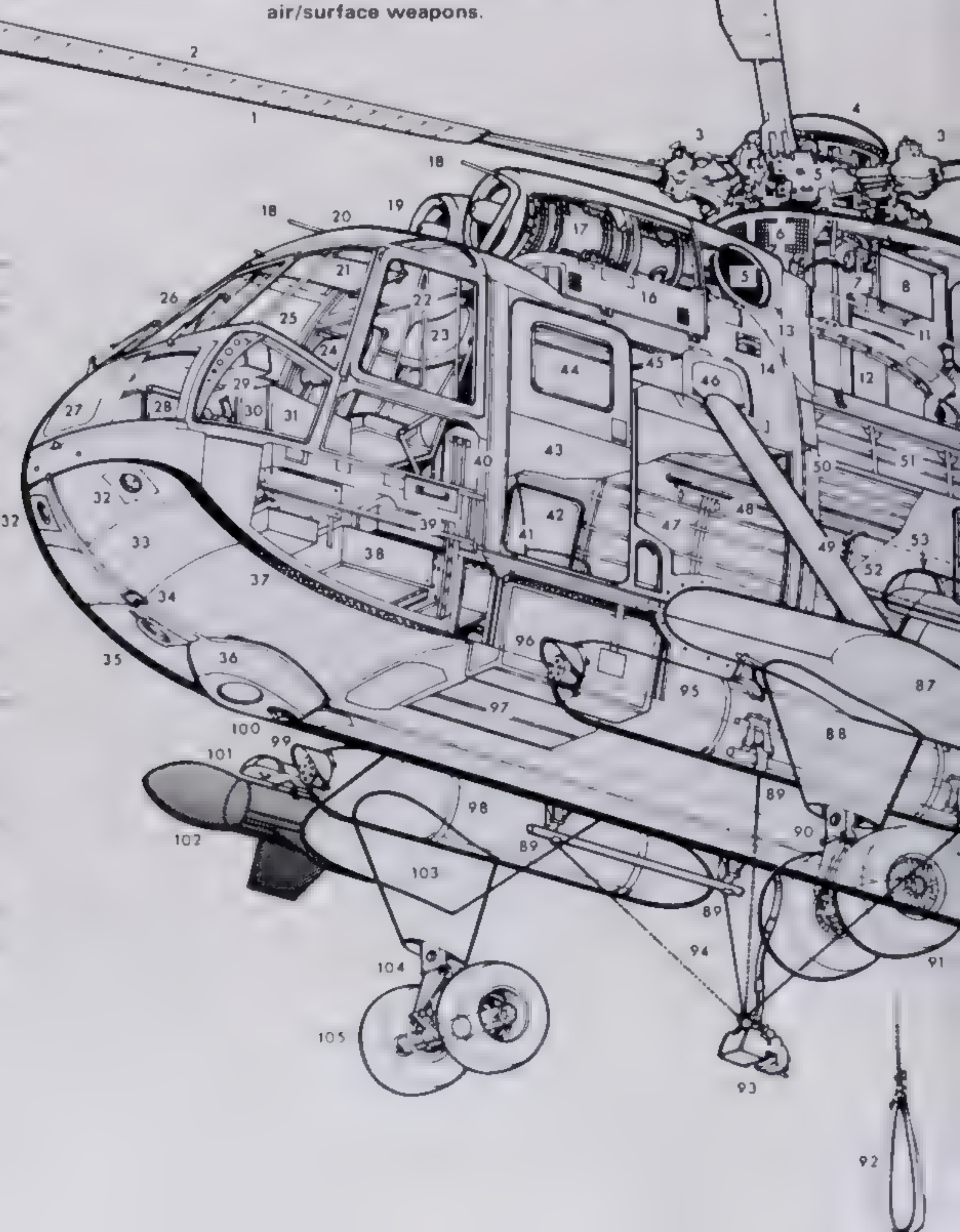
continued on page 222



Above: Three-view of Sea King (Commando differs significantly)



Above: Though its airframe is generally similar to that of the Sea King, the Commando is a totally different helicopter for land warfare. Here the simple fixed landing gears are prominent. This design change, and removal of the comprehensive ASW or SAR equipment carried by the Sea King, enables the Commando to carry 28 equipped troops and a wide range of air/surface weapons.





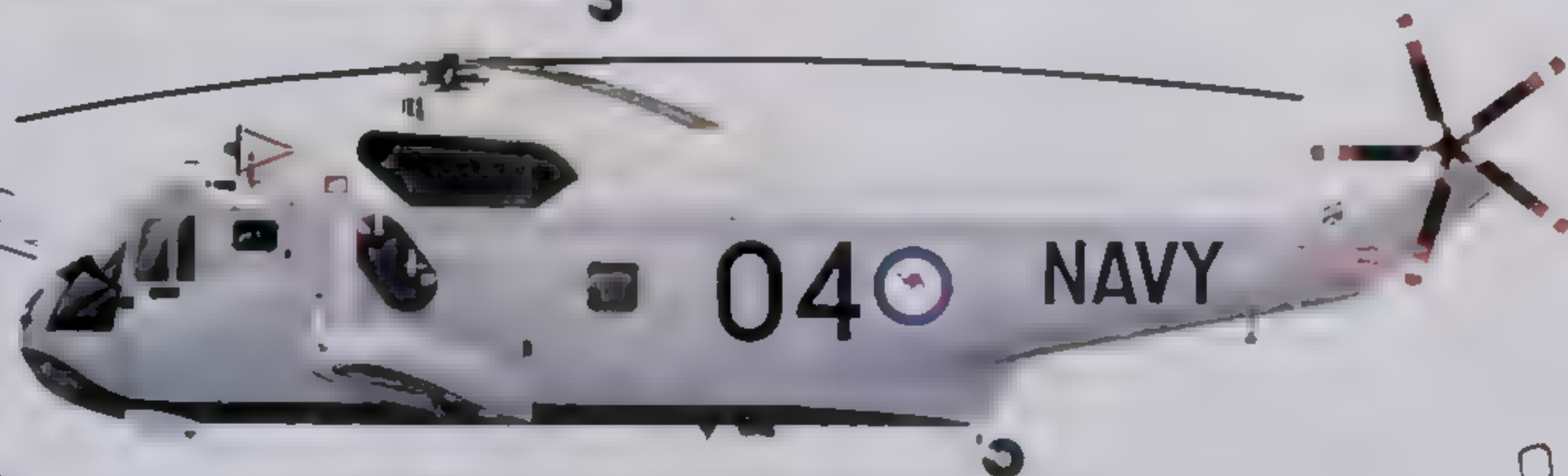
Above: Sea King 41 (SAR version) for West German Marineflieger, MFG.5, Kiel-Holtenau.



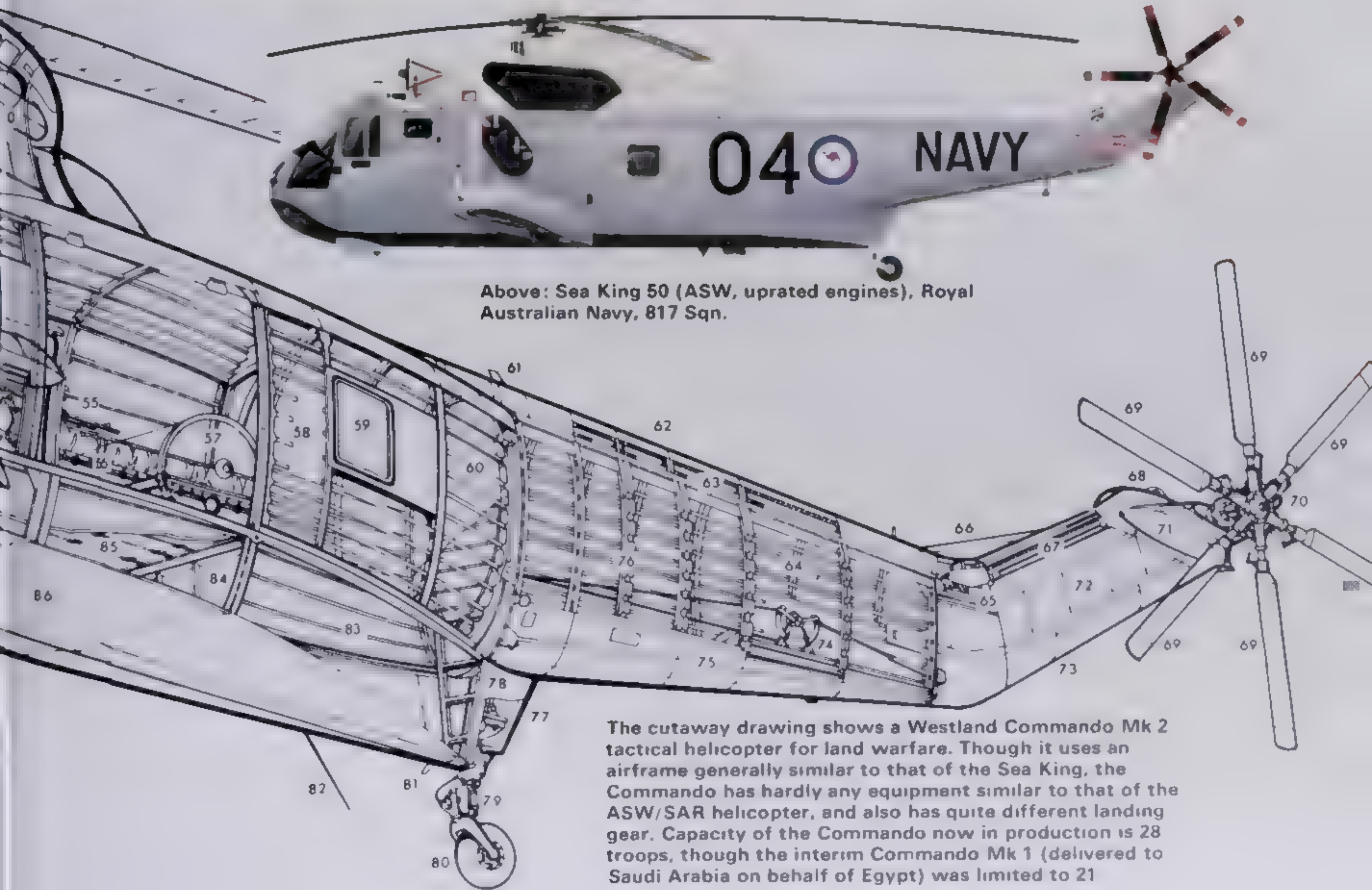
Above: Sea King 43 (SAR), Norwegian Air Force, 331 Sqn. Bodo.



Above: Sea King 45 (ASW) of Pakistan Navy.



Above: Sea King 50 (ASW, uprated engines), Royal Australian Navy, 817 Sqn.



The cutaway drawing shows a Westland Commando Mk 2 tactical helicopter for land warfare. Though it uses an airframe generally similar to that of the Sea King, the Commando has hardly any equipment similar to that of the ASW/SAR helicopter, and also has quite different landing gear. Capacity of the Commando now in production is 28 troops, though the interim Commando Mk 1 (delivered to Saudi Arabia on behalf of Egypt) was limited to 21

Armament

History: Derived from Sikorsky S-61 of 1959, first flight of Sea King

Users: Australia, Belgium, Egypt, W Germany, India, Norway, Pakistan, Saudi Arabia, UK (RAF, Royal Navy)

Development: Sikorsky's S-61 was almost inevitably the subject of a licence agreement between the company and Westland Aircraft, following an association begun in 1947 when a licence was purchased for the S-51. In the case of the S-61, the most immediate significant customer was the Royal Navy, which was searching for an ASW (anti-submarine warfare) helicopter to supplement and eventually replace the Wessex in operation from surface ships. Unlike the US Navy, which chose to regard its ASW helicopters as mere extensions of the all-important ship, the Fleet Air Arm concluded it would be preferable to allow the helicopter to operate independently. The Sea King HAS.1 was thus designed to carry sensors, weapons and a complete tactical centre to hunt down and destroy submerged submarines. The normal equipment for Sea Kings in the ASW role includes dunking sonar, doppler navigator, search radar and an autopilot and weapon system providing for automatic hovering at given heights for a range of other automatic manoeuvres in all weather. The RN bought 58, with 13 more ordered in 1975 and eight in 1976. Total Sea King sales are 150, many of them being for the much simpler SAR (search and rescue) version with seats for up to 22, apart from the crew, and special provision for casualties, cargo and slung loads. The future RAF rescue helicopter is the Sea King HAR.3.

Like the Sea King, the Commando is a purely land-based aircraft with fixed landing gear devoid of floats. It has been optimised to the range/payload needs of tactical operations, and carries much special equipment for use in a wide range of roles. The basic Commando provides accommodation for up to 28 troops in the transport role, or equivalent cargo payload (or 8,000lb, 3,630kg, slung externally). Other roles include logistic support, casualty evacuation, search/rescue or, with any of a wide range of armament fits, air/surface strike. So far no armed Commandos have been ordered (the first sale was a large batch for Egypt, bought by Saudi Arabia in 1973), but various turrets and launchers, manually aimed guns, guided missiles and rocket pods can be fitted. Internal load can include a 105mm gun or Shortland armoured car.



Above: A Royal Navy Sea King HAS.1 dunks its Plessey Type 195 sonar to listen for submarines. Note the twin depressions caused by the split rotor downwash (not heard by the sonar)

Westland/Aérospatiale Lynx

Lynx AH.1, HAS.2 and 2(FN) and HT.3.

Origin: Westland Helicopters, UK, in partnership with Aérospatiale, France

Type: Multi-role helicopter (see text)

Engines: Two 900shp Rolls-Royce Gem 10001 three-shaft turbines (or option, two 750shp P&WC PT6A-34)

Dimensions: Diameter of four-blade main rotor 42ft (12.80m), length overall (rotors turning) 49ft 9in (15.16m), height overall (rotors turning) 12ft (3.66m)

Weights: Empty weight (basic) 5,225lb (2,370kg), empty (equipped for troop transport) 5,641lb (2,558kg), (anti-tank) 6,313lb (2,863kg), (dunking sonar search/strike) 7,218lb (3,274kg), maximum loaded (army) 9,250lb (4,196kg), (navy) 9,500lb (4,309kg)

Performance: Maximum speed 207mph (333km/h), continuous cruise 176mph (284km/h), single-engine cruise 164mph (263km/h), maximum (not vertical) rate of climb 2,370ft (722m)/min, ceiling, well over 25,000ft (7,600m), range (army) 473 miles (761km), (navy) 418 miles (673km), ferry range with cabin tank 861 miles (1,386km)

Armament: See text

History: First flight 21 March 1971 (HAS.2) 25 May 1972, service delivery (Royal Navy) May 1976

Users: Argentina, Belgium, Brazil, France, Netherlands, Qatar, UK (RAF, Royal Navy, Army), Egypt (discussing purchase of large batch plus manufacturing licence and facilities)

Development: Certain to be manufactured in very large numbers over a period greater than ten years, the Lynx is probably the outstanding example



This is a multi-paragraph block of text, likely bleed-through from the reverse side of the page. The text is mostly illegible due to the quality of the scan and the nature of the bleed-through. It appears to contain several paragraphs of descriptive text, possibly related to the helicopter's performance or development.



Above: Three-view of Lynx AH.1 with side view (top) of HAS 2

Facing page, upper: A Lynx HAS.2 of the Royal Navy comes aboard HMS Sheffield, first of the Type 42 surface warships.
 Facing page, lower: Lynx HAS multi-role variant of Royal Netherlands Navy, with Seaspray radar. The final ten Dutch Lynx are fitted with uprated Gem engines and weigh 10,500lb (4763kg).
 Below: The Lynx is the world's most manoeuvrable combat helo, and it has rolled (like this Army AH.1) at over 100° per second and flown backwards at 80mph



Above: Prototype Lynx HAS.2 with AS 12 (now replaced by Sea Skua).



Trainers and utility aircraft



In World War I a junior officer in the Royal Flying Corps, R. R. Smith-Barry, achieved the remarkable feat of working his way to the command of a front-line fighter squadron after having had both legs shattered and been told he had no hope of even returning to flying duty. But his name lives in history because he was so appalled at the quality of young replacement pilots that he said so in public. General Trenchard said, "Go and do something about it", and Smith-Barry laid down such basic principles (which seem obvious today, but were far from obvious then) as the use of specially designed training aircraft, with dual controls, and the employment of instructors trained for the job, instead of combat pilots "resting" from the Front and usually totally disinterested in what they achieved. He also drafted a complete syllabus of flying instruction, and this has remained the basis for all flying instruction ever

since. The methods have changed only in minor ways, though the aircraft have changed dramatically over the years.

Many air forces use aircraft for contrasting purposes, and it is difficult to be dogmatic. Modern jet trainers are quite capable of being used as attack aircraft and even as "fighters" provided the hostile aircraft are not beyond their capability. Such aircraft as the T-28 and Magister have been in the news far more often when engaged in warfare than when in the uncertain hands of pupils, while the Alpha Jet, Hawk, Saab 105 and several Aermacchi models are available equipped primarily as multi-role attack weapon platforms, with training a secondary role (in fact, some are single-seaters). Again, some Supporters are not used in the training role but as utility machines, while many aircraft we have categorised as "utility" spend their lives being



The Japanese are good at doing things that are different, smaller and more sophisticated than rival products. The Mitsubishi MU-2 flies like a Spitfire and can accommodate not only Japanese but also Texans, in whose state many are assembled.

utilised solely for the purpose of training crews.

The utility category is a valid one, though it is difficult to define. One could argue that many retired combat aircraft serve in the utility role; but the purpose-designed utilities, which are the ones in this chapter, are often derived from a civil lightplane, a civil transport or other categories, but almost without exception the result today is a vehicle that can go almost anywhere, be maintained and refuelled at any airfield, and can fly simple transport duties or carry almost any kind of equipment or special installation provided it is not large or heavy enough to warrant a true Transport category aircraft. Most of the utility missions are simply taking a handful of people to look at something or meet someone, but there is far more to it than this.

Subdivisions include: the Forward Air Control (FAC) aircraft, which played a central role in

directing attacks of other aircraft on invisible ground targets in South-East Asia; the observation aircraft, which was an earlier form of the same species and borders on the edges of the category labelled Reconnaissance; the psy-war (psychological warfare) aircraft, equipped with loudspeakers, leaflet dispensers and other devices to persuade or intimidate those on the ground below; lightplanes, too small to be termed transports, engaged in local supply transport (often by parachuting supply containers or bundles); a very wide range of electronic and radio duties, many of which grossly modify the appearance of the aircraft; executive-type duties, notably by air attaches; ambulance duties; missile-site inspection and communications; and a diverse spectrum of para-military duties concerned with civil law and order, forestry patrol, coastal patrol, survey and similar tasks.

Aermacchi M.B. 326 and 339

M.B.326 and 326 GB and GC (AT-26 Xavante), 326K (Atlas Impala), 326L and M.B.339

Origin
Type
Engine

Dimensions
Weights
Performance

Armament: Six underwing pylons for load of up to 4 000 lb (1 814 kg)

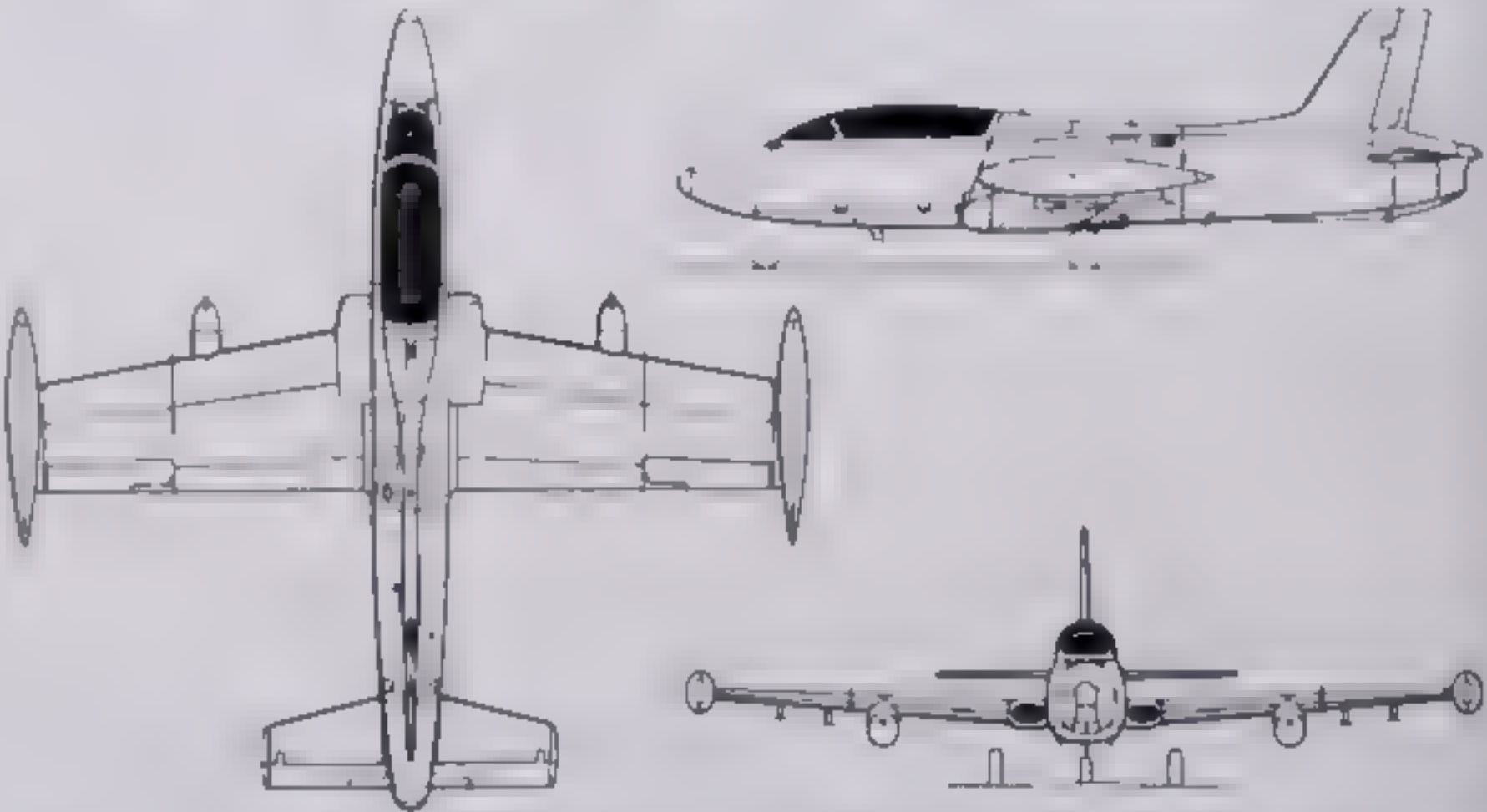
History: First flight 10 December 1966

Users: Argentina, Australia, Brazil, Canada, Chile, Colombia, Ecuador, France, Germany, Greece, India, Italy, Israel, Japan, Korea, Kuwait, Lebanon, Libya, Malaysia, Mexico, Morocco, Oman, Pakistan, Peru, Portugal, Saudi Arabia, Singapore, South Africa, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Arab Emirates, United Kingdom, United States, Venezuela, West Germany, Yugoslavia

Development: The most successful Italian military aircraft since the Second World War, the M.B.326 was developed from the M.B.324, a light attack aircraft. The M.B.326 was designed to be a multi-role aircraft, capable of both attack and reconnaissance. It was designed to be a simple, rugged aircraft that could be operated from unimproved runways. The M.B.326 was designed to be a low-cost aircraft, with a price tag of around \$10 million. The M.B.326 was designed to be a simple, rugged aircraft that could be operated from unimproved runways. The M.B.326 was designed to be a low-cost aircraft, with a price tag of around \$10 million.



Above: First prototype M.B.339, a completely redesigned aircraft with modern stepped cockpits. Aermacchi hope to sell to many air forces by being cheaper than the opposition.



Above: Three-view of typical M.B.326G with wing gun pods.

Below: The EMBRAER AT-26 Xavante of the Brazilian Air Force is the M.B.326GB built under licence. By spring 1977 over 120 had been delivered to attack, recce and training units.



Aero L-29 Delfin

L-29 Delfin (NATO name "Maya")

Origin: Aero Vodochody national corporation, Czechoslovakia
Type: Basic trainer
Engine: One 1 960lb (890kg) thrust M 701 single-shaft turbofan
Dimensions: Span 31ft 0in (9.43m), length 31ft 11in (9.73m), height 10ft 3in (3.13m)
Weights: Empty 5,021lb (2,280kg), maximum loaded 7,160lb (3,250kg)
Performance: Maximum speed at 16 400ft (5 000m) 417mph (671km/h), initial climb 2 755ft (840m)/min, service ceiling 36 090ft (11 000m), maximum range on internal fuel 397 miles (640km)
Armament: Two wing hardpoints on which can be attached two 7.62mm guns, small tanks, or various other loads including bombs of up to 220lb (100kg)
History: First flight 5 April 1959, service delivery 1963, final delivery 1974
Users: Bulgaria, Czechoslovakia, Egypt, E. Germany, Guinea, Hungary, Indonesia, Iraq, Nigeria, Romania, Soviet Union, Syria, Uganda
Development: Designed by a team led by K. Tomas and the late Z. Rubicki, the L-29 was Czechoslovakia's submission in 1960 as the standard Warsaw



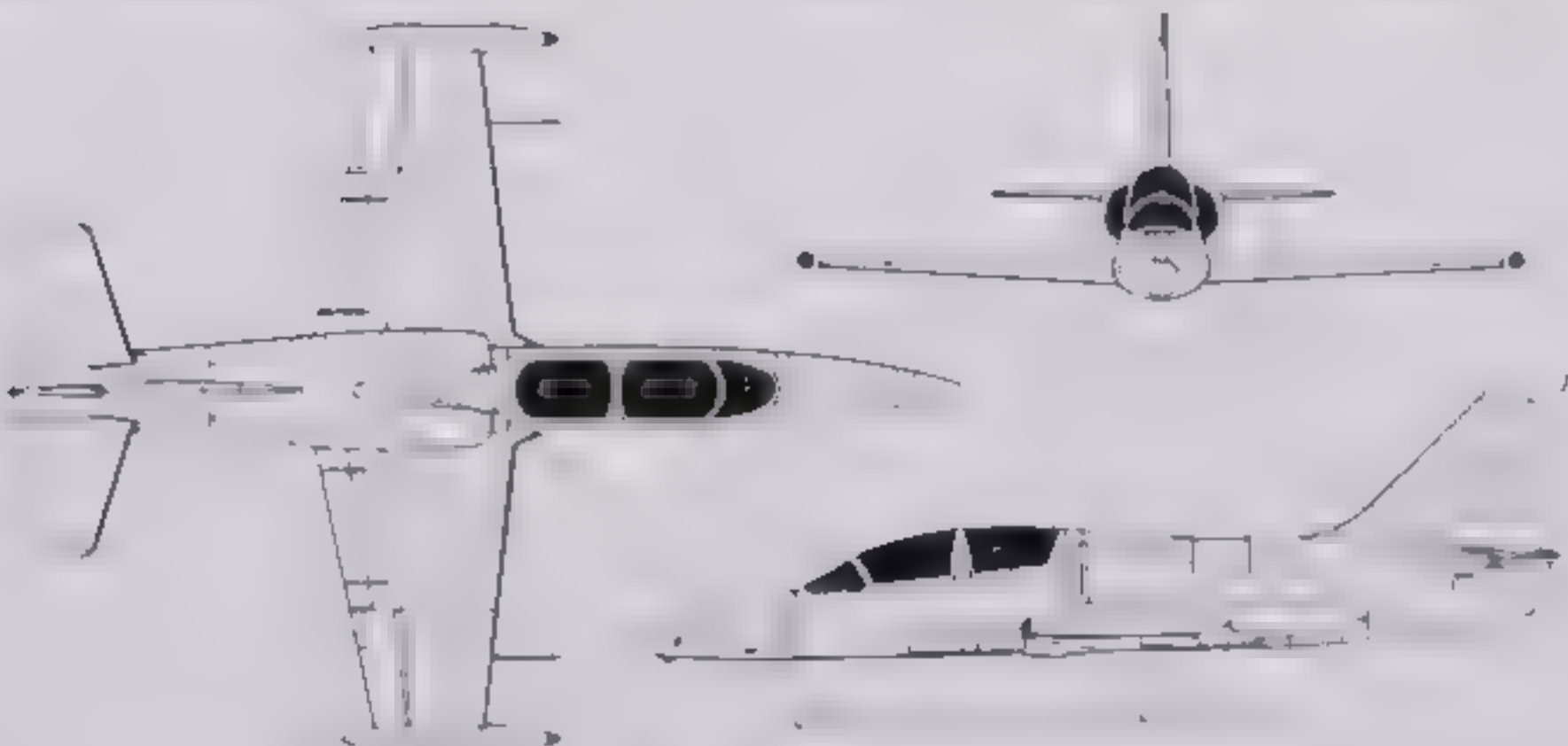
Above: Three-view of L-29 (plus side view of L-29A Akrobat single-seat version for aerobatic demonstration).

Pact trainer. It won, and though disgruntled Poland went ahead with the TS-11 Iskra, other Communist countries and several other air forces adopted this trainer and more than 3 000 were delivered. Two variants, not built in quantity, were the L-29A single-seat aerobatic version and the L-29R for counter-insurgency operations with nose cameras and under-wing stores.

Aero L-39 Albatros

L-39, L-39Z

Origin: Aero Vodochody national corporation, Czechoslovakia
Type: Basic and advanced trainer
Engine: One 3 792 (1 720kg) thrust Walter Titan two-shaft turbofan (licence built Ivchenko AI-25-T)
Dimensions: Span 31ft 0in (9.43m), length 40ft 5in (12.32m), height 15ft 5in (4.72m)
Weights: Empty 7 341lb (3 330kg), maximum loaded 10 141lb (4 600kg)
Performance: Maximum speed at 16 400ft (5 000m) 466mph (750km/h), Mach limit in dive 0.80, cruise speed at 5 000m, up to 423mph (700km/h), initial climb 4 330ft (1 320m)/min, service ceiling 37 075ft (11 300m), max range at 5 000m on internal fuel 565 miles (910km)
Armament: Provision for light external load of tanks, bombs, rockets or 7.62mm gun pods (weight unstated) on two hardpoints under wings; internal fittings confined to gun-camera and sight
History: First flight 4 November 1968, service delivery late 1973
Users: Czechoslovakia, Iraq, Soviet Union (other Warsaw Pact countries in due course)



Above: Three-view of L-39 Albatros.

part of an integrated pilot-training system which includes a flight simulator, ejection-seat trainer and mobile automatic test equipment. The L-39Z was under development in 1973 as a light attack version which, like all L-39s, can be operated from unpaved surfaces. Iraq is believed to have ordered both models.

Development: Designed by a team led by Jan Vlcek, the L-39 is succeeding well in its task of replacing the L-29, and in 1972 it was accepted in principle as the future trainer of all Warsaw Pact countries except Poland. It forms

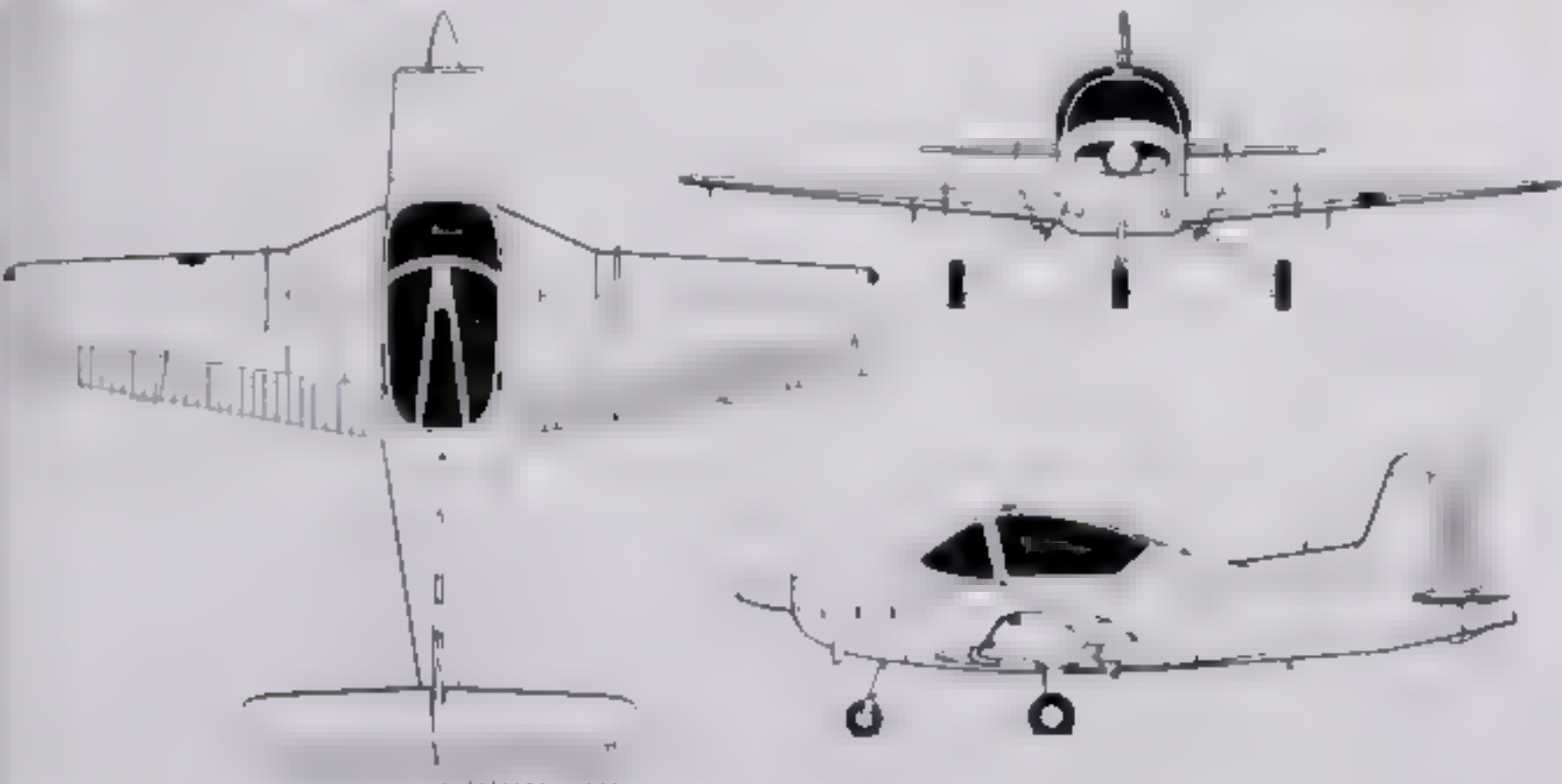
Below: The shapely turbofan-powered Albatros marks a great advance over the mass-produced Delfin. It will be interesting to see how well it does in the world trainer market.



Aerospace Airtrainer CT-4

CT4 and CT4B

Origin: NZ Aerospace Industries, Hamilton, New Zealand
Type: Primary and aerobatic trainer
Engine: One 210hp Rolls Royce Continental IO-360-H flat-six
Dimensions: Span 26ft 0in (7.92m), length 23ft 2in (7.06m), height 8ft 6in (2.59m)
Weights: Empty 1,460lb (662kg), maximum loaded 2,400lb (1088kg)
Performance: Maximum speed 178mph (286km/h), initial climb 1,350ft (411m)/min, take-off or landing over 50ft (15m) about 1,200ft (366m), service ceiling 17,900ft (5455m), max range with reserves (no tip tanks) 815 miles (1311km, 5hr 47min)
History: First flight (Victa Airtourer) 31 March 1959, (CT4) 23 February 1972, service delivery (Thai) 23 October 1973
Users: Australia, Hong Kong, Indonesia, New Zealand, Singapore, Thailand



Above: Three-view of standard Airtrainer CT4.



One of the 37 CT4 Airtrainers of the Royal Australian Air Force. An aggressive development and market research policy should keep this attractive machine selling.

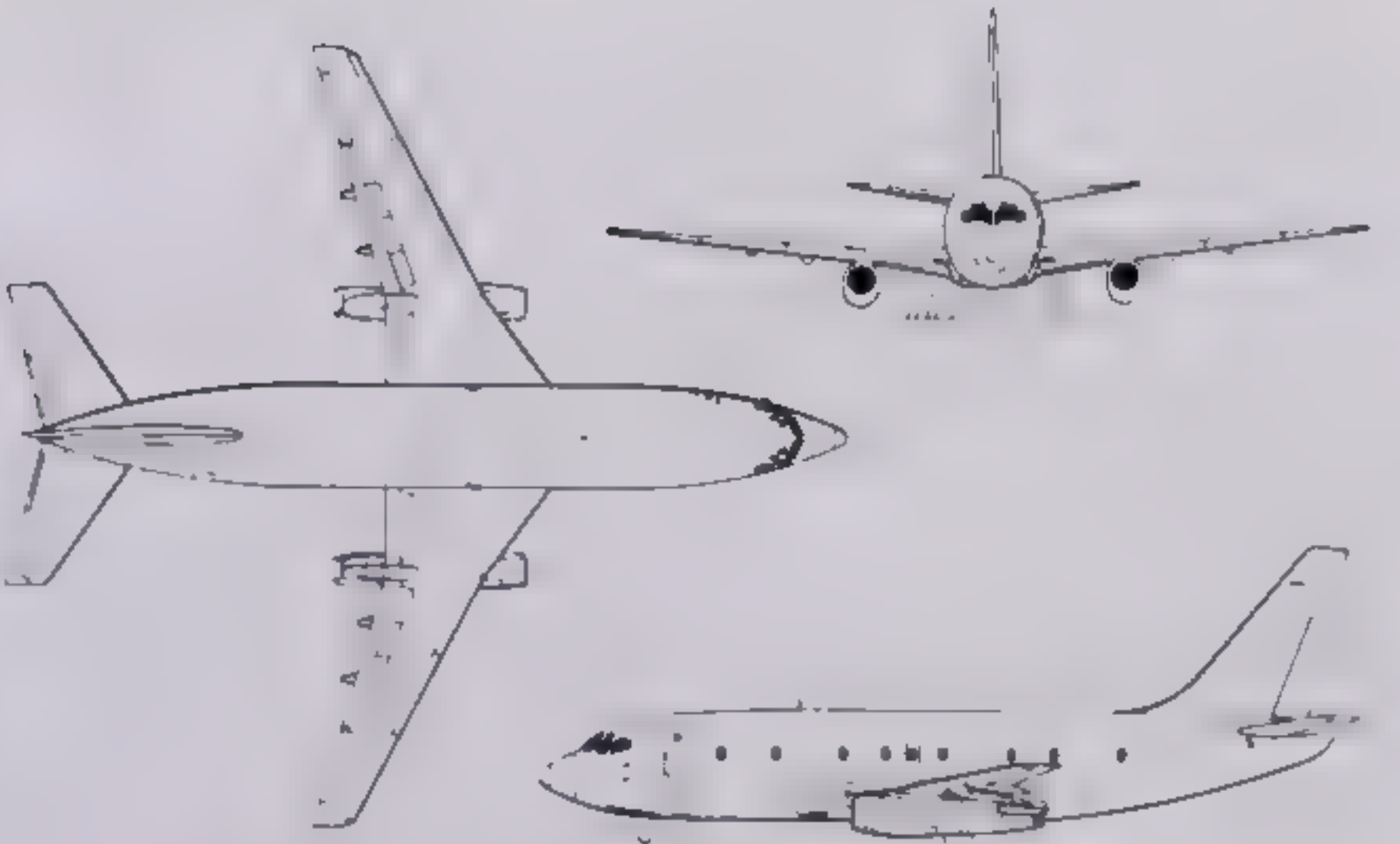
Development: In 1953 Australian Henry Miler won a lightplane design contest held by the Royal Aero Club. The eventual result was the four seat Airtourer, made by Victa in Australia, which in turn led to one of the world's better lightplane trainers, produced by the amalgamated NZ industry. The Airtrainer is all-metal, has electric flaps, a constant-speed propeller and optional third seat. Armed versions remained paper studies as this book went to press, as did a FAC (forward air control) version and more powerful models, one having a turboprop. In 1976 several air forces were negotiating orders, but Aerospace terminated a NZ\$1.5 million deal "for a Swiss flying club" because, according to unofficial reports, the destination was Rhodesia.

Boeing T-43A and 737

T-43A

Origin: The Boeing Company, USA
Type: Navigational trainer
Engines: Two 14,500lb (6575kg) thrust Pratt & Whitney JT8D-9 two shaft turbofans
Dimensions: Span 93ft (38.35m), length 100ft (30.48m), height 37ft (11.28m)
Weights: Empty, about 62,000lb (28,123kg), maximum 115,500lb (52,390kg)
Performance: Maximum speed 586mph (943km/h), maximum cruising speed 576mph (927km/h) at 22,600ft (6890m), economical cruise, Mach 0.7 at 35,000ft (10,670m), initial rate of climb at gross weight 3,750ft (1143m)/min, range with military reserves 2,995 miles (4820km), endurance 6hr
Armament: None
History: First flight (737-100) 9 April 1967, (T-43A) 10 April 1973
User: US Air Force (Air Training Command, though the Navigator Training Wing at Mather AFB is a joint AF/Navy/Marines/Coast Guard unit), basic 737 is used by Brazilian and Venezuelan AF

Development: Derived from the commercial 737-200, the T-43A is the world's most advanced navigational trainer. Though the "baby" of current Boeing production aircraft, it is nevertheless much more capacious and powerful than any Superfortress, and the fleet of 19 used at Mather Air Force Base, California, have replaced 77 Convair T-29 crew trainers previously used. This has been possible because of the T-43's much higher utilisation and ability to carry 12 trainee navigators (each with a complete aircraft instrument and systems station), four proficiency students and three instructors, apart from the T-43A crew. In addition the T-43A is used in conjunction with a comprehensive computer-based mission simulation

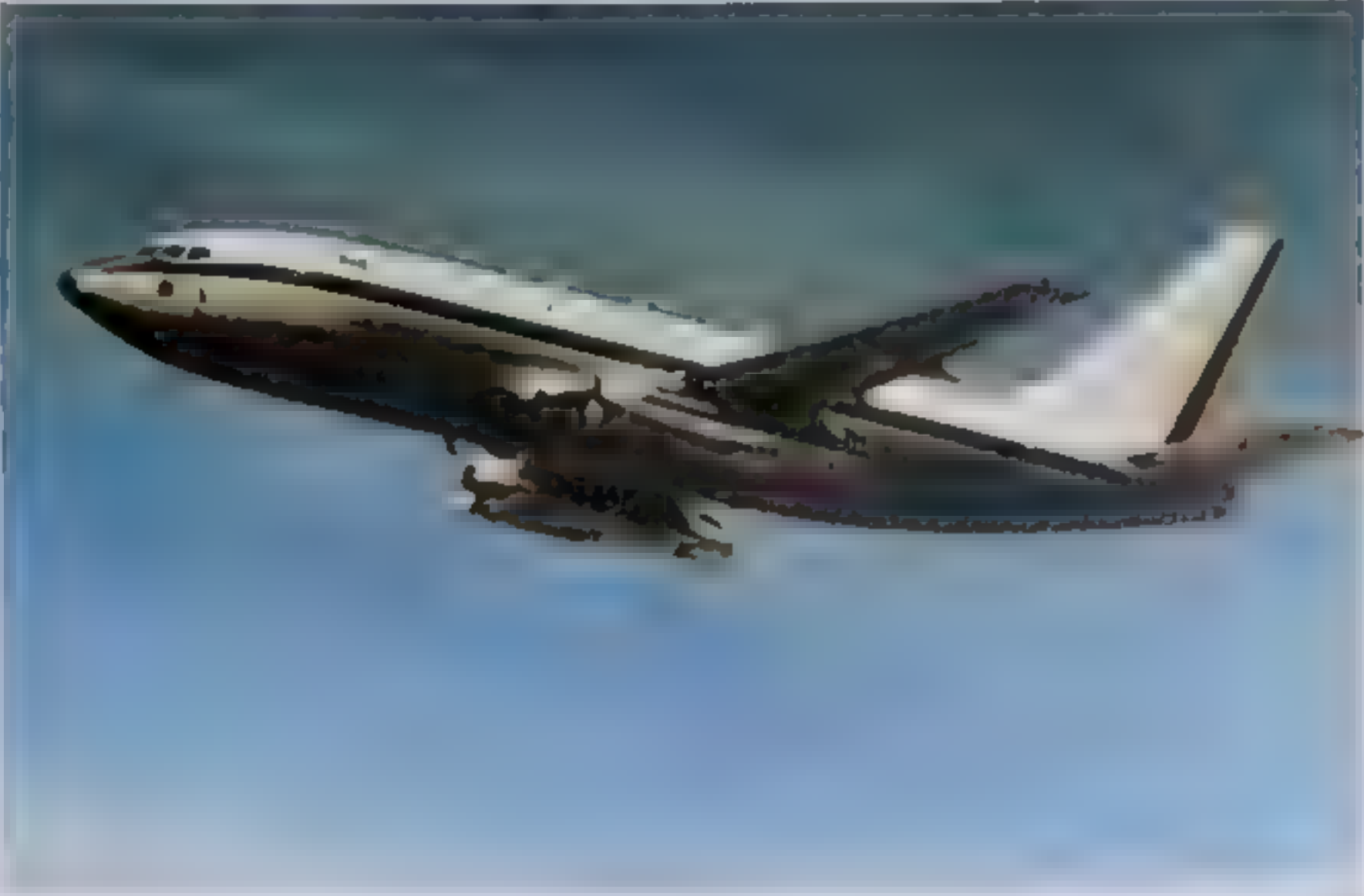


Above: Three-view of Boeing T-43A

system which enables every kind of combat mission up to Mach 2 at 70,000ft (21,340m) altitude to be "flown" on the ground. Each T-43A carries complete inertial navigational systems, Loran and other radio systems, and provision for celestial and even dead-reckoning navigation techniques. Mission endurance is increased by a 667 gal (3027 litre) tank in the rear cargo compartment.

Below, left: A Boeing T-43A could be voted the aircraft least likely to fly into a mountain through a navigation error.

Below: The Brazilian Air Force is one of the customers for the VIP executive 737-200, which can fly 3,857 miles (6207km) and operate from unpaved airstrips.



Beech trainers

T-34C Turbo Mentor

Origin: Beech Aircraft Corporation, Wichita, USA
Type: Primary Trainer
Engine: One 715hp Pratt & Whitney Canada PT6

Dimensions: Span 33ft 4in (10.2m) length 28ft 10in (8.8m)

Weights: Empty 2 630lb (1193kg), normal 3 100lb (1 407kg)

Performance: Maximum speed 250kts (463km/h), initial climb 1 275ft (338m)/min, service ceiling 12 000ft (3 658m), 74% max. power

Armament: None, but Pave Coin (Bonanza) can carry up to 1 180lb of external loads up to 1 180lb, can carry 100lb of bombs and FFAR rocket

History: First flight 21 September 1976, first delivery September 1976

Users: Morocco, US Navy

Development: In 1973 Beech was awarded a US Navy contract to rebuild a T-34B (see below) as a turboprop trainer with upgraded equipment. After exhaustive trials with the resulting YT-34C rebuilds the Navy placed orders for 256 by late 1976, with further batches likely to bring the total to the requirement of 400. The third batch of 98 were priced at \$36.5 million. The airframe has been modified in many details compared with the T-34B to give a fatigue life of 16 000hr at greatly increased weights.

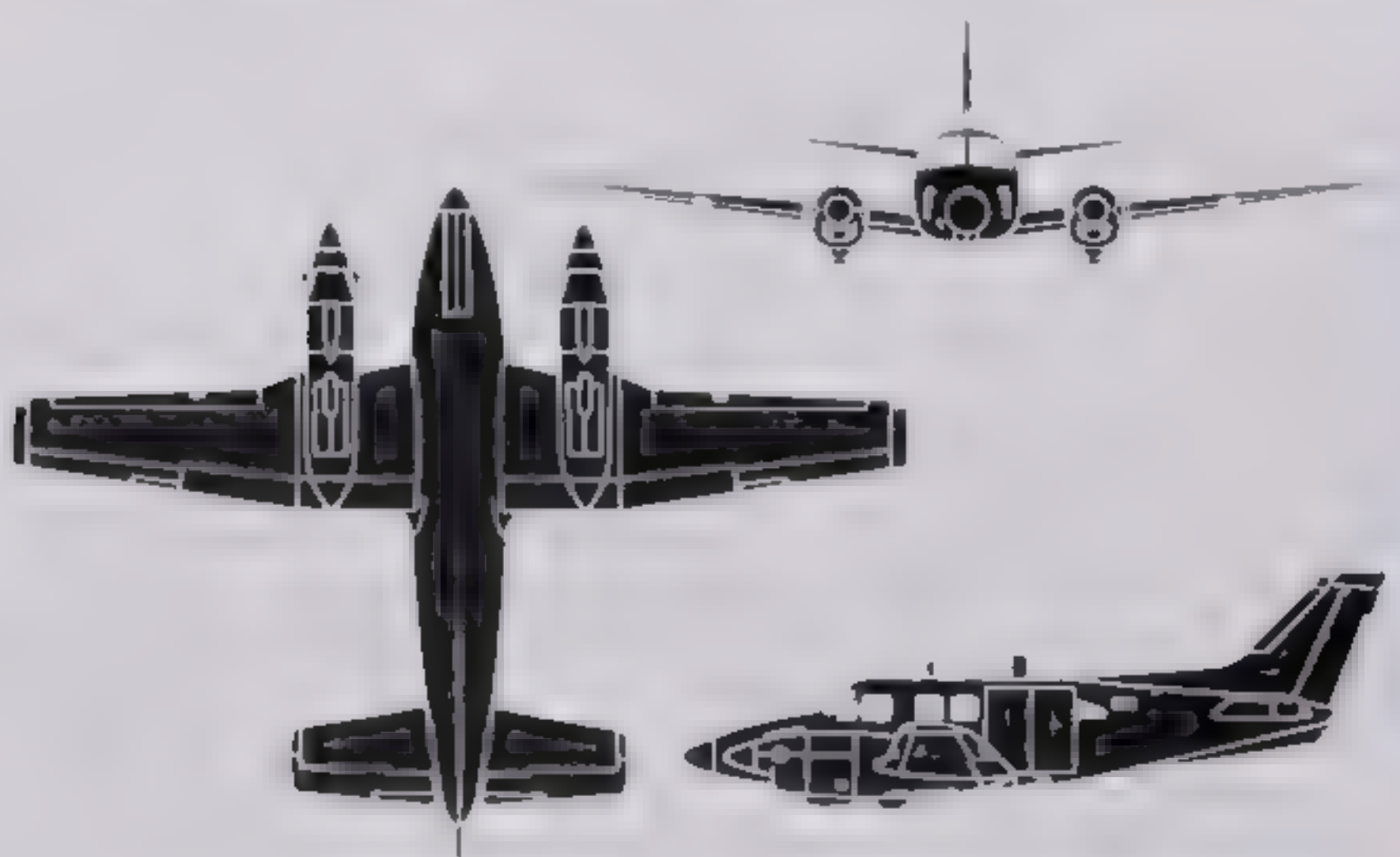
T-24A and T-34B Mentor

Constructor's designation, Model 45.

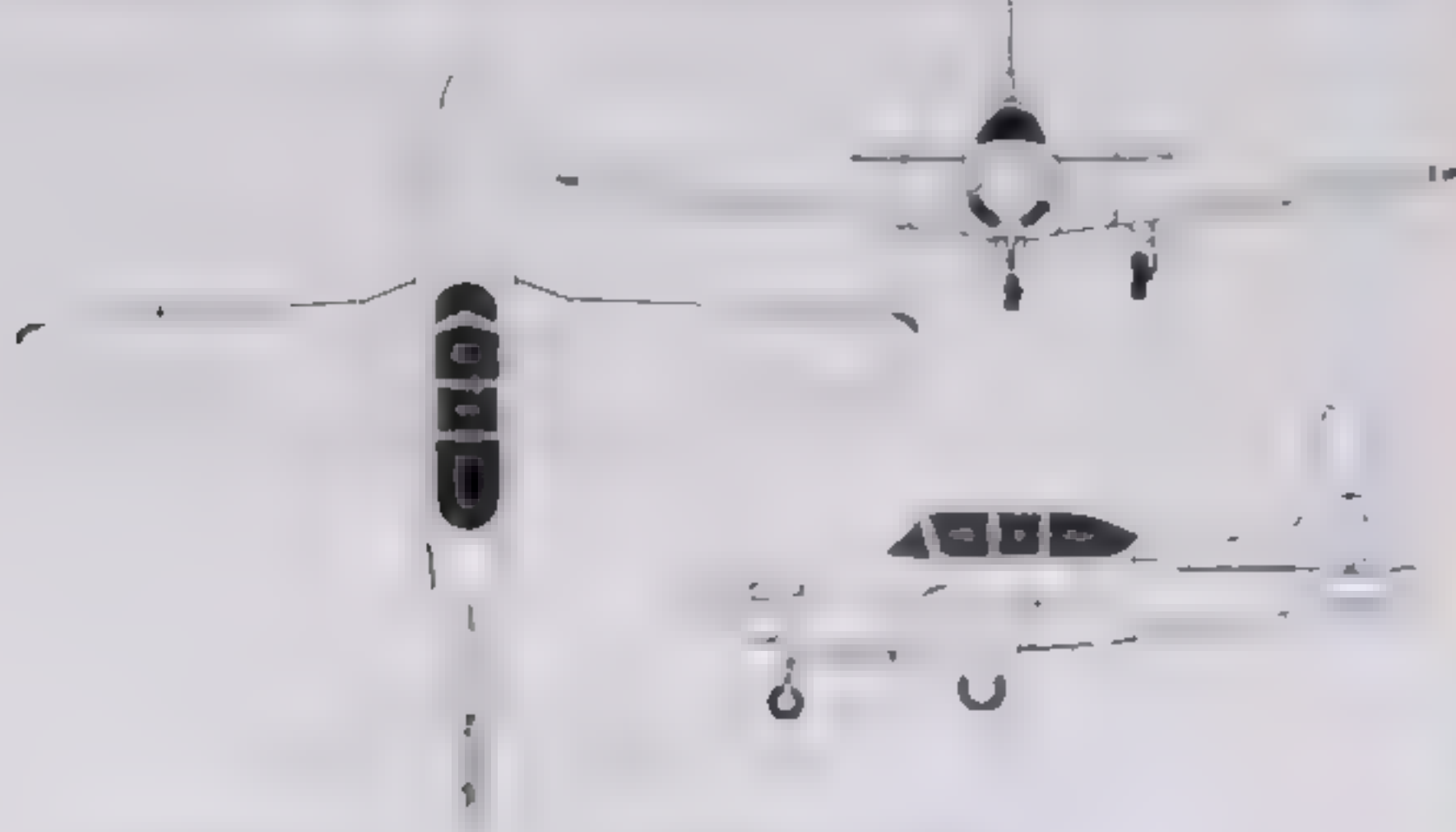
In March 1953 the US Air Force selected the trim T-34 as its standard primary trainer. Powered by the 225hp Continental O-470-13 flat-six, 450 were delivered to the Air Force and 423 basically similar T-43B Mentors to the US Navy. Beech supplied further batches to Argentina, Chile, Colombia, Indonesia, Japan, Mexico, Morocco, Peru, Salvador, Turkey, Venezuela and Uruguay. CanCar built 100 for the RCAF and USAF. Fuji built 140 for the Japan Air SDF and 36 for the Philippine AF, and 75 were built at Cordoba for the Argentine AF. Subsequently aircraft were built for, or transferred to, Greece, Saudi Arabia, Spain (designation E-17) and Turkey. Last delivery 1962. The Fuji LM-1 and -2 Nikko of the Ground SDF and KM-2 and -2B of the Japan Maritime SDF are trainers or 4/5 seat liaison aircraft based on the T-34A.

Beech 18

This twin-engine aircraft began life as a civil transport in 1937, but when it passed out of production 32 years later (an all-time record) 5 204 out of a total of 7 091 had been military. There were more than 22 military variants, the most numerous class at present being communications and training.



Above: Three view of U-21 Ute family (Queen Air, King Air mix)



Above: Three-view of the T-34C Turbo Mentor

Of the Japanese armed forces more than 400 are used by other countries, as navigation trainers, photographic and mapping aircraft and as utility transport.

Beech Bonanza

Produced as a lightplane since 1947, the Bonanza has been built in large numbers in military versions, all with conventional tail and mostly in the A36 family with 285hp IO-520 engine. The US Air Force bought armed Pave Coin versions, as well as various QU-22 Pave Eagle electronic reconnaissance platforms. Users of the F33 series of trainers include Iran, Mexico and Spain (designation E-24).

Beech Musketeer/Sundowner

Military variants of this popular family, with fixed landing gear, are in service with Canada (designation CT-134) and Mexico.

Beech Baron

This mass-produced twin is the US Army's standard instrument trainer (designation T-42A Cochise), other military customers include Spain and Turkey.

Beech B99 Airliner

This twin-turboprop transport is used by the Chilean AF for search and rescue, and by the Royal Thai Army.

Beech Queen Air

Powered by two Lycoming GO-480, GSO-480 or IGSO-480 flat-sixes, this attractive machine is used by the US Army (six U-8 Seminole versions) and the air forces of Uruguay and Venezuela and the Japan Maritime SDF.

Beech King Air

Powered by two 550 or 620hp PT6A turboprops, the first military King Air versions were the US Army's U-21 Ute (nine sub-types for transport, VIP use by attaches and electronic reconnaissance) and US Air Force U-21G. The VC-6B is a VIP Presidential model, and the T-44A is in quantity production as the US Navy multi-engine pilot trainer replacing the Grumman TS-2A and -2B. The U-21F is the US Army version of the pressurised King Air 100 (680hp PT6 engines).

Beech Super King Air

Much larger, powered by 750hp PT6 engines and distinguished by a T-tail, this fine pressurised transport is widely used by the US forces. Off the shelf purchases were made of the C-12A, with pax/convertible interiors and two-pilot flight decks, to the Air Force and Army, the latter bestowing the name Huron. The RU-21J is a grossly modified electronic platform, most examples of which have grotesque vertical dipole aerials along the wings and behind the tail for use in the Army Celly Lancer programme.



Above: The US Navy plans eventually to buy or convert 278 T-34C Turbo Mentors. They are extremely fully equipped.



Above: Two variants of the big pressurised Super King Air, the C-12A Huron of the Army and the VIP C-12A of the Air Force

Canadair CL-41 Tutor

CL-41A (CT-114 Tutor), CL-41G and CL-41R

Origin: Canadair Ltd., Montreal, Canada

Type: basic trainer, (G) light attack aircraft

Engine: one General Electric J85 single shaft turbojet, (A) J85 CAN-40 of 2 633lb (1195kg) thrust, (G) J85 J4 of 2,950lb (1340kg) thrust

Dimensions: Span 36ft 6in (11.13m), length 32ft 0in (9.75m), height 9ft 9½in (2.84m)

Weights: Empty (A) 4,895lb (2220kg), (G) 5,296lb (2400kg), maximum loaded (A) 7,397lb (3355kg), (G) 11,288lb (5131kg)

Performance: Maximum speed at 28 500ft (8700m) (A) 498mph (801 km/h), (G) 480mph (774km/h), initial climb (A) 4,220ft (1286m)/min, service ceiling (both) 43 000ft (13,100m), maximum range (A) 944 miles (1520m), (G, six drop tanks) 1,340 miles (2157km)

Armament: (G) six wing hardpoints for up to 4 000lb (1815kg) of assorted ordnance

History: First flight 13 January 1960, service delivery (A) 29 October 1963, (G) 1967, final delivery 1968

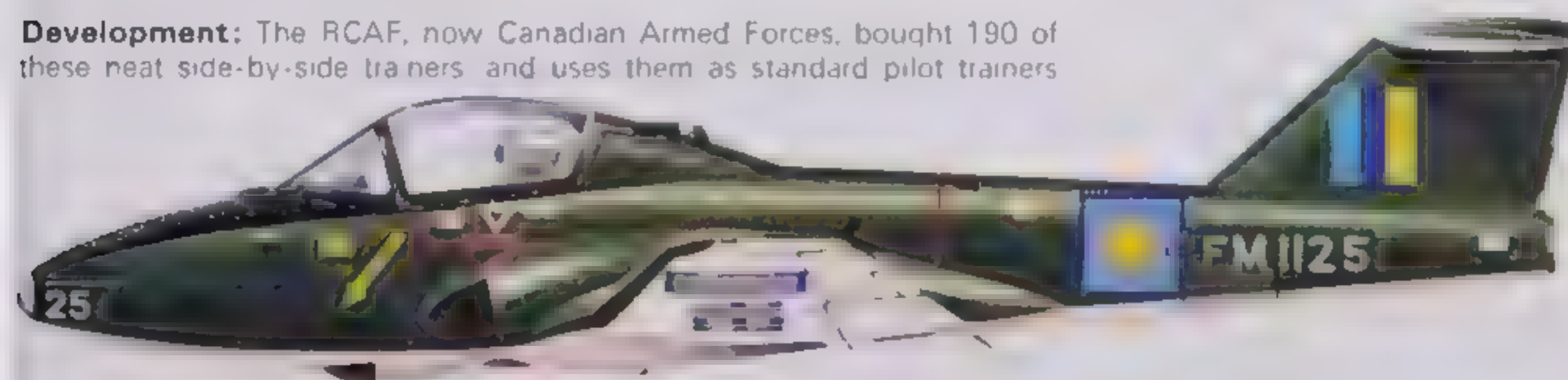
Users: Canada, Malaysia, Netherlands

Development: The RCAF, now Canadian Armed Forces, bought 190 of these neat side-by-side trainers and uses them as standard pilot trainers



Three-view of Canadair CL-41A Tutor.

designated CT-114 Tutor. They equip the Snowbirds aerobatic team. The CL-41R was a one-off systems trainer with Nasarr (CF-104) radar in a long conical nose. Malaysia has two squadrons of the more powerful G model, calling it Tebuan (Wasp). RNethAF pilot training is done on Tutors in Canada.



Above: CL-41G Tebuan of Royal Malaysian Air Force Nos 6 and 9 Sqs.

Below: The 85 CT-114 Tutors of the Canadian Armed Forces are assigned mainly to No 2 Flying School at Moose Jaw, Saskatchewan, where pupils are trained to wings standard. The environment—which looks rather like this for as far as the eye can see—is familiar to thousands of former WW2 aircrew.



CASA C-101

C-101 (to receive EdA designation and name later)

Origin: Construcciones Aeronauticas SA, Getafe, Spain

Type: Basic and advanced trainer, attack and reconnaissance aircraft

Engine: One Garrett-AirResearch TFE 731-3 two-shaft turbofan rated at 3,700lb (1678kg)

Dimensions: Span 34ft 9½in (10.60m), length 40ft 2½in (12.25m), height 14ft 1½in (4.30m)

Weights: Empty (fully equipped) 6 570lb (2980kg), maximum loaded 10 360lb (4700kg)

Performance: (Estimated at 4700kg) maximum speed at 25 000ft (7620m) 494mph (795km/h), limiting Mach number 0.80, initial climb 3 660ft (1116m)/min, service ceiling 43 000ft (13 100m), take off over 50ft (15m) 2,950ft (900m) (landing distance 2 165ft, 660m), maximum range (presumably with external fuel) 1 883 miles (3030km)

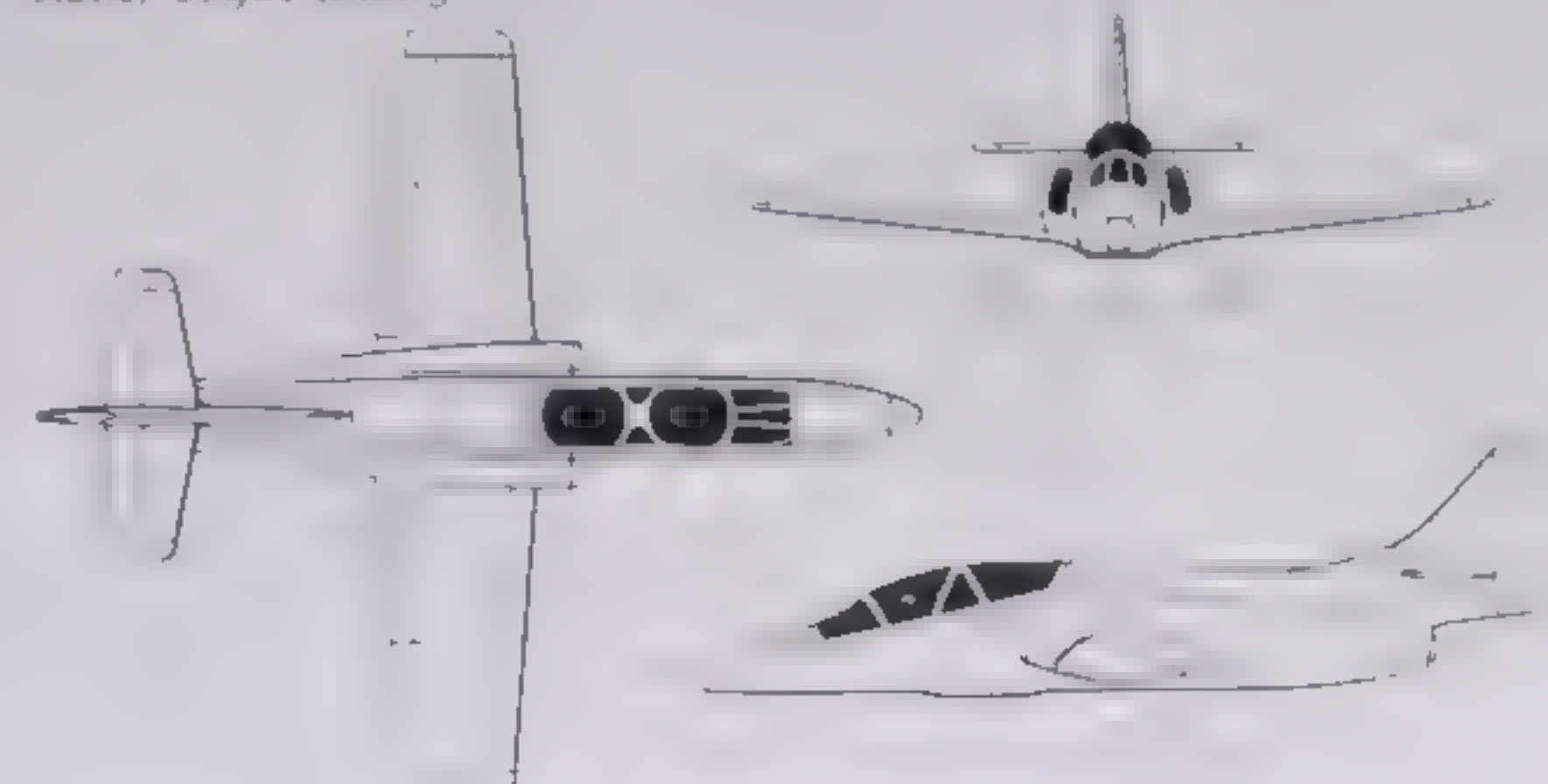
Armament: Seven hardpoints for total external load of 4 740lb (2150kg) including multisensor reconnaissance pods, gunpods or ECM payloads

History: Development contract September 1975, first flight scheduled for 1977

User: Spain

Development: The EdA (Ejército del Aire, Spanish AF) is funding

cost of 1,297 million pesetas the development of this completely new multi-role trainer and tactical aircraft to replace the E-14 and C-10 versions of the Saeta. Unlike other aircraft in this category the C-101 uses an off-the-shelf civil turbofan of high by-pass ratio, which will reduce noise and smoke, and give good fuel economy at all heights, at the cost of reduced speeds and poor high-altitude performance. Northrop and MBB are participating in the programme, mainly through provision of design and development skills and facilities. Six prototypes are being built, two for static/fatigue testing.



Three-view of CASA C-101 (provisional)

Cessna 172

T-41 Mescalero, Reims FR 172 Rocket

Origin: Cessna Aircraft, Wichita, USA, licence production by Reims Aviation, France

Type: basic pilot trainer

Engine: One flat-six piston engine (T-41A) 180hp Continental (T-41B-C-D) 210hp Continental

Dimensions: Span 35ft 10in (10.97m), length 28ft 10in (8.79m), height 8ft 9in (2.71m)

Weights: Empty (A) 1,043kg (2,300lb), maximum loaded (A) 1,043kg (2,300lb) (T-41B)

Performance: Maximum speed (A) 142kts (262km/h), initial climb (A) 645ft (196m)/min, service ceiling (A) 13,100ft (3,995m), range with max fuel at 10,000ft (3,048m) (A) 1,010 miles (1,625km)

History: First flight (civil 172) 1955 (T-41A) August 1964

Users: Argentina, Colombia, Ecuador, Honduras, Peru, Philippine, Arab, Singapore, Thailand, Turkey, US Air Force and Army, Pakistan build under licence

Development: In 1964 the US Air Force decided to abandon its piston engine training and give pilot candidates an initial 30hr on piston engine lightplanes. It chose the best-selling Cessna 172/Cardinal basic trainer, which more than 100 have been delivered. Some 240 T-41A aerobers serve at USAF civilian schools, with civil registration duplicating the USAF serial numbers. The US Army's 255 T-41B Mescaleros fly training and installation support missions, while 12 T-41C cadet trainers serve at the USAF Academy at Colorado Springs. Colombia was first to order the equally powerful T-41D with constant-speed propeller. Pakistan plans to build 60 T-41Ds a year at a new factory near Campbellpur.



Above: That is not a pylon on this T-41A but the reflection of the civil/military serial number in the polished wing.



Above: Three-view of Cessna T-41 (FR 172 similar).

Fuji T1

T1A, T1B and T1C

Origin: Fuji Heavy Industries, Utsunomiya, Japan

Type: Basic and intermediate trainer

Engine: (A) one 4,000lb (1,814kg) thrust Rolls-Royce (Bristol) Orpheus 805 single-shaft turbojet, (B) one 2,645lb (1,200kg) IHI J3-3 single-shaft turbojet, (C) one 3,087lb (1,400kg) J3-7

Dimensions: Span 34ft 5in (10.50m), length 39ft 9in (12.12m), height 11ft (3.35m)

Weights: Empty (A) 6,078lb (2,755kg), (B) 6,261lb (2,840kg), maximum loaded (A) 11,025lb (5,000kg), (B) 9,678lb (4,390kg)

Performance: Maximum speed (A) 575mph (925km/h), (B) 518mph (834km/h), initial climb (A) 6,500ft (1,981m)/min, (B) 4,724ft (1,440m)/min, service ceiling (A) 47,244ft (14,400m), (B) 39,370ft (12,000m), max range (both, clean) about 800 miles (1,290km)

Armament: Provision for one 0.5in Browning machine gun in nose, two wing hardpoints for tanks, Sidewinder air-to-air missiles or other loads to total of 1,500lb (680kg)

History: First flight 19 January 1958, (B) 17 May 1960 service delivery (A) 1960, final delivery 1963

User: Japan

Development: The first jet aircraft designed in Japan after World War II, the T1 was ordered in 1956 as a replacement for the T-6 as standard basic trainer of the Air Self-Defence Force. The tandem cockpit was made similar to that of the Lockheed T-33 but the airframe was based on that of



Above: Three-view of Fuji T1 (all basically similar).

the F-86 Sabre. The home-grown J3 engine was not ready so production began with the imported Orpheus engine, giving considerably higher performance. Styled T1F2 by Fuji and T1A by the JASDF, 40 were delivered in 1960-61. The T1F1, the original planned version with the J3 turbojet, eventually became available and 20 were delivered as the T1B in 1962-3. Only one example was flown of the T1F3 (T1C).

Below: Short Japanese heads barely show in this T1A of the 13th Training Wing, JASDF, at Hyakuri AB.



Hawker Siddeley 125

HS 125 and Dominie

Origin: Hawker Siddeley Aviation, UK
Type: Navigation/pilot/radio trainer, communications and utility
Engines: Two Rolls-Royce Viper single-shaft turbojets, (Series 3 and 400 Dominie) 3,000lb (1,360kg) Viper 301, (Series 600) 3,750lb (1,701kg) Viper 601-22
Dimensions: Span 47ft 0in (14.33m), length (600), 50ft 6in (15.31m) (others) 47ft 5in (14.45m), height (600) 17ft 3in (5.26m), (others) 16ft 6in (5.03m)
Weights: Empty (600) 12,530lb (5,683kg), (others, typical) 10,100lb (4,581kg), maximum loaded (600) 25,000lb (11,340kg) (others, typical) 21,200lb (9,615kg)
Performance: Maximum speed (600) 368mph indicated, equivalent to about 570mph (917km/h) or Mach 0.78, cruising speed (600 at 28,000ft 853km/h) 522mph (840km/h), initial climb (600) 4,900ft (1,493m)/min (others) 4,000ft (1,219m)/min service ceiling (typical) 41,000ft (12,500m) range with max payload of 2,000lb (907kg) and max fuel (600) reserves 1,796 miles (2,891km)
History: First flight (125 prototype) 13 August 1962, first delivery September 1964 first flight (600) 21 January 1971
Users: (Military) Argentina, Australia, Brazil, Ghana, Malaysia, S Africa (name Mercurius) and UK (RAF)

Development: Sold widely as a civil business transport, the HS 125 blends interior space, high performance, good field length and low costs have suited it to many military duties. The Dominie T1 is the navigation trainer derived from the Series 2, with lengthened belly, fitting to accommodate Decca doppler. The RAF uses two communications sub-type: the Dominie CC1 derived from the Series 400 and CC2 based on the more capable 600. Most export military models are early-series communications aircraft. The fan-engined 125-700 has not yet been bought for military duties.



Above: Not a Dominie but a Royal Air Force HS 125 CC.2, a long-body (600 series) communications aircraft of 32 Sqn



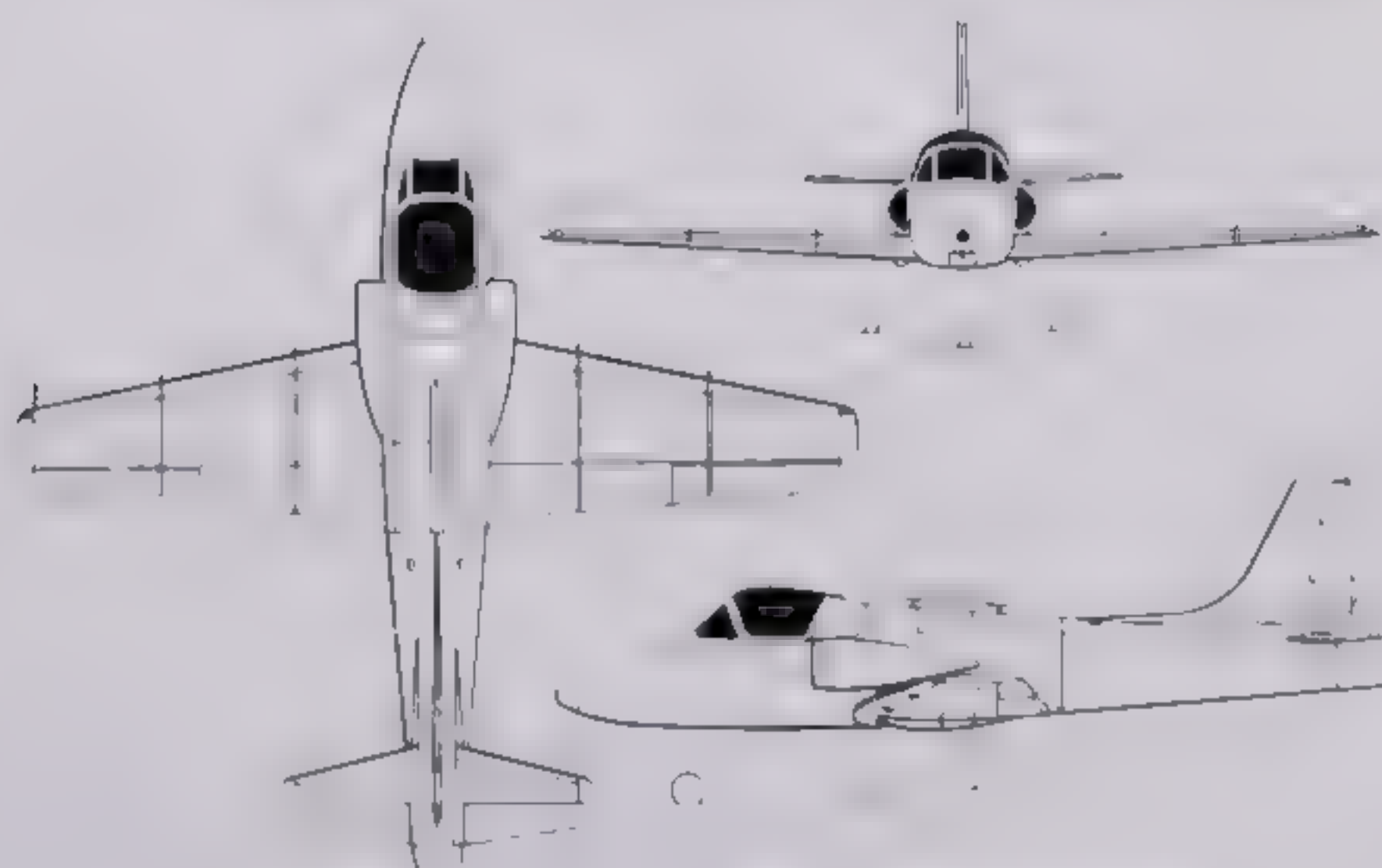
Above: Three-view of Dominie T.1 (125-400 similar).

Hindustan HJT-16 Kiran

HJT-16 Kiran I and II

Origin: Hindustan Aerospace, Bangalore, India
Type: (I) basic trainer, (II) weapon trainer and light attack
Engine: (Mk I) one 2,500lb (1,134kg) thrust Rolls-Royce Viper 11 single shaft turbojet made under licence by HAL, (Mk II) one Rolls-Royce (Bristol) Orpheus 701 single-shaft turbojet made under licence by HAL and de-rated to 3,400lb (1,542kg)
Dimensions: Span 35ft 1 1/2in (10.70m), length 34ft 9in (10.60m), height 11ft 11in (3.63m)
Weights: Empty (I) 5,644lb (2,560kg) (II) 5,909lb (2,680kg), maximum loaded (I) 9,039lb (4,100kg)
Performance: (I) maximum speed at low level (speed at 30,000ft, 9,144m similar) 432mph (695km/h), max cruise 201mph (324km/h), time to reach ceiling of 30,000ft (9,144m) 20min, range at 30,000ft (9,144m) 464 miles (747km)
Armament: (I) none to 68th aircraft, subsequently one hardpoint under each wing for tank, gun pod, bomb or other store to total external load of 1,000lb (454kg); (II) two 7.62mm machine guns in nose, and two hardpoints beneath each wing (inners only equipped for tanks) to total external load of 2,000lb (907kg)
History: First flight 4 September 1964 service delivery March 1968 first flight (II) 6 January 1977
User: Indian AF and Navy

Development: Development of this side-by-side trainer was authorised in 1959, but the programme was delayed by priority accorded the HF-24 fighter. It is a very simple aircraft, but has ejection seats (Martin-Baker H4).



Above: Three-view of Kiran Mk I (Mk II basically same).

a hydraulic system and slight pressurization. The Mk II was planned to be powered by HAL's own HJE-2500 turbojet, but as this is developing slowly the decision was taken in 1975 to fit the Orpheus as already in production for the HAL-built Gnat and Ajeet. The first flight of a Mk I scheduled for October 1976, took place as this book went to press. Meanwhile, production of the Mk I was continuing, with 119 delivered of a total requirement for 180.

Below: Another HJT-16 Kiran Mk I comes off the line at the Bangalore factory. By mid-1977 about 140 had been delivered.



Hawker Siddeley Hawk

P.1182 Hawk T.1

Origin
Type
Engine

Dimensions

Weights

Performance

Armament

History
User

Development



Above: Most of the 175 Hawks for RAF Training Command are finished in this colour scheme, for pilot training.



Above: Three-view of Hawk with centreline gun and rockets.

Below: Several air forces are negotiating for multi-role tactical Hawks with two extra outboard pylons and total external weapon load of 5,600lb (2540kg).



Ilyushin Il-28

Il-28, 28R, 28T and 28U

Origin: Design bureau of Sergei V. Ilyushin, Czechoslovakia (as B 228) and China

Type: Three-seat bomber and ground attack (28R) reconnaissance torpedo carrier, (28U) dual trainer

Engines: (Pre-1950) two 5 005 b (2270kg) thrust Klimov RD 45 (post-1950) two 5 952lb (2700kg) thrust Klimov VK 1 single stage centrifugal turbojet

Dimensions: Span (without tip tanks) 70ft 4½in (21.45m), length 57ft 10½in (17.65m), height 22ft 6in (6.8m)

Weights: Empty 28,417lb (12,890kg), maximum loaded 41,000lb (21,000kg)

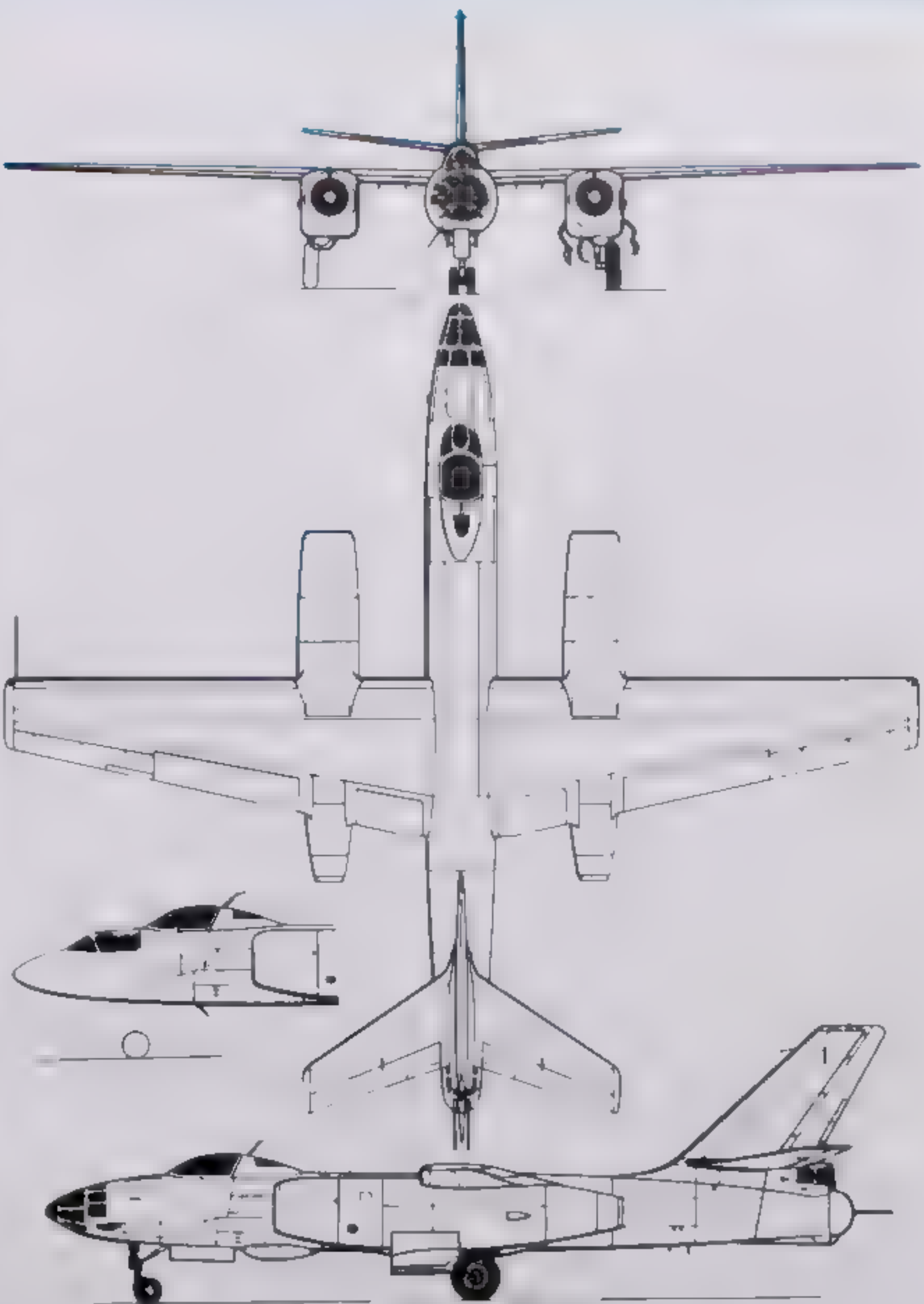
Performance: Maximum speed 559mph (900km/h), initial climb 2,953ft (900m)/min, service ceiling 40,355ft (12,300m), range with bomb load 684 miles (1100km)

Armament: (Il-28 typical) two 23mm NR 23 cannon fixed in nose and two NR 23 in powered tail turret internal bomb capacity of 2 205lb (1000kg) with option of carrying double this load or external load (such as two 400mm light torpedoes)

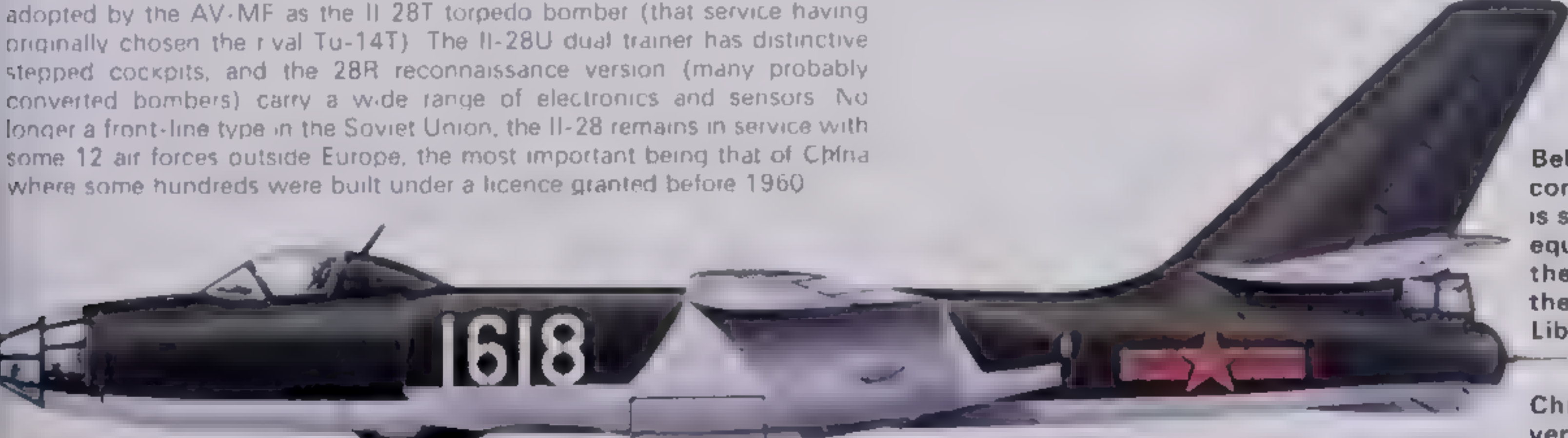
History: First flight (prototype) 8 August 1948, (production Il-28) early 1950 service delivery 1950 final delivery (USSR) about 1960, (China) after 1968

Users: Afghanistan, Algeria, Bulgaria, China, Cuba (not operational), Czechoslovakia (-28U), Egypt, Finland (target tug), E. Germany (U), Indonesia, Iraq, N. Korea, Nigeria, Poland (U), Romania (U), Somalia, Soviet Union, S. Yemen, Syria, Vietnam, Yemen Arab

Development: After World War II the popular media in the West published a succession of indistinct photographs, drawings and other pictures purporting to show Soviet jet aircraft. Apart from the MiG 9 and Yak-15 (and after 1951, MiG 15) all were fictitious and by chance none happened to bear much resemblance to aircraft that actually existed. Thus whereas the 1950 *Jane's* published a drawing and "details" of an Ilyushin four jet bomber it knew nothing of the extremely important Il-28 programme then coming to the production stage. Roughly in the class of the Canberra, the Il-28 prototype flew on two RD-10 (Jumo 004 development) turbojets but the much superior British Nene was quickly substituted and, in VK-1 form, remained standard in the 10 000 or more subsequent examples. Unusual features are the sharply swept tail surfaces, the single-wheel main gears retracting in bulges under the jetpipes, the fixed nose cannon and the rear turret manned by the radio operator. Known to NATO as "Beagle", it equipped all the Warsaw Pact light bomber units in 1955-70 and was also adopted by the AV-MF as the Il-28T torpedo bomber (that service having originally chosen the rival Tu-14T). The Il-28U dual trainer has distinctive stepped cockpits, and the 28R reconnaissance version (many probably converted bombers) carry a wide range of electronics and sensors. No longer a front-line type in the Soviet Union, the Il-28 remains in service with some 12 air forces outside Europe, the most important being that of China where some hundreds were built under a licence granted before 1960.



Above: Three-view of Il-28 with inset showing -28U. Many examples in second-line service with Warsaw Pact air forces now carry additional electronics aeriels.



Above: Il-28T of Navy of Chinese People's Republic.

Below: The Il-28U conversion trainer is still standard equipment with the Air Force of the People's Liberation Army of

China. The bomber version is used by the Navy.



Lockheed T-33

T-33A, T-1A SeaStar, CL-30 Silver Star

Origin Lockheed T-33A, T-1A SeaStar, CL-30 Silver Star
Type T-33A, T-1A SeaStar, CL-30 Silver Star
Engine T-33A, T-1A SeaStar, CL-30 Silver Star

Dimensions: Span (without tip tanks) 38ft 10in (11.85m), length 27ft 10in (8.49m), height 11ft 10in (3.61m)

Weights: Empty 11,905lb (5,400kg), maximum loaded 17,566lb (7,966kg)

Performance Maximum speed 668mph (1,075km/h), initial climb 10,500ft (3,200m)/min, service ceiling 51,000ft (15,545m), maximum range (all) about 15 miles (21.65km)

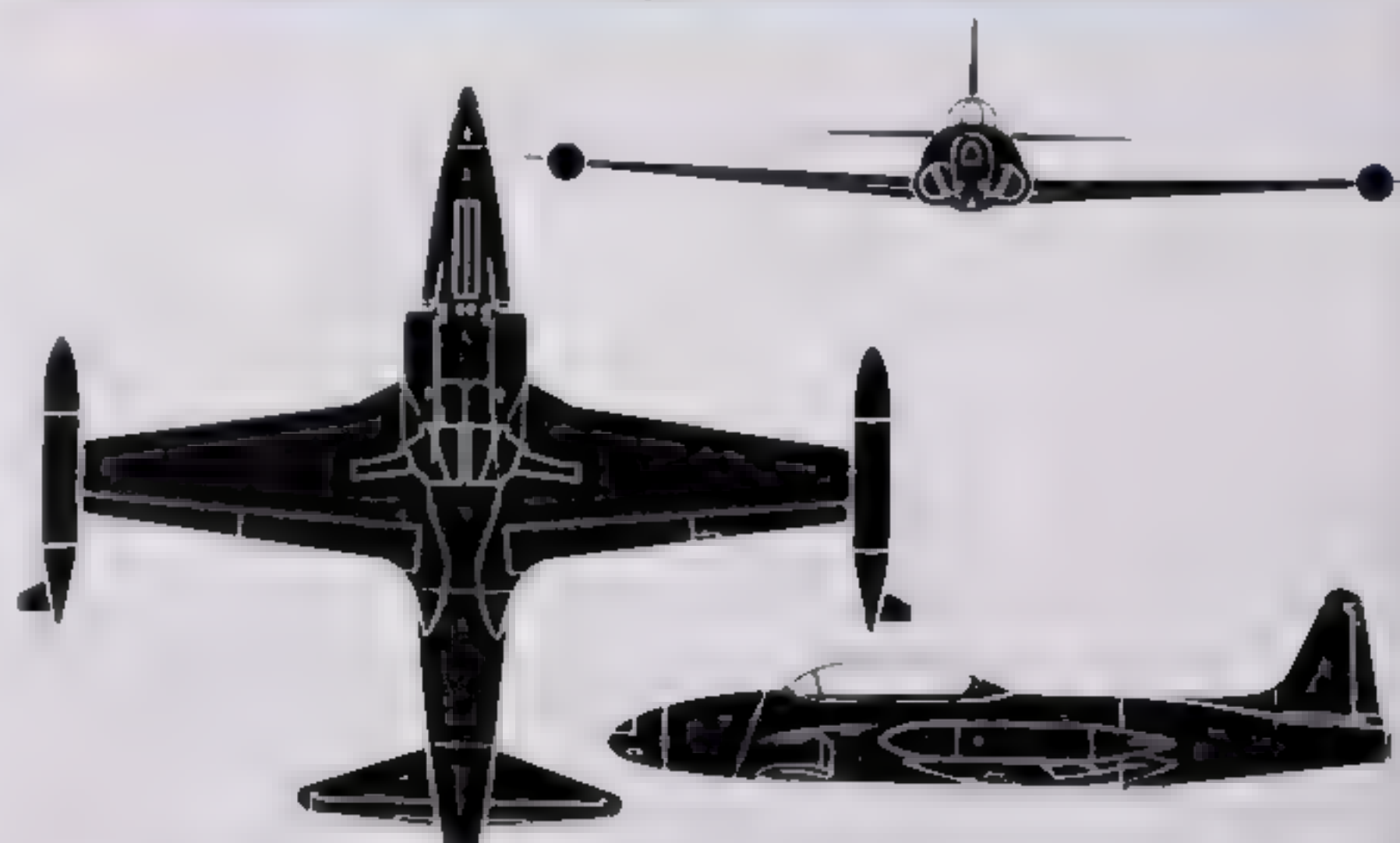
Armament One 37mm cannon under right side of nose and one 23mm cannon under left side of nose

History: First flight (TF-80C) 22 March 1948 last delivery (Japan) 1959
Users Ecuador, Ethiopia, France, Greece, Guatemala, Honduras, Indonesia, Iran, Pakistan, Peru, Philippines, Portugal, Spain, Taiwan, Thailand, Turkey

Development: The world's first jet trainer was produced by adding almost three feet to an F-8U Shooting Star fuselage to accommodate a second seat with a clamshell canopy covering both. Lockheed built 1,000 T-33As for the US Navy and 1,000 T-33As for the US Air Force. The T-33A was the first jet trainer to be produced in large numbers.



Above: Lockheed T-33A of Japan Air Self-Defence Force.



Above: Three-view of Lockheed T-33A (RT has blunt nose)

Mikoyan/Gurevich

MiG-15UTI

MiG-15 and -15bis (Lim-2, S-103),
MiG-15UTI (SBLim-1, CS-102, F-2)

Origin: Design, the Mikoyan/Gurevich bureau, Soviet Union licence production as described in text

Type: (-15) single seat fighter, (-15UTI) dual-control trainer

Engine: (-15) one 5,005 lb (2,270 kg) thrust RD-45F single-shaft centrifugal turbojet, (-15bis and most -15UTI) one 5,952 lb (2,700 kg) VK-1 of same layout (later -15bis) 6,990 lb (wet rating) VK-1A

Dimensions: Span 33ft 0in (10.08m), length (-15, -15bis) 36ft 3in (11.05m), (-15UTI) 32ft 11in (10.04m), height (-15, -15bis) 11ft 10in (3.61m), (-15UTI) 12ft 11in (3.7m)

Weights: Empty (all) close to 8,820 lb (4,000 kg), maximum loaded (-15) 17,566 lb (7,966 kg), (-15bis and variants) 11,270 lb (5,115 kg), (-15UTI) 11,905 lb (5,400 kg) (10,692 lb clean)

Performance: Maximum speed (-15) 668 mph (1,075 km/h), (-15bis) 684 mph (1,100 km/h), (-15UTI) 630 mph (1,015 km/h), initial climb (-15, -15UTI) 10,500 ft (3,200 m)/min, (-15bis) 11,480 ft (3,500 m)/min, service ceiling (-15, -15bis) 51,000 ft (15,545 m), (-15UTI) 47,980 ft (14,625 m), range (at height, with slipper tanks) 885 miles (1,424 km)

Armament: (-15 as first issued) one 37mm cannon under right side of nose and one 23mm cannon under left side of nose (-15, -15bis and variants) one 37mm cannon with 40 rounds under right and two 23mm each with 80 rounds under left, with two underwing hard-points for slipper tanks or stores of up to 1,102 lb (500 kg), (-15UTI) single 23mm with 80 rounds or 12.7mm BK-E with 150 rounds under left side, plus same underwing options

History: First flight 30 December 1947 (MiG-15UTI) 1948 first delivery August 1948 final delivery probably 1953 (USSR) and about 1954 (Poland)

Users: Albania, Algeria, Angola, Bulgaria, China, Cuba, Czech Republic, East Germany, Finland, East Germany, Guinea, Hungary, India, Indonesia, Iran, Iraq, Japan, Korea, Laos, North Korea, North Vietnam, Pakistan, Poland, Portugal, Romania, South Korea, South Vietnam, Taiwan, Thailand, Turkey, USSR, USA, Vietnam, Yugoslavia



Above: Three-view of MiG-15UTI (without tanks)

Development: No combat aircraft in history has had a bigger impact on the air scene than the MiG-15. Its existence was unsuspected in the West until American fighter pilots suddenly found themselves confronted by all-weather silver fighters which could fly faster, climb and dive faster and turn more tightly. Gradually the whole story, and the start of the world pre-eminence of the Mikoyan-Gurevich bureau, could be traced back to the decision of the British government to send to the Soviet Union the latest British turbojet, the Rolls-Royce Nene (long before the Nene was used in any British service aircraft). At one stroke this removed the very serious lack of a suitable engine for the advanced fighter the bureau were planning, and within eight months the prototype MiG-15 had flown and the Nene was frantically being put into production (without a licence) in slightly modified form as the RD-45. The original MiG-15 owed a lot to the Ta-183 and other German designs, but the production machine had a lower tailplane, an anhedral, wing fences, and other changes. Notable features were the extensive use of high-quality welding and the quick-detach package housing the two (later three) heavy cannon. Production rapidly outstripped that of any other aircraft at that time, at least 8,000 being built in the Soviet Union in about five years, plus a further substantial number at Mielec, Poland, as the Lim-2 and at the newly established Vodochody works near Prague as the S-103 (S = stihac, fighter). The two satellite countries also made the UTI trainer under an extension of their original licences, finally producing several thousand trainers by rebuilding MiG-15 fighters phased out of front-line service after 1954. In 1958 the Chinese plant at Shenyang began licence-production of the MiG-15UTI as the F-2. Most MiG-15 fighters were of the more powerful 15bis type with perforated flaps and redesigned rear-fuselage airbrakes; small numbers were made of a night fighter version with simple AI radar and of a ground-attack version with large ordnance carriers inboard of the drop tanks (the latter being originally of the slipper type but after 1952 often being carried below the wing on braced pylons). Known to NATO as "Fagot" (the trainer being "Mongol"), the MiG-15 saw considerable combat in Korea but suffered from the inexperience of its hastily trained Chinese and Korean pilots. As late as 1960 it was still used as a fighter by 15 countries and in 1977 the UTI trainer was still a standard type in the Soviet Union, and in all the countries listed earlier. Very few air forces today use the single-seater.



Above: A MiG-15UTI of the Ilmavoimat (Finnish Air Force) No 31 Sqn gets rolling on hard-packed snow

North American (Rockwell)

T-28 Trojan/Fennec

T-28A Trojan and 16 developments
(see text)

Origin: North American Aviation, Inglewood, USA, see text for subsequent developments

Type: (T-28A) basic trainer, see text

Engine: (T-28A) one 800hp Wright R-1300-1A, see text for later

Dimensions: span 40ft 0in (12.19m) to 40ft 7½in (12.35m), length 32ft 10in (10.00m) to 33ft 8in (10.26m), height 12ft 8in (3.86m) (YAT-28E 14ft 9in T-CH-1 12ft 0in)

Weights: Empty (A) 5,700lb, (D) 6,521lb (2,958kg), (T-CH-1) 5,750lb (2,608kg), maximum loaded (A) 8,000lb (3,629kg), (D) 8,495lb (3,853kg), (E) 12,000lb (5,443kg), (T-CH-1) 9,200lb (4,173kg)

Performance: Maximum speed (A) 283mph, (D) 380mph (611km/h), (T-CH-1) 368mph (592km/h), typical range (D) 500 miles (805km) with full weapon load, 1,000 miles (1,610km) with maximum

Armament: See text

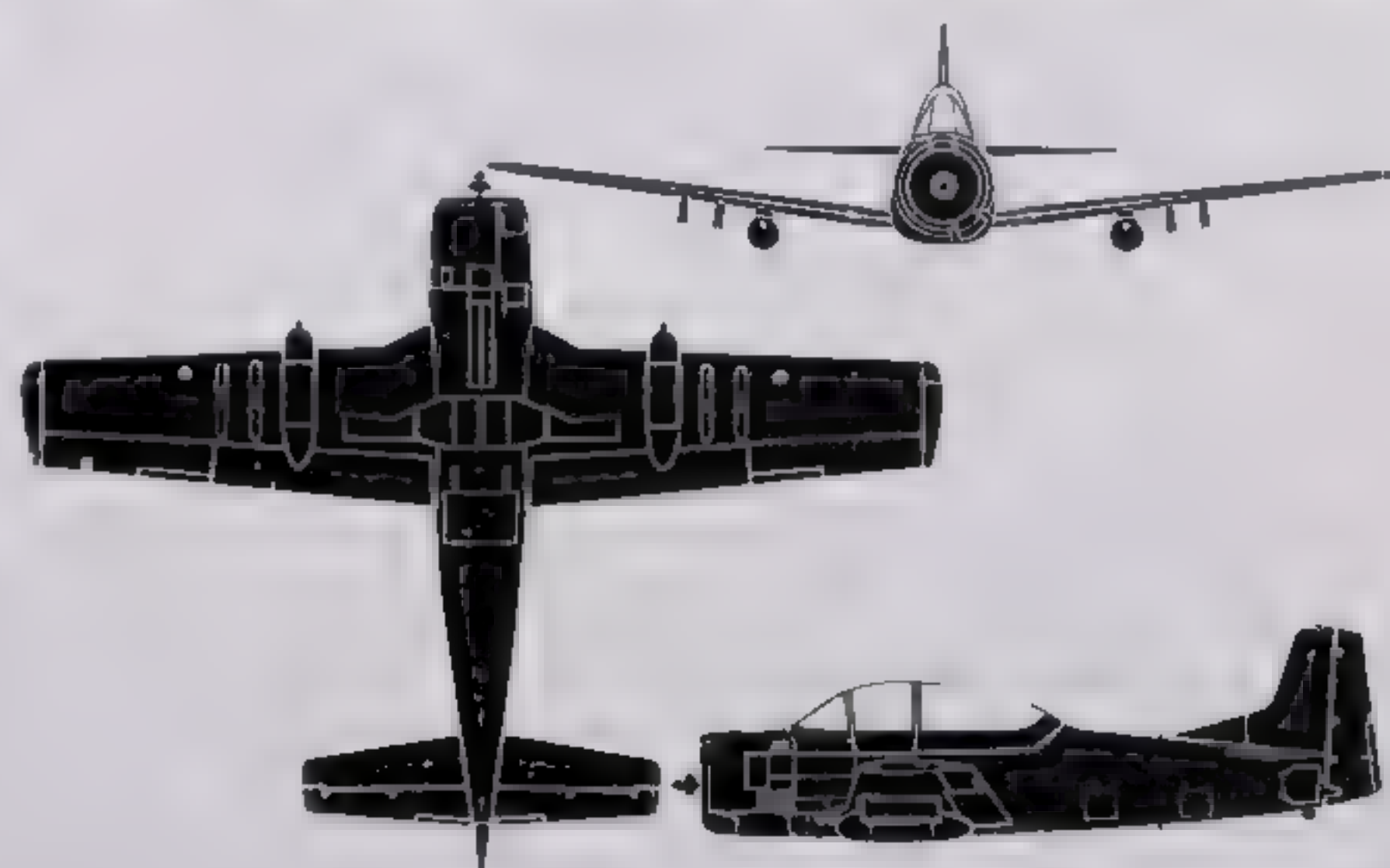
History: First flight (A) 26 September 1949, see text

Users: Argentina, Bolivia, Brazil, Dominica, Ecuador, Ethiopia, Haiti, South Korea, Mexico, Morocco, Nicaragua, Philippines, Taiwan, Thailand, Vietnam, Zaire

Development: The NA-159 won the contest to replace the T-6 as the standard USAF basic trainer, and as the T-28A Trojan was delivered from January 1950. By 1953 1,194 of these tandem nosewheel aircraft had been delivered, and in 1955-7 the Columbus factory delivered 489 T-28Bs to the US Navy with 1,425hp R-1820-86 Cyclone, followed by 299 T-28Cs with arrester hooks. Hamilton made T-28R Nomads, while Pacair built Nomads for desert warfare and reconnaissance, leading to the Sud Aviation Fennec (desert rat) used in Algeria with two twin 0.5in gun pods and two 1,000lb (454kg) bombs. In 1961-69 the original maker received 13 USAF contracts to convert As to T-28D Mutual Aid Program attack configuration, with R-1300 engine but six pylons for a 4,000lb (1,814kg) weapon load. Fairchild Republic converted 72 more. In 1963-68 YAT-28E attack prototypes powered by the 2,450hp Lycoming T55 turboprop carried up to 7,000lb (3,175kg) on 12 pylons. In Taiwan the Aircraft Industry Development Centre used the T-28 as the basis for the T-CH-1 trainer and light attack aircraft with a 1,450hp Lycoming T53-L-701 turboprop, derived from a prototype flown on 23 November 1973.



Above: Many of the B and low-powered A trainers were rebuilt into the T-28D attack version with gun pods and weapon pylons.



Above: Three-view of typical T-28D (Fennec similar).

Northrop T-38 Talon

T-38A

Origin: Northrop Corporation, Hawthorne, USA

Type: Ail-through (basic and advanced) trainer

Engines: Two General Electric J85-5 single-shaft turbojets each rated at 2,680lb (1,216kg) thrust dry and 3,850lb (1,748kg) with maximum afterburner

Dimensions: Span 25ft 3in (7.70m), length 46ft 4½in (14.13m), height 12ft 10½in (3.92m)

Weights: Empty, not stated; maximum loaded 12,093lb (5,485kg)

Performance: Maximum speed at 36,000ft (11,000m) 812mph (1,305 km/h, Mach 1.23), initial climb 30,000ft (9,144m)/min, service ceiling 53,600ft (16,335m), range with max fuel (all internal) 1,093 miles (1,760km)

Armament: No provision

History: First flight 10 April 1959, first operational duty 17 March 1961, final delivery January 1972

Users: W Germany (Luftwaffe, in USA), US (Air Force, Navy and NASA)

Development: Northrop's study for a light fighter, which led to the F-5, also generated a tandem-seat trainer, the N-156T. After two years of company funding six prototypes were ordered in May 1956, and when production finished 1,187 had been delivered. This was an exceptional number of supersonic aircraft of one design for one customer, but today —

despite a good record of safety and reliability — the T-38 is judged to have been a mistake, and no air force has plans to buy supersonic trainers of this kind. Control surfaces are fully powered, but the only external load carried is a travel pod for spares and crew baggage. The Luftwaffe bought 46 retaining US markings, the Navy five and NASA 24 for Astronaut practice.



Above: Three-view of T-38A without centreline pod

Below: Though it was probably a procurement error, the T-38A Talon is a pleasant aircraft with a fine safety record. Here is No 6 from the USAF Thunderbirds display team.



Potez (Aérospatiale) Magister

CM 170, CM 170-2, CM 175

Origin
Type
Engines

Dimensions

Weights: Empty (170-175) about 4,740 lb (2,150 kg)

Performance

Armament

History: First flight 23 July 1952 (production CM 170) 29 February 1951

Users

Development

The Potez Magister is a French two-seater aircraft designed by Potez. It was first flown in 1952 and entered production in 1951. The aircraft is a light aircraft, designed for training and reconnaissance. It has a high-wing configuration and a tailwheel landing gear. The aircraft is powered by a single engine. The Magister has been used by several countries, including France, Algeria, and the United States. It is a simple and reliable aircraft, well-suited for its role as a trainer and reconnaissance aircraft.



Above: Magister built by Flugzeugunion Sud for Luftwaffe

trend in the Six Day War in June 1967 of using this nimble
roles. Subsequently Magisters were popular in local



Above: Three-view of basic CM 170 Magister



Above: Many years ago Potez-Air Fouga supplied 28 Magisters to the Force Aérienne Algérienne (Algerian AF), where they are used in the armed trainer role. This is one of a dozen countries where the trim little French trainer is in fact serving in a tactical combat role, though it would be of value only in a minor dispute or in counter-insurgency operations.

Rockwell International

T-39 Sabreliner

T-39A, B, D and F, CT-39E and G

Origin: Rockwell International Sabreliner Division, El Segundo, USA
Type: Sabreliner

Engines: (A to D and F) two 3,000 lb (1,360 kg) Pratt & Whitney J60-3 engine shaft turbojets (E and G) two 3,300 lb (1,497 kg) JT12A-8 (commercial version, updated). (Sea Sabre 75) two 6,500 lb (2,948 kg) Avco Lycoming ALF 502 geared turbofan

Dimensions: Span 44 ft 5 in (13.53 m), length (A-F) 43 ft 9 in (13.33 m), (G) 48 ft 4 in (14.73 m), (Sea) suggested 56 ft, height 16 ft 0 in (4.88 m)

Weights: Empty, equipped (A-F) 9,257 lb (4,200 kg), (G) 11,250 lb (5,103 kg), maximum loaded (A-D, F) 17,760 lb (8,055 kg), (E-G) 20,172 lb (9,150 kg), (Sea) about 30,000 lb

Performance

Service ceiling or max certified altitude 45,000 ft (13,715 m), max range with typical load and reserves (all) 2,000 miles (3,218 km)

History: First flight (NA-246 prototype) 16 September 1958 (T-39A) 30 June 1961

Users: Argentina (LADE), US (Air Force, Navy)

Development: North American Aviation built the NA-246 as a private venture to meet the Air Force's OX requirement for a 'utility and business trainer' and in 1961-64 delivered 143 T-39As now classified as T-39A



Above: One of the 143 original T-39A Sabreliners of the US Air Force. Some have now been converted in various ways



Above: Three-view of T-39A (most variants have more windows)

Rockwell International

T-2 Buckeye

T-2A (T2J-1), B, C, D and E

Origin: Rockwell International
Type: Basic trainer and light attack
Engine: (A) one 3400hp (2500kW) J44-40-100 turbojet
(B) two 3400hp (2500kW) J44-40-100 turbojets
(C) two 3400hp (2500kW) J44-40-100 turbojets
(D) two 3400hp (2500kW) J44-40-100 turbojets
(E) two 3400hp (2500kW) J44-40-100 turbojets

Dimensions: Span (B) 35ft 10in (10.97m), length (B) 28ft 10in (8.79m), height (B) 10ft 10in (3.30m)

Weights: Empty (C) 8,115lb (3680kg), maximum loaded (C) 11,000lb (4989kg)

Performance: Maximum speed (B) 540mph (869km/h), initial climb (C) 6,200ft (1890m)/min, service ceiling (C) 40,400ft (12,315m), max range (C) 1,100mi (1,770km)

Armament: (See text for E) total external load up to 6,400lb (2900kg) carried on two underwing hardpoints, with provision for gun or rocket

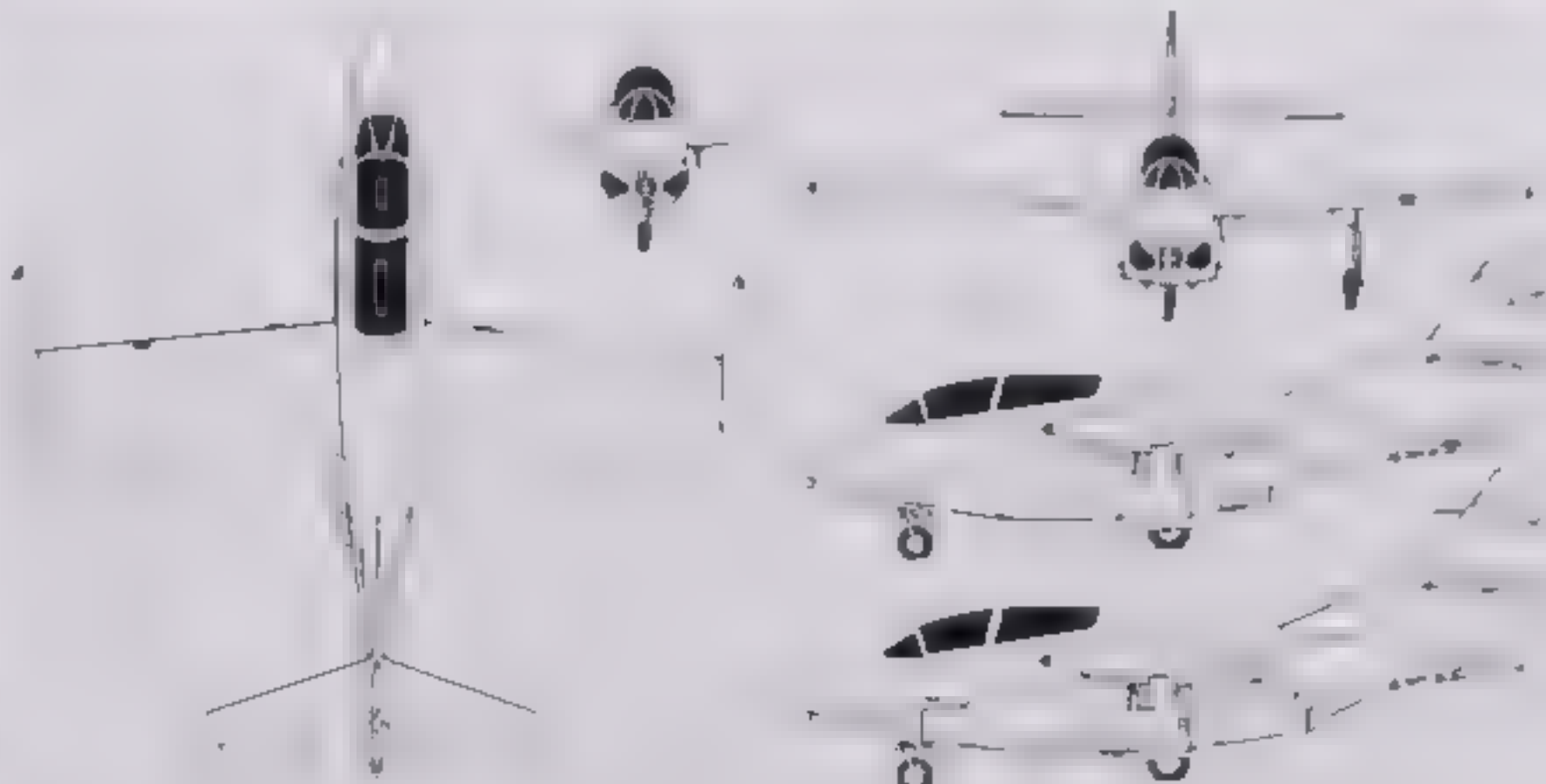
History: First flight 31 January 1958 (B) 30 August 1962
1968

Users: Greece, Morocco, US Navy, Venezuela

Development: North American Aviation's Columbus D-108 competed for the standard US Navy jet trainer, and drew orders for 1,100 Buckeyes, later redesignated T-2A. The T-2B was ordered in 1965-69, almost doubled the installed thrust of the T-2A. The Training Command model T-2C was ordered in 1968-75. Venezuela uses the D without carrier protection, the second batch of 12 having the attack kit of the E. The T-2E was built for the Royal Hellenic AF in 1976-7, has six stores, a total external load of 3,500lb (1,587kg), and tanks protected against small-arms fire.



Above: Rockwell International T-2D Buckeye of Venezuelan AF



Above: Three-view of T-2C (upper side and scrap front views depict single-engined T-2A).

Below: This pair from VT-4 represent 231 US Navy T-2C Buckeyes, which are the standard carrier-indoctrination trainers



Saab 105

SK 60A, B and C, Saab 105O and 105G

Origin: Saab-Scania, I
Type
Engines: (SK 60) two 164

Dimensions: Span 31ft 2 in (9.50m) length (SK 60)

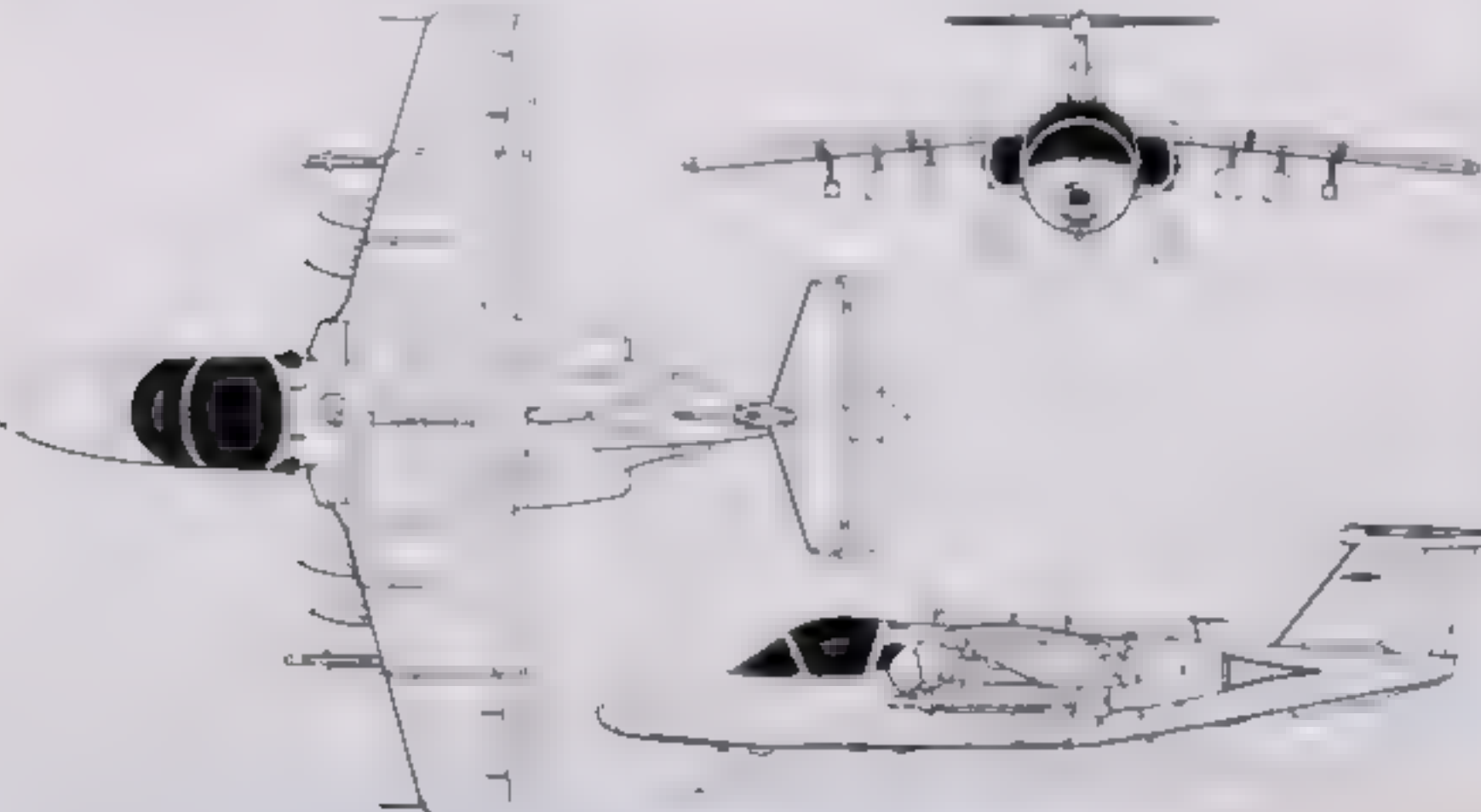
Weights: Empty (SK 60) 5,534lb (2,510kg). (G)

Performance: Maximum speed (clean, low level) (SK 60)

Armament

History: First flight 29 June 1963. (production SK 60A) 27 August 1964

Users: Austria, Sweden



Above: Three-view of basic SK 60A, with Sidewinders.



Above: It has long been Flygvapen policy to disperse combat units to straight public highways. These four SK 60Cs from F20 at Uppsala have recce noses and full weapons capability

Development: Developed as a private venture, a rare thing for Saab, this great side-by-side trainer and multi-role aircraft has hydraulically boosted elevators and ailerons, ejection seats replaceable by four fixed seats for liaison duties, and a wide range of avionics and weapon options. The 150 SK 60s of the Royal Swedish AF comprise A (basic trainers), B (weapon training and strike) and C (reconnaissance) sub types. Austria uses the 105. The G has airframe modifications, heavier and more varied weapon loads and a precision nav/attack system.

Below: Another multi-role Saab 105 wing is F21 at Lulea. Many of their aircraft are SK 60C models, but this formation is of the basic series without reconnaissance cameras in the nose.



Saab Supporter

Supporter (formerly MFI-17)

Origin: Saab-Scania, Linköping, Sweden

Type: Light trainer and multi-role tactical aircraft

Engine: One 200hp Avco Lycoming IO-360-A1B6 flat-six

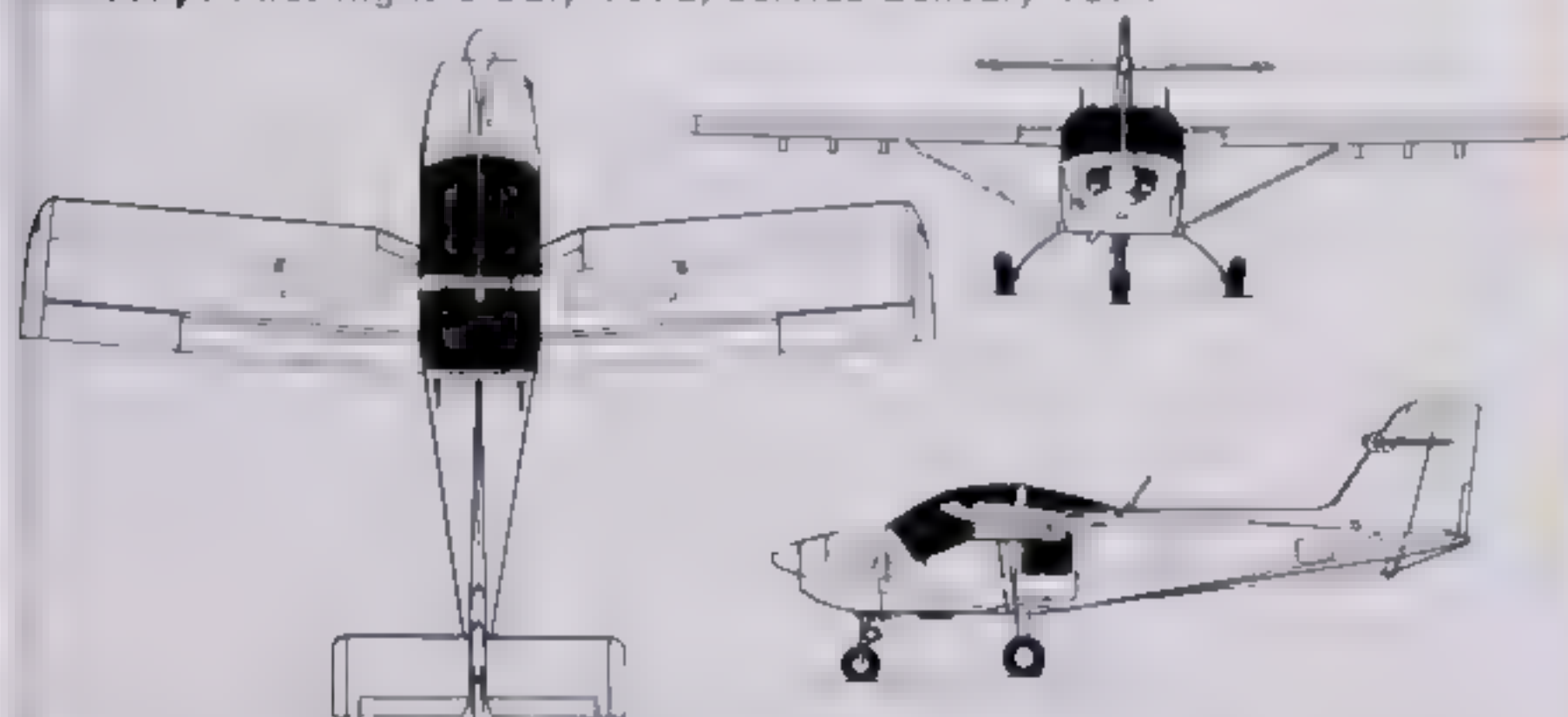
Dimensions: Span 29ft 0½in (8.85m), length 22ft 11½in (7.00m), height 8ft 6½in (2.60m)

Weights: Empty, equipped 1,424lb (646kg) maximum loaded 2,645lb (1,200kg)

Performance: (at 2,480lb) maximum speed 146mph (236km/h), initial climb 807ft (246m); service ceiling 13,450ft (4,100m), typical take-off or landing over 50ft (15m) 1,280ft (390m), maximum range with reserves 672 miles (1,080km)

Armament: Six underwing hardpoints, inners stressed to 330lb (150kg) each, and remainder to 220lb (100kg) for bombs, gun packs, guided missiles, supply packs or other stores

History: First flight 6 July 1972, service delivery 1974



Above: Three-view of Saab Supporter.

Users: Denmark Pakistan

Development: Developed from the civil Safari (formerly MFI 15) which in turn was an enlarged version of the 1958 MFI-9B Mili-Trainer/I made by Malmö Flygindustri, the Supporter is possibly the cheapest military aircraft in the world today. Yet it is extremely capable and versatile and can serve for training, supply dropping, FAC, reconnaissance, attack, liaison, anti-tank or target towing. Pakistan has 45 and the Royal Danish AF will have 32, designated T-17, when this book is published.



Above: One of the 32 T-17 Supporters delivered to the Flyvevæbnet (Royal Danish Air Force) to replace Chipmunks as trainers and KZ.VIIs and Super Cubs in the observation role.

Scottish Aviation SA-3 Bulldog

Bulldog Series 100 and Series 120

Origin: Scottish Aviation, Prestwick, UK

Type: Primary trainer

Engine: One 200hp Avco Lycoming IO-360-A1B6 flat-six

Dimensions: Span 33ft 0in (10.06m); length 23ft 3in (7.09m); height 7ft 5½in (2.28m)

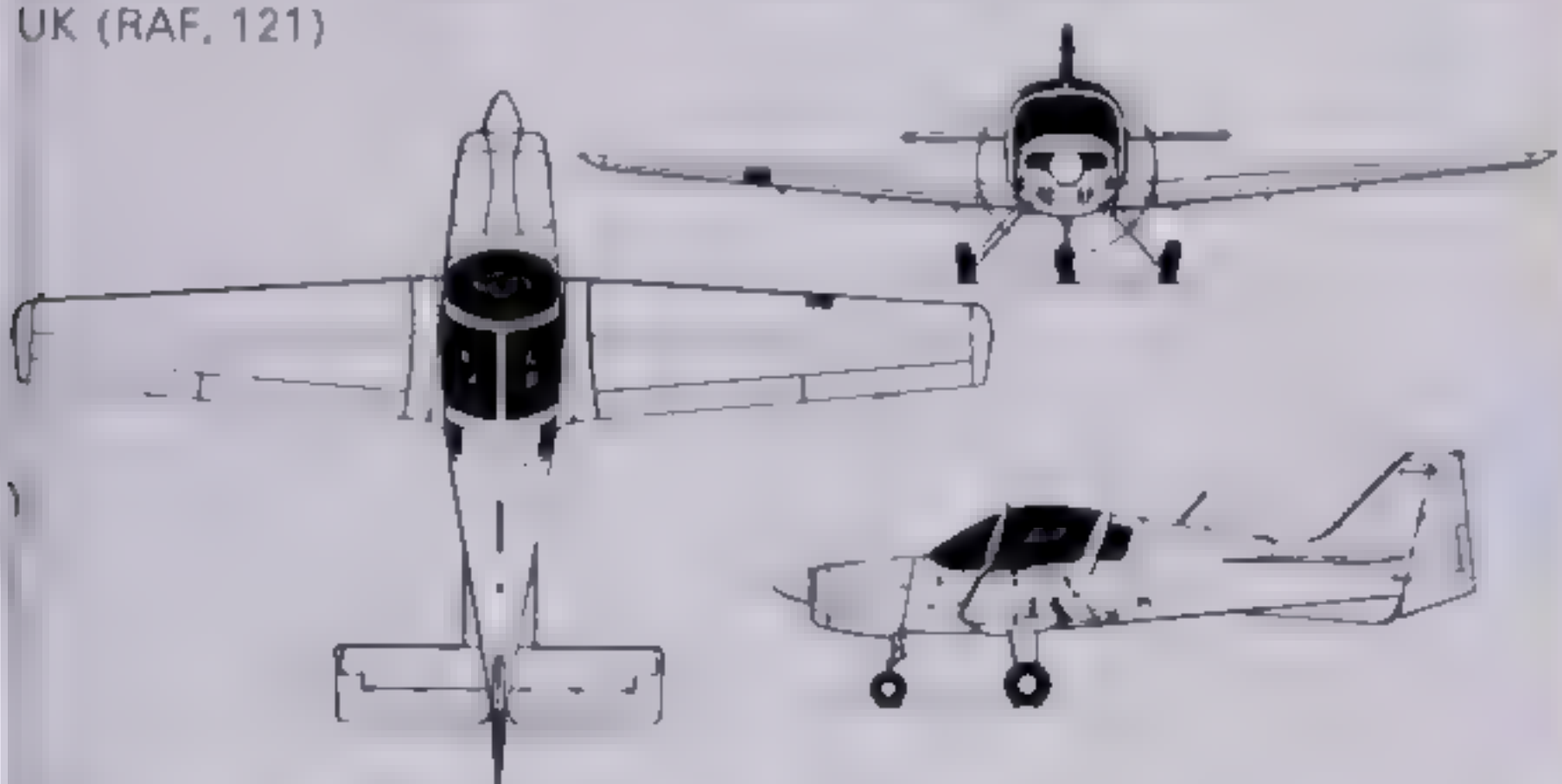
Weights: Empty 1,475lb (669kg), maximum loaded 2,350lb (1,066kg)

Performance: Maximum speed 150mph (241km/h), initial climb 1,034ft (315m)/min; service ceiling 16,000ft (4,875m); take-off or landing over 50ft (15m) within 1,400ft (425m); max range without reserves 621 miles (1,000km)

Armament: Optional four underwing hardpoints for total external load of 640lb (290kg)

History: First flight 19 May 1969, (production Series 100) 22 July 1971, (Series 120) 30 January 1973

Users: Ghana (Series 122), Jordan (125), Kenya (103), Lebanon (126), Malaysia (102), Nigeria (123), Sweden (101, designation SK 61) and UK (RAF, 121)



Above: Three-view of Scottish Aviation Bulldog

Development: Produced by defunct Beagle Aircraft in 1968 as a military trainer version of the Pup, the Bulldog is widely considered to be the best light piston-engined trainer in the world today — a view reinforced by its early selection by Sweden instead of a competing Swedish machine. A third seat or 120lb (54kg) baggage can be accommodated, as can a glider towing hook and a wide range of avionics and special fittings. The first 98 were Series 100, and since 1973 a further 183 have been ordered of the 120 series restressed for semi-aerobatic manoeuvres at maximum weight. In August 1976 the first flight took place of the lengthened and cleaned-up Series 200 (civil name Bullfinch) with retractable landing gear and optional fourth seat.



Above: This pretty picture was taken near Little Rissington, Gloucestershire, soon after the first Bulldog T.1 trainers had been delivered to the Central Flying School. CFS has now closed and most RAF Bulldogs are at 2 FTS, Leeming, Yorks. Many others are serving with the University Air Squadrons throughout the United Kingdom. British Bulldogs are unarmed, but the aircraft has been cleared for SNEB 68mm rockets and dropping of free-fall stores, while Swedish Bulldogs (left) are operational with Bofors Bantam wire-guided missiles.



Above: SK 61 Bulldog of F5 at Ljungbyhed, Sweden

SIAI/Marchetti SF.260

SF.260M, W and SW

Model	Origin	Type	Engine	Dimensions
1	USA	1	1	1
2	USA	1	1	1
3	USA	1	1	1
4	USA	1	1	1
5	USA	1	1	1
6	USA	1	1	1
7	USA	1	1	1
8	USA	1	1	1
9	USA	1	1	1
10	USA	1	1	1
11	USA	1	1	1
12	USA	1	1	1
13	USA	1	1	1
14	USA	1	1	1
15	USA	1	1	1
16	USA	1	1	1
17	USA	1	1	1
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94	USA	1	1	1
9				

Weights: Empty (M, equipped) 1 664

Performance

Performance

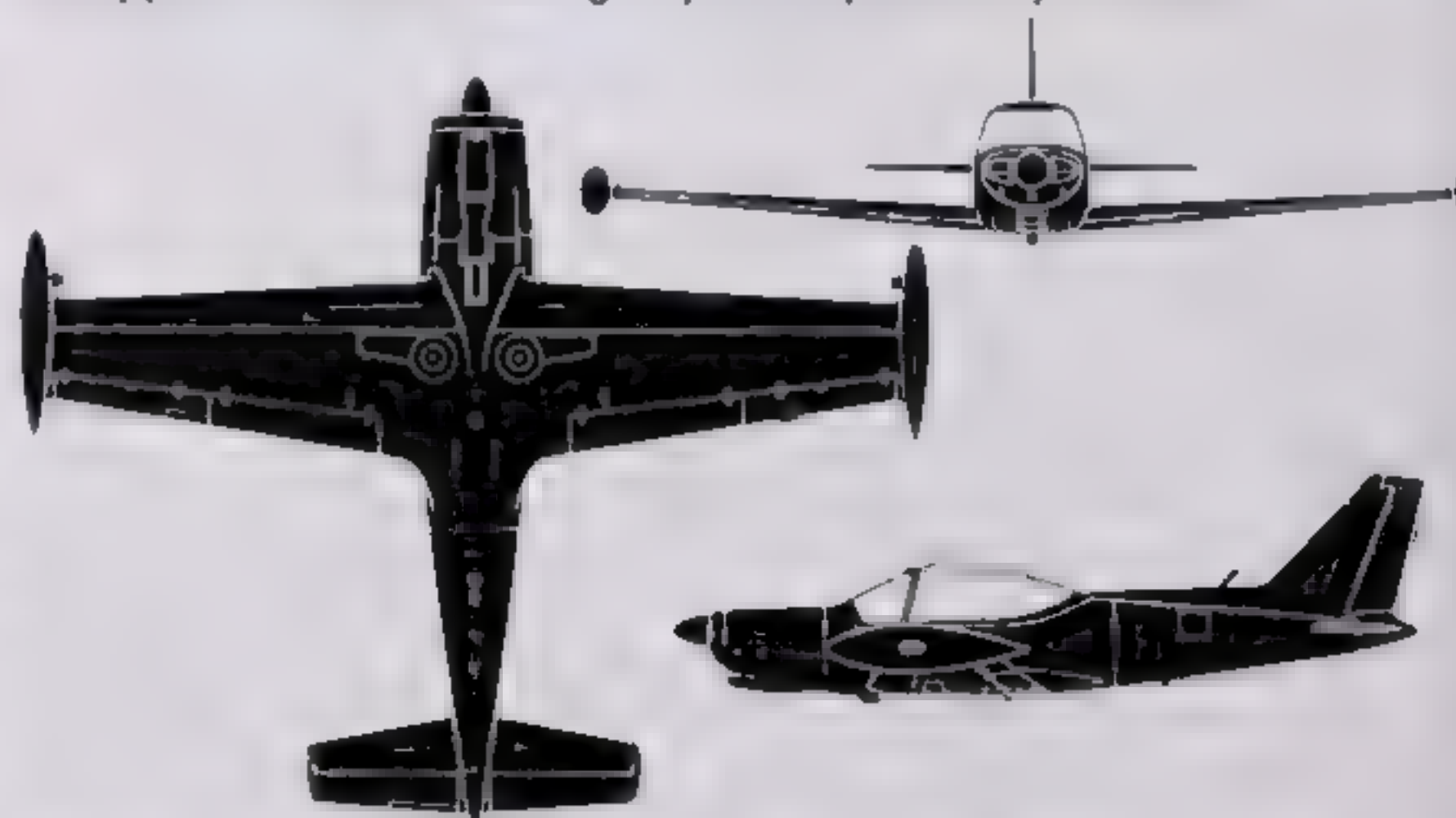
Armament: (W) one underwing hardpoint on each side rated at 330lb

History: First flight (MX) 10 October 1970. (W) May 19

Users: Belgium (M), Burma (M), Dubai (W), Ireland

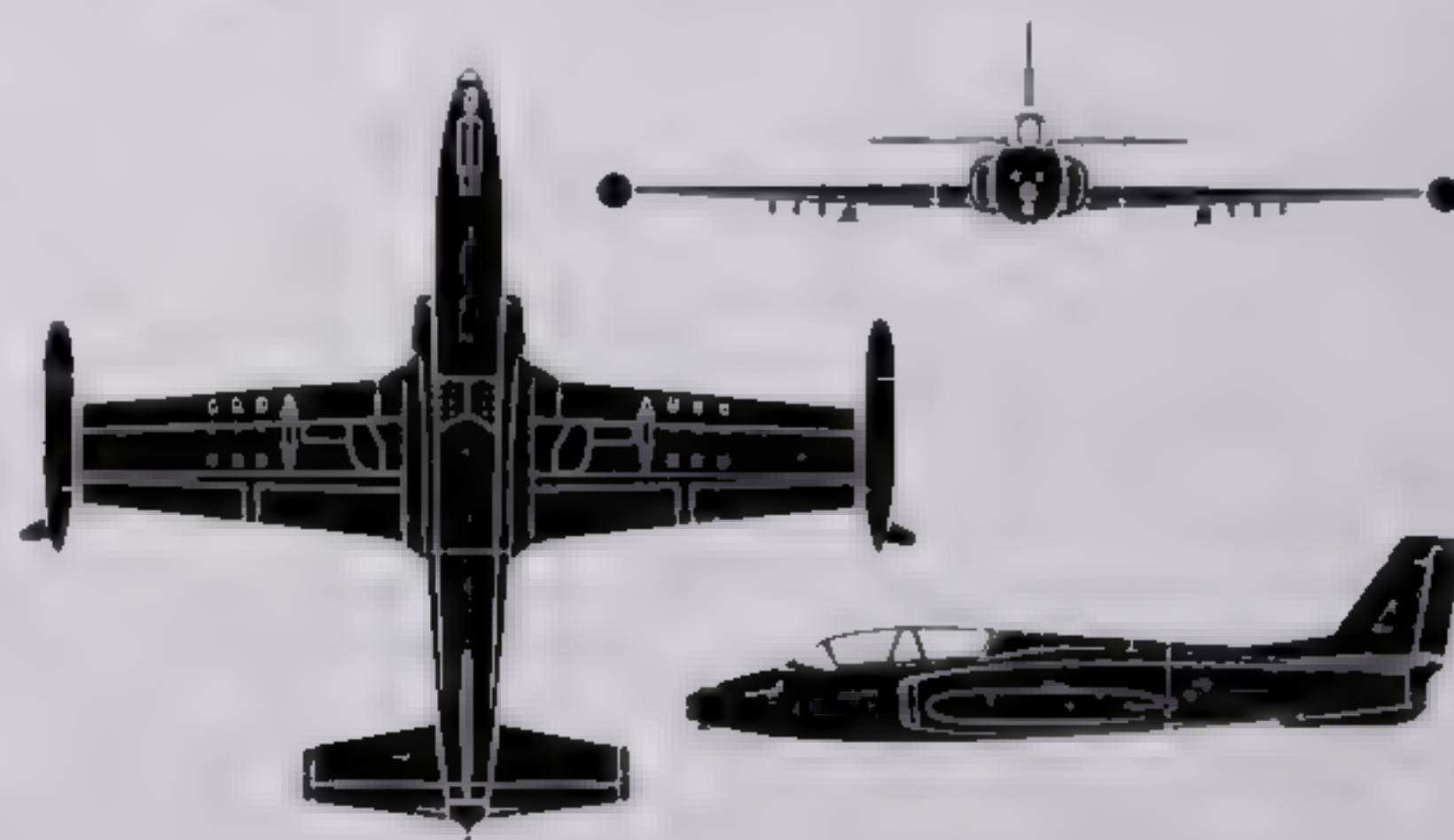
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Above: SIAI-Marchetti SF.260WP light attack aircraft of the Philippine Air Force with gun pods replaced by rockets



Above: Three-view of SIAI-Marchetti 260M without pylons.

... and an air conditioning system is an option. The Jastreb (Hawk) uses a similar airframe, with local strengthening for the more powerful engine and heavier external stores. Again Zambia has received an export version but without the optional cabin pressurization and self-contained engine-start system. Jastrebs can carry cameras in the fuselage and in the nose of the tip tanks, and also tow an aerial target.



Above: Three-view of Soko G2-A Galeb (TJ-1 similar).

Below: The Yugoslav Air Force has about 60 G2-A Galebs, and is receiving a dual trainer/attack model designated TJ-1. About 150 single-seat Jastrebs include the RJ-1 reconnaissance version.

Soko Galeb and Jastreb

G2-A, Galeb, J-1 Jastreb

Origin: "Soko" Metaloprerađivačka Industrija, Yugoslavia

Type: Gaeb) dual armed trainer Jastreb single seat attack

Engine: One Rolls-Royce Turbomeca single-shaft turbojet. (G) 2 500lb (1134kg) thrust Mk 22-6 (J) 3 000lb (1360kg) Mk 531

Dimensions: Span (excluding tie links) 34ft 4½ in (10.47m), (J) 34ft 8 in length 33ft 11 in (10.34m), (J) 35ft 1½ in, height 10ft 9 in (3.28m), (J) 11ft 11½ in

Weights: Empty 5,776 lb (2620kg), (J) 6 217lb maximum loaded 9 210lb (4178kg), (G, clean, fully aerobic) 7 438lb (J) 10 287lb

Performance: Maximum speed 505mph (812km/h), (J) 510mph, initial climb 4 500ft (1370m)/min, service ceiling 39,375ft (12 000m), range (h) max fuel 770 miles (1240km) (J) 945 miles

Armament: (G) 12 7mm guns in nose, each with 80 rounds, underwing pylons for two 220lb (100kg) bombs, or light loads of rockets (J) three 12 7mm in nose, each with 135 rounds, eight underwing hardpoints, two furthest inboard carrying stores of 551lb (250kg), the rest single 127mm rocket.

History: First flight (G) May 1961, service delivery (G) 1965

Users: Libya, Yugoslavia, Zambia

Development: The first Yugoslav jet to go into production, the tandem-seat Galeb (Seagull) has been fully developed and built in modest numbers for the Yugoslav Air Force and Zambia. Pupil and instructor sit in Folland



WSK-Mielec TS-11 Iskra

TS-11 Iskra (Spark) Series 100, 200 and single-seater.

Origin: WSK-PZL-Mielec, Mielec, Poland
Type: Basic trainer and light attack aircraft
Engine: One 2 205lb (1 000 kg) thrust SO-1 or SO-3 turbojet
Dimensions: Span 35ft (10.7m), length 30ft 7in (11.15m), height 11ft 5½ in (3.50m)
Weights: Empty 5 011 lb (2 272 kg), maximum 11 840 lb (5 369 kg) (3840kg)



Above: Three-view of WSK-Mielec TS-11 Iskra 200.



Above: TS-11 in original form but with nose cannon.

Performance: maximum speed (trainer at 3800kg at 16 400ft 5000m 44 mph (72 km/h), (single-seater at 2 020 kg at 16 400ft 36 100ft (11 000m), take-off or landing over 50ft (15m) with n 3 900ft (1 190m) range with max fuel 776 miles (125 km)
Armament: (Series 100, 200 and single-seater) 23mm gun in starboard side of nose, with gun camera, plus four underwing hardpoints for gun pods, rockets or bombs up to 220lb (100kg)
History: First flight 5 February 1960, service delivery March 1961
Users: India, Poland
Development: Rejected in favour of the Czech L 29 as standard trainer for the Warsaw Pact, the sleek TS-11 was nevertheless put into production as standard trainer of its home air force. Most are tandem seat dual trainer in natural metal finish, but limited numbers have been made for the Polish AF of a camouflaged single-seat attack version. The Iskra 100 carries the same armament provisions, and in 1975 production switched to the 200 with long life SO-3 engine and other changes. India bought 50 in 1975, and by 1977 about 400 Iskras had been built.

Yakovlev Yak-18

Yak-18, 18U, 18A, 18P, 18PM and 18T

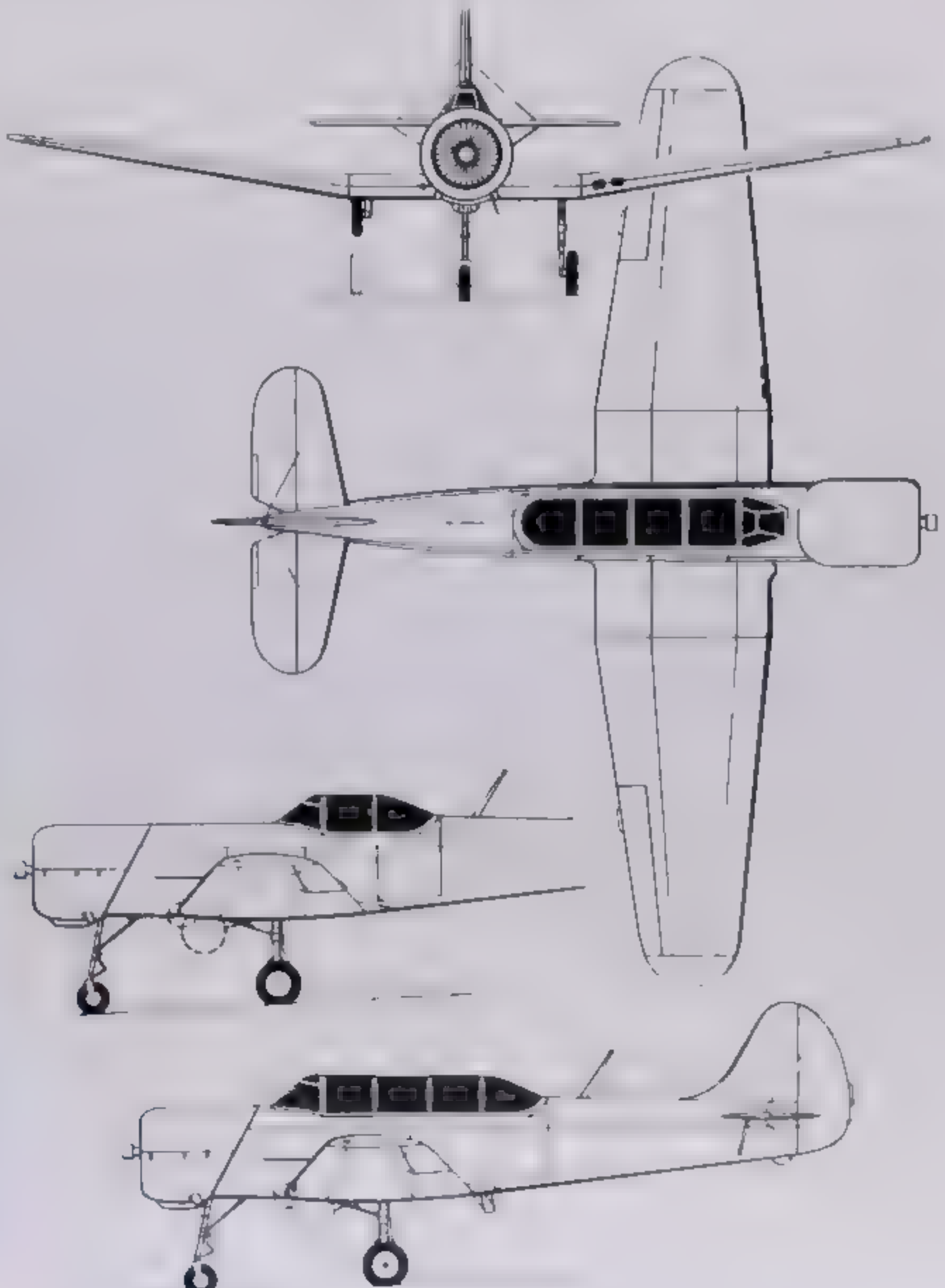
Origin: The design bureau of Alexander Yakovlev, Soviet Union
Type: Primary trainer, see text for other roles
Engine: (18, 18U) one 160hp Shvetsov M-11FR five-cylinder radial (18A, P, PM) one 300hp Ivchenko AI-14RF nine-cylinder radial, (T) one 360hp Vedenev M-14P nine-cylinder radial
Dimensions: Span (18) 33ft 9½ in, (18U, A and P) 34ft 9½ in (10.60m) (T) 36ft 7½ in (11.16m), length (18, 18U) 26ft 5½ in (8.07m), (18A, P and T) 27ft 4½ in (8.35m), height (18, PM) about 7ft (2.13m), (U, A, P, T) 11ft 0in (3.35m)
Weight: Empty (18) about 1 800lb, (U, A, P) typically 2,259lb (1025kg) (T) 2,450lb, maximum loaded (18) 2,469lb (1120kg), (U, A, P) 2,910lb (1320kg), (T) 3 637lb (1650kg)
Performance: Maximum speed (18) 133mph (215km/h), (U) 143mph (230km/h), (U, A, P, T) 186mph (300km/h), range with max fuel (A) 435 miles (700km), (T) 560 miles (900km)
History: First flight 1946, (U) 1955, (A) 1956, (P) probably 1959, (PM) probably 1965, (T) 1967
Users: China, E Germany, N Korea, Mongolia, Romania, S Yemen, Soviet Union, Syria, probably others

Development: Most of the primary training of the Warsaw Pact air forces (apart from Poland, which uses the TS-8 Bies) is done in civil or paramilitary schools, and it is to these that the bulk of some 8 000 of these

popular and robust aircraft have been supplied. The original Yak-18 had an M-11 with helmeted cowling and tailwheel type landing gear with the main legs retracted to the rear. The 18U introduced a long nose wheel with forward retracting main legs, while the much more powerful A was cleaned up and had a variable-pitch propeller. The P and PM were refined single-seat aerobatic versions, the PS being a tailwheel derivative leading to today's Yak 50. The T has a wider centre section and a new stressed-skin fuselage with four-seat cabin, suitable for training, ambulance, patrol and similar duties, with wheel, ski or float landing gear.



Above: Standard basic trainer of the Air Force of the Chinese People's Liberation Army, this version of Yak-18 has a tail unlike that of standard Soviet sub-types



Above: Three-view of Yak-18A with extra side view of the Yak-18P single-seater. Today there are many models with nosewheel or tailwheel and one to four seats

Aeritalia AM.3C

AM.3C, Bosbok

Origin: Italy
Type: Light transport
Engine: One 340hp Piaggio (Lycoming licence) GSO-480 B1B6 flat four
Dimensions: Span 41ft 5½in (12.64m), length 29ft 3½in (8.93m), height 11ft 10in (3.61m)
Weights: Empty 2,380lb (1,080kg), maximum loaded 3,750lb (1,700kg)
Performance: Maximum speed (clean at 8,000ft, 2,440m) 173kph (107mph), 8km/h, service ceiling 27,550ft (8,400m), takeoff or landing over 50ft (15.24m), climb rate 1,500ft/min (457m/min), range 1,100mi (1,770km)
Armament: Two underwing hardpoints each rated at 375lb (170kg), for a wide range of ordnance and other loads including Matra twin-gun pods and Miniguns, retarded or cluster bombs
History: First flight 12 May 1970, first delivery 1973
Users: Italy, Rwanda, Africa

Development: The first modern Italian utility aircraft was the stressed LASA 60 transport designed by Lockheed-Georgia in 1959 and produced by Aermacchi as the AL 60 and by LASA in Mexico. The LASA-60 serves with the Mexican AF while the more powerful (400hp) tailwheel AL 60 made in Italy serves in Rhodesia (named Trojan) and the Central African Republic. The high-lift wing of this aircraft was used in the small-bodied AM 3C, built to meet an Italian AF requirement and to replace the O 19 in the Italian Army. The largest customer was the S African AF, whose 50 Bosboks equip 41 Sqn at Johannesburg and 42 at Potchefstroom. From the Bosbok Atlas Aircraft has developed the eight-seat C4M Kudu, being evaluated by the S African AF.



Above: Three-view of Aeritalia AM.3C.



Above: Bosbok of the SAAF (without underwing stores).

Below: A South African Bosbok on test in Italy before delivery. These powerful aircraft have proved extremely versatile, and are used for FAC, observation, casualty evacuation, supply dropping, tactical transport and armed close support. The high-capacity Kudu is now reported to be in production.



Antonov (WSK) An-2

An-2 (many variants) and WSK-Mielec An-2 (NATO name "Colt")

Origin: The bureau of Oleg K. Antonov, Soviet Union, today made only by WSK-Mielec, Poland, and State Industry of China

Type: STOL transport

Engine: One 1,000hp Shvetsov ASh-62IR nine-cylinder radial, (since 1960) one 1,000hp WSK-Kalisz ASz-62IR

Dimensions: Span 59ft 8½ in (18 18m) (lower wing 46ft 8½ in, 14 24m) length 41ft 9½ in (12 74m), height 13ft 1½ in (4 00m)

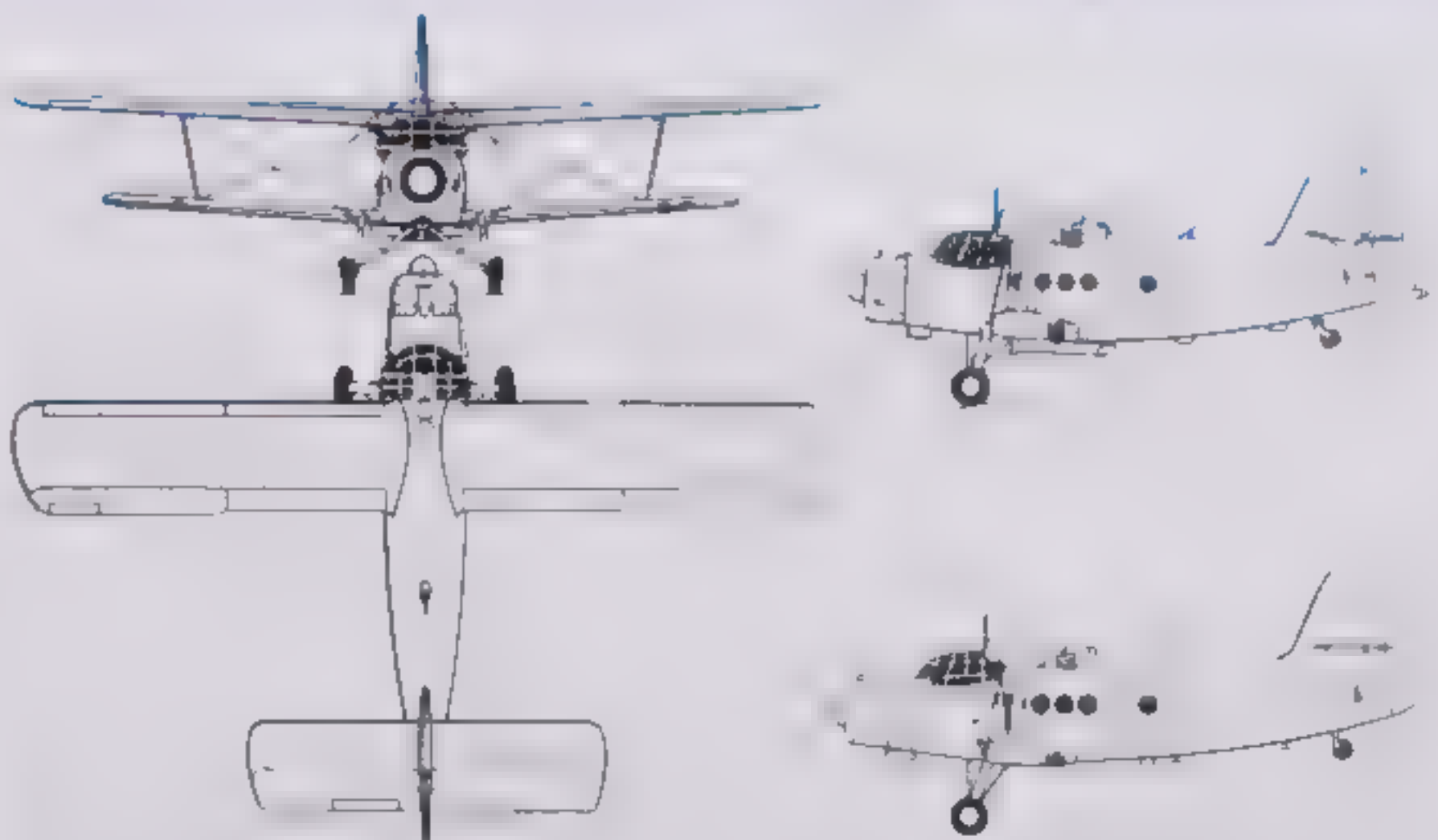
Weights: (2P) Empty 7,605lb (3450kg), maximum loaded 12,125lb (5500kg)

Performance: Maximum speed (P, 11,574lb, 5250kg) 160mph (258 km/h), typical cruise 115mph (185km/h), min safe speed 56mph (90km/h) service ceiling 14,425ft (4400m), typical takeoff or landing over 35ft (10 7m) on grass 1,050ft (320m), range with 1,102lb (500kg) payload 560 miles (900km)

History: First flight 31 August 1947, service delivery (An-2) July 1948 (Fong Chou) December 1957, (WSK) 1960

Users: 44 countries including the following air forces or other military operators Afghanistan, Algeria, Bulgaria, China, Cuba, Egypt, Ethiopia E. Germany, Hungary, Iraq, N. Korea, Mali, Mongolia, Poland, Romania Somalia, Soviet Union, Sudan, Syria, Tanzania, Tunisia, Vietnam

Development: When this bulky biplane appeared in 1947 it appeared to be an obsolete mistake. So unmistakable was it that it has been manufactured in larger quantities than any other single type of aircraft since World War II.



Above: Three-view of An-2M, with side view of standard An-2

Soviet production had topped 5,000 when responsibility was passed to Poland in 1960, and a few hundred additional aircraft with angular tails were made in the Soviet Union in 1964–70. Polish output was continuing in 1977 with the nation's total well over 7,500, and the Chinese Fong Chou output is thought to exceed 5,000, making a combined figure of something over 18,000. Of these perhaps one-quarter are military, and many serve in para-military roles. Versions are numerous, and have different designations in the Soviet Union and Poland, but the chief military roles are paratroop and aircrew training, supply of frontier posts and general transport/casevac duties.

Britten-Norman Defender

BN-2A Defender

Origin: Britten-Norman (Bembridge) Ltd, UK (member of Fairey Group which has production line at Gosselies, Belgium, third production line at IRMA, Romania)

Type: Multi-role light military aircraft

Engines: Two 300hp Lycoming IO-540-K1B5 six-cylinder direct-injection horizontally-opposed

Dimensions: Span (extended raked tips) 53ft (16 15m), length 35ft 8in (10 86m), height 13ft 8½ in (4 18m)

Weights: Empty 3,708lb (1682kg), loaded 6,600lb (2993kg)

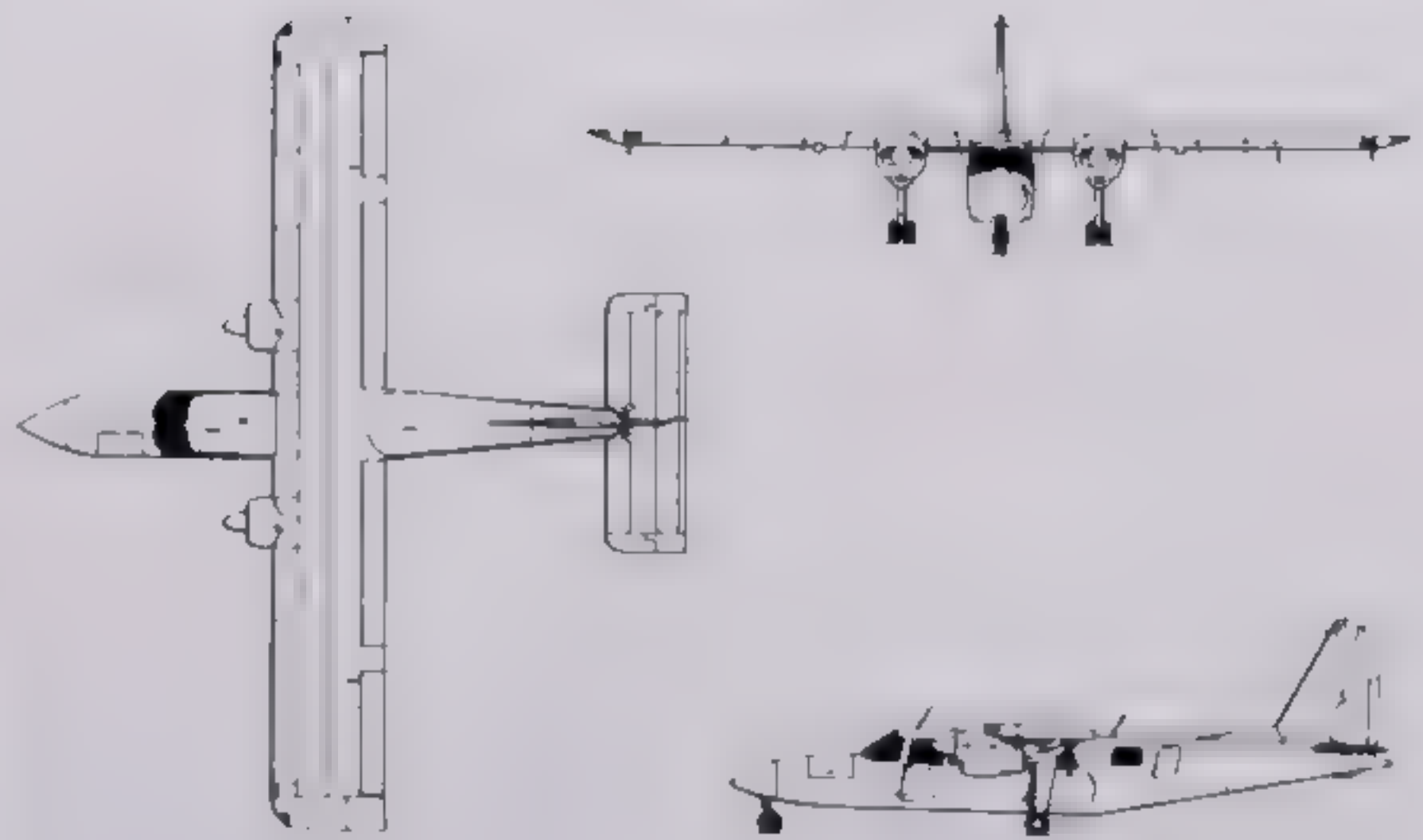
Performance: Maximum speed (clean) 176mph (283km/h), (pylons loaded) 168mph (270km/h), initial climb (clean) 1,300ft (396m)/min service ceiling 17,000ft (5180m), range (maximum combat load) 375 miles (603km), (maximum fuel) 1,723 miles (2772km)

Armament: Weapons normally all carried on four underwing hardpoints inner pair stressed to 700lb (318kg) and outers to 450lb (204kg). Typical loads include 7 62mm Minigun pods, twin 7 62mm machine guns, bombs up to 500lb (227kg), Matra or SURA rocket dispensers, wide variety of grenades and markers, loud hailer, guided missiles or 50gal (227 litre) drop tanks.

History: First flight (civil Islander) 13 June 1965, (Defender) March 1971

Users: Abu Dhabi, Belgium, Brazil, Egypt, Ghana, Guyana, Hong Kong India (Navy), Iraq, Israel, Jamaica, Lesotho, Liberia, Malagasy, Mauritania, Mauritius, Mexico, Nigeria, Oman, Panama, Philippines, Qatar, Rhodesia, Rwanda, Thailand, Turkey Zaire, Zambia

Development: The Britten-Norman Islander has become one of the dominant light utility transports all over the world and it was natural that the enterprising company should have produced a military version. Roles for which the Defender has been planned include internal security, long-range patrol, search/rescue, forward air control, battlefield supply and casualty evacuation, reconnaissance and mapping, and indeed all forms of light limited-war combat, including ground attack, anti-ship and psy-war



Above: Three-view of basic BN-2A Islander.

(psychological) operations. One of the major options is nose-mounted weather radar, providing a marine search capability. Within a year of first flight the Defender had been cleared for most kinds of ordnance dropping, rocket launching and gun firing, including firing of beam guns in the fuselage. Among the options already available are nose radar, various reconnaissance sensors, skis or amphibious floats, by 1978 a turboprop version powered by the 650hp Avco Lycoming LTP101 is expected to be in production. In 1976 marketing began on a proposed military Trislander (the larger three-engined model), with 9-hr endurance and ability to carry 17 troops or 3,745lb loads.

Below: As there are Defenders already flying with air forces all over the world, Hughes Helicopters has caused confusion by giving the same name to its armed 500M helicopter.



Cessna U-17

Model 185 and U-17

Origin: Cessna Aircraft Company, Wichita, USA

Type: Utility transport

Engine: One 300hp Continental IO 520D flat six (U 17C) 230hp (O 470)

Dimensions: Span 35ft 10in (10.92m), length overall 25ft 7½in (7.81m), height overall 7ft 9in (2.36m)

Weights: Empty 1,600lb (725kg), maximum loaded 3,350lb (1519kg)

Performance: Maximum cruise 169mph (272km/h), range on standard tankage, no reserve 660 miles (1062km)

History: First flight (civil 185) July 1960, service delivery (U 17A) 1963

Users: (185 Skywagon bought from Cessna) Honduras, Indonesia (army), Iran (army), Jamaica, Paraguay, Peru, S Africa (U 17) Bolivia, Costa Rica, Greece, S Korea, Laos, Panama, Turkey, Vietnam (206) Guatemala, Israel (207) Argentina (army), Indonesia

Development: Cessna's Model 185 Skywagon is a capable and powerful utility machine carrying up to six people with an optional glassfibre cargo pack under the fuselage. In 1963 the USAF bought 169 U-17A versions for MAP issue to friendly air forces, followed by 136 U 17B and a few U 17C



Above: Three-view of Cessna U-17A/Model 185.

with less power. The 206 Stationair has tricycle gear, swept fin and large side cargo doors. The 207 Turbo-Skywagon is similar but longer, with seventh seat.

Cessna O-2

O-2, Reims FTB 337G

Origin: Cessna Aircraft, Wichita, USA, licence built by Reims Aviation, France

Type: (O 2A) FAC, (O-2B) psy war, (FTB 337G) multi role

Engines: Two (front and rear) 210hp Continental IO-360C flat sixes (Reims) two 225hp Rolls Royce Continental TSIO 360D

Dimensions: Span 38ft 2in (11.63m) (Reims 39ft 8½in 12.1m), length 29ft 9in (9.07m), height 9ft 2in (2.79m)

Weights: Empty (A) 2,848lb (1291kg), (Reims) 3,206lb (1454kg), maximum loaded 4,630lb (2100kg) (USAF overload 5,400lb, 2450kg)

Performance: Maximum cruise (O-2) 195mph (314km/h), (Reims) 230mph (370km/h), takeoff or landing over 50ft (15m) 1,675ft (510m) (Reims 1,200ft, 366m), max range with max fuel (typical) 1,325 miles (2132km)

History: First flight (civil Skymaster) February 1961, (O-2A) 1967

User: Benin, Ecuador, Gabon, Haute Volta, Iran, Ivory Coast, Malagasy



Mauritania, Niger, Senegal, US (AF), Venezuela

Development: Produced to meet a market for a twin engine aircraft that could be flown by private pilots not possessing a twin rating, the push/pull Skymaster is a popular and safe aircraft seating up to six. The USAF selected it in 1966 to provide a superior Forward Air Control aircraft to replace the O-1A Bird Dog. By early 1969 346 had been delivered, and 12 were sold to the Imperial Iranian AF. The USAF also bought 164 O-2B psychological warfare O-2Bs. Most O-2s have side by side dual controls and the O-2A has special electronics and four wing pylons for rockets, Minigun packs, flares and other loads. Cessna dropped the larger turboprop O-2TT. Reims has built over 100 of the more powerful FTB 337G (previously called Milrole), some with radar and/or infra-red pods.



Above: Three-view of O-2A (337G similar).

Left: Cessna O-2A of the US Air Force, with supply containers.

Cessna U-3

U-3A (formerly L-27A) and U-3B

Origin: Cessna Aircraft Company, Wichita, USA

Type: Administrative liaison and cargo transport

Engines: Two 240hp Continental O 470M flat-sixes, (U-3B) two 285hp Continental IO 520M

Dimensions: Span (over tanks) 36ft 11in (11.25m), length overall (U-3A) 27ft 1in (8.25m), (U-3B and modern 310) 31ft 11½in (9.73m), height overall 10ft 8in (3.25m)

Weights: Empty (U-3B, modern 310) 3,337lb (1514kg), maximum loaded (U-3A) 4,700lb (2132kg), (U-3B and modern 310) 5,500lb (2494kg)

Performance: Maximum cruise 223mph (359km/h), range, depending on sub type and fuel, up to 1,740 miles (2800km)

History: First flight (310) 3 January 1953 (L-27A) 1957, (U 3B) 1960

Users: Tanzania, Thailand, USA (AF), Zaïre

Development: The Cessna 310 was one of the first of the modern light twins, with usual seating for five. In 1957 the USAF bought 80 off the shelf as L-27A utility (liaison/cargo) aircraft, subsequently doubling this order. In 1962 these were redesignated U 3A, and 36 U 3Bs were added similar to the later and more powerful 310E with swept fin and 'all-weather'.

Zaïre bought 15 new 310s in 1975. Armée de l'Air communications the Bretagne and Istres test bases is provided by 12 Cessna 411 with 340hp GTSIO 520 engines. Malaysia uses 10 Model 402s (300hp IO 520E) for multi-engine pilot training, and two for photo liaison. Other users of the 402 include Finland.

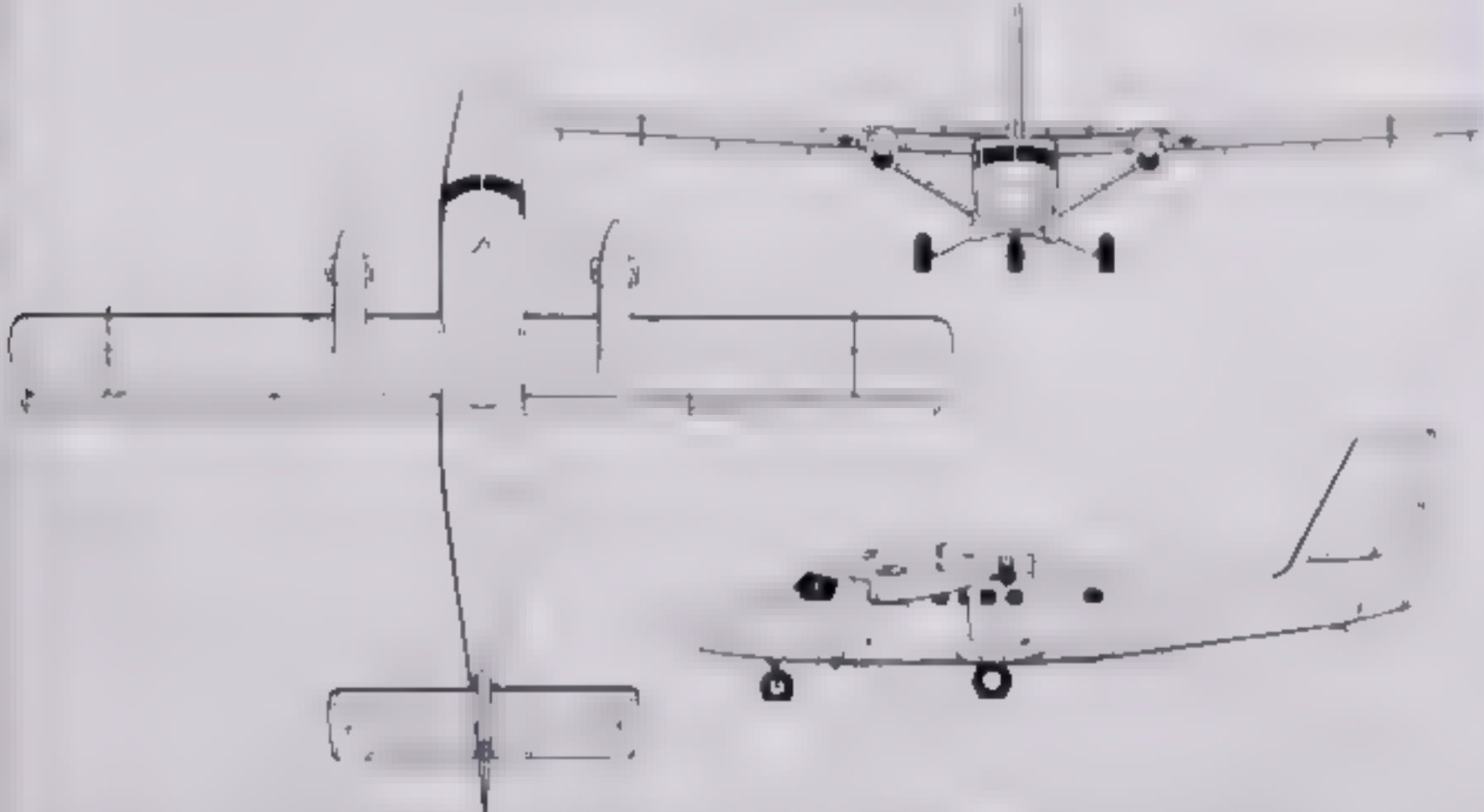


Above: In service since 1957, the U-3A is known throughout the US Air Force as the "Blue Canoe". Most military Cessna twins are later, larger and more powerful.

DH Canada Twin Otter

DHC-6, CC-138, UV-18A

Origin: The de Havilland Aircraft of Canada Downsview, Canada
Type: Multi-role transport
Engines: Two Pratt & Whitney Canada turboprops (Series 10 579ehp PT6A-20, (300) 652ehp PT6A 2
Dimensions: Span 65ft (19.81m), length 51ft 9in (15.77m) (skis and seaplane 48ft, 14.7m), height 19ft 6in (5.94m) (seaplane higher
Weights: Empty (typical military 100, 200) 6 500lb (2950kg), (300) 7 000lb (3175kg), maximum loaded (100, 200 seaplane) 11 600lb (5261kg), (300) 12 500lb (5670kg)
Performance: Maximum cruise (300) 210mph (338km/h) range (300) 1,103 miles (1775km) with 2,131lb (966kg) payload and wing tank 115 miles (185km) with 4,420lb (2004kg) payload
History: First flight 20 May 1965, customer delivery (100) July 1966 (300) spring 1969
Users: (Military) Argentina Canada (CC-138) Chile Ecuador Jamaica Norway Panama Paraguay Peru, United States (UV-18A)



Above: Three-view of DHC-6 Twin Otter 300 (long nose).



Above: Ten of the 12 Twin Otters supplied to the Fuerza Aerea del Peru (Peruvian Air Force) are seaplanes. They are used by Grupo Aéreo 42, based at Iquitos.

Development: Funded entirely by the company when it was a member of the British Hawker Siddeley Group, the Twin Otter is a twin-turboprop outgrowth of the DHC-3 Otter (with single R-1340 Wasp the Otter still serves as the US U-1 and in many other air forces). Unpressurized and simple, the Twin Otter has good STOL performance which is further improved in the current 300S model announced in 1973. Seating can be provided for 20 passengers, and options include wheels, skis or floats, and several special-role kits

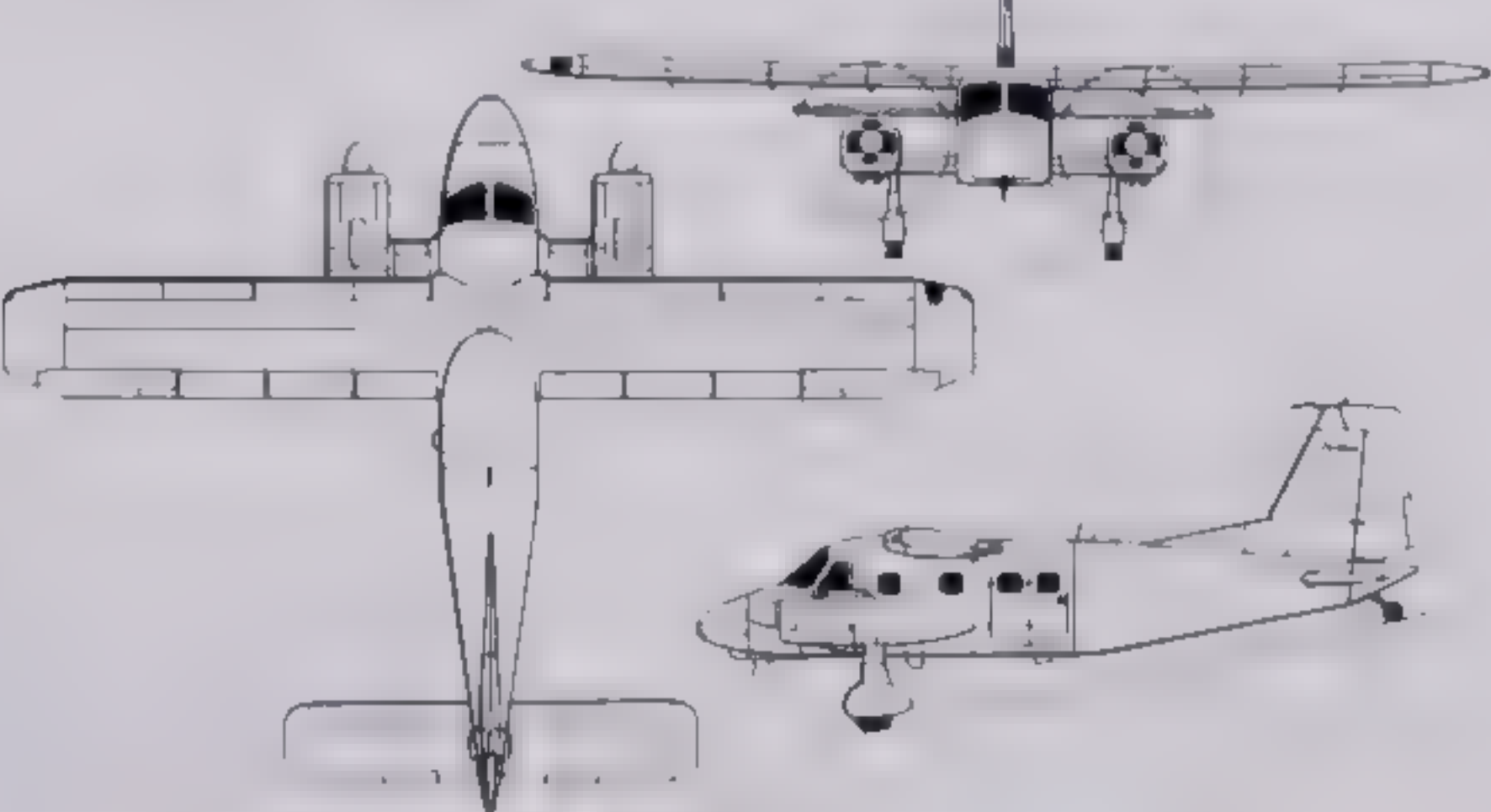
Dornier Do 28D Skyservant

Do 28D-1 and D-2

Origin: Dornier GmbH, Friedrichshafen, West Germany
Type: Multi-role transport
Engines: Two 380hp Lycoming IGSO-540-A1E flat-sixes
Dimensions: Span 51ft 0½in (15.55m), length 37ft 5½in (11.41m), height 12ft 9½in (3.90m)
Weights: Empty 5,080lb (2304kg), maximum loaded 8 853lb (4015kg)
Performance: Typical cruising speed at 10,000ft (3050m) 170mph (273km/h), min-control speed, full flap and power 40mph (65km/h) range with max fuel 1,831 miles (2950km)
History: First flight 23 February 1966 service delivery summer 1967 (military type approval January 1970)
Users: W Germany, Israel, Morocco, Nigeria, Somalia, Turkey, Zambia

Development: Dornier's first post-war aircraft was the Do 27 built in Spain, a very attractive STOL machine built in large numbers for many air forces. Spanish production being designated CASA 127. The twin-engined Do 28 also sold well (120) in military forms, but despite its type-number the Do 28D is an entirely new design, considerably larger and more powerful. Most Skyservants are of the Do 28D-2 model with many refinements and greater weight. Up to 13 passengers can be carried, and there are

various role kits but the wing pylons have been used only for auxiliary tanks. By 1977 sales exceeded 200, including 101 for the Luftwaffe and 20 for the Marineflieger.



Above: Three-view of Do 28D-2 Skyservant.

Below: The West German Marineflieger (Naval AF) has 20 Skyservants (left), while every Luftwaffe Geschwader has four (right).



Douglas DC-3/C-47

C-47 and AC-47, R4D, C-53, Dakota, C-117, L2D and Li-2

Origin: Douglas Aircraft Company, Santa Monica, USA, later built by many other US plants, and licence built in Japan and Soviet Union

Type: Utility transport (formerly also paratroop/glider tug) AC-47 ground weapon platform

Engines: Usually two 1,200hp Pratt & Whitney R-1830 90D or -92 Wasp 14-cylinder two row radials, (C-117D) two 1,530hp Pratt & Whitney R 1820-80 Cyclone nine-cylinder radials, (Li-2) two 1,200hp Pratt & Whitney (Cyclone-derived) nine cylinder radials

Dimensions: Span 95ft (28.96m), length 64ft 5½in (19.64m), height 16ft 11in (5.16m)

Weights: empty about 16,970lb (7700kg), loaded, about 25,200lb (11,432kg) overload limit 33,000lb (14,969kg)

Performance: Maximum speed, about 230mph (370km/h), initial climb about 1,200ft (366m)/min, service ceiling 23,000ft (7,000m), maximum range 2,125 miles (3,420km)

Armament: (AC-47) usually three 7.62mm Miniguns, many other types of armament in other versions, but none usually fitted

History: First flight (DST) 17 December 1935, first service delivery (C-47) October 1938

Users: Argentina, Australia, Bolivia, Brazil, Bulgaria (Li-2), Burma, Canada, Central African Rep., Chad, Chile, Colombia, Congo, Dahomey, Denmark, Dominica, Ecuador, Ethiopia, Finland, France, Gabon, Germany (E, Li-2, W, C-47), Greece, Guatemala, Haiti, Haute-Volta, Honduras, India, Indonesia, Israel, Italy, Ivory Coast, Khmer, Laos, Libya, Malagasy, Malawi, Mali, Mauritania, Mexico, Morocco, New Zealand, Nicaragua, Niger, Nigeria, Norway, Oman, Pakistan, Panama, Papua/NG, Paraguay, Peru, Philippines, Poland (Li-2), Portugal, Rhodesia, Rwanda, Salvador, Senegal, Somalia, S. Africa, Soviet Union (Li-2), Spain, Sri Lanka, Sweden, Syria, Taiwan, Thailand, Togo, Turkey, Uganda, Uruguay, Venezuela, Vietnam (Li-2, C-47), Yemen Arab AF, Yugoslavia, Zaïre, Zambia

Development: When, in 1935, Douglas designer Arthur E. Raymond planned the Douglas Sleeper Transport (DST) as an enlarged and improved



Above: This C-47 of Flyvevæbnet (Royal Danish AF) 721 Sqn is being replaced by a C-130.



Above: Three-view of typical C-47 with wing supply racks.

DC-2, he little thought that, as well as becoming the worldwide standard airliner of its day, it would be by far the most widely used military transport in history. During World War II there were numerous versions, some civil aircraft impressed into military use, some paratroopers and tugs and the vast majority utility C-47 versions with a strong cargo floor and large double doors. Oddities included a glider and a twin-float amphibian. US military production totalled 10,048 by June 1945, followed by small batches of redesigned Super DC-3 versions including the R4D-8 and C-117. Showa and Nakajima in Japan built about 571 of the L2D family and in the Soviet Union production of the Li-2 (with door on the right) is estimated to have exceeded 2,700. Many hundreds of these aircraft, most of them C-47s, remain in daily use in almost every air force (the RAF retired its last in 1970). Many serve as platforms for research projects and counter-measures and in Vietnam the AC-47 – called “Puff the Magic Dragon” – was developed in several versions to deliver suppressive fire against ground targets. Other important variants are the EC-47 series used for multi-spectral sensing and electronic reconnaissance.

In 1977 the DC-3 military versions—most of them rebuilt or otherwise updated C-47s built during World War II—were still among the most widely used military aircraft, though their numbers are gradually being thinned. There are too many different designations to list above; for example, C-47 and C-53 names were Skytrain and Skytrooper, the AC-47 name was Dragon Ship (also “Spooky”) and Canadian examples are now known as CC-129.

Embraer Bandeirante and variants

EMB-110 (versions, see text) and EMB-111

Origin: EMBRAER, São José dos Campos, Brazil

Type: Multi-role transport, (111) maritime patrol

Engines: Two Pratt & Whitney Canada turboprops, (110) 680shp PT6A-27, (111) 750shp PT6A-34

Dimensions: Span 50ft 3in (15.32m), (111, over tip tanks) 51ft 5½in (15.69m), length 46ft 8½in (14.23m), (111) 48ft 3½in (14.71m), height 13ft 6½in (4.13m), (111) 15ft 6½in (4.73m)

Weights: Empty, equipped 7,451lb (3,380kg), (111, basic operating with crew of six) 9,259lb (4,200kg), maximum loaded 12,345lb (5,600kg), (111) 13,558lb (6,150kg)

Performance: (Max wt) maximum speed at 15,000ft (4,575m) 270mph (434km/h), cruise at 15,000ft 262mph (422km/h), (111 cruise at 9,845ft 3,000m, estimated) 288mph (464km/h), takeoff over 50ft (15m) 1,770ft

(540m), (111) 2,740ft (835m), range 1,379 miles (2,220km) with max fuel, 153 miles (246km) with max payload, (111) over 2,000 miles (3,220km) with 8hr endurance

Armament: (111 only) see text

History: First flight 26 October 1968, (111) scheduled for July 1977, service delivery 9 February 1973

Users: Brazil, Chile, Uruguay (and 17 non-military customers)

Development: Designed under the direction of former French constructor Max Holste, this excellent unpressurized twin-turboprop is one of the first latin-American aircraft owing nothing to any foreign design to have sustained a successful programme. By 1977 sales had reached nearly 200, the main military versions being the EMB 110 12-seat transport, (Brazilian AF C-95), 110A navaid/ILS calibration (EC-95), 110B mapping and photogrammetry (RC-95), 110K cargo and 111. The EMB-111 is a redesigned model for the Comando Costeiro (Coastal Command) with extra fuel, more power, APS-128 nose radar, LN-33 inertial system, and extensive equipment including provision for all-weather marking and surface attack with eight 5in (127mm) rockets or other weapons.



Above: Three-view of EMB-110



Above: A FAB C-95 of the kind which formed the foundation of the production programme, which in turn has turned EMBRAER into one of the world's most capable planemakers.

GAF Mission Master

N22B and N24 (basic civil name, Nomad)

Origin: Government Aircraft Factories, Fishermen's Bend, Australia

Type: Multi-role transport

Engines: Two 400shp Allison 250-B17B single-shaft turboprops

Dimensions: Span 54ft 0in (16.46m), length 41ft 2½in (12.56m) (N24 47ft 1½in, 14.36m), height 18ft 1½in (5.52m)

Weights: Empty 4,451lb (2019kg) (N24 4,549lb, 2063kg), maximum loaded 8,500lb (3855kg)

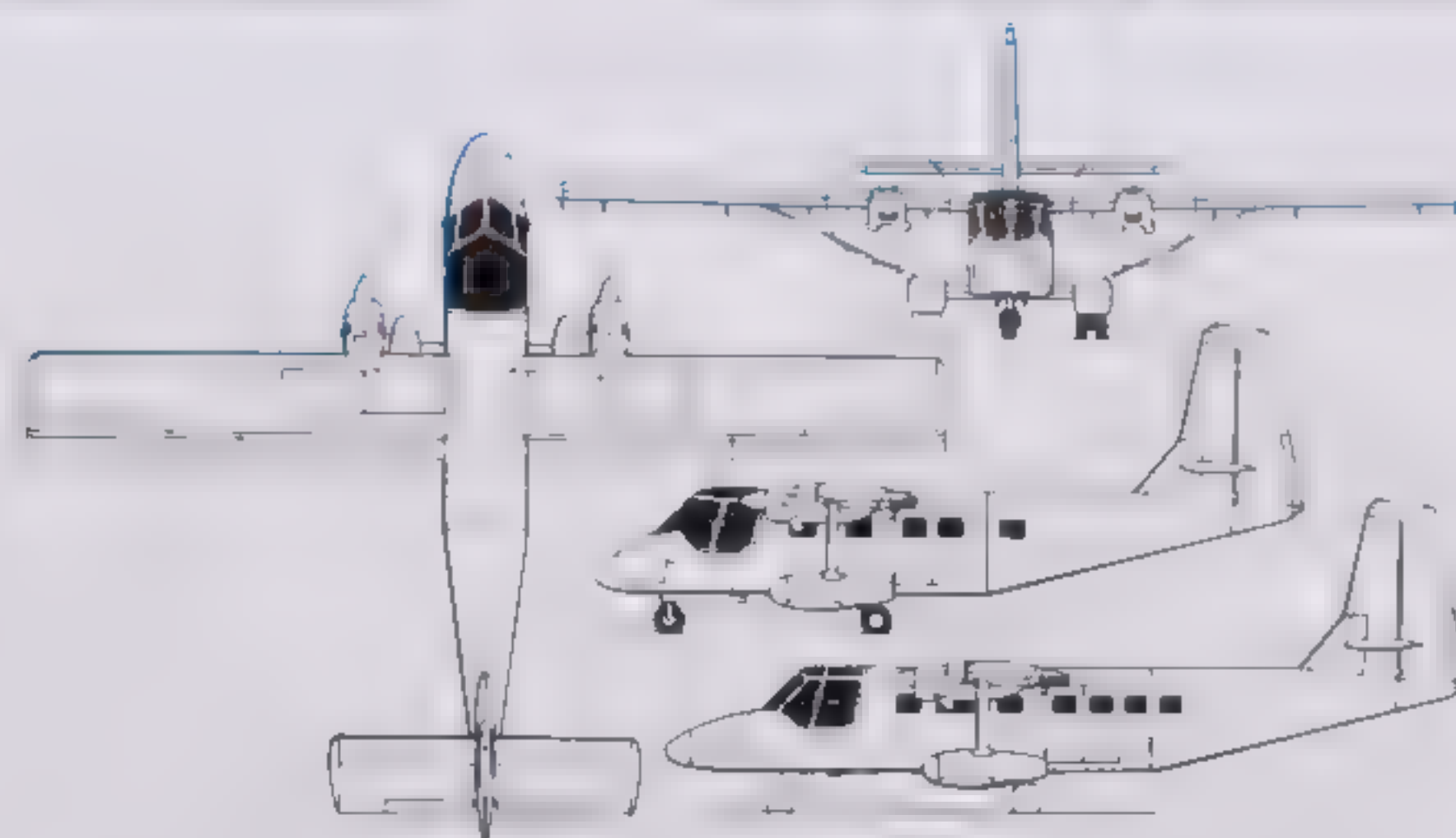
Performance: (Both, max wt, sea level) normal cruise 193mph (311km/h) STOL takeoff over 50ft (15m) 1,260ft (384m), STOL landing over 50ft (15m) 635ft (194m), max range with 45min reserves 668 miles (1074km) (at 10,000ft, 3050m) 840 miles (1352km)

Armament: Four wing hardpoints each rated at 500lb (227kg) for gun pods, bombs, rockets or other stores

History: First flight (Nomad) 23 July 1971 Mission Master service delivery April 1975

Users: Australia, Indonesia, Philippines

Development: One of the world's newest utility aircraft, the GAF Nomad was designed to appeal equally to civil and military customers, with emphasis on the ability to operate reliably in austere environments away



Above: Three-view of N22B, with side view (bottom) of N24.

from proper airfields. The Nomad is built in two forms, the main production model being the N22B called Mission Master in military guise. Unpressurized, it seats up to 12 plus two pilots, and can carry seat armour, self-sealing tanks and various night-vision and other sensors. The stretched N24 seats 15 passengers but had not achieved military sales by mid-1977.

Mitsubishi MU-2

MU-2 sub-types, see text.

Origin: Mitsubishi Heavy Industries, Nagoya, Japan

Type: Multi-role transport

Engines: Two Garrett-AiResearch TPE331 single-shaft turboprops, (A to E) 605shp TPE331-25, (F, G) 665shp TPE331-1, (J, K, M) 724shp TPE331-251, (L) -251 engines re-rated at 776shp

Dimensions: Span (over tip tanks) 39ft 2in (11.94m), length (A to E) 33ft 3in (10.13m), (with nose radar) about 36ft, (J, L) 39ft 5in (12.01m), height 12ft 11in (3.94m), (J, L) 13ft 8in (4.17m)

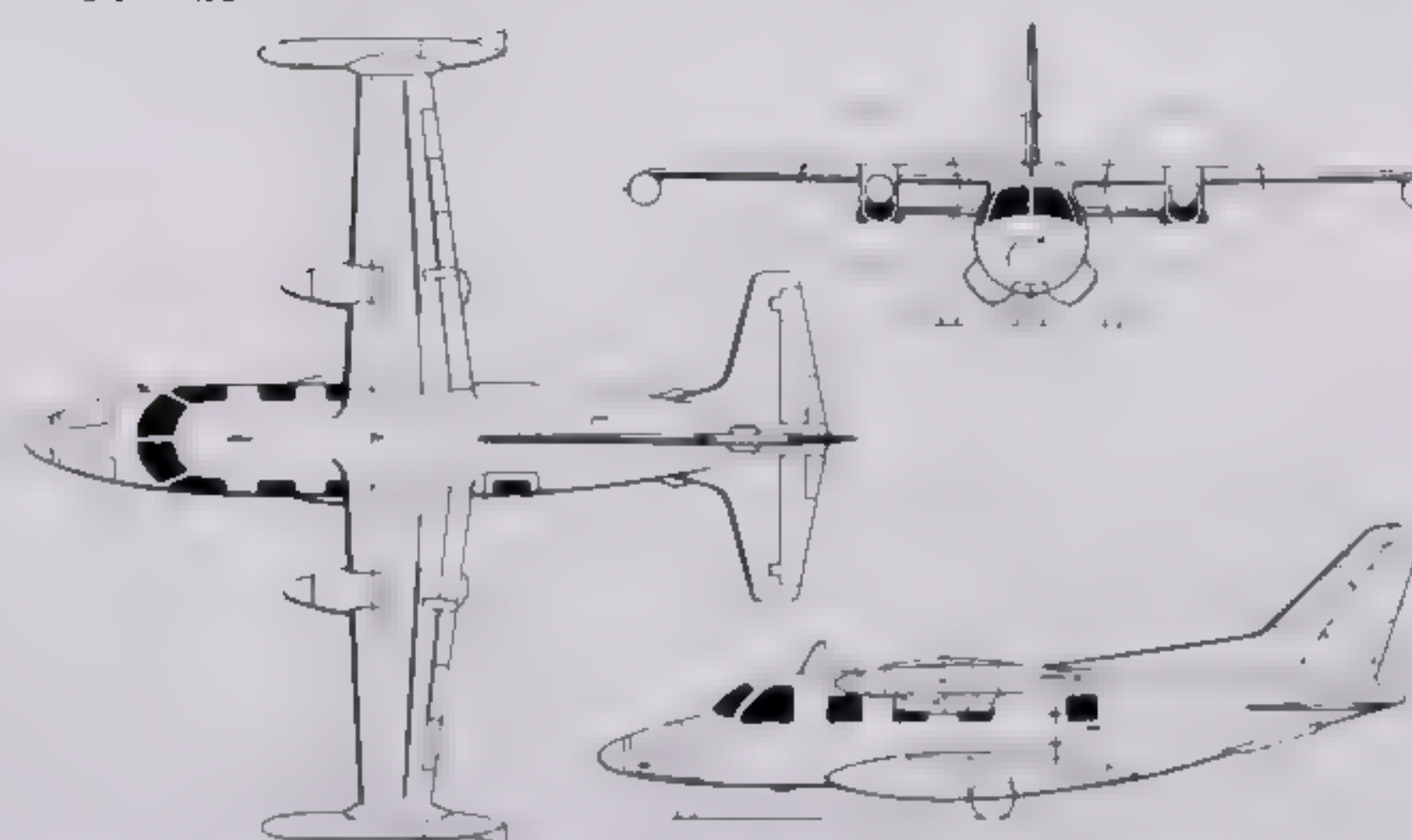
Weights: Empty (early models) 5,700lb (2586kg), (K) 5,920lb (2685kg) (J) 6,800lb (3084kg), maximum loaded (early) 9,350lb (4240kg), (K) 9,920lb (4500kg), (J) 10,800lb (4900kg)

Performance: Max cruising speed (typical, 15,000ft, 4575m) 363mph (584km/h); service ceiling (typical) 33,200ft (10,120m), takeoff or landing over 50ft (15m) (typical) 1,700ft (518m), max range at 25,000ft (7620m) with max fuel and 30min reserve (typical) 1,680 miles (2705km)

History: First flight 14 September 1963, service delivery (C) 30 June 1967

User: Japan

Development: This compact high-wing machine is technically clever with full pressurization, anti-icing, chem-milled skins, spoilers and double-slotted Fowler flaps. Though small, and not suited to bulky loads, it has exceptional performance. By 1977 about 500 had been sold, mostly to civil customers outside Japan. Main military variants are the MU-2C with



Above: Three-view of MU-2L, cleared to 11,575lb (5250kg).

cameras, guns, rockets and bombs, designated LR-1 by the JGSDF, the radar-equipped E used by the JASDF for search and rescue, and the derived K still in production, the J used by the JASDF for flight calibration, and the similar L cleared for use at 11,575lb (5250kg) but not yet in military operation.

Below: The radar-equipped MU-2K, used by the JASDF as a tactical rescue and liaison aircraft.



Pilatus/Fairchild Porter and Peacemaker

PC-6/A and B Turbo-Porter and Fairchild AU-23 Peacemaker

Origin

Type

Engine

Dimensions

Weights

Performance

Armament

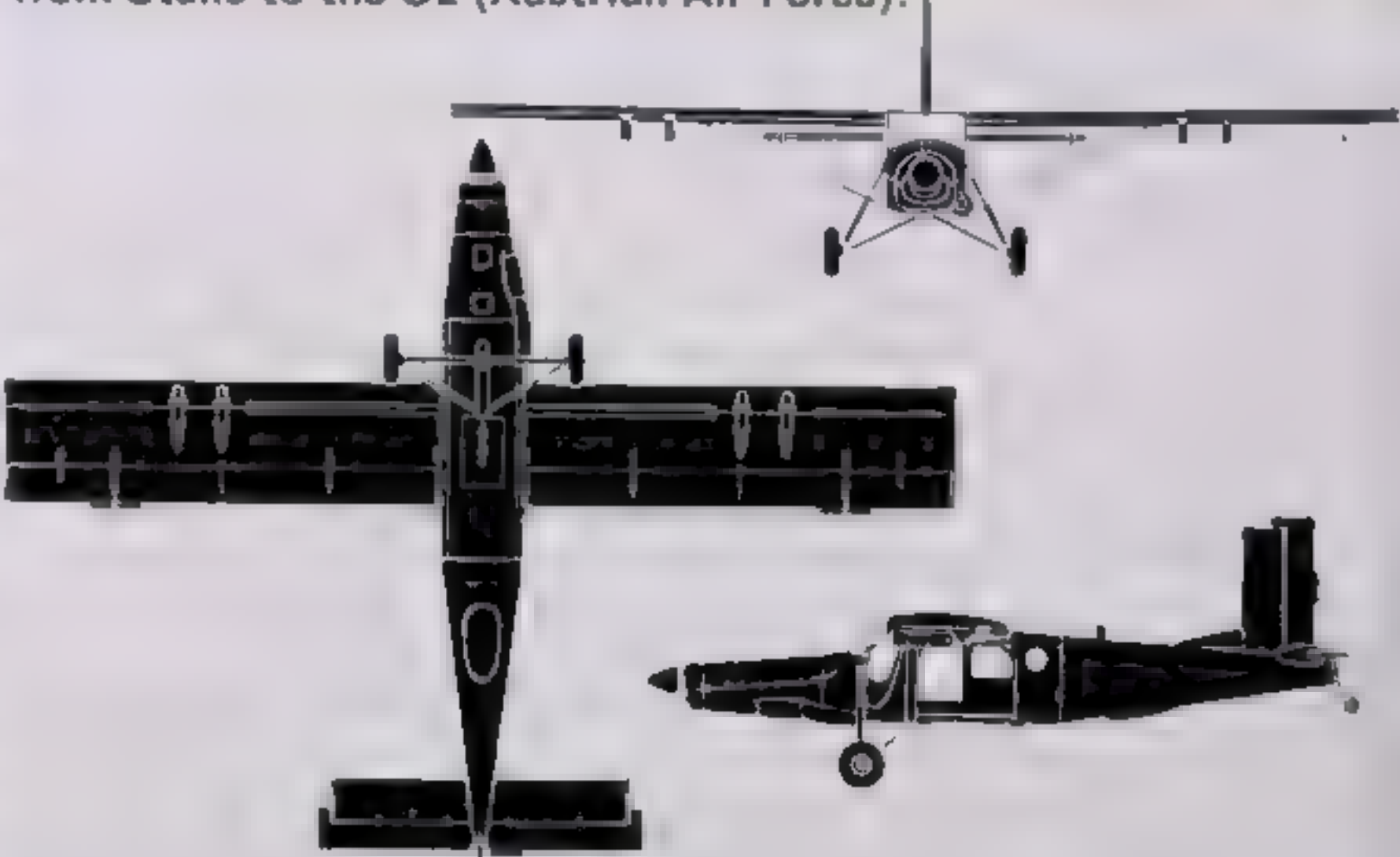
History

Users

Development: In 1961, the first computer program was developed for the purpose of simulating the behavior of a system. This program was written in the Fortran language and was run on the IBM 704 computer. The program was designed to simulate the behavior of a system that was being studied in a laboratory. The results of the simulation were compared with the results of the laboratory experiments, and it was found that the simulation was in good agreement with the experimental results. This was a significant achievement, as it demonstrated that the simulation could be used to predict the behavior of a system before it was built. This was the first time that a computer program had been used to simulate the behavior of a system, and it marked the beginning of the development of simulation as a tool for engineering design.



Above: Fourth of 12 PC-6/B2 Turbo-Porters delivered in 1976 from Stans to the OL (Austrian Air Force).



Above: Three-view of Fairchild AU-23A Peacemaker.

Below: Powered by a different engine from the Swiss-built aircraft, a Peacemaker makes one of its amazing STOL takeoffs.



Rockwell International

OV-10 Bronco

OV-10A to -10E

Origin: Rockwell International Corp, USA

Type: (Except B(Z)) two-seat multi-role counter insurgency, (B) target tug

Engines: (Except B(Z)) two 715ehp Allison T76 410/411 shaft centrifugal turboprops, (B(Z)) as other versions plus General Electric J85-4 turbojet of 2 950lb (1338kg) thrust above fuselage

Dimensions: Span 40ft (12.19m), length (except E) 41ft 7in (12.67m) (D) 44ft (13.4m), height 15ft 2in (4.62m)

Weights: Empty (A) 6,969lb (3161kg) maximum loaded (A) 14,466 lb (6563kg)

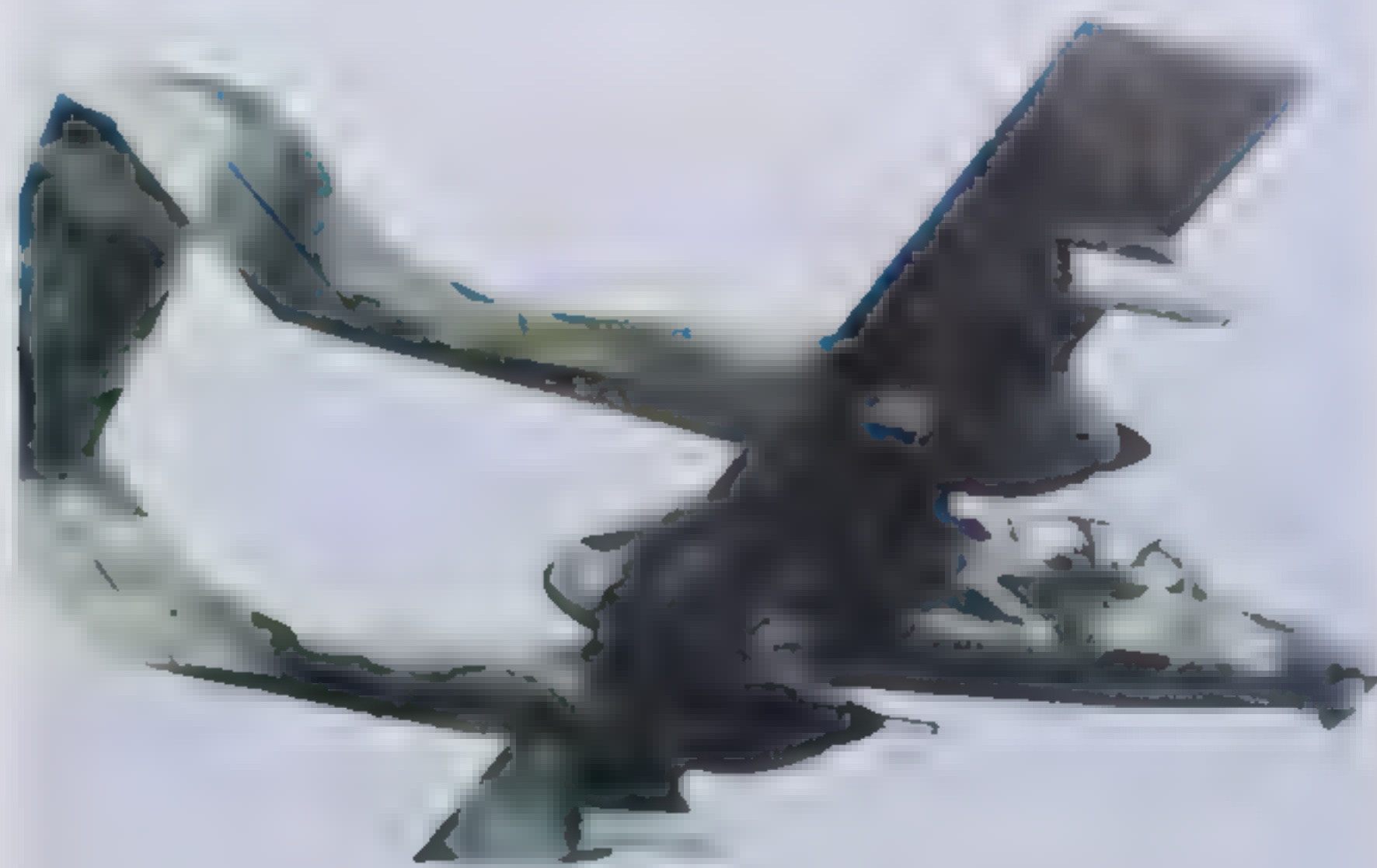
Performance: Maximum speed (A, sea level, clean) 281mph (452km/h) initial climb 2,300ft (700m)/min, (B(Z)) 6,800ft/min, service ceiling 30,000ft (9150m), range with maximum weapon load, about 600 miles (960km), ferry range at 12,000lb gross, 1,428 miles (2300km)

Armament: Four 7.62mm M60C machine guns in sponsons, 1,200lb (544kg) hardpoint on centreline and four 600lb (272kg) points under sponsons, one Sidewinder missile rail under each wing, (OV-10D) as other versions plus three barrel 20mm cannon in remotely aimed ventral power turret

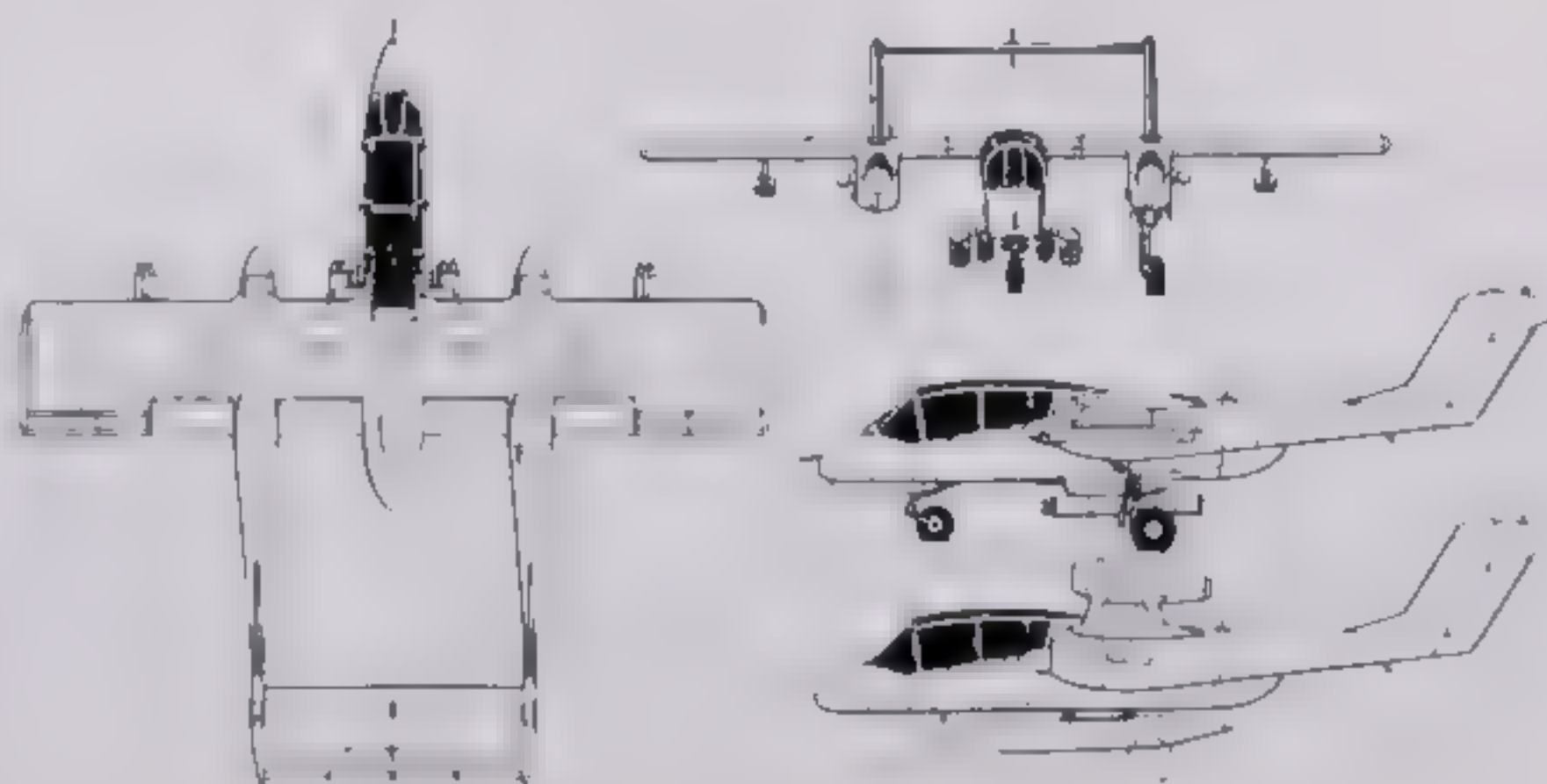
History: First flight 16 July 1965 (production OV-10A) 6 August 1967 (YOV-10D) 9 June 1970

Users: W Germany, Indonesia, S Korea, Thailand, USA (Air Force, Marine Corps), Venezuela

Development: Recognising that no US aircraft was tailored to the urgent task of fighting Co-In (counter-insurgency) operations, or "brush-fire wars", the US Department of Defense in 1960 began study of the problem and in 1962 issued a joint USAF/Navy/Marine Corps specification for a Light Armed Reconnaissance Aircraft (LARA). The winner, in 1964, was the Bronco. Designed to operate from short rough strips (or on floats or skis) it can carry a wide range of tactical equipment and weapons, including doppler radar, TV reconnaissance, five paratroops or two casualties. The OV-10A was ordered in October 1966 and by 1969 the Marine Corps had 114, while the USAF were using 157 for Forward Air Control (FAC) duties in Vietnam. In 1969 Pave Nail Conversion of 15 aircraft fitted them with laser rangefinders, stabilized night sighting system, Loran and other devices for night FAC, attack or target illumination for other aircraft. The B and jet-boosted B(Z) are used by the Luftwaffe. The OV-10D has Night Observation Gunship (NOGS) equipment, with long-nosed IR sensor, cannon turret and, as a conversion, 1,000ehp T76 engines



Above: Most sophisticated Bronco is the NOGS (Night Observation/Gunship System), the Marine Corps YOV-10D.



Above: Three-view of OV-10A with side view (bottom) of OV-10B(Z).



Above: Rockets ripple away from launchers on the body pylons of an OV-10D Bronco of the US Air Force.

Below: Fastest of the Bronco family, the jet-boosted OV-10B(Z) is used by West Germany as a target tug.



Short Skyvan 3M

Skyvan Series 3M

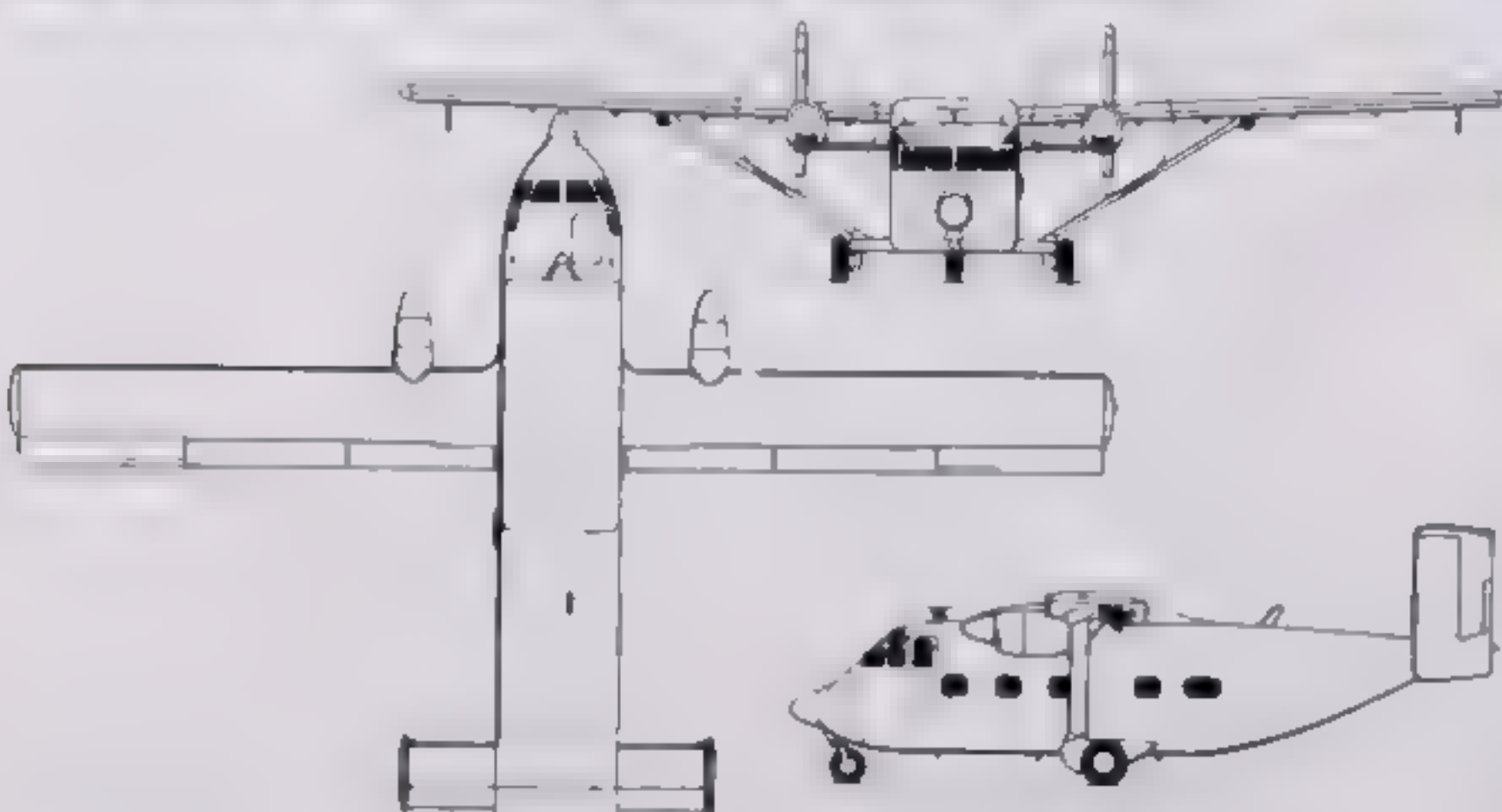
Origin
Type
Engines

Dimensions: Span 64ft 11in (19.8m)

Weights

Performance

History: First flight (SC 7) 17 January 1963 (3M) early 1960s
Users: (Military) Argentina, Austria, Ecuador, Israel, Italy, Mexico, Norway, Oman, Saudi Arabia, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States, Venezuela



Above: Three-view of Short Skyvan 3M with radar.



Above: A Skyvan 3M of the Sultan of Oman's Air Force No 2 Sqn, based at Seeb. This unit has 15 such aircraft, which do not have radar (in such a climate it is hardly needed).

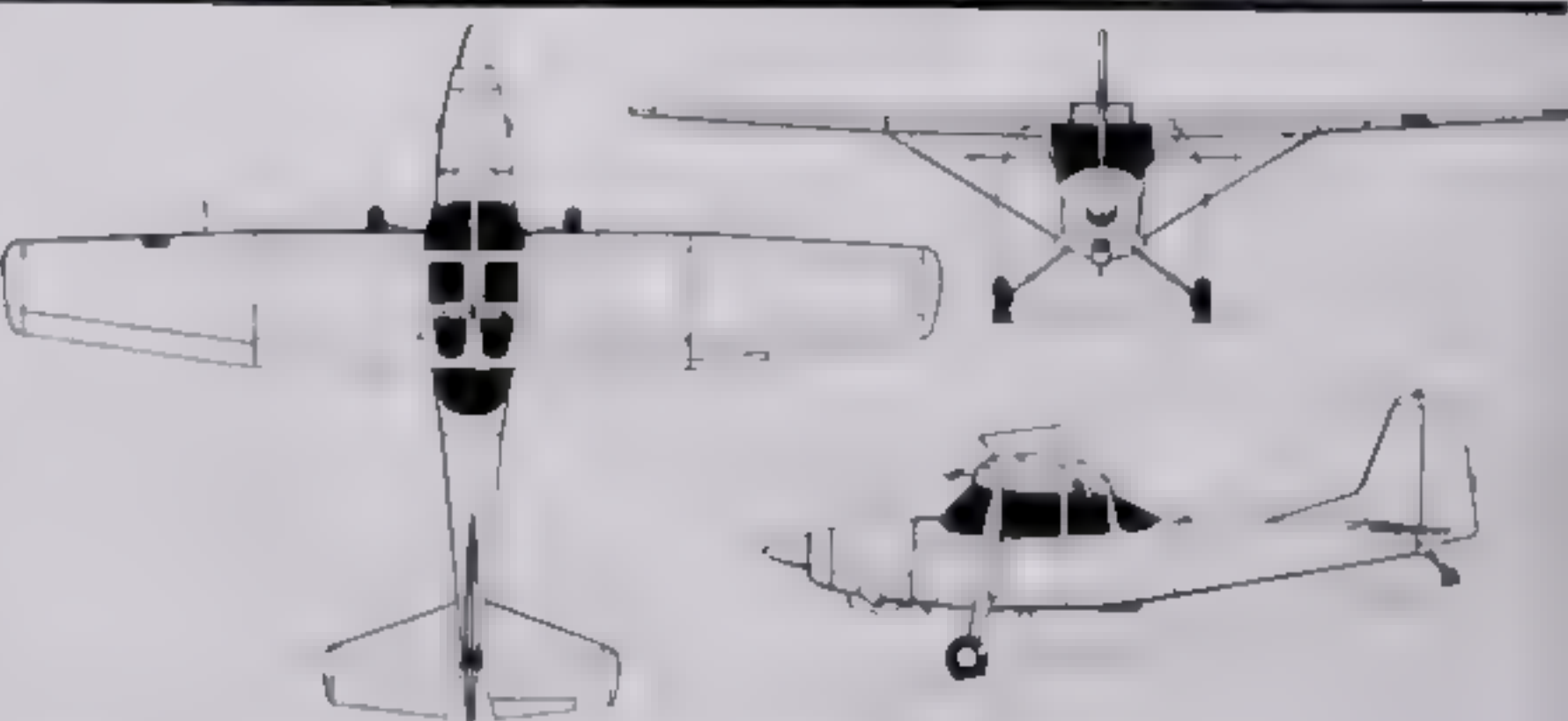
Development: First flown as a piston engined civil aircraft, the Skyvan has been developed in several versions of which one of the most successful is the military Srs 3M. This is a robust all metal, unpressurized airlifter with excellent STOL performance, even in hot/high conditions. It has a full complement of equipment for heavy dropping, loads can include up to 12,000lb (5,443kg) and is fitted with a variety of equipment available for various armaments, multi-sensor reconnaissance, search, rescue and other duties.

SIAl/Marchetti SM.1019

SM.1019E1

Origin: SIAl Marchetti SpA, Sesto Calende, Italy
Type: Two seat STOL utility (see text)
Engine: One 400shp Allison 250 B17 turboprop
Dimensions: Span 36ft 0in (10.972m), length overall 27ft 11½in (8.52m), height overall 9ft 4½in (2.86m)
Weights: Empty 1,521lb (690kg), maximum loaded 3,196lb (1,450kg)
Performance: Maximum cruise 186mph (300km/h), takeoff or landing over 50ft (15m) 722ft (220m), tactical radius in attack mission 69 miles (111km), maximum range in reconnaissance mission cruise at 9,845ft (3,000m) 832 miles (1,340km)
Armament: One hardpoint under each wing for light attack loads such as Minigun pod, rocket pod or reconnaissance pod
History: First flight 24 May 1969, first delivery 1975
User: Italy (army)

Development: One of the latest light tactical aircraft, the SM 1019E differs from the SM 1019 in having a gas turbine engine, giving improved performance. It is also fitted with a new avionics suite, including a new radar, the 1019E1, as delivered in a series of 100 to the Italian



Above: Three-view of SIAl-Marchetti SM.1019E.

Aviazione Leggera dell'Esercito, has electronic, photo and navigation equipment for day/night reconnaissance and also flies IAC, helicopter escort, cargo or training missions. The avionics are comprehensive and include ADF, IFF, Omega VLF, Tacan and ILS, as well as several communications radios.

Below: New SIAl-Marchetti SM.1019E1 turboprop STOL tactical aircraft awaiting collection by the Italian Army (ALE).



Glossary of Terms

The following are some of the terms used in this book which are not part of common language

AAM Air/air missile fired from one aircraft against another
AB Airbase, airfield forming permanent home of military unit(s)
Active Describes device emitting radiation, sound waves or other phenomena
ADC US Air Force Aerospace Defense Command
ADD Soviet Union Aviatsiya Dal Nevo Deistviya, long-range bombers
Aéronavale French naval aviation
AFB US Air Force base
Afterburner Device added to rear of jet engine (turbofan or turbo jet) in which extra fuel is burned to give extra thrust in UK often called reheat jetpipe
AGM Air/ground missile (US designation system)
AIL US electronics, piloted aircraft countermeasures
AIM Air intercept missile (ie, AAM US designation system)
ALQ US electronics, piloted aircraft countermeasures, special or combination of purposes
ALR US electronics, piloted aircraft countermeasures, passive detecting
AMI Aeronautica Militare Italiano (Italian air force)
AMST Advanced Medium STOL Transport
ANG US Air National Guard
APG US electronics, piloted aircraft radar, fire control
APQ US electronics, piloted aircraft radar, combination of purposes
APS US electronics, piloted aircraft radar search range/bearing
APY Special designation for AWACS radar
Armée de l'Air French air force
ASM Air surface missile
ASW Anti-submarine warfare
ATO Assisted takeoff
Attack Combat mission flown against tactical surface targets on land or sea
Avionics Aviation electronics, ie electronic systems carried in aircraft
AV-MF Soviet Union Aviatsiya Voenno morskovo Flota, naval air force
AWACS Airborne Warning And Control System
AWG US electronics, piloted aircraft, armament, fire control
BLC Boundary-layer control method of increasing lift, reducing drag or improving control by blowing air under pressure from thin slits in aircraft skin
Bomber Term virtually defunct originally meant large aircraft that dropped bombs, but today this is done by virtually all combat aircraft, while role of modern bomber is to carry stand off missiles within range of enemy strategic targets
CAF Canadian Armed Forces
CAP Combat Air Patrol (in defence of fleet at sea)
CAS Close air support
Casevac Casualty evacuation (assumed from land battle)
CFS Central Flying School (UK)
Close support Combat mission flown against surface targets hindering friendly ground forces

Code name NATO compiles a list of invented names, easily remembered and unambiguously transmitted quickly by radio, for Soviet and Chinese aircraft missiles and radars. This is because the true designations are either not known or complex. Also called 'reporting name'
Coin Counter insurgent, designed for use against unsophisticated enemy in 'limited' or 'brushfire' conflict
Countermeasures Equipment and operations intended to nullify or confuse hostile sensors, such as chaff, jammers, decoys, could even include a smokescreen
Chaff Small metal shivers, such as strips cut from roll of aluminium foil, scattered by million by aircraft in hostile airspace to blanket enemy radars
Cruise missile Missile that flies to its target like aircraft, with sustained propulsion and lifted by wing
CW Continuous-wave e m radiation (as distinct from pulse)
Decoy Small aircraft or other object arranged to look like attack or bomber aircraft on hostile radars or other sensors, to confuse or divert enemy defences
Delta wing Shaped like triangle in plan, with sharply swept leading edge and almost straight trailing edge
Development aircraft Aircraft built for flight testing to eliminate faults or deficiencies, may later be brought up to production standard
Drogue Device trailed on cable hose or other filament from aircraft (especially on FR hose, to keep latter steady, pull it out from hose reel and provide receptacle for receiver's probe)
ECM Electronic countermeasures equipment or techniques intended to nullify or confuse enemy electronic sensors such as radars
EdA Ejercito del Aire, Spanish air force
Elint Electronic intelligence, mission flown to find out maximum about enemy (or potential enemy) e m emissions such as radars, communications, IR lasers and ECM
E m radiation Electromagnetic radiation which includes visible light, radio waves (including radar microwaves) and IR (heat)
ESM Electronic support measures (sometimes, electronic surveillance measures), including ECM, Elint and related activities
EVS Electro-optical viewing system
EW Electronic warfare, all use of e m radiation in war
FA Soviet Union Frontovaya Aviatsiya, frontal aviation ie tactical air force
FAC Forward Air Controller experienced airborne manager of tactical air strike on surface target
FFAR Folding fin (or free flight) aircraft (or air to air) rocket
Fighter Combat aircraft intended primarily for destruction of enemy aircraft in the air
FIS Fighter Interceptor Squadron (USAF)
FIW Fighter Interceptor Wing (USAF)
FLIR Forward looking infra red
Flying Boom Boeing patented method of FR using rigid

telescopic boom (pipe) steered into receiver's receptacle by operator in tanker
FR Flight refuelling
FTS Flying training school
g Acceleration due to Earth gravity on Earth's surface bodies experience 1g and appear to weigh their normal amount, but in free fall they experience 0g while in air combat they may be accelerated in tight turns and dive pull outs to 6g or more modern fighters being designed to withstand up to 12g
GAU Gun, aircraft unit (US designation system)
Hardpoint Part of aircraft strengthened to carry pylon
Head-up-display Optical device which projects information and/or pictures on pilot's windscreen focussed at infinity so that he can simultaneously search sky/ground ahead
Homing Able to steer automatically towards target or other objective
HUD Head-up display
IA-PVO Soviet Union Istrebitel naya Aviatsiya Protivo vozdušnaya Oborona, interceptor force of air defence force
Inertial Deriving information from gyros and accelerometers of extreme accuracy, able to calculate exact position of aircraft or other vehicle without external information
Interceptor Fighter designed for finding and destroying hostile aircraft even in bad weather or at night
IR Infra red (ie heat)
Jammer EW device emitting such powerful e m radiation as to blot out hostile reception on same wavelength(s)
JASDF Japan Air Self-Defence Force
Jato, JATO Jet assisted takeoff (a misnomer because it invariably means rocket-assisted)
JMSDF Japan Maritime Self-Defence Force (includes naval air arm)
LABS Low altitude bombing system method of delivering nuclear bombs at low level by 'tossing' them much higher than the aircraft to increase time available for latter to get clear
Laser Device for emitting beam of coherent (waves in phase) light, which can be made intense focussed precisely and used with receiver at same wavelength
LLTV, LLLTV Low-light TV, low light level TV, TV able to be used in poor visibility (but not on pitch-black night)
Loiter To cruise at economical low airspeed
LRMTS Laser ranger and marked target seeker
MAC US Air Force Military Airlift Command
Mach Scale used for speed of fast aircraft, Mach 1.0 is speed of sound about 760mph (1223km/h) at sea level (more on hot day, less in winter) and 660mph (1062km/h) at high altitude, thus Mach 1.6 is 60 per cent faster and Mach 2.0 twice as fast
MAD Magnetic anomaly detector sensitive device able to detect local irregularity in Earth's magnetic field due to presence of submerged submarine
Marked target Target illuminated by e m radiation (usually laser) so that attacking aircraft can spot it and missile home on to it
Multi-sensor Equipped with radar,

optical cameras and IR to give fullest picture of hostile target eg to defeat camouflage
NAS Naval Air Station
NATO North Atlantic Treaty Organization
Passive Receiving but not emitting (human ear is example, but in aircraft usually concerned with e m radiation)
PD Pulse doppler, kind of radar
Pod Streamlined container carrying equipment outside aircraft
Pre production aircraft Batch of aircraft built in production tooling but still subject to progressive modifications as result of flight testing and engineering development
Pressurized Aircraft interior inflated to pressure which (though lower than at sea level) is higher than surrounding air at high altitude, for comfort of crew
PRF Pulse-recurrence frequency (radar)
Probe Long boom (rod) projecting ahead of aircraft especially pipe for taking fuel from FR drogue
Profile Combat mission drawn with distance horizontally and height vertically, attack mission may be high as much as possible being flown at high altitude to save fuel but attack being at treetop height to escape radar detection
Prototype First aircraft built of new type strictly does not apply to 'one-off' experimental aircraft but only to type intended later to be mass produced, after development is completed
Pulse radiation E m emission in form of succession of bursts, as distinct from CW
PVO See IA PVO
Pylon Attachment for external store, usually embodying means for release and positive ejection to throw clear of aircraft
Radar System using e m radiation to detect, locate and track targets
RAF Royal Air Force (UK air force)
RB Robot, Swedish designation for guided missiles
Receiver In FR, the aircraft accepting fuel
Reporting name See code name
Retarded bomb Bomb fitted with airbrakes and/or parachute to allow low flying attack aircraft to get clear of burst
RNAS Royal Naval Air Station
RPV Remotely piloted vehicle aircraft (usually small) flown by human pilot not carried on board
SAC US Air Force Strategic Air Command
SAM Surface/air missile, fired from ground or ship to destroy aircraft
Semi-active Guidance system using passive e m receiver to home on to target illuminated by external radiation source
Sensor Device for detecting, and often recording e m and other radiation (optical, radar, IR, magnetic, sound)
Signature Precise characteristics of e m or other signal, such as radar echo from particular type of aircraft, which when analysed acts as kind of fingerprint
Slab tailplane Made in one piece without separate elevators (US stabilizer)
SLAR Side looking aircraft (or airborne) radar
Sonobuoy Acoustic device for detecting submerged submarines

by sound active sonobuoy sends out intense pulses of sound and listens for reflection passive sonobuoy listens for sound generated by submarine itself
Sqdn Squadron
SRS Strategic Reconnaissance Squadron
SAW Strategic Reconnaissance Wing
Stand-off missile ASM designed to hit target when launched from bomber outside range of enemy defences
STOL Short takeoff and landing fixed wing aircraft able to use small rough fields or clearings
Store Any object carried on external pylon
Strategic Concerned with attacks on enemy heartlands (cities, industries etc) as distinct from targets associated with surface battles
Subsonic Below the local speed of sound
Supersonic Above the local speed of sound
Sweep Angle at which wings are raked back (often measured at 1/2 chord ie one quarter of way back from leading edge)
TAC US Air Force Tactical Air Command
Tactical Concerned with local surface warfare on land or sea
Terrain-following radar Equipment able to guide attack aircraft safely at lowest possible height across enemy territory, missing hills or telephone poles even at night or in fog
TFS Tactical Fighter Squadron
TFW Tactical Fighter Wing
Turbofan Gas turbine jet engine with large fan (similar to multi-blade ducted propeller) generating half or more of total thrust rest coming from hot core jet that drives fan
Turbojet Gas-turbine jet engine of simplest kind generating single jet of hot gas
Turboshaft Gas-turbine engine delivering all power through shaft, for driving helicopter or other shaft drive vehicle
TV Television, which in combat aircraft is usually not just passive as in household receiver but active system taking own pictures
UHF Ultra high frequency e m radiation
USAF United States Air Force
USMC United States Marine Corps
USN United States Navy
VAQ US Navy tactical EW (jammer) squadron
Variable geometry Aircraft able to change shape especially by varying sweep angle of wings
VAW US Navy tactical electronic surveillance squadron
VF US Navy fighter squadron
VHF Very high frequency e m radiation (lower frequencies than UHF)
VMA US Marine Corps tactical attack squadron
VMF US Marine Corps fighter squadron
VTOL Vertical takeoff and landing
VTA Soviet Union Voenno transportnaya Aviatsiya, transport aviation
VVS Soviet Union Voenno vozdušnyye Sily combined air forces
Wire guidance Guidance by electrical signals transmitted from operator through fine wires unreeled from missile in flight

Aircraft Designations

The following are some of the designations used for modern military aircraft:

A (US, France, Sweden) attack
AC (US) attack aircraft based on transport
AEW (UK) airborne early warning
AH (US) attack helicopter
AH (UK) Army helicopter
AJ (Sweden) attack, secondary role fighter
AT (US) attack aircraft based on transport
AV (US) attack, V/STOL
B (US, UK, Soviet Union) bomber
C (US, Japan) transport
C (Spain) fighter
CC (UK) communications and VIP transport
CC (Canada) transport
CF (Canada) fighter
CH (US) transport helicopter
CH (Canada) helicopter
CP (Canada) patrol
CT (Canada) trainer
D (Soviet Union) long range
DC (US) director aircraft (of drones or RPVs) based on transport
DF (US) director aircraft based on fighter
E (US) electronic missions
E (France, Spain) trainer
EA (US) electronic aircraft based on attack aircraft
EB (US) electronic aircraft based on bomber
EC (US) electronic aircraft based

on transport
EF (US) electronic aircraft based on fighter
EKA (US) electronic and tanker aircraft based on attack aircraft
EP (US) electronic aircraft based on patrol aircraft
F (US, UK) fighter
F (Soviet Union) boosted augmented thrust
FB (US) literally 'fighter bomber', actually applied to a strategic bomber based on a tactical bomber
FG (UK) fighter, ground attack
FGA (UK) fighter ground attack (superseded)
FGR (UK) fighter ground attack reconnaissance
FRS (UK) fighter reconnaissance strike
FST (Mitsubishi) fighter supersonic trainer
G (Aeritalia) designed by Giuseppe Gabrielli
GA (UK) ground attack
GR (UK) ground attack reconnaissance
H (US) helicopter
HAR (UK) helicopter rescue
HAS (UK) helicopter anti-submarine
HC (UK) helicopter transport
HC (US) rescue or recovery aircraft based on transport (fixed wing)
HCC (UK) helicopter communications
HF (HAL, India) Hindustan, Fighter

HH (US) rescue or recovery helicopter
HJT (Hindustan, India) Hindustan Jet Trainer
HT (UK) helicopter trainer
HU (UK) helicopter, Commando assault
J (Sweden) fighter
JA (Sweden) fighter, secondary role attack
K (UK) tanker
KA (US) tanker based on attack aircraft
KB (US) tanker based on bomber
KC (US) tanker based on transport
L (Soviet Union) fitted with search radar
LC (US) transport with wheel/ski landing gear
LP (US) patrol aircraft with wheel/ski landing gear
LR (Japan) liaison reconnaissance
M (Soviet Union) modified
MB (Aermacchi) designed by Ermanno Bazzocchi
MR (UK) maritime reconnaissance
NB (US) bomber converted for special testing
NF In F 5 programme Netherlands (built by Netherlands Canada)
NKC (US) transport converted for special testing
O (US) observation (ie battlefield reconnaissance FAC)
OH (US, Japan) observation helicopter
OV (US) observation V/STOL
P (US, Japan) patrol measuring ocean surveillance
P (Soviet Union) interceptor
PM (Soviet Union) special

aerobatic version
PQM (US) pilotless RPV developed from lighter
PS (Japan) patrol, anti submarine
QB (US) remotely piloted bomber (possibly with safety pilot)
QF (US) remotely piloted fighter (possibly with safety pilot)
R (US, UK, Soviet Union) reconnaissance
RA (US) reconnaissance aircraft based on attack aircraft
RB (US) reconnaissance aircraft based on bomber
RC (US) reconnaissance or special mission aircraft based on transport
RF (US) reconnaissance aircraft based on fighter
RTF (US) reconnaissance aircraft based on dual fighter trainer
RP (US) mapping/survey aircraft based on patrol aircraft
RU (US) special mission aircraft based on utility aircraft
S (UK) strike
S (US) anti submarine
SF (Sweden) unarmed reconnaissance
SF In F 5 programme Spain
SF (Savoia Marchetti) designed by Stelio Frati
SH (US) anti submarine helicopter
SH (Sweden) armed ocean surveillance
Sk (Sweden) trainer
SP (US) anti submarine aircraft based on patrol aircraft
SPS (Soviet Union) fitted with blown flaps
SR (US) strategic reconnaissance
T (US, UK) trainer

T (Soviet Union) torpedo equipped
T (Spain) transport
TA (US) trainer based on attack aircraft
TE (US) trainer version of electronic aircraft
TF (US, Sweden) trainer version of fighter
TH (US) trainer helicopter
U (US) utility
U (Soviet Union, strictly the sound 'oo' represented by a character resembling a capital Y) trainer
UH (US) utility helicopter
US (US) utility (ie transport) aircraft based on anti submarine aircraft
UTI (Soviet Union) fighter trainer
VC (US) executive VIP transport
VH (US) executive VIP helicopter
WB (US) weather aircraft based on bomber
WC (US) weather aircraft based on transport
WP (US) weather aircraft based on patrol aircraft
WU (US) weather aircraft based on utility (hardly utility the U 2)
X (US) experimental research aircraft
XFV (US) experimental fighter V/STOL
YAH (US) service test attack helicopter
YC (US) service test transport
YF (US) service test fighter
YO (US) service test observation
YUH (US) service test utility helicopter

Hawker Siddeley
Andover 167 HS 748 Coast
guarder 167 167
Buccaneer 73 74 75 NA 39 73
S 1 2 2A 2B 73 73 74 74
S 2C S 2D 73 74 S 50 73 74
Gnat 22 22 FO141 Gnat F1 22
Gnat T1 22
Harrier 76 78 AV 8A (Matador)
76 76 77 78 AV 8B see
McDonnell Douglas GR 1 76
78 GR 3 76 76 78 78
Kestrel 78 P 1127 78 Sea
Harrier FRS 1 76 76 78
TAV BA 76 T 4 76
Hawk 234 234 P 1182 Hawk
T1 234
HS125 233 Dominant CC 1 233
CC 2 233 233 T1 233 233
Hunter 78 79 HO79 78 79
FGA 9 79 FGA 11 79 F1
78 F 2 79 F 4 79 F 5 79
F 6 79 F 51 79 F 56 79
F 74B 79 FR10 79 P 1067
78 T 7 79 T 6 79
Nimrod 139 AEW 140 140
HS 801 Nimrod MR 1 and 2
139 139 140 140 141 R 1
139 140
Sea Hawk 78 79 1 to 6 50 and
100 1 78 79 FB 3 79 FB 5
79 FGA 4 79 FGA 6 79 F 1
79 F 2 79 F 100 79 79
F 101 79 P 1040 76
Shackleton 138 AEW 2 138
138 MR 1 1A 138 MR 2
138 MR 3 138 MR 3 Series 3
138 T 4 138
Vulcan 113 113 B 1 113 B 1A
113 B 2 113 113 SR 2 113

Hawkeye see Grumman E 2
Haze see Mi 8
Hernemann Ed 62 84
Hercules see Lockheed C 130
HerkyBird see Lockheed C 130
HF 24 see HAL
Hind see M Mi 24
H p see Mil Mi 8
Hirundo see Agusta A109
HJT 16 see HAL
Holand see Netherlands
Honduras 58 168 198 232 236
246 248
Hong Kong 194 229 245
Hook see Mil Mi 6
Hoplite see Mil Mi 2
Hormone see Kamov Ka 25
Hornet see McDonnell Douglas F 18
Hound see Mil Mi 4
H P 80 see Handley Page
HS 125 see Hawker Siddeley
HS 748 see Hawker Siddeley
Huey family see Bell
HueyCobra see Bell 209
Hughes
AH 64 206 Model 77 206
YAH 64 206 206
OH 6 Cayuse 205 369HM 205
500M 205 Defender 205
NH 500M 205 OH 6A 205
205 RACA 200 205
TH 65 Osage 205 Model 269
205 Model 300 205 Model
300C 205 TH 65A 205 205
Hungary 33 34 102 155 210 211
227 236 245
Hunter see Hawker Siddeley
Hunting
Jet Provost 55
Percival Provost 55
Huskie see Kaman H 43

IA58 see FMA
IAI
Kfir 24 24 Kfir C 2 24 24
Noshor 24
IAI 201 Arava 168 168
Ice and 165
Ikeda Kenji 88
Il 28 see Ilyushin
Il 38 see Ilyushin
Il 76 see Ilyushin
Ilyushin
Il 28 Beagle 235 236 28R
235 28T 235 235 28U 235
235
Il 38 May 142 142
Il 76 Candid 169 169
Il 22 23 38 39 40 56 78 79
102 134 142 153 160 163
167 189 194 198 210 211
214 222 233 243 245 248
Indonesia 33 34 38 42 126 153
160 165 170 194 198 200
210 211 213 214 227 229
230 235 236 246 248 249
251 252
Intruder see Grumman A 6A
Iran 14 16 19 20 26 44 145
156 165 170 184 195 198
199 200 202 204 214 216
216 246
Iraq 10 33 34 36 38 55 78 102
126 153 155 170 194 210
211 219 227 235 216 245
Ireland 194 238 242
Iroquois see Bell Huey family
Iskra see VSK Meteor TS 11
Israel 12 24 26 30 59 84 120
156 158 168 170 186 188
194 195 200 202 204 213
214 216 238 245 246 247
248 250
Italy 54 80 91 135 137 152 163
170 198 200 202 204 205
213 214 226 242 244 248
252
Ivory Coast 155 194 197 246 248

J
Jaguar see SEPECAT
Jaguar M 61
Jainica 198 200 212 245 246
247

Japan 26 28 30 42 57 80 87
137 143 144 148 175 19H
200 202 203 205 213 214
230 232 236 249
Jastreb see Soko
Jet Provost see Hunting
JetRanger see Bell Kiowa
Jordan 44 58 80 160 170 194
218 241
Jurom VTI CIAR 93 Orao 82 82
K
Ka 20 see Kamov
Ka 25 see Kamov
Kaman
H 43 Huskie 207 HH 2C 207
HH 2D 207 HL2K 1 (UH 2A
Seasprite) 207 SH 2D 207
SH 2F 207 207
Kampuchea 33 62 163 164 200
238
Kamov
Ka 20 Harp 208
Ka 25 Hormone A 208 208
Kamov Nikolai 208
Kawasaki
C 1A 175 175
P 2J 143 143 GK 210 143
KC 135 see Boeing C 135
Kenya 44 45 55 78 160 162 194
198 240
Kfir see IAI
Khmer 210 248
King Air see Beech
Kiowa see Bell
Kran see HAL HJT 16
Korea North 33 34 38 40 102
210 211 235 236 243 245
Korea South 26 42 44 164 167
200 205 236 237 246 251
Kuwait 8 10 12 55 78 84 160
170 196 197 200

L
L 29 see Aero
L 39 see Aero
Labrador see Boeing Vertol H 46
Lansen see Saab
Laos 155 194 213 237 246 248
Lebanon 12 78 194 200 238 241
Lesotho 245
Liberia 194 202 245
Libya 10 12 36 44 128 170 195
198 211 236 242 248
Lightning see BAC
Lockheed
C 5A Galaxy 173 173
C 130 Hercules 170 172 AC
130H 170 170 172 C 130A
170 172 C 130B 172 172
C 130E 172 C 130H 171
172 172 C 130K (C 1) 172
172 DC 130 172 172
EC 130 172 HC 130 172
KC 130 171 172 LC 130
172 WC 130 172 YC 130
170
C 141 StarLifter 174 174
C 141A 174 174 C 141B
174 174 YC 141B 174
EC 121 Warning Star 189
C 121C 189 EC 121C to
EC 121G 179 EC 121H 179
179 EC 121I to EC 121T 179
NC 121K 179 RC 121G 179
F 104 Starfighter 80: CF 104
80 81 F 104A 80 F 104B to
F 80 F 104G 14 80 81
F 104J 80 F 104S 80 80
81 QF 104 80 RF 104 80
81 RTF 104 80 TF 104 80
XF 104 80
P 2 Neptune 144 P 2A to H
144 LP 2H 144 SP 2H 144
144 XP2V 1 144
P 3 Orion 145 CP 140 Aurora
145 EP 3A 145 P 3A B 145
P 3C 145 145 P 3F 145
145 RP 3D 145 WP 3A 145
QT 2 120 Q Star 120 X 268
120 YD 3A 120 120
S 3A Viking 146 146 147 148
148 US 3A 146 148
SR 71 Blackbird 122 A 11
122 SR 71A 122 122 SR
71B C 122 YF 12A 122 122
T 33 236 CL 30 SilverStar 236
T 1A 236 T 33A 236 236
RT 33 236
U 2 121 EPX 121 HASPU 2
121 U 2A 121 U 2B 121
121 U 2C 121 U 2CT 121
U 2D 121 U 2R 121
Lynx see Aerospatiale Westland

M
M 4 see Myasishchev
M 12 see Beriev
McDonnell Douglas
A 4 Skyhawk 84 A 4A to A 4S
84 A 4A 84 A 4B 84 A 4C
84 A 4E 84 A 4F 84 A 4H
84 A 4M Skyhawk II 84 85
A 4N 84 TA 4 84 TA 4J
84 85 TA 4S 84 XAUD 1 84
AV 8B 83 83 YAV 8B 83
C 9A Nightingale 176 176 C 9B
Skytrain II 176 176 VC 9C
176
C 133 Cargomaster 177 C 133A
177 C 133B 177 177
EA 3 Skywarrior 190 A 3B
190 A3D 1 2 2P 2Q
EA 3B 190 A3D 2T (TA
3B) 190 190 EKA 3B 190
190
EB 66 Destroyer 190 B 66 190
EB 66E 190 190 RB 66A B
C 190 WB 66D 190
F 4 Phantom II 26 28 26 27
28 32 EF 4 26 28 F 4A to
F 4S 26 27 28 28 F 4B 27
F 4E 27 F 4EF 28 F 4EJ
28 28 F 4M 27 FGR 2 27
28 FG 1 28 QF 4 26 28

RF 4 28 28 XF4H 1 26
F 15 Eagle 14 29 30 29 30
32 40 F 15A 29 30 30 31
TF 15A 29 30 30 YF 15 31
F 18 Hornet (Cobra) 20 32 32
F 101 Voodoo 25 25 CF 101
25 25 F 101 A B C 25
F 101F 25 RF 101 A B 25
RF 101C 25 25 RF 101 D F
25 RF 101G 25 25 RF
101H 25 25 TF 101F 25
XF 88 25
Model 260 133 133
YC 15 178 178
Maestro see Yakovlev Yak 28J
Magister see Potez (Aerospatiale
Mail see Beriev M 12
Malagasey Republic 194 198 245
246 248
Malawi 248
Malaysia 42 44 160 167 170
198 200 202 214 231 233
241
Mali 33 210 236 245 248
Mata 198 202
Mandrake see Yakovlev Yak 26
Mangrove see Yakovlev Yak 27
Martin
AM 1 Mauler 62
GD B 57 56 57 B57A to B57G
Night Intruder 57 B 57B 57
57 RB 57A D 57 RB 57F
57 57
XB 51 67
Marut see HAL HF 24
Matador see Hawker Siddeley
Harrier
Mauver see Martin AM 1
Mauretania 245 248 252
Maunibus 245
May see Ilyushin II 38
Maya see Aero L 29
MB 326 see Aeromach
MB 339 see Aeromach
MBB
BO 105 209 209 105C 209
105VBH 209 B0106 209 209
Memor see Beech T 34
Mescalero see Cessna 172
Mexico 168 194 198 200 202
230 236 237 245 248 252
Mi 2 see Mi
Mi 4 see Mi
Mi 6 see M
Mi 8 see M
Mi 10 see Mi
Mi 24 see Mi
Midge see Folland Fo 139
Mig 15 see Mikoyan Gurevich
Mig 15UTI see Mikoyan Gurevich
Mig 17 see Mikoyan Gurevich
Mig 19 see Mikoyan Gurevich
Mig 21 see Mikoyan Gurevich
Mig 23 see Mikoyan Gurevich
Mig 25 see Mikoyan Gurevich
Mig 27 see Mikoyan Gurevich
Mikoyan Gurevich
Mig 15UTI 236 236 Mig 15
15bis Fagot (Lim 2 S 103)
236 Mig 15UTI Mongol
(SBLM 1 CS 102 F 2) 236
236
Mig 17 'Fresco' 33 33 17P
33 17F (Lim 5P 5M S 104
F 4) 33 17PF 33 17PFU
33
Mig 19 Farmer 34 36 195
34 36 195F (Lim 7 S 105
F 6) 34 34 35 36 36 102
19PF 34 36 36 19PM 34
35 36 36 1 350 34
Mig 21 'Fishbed' 38 39 40
21bis 38 39 40 21F (S
107) 38 38 39 39 21FA
38 21F2 (HAL Type 77) 38
40 21M 38 40 21MF 38
38 40 21PFM 38 40 40
21PFMA 38 40 21PFS 38
21R 38 40 21SMT 38 39
40 21U 38 40 21UM 38
40 21US 38 40 E 5 38
F B 38 39 41
Mig 23 Flogger 36 37 37
23S 36 37 37 23U 36 37
37
Mig 25 Foxbat 40 40 41
25R 40
25U 40
Mig 27 Flogger D and F 86
86
Mil
Mil Mi 1 210
Mil (WSK Swidnik) Mi 2 (V 2)
Hoplite 210 210
Mi 4 Hound 210 210
Mi 6 Hook 211 211
Mi 8 Hip Haze 211 211
Mi 10 Harke 211 Mi 10K 211
Mi 24 Hind A 212 212
Hind B 212 Hind D 212
Milan see Dassault Breguet
Mirage II see Dassault Breguet
Mirage III see Dassault Breguet
Mirage IVA see Dassault Breguet
Mirage 2000 see Dassault Breguet
Mirage F2 see Dassault Breguet
Missiles
Acrid 40 41 48
ALCM 110 110 115
Aikuli 33 35 36 47 47
Anab 47 47 48 48
Aper 36
Aphid 36
ARM 2H
AS 6 114 114
AS 11 196 218 218
AS 12 114 115 137 139 19F
218 223
AS 20 60
AS 30 54
Ash 49 49 51
Atoll 36 38 38 39 40 51
Bryant 24
Blue Street 113 113
Condor 72

Exocet 195 195
Falcon 17 18 25 46 96 98
Firebreak B 8
Harm 25
HOT 196 198 209 209
Hound Dog 110
Jumbo 93
Kangaroo 127 127
Kennel 126
Kerry 86 104
Kipper 126
Kitchen 128
Martel 74 135 139
Matra 550 Magic 9 10 99 100
Matra R530 10 10 12
Matra Super 530 9 10
Maverick 27 65 105 172
Nord SS 11M 134
Petrel 136
Phoenix 19 19 20 20
Rb 04C 125
Rb 04E 96 98
Rb 05A 96 98
Red Top 8
S 24 40
Sea Killer 201 214
Sea Skua 223
Shrike 28
Sidewinder 10 12 14 15 19
19 26 30 32 33 42 42 44
45 46 78 80 80 87 91 96
98 125 207 232 251
Skyflash 91 96
Sparrow 19 19 29 30 32 80
80 95 207
SRAM 68 68 110 110 115
TOW 196 198 199 199 205
206 209
Mission Master see GAF
Mitsubishi
F 1 87 88 F 1 87 88 89
FST 2 Kai 87 88 T 2 B7
87 88 T 2A B7 88 88 89
XT 2 87
MU 2 249 2C(LR 1) 249 2E
249 2K 249 249 2J 249
2L 249 249
Model 260 see McDonnell Douglas
Model 500M see see Hughes DH 6
Model 698 see Grumman
Model 1041 see Boeing
Mohawk see Grumman OV 1
Mongol see Mikoyan Gurevich
Mig 15UTI
Mongol see Mikoyan Gurevich
Mig 21U
Mongolia 155 210 236 243 245
Morocco 10 33 44 170 197 198
200 202 230 236 237 238
239 242 247 248
Moss see Tupolev Tu 126
MRCA see Panavia Tornado
MU 2 see Mitsubishi
Muskeeter see Beech
Myasishchev
M 4 Bison A B C 123 123

N
Nepal 167 197 252
Neptune see Lockheed P 2
Nesher see IAI
Netherlands 12 14 16 44 80 135
144 165 194 200 209
213 218 222 231 236
New Zealand 55 56 84 145 167
170 198 200 218 229 248
Nicaragua 168 205 213 236 237
248
Niger 248
Nigeria 33 38 165 170 197 209
227 235 236 241 245 247
248
Nightingale see McDonnell Douglas
C 9
Night Intruder see Martin B 57G
Nimrod see Hawker Siddeley
North American
F 86 Sabre 42 42 AF 1E 42
CA 26 (Avon Sabre) 42 CL
13 (Drenda Sabre) 42 F 86A
to E 42 F86F 42 42 F 86G
to L 42 XJF 1 42 XP 86 42
YF 86D 42
F 100 Super Sabre 90 DF 100F
90 F 100A B 90 F 100C 90
90 F 100D 90 90 F 100E
90 F 100F 90 90 YF 100
31
T 28 Trojan Fennec 237 T 28A
237 B 237 C 237 T 28D
237 237 T 28R Nomad 237
Nomad 237 YAT 28E 237
Northrop
A 9A 66
F 5 Freedom Fighter 43 45
F 5A 43 43 45 45 F 5B E
F 43 44 45 45 CF 5A D
43 43 NF 5A B 43 45 45
N 156F 44 RF 5A E G 43
SF 5A B 43
F 5E Tiger II 43 43 44 44 45
F 17 32 YF 17 14 32 32
P 530 14
F 18 Hornet 20 32 32
F 18 Cobra 32
T 38 Talon 44 237 T 38A 237
237 XT 38 44
Norway 12 14 16 44 80 145
170 200 214 222 247 248

O
O 2 see Cessna
OH 6 see Hughes
Oman 55 78 99 100 160 200
202 245 248 252
Orao see Jurom VTI CIAR 93
Orion see Lockheed P 3
Orion Hughes TH 5h
Orion Grumman
OV 10 see Rockwell International
P
P 2 see Lockheed
P 3 see Lockheed
Pakiet see Fairchild C 119

Pakistan 12 34 42 44 57 58 R
105 135 145 170 194 197
198 205 211 214 222 236
236 241 248
Panama 57 200 245 246 247 248
Panav
Tornado 91 92 Model 200
MRCA 91 AD 91 S 91 T 91
Papua New Guinea 248
Paraguay 198 246 247 248
Peacemaker see Fairchild AJ 21
Percival Provost see Hunting
Peru 12 42 44 56 58 78 102
104 137 162 165 170 194
198 200 211 230 232 246
246 247 248 250
Petter Teddy 8 22
Phantom II see McDonnell Douglas
F 4
Philippines 42 44 165 170 200
205 209 211 232 236 237
242 245 248 249
Piatas
Porter 250 PC 6 A 250 PC 6 B
Turbo Porter 250 250 AJ 23
Peacemaker 250 250
Poland 33 34 36 86 102 104
153 155 210 211 235 236
243 245 248
Porter see Piatas
Portugal 42 54 58 144 156 160
194 197 236 248
Potez (Aerospatiale)
Magister 238 CM 170 238 238
CM 170 2 238 CM 175 238
Provider see Fairchild C 123
Prowler see Grumman EA 6A
Pucara see FMA IA58
Puma see Aerospatiale Westland

Q
Qater 78 196 222 245
QT 2 see Lockheed
Queen Air see Beech
R
RA 5C see Rockwell
Republic (Fairchild)
F 105 Thunderchief 94 F 105B
94 F 105D 94 94 F 105F
Thud 94 94 F 105G 94
94 YF 105A 94
Rhodesia 33 34 38 82 102 155
194 210 211 227 235 236
243 245
Rockets
Burrup 50
Genie 18 25
FFAR 17 42 50 206
HVAR 135
SEP 844 12
SNEB 68mm 23 59 78
Sora 56
Zuni 50 137
Rockwell
RA 5C Vigilante 124 124 A 5A
124 A 5B 124 YA3J 1 124
Rockwell International
B 1 115 116 115 117
XFV 12 95 XFFV 12A 95 95
T 2 Buckeye 239 T 2A(T2J 1)
239 T 2B 239 T 2C 239
239 T 2D 239 239 T 2E 239
T 39 Sabreliner 238 CT 39E G
238 Sea Sabre 75 238
T 39A 238 238 T 39B D F
239
OV 10 Bronco 251 OV 10A
251 251 OV 10B 251 OV
10B(2) 251 251 OV 10D
251 251 YOV 10D 251 251
Romania 33 34 38 82 102 155
194 210 211 227 235 236
243 245
Rwanda 194 238 244 245 248

S
707 see Boeing
717 see Boeing
737 see Boeing
747 see Boeing
S 2 see Grumman
S 3A see Lockheed
S 58 see Sikorsky
S 61 see Sikorsky
S 64 see Sikorsky
S 65 see Sikorsky
S 70 see Sikorsky
Saab
32 Lansen 125 A32A 125 125
J32B 125 125 S32C 125
125
35 Draken 46 35XS 46 J35A
46 46 J35B D 46 J35F 46
46 RF 35 46 S35JE 46
SK35C 46 TF 35XD 46 46
37 Viggen 96 98 AJ37 96
96 97 98 98 JA37 96 96
SF37 96 SH37 96 SK37 96
96 98
105 240 Sk60A 240 240
SK60B 240 SK60C 240 240
Saab 1050 240 Saab 105G
240
Supporter 241 241 MF1 17
241 T 17 241 241
Sabre see North American F 86
Saberliner see Rockwell International
T 39
Salvador 168 230 238 248
Saudi Arabia 8 10 12 26 30 42
44 55 156 164 170 194
200 202 222 230 242
252
SGS 2 32 12
Scooter see McDonnell Douglas
A 4
Scottish Aviation
SA 3 Bulldog 241 241 Series
100 241 Series 120 241
Series 200 241 SK 61 241
11 241
Scout see Westland
Scout see Sikorsky S 64
Sea Cobra see Bell 211

S
S 2 see Cessna
OH 6 see Hughes
Oman 55 78 99 100 160 200
202 245 248 252
Orao see Jurom VTI CIAR 93
Orion see Lockheed P 3
Orion Hughes TH 5h
Orion Grumman
OV 10 see Rockwell International
P
P 2 see Lockheed
P 3 see Lockheed
Pakiet see Fairchild C 119

S
707 see Boeing
717 see Boeing
737 see Boeing
747 see Boeing
S 2 see Grumman
S 3A see Lockheed
S 58 see Sikorsky
S 61 see Sikorsky
S 64 see Sikorsky
S 65 see Sikorsky
S 70 see Sikorsky
Saab
32 Lansen 125 A32A 125 125
J32B 125 125 S32C 125
125
35 Draken 46 35XS 46 J35A
46 46 J35B D 46 J35F 46
46 RF 35 46 S35JE 46
SK35C 46 TF 35XD 46 46
37 Viggen 96 98 AJ37 96
96 97 98 98 JA37 96 96
SF37 96 SH37 96 SK37 96
96 98
105 240 Sk60A 240 240
SK60B 240 SK60C 240 240
Saab 1050 240 Saab 105G
240
Supporter 241 241 MF1 17
241 T 17 241 241
Sabre see North American F 86
Saberliner see Rockwell International
T 39
Salvador 168 230 238 248
Saudi Arabia 8 10 12 26 30 42
44 55 156 164 170 194
200 202 222 230 242
252
SGS 2 32 12
Scooter see McDonnell Douglas
A 4
Scottish Aviation
SA 3 Bulldog 241 241 Series
100 241 Series 120 241
Series 200 241 SK 61 241
11 241
Scout see Westland
Scout see Sikorsky S 64
Sea Cobra see Bell 211

S
707 see Boeing
717 see Boeing
737 see Boeing
747 see Boeing
S 2 see Grumman
S 3A see Lockheed
S 58 see Sikorsky
S 61 see Sikorsky
S 64 see Sikorsky
S 65 see Sikorsky
S 70 see Sikorsky
Saab
32 Lansen 125 A32A 125 125
J32B 125 125 S32C 125
125
35 Draken 46 35XS 46 J35A
46 46 J35B D 46 J35F 46
46 RF 35 46 S35JE 46
SK35C 46 TF 35XD 46 46
37 Viggen 96 98 AJ37 96
96 97 98 98 JA37 96 96
SF37 96 SH37 96 SK37 96
96 98
105 240 Sk60A 240 240
SK60B 240 SK60C 240 240
Saab 1050 240 Saab 105G
240
Supporter 241 241 MF1 17
241 T 17 241 241
Sabre see North American F 86
Saberliner see Rockwell International
T 39
Salvador 168 230 238 248
Saudi Arabia 8 10 12 26 30 42
44 55 156 164 170 194
200 202 222 230 242
252
SGS 2 32 12
Scooter see McDonnell Douglas
A 4
Scottish Aviation
SA 3 Bulldog 241 241 Series
100 241 Series 120 241
Series 200 241 SK 61 241
11 241
Scout see Westland
Scout see Sikorsky S 64
Sea Cobra see Bell 211

S
707 see Boeing
717 see Boeing
737 see Boeing
747 see Boeing
S 2 see Grumman
S 3A see Lockheed
S 58 see Sikorsky
S 61 see Sikorsky
S 64 see Sikorsky
S 65 see Sikorsky
S 70 see Sikorsky
Saab
32 Lansen 125 A32A 125 125
J32B 125 125 S32C 125
125
35 Draken 46 35XS 46 J35A
46 46 J35B D 46 J35F 46
46 RF 35 46 S35JE 46
SK35C 46 TF 35XD 46 46
37 Viggen 96 98 AJ37 96
96 97 98 98 JA37 96 96
SF37 96 SH

Hawker Siddeley
Hawker
Hawk see Hawker Siddeley
Horse see Sikorsky S 58
Sea King see Sikorsky S 61
Westland
Sea Knight see Boeing Vertol H 46
Seaspire see Kaman H 43
Sea Stallion see Sikorsky S 65
Senegal 194 196 248
SEPECAT
Jaguar 99 100 A 99 99 100
100 E 99 99 GR 1 99 99
100 100 101 International
99 100 T 2 94
SF 260 see SIAI Marchetti
Shackleton see Hawker Siddeley
Shadow see Fairchild C 119
Shenyang
F 9 Fantan A 102 103 102
103
Shenyang
SS 2 (PS 1) 148 149 149
PX S 148 SS 2A (US 1
PS 1 Ka) 148 149 149
Short
Skyvan 3M 252 252
S 1 A 1
SF 260 242 260M 242 242
260W 242 260WP 242
260SW 242
SM 1019 252 SM 1019E1 252
252
Serra Leone 205
Sikorsky
R 4 214
S 58 213 213 CH 34A C
Choctaw 213 H 34 213
H55 1 213 219 S 58T 213
SH 34G 213 SH 34J Seabat
213 LH 34D E Seahorse 213
213 VH 34D 213
S 61 214 222 HH 3A 214
RH 3A 214 S 61R (CH 3E)
214 214 SH 3A 214 SH 30
Sea King 214 214 SH 3H
214
S 64 215 215 CH 54A Tarhe
215 215 CH 5HB 215
CH 53A Sea Stallion 216 216
CH 53D 216 216 CH 53E
216 RH53D 216 S 65 216
UH 60A 217 217 S 70 217
YUH 60A 217 217
Singapore 26 44 55 78 84 170
194 229 232 242 252
Sikorsky Igor, 214
Skyhawk see McDonnell Douglas
Skyraider see Douglas A 1
Skyserpent see Dornier Do 28D
Skytrain II see McDonnell Douglas
C 9
Skyvan see Short
Skywarrior see McDonnell Douglas
SM 1019 see SIAI Marchetti
Soko
Galeb 242 G2 A 242 242
Jastreb 242 J 1 242 TJ 1
242

Soma 33 38 155 200 211 235
236 245 247 248
South Africa 10 12 42 56 73
138 170 179 194 195 197
218 226 233 244 246 248
South Yemen 33 38 55 155 211
235 236 243 252
Soviet Union 33 34 36 37 38 40
47 49 51 86 102 104 105
114 123 126 127 128 129
132 142 153 154 155 169
191 209 210 211 212 227
235 236 243 245 248
Spain 10 12 26 44 78 80 145
160 170 194 198 199 200
202 204 205 214 230 231
236 248
SR 71 see Lockheed
Sri Lanka 33 55 198 202 236 248
SS 2 see Shin Meiwa
Starfighter see Lockheed F 104
Starliner see Lockheed C 141
Stratocaster see BAC
Stratofortress see Boeing B 52
Stratofreighter see Boeing C 97
Stratolifter see Boeing C 135
Stratotanker see Boeing KC 135
Su 7 see Sukhoi
Su 9 see Sukhoi
Su 11 see Sukhoi
Su 15 see Sukhoi
Su 17 see Sukhoi
Su 20 see Sukhoi
Su 22 see Sukhoi
Sudan 33 38 55 153 155 211
245 250
Sukhoi
Su 7 Fitter 102 103 Su 7BM
102 103 Su 7BMK 102 102
Su 7U 'Moujik', 102 102
Su 9 Fishpot B', 47 47 SU
9U 47
Su 11 'Fishpot C', 47 47
Su 15 Flagon', 48 48 Su 15U
48
Su 17 'Fitter C', 104
Su 19 'Fencer', 105 105
Su 20 104 104
Su 22 104
Sukhoi Pavel O 47
Sundowner see Beech
Super Frelon see Aerospatiale
Super King Air see Beech
Super Mirage see Dassault Breguet
Super Mystere see Dassault
Super Puma see Aerospatiale
Super Sabre see North American
F 100
Supporter see Saab
Sweden 46 56 96 125 170 194
200 202 203 240 241 248
Switzerland 12 44 78 194
Syria 33 36 38 86 102 132 170
195 208 210 211 227 235
236 243 245 248

T
T 2 see Rockwell International
T 24A see Beech

T 28 see North American
T 33 see Lockheed
T 34B see Beech
T 38 see Northrop
T 39 see Rockwell International
T 41 see Cessna 172
T 43A see Boeing
Taiwan 25 42 44 57 80 90 121
137 156 163 164 198 200
205 236 237 248
Taily ho see Northrop T 38
Talon see Northrop T 38
Tanzania 33 34 38 160 198 202
236 245 246
Tchaika see Beriev M 12
Teledyne Ryan
BQM 34B Firebee 172
TH 55 see Hughes
Thailand 42 44 58 137 164 167
200 202 204 213 229 230
232 236 237 242 245 246
248 250 251 252
Thud see Republic (Fairchild)
F 105
Thunderchief see Republic (Fairchild)
F 105
Tiger II see Northrop F 5E
Togo 162 194 197 226 238 248
Tomcat see Grumman F 14
Tornado see Panavia
TOW Cobra see Bell 209
Tracer see Grumman S 2
Tracker see Grumman S 2
Transal
C 160 179 179 C 160D 179
179 C 160F 179 179
C 1602 179
Trojan see North American T 28
TS 11 see WSK Mielec
Tu 16 see Tupolev
Tu 20 see Tupolev
Tu 22 see Tupolev
Tu 24 see Tupolev Tu 126
Tu 26 see Tupolev
Tu 95 see Tupolev Tu 20
Tu 105 see Tupolev Tu 22
Tu 126 see Tupolev
Tunisia 42 44 152 194 197 226
242 245
Tupolev
Tu 16 Badger A to G', 126 126
Tu 88 126
Tu 20 Bear A to F', 127 127
Tu 22 Blinder A to D', 47, 128
128 132 Tu 105 128
Tu 24 see Tu 126
Tu 26 'Backfire A and B', 114
114
Tu 28P 49 49
Tu 88 see Tu 16
Tu 95 see Tu 20
Tu 105 see Tu-22
Tu 126 Moss', 126 126
Turbo Mentor see Beech F34C
Turkey 14 16 17 26 44 58 80
90 137 170 179 198 200
202 230 232 236 245 246
247 248
Tutor see Canadair CL 41

Twin Otter see DH Canada DHC 6
U
U 2 see Lockheed
U 3 see Cessna
Uganda 33 34 38 160 200 202
218 227 236 238 248
UH 1 see Bell 'Huey' family
UH 60A see Sikorsky
United Arab Emirates 200 202
United Kingdom 8 22 26 28 55
73 78 91 99 100 135 138
139 166 167 170 172 196
197 198 218 219 222 223
233 234 241
United States 14 16 17 20 25 26
28 30 32 42 44 50 57 58
62 64 68 70-72 78 80 83
84 90 94 95 105 110 115
120 121 122 124 133 137
144 145 146 156 158 159
162 163 164 170 173 174
176 177 178 184 186 189
190 199 200 202 203 204
205 206 207 214 216 217
229 230 232 237 238 239
246 247 250 251
Uruguay 137 165 200 230 236
248
V
Vantour see Dassault
Venezuela 12 42 55 56 137 160
164 170 194 200 202 229
230 239 248 251
Victor
Airtourer 229
Victor see Handley Page H P 80
Vietnam 33 34 38 44 57 58 62
102 155 163 164 170 210
211 235 236 237 245 246
248
Viggen see Saab
Vigilante see Rockwell RA 5C
Viking see Lockheed S 3A
Voodoo see McDonnell Douglas
F 101
Vought
A 7 Corsair II 105 106 A 7A
105 106 106 A 7B C 105
106 A 7D 105 105 106
A 7E 105 106 106 TA 7C
105 106 YA 7H 106
F 8 Crusader 50 50 DF 8 50
F 8A to J 50 FR 8 50
QF 8 50 XF 8U1 50
Voyageur see Boeing Vertol H 46
VTI/CIAR 93 see Juroim
Vulcan see Hawker Siddeley
Vultur see Breguet Br 960
W
Warning Star see Lockheed EC 121
Wasp see Westland
Wessex see Westland
Westland
Commando 220 220 221 222
MK1, 220 MK2 220 221
Gazelle, 196 196 AH 1 (SA

341B), 196 196 HCC 4 (SA
341E), 196 HT 2 (SA 341C)
196 HT 3 (SA 341D) 196
196
Lynx 222 223 2(FN) 222 223
AH 1 222 223 223 HAS 2
222 223 223 HT 3 222 223
Puma 197 197 HC 1 197
Scout, 218 218 AH 1 218 218
Sea King 200 220 221 HAS 1
222 222 H
HAR 3 222 MK 41 50 220
MK41 221 MK43 221
MK45 221 MK50 221
Wasp 218 HAS 1 218 218
Wessex 219 219 CC 4 219
HAS 1 219 HAS 3 'Carnel',
219 219 HAS 31, 219 219
HC 2 219 219 HU 5 219 219
WSK Mielec
TS 11 Iskra 243 243 Series 100
243 Series 200 243 243

X
X 26 see Lockheed
X 51 see Martin
XFV 12 see Rockwell International
Y
Yak 18 see Yakovlev
Yak 26 see Yakovlev
Yak 28P see Yakovlev
Yak 36 see Yakovlev
Yakovlev
Yak 18 243 243 18A 243 243
18P 243 243 18PM 243
18T, 243, 18U 243
Yak 26 'Mandrake' 129 129
Yak 27P 'Mangrove' 129 129
Yak 28P 'Firebar', 51, 51 Yak
28R, 51, Yak 28U 'Maestro',
51
Yak 36 'Forger A', 107, 107
Yak 36U 'Forger B', 106 107
Freehand', 107
YC-14 see Boeing
YC 15 see McDonnell Douglas
Yemen Arab 33 38 200 210 235
248
YF 12A see Lockheed SR 71
YO 3 see Lockheed
Yugoslavia 22 38 42 82 153 194
196 200 210 211 236 242
248
YUH 61A see Boeing Vertol
Z
Zaire 12 160 162 170 194 197
226 237 242 245 246 248
Zambia 160 162 167 194 200
211 226 242 245 247 248

Picture credits

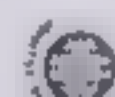
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British Aerospace Corporation BAC
All photographs credited USAF are "official United States
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Jacket front: General Dynamics End papers: MOD Half-title (page 1)
Saab-Scania Full-title (pages 2-3) Lockheed Credits (pages 4-5) Mitsubishi
Page 6-7: Saab Scania 8: MOD 9-13: Dassault Breguet 14-16: General
Dynamics 17: USAF 18: top, via H R Muir, bottom, Jay Miller, via Roger Wright
collection 18-21: Grumman 22: top, Brian M Service, via *Flight International*
library, bottom, William Green collection 23: William Green collection 24: Israeli
Aircraft Industries 25: RCAF 26-27: USAF 28: Top, Roger Wright, bottom
Picciardi Aircraft Slides, 434 Arbutus Avenue, Horsham, P.A., 19044, USA 29-30:
McDonnell Douglas 31: left, Popperfoto, right, McDonnell Douglas 32: McDonnell
Douglas 33, left, Novosti, right, Bill Gunston collection 34-35: Pakistan AF
36-37: *Flight International* library 39: William Green collection 40: top, E and
TV Films, bottom, William Green collection 41: *Flight International* library
42: Japanese ASDF 43: USAF 44: US Navy 45: USAF 46: Saab Scania
47: top, E and TV Films, bottom, William Green collection 48: top and centre
Novosti bottom, Tass 49: E and TV Films 50: LVT Aerospace 51: William Green
collection 52-53: McDonnell Douglas 54: *Flight International* library 55: BAC
56: MOD 57: Martin/General Dynamics 58: Cessna 60: Champlong Arepi
Agency 61: Dassault Breguet 62: USAF 63: top, William Green collection
bottom, J W R Taylor collection 64: USAF 65, left USAF, right Fairchild
66: Fairchild 67: General Dynamics 68: top Grumman via Bill Gunston collection
centre and bottom, General Dynamics 70-71: US Navy 72: top H Levy via
Flight International library others, US Navy 73: top and centre MOD bottom
Hawker Siddeley Aviation 75: MOD 76: Hawker Siddeley Aviation 77: USMC
78: MOD 79: centre left and right Hawker Siddeley Aviation bottom, William
Green collection 80 81: top, Lockheed bottom left, *Flight International* library
bottom right USAF 82: J W R Taylor collection 83: Bill Gunston collection
84 85: McDonnell Douglas 86: top Tass, bottom *Flight International* library
87: Mitsubishi 88 89: Roger Wright 90: top USAF bottom Turkish AF 91:
Panavia 92 93: BAC 94: top via H R Muir bottom Photo 95: Rockwell
97 98: Saab Scania 99 101: BAC 102: E and TV Films 103: top *Flugreview &*
Flugwelt bottom left Interinfo bottom right E and TV Films 104: Interinfo
105: USAF 106: LVT Aerospace 107: MOD 108 109: Boeing 110: top

Boeing, centre, USAF, bottom, Bill Gunston collection 111: top, USAF, bottom,
Photo, 112: Dassault Breguet 113: Hawker Siddeley Aviation 114: J W R
Taylor collection 115-117: Rockwell 118-119: Lockheed 120: top, Grumman
bottom, William Green collection 121: top, USAF, centre and bottom, via H R
Muir 122: top, D Jenkins, via H R Muir, bottom, USAF 123: top, MOD, bottom,
E and TV Films 124: US Navy 125: Saab-Scania 126: MOD 127: top, E and
TV Films, bottom, MOD 128: E and TV Films 129: top, William Green collection,
bottom E and TV Films 130-131: Lockheed 132: top, J W R Taylor collection,
bottom Tass 133: top McDonnell Douglas bottom US Navy 134: Champlong
Arepi Agency 135: Breguet 136: RCAF 137: Canadian Armed Forces 138:
Hawker Siddeley Aviation 139-141: MOD 143-144: Roger Wright 145-148:
Lockheed 149: top, Mitsuo Shibata, bottom, Roger Wright 150: MOD 152:
Aeritalia 153: E and TV Films 154: J W R Taylor collection 155: William Green
collection 156: top and below, Boeing, centre, RCAF, bottom, USAF 157: top,
Boeing, bottom, Roger Wright 158: Roger Wright 159: Boeing 160: left, CASA,
right above and below, de Havilland Aircraft Co 161-162: de Havilland Aircraft Co
163-164: USAF 165: Fokker VFW 166: MOD 167: Hawker Siddeley Aviation
168: Israel Aircraft Industries 169: top, US Navy; bottom, Novosti 171: via
H R Muir 172: top, USAF, bottom, MOD 173: top, USAF, bottom, Rolls Royce
174: top, USAF, bottom, Roger Wright 175: top, Kawasaki; bottom, Roger Wright
176-178: McDonnell Douglas 179: top, Rolls-Royce, bottom, Transal 180-181:
Grumman 182-185: Boeing 187: Grumman 188: top, Brian Mackenzie Service,
via *Aeroplane Monthly*, bottom, Grumman 189: Aviation Photo News 190: top
USAF, bottom, Roger Wright 191: E and TV Films 192-193: MOD 194: top
Aerospatiale, bottom, Rolls Royce 195: Aerospatiale 196-197: Rolls Royce
198: top, Agusta bottom, MOD 199: Bell 200: left, Agusta, right Swedish Army
201: top, US Army, bottom left, Sistel, right centre, US Navy; bottom, US Army
202: top, Canadian Armed Forces, bottom, Bell 203: left top and bottom, Boeing
Vertol top right, Kawasaki bottom right, Rolls Royce 204: Boeing Vertol 205-207:
Hughes 208: top, William Green collection, bottom, MOD 209: MBB 211:
Finland Defence Department 213: Sikorsky 214: top USAF, centre, Sistel
bottom, US Navy 215: US Army 216: top, David Windsor bottom, USMC
215, Sikorsky 218-219, MOD 220: Westland 222: left and top, Westland
bottom right, Bill Gunston collection 223: MOD 224-225: Japanese ASDF
226, top, Rolls Royce bottom, Brazilian AF 227: Aero 228: top, Dassault
Breguet bottom, Dornier 229: top, NZ Aerospace Industries, bottom Boeing
230, Beech 231: RCAF 232: top, Cessna bottom, Roger Wright 233: top
Hawker Siddeley Aviation bottom, Industan Aerospace 234: top, Hawker
Siddeley Aviation bottom, Industrial Photo Laboratories 235: China News Agency
236, top, Roger Wright, bottom, Finland Defence Department 237: top USAF
bottom, via H R Muir 238: top Fouga et cie bottom, USAF 239: Rockwell
240: Saab Scania 241: top, Saab Scania, bottom MOD 242: top *Flight*
International library bottom, "Soko" Metalopreradivacha Industrija 243: top
Flight International library bottom Camera Press 244: Aeritalia 245: Britten
Norman 246: top, USAF bottom Cessna 247: top, de Havilland Aircraft Co
bottom, Dornier 248: Brazilian AF 249: Mitsubishi 250: top Pilatus bottom,
Fairchild 251: top Rockwell, bottom and centre USAF 252: top Short Bros
and Harland bottom SIAI Marchetti







The Author

A former World War II RAF pilot and flying instructor, Bill Gunston has spent most of his working life accumulating the wide-ranging and detailed information included in *The Illustrated Encyclopedia of the World's Modern Military Aircraft*, a book which no author is better qualified to write than he. Since leaving the service in 1948 he has acted as an adviser to several major aviation companies and become one of the most respected of authors on scientific and aviation subjects, a frequent broadcaster, author of numerous books, including the Salamander title, *The Encyclopedia of the World's Combat Aircraft*, and a regular contributor to leading international aviation and defense periodicals. He is a former technical editor of *Flight International* and technology editor of *Science Journal*.

Amongst the many authoritative journals for which he has carried out assignments are *Battle*, *Aeroplane Monthly*, *Aircraft* (Australia); *Aviation Magazine* (France), *Aerospace International*, *Aircraft Production*, *Flying World Airmews* and the *Journal of the Royal United Services Institute*. He is an assistant compiler of *Jane's All the World's Aircraft Annual* and has contributed to *Brassey's Annual* and *Defence Yearbook*, *Aircraft Annual* and *Salamander's Soviet War Machine* and *The Encyclopedia of Air Warfare*.

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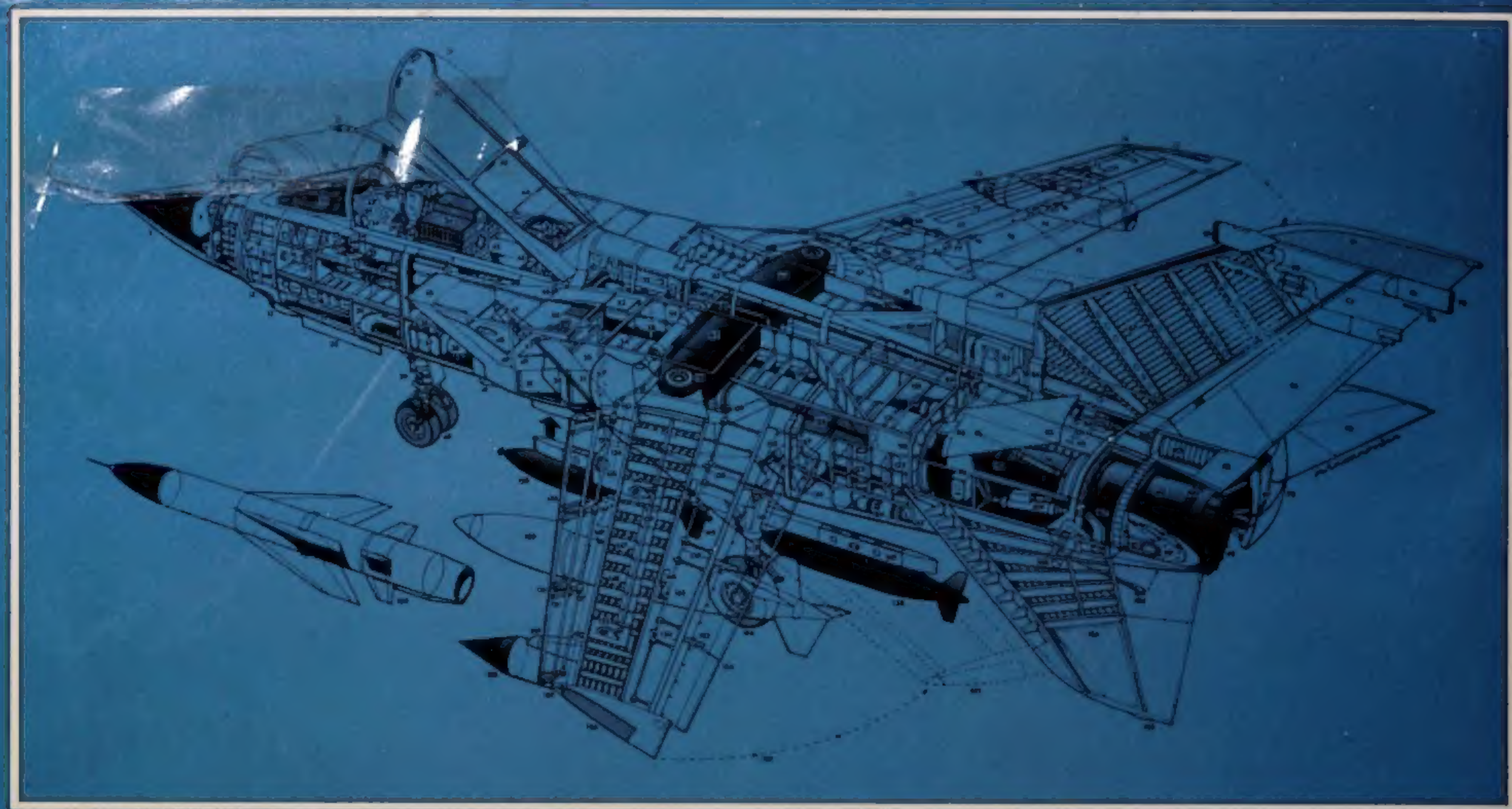
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22